



**MARITIME AND PORT AUTHORITY OF  
SINGAPORE  
SHIPPING CIRCULAR TO SHIPOWNERS  
NO. 14 OF 2011**

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18 JULY, 2011

**Applicable to:** This circular is for the attention of ship owners, managers, operators, agents, masters, crew members and surveyors

**RESOLUTIONS ADOPTED BY THE 89TH SESSION OF THE MARITIME  
SAFETY COMMITTEE (MSC 89) OF IMO**

Refer to Shipping circular no. 15 of 2011 for *Guidance on shipboard operational matters: Circulars approved by the 89th Session of the Maritime Safety Committee (MSC 89) of IMO*

1. This circular informs the Shipping Community of the resolutions adopted by MSC 89, held from 11–20 May 2011 and urges the Community to prepare for the implementation of these resolutions.
2. The resolutions of interest to the shipping community are as follows:
  - a. Amendments to SOLAS (MSC.317(89));
  - b. Amendments to the IMSBC Code (MSC.318(89));
  - c. Amendments to the IS Code, 2008 (MSC.319(89));
  - d. Amendments to the LSA Code (MSC.320(89)); and
  - e. Amendments to the Revised Recommendation on Testing of LSA (Resolution MSC.81(70)) (MSC.321(89) and MSC.323(89));
  - f. Implementation of Best Management Practice (BMP) Guidance (MSC.324(89))

In addition, various Unified Interpretations (UI) relating to provisions in SOLAS and other instruments were approved.

3. MSC 89 also approved the following draft Assembly resolutions of interest to the shipping community, which will be adopted by the 27<sup>th</sup> session of the Assembly (A27) in Nov 2011:

- a. the draft *Revised Recommendations for entering enclosed spaces aboard ships*;
- b. the new draft *International Code on Enhanced Programme of Inspections during Surveys of Bulk Carriers and Oil Tankers, 2011* (2011 ESP Code);
- c. the draft *Code of Safe Practice for Ships Carrying Timber Deck Cargoes, 2011* (2011 TDC Code), revising and updating the original code adopted in 1973;
- d. the draft IMO/World Meteorological Organization (WMO) *Worldwide Met-ocean information and warning service guidance* document;
- e. the draft *Revised procedures for Port State Control, 2011*; and
- f. the draft revised *Survey Guidelines under the Harmonized System of Survey and Certification, 2011*, which include references to relevant amendments to statutory instruments entering into force up to and including 31 December 2011.

The draft Assembly resolutions are included in this shipping circular for advance information. Once adopted by A27, a new shipping circular will be issued.

4. The following summarises the key points relating to the resolutions:

**a. Lifeboat Release and Retrieval Systems (RRS)**

- [MSC.317\(89\)](#) – Adoption of amendments to the International Convention for the Safety of Life at Sea, 1974, as amended
- [MSC.320\(89\)](#) – Adoption of amendments to the LSA Code
- [MSC.321\(89\)](#) – Adoption of amendments to the *Revised Recommendation on Testing of LSA* (Resolution MSC.81(70))

MSC 89 adopted the above resolutions as a “package” that includes two related circulars:

- [MSC.1/Circ.1392](#) – Guidelines for Evaluation and Replacement of Lifeboat Release and Retrieval Systems
- [MSC.1/Circ.1393](#) – Early Application of new SOLAS regulation III/1.5

[MSC.317\(89\)](#) introduces a new regulation III/1.5 for all lifeboat on-load release mechanisms on board new and existing cargo and passenger ships to comply with the new requirements in Chapter IV of the LSA Code relating to lifeboat on-load release mechanisms adopted by resolution [MSC.320\(89\)](#). The *Recommendation on the Testing of LSA* is consequentially amended by resolution [MSC.321\(89\)](#) to ensure that it is consistent with the amended LSA Code. For existing ships, evaluation and assessment are to be carried out in accordance with [MSC.1/Circ.1392](#) *Guidelines for Evaluation and Replacement of*

*Lifeboat Release and Retrieval Systems.* As an interim risk mitigation measure, MSC 86 had in Jun 2009 approved circular [MSC.1/Circ.1327](#) *Guidelines for the Fitting and Use of Fall Preventer Devices (FPDs).*

The new SOLAS regulation enters into force on 1 Jan 2013. Ships shall comply with the amended LSA Code not later than the first scheduled dry-docking after 1 Jul 2014, but not later than 1 Jul 2019. [MSC.1/Circ.1393](#) encourages the early application of the new regulation by SOLAS Contracting Governments, and MPA urges shipowners to comply as soon as practicable on existing ships, and new ships constructed before 1 Jul 2014.

*Shipowners are to take note of the following points:*

1. *Assessment of models: Before 1 Jul 2013, manufacturers are required to assess their models of RRS and complete the assessment and performance tests as required, so as to determine which of the models are compliant, compliant after modification or non-compliant. The results will be reported to IMO, and posted on the GISIS website. Shipowners should refer to the list in order to determine if the models installed on their ships comply with the amended LSA Code. Any release mechanisms that are not compliant or not in the list of compliant models are to be replaced with one which is in the list of compliant models.*
2. *One-time on board follow-up examination: The one-time on board examination is a requirement for all release mechanisms installed on the ship, even if the model is compliant. Shipowners should contact our ROs to make arrangement for the examination.*
3. *Fall Preventer Devices (FPDs): Although it is not mandatory, MPA strongly urges shipowners that suitable FPDs be fitted in accordance with [MSC.1/Circ.1327](#), as properly fitted FPDs would help prevent accidents, pending evaluation of the RRS. They should seek further advice from our ROs on suitable FPDs.*
4. *Early application (MSC.1/Circ.1393): For ships constructed on or after 20 May 2011 but before 1 July 2014, it is acceptable to MPA that they are encouraged to comply with the amended LSA Code, and that at the earliest opportunity, approval processes for compliant RRS may be initiated, in accordance with MSC.1/Circ.1393.*

**b. Amendments to the IMSBC Code**

- [MSC.318\(89\)](#) – Adoption of amendments to the International Maritime Solid Bulk Cargoes (IMSBC) Code

This is the first set of amendments since the IMSBC Code was adopted. The Code will be regularly amended every two years, similar to the IMDG Code.

*Ships carrying solid bulk cargoes should take note of the amendments and implement the amendments accordingly. The amendments enter into force on 1 Jan 2013, but voluntary application of the amendments from 1 Jan 2012 is acceptable to MPA.*

**c. Amendments to the 2008 IS Code**

- [MSC.319\(89\)](#) – Adoption of amendments to the International Code on Intact Stability, 2008 (2008 IS Code)

The amendments concern Part B of the IS Code, which is non-mandatory, and takes into consideration the application of the new 2009 MODU Code: MODUs constructed on or after 1 Jan 2012 should apply intact stability requirements found in the 2009 MODU Code.

**d. Amendments to the Revised Recommendation on Testing of LSA (Resolution MSC.81(70))**

- [MSC.323\(89\)](#) – Adoption of amendments to the Revised Recommendation on Testing of LSA (Resolution MSC.81(70))

This set of amendments to the *Revised Recommendation on Testing of LSA* is not related to lifeboat release mechanisms (see MSC.321(89)), but concerns other life saving appliances.

**e. Unified Interpretations (UI)**

In addition to the adoption of resolutions, MSC 89 also approved the following Unified Interpretations of SOLAS; applicable date is 20 May 2011.

- [MSC.1/Circ.1397](#) – Unified interpretation of SOLAS regulation III/15.1 concerning openings on the ship's side in the area in way of marine evacuation systems
- [MSC.1/Circ.1398](#) – Unified interpretation of SOLAS regulation II-1/29, concerning mechanical, hydraulic and electrical independency and failure detection and response of steering control systems.

The Unified Interpretations and clarifications are acceptable to MPA and should be followed and applied in accordance with the recommended application date mentioned in the circulars. Shipowners are further advised to approach the nine approved classification societies to seek further guidance.

**f. Implementation of Best Management Practices (BMP)**

- [MSC.324\(89\)](#) – Implementation of Best Management Practices (BMP) Guidance

The resolution strongly urges all those concerned to take action to ensure that as a minimum and as recommended in the Best Management Practices:

- (a) ships' masters receive updated information before and during sailing through the defined High Risk Area;
- (b) ships register with the Maritime Security Centre Horn of Africa (MSCHOA) and report to United Kingdom Maritime Trade Operations (UKMTO) Dubai; and
- (c) ships effectively implement all recommended preventive, evasive and defensive measures.

*MPA has previously issued advisory shipping circulars, No. 6 of 2011 and No. 14 of 2010. Shipowners should refer to these circulars on BMP.*

5. Queries relating to this circular should be directed to Mr Ong Hua Siong (Tel: 6375-6210). For queries on compliance with the respective resolutions, please contact the vessel's classification society.

CHEONG KENG SOON  
DIRECTOR OF MARINE  
MARITIME AND PORT AUTHORITY OF SINGAPORE

**RESOLUTION MSC.317(89)**  
**(adopted on 20 May 2011)**

**ADOPTION OF AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR  
THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED**

THE MARITIME SAFETY COMMITTEE,

RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,

RECALLING FURTHER article VIII(b) of the International Convention for the Safety of Life at Sea (SOLAS), 1974 (hereinafter referred to as "the Convention"), concerning the amendment procedure applicable to the Annex to the Convention, other than to the provisions of chapter I thereof,

HAVING CONSIDERED, at its eighty-ninth session, amendments to the Convention, proposed and circulated in accordance with article VIII(b)(i) thereof,

1. ADOPTS, in accordance with article VIII(b)(iv) of the Convention, amendments to the Convention, the text of which is set out in the Annex to the present resolution;
2. DETERMINES, in accordance with article VIII(b)(vi)(2)(bb) of the Convention, that the said amendments shall be deemed to have been accepted on 1 July 2012, unless, prior to that date, more than one third of the Contracting Governments to the Convention or Contracting Governments the combined merchant fleets of which constitute not less than 50% of the gross tonnage of the world's merchant fleet, have notified their objections to the amendments;
3. INVITES SOLAS Contracting Governments to note that, in accordance with article VIII(b)(vii)(2) of the Convention, the amendments shall enter into force on 1 January 2013 upon their acceptance in accordance with paragraph 2 above;
4. REQUESTS the Secretary-General, in conformity with article VIII(b)(v) of the Convention, to transmit certified copies of the present resolution and the text of the amendments contained in the Annex to all Contracting Governments to the Convention;
5. FURTHER REQUESTS the Secretary-General to transmit copies of this resolution and its Annex to Members of the Organization which are not Contracting Governments to the Convention.

ANNEX

**AMENDMENTS TO THE INTERNATIONAL CONVENTION  
FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED**

**CHAPTER III  
LIFE-SAVING APPLIANCES AND ARRANGEMENTS**

**Regulation 1 – Application**

The following new paragraph 5 is added after the existing paragraph 4:

"5 Notwithstanding paragraph 4.2, for all ships, not later than the first scheduled dry-docking after 1 July 2014, but not later than 1 July 2019, lifeboat on-load release mechanisms not complying with paragraphs 4.4.7.6.4 to 4.4.7.6.6 of the Code shall be replaced with equipment that complies with the Code.\*

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\* Refer to the Guidelines for evaluation and replacement of lifeboat release and retrieval systems (MSC.1/Circ.1392)."

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**RESOLUTION MSC.320(89)**  
**(adopted on 20 May 2011)**

**ADOPTION OF AMENDMENTS TO THE  
INTERNATIONAL LIFE-SAVING APPLIANCE (LSA) CODE**

THE MARITIME SAFETY COMMITTEE,

RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,

NOTING resolution MSC.48(66), by which it adopted the International Life-Saving Appliance Code (hereinafter referred to as "the LSA Code"), which has become mandatory under chapter III of the International Convention for the Safety of Life at Sea, 1974 (hereinafter referred to as "the Convention"),

NOTING ALSO article VIII(b) and regulation III/3.10 of the Convention concerning the procedure for amending the LSA Code,

HAVING CONSIDERED, at its eighty-ninth session, amendments to the LSA Code, proposed and circulated in accordance with article VIII(b)(i) of the Convention,

1. ADOPTS, in accordance with article VIII(b)(iv) of the Convention, amendments to the LSA Code, the text of which is set out in the Annex to the present resolution;
2. DETERMINES, in accordance with article VIII(b)(vi)(2)(bb) of the Convention, that the amendments shall be deemed to have been accepted on 1 July 2012, unless prior to that date, more than one third of the Contracting Governments to the Convention or Contracting Governments the combined merchant fleets of which constitute not less than 50% of the gross tonnage of the world's merchant fleet, have notified their objections to the amendments;
3. INVITES Contracting Governments to note that, in accordance with article VIII(b)(vii)(2) of the Convention, the amendments shall enter into force on 1 January 2013 upon their acceptance in accordance with paragraph 2 above;
4. REQUESTS the Secretary-General, in conformity with article VIII(b)(v) of the Convention, to transmit certified copies of the present resolution and the text of the amendments contained in the Annex to all Contracting Governments to the Convention;
5. FURTHER REQUESTS the Secretary-General to transmit copies of this resolution and its Annex to Members of the Organization which are not Contracting Governments to the Convention.



ANNEX

**AMENDMENTS TO THE INTERNATIONAL LIFE-SAVING  
APPLIANCES (LSA) CODE**

**CHAPTER IV  
SURVIVAL CRAFT**

1 In paragraph 4.4.7.6, the following new subparagraphs .2 to .6 are inserted after the existing subparagraph .1:

- "2 notwithstanding subparagraph .7.2 the mechanism shall only open when the release mechanism is operated with the boat fully waterborne or, if the boat is not waterborne, by multiple, deliberate and sustained action which shall include the removal or bypassing of safety interlocks designed to prevent premature or inadvertent release;
  - .2.1 the mechanism shall not be able to open due to wear, misalignment and unintended force within the hook assembly or operating mechanism, control rods or cables as may be connected to, or form part of the hook assembly and with trim of up to 10° and a list of up to 20° either way; and
  - .2.2 the functional criteria of 4.4.7.6.2 and 4.4.7.6.2.1 apply for the range of loads, representing 0% to 100% of the safe working load of the lifeboat release and retrieval system for which it may be approved;
- .3 unless a release mechanism is of the load over centre type, which is held fully closed by the weight of the lifeboat, the hook assembly shall be designed so that the moveable hook component is kept fully closed by the hook locking parts capable of holding its safe working load under any operational conditions until the hook locking part is deliberately caused to open by means of the operating mechanism. For designs utilizing the tail of the movable hook component and cam either directly or indirectly securing the tail of the movable hook component, the hook assembly shall continue to be closed and hold its safe working load through rotation of the cam of up to 45 degrees in either direction, or 45 degrees in one direction if restricted by design, from its locked position;
- .4 to provide hook stability, the release mechanism shall be designed so that, when it is fully reset in the closed position, the weight of the lifeboat does not cause any force to be transmitted to the operating mechanism;
- .5 locking devices shall be designed so that they can not turn to open due to forces from the hook load; and
- .6 if a hydrostatic interlock is provided, it shall automatically reset upon lifting the boat from the water."

2 In paragraph 4.4.7.6, the existing subparagraph .2 is replaced by the following:

"7 the mechanism shall have two release capabilities: normal (off-load) release capability and on-load release capability:

.7.1 normal (off-load) release capability shall release the lifeboat when it is waterborne or when there is no load on the hooks, and not require manual separation of the lifting ring or shackle from the jaw of the hook; and

.7.2 on-load release capability shall release the lifeboat with a load on the hooks. This release mechanism shall be provided with a hydrostatic interlock unless other means are provided to ensure that the boat is waterborne before the release mechanism can be activated. In case of failure or when the boat is not waterborne, there shall be a means to override the hydrostatic interlock or similar device to allow emergency release. This interlock override capability shall be adequately protected against accidental or premature use. Adequate protection shall include special mechanical protection not normally required for off-load release, in addition to a danger sign. The protection shall be deliberately destroyed by applying a suitable minimum force, for instance by breaking a protection glass or translucent cover. A label or thin wire seal is not considered sufficiently robust. To prevent a premature on-load release, on-load operation of the release mechanism shall require multiple, deliberate and sustained action or actions by the operator;"

3 In paragraph 4.4.7.6, the existing subparagraph .3 is renumbered as subparagraph .8 and the words "without excessive force" are replaced by the words ", and any indicators shall not indicate the release mechanism is reset".

4 In paragraph 4.4.7.6, the following new subparagraph .9 is inserted after the renumbered subparagraph 8:

"9 all components of the hook unit, release handle unit, control cables or mechanical operating links and the fixed structural connections in a lifeboat shall be of material corrosion resistant in the marine environment without the need for coatings or galvanizing. Design and manufacturing tolerances shall be such that anticipated wear throughout the service life of the mechanism shall not adversely affect its proper functioning. Mechanical operating links such as control cables shall be waterproof and shall have no exposed or unprotected areas;"

5 In paragraph 4.4.7.6, the existing subparagraphs .4 to .8 are renumbered as subparagraphs .10 to .14, respectively.

6 In paragraph 4.4.7.6, in the renumbered subparagraph .10, the word "clearly" is replaced by the word "unambiguously".

7 In paragraph 4.4.7.6, in the renumbered subparagraph .14, the words "the load-bearing components of the release mechanism and" are added at the beginning and the words "of the release mechanism" are deleted.

8 In paragraph 4.4.7.6, the following new subparagraphs .15 and .16 are inserted after the renumbered subparagraph .14:

" .15 a hydrostatic interlock shall be designed for a factor of safety of not less than 6 times maximum operating force based on the ultimate strength of the materials used;

.16 the operating cables shall be designed for a factor of safety of not less than 2.5 times maximum operating force based on the ultimate strength of the materials used; and".

9 In paragraph 4.4.7.6, the existing subparagraph .9 is renumbered as subparagraph .17 and in the renumbered subparagraph .17, the references to paragraphs "4.4.7.6.2.2 and 4.4.7.6.3" are replaced by the references to paragraphs "4.4.7.6.7, 4.4.7.6.8 and 4.4.7.6.15".

10 In paragraph 4.4.7.6, the referenced subparagraph .9 is replaced by .17.

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**RESOLUTION MSC.321(89)**  
**(adopted on 20 May 2011)**

**ADOPTION OF AMENDMENTS TO THE REVISED RECOMMENDATION ON TESTING OF  
LIFE-SAVING APPLIANCES (RESOLUTION MSC.81(70)), AS AMENDED**

THE MARITIME SAFETY COMMITTEE,

RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,

RECALLING ALSO resolution A.689(17) entitled "Testing of life-saving appliances", by which the Assembly, at its seventeenth session, adopted the Recommendation on testing of life-saving appliances,

RECALLING FURTHER that the Assembly, when adopting resolution A.689(17), authorized the Committee to keep the Recommendation on testing of life-saving appliances under review and to adopt, when appropriate, amendments thereto,

NOTING resolution MSC.81(70), by which, at its seventieth session, it adopted the Revised recommendation on testing of life-saving appliances, introducing more precise provisions for the testing of life-saving appliances based on the requirements of the International Life-Saving Appliances (LSA) Code,

RECOGNIZING the need to appropriately align the relevant provisions of the Revised recommendation on testing of life-saving appliances with the associated amendments to the LSA Code adopted by resolution MSC.320(89),

HAVING CONSIDERED, at its eighty-ninth session, proposed amendments to the Revised recommendation on testing of life-saving appliances, prepared by the Sub-Committee on Ship Design and Equipment at its fifty-fifth session,

1. ADOPTS amendments to the Revised recommendation on testing of life-saving appliances (resolution MSC.81(70)), the text of which is set out in the Annex to the present resolution;
2. RECOMMENDS Governments to apply the annexed amendments when testing life-saving appliances.

ANNEX

**AMENDMENTS TO THE REVISED RECOMMENDATION ON TESTING OF  
LIFE-SAVING APPLIANCES (RESOLUTION MSC.81(70)), AS AMENDED**

**PART 1  
PROTOTYPE TESTS FOR LIFE-SAVING APPLIANCES**

- 1 The existing paragraphs 6.9.3 and 6.9.4 are replaced by the following:

"6.9.3 With the operating mechanism disconnected it should be demonstrated when the lifeboat is loaded with its full complement of persons and equipment and towed at speeds of 5 knots that the moveable hook component stays closed. Furthermore, with the operating mechanism connected, it should be demonstrated that the lifeboat when loaded with its full complement of persons and equipment when towed at speeds of 5 knots can be released. Both of the above should be demonstrated as follows as follows:

- .1 a force equal to 25% of the safe working load of the hook should be applied to the hook in the lengthwise direction of the boat at an angle of 45° to the vertical. This test should be conducted in the aftward as well as the forward direction;
- .2 a force equal to the safe working load of the hook should be applied to the hook in an athwartships direction at an angle of 20° to the vertical. This test should be conducted on both sides; and
- .3 a force equal to the safe working load of the hook should be applied to the hook in a direction halfway between the positions of tests 1 and 2 (i.e. 45° to the longitudinal axis of the boat in plan view) at an angle of 33° to the vertical. This test should be conducted in four positions.

There should be no damage as a result of these tests.

- 6.9.4 A release mechanism should be conditioned and tested as follows:

- .1 the lifeboat release and retrieval system and the longest used connection cable/linkage associated with the system should be mounted and adjusted according to instructions from the original equipment manufacturer and then loaded to 100% of its safe working load and released. Load and release should be repeated 50 times. During the 50 releases, the lifeboat release and retrieval system should be released simultaneously from each fall to which it is connected without any binding or damage to any part of the lifeboat release and retrieval system. The system should be considered as "failed" if any failure during the conditioning or unintended release occurs when load is applied but the system has not yet been operated;
- .2 the lifeboat release and retrieval system should then be disassembled, the parts examined and wear recorded. The release and retrieval system should then be reassembled;

- .3 the hook assembly, whilst disconnected from the operating mechanism, should then be tested 10 times with cyclic loading from zero load to 1.1 times the safe working load, at a nominal 10 seconds per cycle; unless the release mechanism has been specifically designed to operate as an off-load hook with on-load capability using the weight of the boat to close the hook, in this case the cyclic load should be from no more than 1% to 1.1 times the SWL. For cam-type designs, the test should be carried out at an initial cam rotation of 0° (fully reset position), and repeated at 45° in either direction, or 45° in one direction if restricted by design. The specimen should remain closed during the test. The system should be considered as "failed" if any failure during this test or any unintended release or opening occurs; and
- .4 the cable and operating mechanism should then be reconnected to the hook assembly; and the lifeboat release and retrieval system should then be demonstrated to operate satisfactorily under its safe working load. The actuation force should be no less than 100 N and no more than 300 N, if a cable is used it should be the maximum length specified by the manufacturer, and secured in the same manner it would be secured in the lifeboat. The demonstration should verify that any interlocks, indicators and handles are still functioning and are correctly positioned in accordance with the operation and safety instruction from the original equipment manufacturer. The release mechanism is deemed to have passed the testing under paragraph 6.9.4 when the tests have been conducted successfully. The system should be considered as "failed" if any failure during this test or any unintended release or opening occurs.

6.9.5 A second release mechanism should be tested as follows:

- .1 the actuation force of the release mechanism should be measured loaded with 100% of its safe working load. The actuation force should be no less than 100 N and no more than 300 N. If a cable is used, it should be of the maximum length specified by the manufacturer, and secured in the same manner it would be secured in a lifeboat. The demonstration should verify that any interlocks, indicators and handles are still functioning and are correctly positioned in accordance with the operation and safety instruction from the original equipment manufacturer; and
- .2 the release mechanism should be mounted on a tensile strength testing device. The load should be increased to at least six times the working load of the release mechanism without failure of the release mechanism."

2 The existing paragraphs 6.9.5 and 6.9.6 are renumbered as 6.9.6 and 6.9.7, respectively.

3 In paragraph 6.11.3, the referenced paragraph number "6.9.4" is replaced with "6.9.3".

4 In paragraphs 7.1.1 and 7.4.1, the referenced paragraph numbers "6.9.5" and "6.9.6" are replaced by "6.9.6" and "6.9.7", respectively.

5 In paragraphs 7.2.1, 7.3.1, 7.5 and 7.6, the referenced paragraph numbers "6.9.1 to 6.9.4" are replaced by "6.9.1 to 6.9.5".

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MSC.1/Circ.1392  
27 May 2011

## **GUIDELINES FOR EVALUATION AND REPLACEMENT OF LIFEBOAT RELEASE AND RETRIEVAL SYSTEMS**

1 The Maritime Safety Committee, at its eighty-ninth session (11 to 20 May 2011), approved the Guidelines for evaluation and replacement of lifeboat release and retrieval systems, set out in the annex, as per SOLAS regulation III/1.5, following the recommendations made by the Sub-Committee on Ship Design and Equipment, at its fifty-fifth session, and the *Ad Hoc* Working Group on Lifeboat Release Hooks (16 to 18 March 2011).

2 Member Governments are invited to use the annexed Guidelines when applying SOLAS regulation III/1.5, as adopted by resolution MSC.317(89), and to bring them to the attention of all parties concerned.

3 Member Governments, shipowners and manufacturers of lifeboat release and retrieval systems are also strongly urged, pending the entry into force of SOLAS regulation III/1.5, to use the annexed Guidelines to evaluate existing lifeboat release and retrieval systems at the earliest available opportunity.\*

4 Member Governments are strongly urged to ensure that all ships fitted with on-load release systems for lifeboats, are equipped with fall preventer devices as per paragraph 6 of these Guidelines at the earliest available opportunity.

5 Member Governments are encouraged to consider the results of evaluations reported to the Organization by other Member Governments on types of existing lifeboat release and retrieval systems.

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\* Reference is made to MSC.1/Circ.1393 on Early application of new SOLAS regulation III/1.5.





## ANNEX

### GUIDELINES FOR EVALUATION AND REPLACEMENT OF LIFEBOAT RELEASE AND RETRIEVAL SYSTEMS

#### General

1 New SOLAS regulation III/1.5, which is expected to enter into force on 1 January 2013, requires that for all ships, on-load release mechanisms\* not complying with paragraphs 4.4.7.6.4 to 4.4.7.6.6 of the LSA Code, as amended by resolution MSC.320(89) (hereinafter called "the LSA Code"), be replaced or modified not later than the next scheduled dry-docking after 1 July 2014, but not later than 1 July 2019.

2 Considering that paragraphs 4.4.7.6.4 to 4.4.7.6.6 of the LSA Code represent important safety improvements, manufacturers should carry out a self assessment of their types of existing lifeboat release and retrieval systems in accordance with these Guidelines at the earliest available opportunity.

3 An Administration, or a recognized organization acting on its behalf, should carry out a design review to check that the type of existing lifeboat release and retrieval systems comply with paragraphs 4.4.7.6.4 to 4.4.7.6.6 of the LSA Code and should witness the performance test to check that it is performed in accordance with appendix 1 of these Guidelines. This evaluation should be completed not later than 1 July 2013 and the report should be submitted in accordance with paragraph 14 below.

4 Administrations, or recognized organizations acting on their behalf, should, when applying SOLAS regulation III/1.5, ensure that an evaluation of the type of existing lifeboat release and retrieval system is undertaken, for compliance with paragraphs 4.4.7.6.4 to 4.4.7.6.6 of the LSA Code, in accordance with these Guidelines.

5 A flowchart of the lifeboat release and retrieval system evaluation process, is set out in appendix 2.

6 On each ship, fall preventer devices in accordance with the Guidelines for the fitting and use of fall preventer devices (FPDs) (MSC.1/Circ.1327) should be employed for each existing lifeboat release and retrieval system until the system is:

- .1 found compliant with the LSA Code; or
- .2 modified and found compliant with the LSA Code; or
- .3 found compliant with paragraphs 4.4.7.6.4 to 4.4.7.6.6 of the LSA Code and paragraphs 16 and 17 (overhaul examination) of these Guidelines; or
- .4 modified and found compliant with paragraphs 4.4.7.6.4 to 4.4.7.6.6 of the LSA Code and paragraphs 16 and 17 (overhaul examination) of these Guidelines; or
- .5 replaced by a new lifeboat release and retrieval system.

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\* For the purpose of these Guidelines, the expression "on-load release mechanism" has been replaced by "lifeboat release and retrieval system" (see paragraph 9.1).

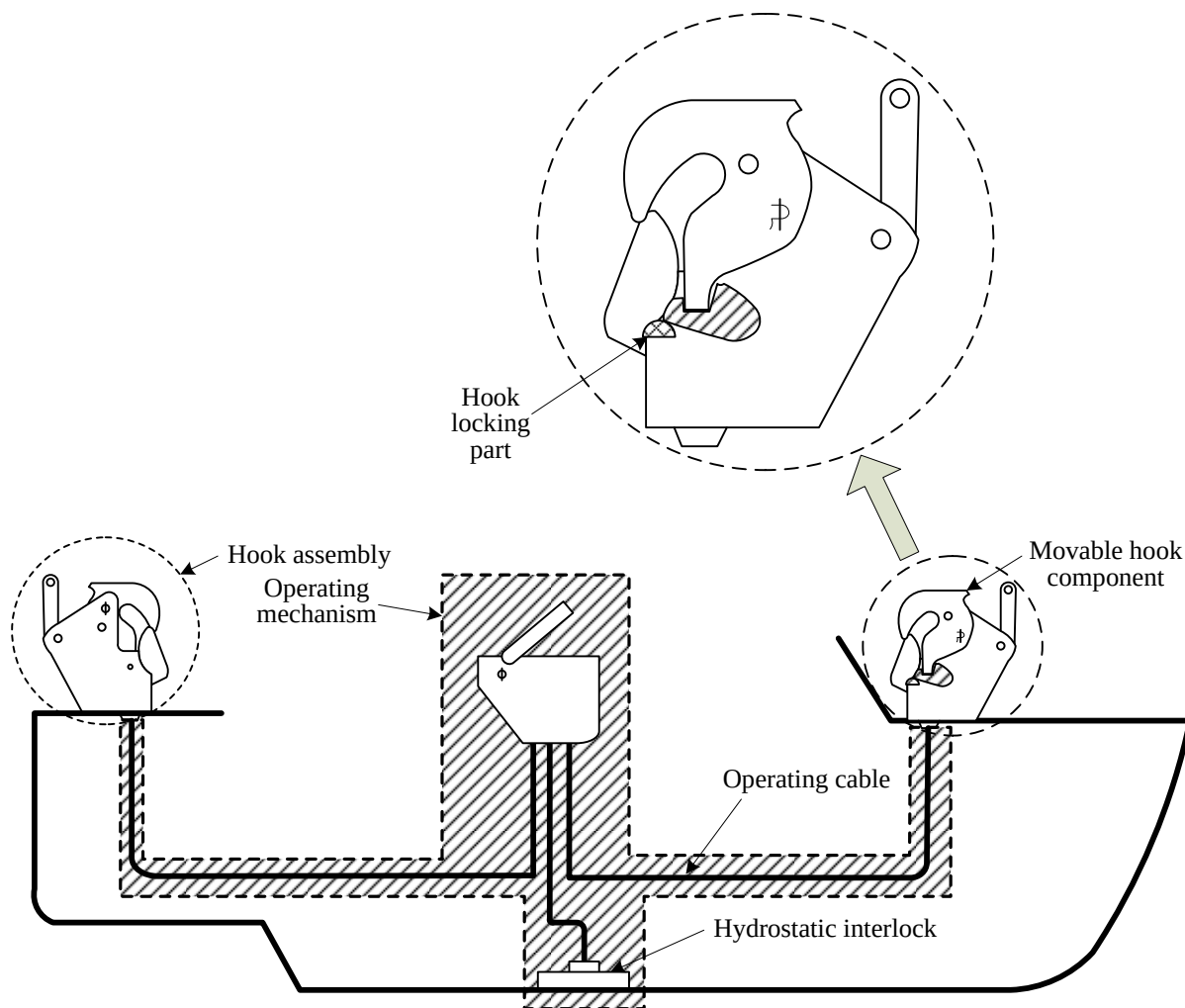
## Modifications

7 A lifeboat release and retrieval system that has been determined to be non-compliant in accordance with these Guidelines may be modified to comply with the requirements of the revised paragraphs 4.4.7.6.4 to 4.4.7.6.6 of the LSA Code and the requirements of the existing applicable Code, provided that the modified release and retrieval system is evaluated in accordance with these Guidelines.

8 A type of lifeboat release and retrieval system that, after modification, complies with the requirements of the revised paragraphs 4.4.7.6.4 to 4.4.7.6.6 of the LSA Code and the requirements of the existing applicable Code should be identified as a system compliant after modification and reported as such. The report should include both the identification of the original type and the modified type.

## Definitions

9 For the purpose of these Guidelines, the definitions given hereunder should apply, in accordance with the following figure.



**Lifeboat release and retrieval system**

9.1 *Lifeboat release and retrieval system* is the means by which the lifeboat is connected to, and released from, the lifeboat falls for lowering, launch and retrieval. It comprises the hook assembly and operating mechanism.

9.2 *Hook assembly* is the mechanism, attached to the lifeboat, which connects the lifeboat to the lifeboat falls.

9.3 *Movable hook component* is that part of the hook assembly in direct contact with the connection with the lifeboat falls which moves to enable release from the falls.

9.4 *Hook locking part* is the component(s) within a hook assembly which holds the movable hook component in the closed position until activated by the operating mechanism to release the hook. This activation may be performed through other components within the hook assembly.

9.5 *Operating mechanism* is the means by which the operator activates the opening, or release, of the movable hook component. It includes the operating handle, linkages/cables and hydrostatic interlock, if fitted.

9.6 *Type*, in relation to the design of a lifeboat release and retrieval system, means an identical lifeboat release and retrieval system of given safe working load, make and model (thus any change to the materials of construction, design arrangement or dimensions constitutes a change of type).

9.7 *On-load release* is the action of opening the lifeboat release and retrieval system whilst there is load on the hook assemblies.

9.8 *Evaluation* is a design review and a performance test of a type of lifeboat release and retrieval system.

9.9 *Manufacturer*, with respect to existing lifeboat release and retrieval systems, is:

- .1 the original equipment manufacturer; or
- .2 a manufacturer of lifeboat release and retrieval systems who has taken on the responsibility for a range or type of lifeboat release and retrieval system; or
- .3 any other person or entity which has taken responsibility for a range or type of lifeboat release and retrieval system when the original manufacturer no longer exists or supports the equipment.

9.10 *Modifications* are changes to the design of an approved lifeboat release and retrieval system which may affect compliance with the original approval requirements or the prescribed conditions for the use of the product.

9.11 *New lifeboat release and retrieval system* is a lifeboat release and retrieval system that has been approved in accordance with paragraph 4.4.7.6 of chapter IV of the LSA Code, as amended by resolution MSC.320(89).

9.12 *Existing lifeboat release and retrieval system* is a lifeboat release and retrieval system that has not been approved in accordance with paragraph 4.4.7.6 of chapter IV of the LSA Code, as amended by resolution MSC.320(89).

9.13 *Company* means company as defined in SOLAS regulation IX/1.2.

## **Design review**

10 Documentation and information for each type of lifeboat release and retrieval system should be submitted to the Administration, or recognized organization acting on its behalf, in order that an assessment can be carried out to determine compliance with paragraphs 4.4.7.6.4 to 4.4.7.6.6 of the LSA Code. The manufacturer should submit the approval certificate, along with all associated supporting design calculations, plans and testing documentation to the Administration or recognized organization acting on its behalf. The design information should include the specification and the installation instructions for the complete operating system as well as all safety instructions regarding the operating system and any interlocks provided. Any submission for testing of a lifeboat release and retrieval system that cannot be supported with the above-mentioned information should not be eligible for testing against the requirements of the LSA Code.

11 If the outcome of the design review is non-compliance with the applicable paragraphs of the LSA Code, the lifeboat release and retrieval system should be replaced or modified to be made compliant.

## **Performance test**

12 After a successful completion of the design review, a performance test should be conducted by the manufacturer for each type of lifeboat release and retrieval systems for compliance with paragraphs 4.4.7.6.4 to 4.4.7.6.6 of the LSA Code, using the test specified in appendix 1 to these Guidelines. The performance test should be witnessed by the Administration or a recognized organization acting on its behalf.

13 Should any part of the lifeboat release and retrieval system fail at any stage during the test specified in paragraphs 1 to 4 of appendix 1, this type of lifeboat release and retrieval system should be deemed to be non-compliant and reported as such.

## **Reporting of the results of evaluation of existing lifeboat release and retrieval system**

14 The Administration should report the results of each type of existing lifeboat release and retrieval system evaluation carried out in accordance with these Guidelines to the Organization, based on the reporting procedure, as set out in appendix 3.

15 Depending on the outcome of the evaluation, every lifeboat release and retrieval system should be categorized as being either compliant, compliant after modification or non-compliant. Thereafter:

- .1 systems categorized as being compliant, or compliant after modification, may remain in service; and
- .2 every system categorized as being non-compliant should be replaced with a new system or modified to be made compliant.

## **One-time follow-up overhaul examination**

16 Not later than the first scheduled dry-docking after 1 July 2014, every lifeboat release and retrieval system of a type found to be compliant in respect of the existing lifeboat release and retrieval system evaluation should be subject to an overhaul examination according to annex 1 to the Measures to prevent accidents with lifeboats (MSC.1/Circ.1206/Rev.1) by the manufacturer or by one of their representatives. The examination also includes verification that

the system examined is of the same type as the system that passed the evaluation and is suitable for the ship.

17 The scope of the overhaul examination should also include a detailed assessment of the condition of the components of the lifeboat release and retrieval system to observe the extent of wear, corrosion, erosion and other types of material degradation that may have occurred. Upon satisfactory completion of the overhaul examination, the manufacturer or one of their representatives should issue a factual statement to confirm this, for retention on board.

### **Procedure for replacement of non-compliant lifeboat release and retrieval systems**

18 The procedure outlined below should be followed in all cases where a lifeboat is to be fitted with replacement lifeboat release and retrieval systems with on-load release capability. It is noted that every lifeboat, complete with lifeboat release and retrieval system, is type-approved at manufacture and it is important to recognize that a lifeboat which is retro-fitted with a replacement lifeboat release and retrieval system to the satisfaction of the Administration should be regarded as offering a level of safety which is higher than that of the original installation.

19 Companies should, where possible, select replacement equipment acceptable to the lifeboat manufacturer. However, in cases where the lifeboat manufacturer is unable to offer a suitable replacement lifeboat release and retrieval system, the Company may select an alternative lifeboat release and retrieval system, with the agreement, if possible, of the lifeboat manufacturer.

20 The replacement equipment should be approved by the Administration or a recognized organization acting on its behalf, under the provisions of the LSA Code. Prior to the installation commencing, the Company should submit to the Administration, or a recognized organization acting on its behalf, for review and approval, as a minimum the following information:

- .1 the proposed replacement equipment including approval certification;
- .2 the engineering analysis of the replacement installation including:
  - .1 drawings of the original lifeboat release and retrieval system arrangement;
  - .2 detailed drawings showing clearly the proposed changes (e.g., position of suspension, lifeboat release and retrieval system, fixed structural connections of the release mechanism, link plates, including materials used for nuts and bolts with regard to strength and corrosion resistance); and
  - .3 if the drawings show that forces and/or force couples will change and/or the lifeboat release and retrieval system fixed structural connections of the release mechanism will change, calculation of static forces including a safety factor of 6, according to the LSA Code, from lifeboat release and retrieval system into lifeboat structure, including tension and shear forces in bolts, link plates, welds and keel shoe(s);
- .3 considering that a lifeboat release and retrieval system does not consist just of the hook assemblies themselves, but also of release handles, cabling, etc., in the lifeboat, the evaluation of a replacement hook assembly other than that originally provided in the lifeboat should include such factors as loadings of the release handle on the console, efficiency of any hydrostatic interlock in light and

loaded conditions, whether the size/configuration of the replacement equipment would affect the stability or seating space of the lifeboat, and its compatibility with its launching appliance;

- .4 amended operating and training manuals; and
- .5 identification of the person(s) responsible for design appraisal, installation work and post-installation testing and evidence of their competence.

21 The Administration, or a recognized organization acting on its behalf, may allow that hook fixed structural connections of the release mechanism and supporting structure which are not made of material corrosion resistant in the marine environment, as required by paragraph 4.4.7.6.9 of the LSA Code, need not be replaced if they are in a good condition and installed in a sheltered position inside the lifeboat.

22 A copy of the engineering drawing(s) approved by the Administration, or by the recognized organization acting on its behalf, should be used during installation and testing and retained on board.

23 The installation should be carried out by the manufacturer or by one of their representatives. All work carried out should be witnessed by the Administration, or by a recognized organization acting on its behalf. Valid operating and safety instructions should be posted at the operating position and adjacent to the lifeboat release and retrieval system(s).

24 Post-installation testing should be carried out by the manufacturer or by one of their representatives and comprise the following:

- .1 1.1 x load and simultaneous release test according to the Revised recommendation on testing of life-saving appliances (resolution MSC.81(70)), part 2, paragraph 5.3.1, or an equivalent method acceptable to the Administration;
- .2 load test according to the Revised recommendation on testing of life-saving appliances (resolution MSC.81(70)), part 2, paragraph 5.3.4, as amended by resolution MSC.226(82), if the fixed structural connections of the release mechanism of the lifeboat is modified; and
- .3 if the lifeboat is also a rescue boat and/or is installed on a cargo ship of 20,000 gross tonnage or above, the 5 knots installation test should be carried out, in accordance with the Revised recommendation on testing of life-saving appliances (resolution MSC.81(70)), part 2, paragraph 5.4.

25 All tests should be witnessed by the Administration, or by a recognized organization acting on its behalf, which should also verify that the installation complies in all respects with the documentation submitted by the Company and approved by the Administration, or a recognized organization acting on its behalf.

26 Following completion of installation testing, the Administration, or a recognized organization acting on its behalf, should issue a Statement of Acceptance, using the template set out in appendix 4, to the Company, for retention on board.

## APPENDIX 1

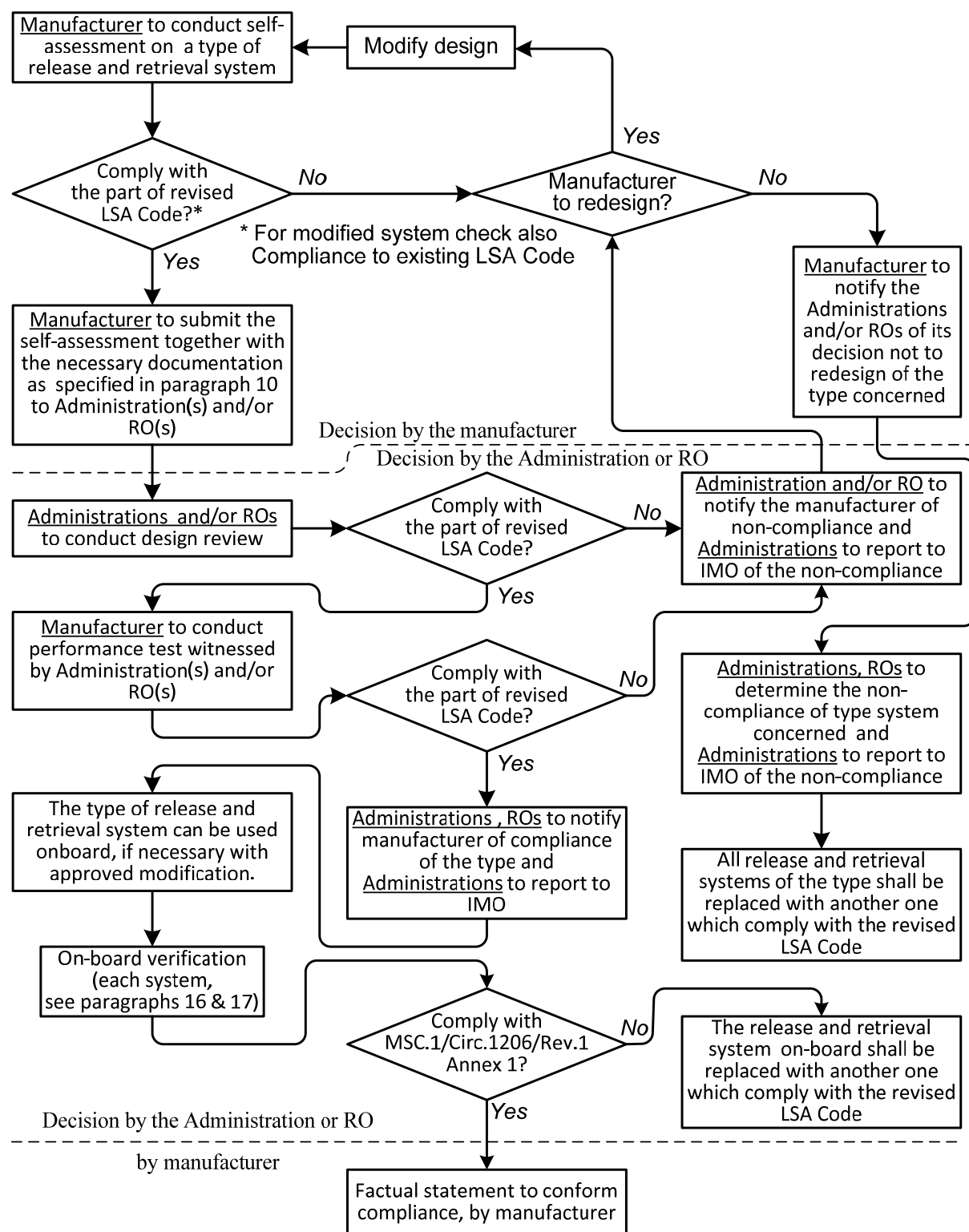
### TEST REQUIREMENTS FOR THE EVALUATION OF LIFEBOAT RELEASE AND RETRIEVAL SYSTEMS

A release and retrieval system should be conditioned and tested as follows:

- .1 the lifeboat release and retrieval system and the longest used connection cable/linkage associated with the system should be mounted and adjusted according to instructions from the original equipment manufacturer and then loaded to 100% of its safe working load and released. Load and release should be repeated 50 times. During the 50 releases, the lifeboat release and retrieval system should be released simultaneously from each fall to which it is connected without any binding or damage to any part of the lifeboat release and retrieval system. The system should be considered as "failed" if any failure during the conditioning or unintended release occurs when load is applied but the system has not yet been operated;
- .2 the lifeboat release and retrieval system should then be disassembled, the parts examined and wear recorded. The release and retrieval system should then be reassembled;
- .3 the hook assembly, whilst disconnected from the operating mechanism, should then be tested 10 times with cyclic loading from zero load to 1.1 times the safe working load, at a nominal 10 seconds per cycle; unless the release and retrieval system has been specifically designed to operate as an off-load hook with on-load capability using the weight of the boat to close the hook, in this case the cyclic load should be from no more than 1% to 1.1 times the SWL; and
- .4 the cable and operating mechanism should then be reconnected to the hook assembly; and the lifeboat release and retrieval system should then be demonstrated to operate satisfactorily under its safe working load. The actuation force should be no less than 100 N and no more than 300 N, if a cable is used it should be the maximum length specified by the manufacturer, and secures in the same manner it would be secured in the lifeboat. The demonstration should verify that any interlocks, including hydrostatic interlocks, where fitted, indicators and handles are still functioning and are correctly positioned in accordance with the operation and safety instruction from the original equipment manufacturer. The release and retrieval system is deemed to have passed the testing under this appendix when the tests have been conducted successfully. The system should be considered as "failed" if any failure during this test or any unintended release or opening occurs.



## APPENDIX 2

EXISTING LIFEBOAT RELEASE AND RETRIEVAL SYSTEM  
EVALUATION PROCESS FLOW CHART

### APPENDIX 3

#### INFORMATION ON THE EVALUATION OF EXISTING LIFEBOAT RELEASE AND RETRIEVAL SYSTEMS TO BE REPORTED

The following information should be provided for each lifeboat release and retrieval system:

|                                                                                                                                                         |                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Manufacturer's Details                                                                                                                                  | <b>Name</b>                                                     |
|                                                                                                                                                         | <b>Address</b>                                                  |
|                                                                                                                                                         | <b>E-mail address</b>                                           |
| Lifeboat release and retrieval system                                                                                                                   | <b>Type (see paragraph 9.6) and identification</b>              |
| In case of modification                                                                                                                                 | <b>Original type and identification</b>                         |
|                                                                                                                                                         | <b>Details of modification</b>                                  |
| Specification of type (e.g., Maximum Safe Working Load (SWL))                                                                                           |                                                                 |
| Details of the Administration, or recognized organization acting on its behalf, undertaking the evaluation of the lifeboat release and retrieval system | <b>Name</b>                                                     |
|                                                                                                                                                         | <b>Address</b>                                                  |
|                                                                                                                                                         | <b>E-mail address</b>                                           |
| Evaluation report details                                                                                                                               | <b>No.</b>                                                      |
|                                                                                                                                                         | <b>Date</b>                                                     |
| Evaluation result                                                                                                                                       | <b>Compliant / Non-compliant / Compliant after modification</b> |
| Report information                                                                                                                                      | <b>Link to the relevant report (url)</b>                        |
| Reported by                                                                                                                                             | <b>Name of the Administration</b>                               |

#### APPENDIX 4

### STATEMENT OF ACCEPTANCE OF THE INSTALLATION OF REPLACEMENT RELEASE AND RETRIEVAL SYSTEM TO AN EXISTING LIFEBOAT

*Issued in accordance with the provisions of regulation I/5 of the International Convention for the Safety of Life at Sea (SOLAS), 1974, as amended, under the authority of [Administration]\**

Name of ship:  
Port of registry:  
IMO Number:

Lifeboat details:

Replacement release and retrieval system details:

| Lifeboat identity | Lifeboat serial number | Release and retrieval system serial number (fwd) | Release and retrieval system serial number (aft) |
|-------------------|------------------------|--------------------------------------------------|--------------------------------------------------|
| No.1 (Stbd)       |                        |                                                  |                                                  |
| No.2 (Port)       |                        |                                                  |                                                  |

The above release and retrieval system has been installed and tested under the supervision of the [Administration or a recognized organization authorized to act on its behalf]\*, as documented in Survey report no...; certificate no.... dated ... and [installation] drawing(s) no(s) ... dated ... .

This statement is to confirm that:

- .1 The replacement release and retrieval system meets the relevant requirements of the LSA Code, chapter IV, section 4.4.7.6.
- .2 The replacement release and retrieval system construction and the equipment of the above-mentioned ship was found to comply with the provisions of SOLAS regulation III/4 when tested in accordance with the Revised recommendation on testing of life-saving appliances (resolution MSC.81(70)), part 2, section 5.3.1. [The test required by paragraph 5.3.4 is waived as impracticable for this replacement procedure.]\*
- .3 The validity of the relevant Safety Certificate is not affected by the installation of the replacement release and retrieval system.
- .4 The installation of the replacement release and retrieval system offers a level of safety which is at least as effective as the original manufacturer's equipment.

The [Administration, or a recognized organization authorized to act on its behalf]\* certifies that this Statement of Acceptance augments and supersedes the affected sections of the original lifeboat approval certification. The statement must be kept on board the ship with all other relevant documentation at all times.

.....  
(Date)

.....

(Stamp)

---

\* Insert as appropriate.

4 ALBERT EMBANKMENT  
LONDON SE1 7SR  
Telephone: +44 (0)20 7735 7611 Fax: +44 (0)20 7587 3210

MSC.1/Circ.1393  
27 May 2011

## **EARLY APPLICATION OF NEW SOLAS REGULATION III/1.5**

1 The Maritime Safety Committee, at its eighty-ninth session (11 to 20 May 2011), adopted, by resolution MSC.317(89), new SOLAS regulation III/1.5 and, by resolution MSC.320(89), related amendments to chapter IV of the LSA Code, which are expected to enter into force on 1 January 2013. The Committee also approved the associated Guidelines for evaluation and replacement of lifeboat release and retrieval systems (MSC.1/Circ.1392).

2 In light of the provision of the new SOLAS regulation III/1.5, the Committee agreed that:

- .1 for ships constructed on or after 1 July 2014, on-load release and retrieval systems shall comply with the LSA Code, as amended by resolution MSC.320(89); and
- .2 Member Governments are encouraged to ensure that ships constructed on or after 20 May 2011 but before 1 July 2014, on-load release and retrieval systems comply with the LSA Code, as amended by resolution MSC.320(89).

3 In order for parties concerned to take relevant actions with regard to paragraph 2 above, Member Governments are encouraged to initiate, at the earliest opportunity, approval processes for new on-load release and retrieval systems, which comply with the LSA Code, as amended by resolution MSC.320(89).

4 In adopting the aforementioned amendments, the Committee agreed to the recommendation by the Sub-Committee on Ship Design and Equipment, at its fifty-fifth session (21 to 25 March 2011), that parties concerned should be encouraged to take necessary action to evaluate existing lifeboat release and retrieval systems, based on the aforementioned Guidelines, at the earliest available opportunity, pending the entry into force of new SOLAS regulation III/1.5.

5 Member Governments, shipyards and shipowners are invited to take account of this circular and bring it to the attention of all parties concerned. In particular, manufacturers are urged to evaluate existing lifeboat release and retrieval systems at the earliest available opportunity, in accordance with the aforementioned Guidelines.



Ref. T4/3.01

MSC.1/Circ.1327  
11 June 2009

## **GUIDELINES FOR THE FITTING AND USE OF FALL PREVENTER DEVICES (FPDs)**

1 The Maritime Safety Committee, at its eighty-sixth session (27 May to 5 June 2009), approved the Guidelines for the fitting and use of fall preventer devices (FPDs), set out in the annex, following the recommendations made by the Sub-Committee on Ship Design and Equipment, at its fifty-second session.

2 The use of FPDs should be considered as an interim risk mitigation measure, only to be used in connection with existing on-load release hooks, at the discretion of the master, pending the wide implementation of improved hook designs with enhanced safety features.

3 Member Governments are invited to use the annexed Guidelines when approving the use of fall preventer devices (FPDs), and to bring them to the attention of all parties concerned.

\*\*\*



## ANNEX

### GUIDELINES FOR THE FITTING AND USE OF FALL PREVENTER DEVICES (FPDs)

#### 1 Background

1.1 In 1986, on-load release hooks for lifeboats and rescue boats were made mandatory in the SOLAS Convention, in response to Norway's worst offshore accident in March 1980, when the **Alexander Kielland** platform in the North Sea Ekofisk field capsized, killing 123 of the 212 persons on board. These then new SOLAS requirements were considered an important step forward in lifeboat design.

1.2 Some deaths in that accident were attributed to the fact that the lifeboat had no means of release when its weight was on the hook and falls. Therefore, on-load release systems were seen to offer benefits.

1.3 Since the IMO requirements for all ships to be fitted with on-load release systems came into force, there have been a number of serious accidents during drills and servicing.

1.4 Many of these accidents were attributed to either lack of maintenance, poor design or inadequate training. Failures of equipment can result in the premature opening of the on-load hook mechanism, causing the lifeboat to fall from the davits unexpectedly, even with three safety interlocks provided for in the design.

1.5 A number of current designs of on-load release hooks are designed to open under the effect of the lifeboat's own weight and often need to be held closed by the operating mechanism. This means that any defects or faults in the operating mechanism, errors by the crew or incorrect resetting of the hook after being previously operated, can result in premature release.

1.6 A "Fall Preventer Device" (FPD) can be used to minimize the risk of injury or death by providing a secondary alternate load path in the event of failure of the on-load hook or its release mechanism or of accidental release of the on-load hook. However, FPDs should not be regarded as a substitute for a safe on-load release mechanism.

#### 2 Design and operation of FPDs

##### 2.1 *Locking pins*

The following points should be considered when utilizing locking pins as FPDs:

- .1 existing on-load release hooks fitted to ships should **not** be modified by drilling to provide a locking pin insertion point, unless approved by the Administration in accordance with paragraph 4, as this may significantly reduce the strength of the hook;
- .2 locking pins should have clear operational instructions located near the insertion point of the locking pin and be colour coded so that it is clear where the pins are to be inserted;

- .3 locking pins should be designed so that they cannot be inadvertently inserted in the wrong place;
- .4 locking pins should be confirmed to be in place prior to turning out the lifeboat and during descent to the water;
- .5 strict procedures, including a warning notice at the release handle, should be in place to ensure that the locking pin is removed before the release mechanism is activated. The handle of the locking pin should be coloured red or a suitable contrasting safety colour and prominently marked with a warning that it must be removed before activating the release mechanism;
- .6 the removal of the pin should be achievable quickly and easily without posing any risk to the operating crew designated to carry out the task once the lifeboat has reached the water;
- .7 if the removal of the pins requires opening of the lifeboat hatch it should be readily achievable by the operating crew at each device from within the craft;
- .8 once the on-load release hooks have been connected to recover the lifeboat, the locking pins should be re-inserted before the boat is hoisted clear of the water. The locking pins should be designed so that they do not interfere with either the lifting or re-stowing of the lifeboat into the davits; and
- .9 where provided, fall preventer locking pins should not be used for any other purpose and should be fitted to the lifeboat at all times.

## **2.2 *Strops or slings***

Wires or chains should not be used as FPDs, as they do not absorb shock loads. The following points should be considered when synthetic strops or slings are used as FPDs:

- .1 where FPDs are synthetic strops or slings and no modifications are required to the lifeboat, the on-load release hook or launching equipment, a functional test should be carried out. The functional test should demonstrate, to the satisfaction of the Administration, that the equipment performs without interfering in the operation of the lifeboat or launching equipment. Strops or slings should be of resilient fibre in construction;
- .2 the strops or slings should be issued with an appropriate certificate documenting a tensile strength which provides for a factor of safety of at least six, based on the total weight of the lifeboat when loaded with its full complement of persons and equipment. The strops or slings should be inspected before use and thoroughly inspected by ship's crew every six months. The material of the strop or sling should be rot-proof, corrosion-resistant, not be unduly affected by seawater, oil or fungal attack, and UV resistant. The strops or slings should be permanently marked with the date of entry into service;



- .3 strict procedures, including a warning notice at the release handle, should be in place to ensure that the strops or slings are removed before the release mechanism is activated;
- .4 the attachment point of the strop or sling to the on-load release hook and the davit falls block should be clearly marked and designed so that any connection device such as shackles cannot be connected to either the wrong part of the block or the wrong part of the on-load release hook;
- .5 the release of the strops or slings should be achievable quickly and easily without posing any risk to the operating crew designated to carry out the task once the lifeboat has reached the water. If the release of the strops or slings requires opening of the lifeboat hatch it should be readily achievable by the operating crew at each device from within the craft. Once detached, the strops or slings should not interfere with the operation of the on-load release gear or the propeller;
- .6 once the on-load release hooks have been connected to recover the lifeboat, the strops or slings should be reattached to the lifeboat before the boat is hoisted clear of the water. The strops or slings should be designed so that they do not interfere with either the lifting or re-stowing of the lifeboat into the davits;
- .7 a strop or sling used as an FPD should be sized and arranged to allow the transfer of load from the hook mechanism to the strop with minimal movement (drop) of the boat in the event of a release mechanism failure. Should a fall preventer strop or sling be subject to an unintentional dynamic shock loading, then the strop or sling should be replaced and the associated attachment points inspected. In such cases, the Administration should be informed as soon as possible and the master should provide a full report of the circumstances of the incident; and
- .8 where provided, fall preventer strops or slings should not be used for any other purpose and should be fitted to the lifeboat at all times.

### **3 Drills, testing, inspections and maintenance of lifeboats and launching appliances**

3.1 The ship's master or the officer in charge of any lifeboat lowering or lifting operation should ensure that, where provided, lifeboat FPDs are properly in place before commencing any drill, testing, inspection or maintenance where persons are in the lifeboat.

3.2 The ship's operating crew should be familiar with the operation of the FPD fitted to the lifeboat on their ship. The procedure to be followed should be contained in the ISM Code documentation and the ship's training manual.

3.3 Those conducting training drills and drafting ISM Code procedures should take into account that with certain types of ship such as oil, gas or chemical tankers it may not be possible to use an FPD in an abandon ship situation where the release mechanism of the device is not inside the lifeboat. In such cases, the master should take this into account when considering application of paragraphs 2.1.9 or 2.2.8. Where a different procedure is followed during routine drills compared with an abandon ship situation, this should be clearly described in the ISM Code documentation and training manual.

#### **4        Modification of existing approved on-load hooks already fitted to a ship to incorporate FPDs**

The shipowner or original equipment manufacturer should contact the Administration for approval before any modification, such as modifying existing lifeboats and hooks for oil and chemical tankers so that FPDs can be released from within the lifeboat, is made to a hook, lifeboat or davit to accommodate the use of FPDs. Any retesting of any equipment should be agreed and witnessed by the Administration or a recognized organization appointed by them and documented in the relevant approval file.

---

**RESOLUTION MSC.318(89)**  
**(adopted on 20 May 2011)**

**ADOPTION OF AMENDMENTS TO THE INTERNATIONAL MARITIME  
SOLID BULK CARGOES (IMSBC) CODE**

THE MARITIME SAFETY COMMITTEE,

RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,

NOTING resolution MSC.268(85) by which it adopted the International Maritime Solid Bulk Cargoes Code (hereinafter referred to as "the IMSBC Code"), which has become mandatory under chapters VI and VII of the International Convention for the Safety of Life at Sea (SOLAS), 1974 (hereinafter referred to as "the Convention"),

NOTING ALSO article VIII(b) and regulation VI/1-1.1 of the Convention concerning the amendment procedure for amending the IMSBC Code,

HAVING CONSIDERED, at its eighty-ninth session, amendments to the IMSBC Code, proposed and circulated in accordance with article VIII(b)(i) of the Convention,

1. ADOPTS, in accordance with article VIII(b)(iv) of the Convention, amendments to the IMSBC Code, the text of which is set out in the Annex to the present resolution;
2. DETERMINES, in accordance with article VIII(b)(vi)(2)(bb) of the Convention, that the said amendments shall be deemed to have been accepted on 1 July 2012, unless, prior to that date, more than one third of the Contracting Governments to the Convention or Contracting Governments the combined merchant fleets of which constitute not less than 50% of the gross tonnage of the world's merchant fleet, have notified their objections to the amendments;
3. INVITES Contracting Governments to the Convention to note that, in accordance with article VIII(b)(vii)(2) of the Convention, the amendments shall enter into force on 1 January 2013 upon their acceptance in accordance with paragraph 2 above;
4. AGREES that Contracting Governments to the Convention may apply the aforementioned amendments in whole or in part on a voluntary basis as from 1 January 2012;
5. REQUESTS the Secretary-General, in conformity with article VIII(b)(v) of the Convention, to transmit certified copies of the present resolution and the text of the amendments contained in the Annex to all Contracting Governments to the Convention;
6. FURTHER REQUESTS the Secretary-General to transmit copies of this resolution and its Annex to Members of the Organization which are not Contracting Governments to the Convention.

ANNEX

**AMENDMENT 01-11 TO THE INTERNATIONAL MARITIME  
SOLID BULK CARGOES (IMSBC) CODE**

**APPENDIX 1**

**INDIVIDUAL SCHEDULES OF SOLID BULK CARGOES**

**ALUMINIUM FERROSILICON POWDER, UN 1395**

**PRECAUTIONS**

1 In the second sentence, the words "competent Authority" are replaced by the word "Administration".

**ALUMINIUM NITRATE, UN 1438**

**PRECAUTIONS**

2 At the end of the paragraph, the following sentence is added:

"The master and officers are to note that the ship's fixed gas fire-fighting installation will be ineffective on fires involving this cargo and that applying copious amounts of water may be necessary."

**ALUMINIUM SILICON POWDER, UNCOATED UN 1398**

**PRECAUTIONS**

3 In the second sentence, the words "competent Authority" are replaced by the word "Administration".

**AMMONIUM NITRATE UN 1942**

**with not more than 0.2% total combustible substances including any organic substance calculated as carbon, to the exclusion of any other added substance**

**DESCRIPTION**

4 The words "Supporter of combustion. Hygroscopic." are deleted.

5 A new note after Description is added, as follows:

**"Note:**

This schedule should only be used for substances that do not exhibit properties of class 1 when tested in accordance to Test Series 1 and 2 of class 1 (see UN Manual of Tests and Criteria, part 1).".

**HAZARD**

6 At the beginning of the first paragraph, the words "Oxidizer, supports combustion." are added.

7 In the existing first sentence, the words "these materials" are replaced by the words "this cargo".

**STOWAGE & SEGREGATION**

8 In the first sentence, the word "should" is replaced by the word "shall".

**LOADING**

9 In the first sentence, the references "4 and 5" are replaced by the references "4, 5 and 6".

***During loading, the following provisions shall be complied with:***

10 The first bullet related to smoking is deleted.

**PRECAUTIONS**

11 In the first paragraph, the existing third sentence is replaced by the following text:

"The master and officers are to note that the ship's fixed gas fire-extinguishing installation will be ineffective on the fire involving this cargo and that applying copious amount of water may be necessary."

12 In the first paragraph, the last sentence is deleted.

**CARRIAGE**

13 A new second sentence is added, as follows:

"The temperature of this cargo shall be monitored and recorded daily during the voyage to detect decomposition, which may result in heating and oxygen depletion."

**DISCHARGE**

14 The second sentence is replaced by the following text:

"Bunkering of fuel oil shall not be allowed. Pumping of fuel oil in spaces adjacent to the cargo spaces for this cargo, other than the engine-room, shall not be allowed."

**EMERGENCY PROCEDURES**

15 In the provisions regarding Fire in cargo space containing this material, the word "installation" is inserted after the words "Ship's fixed gas fire-extinguishing".

16 In the provisions regarding Fire in cargo space containing this material, the words "Use copious quantities of water" are replaced by the words "Use copious amount of water and isolate the source of heat, if any".

## **AMMONIUM NITRATE BASED FERTILIZER UN 2067**

### **DESCRIPTION**

17 In subparagraph .2, after the word "dolomite", the words "and/or mineral calcium sulphate" are inserted.

### **NOTES:**

18 In note 3, the word "explosive" is deleted.

### **STOWAGE & SEGREGATION**

19 In the fourth sentence, the words "any tank or double bottom" are replaced by the words "any tank, double bottom or pipe".

### **LOADING**

20 In the first sentence, the references "4 and 5" are replaced by the references "4, 5 and 6".

### **PRECAUTIONS**

21 After the first sentence, the following sentence is added:

"The master and officers are to note that the ship's fixed gas fire-extinguishing installation will be ineffective on fires involving this cargo and that applying copious amount of water may be necessary."

### **CARRIAGE**

22 In the second sentence, the words "resulting in spontaneous" are replaced by the words "which may result in".

### **DISCHARGE**

23 The first sentence is replaced by the following text:

"Bunkering of fuel oil shall not be allowed. Pumping of fuel oil in spaces adjacent to the cargo spaces for this cargo, other than the engine-room, shall not be allowed."

24 After the second new sentence, the following text is added:

"Ammonium nitrate based fertilizers are hygroscopic and may cake in overhangs, impairing safety during discharge."

### **EMERGENCY PROCEDURES**

25 In the provisions regarding Fire in cargo space containing this material, the words "Ship's fixed fire-fighting installation" are replaced by the words "Ship's fixed gas fire-extinguishing installation".

26 In the provisions regarding Fire in cargo space containing this material, in the third sentence, after the word "water", the words "and isolate the source of heat, if any" are inserted.

## **AMMONIUM NITRATE BASED FERTILIZER UN 2071**

### **DESCRIPTION**

27 In the second paragraph, a footnote associated with "(see UN Manual of Tests and Criteria, part III, subsection 38.2)" is inserted, as follows:

"\* See also section 5 of Appendix 2 to this Code."

### **HAZARD**

28 In the first paragraph, the word "mixtures" is replaced by the word "cargoes" in the first sentence, and in the last sentence the words "mixtures is subject to the" are replaced by the words "cargoes are subject to an".

### **STOWAGE & SEGREGATION**

29 In the last sentence, after the word "standard,", the words "this cargo shall be stowed" are inserted.

### **LOADING**

30 In the first paragraph, first sentence, the references "4 and 5" are replaced by the references "4, 5 and 6".

#### ***During loading, the following provisions shall be complied with:***

31 A new bullet is added, as follows:

- "• As far as reasonably practicable, combustible securing and protecting materials shall not be used. When wooden dunnage is necessary, only a minimum amount shall be used."

### **PRECAUTIONS**

32 After the first sentence, the following text is added:

"The master and officers are to note that the ship's fixed gas fire-fighting installation will be ineffective on fires involving this cargo and that applying copious amount of water may be necessary."

### **DISCHARGE**

33 The first sentence is replaced by the following text:

"Bunkering of fuel oil shall not be allowed. Pumping of fuel oil in spaces adjacent to the cargo spaces for this cargo, other than the engine-room, shall not be allowed."

34 After the second new sentence, the following sentence is added:

"Ammonium nitrate based fertilizers are hygroscopic and may cake in overhangs, impairing safety during discharge."

#### **EMERGENCY PROCEDURES**

35 In the provisions regarding Fire in cargo space containing this material, the words "Ship's fixed fire-fighting installation" are replaced by the words "Ship's fixed gas fire-extinguishing installation".

#### **AMMONIUM NITRATE BASED FERTILIZER (non-hazardous)**

##### **DESCRIPTION**

36 In subparagraph .2, after the word "dolomite", the words "and/or mineral calcium sulphate" are inserted.

37 In subparagraph .4, a footnote associated with "(see UN Manual of Tests and Criteria, part III, subsection 38.2)" is inserted, as follows:

"\* See also section 5 of Appendix 2 to this Code."

##### **HAZARD**

38 The entire text under this heading is replaced by the following text:

"This cargo is non combustible or has a low fire risk. Even though this cargo is classified as non-hazardous, some of the properties of the ammonium nitrate based fertilizer classified in class 9 under UN 2071 are exhibited when heated strongly. When this cargo is heated strongly, it will decompose and give off toxic gases with the risk of toxic fumes in the cargo space, adjacent spaces and on deck. Monitoring of the cargo temperature may give an early indication of decomposition. Fertilizer dust might be irritating to skin and mucous membranes. It is hygroscopic cargo and will cake if wet."

##### **STOWAGE & SEGREGATION**

39 In the third sentence, the words "any tank or double bottom" are replaced by the words "any tank, double bottom or pipe".

40 In the fourth sentence, after the words "this type", the word "should" is replaced by the word "shall".

41 In the fifth sentence, the word "barrier" is replaced by the word "arrangement".

42 In the last paragraph, the last sentence is replaced by the following text:

"This requirement need not apply if the bulkhead is class A-60 or to short international voyages."



**LOADING**

43 In the first paragraph, first sentence, the references "4 and 5" are replaced by the references "4, 5 and 6".

**PRECAUTIONS**

44 After the first sentence, the following text is added:

"The master and officers are to note that the ship's fixed gas fire-fighting installation will be ineffective on fires involving this cargo and that applying copious amount of water may be necessary."

**CARRIAGE**

45 The second sentence is deleted.

**DISCHARGE**

46 The first sentence is replaced by the following text:

"Bunkering of fuel oil shall not be allowed. Pumping of fuel oil in spaces adjacent to the cargo spaces for this cargo, other than the engine-room, shall not be allowed. Ammonium nitrate based fertilizers are hygroscopic and may cake in overhangs, impairing safety during discharge."

**EMERGENCY PROCEDURES**

47 In the provisions regarding Fire in cargo space containing this material, the words "Ship's fixed fire-fighting installation" are replaced by the words "Ship's fixed gas fire-extinguishing installation".

48 In the provisions regarding Fire in cargo space containing this material, in the third sentence, after the word "water", the words "and isolate the source of heat, if any" are inserted.

**AMMONIUM SULPHATE****HAZARD**

49 At the end of the paragraph, the following text is added:

"This cargo is hygroscopic and will cake if wet."

**LOADING**

50 In the last sentence, the references " 4 and 5" are replaced by the references "4, 5 and 6".

**DISCHARGE**

51 At the beginning of the paragraph, the following text is added:

"Ammonium sulphate is hygroscopic and may cake in overhangs, impairing safety during discharge."

**BARIUM NITRATE, UN 1446**

**PRECAUTIONS**

52 At the end of the paragraph, the following text is added:

"The master and officers are to note that the ship's fixed gas fire-fighting installation will be ineffective on fires involving this cargo and that applying copious amount of water may be necessary."

**BORAX (PENTAHYDRATE CRUDE)**

**DISCHARGE**

53 At the beginning of the paragraph, the following text is added:

"Borax (pentahydrate crude) is hygroscopic and may cake in overhangs, impairing safety during discharge."

**BORAX, ANHYDROUS (crude or refined)**

**LOADING**

54 The references "4 and 5" are replaced by the references "4, 5 and 6".

**DISCHARGE**

55 At the beginning of the paragraph, the following text is added:

"Borax anhydrous (crude or refined) is hygroscopic and may cake in overhangs, impairing safety during discharge."

**BROWN COAL BRIQUETTES**

**HAZARD**

56 The text is replaced by the following text:

"This cargo is easily ignited, is liable to heat spontaneously, may ignite spontaneously and may deplete oxygen in the cargo space."

**APPENDIX**

57 In paragraph 1.1, in the section for Precautions, the words "and opening cargo space enclosures" are deleted.

**CALCIUM NITRATE, UN 1454****PRECAUTIONS**

58 At the end of the paragraph, the following text is added:

"The master and officers are to note that the ship's fixed gas fire-fighting installation will be ineffective on fires involving this cargo and that applying copious amount of water may be necessary."

**DISCHARGE**

59 At the beginning of the paragraph, the following text is added:

"Calcium nitrate is hygroscopic and may cake in overhangs, impairing safety during discharge."

**CALCIUM NITRATE FERTILIZER****LOADING**

60 The references "4 and 5" are replaced by the references "4, 5 and 6".

**CLAY****CLEAN-UP**

61 The text is replaced by "After discharge of this cargo, particular attention shall be given to the bilge wells of the cargo spaces".

**DIAMMONIUM PHOSPHATE (D.A.P.)****LOADING**

62 The references "4 and 5" are replaced by the references "4, 5 and 6".

**DISCHARGE**

63 At the beginning of the paragraph, the following text is added:

"Diammonium phosphate is hygroscopic and may cake in overhangs, impairing safety during discharge."

**DISTILLERS DRIED GRAINS WITH SOLUBLES**

64 After existing schedule on DIRECT REDUCED IRON (C), the new schedule on DISTILLERS DRIED GRAINS WITH SOLUBLES is added, as follows:

**"DISTILLERS DRIED GRAINS WITH SOLUBLES****DESCRIPTION**

A dried blend of coarse grains and condensed distillers solubles that remain after the fermentation of the starch fraction of corn with yeasts and enzymes to produce ethanol and carbon dioxide. Yellowish brown in colour with a cooked corn odour. Moisture content not more than 13% and oil content not more than 11%. This schedule is not applicable to wet distillers grain (WDG) and distillers dried grain (DDG), which are not transported in bulk.

**CHARACTERISTICS**

| ANGLE OF REPOSE | BULK DENSITY (kg/m <sup>3</sup> ) | STOWAGE FACTOR (m <sup>3</sup> /t) |
|-----------------|-----------------------------------|------------------------------------|
| Not applicable  | 450 to 520                        | 1.92 to 2.22                       |
| SIZE            | CLASS                             | GROUP                              |
| Not applicable  | Not applicable                    | C                                  |

**HAZARD**

No special hazards.

This cargo is non-combustible or has a low fire risk.

**STOWAGE & SEGREGATION**

No special requirements.

**HOLD CLEANLINESS**

Clean and dry as relevant to the hazards of the cargo.

**WEATHER PRECAUTIONS**

This cargo shall be kept as dry as practicable. This cargo shall not be handled during precipitation. During handling of this cargo all non-working hatches of the cargo spaces into which this cargo is loaded shall be closed.

**LOADING**

Load in open unconfined areas. Trim in accordance with the relevant provision required under sections 4 and 5 of the Code.

**PRECAUTIONS**

No special requirements.

**VENTILATION**

No special requirements.

**CARRIAGE**

Hatches of the cargo spaces carrying this cargo shall be weathertight to prevent the ingress of water.

**DISCHARGE**

If this cargo has hardened, it shall be trimmed to avoid the formation of overhanging faces, as necessary.

**CLEAN-UP**

No special requirements."

**FERROPHOSPHORUS (including briquettes)****CARRIAGE**

65 The text is replaced by the following text:

"For quantitative measurement of flammable and toxic gases such as Phosphine, which may be evolved from this cargo in accordance with the cargo information, suitable detectors for each gas or combination of gases shall be on board while this cargo is carried. The detectors shall be of certified safe type for use in explosive atmosphere. The concentrations of these gases in the cargo spaces carrying this cargo shall be measured regularly, during the voyage, and the results of the measurements shall be recorded and kept on board."

**FERROSILICON UN 1408****APPENDIX – DETAILED REQUIREMENTS**

66 In the first sentence, the words "competent Authority" are replaced by the word "Administration", in all cases.

**FERROSILICON with 25% to 30% silicon, or 90% or more silicon****LOADING**

67 The second sentence is replaced by the following text:

"As the density of the cargo is extremely high, the tanktop may be overstressed unless the cargo is evenly spread across the tanktop for homogenous weight distribution. Due consideration shall be paid to ensure that the tanktop is not overstressed during voyage and during loading by a pile of the cargo. Refer to the appendix to this schedule."

**APPENDIX – DETAILED REQUIREMENTS**

68 In the first and second sentences, the words "competent Authority" are replaced by the word "Administration".

**FERROUS SULPHATE HEPTAHYDRATE**

69 After existing schedule on FERROUS METAL BORINGS, SHAVINGS, TURNINGS or CUTTINGS UN 2793, a new schedule on FERROUS SULPHATE HEPTAHYDRATE is added, as follows:

**"FERROUS SULPHATE HEPTAHYDRATE****DESCRIPTION**

Pale green crystals. Highly soluble in water. Product commonly referred to as "Copperas".

**CHARACTERISTICS**

| <b>ANGLE OF REPOSE</b> | <b>BULK DENSITY (kg/m<sup>3</sup>)</b> | <b>STOWAGE FACTOR (m<sup>3</sup>/t)</b> |
|------------------------|----------------------------------------|-----------------------------------------|
| Not applicable         | 750 to 1250                            | 0.8 to 1.3                              |
| <b>SIZE</b>            | <b>CLASS</b>                           | <b>GROUP</b>                            |
| Crystals               | Not applicable                         | C                                       |

**HAZARD**

Harmful if swallowed. Causes serious eye irritation. Causes skin irritation.

This cargo is non-combustible or has a low fire-risk.

Tends to caking when damp.

This cargo is highly soluble and will be acidic when wet.

Excessive levels spilt into water systems may result in oxygen depletion from the water.

**STOWAGE & SEGREGATION**

"Separated from" Oxidizing substances.

**HOLD CLEANLINESS**

Clean and dry as relevant to the hazards of the cargo.

**WEATHER PRECAUTIONS**

This cargo shall be kept as dry as practicable. It shall not be handled during precipitation. During handling of this cargo all non-working hatches of the cargo spaces into which this cargo is loaded shall be closed.

**LOADING**

Trim in accordance with the relevant provision required under sections 4 and 5 of the Code.

**PRECAUTIONS**

Avoid contact with eyes and skin. Persons who may be in contact with the product shall wear protective clothing, gloves and eye protection. Typically, this is a non-dusty product, however, in particularly dry conditions, if dust is generated a filter mask shall also be worn. Bilge wells shall be clean, dry and covered, as appropriate, to prevent ingress of the cargo.

**VENTILATION**

The cargo spaces carrying this cargo shall not be ventilated during voyage.

**CARRIAGE**

Hatches of the cargo spaces shall be weathertight to prevent water ingress.

**DISCHARGE**

If this cargo has hardened, it shall be trimmed to avoid the formation of overhangs, as necessary.

**CLEAN-UP**

After discharge of this cargo, the cargo spaces and the bilge wells shall be swept clean and then thoroughly washed out."

**FERTILIZERS WITHOUT NITRATES (non-hazardous)****DISCHARGE**

70 At the beginning of the paragraph, the following text is added:

"Fertilizers without nitrates are hygroscopic and may cake in overhangs, impairing safety during discharge."

**FLY ASH**

71 In the title of the schedule of FLY ASH, the word ", DRY" is added.

**FLY ASH, WET**

72 After existing schedule on FLY ASH, a new schedule on FLY ASH, WET is added, as follows:

**"FLY ASH, WET****DESCRIPTION**

Greyish powder. This cargo is a mixture of the light, finely divided dusty fine powder residue from coal and oil fired power stations and water (not less than 10% of water). Ammonia odour.

**CHARACTERISTICS**

| <b>ANGLE OF REPOSE</b> | <b>BULK DENSITY (kg/m<sup>3</sup>)</b> | <b>STOWAGE FACTOR (m<sup>3</sup>/t)</b> |
|------------------------|----------------------------------------|-----------------------------------------|
| Not applicable         | 900 to 1300                            | 0.77 – 1.11                             |
| <b>SIZE</b>            | <b>CLASS</b>                           | <b>GROUP</b>                            |
| Under 1 mm             | Not applicable                         | A                                       |

**HAZARD**

Wet fly ash is liable to flow if it has sufficiently high moisture content.

It is non-combustible or has a low fire-risk.

**STOWAGE & SEGREGATION**

"Separated from" foodstuffs.

**HOLD CLEANLINESS**

No special requirements.

**WEATHER PRECAUTIONS**

When a cargo is carried in a ship other than specially constructed or fitted cargo ship complying with the requirements in subsection 7.3.2 of this Code, the following provisions shall be complied with:

- .1 The moisture content of the cargo shall be kept less than its TML during voyage.
- .2 Unless expressly provided otherwise in this individual schedule, the cargo shall not be handled during precipitation.

- .3 Unless expressly provided otherwise in this individual schedule, during handling of the cargo, all non-working hatches of the cargo spaces into which the cargo is loaded or to be loaded shall be closed.
- .4 The cargo may be handled during precipitation provided that the actual moisture content of the cargo is sufficiently less than its TML so that the actual moisture content is not liable to be increased beyond the TML by the precipitation.
- .5 The cargo in a cargo space may be discharged during precipitation provided that the total amount of the cargo in the cargo space is to be discharged in the port.

#### **LOADING**

Trim in accordance with the relevant provisions required under sections 4 and 5 of the Code.

#### **PRECAUTIONS**

Bilge wells shall be clean, dry and covered as appropriate, to prevent ingress of the cargo.

#### **VENTILATION**

The cargo spaces carrying this cargo shall not be ventilated during voyage.

#### **CARRIAGE**

The appearance of the surface of this cargo shall be checked regularly during voyage. If free water above the cargo or fluid state of the cargo is observed during voyage, the master shall take appropriate actions to prevent cargo shifting and potential capsize of the ship, and give consideration to seeking emergency entry into a place of refuge.

#### **DISCHARGE**

No special requirements.

#### **CLEAN-UP**

After discharge of this cargo, the bilge wells and the scuppers of the cargo spaces shall be checked and any blockage in the bilge wells and the scuppers shall be removed."

### **GRANULAR FERROUS SULPHATE**

73 After the new schedule on FLY ASH, WET, a new schedule on GRANULAR FERROUS SULPHATE is added, as follows:

#### **"GRANULAR FERROUS SULPHATE**

##### **DESCRIPTION**

Grey to brown granules. Absorbs moisture and is highly soluble in water.



**CHARACTERISTICS**

| ANGLE OF REPOSE | BULK DENSITY (kg/m <sup>3</sup> ) | STOWAGE FACTOR (m <sup>3</sup> /t) |
|-----------------|-----------------------------------|------------------------------------|
| 30° to 45°      | 1100 to 1600                      | 0.63 – 0.9                         |
| SIZE            | CLASS                             | GROUP                              |
| Up to 15 mm     | Not applicable                    | C                                  |

**HAZARD**

Harmful if swallowed. Causes serious eye irritation. Causes skin irritation.

This cargo is non-combustible or has a low fire-risk.

Tends to caking when damp.

It is highly soluble and will be acidic when wet.

Excessive levels spilt into water systems may result in oxygen depletion from the water.

**STOWAGE & SEGREGATION**

"Separated from" oxidizing substances.

**HOLD AND CLEANLINESS**

Clean and dry as relevant to the hazards of the cargo.

**WEATHER PRECAUTIONS**

This cargo shall be kept as dry as practicable. It shall not be handled during precipitation.

During handling of this cargo all non-working hatches of the cargo spaces into which this cargo is loaded shall be closed.

**LOADING**

Trim in accordance with the relevant provision required under sections 4, 5 and 6 of the Code.

**PRECAUTIONS**

Avoid contact with eyes and skin. Persons who may be in contact with the product shall wear protective clothing, gloves and eye protection. Minimize dust generation when loading.

If dust is generated a filter mask shall also be worn.

Bilge wells shall be clean, dry and covered as appropriate, to prevent ingress of the cargo.

**VENTILATION**

The cargo spaces carrying this cargo shall not be ventilated during voyage.

**CARRIAGE**

Hatches of the cargo spaces shall be weathertight to prevent water ingress.

**DISCHARGE**

If this cargo has hardened, it shall be trimmed to avoid the formation of overhangs, as necessary.

**CLEAN-UP**

After discharge of this cargo, the cargo spaces and the bilge wells shall be swept clean and then thoroughly washed out."

## **GYPSUM**

### **WEATHER PRECAUTIONS**

74 In the second and third sentences, the words "handled" and "handling" are replaced by the words "loaded" and "loading", respectively.

## **LEAD NITRATE, UN 1469**

### **PRECAUTIONS**

75 At the end of the paragraph, the following text is added:

"The master and officers are to note that the ship's fixed gas fire-fighting installation will be ineffective on fires involving this cargo and that applying copious amount of water may be necessary."

## **MAGNESIUM NITRATE, UN 1474**

### **PRECAUTIONS**

76 The sentence is replaced by the following text:

"The master and officers are to note that the ship's fixed gas fire-fighting installation will be ineffective on fires involving this cargo and that applying copious amount of water may be necessary."

### **DISCHARGE**

77 At the beginning of the paragraph, the following text is added:

"Magnesium nitrate is hygroscopic and may cake in overhangs, impairing safety during discharge."

## **MAGNESIUM SULPHATE FERTILIZERS**

78 After existing schedule on MAGNESIUM NITRATE, UN 1474, a new schedule on MAGNESIUM SULPHATE FERTILIZERS is added, as follows:

### **"MAGNESIUM SULPHATE FERTILIZERS**

#### **DESCRIPTION**

Powdered fertilizers or fertilizer components containing magnesium sulphate.  
Grey to brown powder. Partially soluble in water and may be dusty.

**CHARACTERISTICS**

| ANGLE OF REPOSE | BULK DENSITY (kg/m <sup>3</sup> ) | STOWAGE FACTOR (m <sup>3</sup> /t) |
|-----------------|-----------------------------------|------------------------------------|
| 30° to 35°      | 850 to 1150                       | 0.87 – 1.18                        |
| SIZE            | CLASS                             | GROUP                              |
| Powder          | Not applicable                    | C                                  |

**HAZARD**

May be harmful if swallowed. May cause skin or eye irritation.

This cargo is non-combustible or has a low fire-risk.

It is partially soluble.

Dusty but may cake if wetted.

**STOWAGE & SEGREGATION**

No special requirements.

**HOLD CLEANLINESS**

Clean and dry as relevant to the hazards of the cargo.

**WEATHER PRECAUTIONS**

This cargo shall be kept as dry as practicable. It shall not be handled during precipitation.

During handling of this cargo all non-working hatches of the cargo spaces into which this cargo is loaded shall be closed.

**LOADING**

Trim in accordance with the relevant provision required under sections 4, 5 and 6 of the Code.

**PRECAUTIONS**

Avoid contact with eyes and skin. Minimize dust generation when loading. Persons who may be exposed to the dust of the cargo shall wear goggles or other equivalent dust eye protection and dust filter mask. Those persons shall wear protective clothing, as necessary. Bilge wells shall be clean, dry and covered as appropriate, to prevent ingress of the cargo.

**VENTILATION**

The cargo spaces carrying this cargo shall not be ventilated during voyage.

**CARRIAGE**

Hatches of the cargo spaces shall be weathertight to prevent water ingress.

**DISCHARGE**

If this cargo has hardened, it shall be trimmed to avoid the formation of overhangs, as necessary.

**CLEAN-UP**

After discharge of this cargo, the cargo spaces and the bilge wells shall be swept clean and then thoroughly washed out."

## **METAL SULPHIDE CONCENTRATES**

### **LOADING**

79 The text ", in particular on smaller ships, i.e. 100 m long or less." is deleted.

## **MINERAL CONCENTRATES**

### **BULK CARGO SHIPPING NAMES**

80 The sentence "All known Bulk Cargo Shipping Names (BCSN) of mineral concentrates are listed above but the list is not exhaustive" after the list of Bulk Cargo Shipping Names is deleted.

### **LOADING**

81 The text ", in particular on smaller ships, i.e. 100 m long or less." is deleted.

## **MONOAMMONIUM PHOSPHATE (M.A.P.)**

### **LOADING**

82 The references "4 and 5" are replaced by the references "4, 5 and 6".

### **DISCHARGE**

83 At the beginning of the paragraph, the following text is added:

"Monoammonium phosphate is hygroscopic and may cake in overhangs, impairing safety during discharge."

## **PHOSPHATE ROCK (calcined)**

### **DISCHARGE**

84 At the beginning of the paragraph, the following text is added:

"Phosphate rock (calcined) is hygroscopic and may cake in overhangs, impairing safety during discharge."

## **POTASH**

### **LOADING**

85 The references "4 and 5" are replaced by the references "4, 5 and 6".

**DISCHARGE**

86 At the beginning of the paragraph, the following text is added:

"Potash is hygroscopic and may cake in overhangs, impairing safety during discharge."

**POTASSIUM CHLORIDE****LOADING**

87 The references "4 and 5" are replaced by the references "4, 5 and 6".

**DISCHARGE**

88 At the beginning of the paragraph, the following text is added:

"Potassium chloride is hygroscopic and may cake in overhangs, impairing safety during discharge."

**POTASSIUM NITRATE UN 1486****LOADING**

89 The references "4 and 5" are replaced by the references "4, 5 and 6".

**PRECAUTIONS**

90 At the end of the paragraph, the following text is added:

"The master and officers are to note that the ship's fixed gas fire-fighting installation will be ineffective on fires involving this cargo and that applying copious amount of water may be necessary."

**DISCHARGE**

91 At the beginning of the paragraph, the following text is added:

"Potassium nitrate is hygroscopic and may cake in overhangs, impairing safety during discharge."

**POTASSIUM SULPHATE****LOADING**

92 The references "4 and 5" are replaced by the references "4, 5 and 6".

## **PYRITES, CALCINED (Calcined Pyrites)**

### **PRECAUTIONS**

93 In the third sentence, the words "lime before loading" are replaced by the words "protective coating such as lime-wash before loading to avoid any potential corrosive reaction between the cargo, water and steel."

## **RASORITE (ANHYDROUS)**

### **DISCHARGE**

94 At the beginning of the paragraph, the following sentence is added:

"Rasorite (anhydrous) is hygroscopic and may cake in overhangs, impairing safety during discharge."

## **SALT**

### **WEATHER PRECAUTIONS**

95 In the second and third sentences, the words "handled" and "handling" are replaced by the words "loaded" and "loading", respectively.

## **SEED CAKE, containing vegetable oil UN 1386 (b)**

### **DESCRIPTION**

96 In the last paragraph, after the first sentence, the following text is added:

"The provisions of this schedule should also not apply to mechanically expelled citrus pulp pellets containing not more than 2.5% oil and 14% oil and moisture combined."

### **PRECAUTIONS**

97 In the fifth sentence the words "it becomes apparent that fire is not liable to take place in the cargo space, to avoid the possibility of ignition of solvent vapours" are replaced by the words "fire is apparent".

### **REMARKS**

98 The first sentence is deleted.

## **SEED CAKE UN 2217**

### **PRECAUTIONS**

99 In the existing fourth sentence, the words "it becomes apparent that fire is not liable to take place in the cargo space, to avoid the possibility of ignition of solvent vapours" are replaced by the words "fire is apparent".

**REMARKS**

100 The first sentence is deleted.

**SEED CAKE (non-hazardous)****DESCRIPTION**

101 At the end of the first paragraph, the following sentence is added:

"The provisions of this schedule also apply to mechanically expelled citrus pulp pellets containing not more than 2.5% oil and 14% oil and moisture combined."

**SODIUM NITRATE, UN 1498****PRECAUTIONS**

102 At the end of the paragraph, the following text is added:

"The master and officers are to note that the ship's fixed gas fire-fighting installation will be ineffective on fires involving this cargo and that applying copious amount of water may be necessary."

**DISCHARGE**

103 At the beginning of the paragraph, the following text is added:

"Sodium nitrate is hygroscopic and may cake in overhangs, impairing safety during discharge."

**SODIUM NITRATE AND POTASSIUM NITRATE MIXTURE UN 1499****LOADING**

104 The references "4 and 5" are replaced by the references "4, 5 and 6".

**PRECAUTIONS**

105 At the end of the paragraph, the following text is added:

"The master and officers are to note that the ship's fixed gas fire-fighting installation will be ineffective on fires involving this cargo and that applying copious amount of water may be necessary."

**DISCHARGE**

106 At the beginning of the paragraph, the following text is added:

"Sodium nitrate and potassium nitrate mixture is hygroscopic and may cake in overhangs, impairing safety during discharge."

## **SULPHUR (formed, solid)**

### **PRECAUTIONS**

107 The third sentence "Holds including trimming plates and tanktops shall be treated with effective, commercially available protective coating or lime-washed to avoid any potential corrosive reaction between sulphur, water and steel" is replaced by the sentence "Holds including trimming plates and tanktops shall be treated with protective coating such as lime-wash to avoid any potential corrosive reaction between sulphur, water and steel."

## **SUPERPHOSPHATE**

### **LOADING**

108 The references "4 and 5" are replaced by the references "4, 5 and 6".

### **DISCHARGE**

109 At the beginning of the paragraph, the following text is added:

"Superphosphate is hygroscopic and may cake in overhangs, impairing safety during discharge."

## **TAPIOCA**

### **LOADING**

110 The references "4 and 5" are replaced by the references "4, 5 and 6".

## **UREA**

### **LOADING**

111 The references "4 and 5" are replaced by the references "4, 5 and 6".

### **DISCHARGE**

112 At the beginning of the paragraph, the following text is added:

"Urea is hygroscopic and may cake in overhangs, impairing safety during discharge."

## **WOOD PELLETS**

### **LOADING**

113 The references "4 and 5" are replaced by the references "4, 5 and 6".



**WOOD PRODUCTS – GENERAL**

114 After existing schedule on WOOD PELLETS, a new schedule on Wood products – General is added, as follows:

**"Wood products – General**

(See Bulk Cargo Shipping Names below)

**LOGS** **PULP WOOD**  
**TIMBER** **ROUNDWOOD**  
**SAW LOGS**

**DESCRIPTION**

This schedule only applies to wood products transported in bulk, i.e. loaded and discharged by such means as elevator or grab and which are not specifically referred to in a separate schedule.

**CHARACTERISTICS**

| <b>ANGLE OF REPOSE</b> | <b>BULK DENSITY (kg/m<sup>3</sup>)</b> | <b>STOWAGE FACTOR (m<sup>3</sup>/t)</b> |
|------------------------|----------------------------------------|-----------------------------------------|
| Not applicable         | 250 to 500                             | 2 to 4                                  |
| <b>SIZE</b>            | <b>CLASS</b>                           | <b>GROUP</b>                            |
| -                      | MHB                                    | B                                       |

**HAZARD**

These cargoes are liable to cause oxygen depletion and increase of carbon dioxide in the cargo space and adjacent spaces.

These cargoes are non-combustible or have a low fire-risk.

**STOWAGE & SEGREGATION**

No special requirements.

**HOLD CLEANLINESS**

No special requirements.

**WEATHER PRECAUTIONS**

No special requirements.

**LOADING**

Trim in accordance with the relevant provisions required under sections 4 and 5 of the Code.

**PRECAUTIONS**

Entry of personnel into cargo and adjacent confined spaces shall not be permitted until tests have been carried out and it has been established that the oxygen level is 21%. If this condition is not met, additional ventilation shall be applied to the cargo hold or adjacent enclosed spaces and remeasuring shall be conducted after a suitable interval.

An oxygen meter shall be worn and activated by all crew when entering cargo and adjacent enclosed spaces.

**VENTILATION**

Ventilation of enclosed spaces adjacent to a cargo hold before entry may be necessary even if these spaces are apparently sealed from the cargo hold.

**CARRIAGE**

No special requirements.

**DISCHARGE**

No special requirements.

**CLEAN-UP**

No special requirements.

**EMERGENCY PROCEDURES**

**SPECIAL EMERGENCY EQUIPMENT TO BE CARRIED**

Self-contained breathing apparatus and an oxygen meter should be available.

**EMERGENCY PROCEDURES**

Nil

**EMERGENCY ACTION IN THE EVENT OF FIRE**

Batten down; use ship's fixed fire-fighting installation if fitted. Exclusion of air may be sufficient to control fire.

**MEDICAL FIRST AID**

Refer to Medical First Aid Guide (MFAG), as amended.

**WOOD PULP PELLETS**

- 115 The existing schedule on WOOD PULP PELLETS is deleted in its entirety.

**APPENDIX 3**

***Properties of solid bulk cargoes***

- 116 In subsection 1.1, the following bulk cargo shipping names are deleted:

CASTOR BEANS  
SODIUM NITRATE

- 117 In subsection 1.1, the following bulk cargo shipping names are added:

GRANULAR FERROUS SULPHATE  
MAGNESIUM SULPHATE FERTILIZERS  
TAPIOCA  
WOOD PELLETS

**APPENDIX 4*****Index***

118 The BCSN "FLY ASH" is replaced by "FLY ASH, DRY".

119 The following rows are added in the table:

| <b>MATERIAL</b>                              | <b>GROUP</b> | <b>REFERENCES</b>                           |
|----------------------------------------------|--------------|---------------------------------------------|
| <b>DISTILLERS DRIED GRAINS WITH SOLUBLES</b> | <b>C</b>     |                                             |
| <b>FERROUS SULPHATE HEPTAHYDRATE</b>         | <b>C</b>     |                                             |
| <b>FLY ASH, WET</b>                          | <b>A</b>     |                                             |
| <b>GRANULAR FERROUS SULPHATE</b>             | <b>C</b>     |                                             |
| <b>LOGS</b>                                  | <b>B</b>     | <b>see Wood Products – General schedule</b> |
| <b>MAGNESIUM SULPHATE FERTILIZERS</b>        | <b>C</b>     |                                             |
| <b>PULP WOOD</b>                             | <b>B</b>     | <b>see Wood Products – General schedule</b> |
| <b>ROUNDWOOD</b>                             | <b>B</b>     | <b>see Wood Products – General schedule</b> |
| <b>SAW LOGS</b>                              | <b>B</b>     | <b>see Wood Products – General schedule</b> |
| <b>TIMBER</b>                                | <b>B</b>     | <b>see Wood Products – General schedule</b> |
| <b>Wood Products – General</b>               | <b>B</b>     |                                             |

120 The words "WOOD PULP PELLETS" and "Pellets, wood pulp" are deleted.

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**RESOLUTION MSC.319(89)**  
**(adopted on 20 May 2011)**

**ADOPTION OF AMENDMENTS TO PART B OF THE INTERNATIONAL CODE ON  
INTACT STABILITY, 2008  
(2008 IS CODE)**

THE MARITIME SAFETY COMMITTEE,

RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,

RECALLING ALSO resolution MSC.267(85) by which it adopted the International Code on Intact Stability, 2008 (2008 IS Code),

NOTING the provisions regarding the procedure for amendments to part B (recommendatory part) of the 2008 IS Code, stipulated in paragraph 27.2 of regulation II-1/2 of the International Convention for the Safety of Life at Sea (SOLAS), 1974 (1974 SOLAS Convention), as amended by resolution MSC.269(85), and in paragraph (16).2 of regulation I/3 of the Protocol of 1988 relating to the International Convention on Load Lines, 1966 (1988 Load Lines Protocol), as amended by resolution MSC.270(85),

RECOGNIZING the need to include a reference to the Code for the Construction and Equipment of Mobile Offshore Drilling Units, 2009 (2009 MODU Code) in the 2008 IS Code,

HAVING CONSIDERED, at its eighty-ninth session, the proposed amendments to part B of the 2008 IS Code, prepared by the Sub-Committee on Stability and Load Lines and on Fishing Vessels Safety, at its fifty-second session,

1. ADOPTS amendments to part B of the 2008 IS Code, the text of which is set out in the Annex to the present resolution;
2. RECOMMENDS Governments concerned to use the amendments to part B of the 2008 IS Code as a basis for relevant safety standards, unless their national stability requirements provide at least an equivalent degree of safety;
3. INVITES Contracting Governments to the 1974 SOLAS Convention and Parties to the 1988 Load Lines Protocol to note that the above amendments to the 2008 IS Code will take effect on 20 May 2011.

ANNEX

**AMENDMENTS TO PART B OF THE INTERNATIONAL CODE ON  
INTACT STABILITY, 2008 (2008 IS CODE)**

**PART B  
RECOMMENDATIONS FOR CERTAIN TYPES  
OF SHIPS AND ADDITIONAL GUIDELINES**

**CHAPTER 2  
RECOMMENDED DESIGN CRITERIA FOR CERTAIN TYPES OF SHIPS**

The existing section 2.6 is replaced by the following:

**"2.6 Mobile offshore drilling units (MODUs)**

For MODUs, constructed:

- .1 on or after 1 January 2012, the provisions of chapter 3 of the 2009 MODU Code, adopted by resolution A.1023(26), should apply;
- .2 before 1 January 2012, but on or after 1 May 1991, the provisions of chapter 3 of the 1989 MODU Code, adopted by resolution A.649(16), should apply; and
- .3 before 1 May 1991, the provisions of chapter 3 of the 1979 MODU Code, adopted by resolution A.414(XI), should apply."

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**RESOLUTION MSC.323(89)**  
**(adopted on 20 May 2011)**

**ADOPTION OF AMENDMENTS TO THE REVISED RECOMMENDATION ON TESTING  
OF LIFE-SAVING APPLIANCES (RESOLUTION MSC.81(70))**

THE MARITIME SAFETY COMMITTEE,

RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,

RECALLING ALSO resolution A.689(17) entitled "Testing of life-saving appliances", by which the Assembly, at its seventeenth session, adopted the Recommendation on testing of life-saving appliances,

RECALLING FURTHER that the Assembly, when adopting resolution A.689(17), authorized the Committee to keep the Recommendation on testing of life-saving appliances under review and to adopt, when appropriate, amendments thereto,

NOTING resolution MSC.81(70), by which, at its seventieth session, it adopted the Revised recommendation on testing of life-saving appliances, recognizing the need to introduce more precise provisions for the testing of life-saving appliances based on the requirements of the International Life-Saving Appliances (LSA) Code,

RECOGNIZING the need to harmonize amendments to the Revised recommendation on testing of life-saving appliances as adopted by resolutions MSC.200(80) and MSC.226(82),

HAVING CONSIDERED, at its eighty-ninth session, proposed amendments to the Revised recommendation on testing of life-saving appliances, prepared by the Sub-Committee on Ship Design and Equipment at its fifty-fourth session,

1. ADOPTS amendments to the Revised recommendation on testing of life-saving appliances (resolution MSC.81(70)), as amended, the text of which is set out in the Annex to the present resolution;

2. RECOMMENDS Governments to apply the annexed amendments when testing life-saving appliances.

ANNEX

**AMENDMENTS TO THE REVISED RECOMMENDATION ON  
TESTING OF LIFE-SAVING APPLIANCES  
(RESOLUTION MSC.81(70), AS AMENDED)**

**PART 1  
PROTOTYPE TESTS FOR LIFE-SAVING APPLIANCES**

**2 – LIFEJACKETS**

1 In paragraph 2.4, in the footnote, the term ":2006" is added after the referenced standard "ISO 12402-7".

2 Paragraph 2.10.4.1 is deleted and the following paragraphs are renumbered accordingly.

3 In the renumbered paragraph 2.10.4.1, the referenced ISO standards are amended as follows:

- .1 ISO 2411:1991 is replaced by ISO 2411:2000;
- .2 ISO 188 is replaced by ISO 188:2007;
- .3 ISO 4674:1977 is replaced by ISO 4674-1:2003 and ISO 4674-2:1998;
- .4 ISO 7854:1984 is replaced by ISO 7854:1995;
- .5 ISO 1421:1977 is replaced by ISO 1421:1998;
- .6 ISO 105-B02:1988 is replaced by ISO 105-B02:1994;
- .7 ISO 105-X12:1995 is replaced by ISO 105-X12:2001; and
- .8 ISO 105-E02:1978 may be replaced by ISO 105-E02:1994.

4 In the renumbered paragraph 2.10.4.3.2, in the fourth sentence, the word "show" is inserted after the word "should".

5 In the renumbered paragraph 2.10.4.5.1, the referenced standard "ISO 9227:1990" is replaced by the standard "ISO 9227:2006".

6 In the renumbered paragraph 2.10.4.6.1, in the sixth sentence, the word "points" is replaced by the words "be pointed".

7 In the renumbered paragraph 2.10.4.6.3, in the chapeau, the words "should be" are inserted after the word "lifejacket" and the word "access" is replaced by the word "assess".

**3 – IMMERSION SUITS, ANTI-EXPOSURE SUITS AND THERMAL PROTECTIVE AIDS**

8 The heading "*Flotation test*" is inserted before paragraph 3.1.7.

9 In paragraph 3.1.18:

- .1 in the first sentence, the reference to paragraph number "2.6.1" is replaced by "2.5.1" and the words "to the parts other than the lifting loop" are added at the end; and
- .2 the following new sentence is inserted between the existing first and second sentences:  
  
"For the lifting loop strength test, a force of not less than 3200 N should be applied."

10 In paragraph 3.3.9, in the second sentence, the word "conductivity" is replaced by the word "conductance" and the term "0.25 W/(m·K)" is replaced by "7,800 W/(m<sup>2</sup>·K)".

#### **4 – PYROTECHNICS – ROCKET PARACHUTE FLARES, HAND FLARES AND BUOYANT SMOKE SIGNALS**

11 In paragraph 4.8.3:

- .1 in subparagraph 4.8.3.1, the word "blown" in the first sentence is replaced by the word "drawn" and the following sentence is inserted at the end of the subparagraph:  
  
"Smoke density should be at least 70% throughout the minimum emission time.";
- .2 in subparagraph 4.8.3.2 and in the Note to the paragraph, the term "8.75 YR 6/14" is replaced by "8.75 R 6/14".

#### **5 – LIFERAFTS – RIGID AND INFLATABLE**

12 Paragraph 5.11 is replaced by the following:

##### **"5.11 Swamp test**

It should be demonstrated that the liferaft, when fully swamped, is capable of supporting its full equipment and the number of persons for which it is to be approved. It should also be demonstrated that the liferaft does not seriously deform in this condition."

13 In section 5.17, the referenced ISO standards are amended as follows:

- .1 ISO 1421 is replaced by ISO 1421:1998;
- .2 ISO 2411 is replaced by ISO 2411:2000;
- .3 ISO 4892-4:1994 is replaced by ISO 4892-4:2004;
- .4 ISO 4892-2 is replaced by ISO 4892-2:2006 with amendment 1:2009;
- .5 ISO 4675 is replaced by ISO 4675:1990;
- .6 ISO 7854 is replaced by ISO 7854:1995;



.7 ISO 5978 is replaced by ISO 5978:1990; and

.8 ISO 3011 is replaced by ISO 3011:1997.

## **6 – LIFEBOATS**

14 In paragraph 6.2.2, the references to paragraph numbers "2.7" and "2.7.6.3" are replaced by "2.6" and "2.6.6.3", respectively.

15 In paragraph 6.2.5, the references to paragraph numbers "2.7.5" and "2.7.8" are replaced by "2.6.5" and "2.6.7", respectively.

## **7 – RESCUE BOATS AND FAST RESCUE BOATS**

16 In paragraph 7.2.11, the words "full tank" are replaced by the words "fully filled fuel tank".

## **8 – LAUNCHING AND EMBARKATION APPLIANCES**

17 In paragraph 8.2.3, the referenced standard "ISO 3768:1976" is replaced by "ISO 9227:2006 – Corrosion tests in artificial atmospheres – Salt spray tests".

## **10 – POSITION INDICATING LIGHTS FOR LIFE-SAVING APPLIANCES**

18 In paragraph 10.1.2, the last sentence is amended to read as follows:

"The interior lights should provide an arithmetic mean luminous intensity of not less than 0.5 cd when measured over the entire upper hemisphere to permit reading of survival instructions and equipment instructions for a period of not less than 12 h."

19 In paragraph 10.2.2:

.1 in the first sentence, the word "and" after the term "-1°C" is deleted and the words ", and the other should be taken from ordinary room condition and operated immersed in fresh water at ambient temperature" are added at the end of the sentence; and

.2 at the beginning of the second sentence, the words "Both lights" are replaced by the words "All of the lights".

20 In section 10.4, the referenced standard "IEC 945: 3rd edition (November 1996)" is replaced by the standard "IEC 60945:2002".

## **13 – SEARCHLIGHTS FOR LIFEBOATS AND RESCUE BOATS**

21 In paragraphs 13.1 to 13.3, the referenced standards "IEC 945" and "IEC 447" are replaced by the standards "IEC 60945:2002" and "IEC 60447:2004", respectively.

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**RESOLUTION MSC.324(89)**  
**(adopted on 20 May 2011)**

**IMPLEMENTATION OF BEST MANAGEMENT PRACTICE GUIDANCE**

THE MARITIME SAFETY COMMITTEE,

RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,

RECALLING ALSO the provisions of resolution A.1026(26) on Piracy and armed robbery against ships in waters off the coast of Somalia,

RECALLING FURTHER that the Committee, in response to resolution A.1002(25) on Piracy and armed robbery against ships in waters off the coast of Somalia, and the discussions of the issue within the UN Security Council, approved revised recommendations to Governments and guidance to shipowners and ship operators, shipmasters and crews on preventing and suppressing acts of piracy and armed robbery against ships<sup>\*</sup>,

NOTING WITH APPRECIATION the development by industry organizations of specific advice in relation to the situation in waters off the coast of Somalia and, in particular, the development and publication of Best Management Practices to Deter Piracy off the Coast of Somalia and in the Arabian Sea Area<sup>\*\*</sup>,

NOTING ALSO that the UN Security Council, through resolution 1950(2010) on the situation in Somalia, welcomed the revision by IMO of its recommendations and guidance on preventing and suppressing piracy and armed robbery against ships; underlined the importance of implementing such recommendations and guidance by all stakeholders, including the shipping industry; and urged States, in collaboration with the shipping and insurance industries, and the IMO, to continue to develop and implement avoidance, evasion and defensive best practices and advisories to take when under attack or when sailing in the waters off the coast of Somalia,

FURTHER NOTING that the UN General Assembly through paragraph 95 of resolution 65/37 on Oceans and the law of the sea, has invited the Assembly to "consider adopting a resolution on commitments to Best Management Practices to avoid, deter or delay acts of piracy",

DESIRING to ensure that shipowners, shipping companies, ship operators, masters and crews take all appropriate and recommended measures to protect their ships and those on board from pirates and armed robbers,

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<sup>\*</sup> MSC.1/Circ.1333 on Recommendations to Governments for preventing and suppressing piracy and armed robbery against ships, and MSC.1/Circ.1334 on Guidance to shipowners and ship operators, shipmasters and crews on preventing and suppressing acts of piracy and armed robbery against ships.

<sup>\*\*</sup> MSC.1/Circ.1337 on Piracy and armed robbery against ships in waters off the coast of Somalia – Best Management Practices to Deter Piracy off the Coast of Somalia and in the Arabian Sea Area developed by the industry.

RECOGNIZING that it is the responsibility of flag States to ensure that ships entitled to fly their flag implement measures to protect themselves against attacks by pirates,

RECOGNIZING ALSO that measures of avoidance, evasion, deterrence and delay remain the best practices of self-protection against piracy,

RECOGNIZING FURTHER that, *inter alia*, effectively implementing the Best Management Practices significantly decreases the risk of successful pirate attacks,

1. STRONGLY URGES all those concerned to take action to ensure that as a minimum and as recommended in the Best Management Practices:

- (a) ships' masters receive updated information before and during sailing through the defined High Risk Area;
- (b) ships register with the Maritime Security Centre Horn of Africa (MSCHOA) and report to United Kingdom Maritime Trade Operations (UKMTO) Dubai; and
- (c) ships effectively implement all recommended preventive, evasive and defensive measures;

2. INVITES Member States to encourage ships flying their flag to implement the Best Management Practices while sailing in the defined High Risk Area;

3. URGES Member States and non-governmental organizations in consultative status to bring this resolution to the attention of all parties concerned.

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MSC.1/Circ.1397  
10 June 2011

## **UNIFIED INTERPRETATION OF SOLAS REGULATION III/15.1**

### **STOWAGE OF MARINE EVACUATION SYSTEMS**

1 The Maritime Safety Committee, at its eighty-ninth session (11 to 20 May 2011), with a view to ensuring a uniform approach towards the application of the provisions of SOLAS regulation III/15.1 concerning the stowage of marine evacuation systems, and following a recommendation made by the Sub-Committee on Ship Design and Equipment at its fifty-fourth session, approved the following unified interpretation:

SOLAS regulation III/15.1 requires that the ship's side shall not have any openings between the embarkation station of the marine evacuation station and the sea level in the lightest seagoing condition. This means no openings, be they permanent openings, recessed promenades or temporary openings such as shell doors, windows or ports, should be allowed in this particular area.

On passenger ships, windows and side scuttles of the non-opening type should be allowed in this area if complying with SOLAS regulation II-2/9.4.1.3.3. On cargo ships, the windows and side scuttles in the area in way of a marine evacuation system, if installed, should only be of the non-opening type.

2 Member Governments are invited to use the above interpretation from 20 May 2011 when applying the relevant provisions of SOLAS regulation III/15.1 and to bring it to the attention of all parties concerned.

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MSC.1/Circ.1398  
10 June 2011

**UNIFIED INTERPRETATION OF SOLAS REGULATION II-1/29**

1 The Maritime Safety Committee, at its eighty-ninth session (11 to 20 May), with a view to ensuring a uniform approach towards the application of the provisions of SOLAS regulation II-1/29, following a recommendation made by the Sub-Committee on Ship Design and Equipment at its fifty-fourth session, approved the annexed unified interpretation concerning mechanical, hydraulic and electrical independency and failure detection and response of steering control systems.

2 Member Governments are invited to use the annexed interpretation from 20 May 2011 when applying the relevant provisions of SOLAS regulation II-1/29 and to bring it to the attention of all parties concerned.

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## ANNEX

### MECHANICAL, HYDRAULIC AND ELECTRICAL INDEPENDENCY AND FAILURE DETECTION AND RESPONSE OF STEERING CONTROL SYSTEMS

#### 1 Scope

The interpretation applies to steering gear control systems, as defined in SOLAS regulation II-1/3.1, for the main and auxiliary steering gear, operable from the navigation bridge, for which SOLAS stipulates two steering gear control systems independent of each other (SOLAS chapter II-1, regulations 29.6.1, 29.7.2, 29.7.3, 29.15 and 29.16).

SOLAS chapter II-1, regulation 29.6.1, reads:

*"Where the main steering gear comprises two or more identical power units, an auxiliary steering gear need not be fitted, provided that:*

- .1 in a passenger ship, the main steering gear is capable of operating the rudder as required by paragraph 3.2 while any one of the power units is out of operation;*
- .2 in a cargo ship, the main steering gear is capable of operating the rudder as required by paragraph 3.2 while operating with all power units; and*
- .3 the main steering gear is so arranged that after a single failure in its piping system or in one of the power units the defect can be isolated so that steering capability can be maintained or speedily regained."*

SOLAS chapter II-1, regulations 29.7.2 and 7.3, read:

*"7 Steering gear control shall be provided:*

- .2 where the main steering gear is arranged in accordance with paragraph 6, by two independent control systems, both operable from the navigation bridge. This does not require duplication of the steering wheel or steering lever. Where the control system consists of a hydraulic telemotor, a second independent system need not be fitted, except in a tanker, chemical tanker or gas carrier of 10,000 gross tonnage and upwards; and*
- .3 for the auxiliary steering gear, in the steering gear compartment and, if power-operated, it shall also be operable from the navigation bridge and shall be independent of the control system for the main steering gear."*

SOLAS chapter II-1, regulations 29.15 and 16, read:

*"15 In every tanker, chemical tanker or gas carrier of 10,000 gross tonnage and upwards and in every other ship of 70,000 gross tonnage and upwards, the main steering gear shall comprise two or more identical power units complying with the provisions of paragraph 6.*

16 Every tanker, chemical tanker or gas carrier of 10,000 gross tonnage and upwards shall, subject to paragraph 17, comply with the following:

- .1 the main steering gear shall be so arranged that in the event of loss of steering capability due to a single failure in any part of one of the power actuating systems of the main steering gear, excluding the tiller, quadrant or components serving the same purpose, or seizure of the rudder actuators, steering capability shall be regained in not more than 45 s after the loss of one power actuating system;
- .2 the main steering gear shall comprise either:
  - .2.1 two independent and separate power actuating systems, each capable of meeting the requirements of paragraph 3.2; or
  - .2.2 at least two identical power actuating systems which, acting simultaneously in normal operation, shall be capable of meeting the requirements of paragraph 3.2. Where necessary to comply with this requirement, interconnection of hydraulic power actuating systems shall be provided. Loss of hydraulic fluid from one system shall be capable of being detected and the defective system automatically isolated so that the other actuating system or systems shall remain fully operational;
- .3 steering gears other than of the hydraulic type shall achieve equivalent standards."

The requirements of SOLAS chapter II-1, regulations 3.1, 3.3, 3.13 and 29; and IEC Publication 60092-204 – Electric and electro-hydraulic steering gear (1987) have been considered, as far as containing requirements for the independency of the control systems.

## **2 Basic requirements**

2.1 Two independent steering gear control systems should be provided and should be so arranged that a mechanical or electrical failure in one of them will not render the other one inoperative.

2.2 The term "steering gear control system" as defined in SOLAS chapter II-1, part A, regulation 3.1, should be understood as "steering control system" covering "the equipment required to control the steering gear power actuating system".

## **3 Separation of control systems and components**

### **3.1 General**

Wires, terminals and the components for duplicated steering gear control systems installed in units, control boxes, switchboards or bridge consoles should be separated as far as practicable. Where physical separation is not practicable, separation may be achieved by means of a fire retardant plate.



### 3.2 Steering wheel or steering lever

All electric components of the steering gear control systems should be duplicated. This does not require duplication of the steering wheel or steering lever.

### 3.3 Steering mode selector switch

If a joint steering mode selector switch (uniaxial switch) is employed for both steering gear control systems, the connections for the circuits of the control systems should be divided accordingly and separated from each other by an isolating plate or by air gap.

### 3.4 Follow-up amplifier

In the case of double follow-up control (see appendix, example 2), the amplifiers should be designed and fed so as to be electrically and mechanically separated. In the case of non-follow-up control and follow-up control, it should be ensured that the follow-up amplifiers are protected selectively (see appendix, example 3).

### 3.5 Additional control systems

Control circuits for additional control systems, e.g., steering lever or autopilot should be designed for all-pole disconnection (see appendix, examples 1, 2 and 3).

### 3.6 Feed-back units and limit switches

The feed-back units and limit switches, if any, for the steering gear control systems should be separated electrically and mechanically connected to the rudder stock or actuator separately.

### 3.7 Hydraulic control components

3.7.1 Hydraulic system components in the power actuating or hydraulic servo systems controlling the power systems of the steering gear (e.g., solenoid valves, magnetic valves) should be considered as part of the steering gear control system and should be duplicated and separated.

3.7.2 Hydraulic system components in the steering gear control system that are part of a power unit may be regarded as being duplicated and separated when there are two or more separate power units provided and the piping to each power unit can be isolated.

## 4 Failure detection and response of control systems

### 4.1 Failure detection

4.1.1 The most probable failures that may cause reduced or erroneous system performance should be detected, and should consider at least the following:

- .1 power supply failure;
- .2 loop failures in closed loop systems, both command and feedback loops (normally short circuit, broken connections and earth faults);

- .3 if programmable electronic systems are used:
  - .1 data communication errors; and
  - .2 computer hardware and software failures; and
- .4 hydraulic locking considering order given by steering wheel or lever.

4.1.2 All failures detected should initiate an audible and visual alarm on the navigation bridge. Hydraulic locking should always be warned individually unless system design makes manual action unnecessary.

**Note:** "Hydraulic locking" includes all situations where two hydraulic systems (usually identical) oppose each other in such a way that it may lead to loss of steering. It can either be caused by pressure in the two hydraulic systems working against each other or by hydraulic "by-pass" meaning that the systems puncture each other and cause pressure drop on both sides or make it impossible to build up pressure.

4.1.3 Alternatively to 4.1.1.2 and 4.1.1.3, depending on the rudder characteristic, critical deviations between rudder order and response should be indicated visually and audibly as steering failure alarm on the navigating bridge.

4.1.4 The following parameters should be monitored:

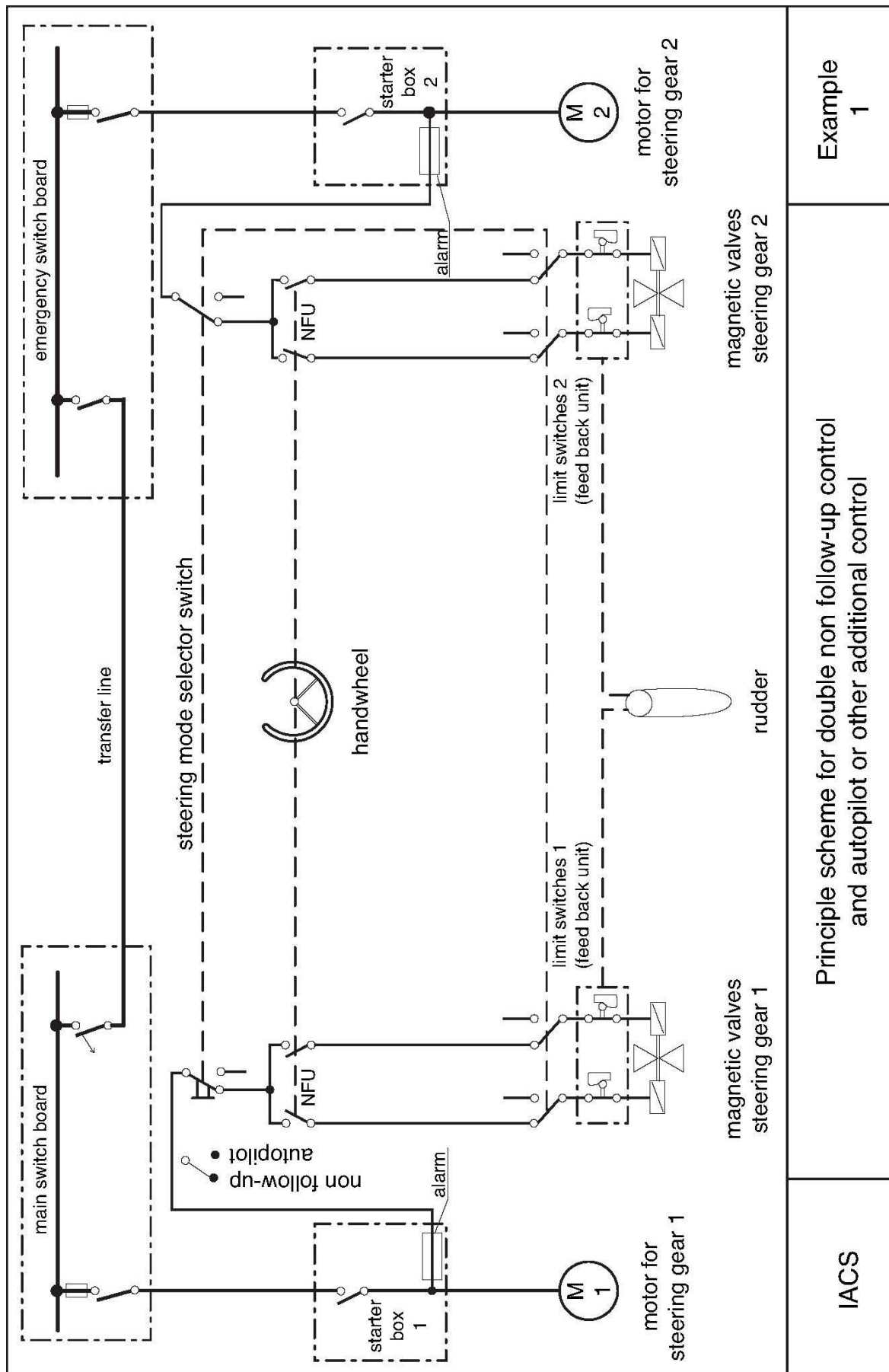
- .1 Direction: Actual rudder position follows the set value.
- .2 Delay: Rudder's actual position reaches set position within acceptable time limits.
- .3 Accuracy: The end actual position should correspond to the set value within the design offset tolerances.

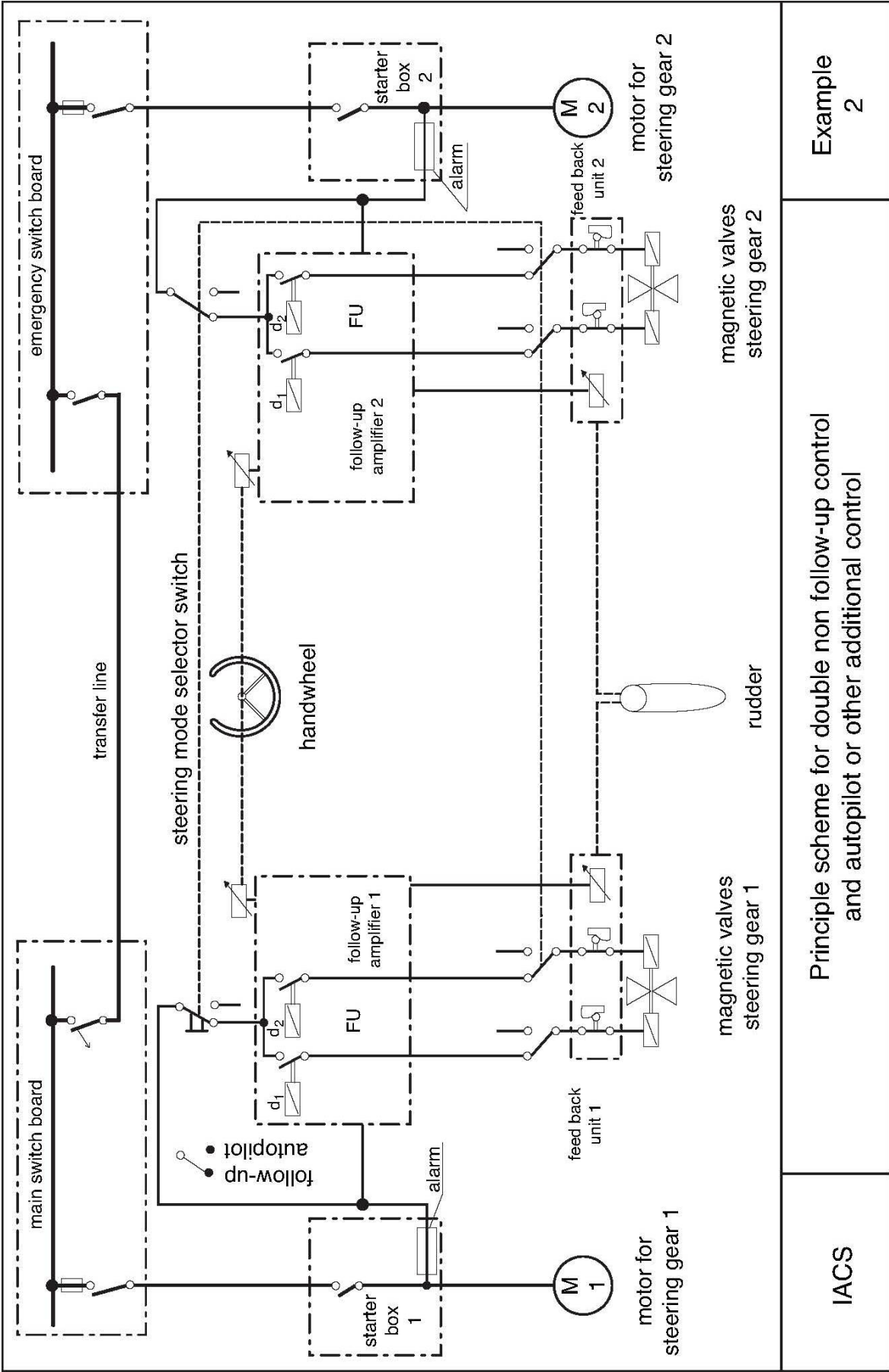
4.2 System response upon failure

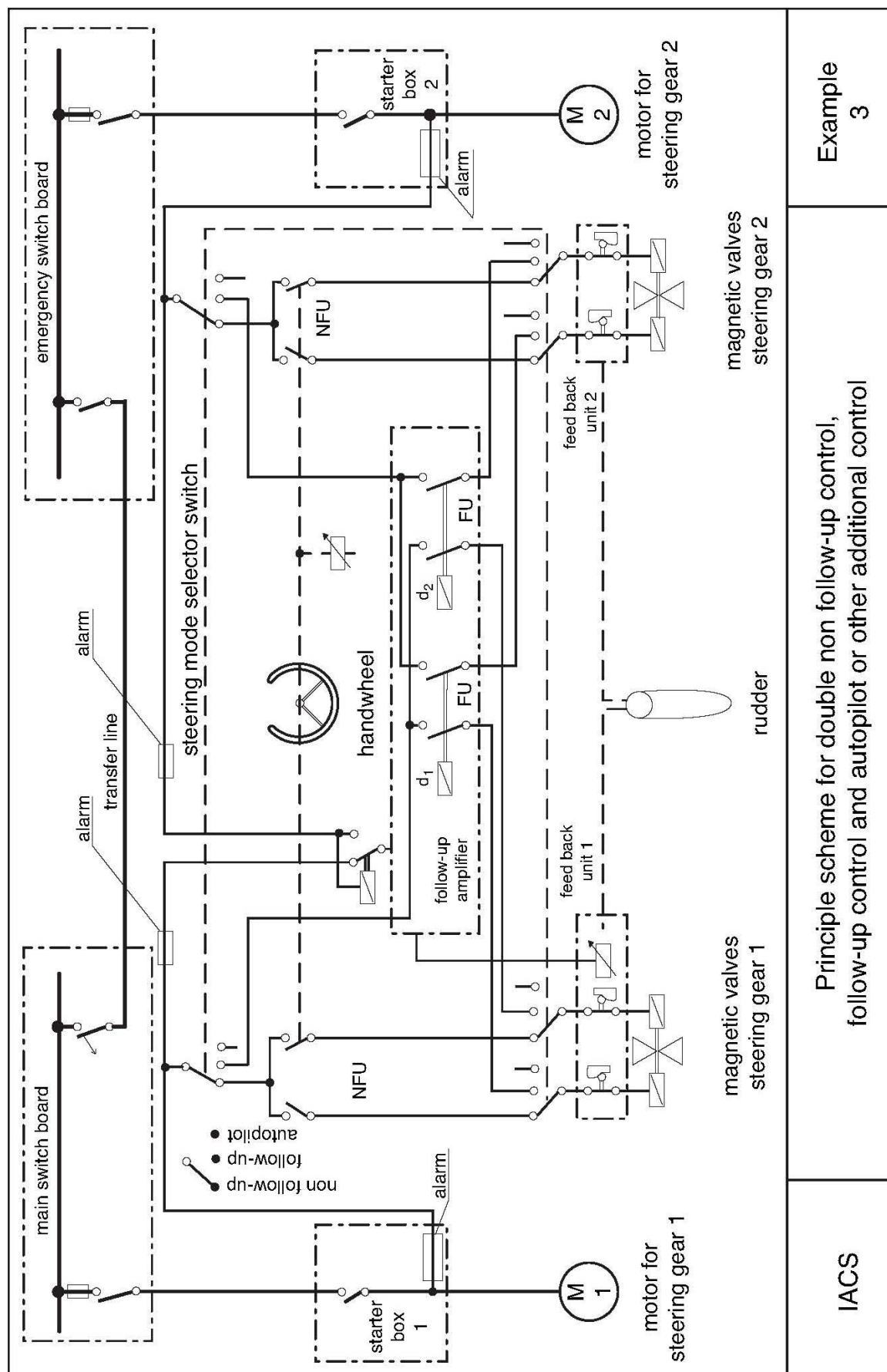
The most probable failures, e.g., loss of power or loop failure, should result in the least critical of any new possible conditions.

## Appendix

Reference should be made to the following examples 1, 2 and 3, which can be regarded as basic design.







**ANNEX 23**

**DRAFT ASSEMBLY RESOLUTION**

**ADOPTION OF THE REVISED RECOMMENDATIONS FOR ENTERING  
ENCLOSED SPACES ABOARD SHIPS**

THE ASSEMBLY,

RECALLING Article 15(j) of the Convention on the International Maritime Organization concerning the functions of the Assembly in relation to regulations and guidelines concerning maritime safety,

RECALLING ALSO that it adopted, by resolution A.864(20), the Recommendations for entering enclosed spaces aboard ships, incorporating therein recommendations for entering cargo spaces, tanks, pump-rooms, fuel tanks, cofferdams, duct keels, ballast tanks and similar enclosed spaces,

BEING CONCERNED about at the continued loss of life resulting from personnel entering shipboard spaces in which the atmosphere is oxygen-depleted, oxygen enriched, toxic or flammable,

BEING AWARE of the work undertaken in this regard by the International Labour Organization, Governments and segments of the private sector,

HAVING CONSIDERED the recommendation made by the Maritime Safety Committee at its eighty-ninth session,

1. ADOPTS the Revised Recommendations for entering enclosed spaces aboard ships, set out in the Annex to the present resolution;
2. INVITES Governments to bring the annexed Revised Recommendations to the attention of shipowners, ship operators and seafarers, urging them to apply the Revised Recommendations, as appropriate, to all ships;
3. REQUESTS the Maritime Safety Committee to keep the Revised Recommendations under review and amend them, as necessary;
4. REVOKES resolution A.864(20).

## ANNEX

### REVISED RECOMMENDATIONS FOR ENTERING ENCLOSED SPACES ABOARD SHIPS

#### PREAMBLE

The objective of these Recommendations is to encourage the adoption of safety procedures aimed at preventing casualties to ships' personnel entering enclosed spaces where there may be an oxygen deficient, oxygen enriched, flammable and/or toxic atmosphere.

Investigations into the circumstances of casualties that have occurred have shown that accidents on board ships are in most cases caused by an insufficient knowledge of, or disregard for, the need to take precautions rather than a lack of guidance.

The following practical Recommendations apply to all types of ships and provide guidance to ship operators and seafarers. It should be noted that on ships where entry into enclosed spaces may be infrequent, for example, on certain passenger ships or small general cargo ships, the dangers may be less apparent, and accordingly there may be a need for increased vigilance.

The Recommendations are intended to complement national laws or regulations, accepted standards or particular procedures which may exist for specific trades, ships or types of shipping operations.

It may be impracticable to apply some recommendations to particular situations. In such cases, every endeavour should be made to observe the intent of the Recommendations, and attention should be paid to the risks that may be involved.

#### 1 INTRODUCTION

The atmosphere in any enclosed space may be oxygen deficient or oxygen enriched and/or contain flammable and/or toxic gases or vapours. Such unsafe atmospheres could also subsequently occur in a space previously found to be safe. Unsafe atmospheres may also be present in spaces adjacent to those spaces where a hazard is known to be present.

#### 2 DEFINITIONS

2.1 *Enclosed space* means a space which has any of the following characteristics:

- .1 limited openings for entry and exit;
- .2 inadequate ventilation; and
- .3 is not designed for continuous worker occupancy,

and includes, but is not limited to, cargo spaces, double bottoms, fuel tanks, ballast tanks, cargo pump-rooms, cargo compressor rooms, cofferdams, chain lockers, void spaces, duct keels, inter-barrier spaces, boilers, engine crankcases, engine scavenge air receivers, sewage tanks, and adjacent connected spaces. This list is not exhaustive and a list should be produced on a ship-by-ship basis to identify enclosed spaces.

2.2 *Adjacent connected space* means a normally unventilated space which is not used for cargo but which may share the same atmospheric characteristics with the enclosed space such as, but not limited to, a cargo space accessway.

2.3 *Competent person* means a person with sufficient theoretical knowledge and practical experience to make an informed assessment of the likelihood of a dangerous atmosphere being present or subsequently arising in the space.

2.4 *Responsible person* means a person authorized to permit entry into an enclosed space and having sufficient knowledge of the procedures to be established and complied with on board, in order to ensure that the space is safe for entry.

2.5 *Attendant* means a person who is suitably trained within the safety management system, maintains a watch over those entering the enclosed space, maintains communications with those inside the space and initiates the emergency procedures in the event of an incident occurring.

### **3 SAFETY MANAGEMENT FOR ENTRY INTO ENCLOSED SPACES**

3.1 The safety strategy to be adopted in order to prevent accidents on entry to enclosed spaces should be approached in a comprehensive manner by the company.

3.2 The company should ensure that the procedures for entering enclosed spaces are included among the key shipboard operations concerning the safety of the personnel and the ship, in accordance with paragraph 7 of the International Safety Management (ISM) Code.

3.3 The company should elaborate a procedural implementation scheme which provides for training in the use of atmospheric testing equipment in such spaces and a schedule of regular onboard drills for crews.

3.3.1 Competent and responsible persons should be trained in enclosed space hazard recognition, evaluation, measurement, control and elimination, using standards acceptable to the Administration.

3.3.2 Crew members should be trained, as appropriate, in enclosed space safety, including familiarization with onboard procedures for recognizing, evaluating and controlling hazards associated with entry into enclosed spaces.

3.4 Internal audits by the company and external audits by the Administration of the ship's safety management system should verify that the established procedures are complied with in practice and are consistent with the safety strategy referred to in paragraph 3.1.

### **4 ASSESSMENT OF RISK**

4.1 The company should ensure that a risk assessment is conducted to identify all enclosed spaces on board the ship. This risk assessment should be periodically revisited to ensure its continued validity.

4.2 In order to ensure safety, a competent person should always make a preliminary assessment of any potential hazards in the space to be entered, taking into account previous cargo carried, ventilation of the space, coating of the space and other relevant factors. The competent person's preliminary assessment should determine the potential for the presence of an oxygen deficient, oxygen enriched, flammable or toxic atmosphere. The competent person should bear in mind that the ventilation procedures for an adjacent connected space may be different from the procedures for the ventilation of the enclosed space itself.



4.3 The procedures to be followed for testing the atmosphere in the space and for entry should be decided on the basis of the preliminary assessment. These will depend on whether the preliminary assessment shows that:

- .1 there is minimal risk to the health or life of personnel entering the space; or
- .2 there is no immediate risk to health or life but a risk could arise during the course of work in the space; or
- .3 a risk to health or life is identified.

4.4 Where the preliminary assessment indicates minimal risk to health or life or potential for a risk to arise during the course of work in the space, the precautions described in sections 5, 6, 7 and 8 should be followed, as appropriate.

4.5 Where the preliminary assessment identifies a risk to life or health, if entry is to be made, the additional precautions specified in section 9 should also be followed.

4.6 Throughout the assessment process, there should be an assumption that the space to be entered is considered to be hazardous until positively proved to be safe for entry.

## **5 AUTHORIZATION OF ENTRY**

5.1 No person should open or enter an enclosed space unless authorized by the master or the nominated responsible person and unless the appropriate safety procedures laid down for the particular ship have been followed.

5.2 Entry into enclosed spaces should be planned and the use of an entry permit system, which may include the use of a checklist, is recommended. An Enclosed Space Entry Permit should be issued by the master or the nominated responsible person, and completed by the personnel who enter the space prior to entry. An example of the Enclosed Space Entry Permit is provided in the appendix.

## **6 GENERAL PRECAUTIONS**

6.1 Entry doors or hatches leading to enclosed spaces should at all times be secured against entry, when entry is not required.

6.2 A door or hatch cover which is opened to provide natural ventilation of an enclosed space may, wrongly, be taken to be an indication of a safe atmosphere and therefore, an attendant may be stationed at the entrance or the use of a mechanical barrier, such as a rope or chain positioned across the opening with an attached warning sign, could prevent such accidental entry.

6.3 The master or the responsible person should determine that it is safe to enter an enclosed space by ensuring that:

- .1 potential hazards have been identified in the assessment and as far as possible isolated or made safe;
- .2 the space has been thoroughly ventilated by natural or mechanical means to remove any toxic or flammable gases and to ensure an adequate level of oxygen throughout the space;

- .3 the atmosphere of the space has been tested as appropriate with properly calibrated instruments to ascertain acceptable levels of oxygen and acceptable levels of flammable or toxic vapours;
- .4 the space has been secured for entry and properly illuminated;
- .5 a suitable system of communication between all parties for use during entry has been agreed and tested;
- .6 an attendant has been instructed to remain at the entrance to the space whilst it is occupied;
- .7 rescue and resuscitation equipment has been positioned ready for use at the entrance to the space and rescue arrangements have been agreed;
- .8 personnel are properly clothed and equipped for the entry and subsequent tasks; and
- .9 a permit has been issued, authorizing entry.

The precautions in subparagraphs .6 and .7 may not apply to every situation described in this section. The person authorizing entry should determine whether an attendant and the positioning of rescue equipment at the entrance to the space are necessary.

6.4 Only trained personnel should be assigned the duties of entering, functioning as attendants or functioning as members of rescue teams. Ships' crews with rescue and first aid duties should be drilled periodically in rescue and first aid procedures. Training should include as a minimum:

- .1 identification of the hazards likely to be faced during entry into enclosed spaces;
- .2 recognition of the signs of adverse health effects caused by exposure to hazards during entry; and
- .3 knowledge of personal protective equipment required for entry.

6.5 All equipment used in connection with entry should be in good working condition and inspected prior to use.

## **7 TESTING THE ATMOSPHERE**

7.1 Appropriate testing of the atmosphere of a space should be carried out with properly calibrated equipment by persons trained in the use of the equipment. The manufacturers' instructions should be strictly followed. Testing of the space should be carried out before any person enters the space and at regular intervals thereafter until all work is completed. Where appropriate, the testing of the space should be carried out at as many different levels as is necessary to obtain a representative sample of the atmosphere in the space. In some cases it may be difficult to test the atmosphere throughout the enclosed space without entering the space (e.g., the bottom landing of a stairway) and this should be taken into account when assessing the risk to personnel entering the space. The use of flexible hoses or fixed sampling lines, which reach remote areas within the enclosed space, may allow for safe testing without having to enter the space.

7.2 For entry purposes, steady readings of all of the following should be obtained:

- .1 21% oxygen by volume by oxygen content meter;

**Note:** National requirements may determine the safe atmosphere range.

- .2 not more than 1% of lower flammable limit (LFL) on a suitably sensitive combustible gas indicator, where the preliminary assessment has determined that there is potential for flammable gases or vapours; and
- .3 not more than 50% of the occupational exposure limit (OEL)\* of any toxic vapours and gases.

If these conditions cannot be met, additional ventilation should be applied to the space and re-testing should be conducted after a suitable interval.

7.3 Any gas testing should be carried out with ventilation to the enclosed space stopped, and after conditions have stabilized, in order to obtain accurate readings.

7.4 Where the preliminary assessment has determined that there is potential for the presence of toxic gases and vapours, appropriate testing should be carried out, using fixed or portable gas or vapour detection equipment. The readings obtained by this equipment should be below the occupational exposure limits for the toxic gases or vapours given in accepted national or international standards, in accordance with paragraph 7.2. It should be noted that testing for flammability or oxygen content does not provide a suitable means of measuring for toxicity, nor *vice versa*.

7.5 It should be emphasized that the internal structure of the space, cargo, cargo residues and tank coatings may also present situations where oxygen deficient areas may exist, and should always be suspected, even when an enclosed space has been satisfactorily tested as being suitable for entry. This is particularly the case for spaces where the path of the supply and outlet ventilation is obstructed by structural members or cargo.

## 8 PRECAUTIONS DURING ENTRY

8.1 The atmosphere should be tested frequently whilst the space is occupied and persons should be instructed to leave the space should there be a deterioration in the conditions.

8.2 Persons entering enclosed spaces should be provided with calibrated and tested multi-gas detectors that monitor the levels of oxygen, carbon monoxide and other gases as appropriate.

8.3 Ventilation should continue during the period that the space is occupied and during temporary breaks. Before re-entry after a break, the atmosphere should be re-tested. In the event of failure of the ventilation system, any persons in the space should leave immediately.

8.4 Particular care should be exhibited when working on pipelines and valves within the space. If conditions change during the work, increased frequency of testing of the atmosphere should be performed. Changing conditions that may occur include increasing

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\* It should be noted that the term Occupational Exposure Limit (OEL) includes the Permissible Exposure Limit (PEL), Maximum Admissible Concentration (MAC) and Threshold Limit Value (TLV) or any other internationally recognized terms.

ambient temperatures, the use of oxygen-fuel torches, mobile plant, work activities in the enclosed space that could evolve vapours, work breaks, or if the ship is ballasted or trimmed during the work.

8.5 In the event of an emergency, under no circumstances should the attending crew member enter the space before help has arrived and the situation has been evaluated to ensure the safety of those entering the space to undertake rescue operations. Only properly trained and equipped personnel should perform rescue operations in enclosed spaces.

## **9 ADDITIONAL PRECAUTIONS FOR ENTRY INTO A SPACE WHERE THE ATMOSPHERE IS KNOWN OR SUSPECTED TO BE UNSAFE**

9.1 Spaces that have not been tested should be considered unsafe for persons to enter. If the atmosphere in an enclosed space is suspected or known to be unsafe, the space should only be entered when no practical alternative exists. Entry should only be made for further testing, essential operation, safety of life or safety of a ship. The number of persons entering the space should be the minimum compatible with the work to be performed.

9.2 Suitable breathing apparatus, e.g., of the air-line or self-contained type, should always be worn, and only personnel trained in its use should be allowed to enter the space. Air-purifying respirators should not be used as they do not provide a supply of clean air from a source independent of the atmosphere within the space.

9.3 Persons entering enclosed spaces should be provided with calibrated and tested multi-gas detectors that monitor the levels of oxygen, carbon monoxide and other gases as appropriate.

9.4 Rescue harnesses should be worn and, unless impractical, lifelines should be used.

9.5 Appropriate protective clothing should be worn, particularly where there is any risk of toxic substances or chemicals coming into contact with the skin or eyes of those entering the space.

9.6 The advice in paragraph 8.5 concerning emergency rescue operations is particularly relevant in this context.

## **10 HAZARDS RELATED TO SPECIFIC TYPES OF SHIPS OR CARGO**

### **10.1 Dangerous goods in packaged form**

10.1.1 The atmosphere of any space containing dangerous goods may put at risk the health or life of any person entering it. Dangers may include flammable, toxic or corrosive gases or vapours that displace oxygen, residues on packages and spilled material. The same hazards may be present in spaces adjacent to the cargo spaces. Information on the hazards of specific substances is contained in the International Maritime Dangerous Goods (IMDG) Code, the Emergency Procedures for Ships Carrying Dangerous Goods (EMS) and Material Safety Data Sheets (MSDS)\*. If there is evidence or suspicion that leakage of dangerous substances has occurred, the precautions specified in section 9 should be followed.

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Refer to the Recommendations for material safety data sheets (MSDS) for MARPOL Annex I oil cargo and oil fuel (resolution MSC.286(86)).

10.1.2 Personnel required to deal with spillages or to remove defective or damaged packages should be appropriately trained and wear suitable breathing apparatus and appropriate protective clothing.

## **10.2 Liquid bulk**

The tanker industry has produced extensive advice to operators and crews of ships engaged in the bulk carriage of oil, chemicals and liquefied gases, in the form of specialist international safety guides. Information in the guides on enclosed space entry amplifies these Recommendations and should be used as the basis for preparing entry plans.

## **10.3 Solid bulk**

On ships carrying solid bulk cargoes, dangerous atmospheres may develop in cargo spaces and adjacent spaces. The dangers may include flammability, toxicity, oxygen depletion or self-heating, as identified in the shipper's declaration. For additional information, reference should be made to the International Maritime Solid Bulk Cargoes (IMSBC) Code.

## **10.4 Use of Nitrogen as an inert gas<sup>\*</sup>**

Nitrogen is a colourless and odourless gas that, when used as an inert gas, causes oxygen deficiency in enclosed spaces and at exhaust openings on deck during purging of tanks and void spaces and use in cargo holds. It should be noted that one deep breath of 100% nitrogen gas will be fatal.

## **10.5 Oxygen-depleting cargoes and materials**

A prominent risk with such cargoes is oxygen depletion due to the inherent form of the cargo, for example, self-heating, oxidation of metals and ores or decomposition of vegetable oils, fish oils, animal fats, grain and other organic materials or their residues. The materials listed below are known to be capable of causing oxygen depletion. However, the list is not exhaustive. Oxygen depletion may also be caused by other materials of vegetable or animal origin, by flammable or spontaneously combustible materials and by materials with a high metal content, including, but not limited to:

- .1 grain, grain products and residues from grain processing (such as bran, crushed grain, crushed malt or meal), hops, malt husks and spent malt;
- .2 oilseeds as well as products and residues from oilseeds (such as seed expellers, seed cake, oil cake and meal);
- .3 copra;
- .4 wood in such forms as packaged timber, round wood, logs, pulpwood, props (pit props and other propwood), woodchips, woodshavings, wood pellets and sawdust;
- .5 jute, hemp, flax, sisal, kapok, cotton and other vegetable fibres (such as esparto grass/Spanish grass, hay, straw, bhusa), empty bags, cotton waste, animal fibres, animal and vegetable fabric, wool waste and rags;
- .6 fish, fishmeal and fishscrap;

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<sup>\*</sup> Refer to the Guidelines on tank entry for tankers using nitrogen as an inerting medium (MSC.1/Circ.1401).

- .7 guano;
- .8 sulphidic ores and ore concentrates;
- .9 charcoal, coal, lignite and coal products;
- .10 direct reduced iron (DRI);
- .11 dry ice;
- .12 metal wastes and chips, iron swarf, steel and other turnings, borings, drillings, shavings, filings and cuttings; and
- .13 scrap metal.

#### **10.6 Fumigation**

When a ship is fumigated, the detailed recommendations contained in the Recommendations on the safe use of pesticides in ships (MSC.1/Circ.1358) should be followed. Spaces adjacent to fumigated spaces should be treated as if fumigated.

#### **11 CONCLUSION**

Failure to observe simple procedures can lead to persons being unexpectedly overcome when entering enclosed spaces. Observance of the principles and procedures outlined above will form a reliable basis for assessing risks in such spaces and for taking necessary precautions.

## APPENDIX

### EXAMPLE OF AN ENCLOSED SPACE ENTRY PERMIT

This permit relates to entry into any enclosed space and should be completed by the master or responsible person and by any persons entering the space, e.g., competent person and attendant.

| <b>GENERAL</b>                                                                                                                             |                                  |                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|------------------------------------------|
| Location/name of enclosed space                                                                                                            |                                  |                                          |
| Reason for entry .....                                                                                                                     |                                  |                                          |
| This permit is valid                                                                                                                       | from: _____ hrs<br>to: _____ hrs | Date .....<br>Date .....<br>(See Note 1) |
| <b>SECTION 1 – PRE-ENTRY PREPARATION</b><br>(To be checked by the master or nominated responsible person)                                  |                                  |                                          |
|                                                                                                                                            | Yes                              | No                                       |
| • Has the space been thoroughly ventilated by mechanical means?                                                                            | .....                            | .....                                    |
| • Has the space been segregated by blanking off or isolating all connecting pipelines or valves and electrical power/equipment?            | .....                            | .....                                    |
| • Has the space been cleaned where necessary?                                                                                              | .....                            | .....                                    |
| • Has the space been tested and found safe for entry? (See note 2)                                                                         | .....                            | .....                                    |
| • Pre-entry atmosphere test readings:                                                                                                      |                                  |                                          |
| - oxygen .....% vol (21%)*                                                                                                                 | By:                              |                                          |
| - hydrocarbon .....% LFL (less than 1%)                                                                                                    |                                  |                                          |
| - toxic gases ..... ppm (less than 50% OEL of the specific gas)                                                                            | Time:                            |                                          |
|                                                                                                                                            | (See note 3)                     |                                          |
| • Have arrangements been made for frequent atmosphere checks to be made while the space is occupied and after work breaks?                 | .....                            | .....                                    |
| • Have arrangements been made for the space to be continuously ventilated throughout the period of occupation and during work breaks?..... | .....                            | .....                                    |
| • Are access and illumination adequate?                                                                                                    | .....                            | .....                                    |

\* Note that national requirements may determine the safe atmosphere range.

|                                                                                                                               | Yes | No |
|-------------------------------------------------------------------------------------------------------------------------------|-----|----|
| • Is rescue and resuscitation equipment available for immediate use by the entrance to the space?                             | "   | "  |
| • Has an attendant been designated to be in constant attendance at the entrance to the space?                                 | "   | "  |
| • Has the officer of the watch (bridge, engine-room, cargo control room) been advised of the planned entry?                   | "   | "  |
| • Has a system of communication between all parties been tested and emergency signals agreed?                                 | "   | "  |
| • Are emergency and evacuation procedures established and understood by all personnel involved with the enclosed space entry? | "   | "  |
| • Is all equipment used in good working condition and inspected prior to entry?                                               | "   | "  |
| • Are personnel properly clothed and equipped?                                                                                | "   | "  |

**SECTION 2 – PRE-ENTRY CHECKS**  
(To be checked by each person entering the space)

|                                                                                                                                                            | Yes | No |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|
| • I have received instructions or permission from the master or nominated responsible person to enter the enclosed space                                   | "   | "  |
| • Section 1 of this permit has been satisfactorily completed by the master or nominated responsible person                                                 | "   | "  |
| • I have agreed and understand the communication procedures                                                                                                | "   | "  |
| • I have agreed upon a reporting interval of ..... minutes                                                                                                 | "   | "  |
| • Emergency and evacuation procedures have been agreed and are understood                                                                                  | "   | "  |
| • I am aware that the space must be vacated immediately in the event of ventilation failure or if atmosphere tests show a change from agreed safe criteria | "   | "  |



| <b>SECTION 3 – BREATHING APPARATUS AND OTHER EQUIPMENT</b><br>(To be checked jointly by the master or nominated responsible person and the person who is to enter the space) |       |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
|                                                                                                                                                                              | Yes   | No    |
| • Those entering the space are familiar with any breathing apparatus to be used                                                                                              | ..... | ..... |
| • The breathing apparatus has been tested as follows:                                                                                                                        |       |       |
| - gauge and capacity of air supply                                                                                                                                           | ..... | ..... |
| - low pressure audible alarm if fitted                                                                                                                                       | ..... | ..... |
| - face mask – under positive pressure and not leaking                                                                                                                        | ..... | ..... |
| • The means of communication has been tested and emergency signals agreed                                                                                                    | ..... | ..... |
| • All personnel entering the space have been provided with rescue harnesses and, where practicable, lifelines                                                                | ..... | ..... |

Signed upon completion of sections 1, 2 and 3 by:

Master or nominated responsible person ..... Date ..... Time .....

Attendant ..... Date ..... Time .....

Person entering the space ..... Date ..... Time .....

| <b>SECTION 4 – PERSONNEL ENTRY</b><br>(To be completed by the responsible person supervising entry)   |      |            |
|-------------------------------------------------------------------------------------------------------|------|------------|
| Names .....                                                                                           |      |            |
| Time in ..... Time out .....                                                                          |      |            |
| <b>SECTION 5 – COMPLETION OF JOB</b><br>(To be completed by the responsible person supervising entry) |      |            |
| • Job completed                                                                                       | Date | Time ..... |
| • Space secured against entry                                                                         | Date | Time ..... |
| • The officer of the watch has been duly informed                                                     | Date | Time.....  |

Signed upon completion of sections 4 and 5 by:

Responsible person supervising entry ..... Date ..... Time .....

THIS PERMIT IS RENDERED INVALID SHOULD VENTILATION OF THE SPACE STOP  
OR IF ANY OF THE CONDITIONS NOTED IN THE CHECKLIST CHANGE

**Notes:**

- 1      The permit should contain a clear indication as to its maximum period of validity.
  - 2      In order to obtain a representative cross-section of the space's atmosphere, samples should be taken from several levels and through as many openings as possible. Ventilation should be stopped for about 10 min before the pre-entry atmosphere tests are taken.
  - 3      Tests for specific toxic contaminants, such as benzene or hydrogen sulphide, should be undertaken depending on the nature of the previous contents of the space.
-

**ANNEX 14**

**DRAFT ASSEMBLY RESOLUTION**

**ADOPTION OF THE INTERNATIONAL CODE ON THE ENHANCED  
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS  
AND OIL TANKERS, 2011 (2011 ESP CODE)**

THE ASSEMBLY,

RECALLING Article 15(j) of the Convention on the International Maritime Organization concerning the functions of the Assembly in relation to regulations and guidelines concerning maritime safety and the prevention and control of marine pollution from ships,

RECALLING ALSO resolution A.744(18) by which the Assembly adopted the Guidelines on the enhanced programme of inspections during surveys of bulk carriers and oil tankers (hereinafter referred to as "the Guidelines"),

NOTING the 1994 Conference of Contracting Governments to the Convention for the Safety of Life at Sea (SOLAS), 1974 (hereinafter referred to as "the Convention") by which it adopted regulation XI-1/2 of the Convention to make the provisions of the Guidelines mandatory under the Convention,

RECALLING FURTHER resolutions MSC.49(66), MSC.105(73), MSC.125(75), MSC.144(77), MSC.197(80), MSC.261(84) and resolution 2 of the 1997 Conference of Contracting Governments to the Convention, by which amendments to the Guidelines were adopted by the Maritime Safety Committee and the Conference of Contracting Governments to the Convention, in accordance with article VIII(b) and regulation XI-1/2 of the Convention,

RECOGNIZING that the numerous amendments made to the Guidelines necessitate a comprehensive revision in order to ensure effective implementation of their provisions and to maintain the highest practical level of safety,

HAVING CONSIDERED the recommendations made by the Maritime Safety Committee at its eighty-ninth session,

1. ADOPTS the International Code on the Enhanced Programme of Inspections during Surveys of Bulk Carriers and Oil Tankers, 2011 (2011 ESP Code), the text of which is set out in the Annex to the present resolution;
2. INVITES Contracting Governments to the Convention to note that the 2011 ESP Code will take effect upon entry into force of the associated amendments to chapter XI-1 of the Convention;
3. REQUESTS the Secretary-General of the Organization to transmit certified copies of the present resolution and the text of the 2011 ESP Code, contained in the Annex, to all Contracting Governments to the Convention after the aforementioned amendments to chapter XI-1 of the Convention have been adopted;
4. ALSO REQUESTS the Secretary-General of the Organization to transmit copies of the present resolution and the text of the Code contained in the Annex to all Members of the Organization which are not Contracting Governments to the SOLAS Convention after the aforementioned amendments to chapter XI-1 of the Convention have been adopted;

5. FURTHER REQUESTS the Maritime Safety Committee to keep the Code under review and update it as necessary, in the light of experience gained in its application.

ANNEX

**INTERNATIONAL CODE ON THE ENHANCED PROGRAMME OF INSPECTIONS  
DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS, 2011  
(2011 ESP CODE)**

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**ANNEX B**

**CODE ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF OIL TANKERS**

**Part A**

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- 1.1 Application
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**2 Renewal survey**

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- 2.4 Extent of overall and close-up surveys
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**5 Preparations for survey**

- 5.1 Survey programme
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- 5.6 Survey planning meeting

**6 Documentation on board**

- 6.1 General
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**7 Procedures for thickness measurements**

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|          |                                                                                                          |
|----------|----------------------------------------------------------------------------------------------------------|
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- 2 Renewal survey**
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- 3 Annual survey**
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- 4 Intermediate survey**
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**5 Preparations for survey**

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**6 Documentation on board**

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- Annex 6A Survey programme
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## ANNEX A

### CODE ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING THE SURVEYS OF BULK CARRIERS

#### Part A

### CODE ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS HAVING SINGLE-SIDE SKIN CONSTRUCTION

#### 1 General

##### 1.1 *Application*<sup>1</sup>

**1.1.1** The Code should apply to all self-propelled bulk carriers of 500 gross tonnage and above other than double-side skin bulk carriers as defined in paragraph 1.2.1 of part B of annex A.

**1.1.2** The Code should apply to surveys of hull structure and piping systems in way of cargo holds, cofferdams, pipe tunnels, void spaces, fuel oil tanks, within the cargo length area and all ballast tanks.

**1.1.3** The Code contains the minimum extent of examination, thickness measurements and tank testing. The survey should be extended when substantial corrosion and/or structural defects are found and include additional close-up survey when necessary.

**1.1.4** Ships subject to compliance with regulation XII/6.1 of the Convention should be subject to the additional thickness measurement guidance contained in annex 11.

**1.1.5** Ships subject to compliance with resolution MSC.168(79) should be subject to the additional thickness measurement guidance contained in annex 15.

**1.1.6** For bulk carriers with hybrid cargo hold arrangements, e.g., with some cargo holds of single-side skin and others of double-side skin, the requirements of part B of annex A apply to cargo holds of double-side skin and associated wing spaces.

**1.1.7** The surveys should be carried out during the surveys prescribed by regulation I/10 of the 1974 SOLAS Convention, as amended.

##### 1.2 *Definitions*

**1.2.1** *Bulk carrier* is a ship which is constructed generally with single deck, topside tanks and hopper side tanks in cargo spaces, and is intended primarily to carry dry cargo in bulk and includes combination carriers.<sup>2</sup>

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<sup>1</sup> The intention of the Code is to ensure that an appropriate level of review of plans and documents is conducted and consistency in application is attained. Such evaluation of survey reports, survey programmes, planning documents, etc., should be carried out at the managerial level of the Administration or organization recognized by the Administration.

<sup>2</sup> For single-side skin combination carriers additional requirements are specified in the Guidelines on the enhanced programme of inspections during surveys for oil tankers set out in part B of annex B.

**1.2.2** *Ballast tank* is a tank which is used for salt water ballast, or, where applicable, a space which is used for both cargo and salt water ballast will be treated as a ballast tank when substantial corrosion has been found in that space.

**1.2.3** *Spaces* are separate compartments including holds, tanks, cofferdams and void spaces bounding cargo holds, decks and the outer hull.

**1.2.4** *Overall survey* is a survey intended to report on the overall condition of the hull structure and determine the extent of additional close-up surveys.

**1.2.5** *Close-up survey* is a survey where the details of structural components are within the close visual inspection range of the surveyor, i.e. normally within reach of hand.

**1.2.6** *Transverse section* is the cross section of the hull perpendicular to the ship's centerline and includes all longitudinal members such as plating, longitudinals and girders at the deck, side and bottom, inner bottom and hopper side plating, longitudinal bulkheads, and bottom plating in top wing tanks.

**1.2.7** *Representative spaces* are those which are expected to reflect the condition of other spaces of similar type and service and with similar corrosion prevention systems. When selecting representative spaces account should be taken of the service and repair history on board and identifiable critical and/or suspect areas.

**1.2.8** *Suspect areas* are locations showing substantial corrosion and/or are considered by the surveyor to be prone to rapid wastage.

**1.2.9** *Substantial corrosion* is an extent of corrosion such that assessment of corrosion pattern indicates wastage in excess of 75% of allowable margins, but within acceptable limits. For ships built under the IACS Common Structural Rules, substantial corrosion is an extent of corrosion such that the assessment of the corrosion pattern indicates a gauged (or measured) thickness between  $t_{net} + 0.5$  mm and  $t_{net}$ .

**1.2.10** *A corrosion prevention system* is normally considered a full hard protective coating. Hard protective coating should usually be epoxy coating or equivalent. Other coating systems, which are neither soft nor semi-hard coatings, may be considered acceptable as alternatives provided that they are applied and maintained in compliance with the manufacturer's specifications.

**1.2.11** *Coating condition* is defined as follows:

GOOD condition with only minor spot rusting;

FAIR condition with local breakdown of coating at edges of stiffeners and weld connections and/or light rusting over 20% or more of areas under consideration, but less than as defined for POOR condition; and

POOR condition with general breakdown of coating over 20% or more of areas or hard scale at 10% or more of areas under consideration.

**1.2.12** *Critical structural areas* are locations which have been identified from calculations to require monitoring or from the service history of the subject ship or from similar or sister ships to be sensitive to cracking, buckling or corrosion which would impair the structural integrity of the ship.

**1.2.13** *Cargo length area* is that part of the ship which includes cargo holds and adjacent areas including fuel tanks, cofferdams, ballast tanks and void spaces.

**1.2.14** *Intermediate survey* is a survey carried out either at the second or third annual survey or between these surveys.

**1.2.15** *A prompt and thorough repair* is a permanent repair completed at the time of survey to the satisfaction of the surveyor, therein removing the need for the imposition of any associated condition of classification or recommendation.

**1.2.16** *Convention* means the International Convention for the Safety of Life at Sea, 1974, as amended.

**1.2.17** *Specially considered* (in connection with close-up surveys and thickness measurements) means sufficient close-up inspection and thickness measurements are taken to confirm the actual average condition of the structure under the coating.

### **1.3 Repairs**

**1.3.1** Any damage in association with wastage over the allowable limits (including buckling, grooving, detachment or fracture), or extensive areas of wastage over the allowable limits, which affects or, in the opinion of the Administration, will affect the ship's structural, watertight or weathertight integrity, should be promptly and thoroughly (see 1.2.15) repaired. Areas to be considered include:

- .1 side structure and side plating;
- .2 deck structure and deck plating;
- .3 bottom structure and bottom plating;
- .4 inner bottom structure and inner bottom plating;
- .5 inner side structure and inner side plating;
- .6 watertight or oiltight bulkheads;
- .7 hatch covers or hatch coamings; and
- .8 items in 3.3.10.

For locations where adequate repair facilities are not available, the Administration may allow the ship to proceed directly to a repair facility. This may require discharging the cargo and/or temporary repairs for the intended voyage.

**1.3.2** Additionally, when a survey results in the identification of corrosion or structural defects, either of which, in the opinion of the Administration, will impair the ship's fitness for continued service, remedial measures should be implemented before the ship continues in service.

### **1.4 Surveyors**

For bulk carriers of 20,000 tons deadweight and above, two surveyors should jointly carry out the first scheduled renewal survey after the bulk carrier passes 10 years of age (i.e. third renewal survey), and all subsequent renewal surveys and intermediate surveys.

[On bulk carriers of 100,000 tons deadweight and above, the intermediate survey between 10 and 15 years of age should be performed by two surveyors.] If the surveys are carried out by a recognized organization, the surveyors should be exclusively employed by such recognized organizations.

## **1.5      *Thickness measurements and close-up surveys***

In any kind of survey, i.e. renewal, intermediate, annual or other surveys having the scope of the foregoing ones, thickness measurements of structures in areas where close-up surveys are required should be carried out simultaneously with close-up surveys.

## **2          *Renewal survey***

### **2.1      *General***

**2.1.1** The renewal survey may be commenced at the fourth annual survey and be progressed during the succeeding year with a view to completion by the fifth anniversary date.

**2.1.2** As part of the preparation for the renewal survey, the survey programme should be dealt with in advance of the survey. The thickness measurement should not be held before the fourth annual survey.

**2.1.3** The survey should include, in addition to the requirements of the annual survey, examination, tests and checks of sufficient extent to ensure that the hull and related piping, as required in 2.1.5, is in a satisfactory condition and is fit for its intended purpose for the new period of validity of the Cargo Ship Safety Construction Certificate, subject to proper maintenance and operation and to periodical surveys being carried out at the due dates.

**2.1.4** All cargo holds, ballast tanks, pipe tunnels, cofferdams and void spaces bounding cargo holds, decks and outer hull should be examined, and this examination should be supplemented by thickness measurement and testing as required by 2.6 and 2.7, to ensure that the structural integrity remains effective. The examination should be sufficient to discover substantial corrosion, significant deformation, fractures, damages or other structural deterioration, that may be present.

**2.1.5** All piping systems within the above spaces should be examined and operationally tested under working pressure to the attending surveyor's satisfaction to ensure that the tightness and condition remain satisfactory.

**2.1.6** The survey extent of ballast tanks converted to void spaces should be specially considered in relation to the requirements for ballast tanks.

**2.1.7** Concurrent crediting to both intermediate survey and renewal survey for surveys and thickness measurements of spaces should not be acceptable.

### **2.2      *Dry dock survey***

**2.2.1** A survey in dry dock should be a part of the renewal survey. There should be a minimum of two inspections of the outside of the ship's bottom during the five-year period of the certificate. In all cases, the maximum interval between bottom inspections should not exceed 36 months.

**2.2.2** For ships of 15 years of age and over, inspection of the outside of the ship's bottom should be carried out with the ship in dry dock. For ships of less than 15 years of age, alternate inspections of the ship's bottom not conducted in conjunction with the renewal survey may be carried out with the ship afloat. Inspection of the ship afloat should only be carried out when the conditions are satisfactory and the proper equipment and suitably qualified staff is available.

**2.2.3** If a survey in dry dock is not completed in conjunction with the renewal survey or if the 36 month maximum interval referred to in 2.2.1 is not complied with, the Cargo Ship Safety Construction Certificate should cease to be valid until a survey in dry dock is completed.

**2.2.4** The overall and close-up surveys and thickness measurements, as applicable, of the lower portions of the cargo holds and ballast tanks should be carried out in accordance with the applicable requirements for renewal surveys, if not already performed.

**Note:** Lower portions of the cargo holds and ballast tanks are considered to be the parts below light ballast water line.

### **2.3      *Space protection***

**2.3.1** Where provided, the condition of the corrosion prevention system of ballast tanks should be examined. For ballast tanks, excluding double-bottom tanks, where a hard protective coating is found in POOR condition as defined in 1.2.11, and it is not renewed, or where a soft or semi-hard coating has been applied, or where a hard protective coating has not been applied from the time of construction, the tanks in question should be examined at annual intervals. Thickness measurements should be carried out as deemed necessary by the surveyor. When such breakdown of hard protective coating is found in water ballast double-bottom tanks and it is not renewed, where a soft or semi-hard coating has been applied or where a hard protective coating has not been applied from the time of construction, the tanks in question may be examined at annual intervals. When considered necessary by the surveyor, or where extensive corrosion exists, thickness measurement should be carried out.

**2.3.2** Where a hard protective coating is provided in cargo holds and is found in GOOD condition, the extent of close-up surveys and thickness measurements may be specially considered.

### **2.4      *Hatch covers and coamings***

The hatch covers and coamings should be surveyed as follows:

**2.4.1** A thorough inspection of the items listed in 3.3 should be carried out in addition to all hatch covers and coamings.

**2.4.2** Checking of the satisfactory operation of all mechanically operated hatch covers should be made, including:

- .1** stowage and securing in open condition;
- .2** proper fit and efficiency of sealing in closed condition; and
- .3** operational testing of hydraulic and power components, wires, chains, and link drives.



**2.4.3** The effectiveness of sealing arrangements of all hatch covers by hose testing or equivalent should be checked.

**2.4.4** Thickness measurement of the hatch cover and coaming plating and stiffeners should be carried out as given in annex 2.

## **2.5** *Extent of overall and close-up surveys*

**2.5.1** An overall survey of all tanks and spaces should be carried out at the renewal survey. Fuel oil tanks in the cargo length area should be surveyed as follows:

| Renewal survey<br>no.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Renewal survey<br>no.2   | Renewal survey<br>no.3    | Renewal survey<br>no.4 and subsequent |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------------------------|---------------------------------------|
| Age $\leq 5$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $5 < \text{Age} \leq 10$ | $10 < \text{Age} \leq 15$ | $15 < \text{Age}$                     |
| None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | One                      | Two                       | Half, minimum two                     |
| <b>Notes:</b> <ol style="list-style-type: none"> <li>1. These requirements apply to tanks of integral (structural) type.</li> <li>2. If a selection of tanks is accepted to be examined, then different tanks should be examined at each renewal survey, on a rotational basis.</li> <li>3. Peak tanks (all uses) should be examined internally at each renewal survey.</li> <li>4. At renewal survey no.3 and subsequent renewal surveys, one deep tank for fuel oil in the cargo area should be included, if fitted.</li> </ol> |                          |                           |                                       |

**2.5.2** The minimum requirements for close-up surveys at renewal survey are given in annex 1.

**2.5.3** The surveyor may extend the close-up survey as deemed necessary taking into account the maintenance of the spaces under survey, the condition of the corrosion prevention system and where spaces have structural arrangements or details which have suffered defects in similar spaces or on similar ships according to the available information.

**2.5.4** For areas in spaces where hard protective coatings are found to be in a GOOD condition, the extent of close-up surveys according to annex 1 may be specially considered. Refer also to 2.3.2.

## **2.6** *Extent of thickness measurements*

**2.6.1** The minimum requirements for thickness measurements at the renewal survey are given in annex 2. Annex 11 provides additional thickness measurement guidelines applicable to the vertically corrugated transverse watertight bulkhead between cargo holds no.1 and no.2 on ships subject to compliance with regulation XII/6.1 of the Convention. For additional thickness measurement guidelines applicable to the side shell frames and brackets on ships subject to compliance with resolution MSC.168(79), reference is made to 1.1.5 and annex 15.

**2.6.2** Representative thickness measurements to determine both general and local levels of corrosion in the shell frames and their end attachments in all cargo holds and ballast tanks should be carried out. Thickness measurement should also be carried out to determine the corrosion levels on the transverse bulkhead plating. The extent of thickness measurements

may be specially considered provided the surveyor is satisfied by the close-up examination, that there is no structural diminution, and the hard protective coating where applied remains efficient.

**2.6.3** Provisions for extended measurements for areas with substantial corrosion as defined in 1.2.9 are given in annex 10 and may be additionally specified in the survey programme as required in 5.1. These extended thickness measurements should be carried out before the survey is credited as completed. Suspect areas identified at previous surveys should be examined. Areas of substantial corrosion identified at previous surveys should have thickness measurements taken.

**2.6.4** The surveyor may further extend the thickness measurements as deemed necessary.

**2.6.5** For areas in tanks where hard protective coatings are found to be in GOOD condition as defined in 1.2.11, the extent of thickness measurements according to annex 2 may be specially considered by the Administration.

**2.6.6** Transverse sections should be chosen where the largest reductions are suspected to occur or are revealed from deck plating measurements.

## **2.7      *Extent of tank pressure testing***

**2.7.1** All boundaries of water ballast tanks, deep tanks and cargo holds used for water ballast within the cargo length area should be pressure tested. For fuel oil tanks, only representative tanks should be pressure tested.

**2.7.2** The surveyor may extend the tank testing as deemed necessary.

**2.7.3** Boundaries of ballast tanks should be tested with a head of liquid to the top of air pipes.

**2.7.4** Boundaries of ballast holds should be tested with a head of liquid to near the top of hatches.

**2.7.5** Boundaries of fuel oil tanks should be tested with a head of liquid to the highest point that liquid will rise under service conditions. Tank testing of fuel oil tanks may be specially considered based on a satisfactory external examination of the tank boundaries and a confirmation from the master stating that the pressure testing has been carried out according to the requirements with satisfactory results.

**2.7.6** The testing of double-bottom tanks and other spaces not designed for the carriage of liquid may be omitted, provided a satisfactory internal examination together with an examination of the tanktop is carried out.

## **2.8      *Additional renewal survey requirements after determining compliance with regulations XII/12 and XII/13 of the Convention***

**2.8.1** For ships complying with the requirements of regulation XII/12 of the Convention for hold, ballast and dry space water level detectors, the renewal survey should include an examination and a test of the water ingress detection system and of their alarms.

**2.8.2** For ships complying with the requirements of regulation XII/13 of the Convention for the availability of pumping systems, the renewal survey should include an examination and a test of the means for draining and pumping ballast tanks forward of the collision bulkhead and bilges of dry spaces any part of which extends forward of the foremost cargo hold, and of their controls.

### **3 Annual survey**

#### **3.1 General**

The annual survey should consist of an examination for the purpose of ensuring, as far as practicable, that the hull, weather decks, hatch covers, coamings and piping are maintained in a satisfactory condition and should take into account the service history, condition and extent of the corrosion prevention system of ballast tanks and areas identified in the survey report file.

#### **3.2 Examination of the hull**

**3.2.1** Examination of the hull plating and its closing appliances should be carried out as far as can be seen.

**3.2.2** Examination of watertight penetrations should be carried out as far as practicable.

#### **3.3 Examination of weather decks, hatch covers and coamings**

**3.3.1** It should be confirmed that no unapproved changes have been made to the hatch covers, hatch coamings and their securing and sealing devices since the last survey.

**3.3.2** A thorough survey of cargo hatch covers and coamings is only possible by examination in the open as well as closed positions and should include verification of proper opening and closing operation. As a result, at least the hatch covers sets within the forward 25% of the ship's length and at least one additional set, such that all the sets on the ship are assessed at least once in every five-year period, should be surveyed open, closed and in operation to the full extent in each direction at each annual survey, including:

- .1** stowage and securing in open condition;
- .2** proper fit and efficiency of sealing in closed condition; and
- .3** operational testing of hydraulic and power components, wires, chains and link drives.

The closing of the covers should include the fastening of all peripheral, and cross joint cleats or other securing devices. Particular attention should be paid to the condition of hatch covers in the forward 25% of the ship's length, where sea loads are normally greatest.

**3.3.3** If there are indications of difficulty in operating and securing hatch covers, additional sets above those required by 3.3.2, at the discretion of the surveyor, should be tested in operation.

**3.3.4** Where the cargo hatch securing system does not function properly, repairs should be carried out under the supervision of the Administration. Where hatch covers or coamings undergo substantial repairs, the strength of securing devices should be upgraded to comply with annex 13.

**3.3.5** For each cargo hatch cover set, at each annual survey, the following items should be surveyed:

- .1 cover panels, including side plates, and stiffener attachments that may be accessible in the open position by close-up survey (for corrosion, cracks, deformation);
- .2 sealing arrangements of perimeter and cross joints (gaskets for condition and permanent deformation, flexible seals on combination carriers, gasket lips, compression bars, drainage channels and non return valves);
- .3 clamping devices, retaining bars, cleating (for wastage, adjustment, and condition of rubber components);
- .4 closed cover locating devices (for distortion and attachment);
- .5 chain or rope pulleys;
- .6 guides;
- .7 guiderails and track wheels;
- .8 stoppers;
- .9 wires, chains, tensioners and gypsies;
- .10 hydraulic system, electrical safety devices and interlocks; and
- .11 end and interpanel hinges, pins and stools where fitted.

**3.3.6** At each hatchway, at each annual survey, the coamings, with plating, stiffeners and brackets should be checked for corrosion, cracks and deformation, especially of the coaming tops including close-up survey.

**3.3.7** Where considered necessary, the effectiveness of sealing arrangements may be proved by hose or chalk testing supplemented by dimensional measurements of seal compressing components.

**3.3.8** Where portable covers, wooden or steel pontoons are fitted, the satisfactory condition of the following should be confirmed, where applicable:

- .1 wooden covers and portable beams, carriers or sockets for the portable beam, and their securing devices;
- .2 steel pontoons, including close-up survey of hatch cover plating;
- .3 tarpaulins;
- .4 cleats, battens and wedges;
- .5 hatch securing bars and their securing devices;
- .6 loading pads/bars and the side plate edge;

- .7 guideplates and chocks; and
- .8 compression bars, drainage channels and drain pipes (if any).

**3.3.9** The flame screens on vents to all bunker tanks should be examined.

**3.3.10** Bunker and vent piping systems, including ventilators should be examined.

### **3.4** *Examination of cargo holds*

**3.4.1** For bulk carriers of 10 to 15 years of age, the following should be carried out:

- .1 overall survey of all cargo holds;
- .2 close-up examination of sufficient extent, minimum 25% of frames, to establish the condition of the lower region of the shell frames including approximately lower one-third length of the side frame at the side shell and side frame end attachment and the adjacent shell plating in the forward cargo hold. Where this level of survey reveals the need for remedial measures, the survey should be extended to include a close-up survey of all of the shell frames and adjacent shell plating of that cargo hold as well as a close-up survey of sufficient extent of all remaining cargo holds;
- .3 when considered necessary by the surveyor, or where extensive corrosion exists, thickness measurement should be carried out. If the results of these thickness measurements indicate that substantial corrosion is found, the extent of thickness measurements should be increased in accordance with annex 10. These extended thickness measurements should be carried out before the annual survey is credited as completed. Suspect areas identified at previous surveys should be examined. Areas of substantial corrosion identified at previous surveys should have thickness measurements taken;
- .4 where the hard protective coating in cargo holds is found to be in GOOD condition, the extent of close-up surveys and thickness measurements may be specially considered; and
- .5 all piping and penetrations in cargo holds, including overboard piping, should be examined.

**3.4.2** For bulk carriers over 15 years of age, the following should be carried out:

- .1 overall survey of all cargo holds;
- .2 close-up examination of sufficient extent, minimum 25% of frames, to establish the condition of the lower region of the shell frames including approximately lower one-third length of the side frame at the side shell and side frame end attachment and the adjacent shell plating in the forward cargo hold and one other selected cargo hold. Where this level of survey reveals the need for remedial measures, the survey should be extended to include a close-up survey of all of the shell frames and adjacent shell plating of the cargo hold as well as a close-up survey of sufficient extent of all remaining cargo holds;

- .3 when considered necessary by the surveyor, or when extensive corrosion exists, thickness measurement should be carried out. If the results of these thickness measurements indicate that substantial corrosion is found, the extent of thickness measurements should be increased in accordance with annex 10. These extended thickness measurements should be carried out before the annual survey is credited as completed. Suspect areas identified at previous surveys should be examined. Areas of substantial corrosion identified at previous surveys should have thickness measurements taken;
- .4 where hard protective coating is fitted in cargo holds and is found in GOOD condition, the extent of the close-up surveys and thickness measurements may be specially considered; and
- .5 all piping and penetrations in cargo holds, including overboard piping, should be examined.

### **3.5 Examination of ballast tanks**

Examination of ballast tanks should be carried out when required as a consequence of the results of the renewal survey and intermediate survey. When considered necessary by the Administration, or when extensive corrosion exists, thickness measurement should be carried out. If the results of these thickness measurements indicate that substantial corrosion is found, the extent of thickness measurements should be increased in accordance with annex 10. These extended thickness measurements should be carried out before the survey is credited as completed. Suspect areas identified at previous surveys should be examined. Areas of substantial corrosion identified at previous surveys should have thickness measurements taken.

### **3.6 Additional annual survey requirements for the foremost cargo hold of ships subject to regulation XII/9 of the Convention in accordance with the requirements of annex 12**

**3.6.1** Ships subject to regulation XII/9 of the Convention are those meeting all of the following conditions:

- .1 bulk carriers of 150 m in length and upwards of single-side skin construction;
- .2 carrying solid bulk cargoes having a density of 1,780 kg/m<sup>3</sup> and above;
- .3 constructed before 1 July 1999; and
- .4 constructed with an insufficient number of transverse watertight bulkheads to enable them to withstand flooding of the foremost cargo hold in all loading conditions and remain afloat in a satisfactory condition of equilibrium as specified in regulation XII/4.4 of the Convention.

**3.6.2** In accordance with regulation XII/9 of the Convention, for the foremost cargo hold of such ship the additional survey requirements listed in annex 12 should be applied.

### **3.7      *Additional annual survey requirements after determining compliance with regulations XII/12 and XII/13 of the Convention***

**3.7.1**    For ships complying with the requirements of regulation XII/12 of the Convention for hold, ballast and dry cargo space water level detectors, the annual survey should include an examination and a test, at random, of the water ingress detection systems and of their alarms.

**3.7.2**    For ships complying with the requirements of regulation XII/13 of the Convention for the availability of pumping systems, the annual survey should include an examination and a test, of the means for draining and pumping ballast tanks forward of the collision bulkhead and bilges of dry spaces any part of which extends forward of the foremost cargo hold, and of their controls.

## **4            *Intermediate survey***

### **4.1      *General***

**4.1.1**    Items that are additional to the requirements of the annual survey may be surveyed either at the second or third annual survey or between these surveys.

**4.1.2**    The extent of survey is dependent upon the age of the ship as specified in 4.2, 4.3 and 4.4.

**4.1.3**    Concurrent crediting to both intermediate survey and renewal survey for surveys and thickness measurements of space should not be acceptable.

### **4.2      *Bulk carriers 5 to 10 years of age***

#### **4.2.1    *Ballast tanks***

**4.2.1.1** For tanks used for water ballast, an overall survey of representative tanks selected by the surveyor should be carried out. The selection should include fore and aft peak tanks and a number of other tanks, taking into account the total number and type of ballast tanks. If such overall survey reveals no visible structural defects, the examination may be limited to verification that the corrosion prevention system remains efficient.

**4.2.1.2** Where POOR coating condition, corrosion or other defects are found in water ballast tanks or where hard protective coating was not applied from the time of construction, the examination should be extended to other ballast tanks of the same type.

**4.2.1.3** In ballast tanks other than double-bottom tanks, where a hard protective coating is found in POOR condition and it is not renewed, or where soft or semi-hard coating has been applied, or where a hard protective coating was not applied from the time of construction, the tanks in question should be examined and thickness measurements carried out as considered necessary at annual intervals. When such breakdown of coating is found in ballast double-bottom tanks, where a soft or semi-hard coating has been applied, or where a hard protective coating has not been applied, the tanks in question may be examined at annual intervals. When considered necessary by the surveyor, or where extensive corrosion exists, thickness measurements should be carried out.

**4.2.1.4** In addition to the requirements above, suspect areas identified at previous renewal surveys should be overall and close-up surveyed.

#### **4.2.2**    *Cargo holds*

**4.2.2.1** An overall survey of all cargo holds, including close-up survey of sufficient extent, minimum 25% of frames, should be carried out to establish the condition of:

- .1        shell frames including their upper and lower end attachments, adjacent shell plating, and transverse bulkheads in the forward cargo hold and one other selected cargo hold; and
- .2        areas found to be suspect areas at previous surveys.

**4.2.2.2** Where considered necessary by the surveyor as a result of the overall and close-up survey as described in 4.2.2.1, the survey should be extended to include a close-up survey of all of the shell frames and adjacent shell plating of that cargo hold as well as a close-up survey of sufficient extent of all remaining cargo holds.

#### **4.2.3**    *Extent of thickness measurement*

**4.2.3.1** Thickness measurement should be carried out to an extent sufficient to determine both general and local corrosion levels at areas subject to close-up survey as described in 4.2.2.1. The minimum requirement for thickness measurements at the intermediate survey are areas found to be suspect areas at previous surveys.

**4.2.3.2** The extent of thickness measurement may be specially considered provided the surveyor is satisfied by the close-up survey, that there is no structural diminution and the hard protective coatings are found to be in a GOOD condition.

**4.2.3.3** Where substantial corrosion is found, the extent of thickness measurements should be increased in accordance with the requirements of annex 10. These extended thickness measurements should be carried out before the survey is credited as completed. Suspect areas identified at previous surveys should be examined. Areas of substantial corrosion identified at previous surveys should have thickness measurements taken.

#### **4.3**        *Bulk carriers 10 to 15 years of age*

**4.3.1** The requirements of the intermediate survey should be the same extent as the previous renewal survey as required in 2 and 5.1. However, internal examination of fuel tanks and pressure testing of all tanks are not required unless deemed necessary by the attending surveyor.

**4.3.2** In application of 4.3.1, the intermediate survey may be commenced at the second annual survey and be progressed during the succeeding year with a view to completion at the third annual survey in lieu of the application of 2.1.1.

**4.3.3** In application of 4.3.1, an underwater survey may be considered in lieu of the requirements of 2.2.

#### **4.4**        *Bulk carriers exceeding 15 years of age*

**4.4.1** The requirements of the intermediate survey should be to the same extent as the previous renewal survey required in 2 and 5.1. However, internal examination of fuel tanks and pressure testing of all tanks are not required unless deemed necessary by the attending surveyor.



**4.4.2** In application of 4.4.1, the intermediate survey may be commenced at the second annual survey and be progressed during the succeeding year with a view to completion at the third annual survey in lieu of the application of 2.1.1.

**4.4.3** In application of 4.4.1, a survey in dry dock should be part of the intermediate survey. The overall and close-up surveys and thickness measurements, as applicable, of the lower portions of the cargo holds and water ballast tanks should be carried out in accordance with the applicable requirements for intermediate surveys, if not already performed.

**Note:** Lower portions of the cargo holds and ballast tanks are considered to be the parts below light ballast water line.

## **5 Preparations for survey**

### **5.1 Survey programme**

**5.1.1** The owner in co-operation with the Administration should work out a specific survey programme prior to the commencement of any part of:

- .1 the renewal survey; and
- .2 the intermediate survey for bulk carriers over 10 years of age.

The survey programme should be in a written format based on the information in annex 4A. The survey should not commence until the survey programme has been agreed.

**5.1.1.1** Prior to the development of the survey programme, the survey planning questionnaire should be completed by the owner based on the information set out in annex 4B, and forwarded to the Administration.

**5.1.1.2** The survey programme at intermediate survey may consist of the survey programme at the previous renewal survey supplemented by the condition evaluation report of that renewal survey and later relevant survey reports. The survey programme should be worked out taking into account any amendments to the survey requirements after the last renewal survey carried out.

**5.1.2** In developing the survey programme, the following documentation should be collected and consulted with a view to selecting tanks, holds, areas and structural elements to be examined:

- .1 survey status and basic ship information;
- .2 documentation on board, as described in 6.2 and 6.3;
- .3 main structural plans (scantlings drawings), including information regarding use of high-tensile steels (HTS);
- .4 relevant previous survey and inspection reports from both the classification society and the owner;
- .5 information regarding the use of ship's holds and tanks, typical cargoes and other relevant data;

- .6 information regarding corrosion prevention level on the new building; and
- .7 information regarding the relevant maintenance level during operation.

**5.1.3** The submitted survey programme should account for and comply, as a minimum, with the requirements of 2.7 and annexes 1 and 2 for close-up survey, thickness measurement and tank testing, respectively, and is to include relevant information including at least:

- .1 basic ship information and particulars;
- .2 main structural plans (scantling drawings), including information regarding use of high-tensile steels (HTS);
- .3 plan of holds and tanks;
- .4 list of holds and tanks with information on use, protection and condition of coating;
- .5 conditions for survey (e.g., information regarding hold and tank cleaning, gas-freeing, ventilation, lighting, etc.);
- .6 provisions and methods for access to structures;
- .7 equipment for surveys;
- .8 nomination of holds and tanks and areas for close-up survey (see 2.5);
- .9 nomination of sections for thickness measurement (see 2.6);
- .10 nomination of tanks for testing (see 2.7); and
- .11 damage experience related to ship in question.

**5.1.4** The Administration will advise the owner of the maximum acceptable structural corrosion diminution levels applicable to the ship.

**5.1.5** Use may also be made of the Guidelines for technical assessment in conjunction with the planning of enhanced surveys for bulk carriers contained in annex 9. These Guidelines are a recommended tool which may be invoked at the discretion of the Administration, when considered necessary and appropriate, in conjunction with the preparation of the required survey programme.

## **5.2 Conditions for survey**

**5.2.1** The owner should provide the necessary facilities for a safe execution of the survey.

**5.2.2** In order to enable the attending surveyors to carry out the survey, provisions for proper and safe access should be agreed between the owner and the Administration.

**5.2.3** Details of the means of access should be provided in the survey planning questionnaire.

**5.2.4** In cases where the provisions of safety and required access are judged by the attending surveyor(s) not to be adequate, the survey of the spaces involved should not proceed.

**5.2.5** Cargo holds, tanks and spaces should be safe for access. Cargo holds, tanks and spaces should be gas free and properly ventilated. Prior to entering a tank, void or enclosed space, it should be verified that the atmosphere in the tank is free from hazardous gas and contains sufficient oxygen.

**5.2.6** In preparation for survey and thickness measurements and to allow for a thorough examination, all spaces should be cleaned including removal from surfaces of all loose accumulated corrosion scale. Spaces should be sufficiently clean and free from water, scale, dirt, oil residues, etc., to reveal corrosion, deformation, fractures, damages or other structural deterioration as well as the condition of the coating. However, those areas of structure whose renewal has already been decided by the owner need only be cleaned and descaled to the extent necessary to determine the limits of the areas to be renewed.

**5.2.7** Sufficient illumination should be provided to reveal corrosion, deformation, fractures, damages or other structural deterioration as well as the condition of the coating.

**5.2.8** Where soft or semi-hard coatings have been applied, safe access should be provided for the surveyor to verify the effectiveness of the coating and to carry out an assessment of the conditions of internal structures which may include spot removal of the coating. When safe access cannot be provided, the soft or semi-hard coating should be removed.

**5.2.9** The surveyor(s) should always be accompanied by at least one responsible person, assigned by the owner, experienced in tank and enclosed spaces inspection. In addition a backup team of at least two experienced persons should be stationed at the hatch opening of the tank or space that is being surveyed. The back-up team should continuously observe the work in the tank or space and should keep lifesaving and evacuation equipment ready for use.

**5.2.10** A communication system should be arranged between the survey party in the cargo hold, tank or space being examined, the responsible officer on deck and, as the case may be, the navigation bridge. The communication arrangements should be maintained throughout the survey.

### **5.3 Access to structures<sup>3</sup>**

**5.3.1** For overall survey, means should be provided to enable the surveyor to examine the structure in a safe and practical way.

**5.3.2** For close-up surveys of the hull structure, other than cargo hold shell frames, one or more of the following means for access, acceptable to the surveyor, should be provided:

- .1 permanent staging and passages through structures;
- .2 temporary staging and passages through structures;
- .3 lifts and moveable platforms;
- .4 portable ladders;

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<sup>3</sup> Refer to Guidelines on the means of access to structures for inspection and maintenance of oil tankers and bulk carriers (MSC/Circ.686).

- .5 boats or rafts; and
- .6 other equivalent means.

**5.3.3** For close-up surveys of the cargo hold shell frames of bulk carriers less than 100,000 dwt, one or more of the following means for access, acceptable to the surveyor, should be provided:

- .1 permanent staging and passages through structures;
- .2 temporary staging and passages through structures;
- .3 portable ladder restricted to not more than 5 m in length may be accepted for surveys of lower section of a shell frame including bracket;
- .4 hydraulic arm vehicles such as conventional cherry pickers, lifts and movable platforms;
- .5 boats or rafts provided the structural capacity of the hold is sufficient to withstand static loads at all levels of water; and
- .6 other equivalent means.

**5.3.4** For close-up surveys of the cargo hold shell frames of bulk carriers of 100,000 dwt and above, the use of portable ladders should not be accepted and one or more of the following means for access, acceptable to the surveyor, should be provided:

Annual surveys, intermediate survey under 10 years of age and first renewal survey:

- .1 permanent staging and passages through structures;
- .2 temporary staging and passages through structures;
- .3 hydraulic arm vehicles such as conventional cherry pickers, lifts and movable platforms;
- .4 boats or rafts provided the structural capacity of the hold is sufficient to withstand static loads at all levels of water; and
- .5 other equivalent means.

Subsequent intermediate surveys and renewal surveys:

- .1 either permanent or temporary staging and passages through structures for close-up survey of at least the upper part of hold frames;
- .2 hydraulic arm vehicles such as conventional cherry pickers for surveys of lower and middle part of shell frames as alternative to staging;
- .3 lifts and movable platforms;

- .4 boats or rafts provided the structural capacity of the hold is sufficient to withstand static loads at all levels of water; and
- .5 other equivalent means.

**5.3.5** Notwithstanding the above requirements, the use of a portable ladder fitted with a mechanical device to secure the upper end of the ladder is acceptable for the "close-up examination of sufficient extent, minimum 25% of frames, to establish the condition of the lower region of the shell frames, including approximately lower one third length of side frame at side shell and side frame and attachment and the adjacent shell plating of the forward cargo hold" at annual survey, required in 3.4.1.2 and the "one other selected cargo hold" required in 3.4.2.2.

#### **5.4 *Equipment for survey***

**5.4.1** Thickness measurements should normally be carried out by means of ultrasonic test equipment. The accuracy of the equipment should be proven to the surveyor as required.

**5.4.2** One or more of the following fracture detection procedures may be required if deemed necessary by the surveyor:

- .1 radiographic equipment;
- .2 ultrasonic equipment;
- .3 magnetic particle equipment; and/or
- .4 dye penetrant.

**5.4.3** Explosimeter, oxygen-meter, breathing apparatus, lifelines, riding belts with rope and hook and whistles together with instructions and guidance on their use should be made available during the survey. A safety checklist should be provided.

**5.4.4** Adequate and safe lighting should be provided for the safe and efficient conduct of the survey.

**5.4.5** Adequate protective clothing should be made available and used (e.g., safety helmet, gloves, safety shoes, etc.) during the survey.

#### **5.5 *Surveys at sea or at anchorage***

**5.5.1** Surveys at sea or at anchorage may be accepted provided the surveyor is given the necessary assistance from the personnel on board. Necessary precautions and procedures for carrying out the survey should be in accordance with 5.1, 5.2, 5.3 and 5.4.

**5.5.2** A communication system should be arranged between the survey party in the spaces and the responsible officer on deck. This system should also include the personnel in charge of ballast pump handling if boats or rafts are used.

**5.5.3** Surveys of tanks or applicable holds by means of boats or rafts should only be undertaken with the agreement of the surveyor, who should take into account the safety arrangements provided, including weather forecasting and ship response under foreseeable conditions and provided the expected rise of water within the tank does not exceed 0.25 m.

**5.5.4** When rafts or boats will be used for close-up survey the following conditions should be observed:

- .1 only rough duty, inflatable rafts or boats, having satisfactory residual buoyancy and stability even if one chamber is ruptured, should be used;
- .2 the boat or raft should be tethered to the access ladder and an additional person should be stationed down the access ladder with a clear view of the boat or raft;
- .3 appropriate lifejackets should be available for all participants;
- .4 the surface of water in the tank or hold should be calm (under all foreseeable conditions the expected rise of water within the tank should not exceed 0.25 m) and the water level stationary. On no account should the level of the water be rising while the boat or raft is in use;
- .5 the tank, hold or space must contain clean ballast water only. Even a thin sheen of oil on the water is not acceptable; and
- .6 at no time should the water level be allowed to be within 1 m of the deepest under deck web face flat so that the survey team is not isolated from a direct escape route to the tank hatch. Filling to levels above the deck transverse should only be contemplated if a deck access manhole is fitted and open in the bay being examined, so that an escape route for the survey party is available at all times. Other effective means of escape to the deck may be considered.

**5.5.5** Rafts or boats alone may be allowed for inspection of the under deck areas for tanks or spaces, if the depth of the webs is 1.5 m or less.

**5.5.6** If the depth of the webs is more than 1.5 m, rafts or boats alone may be allowed only:

- .1 when the coating of the under-deck structure is in GOOD condition and there is no evidence of wastage; or
- .2 if a permanent means of access is provided in each bay to allow safe entry and exit. This means:
  - .1 access direct from the deck via a vertical ladder and a small platform fitted approximately 2 m below the deck in each bay; or
  - .2 access to deck from a longitudinal permanent platform having ladders to deck in each end of the tank. The platform should, for the full length of the tank, be arranged in level with, or above, the maximum water level needed for rafting of under deck structure. For this purpose, the ullage corresponding to the maximum water level should be assumed not more than 3 m from the deck plate measured at the midspan of deck transverse and the middle length of the tank.

If neither of the above conditions are met, then staging or other equivalent means should be provided for the survey of the under-deck areas.

**5.5.7** The use of rafts or boats alone in 5.5.5 and 5.5.6 does not preclude the use of boats or rafts to move about within a tank during a survey.

## **5.6 Survey planning meeting**

**5.6.1** The establishment of proper preparation and the close co-operation between the attending surveyor(s) and the owner's representatives on board prior to and during the survey are an essential part in the safe and efficient conduct of the survey. During the survey on board safety meetings should be held regularly.

**5.6.2** Prior to commencement of any part of the renewal and intermediate survey, a survey planning meeting should be held between the attending surveyor(s), the owner's representative in attendance, the thickness measurement company operator (as applicable) and the master of the ship or an appropriately qualified representative appointed by the master or company for the purpose to ascertain that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the survey work to be carried out (see also 7.1.2).

**5.6.3** The following is an indicative list of items that should be addressed in the meeting:

- .1 schedule of the vessel (i.e. the voyage, docking and undocking manoeuvres, periods alongside, cargo and ballast operations, etc.);
- .2 provisions and arrangements for thickness measurements (i.e. access, cleaning/descaling, illumination, ventilation, personal safety);
- .3 extent of the thickness measurements;
- .4 acceptance criteria (refer to the list of minimum thicknesses);
- .5 extent of close-up survey and thickness measurement considering the coating condition and suspect areas/areas of substantial corrosion;
- .6 execution of thickness measurements;
- .7 taking representative readings in general and where uneven corrosion/pitting is found;
- .8 mapping of areas of substantial corrosion; and
- .9 communication between attending surveyor(s), the thickness measurement company operator(s) and owner representative(s) concerning findings.

## **6 Documentation on board**

### **6.1 General**

**6.1.1** The owner should obtain, supply and maintain on board documentation as specified in 6.2 and 6.3, which should be readily available for the surveyor. The condition evaluation report referred to in 6.2 should include a translation into English.

**6.1.2** The documentation should be kept on board for the lifetime of the ship.

## **6.2      *Survey report file***

**6.2.1**    A survey report file should be a part of the documentation on board consisting of:

- .1          reports of structural surveys (annex 6);
- .2          condition evaluation report (annex 7); and
- .3          thickness measurement reports (annex 8).

**6.2.2**    The survey report file should be available also in the owner's and the Administration offices, or in the office of the organization recognized by the Administration.

## **6.3      *Supporting documents***

The following additional documentation should be available on board:

- .1          survey programme as required under 5.1 until such time as the renewal survey, or intermediate survey, as applicable, has been completed.
- .2          main structural plans of cargo holds and ballast tanks;
- .3          previous repair history;
- .4          cargo and ballast history;
- .5          inspections by ship's personnel with reference to:
  - .1          structural deterioration in general;
  - .2          leakages in bulkheads and piping;
  - .3          condition of coating or corrosion prevention system, if any. Guidance for reporting is shown in annex 3; and
- .6          any other information that would help to identify critical structural areas and/or suspect areas requiring inspection.

## **6.4      *Review of documentation on board***

Prior to survey, the surveyor should examine the completeness of the documentation on board, and its contents as a basis for the survey.

## **7          *Procedures for thickness measurements***

### **7.1      *General***

**7.1.1**    The required thickness measurements, if not carried out by the recognized organization acting on behalf of the Administration, should be witnessed by a surveyor of the recognized organization. The surveyor should be on board to the extent necessary to control the process.

**7.1.2**    The thickness measurement company should be part of the survey planning meeting to be held prior to commencing the survey.



**7.1.3** Thickness measurements of structures in areas where close-up surveys are required should be carried out simultaneously with close-up surveys.

**7.1.4** In all cases the extent of the thickness measurements should be sufficient as to represent the actual average condition.

**7.1.5** Procedural requirements for thickness measurements are set out in annex 14.

## **7.2      *Certification of thickness measurement company***

The thickness measurements should be carried out by a qualified company certified by an organization recognized by the Administration according to principles stated in annex 5.

## **7.3      *Reporting***

**7.3.1** A thickness measurement report should be prepared and submitted to the Administration. The report should give the location of measurements, the thickness measured as well as corresponding original thickness. Furthermore, the report should give the date when the measurements were carried out, type of measuring equipment, names of personnel and their qualifications and be signed by the operator. The thickness measurement report should follow the principles as specified in the recommended procedures for thickness measurements set out in annex 8.

**7.3.2** The surveyor should review the final thickness measurement report and countersign the cover page.

# **8          *Reporting and evaluation of survey***

## **8.1      *Evaluation of survey report***

**8.1.1** The data and information on the structural condition of the ship collected during the survey should be evaluated for acceptability and continued structural integrity of the ship.

**8.1.2** The analysis of data should be carried out and endorsed by the Administration or recognized organization authorized by the Administration and the conclusions of the analysis should form a part of the condition evaluation report.

## **8.2      *Reporting***

**8.2.1** Principles for survey reporting are shown in annex 6.

**8.2.2** When a survey is split between different survey stations, a report should be made for each portion of the survey. A list of items examined and/or tested (pressure testing, thickness measurements, etc.) and an indication of whether the item has been credited, should be made available to the next attending surveyor(s), prior to continuing or completing the survey.

**8.2.3** A condition evaluation report of the survey and results should be issued to the owner as shown in annex 7 and placed on board the ship for reference at future surveys. The condition evaluation report should be endorsed by the Administration or by the recognized organization on behalf of the Administration.

## ANNEX 1

### REQUIREMENTS FOR CLOSE-UP SURVEY AT RENEWAL SURVEYS

| <b>Age ≤ 5 years</b><br><b>Renewal Survey No.</b><br><b>1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>5 &lt; Age ≤ 10 years</b><br><b>Renewal Survey No.</b><br><b>2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>10 &lt; Age ≤ 15 years</b><br><b>Renewal Survey No.</b><br><b>3</b>                                                                                                                                                                                                                                                                                                                                                                                    | <b>Age &gt; 15 years</b><br><b>Renewal Survey No.</b><br><b>4 and subsequent</b>                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(A) 25% of shell frames in the forward cargo hold at representative positions</p> <p>(A) Selected frames in remaining cargo holds</p> <p>(B) One transverse web with associated plating and longitudinals in two representative water ballast tanks of each type (i.e. topside, or hopper side tank)</p> <p>(C) Two selected cargo hold transverse bulkheads, including internal structure of upper and lower stools, where fitted</p> <p>(D) All cargo hold hatch covers and coamings (plating and stiffeners)</p> | <p>(A) All shell frames in the forward cargo hold and 25% of shell frames in each of the remaining cargo holds including upper and lower end attachments and adjacent shell plating</p> <p>(A) For bulk carriers of 100,000 dwt and above, all shell frames in the forward cargo hold and 50% of shell frames in each of the remaining cargo holds, including upper and lower end attachments and adjacent shell plating.</p> <p>(B) One transverse web with associated plating and longitudinals in each water ballast tank</p> <p>(B) Forward and aft transverse bulkhead in one ballast tank, including stiffening system</p> <p>(C) All cargo hold transverse bulkheads, including internal structure of upper and lower stools, where fitted</p> <p>(D) All cargo hold hatch covers and coamings (plating and stiffeners)</p> <p>(E) All deck plating and under deck structure inside line of hatch openings between all cargo hold hatches</p> | <p>(A) All shell frames in the forward and one other selected cargo hold and 50% of frames in each of the remaining cargo holds, including upper and lower end attachments and adjacent shell plating</p> <p>(B) All transverse bulkheads in ballast tanks, including stiffening system</p> <p>(B) All transverse webs with associated plating and longitudinals in each water ballast tank</p> <p>Areas (C), (D), and (E) as for renewal survey no.2</p> | <p>(A) All shell frames in all cargo holds including upper and lower end attachments and adjacent shell plating.</p> <p>Areas (B)–(E) as for column 3</p> |

- (A) Cargo hold transverse frame.
- (B) Transverse web or watertight transverse bulkhead in water ballast tanks.
- (C) Cargo hold transverse bulkhead platings, stiffeners and girders.
- (D) Cargo hold hatch covers and coamings.
- (E) Deck plating inside line of hatch openings between cargo hold hatches.

See sketches of appendix 3 to annex 8 for areas corresponding to (A), (B), (C), (D) and (E).

See also sketch in annex 15 for zones of side shell frames for ships subject to compliance with resolution MSC.168(79).

**Note:** Close-up survey of transverse bulkheads to be carried out at four levels:

- Level (a) Immediately above the inner bottom and immediately above the line of gussets (if fitted) and shedders for ships without lower stool.
- Level (b) Immediately above and below the lower stool shelf plate (for those ships fitted with lower stools), and immediately above the line of the shedder plates.
- Level (c) About mid-height of the bulkhead.
- Level (d) Immediately below the upper deck plating and immediately adjacent to the upper wing tank, and immediately below the upper stool shelf plate for those ships fitted with upper stools, or immediately below the topside tank.

## ANNEX 2

### REQUIREMENTS FOR THICKNESS MEASUREMENTS AT RENEWAL SURVEYS

| Age ≤ 5 years   | 5 < Age ≤ 10 years                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 10 < Age ≤ 15 years                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Age > 15 years                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1               | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1 Suspect areas | 1 Suspect areas<br>2 Within the cargo length area:<br>two transverse sections of deck<br>plating outside line of cargo<br>hatch openings<br>3 Measurements, for general<br>assessment and recording of<br>corrosion pattern, of those<br>structural members subject to<br>close-up survey according to<br>annex 1<br>4 Wind and water strakes in way<br>of transverse sections<br>considered under point 2 above<br>5 Selected wind and water strakes<br>outside the cargo length area<br>6 See 1.1.5 and annex 15 for<br>additional thickness<br>measurement guidelines<br>applicable to the side shell<br>frames and brackets on ships<br>subject to compliance with<br>resolution MSC.168(79) | 1 Suspect areas<br>2 Within the cargo length area:<br>.1 each deck plate outside line of<br>cargo hatch openings<br>.2 two transverse sections, one of<br>which should be in the amidship<br>area, outside line of cargo hatch<br>openings<br>3 Measurement, for general assessment<br>and recording of corrosion pattern, of<br>those structural members subject to<br>close-up survey according to annex 1<br>4 All wind and water strakes within the<br>cargo length area<br>5 Selected wind and water strakes<br>outside the cargo length area<br>6 See 1.1.5 and annex 15 for additional<br>thickness measurement guidelines<br>applicable to the side shell frames and<br>brackets on ships subject to compliance<br>with resolution MSC.168(79)<br>7 As required by annex 12 for ships<br>subject to compliance with regulation<br>XII/6.1 of the Convention | 1 Suspect areas<br>2 Within the cargo length area:<br>.1 each deck plate outside<br>line of cargo hatch<br>openings<br>.2 three transverse sections,<br>one of which should be in<br>the amidship area, outside<br>line of cargo hatch<br>openings<br>.3 each bottom plate<br>3 Point 3 referred to in column 3<br>4 All wind and water strakes, full<br>length<br>5 See 1.1.5 and annex 15 for<br>additional thickness<br>measurement guidelines<br>applicable to the side shell<br>frames and brackets on ship<br>subject to compliance with<br>resolution MSC.168(79)<br>6 As required by annex 12 for<br>ships subject to compliance<br>with regulation XII/6.1 of the<br>Convention |

### ANNEX 3

## OWNER'S INSPECTION REPORT

### Structural condition

|                                                  |           |         |           |         |         |                     |       |
|--------------------------------------------------|-----------|---------|-----------|---------|---------|---------------------|-------|
| Ship's name: .....                               |           |         |           |         |         |                     |       |
| Owner's inspection report – Structural condition |           |         |           |         |         |                     |       |
| For tank/hold no: .....                          |           |         |           |         |         |                     |       |
| Grade of steel: deck: ..... side: .....          |           |         |           |         |         |                     |       |
| bottom: ..... longitudinal bulkhead: .....       |           |         |           |         |         |                     |       |
| Elements                                         | Cracks    | Buckles | Corrosion | Coating | Pitting | Modification/repair | Other |
|                                                  | condition |         |           |         |         |                     |       |
| Deck:                                            |           |         |           |         |         |                     |       |
| Bottom:                                          |           |         |           |         |         |                     |       |
| Side:                                            |           |         |           |         |         |                     |       |
| Side framing:                                    |           |         |           |         |         |                     |       |
| Longitudinal bulkheads:                          |           |         |           |         |         |                     |       |
| Transverse bulkheads:                            |           |         |           |         |         |                     |       |
| Repairs carried out due to:                      |           |         |           |         |         |                     |       |
| Thickness measurements carried out (dates):      |           |         |           |         |         |                     |       |
| Results in general:                              |           |         |           |         |         |                     |       |
| Overdue surveys:                                 |           |         |           |         |         |                     |       |
| Outstanding conditions of class:                 |           |         |           |         |         |                     |       |
| Comments:                                        |           |         |           |         |         |                     |       |
| Date of inspection: .....                        |           |         |           |         |         |                     |       |
| Inspected by: .....                              |           |         |           |         |         |                     |       |
| Signature: .....                                 |           |         |           |         |         |                     |       |

## ANNEX 4A

### SURVEY PROGRAMME

#### Basic information and particulars

|                                    |
|------------------------------------|
| Name of ship:                      |
| IMO number:                        |
| Flag State:                        |
| Port of registry:                  |
| Gross tonnage:                     |
| Deadweight (metric tonnes):        |
| Length between perpendiculars (m): |
| Shipbuilder:                       |
| Hull number:                       |
| Recognized organization (RO):      |
| RO ship identity:                  |
| Date of build of the ship:         |
| Owner:                             |
| Thickness measurement company:     |

#### **1 Preamble**

##### **1.1 Scope**

**1.1.1** The present survey programme covers the minimum extent of overall surveys, close-up surveys, thickness measurements and pressure testing within the cargo length area, cargo holds, ballast tanks, including fore and aft peak tanks, required by the Code.

**1.1.2** The arrangements and safety aspects of the survey should be acceptable to the attending surveyor(s).

##### **1.2 Documentation**

All documents used in the development of the survey programme should be available on board during the survey as required by section 6.

#### **2 Arrangement of cargo holds, tanks and spaces**

This section of the survey programme should provide information (either in the form of plans or text) on the arrangement of cargo holds, tanks and spaces that fall within the scope of the survey.

#### **3 List of cargo holds, tanks and spaces with information on their use, extent of coatings and corrosion prevention system**

This section of the survey programme should indicate any changes relating to (and should update) the information on the use of the holds and tanks of the ship, the extent of coatings and the corrosion prevention system provided in the survey planning questionnaire.

#### **4 Conditions for survey**

This section of the survey programme should provide information on the conditions for survey, e.g., information regarding cargo hold and tank cleaning, gas freeing, ventilation, lighting, etc.

#### **5 Provisions and method of access to structures**

This section of the survey programme should indicate any changes relating to (and should update) the information on the provisions and methods of access to structures provided in the survey planning questionnaire.

#### **6 List of equipment for survey**

This section of the survey programme should identify and list the equipment that will be made available for carrying out the survey and the required thickness measurements.

#### **7 Survey requirements**

##### **7.1 Overall survey**

This section of the survey programme should identify and list the spaces that should undergo an overall survey for this ship in accordance with 2.5.1.

##### **7.2 Close-up survey**

This section of the survey programme should identify and list the hull structures that should undergo a close-up survey for this ship in accordance with 2.5.2.

#### **8 Identification of tanks for tank testing**

This section of the survey programme should identify and list the cargo holds and tanks that should undergo tank testing for this ship in accordance with 2.7.

#### **9 Identification of areas and sections for thickness measurements**

This section of the survey programme should identify and list the areas and sections where thickness measurements should be taken in accordance with 2.6.1.

#### **10 Minimum thickness of hull structures**

This section of the survey programme should specify the minimum thickness for hull structures of this ship that are subject to survey, according to (a) or (b):

- (a) Determined from the attached wastage allowance table and the original thickness to the hull structure plans of the ship;
- (b) Given in the following table(s):

| Area or location                                 | Original as-built thickness (mm) | Minimum thickness (mm) | Substantial corrosion thickness (mm) |
|--------------------------------------------------|----------------------------------|------------------------|--------------------------------------|
| <b>Deck</b>                                      |                                  |                        |                                      |
| Plating                                          |                                  |                        |                                      |
| Longitudinals                                    |                                  |                        |                                      |
| Longitudinal girders                             |                                  |                        |                                      |
| Cross deck plating                               |                                  |                        |                                      |
| Cross deck stiffeners                            |                                  |                        |                                      |
| <b>Bottom</b>                                    |                                  |                        |                                      |
| Plating                                          |                                  |                        |                                      |
| Longitudinals                                    |                                  |                        |                                      |
| Longitudinal girders                             |                                  |                        |                                      |
| <b>Inner bottom</b>                              |                                  |                        |                                      |
| Plating                                          |                                  |                        |                                      |
| Longitudinals                                    |                                  |                        |                                      |
| Longitudinal girders                             |                                  |                        |                                      |
| Floors                                           |                                  |                        |                                      |
| <b>Ship side in way of topside tanks</b>         |                                  |                        |                                      |
| Plating                                          |                                  |                        |                                      |
| Longitudinals                                    |                                  |                        |                                      |
| <b>Ship side in way of hopper side tanks</b>     |                                  |                        |                                      |
| Plating                                          |                                  |                        |                                      |
| Longitudinals                                    |                                  |                        |                                      |
| <b>Ship side in way of tanks (if applicable)</b> |                                  |                        |                                      |
| Plating                                          |                                  |                        |                                      |
| Longitudinals                                    |                                  |                        |                                      |
| Longitudinal stringers                           |                                  |                        |                                      |
| <b>Ship side in way of cargo holds</b>           |                                  |                        |                                      |
| Plating                                          |                                  |                        |                                      |
| Side frames webs                                 |                                  |                        |                                      |
| Side frames flanges                              |                                  |                        |                                      |
| Upper brackets webs                              |                                  |                        |                                      |
| Upper brackets flanges                           |                                  |                        |                                      |
| Lower brackets webs                              |                                  |                        |                                      |
| Lower brackets flanges                           |                                  |                        |                                      |
| <b>Longitudinal bulkhead (if applicable)</b>     |                                  |                        |                                      |
| Plating                                          |                                  |                        |                                      |
| Longitudinals (if applicable)                    |                                  |                        |                                      |
| Longitudinal girders (if applicable)             |                                  |                        |                                      |
| <b>Transverse bulkheads</b>                      |                                  |                        |                                      |
| Plating                                          |                                  |                        |                                      |
| Stiffeners (if applicable)                       |                                  |                        |                                      |
| Upper stool plating                              |                                  |                        |                                      |
| Upper stool stiffeners                           |                                  |                        |                                      |
| Lower stool plating                              |                                  |                        |                                      |
| Lower stool stiffeners                           |                                  |                        |                                      |



| Area or location                              | Original as-built thickness (mm) | Minimum thickness (mm) | Substantial corrosion thickness (mm) |
|-----------------------------------------------|----------------------------------|------------------------|--------------------------------------|
| <b>Transverse web frames in topside tanks</b> |                                  |                        |                                      |
| Plating                                       |                                  |                        |                                      |
| Flanges                                       |                                  |                        |                                      |
| Stiffeners                                    |                                  |                        |                                      |
| <b>Transverse web frames in hopper tanks</b>  |                                  |                        |                                      |
| Plating                                       |                                  |                        |                                      |
| Flanges                                       |                                  |                        |                                      |
| Stiffeners                                    |                                  |                        |                                      |
| Hatch covers                                  |                                  |                        |                                      |
| Plating                                       |                                  |                        |                                      |
| Stiffeners                                    |                                  |                        |                                      |
| <b>Hatch coamings</b>                         |                                  |                        |                                      |
| Plating                                       |                                  |                        |                                      |
| Stiffeners                                    |                                  |                        |                                      |

**Note:** The wastage allowance tables should be attached to the survey programme.

## 11 Thickness measurement company

This section of the survey programme should identify changes, if any, relating to the information on the thickness measurement company provided in the survey planning questionnaire.

## 12 Damage experience related to the ship

This section of the survey programme should, using the tables provided below, provide details of the hull damages for at least the last three years in way of the cargo holds, ballast tanks and void spaces within the cargo length area. These damages are subject to survey.

### Hull damages sorted by location for this ship

| Cargo hold, tank or space number or area | Possible cause, if known | Description of the damages | Location | Repair | Date of repair |
|------------------------------------------|--------------------------|----------------------------|----------|--------|----------------|
|                                          |                          |                            |          |        |                |
|                                          |                          |                            |          |        |                |
|                                          |                          |                            |          |        |                |
|                                          |                          |                            |          |        |                |
|                                          |                          |                            |          |        |                |

**Hull damages for sister or similar ships (if available) in the case of design-related damage**

| <b>Cargo hold, tank or space number or area</b> | <b>Possible cause, if known</b> | <b>Description of the damages</b> | <b>Location</b> | <b>Repair</b> | <b>Date of repair</b> |
|-------------------------------------------------|---------------------------------|-----------------------------------|-----------------|---------------|-----------------------|
|                                                 |                                 |                                   |                 |               |                       |
|                                                 |                                 |                                   |                 |               |                       |
|                                                 |                                 |                                   |                 |               |                       |
|                                                 |                                 |                                   |                 |               |                       |
|                                                 |                                 |                                   |                 |               |                       |

**13 Areas identified with substantial corrosion from previous surveys**

This section of the survey programme should identify and list the areas of substantial corrosion from previous surveys.

**14 Critical structural areas and suspect areas**

This section of the survey programme should identify and list the critical structural areas and the suspect areas, when such information is available.

**15 Other relevant comments and information**

This section of the survey programme should provide any other comments and information relevant to the survey.

**Appendices**

**Appendix 1 – List of plans**

5.1.3.2 requires that main structural plans of cargo holds and ballast tanks (scantling drawings), including information on regarding use of high tensile steel (HTS) should be available. This appendix of the survey programme should identify and list the main structural plans which form part of the survey programme.

**Appendix 2 – Survey planning questionnaire**

The survey planning questionnaire (annex 4B), which has been submitted by the owner, should be appended to the survey programme.

**Appendix 3 – Other documentation**

This part of the survey programme should identify and list any other documentation that forms part of the plan.

Prepared by the owner in co-operation with the Administration for compliance with 5.1.3:

Date:

*(name and signature of authorized owner's representative)*

Date:

*(name and signature of authorized representative of the Administration)*

## **ANNEX 4B**

### **SURVEY PLANNING QUESTIONNAIRE**

**1** The following information will enable the owner in co-operation with the Administration to develop a survey programme complying with the requirements of the Code. It is essential that the owner provides, when completing the present questionnaire, up-to-date information. The present questionnaire, when completed, should provide all information and material required by the Code.

#### **Particulars**

Ship's name:  
IMO number:  
Flag State:  
Port of registry:  
Owner:  
Recognized organization:  
RO Ship identity:  
Gross tonnage:  
Deadweight (metric tonnes):  
Date of build:

#### **Information on access provision for close-up surveys and thickness measurement**

**2** The owner should indicate, in the table below, the means of access to the structures subject to close-up survey and thickness measurement. A close-up survey is an examination where the details of structural components are within the close visual inspection range of the attending surveyor, i.e. normally within reach of hand.

| Hold/Tank No. | Structure                          | Temporary staging | Rafts | Ladders | Direct access | Other means (please specify) |
|---------------|------------------------------------|-------------------|-------|---------|---------------|------------------------------|
| F.P.          | Fore peak                          |                   |       |         |               |                              |
| A.P.          | Aft peak                           |                   |       |         |               |                              |
| Cargo holds   | Hatch side coamings                |                   |       |         |               |                              |
|               | Topside sloping plate              |                   |       |         |               |                              |
|               | Upper stool plating                |                   |       |         |               |                              |
|               | Cross deck                         |                   |       |         |               |                              |
|               | Side shell, frames and brackets    |                   |       |         |               |                              |
|               | Transverse bulkhead                |                   |       |         |               |                              |
|               | Hopper tank plating                |                   |       |         |               |                              |
|               | Lower stool                        |                   |       |         |               |                              |
|               | Tank top                           |                   |       |         |               |                              |
| Topside tanks | Under deck structure               |                   |       |         |               |                              |
|               | Side shell and structure           |                   |       |         |               |                              |
|               | Sloping plate and structure        |                   |       |         |               |                              |
|               | Webs and bulkheads                 |                   |       |         |               |                              |
| Hopper tanks  | Hopper sloping plate and structure |                   |       |         |               |                              |
|               | Side shell and structure           |                   |       |         |               |                              |
|               | Bottom structure                   |                   |       |         |               |                              |
|               | Webs and bulkheads                 |                   |       |         |               |                              |
|               | Double-bottom structure            |                   |       |         |               |                              |
|               | Upper stool internal structure     |                   |       |         |               |                              |
|               | Lower stool internal structure     |                   |       |         |               |                              |
|               |                                    |                   |       |         |               |                              |
|               |                                    |                   |       |         |               |                              |
|               |                                    |                   |       |         |               |                              |
|               |                                    |                   |       |         |               |                              |
|               |                                    |                   |       |         |               |                              |

| History of bulk cargoes of a corrosive nature (e.g., high sulphur content) |
|----------------------------------------------------------------------------|
|                                                                            |
|                                                                            |
|                                                                            |
|                                                                            |
|                                                                            |
|                                                                            |

### Owner's inspections

3 Using a format similar to that of the table below (which is given as an example), the owner should provide details of the results of their inspections, for the last 3 years – in accordance with the Code – on all CARGO holds and BALLAST tanks and VOID spaces within the cargo area.

| Tank/Hold No.               | Corrosion protection (1) | Coating extent (2) | Coating condition (3) | Structural deterioration (4) | Hold and tank history (5) |
|-----------------------------|--------------------------|--------------------|-----------------------|------------------------------|---------------------------|
| Cargo holds                 |                          |                    |                       |                              |                           |
| Topside tanks               |                          |                    |                       |                              |                           |
| Hopper tanks                |                          |                    |                       |                              |                           |
| Double-side skin tanks      |                          |                    |                       |                              |                           |
| Double-bottom tanks         |                          |                    |                       |                              |                           |
| Upper stools                |                          |                    |                       |                              |                           |
| Lower stools                |                          |                    |                       |                              |                           |
|                             |                          |                    |                       |                              |                           |
| Fore peak                   |                          |                    |                       |                              |                           |
| Aft peak                    |                          |                    |                       |                              |                           |
| Miscellaneous other spaces: |                          |                    |                       |                              |                           |

### Note:

Indicate tanks which are used for oil/ballast.

- 1) HC = hard coating; SC = soft coating;  
SH=semi-hard coating; NP = no protection
- 2) U = upper part; M = middle part;  
L = lower part; C = complete
- 3) G = good; F = fair; P = poor;  
RC = recoated (during the last three years)
- 4) N = no findings recorded; Y = findings recorded, description of findings should be attached to this questionnaire
- 5) DR = Damage & Repair; L = Leakages;  
CV = Conversion (description to be attached to this questionnaire)

|                                                                             |
|-----------------------------------------------------------------------------|
| Name of owner's representative:<br>.....<br>Signature: .....<br>Date: ..... |
|-----------------------------------------------------------------------------|

**Reports of port State control inspections**

|                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| List the reports of port State control inspections containing hull structural related deficiencies and relevant information on rectification of the deficiencies: |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |

**Safety management system**

|                                                                                                |
|------------------------------------------------------------------------------------------------|
| List non-conformities related to hull maintenance including the associated corrective actions: |
|------------------------------------------------------------------------------------------------|

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |

**Name and address of the approved thickness measurement company:**

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |

## **ANNEX 5**

### **PROCEDURES FOR CERTIFICATION OF A COMPANY ENGAGED IN THICKNESS MEASUREMENT OF HULL STRUCTURES**

#### **1 Application**

This guidance applies for certification of a company which intends to engage in the thickness measurement of hull structures of ships.

#### **2 Procedures for certification**

##### *Submission of documents*

**2.1** The following documents should be submitted to an organization recognized by the Administration for approval:

- .1** outline of the company, e.g., organization and management structure;
- .2** experience of the company on thickness measurement of hull structures of ships;
- .3** technicians' careers, i.e. experience of technicians as thickness measurement operators, technical knowledge and experience of hull structure, etc. Operators should be qualified according to a recognized industrial NDT Standard;
- .4** equipment used for thickness measurement such as ultrasonic testing machines and their maintenance/calibration procedures;
- .5** a guide for thickness measurement operators;
- .6** training programmes for technicians for thickness measurement; and
- .7** measurement record format in accordance with recommended procedures for thickness measurements (see annex 8).

##### *Auditing of the company*

**2.2** Upon reviewing the documents submitted with satisfactory results, the company should be audited in order to ascertain that the company is duly organized and managed in accordance with the documents submitted, and eventually is capable of conducting thickness measurement of the hull structure of ships.

**2.3** Certification is conditional upon an on board demonstration of thickness measurement as well as satisfactory reporting.

#### **3 Certification**

**3.1** Upon satisfactory results of both the audit of the company referred to in 2.2 and the demonstration tests referred to in 2.3, the Administration or organization recognized by the Administration should issue a certificate of approval as well as a notice to the effect that the thickness measurement operation system of the company has been certified.

**3.2** Renewal/endorsement of the certificate should be made at intervals not exceeding three years by verification that original conditions are maintained.

**4 Report of any alteration to the certified thickness measurement operation system**

In cases where any alteration to the certified thickness measurement operation system of the company is made, such an alteration should be immediately reported to the organization recognized by the Administration. Re-audit should be made where deemed necessary by the organization recognized by the Administration.

**5 Withdrawal of the certification**

The certification may be withdrawn in the following cases:

- .1** where the measurements were improperly carried out or the results were improperly reported;
- .2** where the surveyor found any deficiencies in the approved thickness measurement operation systems of the company; and
- .3** where the company failed to report any alteration referred to in 4 to the organization recognized by the Administration as required.



## **ANNEX 6**

### **SURVEY REPORTING PRINCIPLES**

As a principle, for bulk carriers subject to the Code, the surveyor should include the following contents in his report for survey of hull structure and piping systems, as relevant for the survey.

#### **1 General**

**1.1** A survey report should be generated in the following cases:

- .1** in connection with commencement, continuation and/or completion of periodical hull surveys, i.e. annual, intermediate and renewal surveys, as relevant;
- .2** when structural damages/defects have been found;
- .3** when repairs, renewals or modifications have been carried out; and
- .4** when condition of class (recommendation) has been imposed or has been deleted.

**1.2** The reporting should provide:

- .1** evidence that prescribed surveys have been carried out in accordance with applicable requirements;
- .2** documentation of surveys carried out with findings, repairs carried out and condition of class (recommendation) imposed or deleted;
- .3** survey records, including actions taken, which should form an auditable documentary trail. Survey reports should be kept in the survey report file required to be on board;
- .4** information for planning of future surveys; and
- .5** information which may be used as input for maintenance of classification rules and instructions.

**1.3** When a survey is split between different survey stations, a report should be made for each portion of the survey. A list of items surveyed, relevant findings and an indication of whether the item has been credited, are to be made available to the next attending surveyor, prior to continuing or completing the survey. Thickness measurement and tank testing carried out is also to be listed for the next surveyor.

#### **2 Extent of the survey**

**2.1** Identification of compartments where an overall survey has been carried out.

**2.2** Identification of locations, in each ballast tank and cargo hold including hatch covers and coamings, where a close-up survey has been carried out, together with information on the means of access used.

**2.3** Identification of locations, in each ballast tank and cargo hold including hatch covers and coamings, where thickness measurement has been carried out.

**Note:** As a minimum, the identification of location of close-up survey and thickness measurement should include a confirmation, with description of individual structural members, corresponding to the extent of requirements stipulated in annex A based on type of periodical survey and the ship's age.

Where only partial survey is required, i.e. 25% of shell frames, one transverse web, two selected cargo hold transverse bulkheads, the identification should include location within each ballast tank and cargo hold by reference to frame numbers.

**2.4** For areas in ballast tanks and cargo holds where protective coating is found to be in good condition and the extent of close-up survey and/or thickness measurement has been specially considered, structures subject to special consideration should be identified.

**2.5** Identification of tanks subject to tank testing.

**2.6** Identification of piping systems on deck and within cargo holds, ballast tanks, pipe tunnels, cofferdams and void spaces where:

- .1 examination including internal examination of piping with valves and fittings and thickness measurement, as relevant, has been carried out; and
- .2 operational test to working pressure has been carried out.

### **3 Result of the survey**

**3.1** Type, extent and condition of protective coating in each tank, as relevant (rated GOOD, FAIR or POOR) including identification of tanks fitted with anodes.

**3.2** Structural condition of each compartment with information on the following, as relevant:

- .1 identification of findings, such as:
  - .1 corrosion with description of location, type and extent;
  - .2 areas with substantial corrosion;
  - .3 cracks/fractures with description of location and extent;
  - .4 buckling with description of location and extent; and
  - .5 indents with description of location and extent;
- .2 identification of compartments where no structural damages/ defects are found. The report may be supplemented by sketches/photos; and
- .3 thickness measurement report should be verified and signed by the surveyor controlling the measurements on board.

#### **4 Actions taken with respect to findings**

**4.1** Whenever the attending surveyor is of the opinion that repairs are required, each item to be repaired should be identified in a numbered list. Whenever repairs are carried out, details of the repairs effected should be reported by making specific reference to relevant items in the numbered list.

**4.2** Repairs carried out should be reported with identification of:

- .1** compartment;
- .2** structural member;
- .3** repair method (i.e. renewal or modification), including:
  - .1** steel grades and scantlings (if different from the original); and
  - .2** sketches/photos, as appropriate;
- .4** repair extent; and
- .5** non-destructive test (NDT)/tests.

**4.3** For repairs not completed at the time of survey, condition of class/recommendation should be imposed with a specific time limit for the repairs. In order to provide correct and proper information to the surveyor attending for survey of the repairs, condition of class/recommendation should be sufficiently detailed with identification of each item to be repaired. For identification of extensive repairs, reference may be made to the survey report.

## ANNEX 7

### CONDITION EVALUATION REPORT

#### Issued upon completion of renewal survey

##### **General particulars**

|                                |                                                   |
|--------------------------------|---------------------------------------------------|
| Ship's name:                   | Class/Administration identity number:             |
|                                | Previous class/Administration identity number(s): |
|                                | IMO number:                                       |
| Port of registry:              | National flag:                                    |
|                                | Previous national flag(s):                        |
| Deadweight<br>(metric tonnes): | Gross tonnage:                                    |
|                                | National:                                         |
|                                | ITC (1969):                                       |
| Date of build:                 | Classification notation:                          |
| Date of major conversion:      |                                                   |
| Type of conversion:            | Owner:                                            |
|                                | Previous owner(s):                                |

**1** The survey reports and documents listed below have been reviewed by the undersigned and found to be satisfactory.

**2** A summary of the survey is attached herewith on sheet 2.

**3** The renewal survey has been completed in accordance with the present Code on (date) .....

|                                          |                   |       |
|------------------------------------------|-------------------|-------|
| Condition evaluation report completed by | Name<br>Signature | Title |
| Office                                   | Date              |       |
| Condition evaluation report verified by  | Name<br>Signature | Title |
| Office                                   | Date              |       |

Attached reports and documents:

- 1)
- 2)
- 3)
- 4)
- 5)

### **Contents of condition evaluation report**

- |        |                                               |                                                                                      |
|--------|-----------------------------------------------|--------------------------------------------------------------------------------------|
| Part 1 | – General particulars:                        | – See front page                                                                     |
| Part 2 | – Report review:                              | – Where and how survey was done                                                      |
| Part 3 | – Close-up survey:                            | – Extent (which tanks/holds)                                                         |
| Part 4 | – Thickness measurements:                     | – Reference to thickness measurement report                                          |
|        |                                               | – Summary of where measured                                                          |
|        |                                               | – Separate form indicating the spaces with substantial corrosion, and corresponding: |
|        |                                               | – thickness diminution                                                               |
|        |                                               | – corrosion pattern                                                                  |
| Part 5 | – Tank/hold corrosion prevention system:      | – Separate form indicating:                                                          |
|        |                                               | – location of coating                                                                |
|        |                                               | – condition of coating (if applicable)                                               |
| Part 6 | – Repairs:                                    | – Identification of spaces/areas                                                     |
| Part 7 | – Condition of class/flag State requirements: |                                                                                      |
| Part 8 | – Memoranda:                                  | – Acceptable defects                                                                 |
|        |                                               | – Any points of attention for future surveys, e.g., for suspect areas                |
|        |                                               | – Extended annual/intermediate survey due to coating breakdown                       |
| Part 9 | – Conclusion:                                 | – Statement on evaluation/verification of survey report                              |

### **Extract of thickness measurements**

Reference is made to the thickness measurement report:

| <b>Position of substantially corroded tanks/areas<sup>1</sup> or areas with deep pitting<sup>3</sup></b> | <b>Thickness diminution [%]</b> | <b>Corrosion pattern<sup>2</sup></b> | <b>Remarks:<br/>(e.g., reference to attached sketches)</b> |
|----------------------------------------------------------------------------------------------------------|---------------------------------|--------------------------------------|------------------------------------------------------------|
|                                                                                                          |                                 |                                      |                                                            |

### **Notes:**

- 1 Substantial corrosion, i.e. 75%–100% of acceptable margins wasted.
- 2 P = Pitting  
C = Corrosion in general
- 3 Any bottom plating with a pitting intensity of 20% or more, with wastage in the substantial corrosion range or having an average depth of pitting of 1/3 or more of actual plate thickness should be noted.

***Tank/hold corrosion prevention system***

| <b>Tank/hold nos.<sup>1</sup></b> | <b>Tank/hold protection<sup>2</sup></b> | <b>Coating condition<sup>3</sup></b> | <b>Remarks</b> |
|-----------------------------------|-----------------------------------------|--------------------------------------|----------------|
|                                   |                                         |                                      |                |

**Notes:**

- 1 All ballast tanks and cargo holds should be listed.
- 2 C = Coating  
NP = No protection
- 3 Coating condition according to the following standard:  
  
GOOD condition with only minor spot rusting.  
  
FAIR condition with local breakdown of coating at edges of stiffeners and weld connections and/or light rusting over 20% or more of areas under consideration, but less than as defined for POOR condition.  
  
POOR condition with general breakdown of coating over 20% or more of areas or hard scale at 10% or more of areas under consideration.  
  
If coating condition POOR is given, extended annual surveys should be introduced. This should be noted in part 8 of the Contents of condition evaluation report.

## **ANNEX 8**

### **RECOMMENDED PROCEDURES FOR THICKNESS MEASUREMENTS**

#### **General**

- 1** These procedures should be used for recording thickness measurements as required by part A of annex A.
- 2** Reporting forms TM1-BC, TM2-BC(i), TM2-BC(ii) TM3-BC, TM4-BC, TM5-BC, TM6-BC, TM7-BC(i) and TM7-BC(ii), set out in appendix 2 should be used for recording thickness measurements and the maximum allowance diminution should be stated. The maximum allowable diminution could be stated in an attached document.
- 3** Appendix 3 contains guidance diagrams and notes relating to the reporting forms and the requirements for thickness measurement.
- 4** The reporting forms should, where appropriate, be supplemented by data presented on structural sketches.

- |            |                                   |
|------------|-----------------------------------|
| Appendix 1 | General particulars               |
| Appendix 2 | Reports on thickness measurement  |
| Appendix 3 | Guidance on thickness measurement |

## Appendix 1

### GENERAL PARTICULARS

Ship's name: .....  
IMO number: .....  
Class/Administration identity number: .....  
Port of registry: .....  
Gross tonnage: .....  
Deadweight: .....  
Date of build: .....  
Classification society: .....

---

Name of company performing thickness measurement: .....  
.....  
Thickness measurement company certified by: .....  
Certificate number: .....  
Certificate valid from: ..... to .....  
Place of measurement: .....  
First date of measurement: .....  
Last date of measurement: .....  
Renewal survey/intermediate survey<sup>4</sup> due: .....  
Details of measurement equipment: .....  
Qualification of operator: .....

---

|                               |                              |
|-------------------------------|------------------------------|
| Report number: .....          | consisting of ..... pages    |
| Name of operator: .....       | Name of surveyor: .....      |
| Signature of operator: .....  | Signature of surveyor: ..... |
| Company official stamp: ..... | Administration: .....        |

Official stamp

---

<sup>4</sup> Delete as appropriate.



## Appendix 2

### REPORTS ON THICKNESS MEASUREMENT

#### Reports on thickness measurement of all deck plating, all bottom shell plating or side shell plating (TM1-BC)

Ship's name..... IMO number..... Class Identity no.....Report no.....

Ship's name: \_\_\_\_\_ IMO number: \_\_\_\_\_ Class society no: \_\_\_\_\_ Report no: \_\_\_\_\_

| STRAKE POSITION |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
|-----------------|---------------|-----------------|-----------------|---|--------------|---|--------------|---|-------------|---|--------------|---|--------------|---|-----------------|---|-----------------------------------|
| PLATE POSITION  | No. or letter | Orig. thk. (mm) | Forward reading |   |              |   |              |   | Aft reading |   |              |   |              |   | Mean diminution |   | Maximum allowable diminution (mm) |
|                 |               |                 | Gauged          |   | Diminution P |   | Diminution S |   | Gauged      |   | Diminution P |   | Diminution S |   | P               | S |                                   |
|                 |               |                 | P               | S | mm           | % | mm           | % | P           | S | mm           | % | mm           | % |                 |   |                                   |
| 12th forward    |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 11th            |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 10th            |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 9th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 8th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 7th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 6th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 5th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 4th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 3rd             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 2nd             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 1st             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| Amidships       |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 1st aft         |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 2nd             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 3rd             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 4th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 5th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 6th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 7th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 8th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 9th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 10th            |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 11th            |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |
| 12th            |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                 |   |                                   |

Operator's signature.....

Notes – see following page

**Notes to report TM1-BC:**

- 1 This report should be used for recording the thickness measurement of:
  - .1 All strength deck plating within the cargo length area.
  - .2 Keel, bottom shell plating and bilge plating within the cargo length area.
  - .3 Side shell plating that is all wind and water strakes within the cargo length area.
  - .4 Side shell plating that is selected wind and water strakes outside the cargo length area.
- 2 The strake position should be clearly indicated as follows:
  - .1 For strength deck, indicate the number of the strake of plating inboard from the stringer plate.
  - .2 For bottom plating, indicate the number of the strake of plating outboard from the keel plate.
  - .3 For side shell plating, give number of the strake of plating below sheerstrake and letter as shown on shell expansion.
- 3 Only the deck plating strakes outside line of openings should be recorded.
- 4 Measurements should be taken at the forward and aft areas of all plates.
- 5 The single measurements recorded should represent the average of multiple measurements.
- 6 The maximum allowable diminution could be stated in an attached document.

**Report on thickness measurement of shell and deck plating (one, two or three transverse sections) (TM2-BC(i))**

Ship's name..... IMO number..... Class Identity no..... Report no.....

| STRENGTH DECK AND SHEERSTRAKE PLATING |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
|---------------------------------------|----------------------------------------------|----------------|--------|---|--------------|---|--------------|---|-----------------------------------------------|---------------|----------------|--------|---|--------------|---|--------------|----------------------------------------------|------------------------|---------------|----------------|--------|---|--------------|---|--------------|---|------------------------|
| STRAKE POSITION                       | FIRST TRANSVERSE SECTION AT FRAME NUMBER.... |                |        |   |              |   |              |   | SECOND TRANSVERSE SECTION AT FRAME NUMBER.... |               |                |        |   |              |   |              | THIRD TRANSVERSE SECTION AT FRAME NUMBER.... |                        |               |                |        |   |              |   |              |   |                        |
|                                       | No. or letter                                | Orig. thk (mm) | Gauged |   | Diminution P |   | Diminution S |   | Max allow. dimin. (mm)                        | No. or letter | Orig. thk (mm) | Gauged |   | Diminution P |   | Diminution S |                                              | Max allow. dimin. (mm) | No. or letter | Orig. thk (mm) | Gauged |   | Diminution P |   | Diminution S |   | Max allow. dimin. (mm) |
|                                       |                                              |                | P      | S | mm           | % | mm           | % |                                               |               |                | P      | S | mm           | % | mm           | %                                            |                        |               |                | P      | S | mm           | % | mm           | % |                        |
| Stringer plate                        |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| 1st strake inboard                    |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| 2nd                                   |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| 3rd                                   |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| 4th                                   |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| 5th                                   |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| 6th                                   |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| 7th                                   |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| 8th                                   |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| 9th                                   |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| 10th                                  |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| 11th                                  |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| 12th                                  |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| 13th                                  |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| 14th                                  |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| centre strake                         |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| sheerstrake                           |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| TOPSIDE                               |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |
| TOTAL                                 |                                              |                |        |   |              |   |              |   |                                               |               |                |        |   |              |   |              |                                              |                        |               |                |        |   |              |   |              |   |                        |

Operator's signature.....

Notes – see following page

**Notes to report TM2-BC(i):**

- 1** This report should be used for recording the thickness measurement of strength deck plating and sheer-strake plating transverse sections:

One, two or three sections within the cargo length area, comprising structural items (1), (2) and (3) as shown on the diagram of typical transverse sections (appendix 3).
- 2** Only the deck plating outside the line of openings should be recorded.
- 3** The topside area comprises deck plating, stringer plate and sheer-strake (including rounded gunwales).
- 4** The exact frame station of measurement should be stated.
- 5** The single measurements recorded should represent the average of multiple measurements.
- 6** The maximum allowable diminution could be stated in an attached document.

**Report on thickness measurement of shell and deck plating (one, two or three transverse sections) (TM2-BC(ii))**

Ship's name..... IMO number..... Class Identity no..... Report no.....

| SHELL PLATING         |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
|-----------------------|----------------------------------------------|----------------|--------|---|--------------|---|--------------|-----------------------------------------------|---------------|----------------|--------|---|--------------|---|----------------------------------------------|------------------------|---------------|----------------|--------|---|--------------|---|--------------|------------------------|
|                       | FIRST TRANSVERSE SECTION AT FRAME NUMBER.... |                |        |   |              |   |              | SECOND TRANSVERSE SECTION AT FRAME NUMBER.... |               |                |        |   |              |   | THIRD TRANSVERSE SECTION AT FRAME NUMBER.... |                        |               |                |        |   |              |   |              |                        |
| STRAKE POSITION       | No. or letter                                | Orig. thk (mm) | Gauged |   | Diminution P |   | Diminution S | Max allow. dimin. (mm)                        | No. or letter | Orig. thk (mm) | Gauged |   | Diminution P |   | Diminution S                                 | Max allow. dimin. (mm) | No. or letter | Orig. thk (mm) | Gauged |   | Diminution P |   | Diminution S | Max allow. dimin. (mm) |
|                       |                                              |                | P      | S | mm           | % |              |                                               |               |                | mm     | % | P            | S |                                              |                        |               |                | mm     | % | mm           | % |              |                        |
| 1st below sheerstrake |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 2nd                   |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 3rd                   |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 4th                   |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 5th                   |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 6th                   |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 7th                   |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 8th                   |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 9th                   |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 10th                  |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 11th                  |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 12th                  |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 13th                  |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 14th                  |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 15th                  |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 16th                  |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 17th                  |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 18th                  |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 19th                  |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| 20th                  |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| keel strake           |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |
| BOTTOM TOTAL          |                                              |                |        |   |              |   |              |                                               |               |                |        |   |              |   |                                              |                        |               |                |        |   |              |   |              |                        |

Operator's signature.....

Notes – see following page

**Notes to report TM2-BC(ii):**

- 1** This report should be used for recording the thickness measurements of shell plating at transverse sections:

One, two or three sections within the cargo length area comprising structural items (4), (5), (6), and (7) as shown on the diagram of typical transverse sections (appendix 3).
- 2** The bottom area comprises keel, bottom and bilge plating.
- 3** The exact frame station of measurement should be stated.
- 4** The single measurements recorded should represent the average of multiple measurements.
- 5** The maximum allowable diminution could be stated in an attached document.

**Report on thickness measurement of longitudinal members (one, two or three transverse sections) (TM3-BC)**

Ship's name..... IMO number..... Class Identity no..... Report no.....

[illegible]

Operator's signature.....

Notes – see following page

**Notes to report TM3-BC:**

- 1** This report should be used for recording the thickness measurement of longitudinal members at transverse sections:

One, two or three sections within the cargo length area, comprising the appropriate structural items (8) to (20) as shown on the diagram of typical transverse sections (appendix 3).
- 2** The exact frame station of measurement should be stated.
- 3** The single measurements recorded should represent the average of multiple measurements.
- 4** The maximum allowable diminution could be stated in an attached document.





***Notes to report TM4-BC:***

- 1**      This report should be used for recording the thickness measurement of transverse structural members, comprising appropriate structural items (23) to (25) as shown on the diagram of typical transverse sections (appendix 3).
- 2**      Guidance for areas of measurement is indicated in appendix 3.
- 3**      The single measurements recorded should represent the average of multiple measurements.
- 4**      The maximum allowable diminution could be stated in an attached document.

## Report on thickness measurement of cargo hold transverse bulkheads (TM5-BC)

Ship's name..... IMO number..... Class Identity no..... Report no.....

[illegible]

Operator's signature.....

Notes – see following page

***Notes to report TM5-BC:***

- 1**      This report form should be used for recording the thickness measurement of cargo hold transverse bulkheads.
- 2**      Guidance for areas of measurement is indicated in appendix 3.
- 3**      The single measurements recorded should represent the average of multiple measurements.
- 4**      The maximum allowable diminution could be stated in an attached document.

## Report on thickness measurement of miscellaneous structural members (TM6-BC)

Ship's name..... IMO number..... Class Identity no..... Report no.....

[illegible]

Operator's signature.....

Notes – see following page

***Notes to report TM6-BC:***

- 1**      This report should be used for recording the thickness measurement of miscellaneous structural members including structural items (28), (29), (30) and (31) as shown on the diagram of typical transverse sections in appendix 3.
- 2**      Guidance for areas of measurement is indicated in appendix 3.
- 3**      The single measurements recorded should represent the average of multiple measurements.
- 4**      The maximum allowable diminution could be stated in an attached document.

## Report on thickness measurement of cargo hold transverse frames (TM7-BC(i))

Ship's name..... IMO number..... Class Identity no..... Report no.....

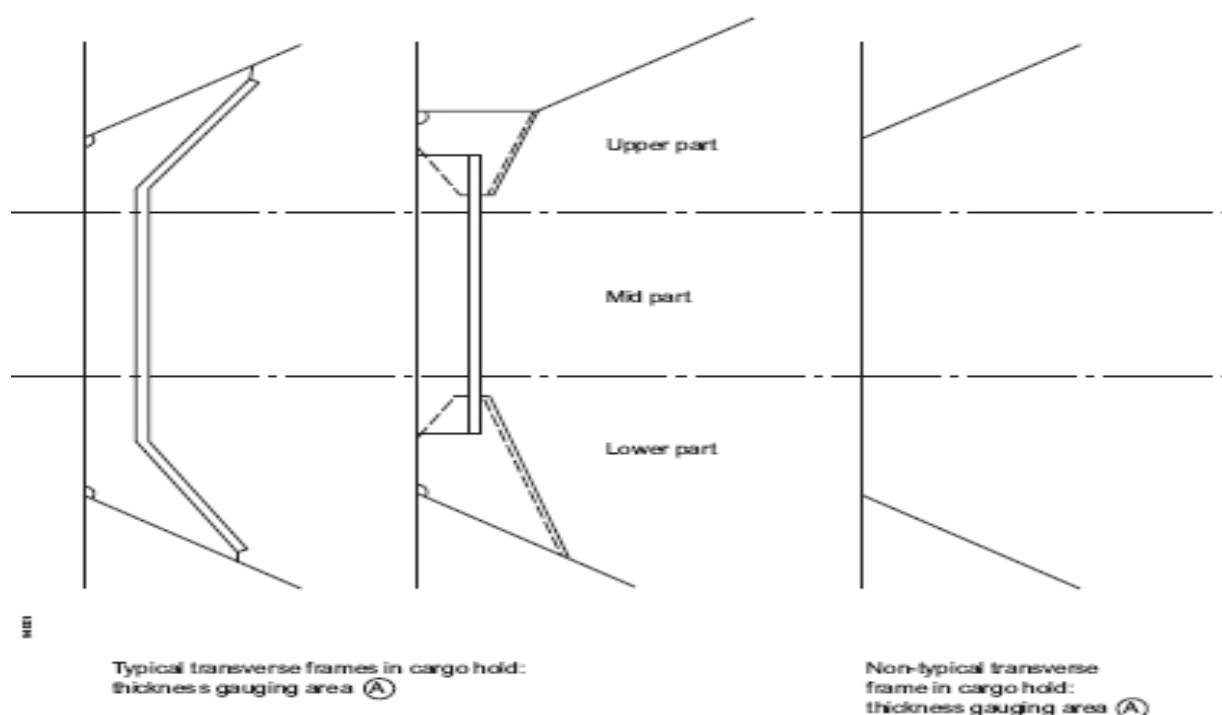
[illegible]

Operator's signature.....

Notes – see following page

**Notes to report TM7-BC(i):**

- 1 This report should be used for recording the thickness measurement of:
  - cargo hold transverse frames; and
  - structural item number (34) as shown on the diagram of typical transverse section indicating longitudinal and transverse members in appendix 3.
- 2 Guidance for areas of measurement is indicated in appendix 3.
- 3 The single measurements recorded should represent the average of multiple measurements.
- 4 The maximum allowable diminution could be stated in an attached document







**Notes on report TM7-BC(ii):**

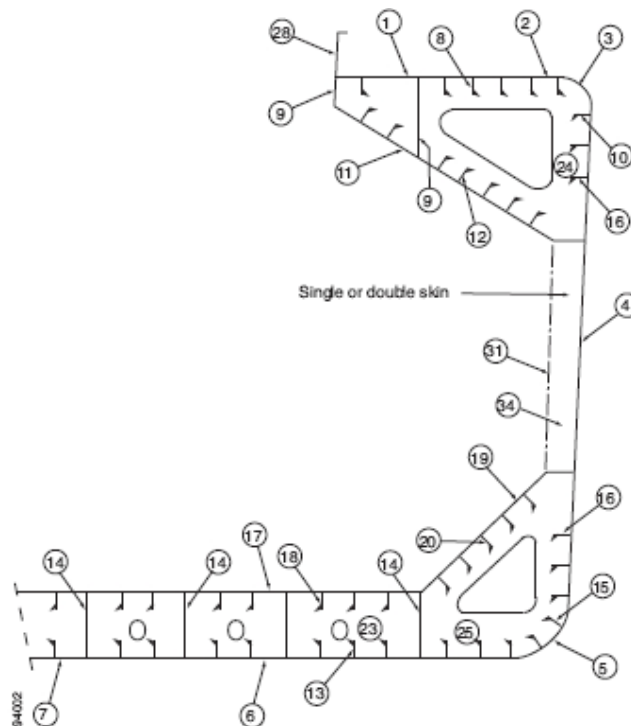
- 1 This report should be used for recording the thickness measurement of:  
Cargo hold transverse frames for application of resolution MSC.168(79).
- 2 Guidance for areas of measurement is provided in annex 15.
- 3 Symbols should be applied as follows:  

|            |   |                                                                                                                                                                   |
|------------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $t_M$      | = | thickness as measured, in mm                                                                                                                                      |
| $t_{REN}$  | = | thickness at which renewal is required in accordance with paragraph 2.1.2 of annex 2 to resolution MSC.168(79)                                                    |
| $t_{COAT}$ | = | $0.75 \cdot$ (thickness, in mm, as required by annex 1 to resolution MSC.168(79), in paragraph 3 for frame webs and in paragraph 4 for upper and lower brackets). |
- 4 The maximum allowable diminution could be stated in an attached document.

### Appendix 3

#### GUIDANCE ON THICKNESS MEASUREMENT

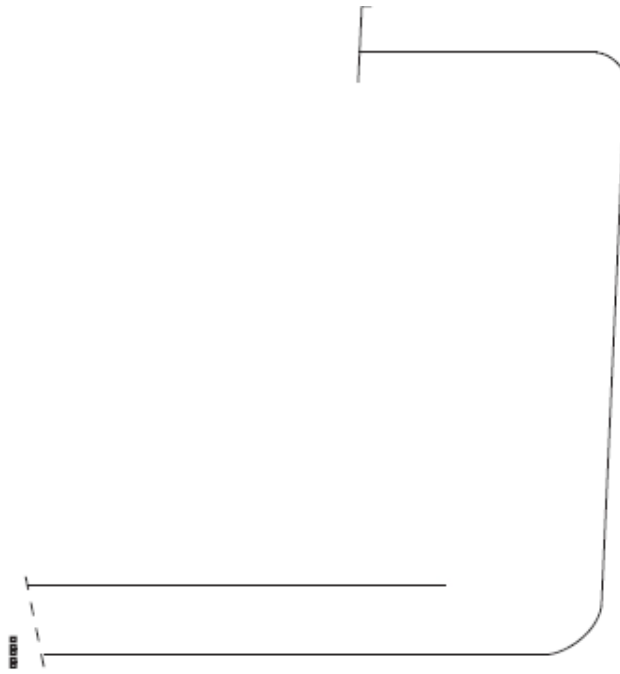
*Typical transverse section indicating longitudinal and transverse members*



| REPORT ON TM3-BC                                                                                                                                                                                                                        | REPORT ON TM3-BC                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>① Strength deck plating</li> <li>② Stringer plate</li> <li>③ Sheerstrake</li> <li>④ Side shell plating</li> <li>⑤ Bilge plating</li> <li>⑥ Bottom shell plating</li> <li>⑦ Keel plate</li> </ul> | <ul style="list-style-type: none"> <li>⑧ Deck longitudinals</li> <li>⑨ Deck girders</li> <li>⑩ Sheerstrake longitudinals</li> <li>⑪ Topside tank sloping plating</li> <li>⑫ Topside tank sloping plating longitudinals</li> <li>⑬ Bottom longitudinals</li> <li>⑭ Bottom girders</li> <li>⑮ Bilge longitudinals</li> </ul> | <ul style="list-style-type: none"> <li>⑯ Side shell longitudinals</li> <li>⑰ Inner bottom plating</li> <li>⑱ Inner bottom longitudinals</li> <li>⑲ Hopper side plating</li> <li>⑳ Hopper side longitudinals</li> <li>㉑</li> <li>㉒</li> </ul> |
| REPORT ON TM4-BC                                                                                                                                                                                                                        | REPORT ON TM6-BC                                                                                                                                                                                                                                                                                                           | REPORT ON TM7-BC                                                                                                                                                                                                                             |
| <ul style="list-style-type: none"> <li>㉓ Double bottom tank doors</li> <li>㉔ Topside tank transverses</li> <li>㉕ Hopper side tank transverses</li> <li>㉖</li> <li>㉗</li> </ul>                                                          | <ul style="list-style-type: none"> <li>㉘ Hatch coamings</li> <li>㉙ Deck plating between hatches</li> <li>㉚ Hatch covers</li> <li>㉛ Inner bulkhead plating</li> <li>㉜</li> <li>㉝</li> <li>㉞</li> </ul>                                                                                                                      | <ul style="list-style-type: none"> <li>㉟ Hold frames or diaphragms</li> </ul>                                                                                                                                                                |

### Transverse section outline

(To be used for longitudinal and transverse members where the typical transverse section is not applicable)



| REPORT ON TM2-RC |                       |
|------------------|-----------------------|
| ①                | Strength deck plating |
| ②                | Stringer plate        |
| ③                | Sheerstrake           |
| ④                | Side shell plating    |
| ⑤                | Bilge plating         |
| ⑥                | Bottom shell plating  |
| ⑦                | Keel plate            |

| REPORT ON TM3-RC |                                            |
|------------------|--------------------------------------------|
| ⑧                | Deck longitudinals                         |
| ⑨                | Deck girders                               |
| ⑩                | Sheerstrake longitudinals                  |
| ⑪                | Topside tank sloping plating               |
| ⑫                | Topside tank sloping plating longitudinals |
| ⑬                | Bottom longitudinals                       |
| ⑭                | Bottom girders                             |
| ⑮                | Bilge longitudinals                        |
| ⑯                | Side shell longitudinals                   |
| ⑰                | Inner bottom plating                       |
| ⑱                | Inner bottom longitudinals                 |
| ⑲                | Hopper side plating                        |
| ⑳                | Hopper side longitudinals                  |
| ㉑                |                                            |
| ㉒                |                                            |

| REPORT ON TM4-RC |                             |
|------------------|-----------------------------|
| ㉓                | Double-bottom tank floors   |
| ㉔                | Topside tank transverse     |
| ㉕                | Hopper side tank transverse |
| ㉖                |                             |
| ㉗                |                             |

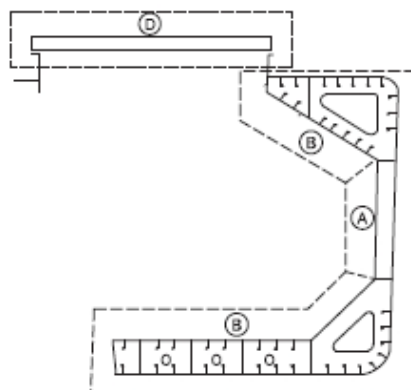
| REPORT ON TM5-RC |                              |
|------------------|------------------------------|
| ㉘                | Hatch coamings               |
| ㉙                | Deck plating between hatches |
| ㉚                | Hatch covers                 |
| ㉛                | Inner bulkhead plating       |
| ㉜                |                              |
| ㉝                |                              |

| REPORT ON TM7-RC |                           |
|------------------|---------------------------|
| ㉞                | Hold frames or diaphragms |

### ***Close-up survey and thickness measurement areas***

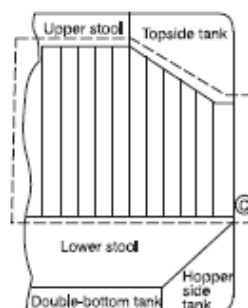
(Typical areas for thickness measurement of cargo hold frames, structural members and transverse bulkheads in association with close-up survey requirements)

Typical transverse section  
Areas (A), (B) and (D)



Thickness to be reported on TM3-BC, TM4-BC, TM6-BC and TM7-BC as appropriate

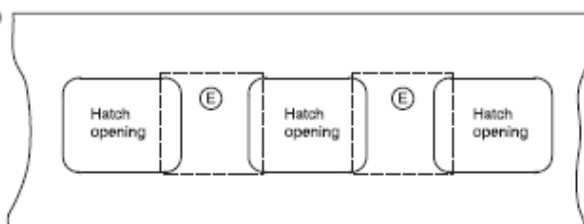
A cargo hold transverse bulkhead  
Area (C)



Thickness to be reported on TM5-BC

Typical areas of deck plating inside line of hatch openings between cargo hold hatches

Area (E)



Thickness to be reported on TM6-BC

94034

## **ANNEX 9**

### **GUIDELINES FOR TECHNICAL ASSESSMENT IN CONJUNCTION WITH THE PLANNING OF ENHANCED SURVEYS FOR BULK CARRIERS – RENEWAL SURVEY HULL**

#### **1 Introduction**

These Guidelines contain information and suggestions concerning technical assessments which may be of use in conjunction with the planning of enhanced renewal surveys of bulk carriers. As indicated in 5.1.5, the Guidelines are a recommended tool which may be invoked at the discretion of an Administration, when considered necessary and appropriate, in conjunction with the preparation of the required survey programme.

#### **2 Purpose and principles**

##### **2.1 Purpose**

The purpose of the technical assessments described in these Guidelines is to assist in identifying critical structural areas, nominating suspect areas and in focusing attention on structural elements or areas of structural elements which may be particularly susceptible to, or evidence a history of, wastage or damage. This information may be useful in nominating locations, areas, holds and tanks for thickness measurement, close-up survey and tank testing.

Critical structural areas are locations which have been identified from calculations to require monitoring or from the service history of the subject ship or from similar or sister ships (if available) to be sensitive to cracking, buckling or corrosion which would impair the structural integrity of the ship.

##### **2.2 Minimum requirements**

These Guidelines may not be used to reduce the requirements of annexes 1 and 2 and paragraph 2.7 of part A for close-up survey, thickness measurement and tank testing, respectively, which are, in all cases, to be complied with as a minimum.

##### **2.3 Timing**

As with other aspects of survey planning, the technical assessments described in these Guidelines should be completed by the owner or operator in co-operation with the Administration well in advance of the commencement of the renewal survey, i.e. prior to commencing the survey and normally at least 12 to 15 months before the survey's completion due date.

##### **2.4 Aspects to be considered**

**2.4.1** Technical assessments, which may include quantitative or qualitative evaluation of relative risks of possible deterioration, of the following aspects of a particular ship may be used as a basis for the nomination of holds, tanks and areas for survey:

- .1** design features such as stress levels on various structural elements, design details and extent of use of high-tensile steel;

- .2 former history with respect to corrosion, cracking, buckling, indents and repairs for the particular ship as well as similar vessels, where available; and
- .3 information with respect to types of cargo carried, protection of tanks, and condition of coating, if any, of holds and tanks.

**2.4.2** Technical assessments of the relative risks of susceptibility to damage or deterioration of various structural elements and areas should be judged and decided on the basis of recognized principles and practices, such as may be found in reference 3.

### **3 Technical assessment**

#### **3.1 General**

**3.1.1** There are three basic types of possible failure which may be the subject of technical assessment in connection with planning of surveys: corrosion, cracks and buckling. Contact damages are not normally covered by the survey plan since indents are usually noted in memoranda and assumed to be dealt with as a normal routine by surveyors.

**3.1.2** Technical assessments performed in conjunction with the survey planning process should, in principle, be as shown schematically in figure 1 which depicts, schematically, how technical assessments can be carried out in conjunction with the survey planning process. The approach is based on an evaluation of experience and knowledge basically related to:

- .1 design; and
- .2 corrosion.

**3.1.3** The design should be considered with respect to structural details which may be susceptible to buckling or cracking as a result of vibration, high stress levels or fatigue.

**3.1.4** Corrosion is related to the ageing process, and is closely connected with the quality of corrosion protection at new building, and subsequent maintenance during the service life. Corrosion may also lead to cracking and/or buckling.

#### **3.2 Methods**

##### **3.2.1 Design details**

**3.2.1.1** Damage experience related to the ship in question and similar ships, where available, is the main source of information to be used in the process of planning. In addition, a selection of structural details from the design drawings should be included.

**3.2.1.2** Typical damage experience to be considered will consist of:

- .1 number, extent, location and frequency of cracks; and
- .2 location of buckles.

**3.2.1.3** This information may be found in the survey reports and/or the owner's files, including the results of the owner's own inspections. The defects should be analysed, noted and marked on sketches.

**3.2.1.4** In addition, general experience should be utilized. For example, figure 2 shows typical locations in bulk carriers which experience has shown may be susceptible to structural damage. Also, reference should be made to reference 3 which contains a catalogue of typical damages and proposed repair methods for various bulk carrier structural details.

**3.2.1.5** Such figures should be used together with a review of the main drawings, in order to compare with the actual structure and search for similar details which may be susceptible to damage. An example is shown in figure 3.

**3.2.1.6** The review of the main structural drawings, in addition to using the above-mentioned figures, should include checking typical design details where cracking has been experienced. The factors contributing to damage should be carefully considered.

**3.2.1.7** The use of high-tensile steel (HTS) is an important factor. Details showing good service experience where ordinary, mild steel has been used may be more susceptible to damage when HTS, and its higher associated stresses, are utilized. There is extensive and, in general, good experience with the use of HTS for longitudinal material in deck and bottom structures. Experience in other locations, where the dynamic stresses may be higher, is less favourable, e.g., side structures.

**3.2.1.8** In this respect, stress calculations of typical and important components and details, in accordance with relevant methods, may prove useful and should be considered.

**3.2.1.9** The selected areas of the structure identified during this process should be recorded and marked on the structural drawings to be included in the survey programme.

## **3.2.2**     *Corrosion*

**3.2.2.1** In order to evaluate relative corrosion risks, the following information is generally to be considered:

- .1        usage of tanks, holds and spaces;
- .2        condition of coatings;
- .3        cleaning procedures;
- .4        previous corrosion damage;
- .5        ballast use and time for cargo holds;
- .6        risk of corrosion in cargo holds and ballast tanks; and
- .7        location of ballast tanks adjacent to heated fuel oil tanks.

**3.2.2.2** Reference 2 gives definitive examples which can be used for judging and describing coating condition, using typical pictures of conditions.

**3.2.2.3** For bulk carriers, reference 3 should be used as the basis for the evaluation, together with the age of the ship and relevant information on the anticipated condition of the ship as derived from the information collected in order to prepare the survey programme.



**3.2.2.4** The various tanks, holds and spaces should be listed with the corrosion risks nominated accordingly.

**3.2.3** *Locations for close-up survey and thickness measurement*

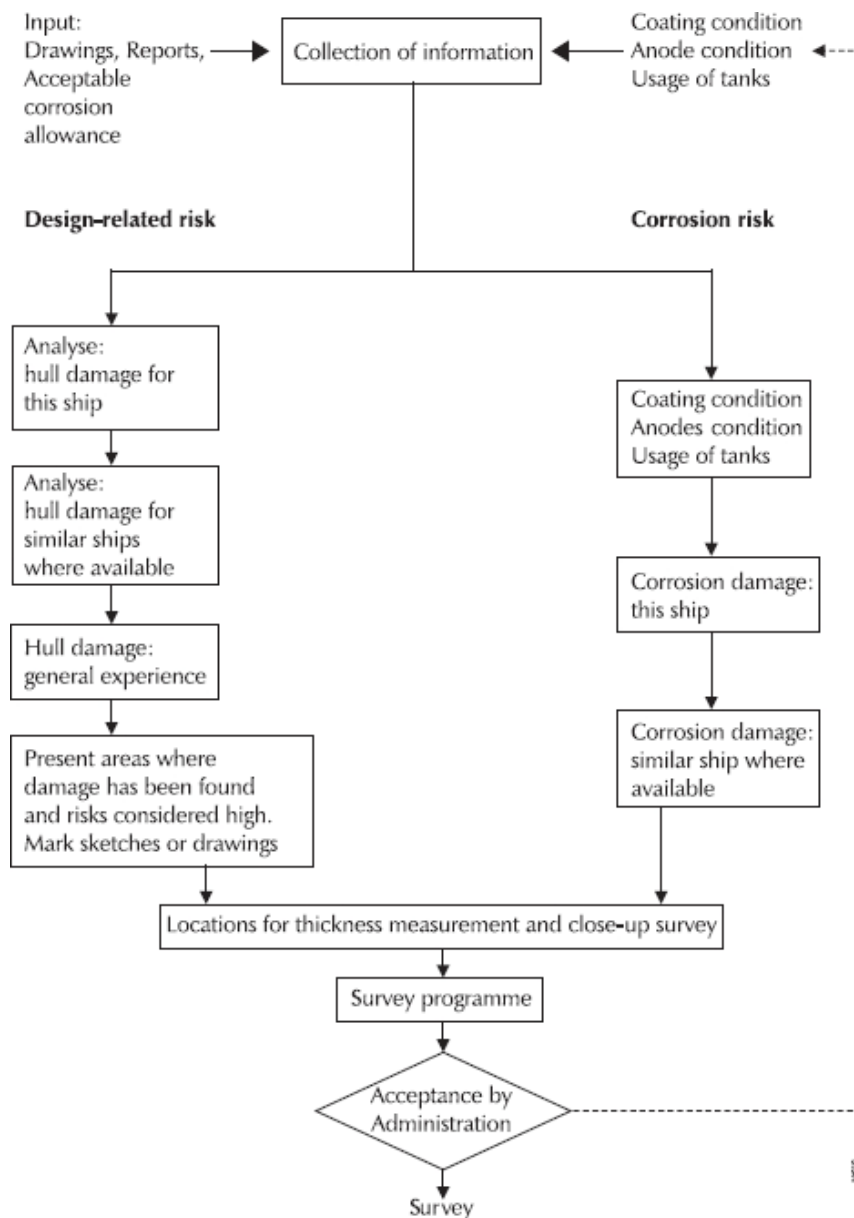
**3.2.3.1** On the basis of the table of corrosion risks and the evaluation of design experience, the locations for initial close-up survey and thickness measurement (sections) may be nominated.

**3.2.3.2** The sections subject to thickness measurement should normally be nominated in tanks, holds and spaces where corrosion risk is judged to be the highest.

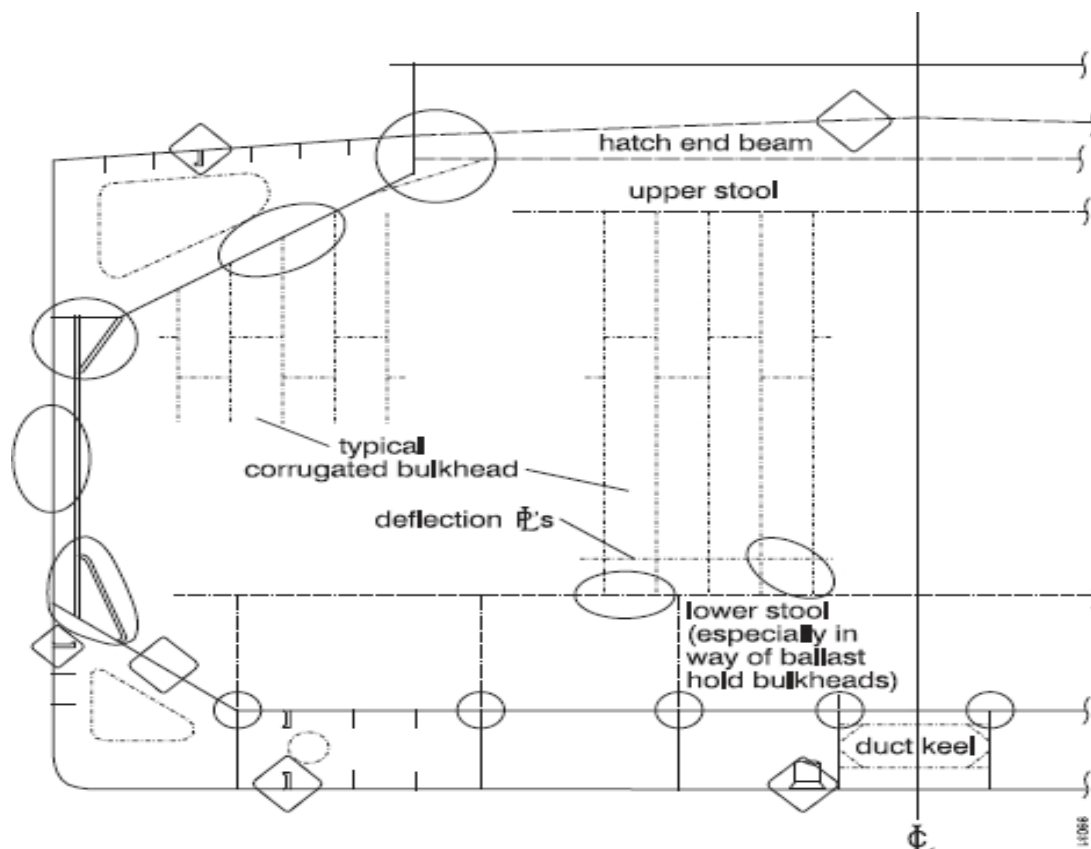
**3.2.3.3** The nomination of tanks, holds and spaces for close-up survey should, initially, be based on highest corrosion risk, and should always include ballast tanks. The principle for the selection should be that the extent is increased by age or where information is insufficient or unreliable.

**References**

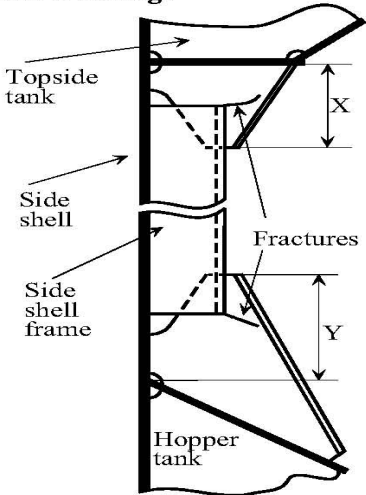
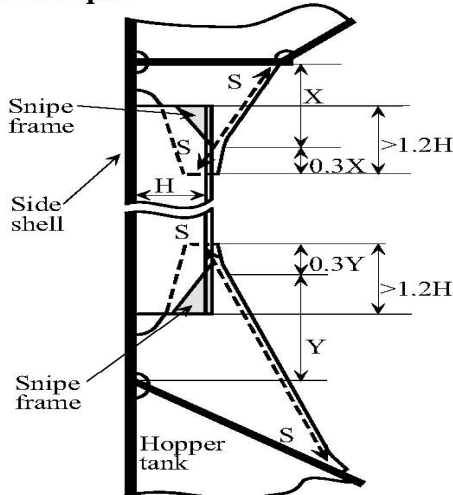
- 1** TSCF, Guidance Manual for the Inspection and Condition Assessment of Tanker Structures, 1986.
- 2** TSCF, Condition Evaluation and Maintenance of Tanker Structures, 1992.
- 3** IACS, Bulk Carriers: Guidelines for Surveys, Assessment and Repair of Hull Structures, 2007.



**Figure 1** – Technical assessment and the survey planning process



**Figure 2** – Typical locations susceptible to structural damage or corrosion

| <b>BULK CARRIERS</b>                                                                                                                                      |                                               | <b>Guidelines for Surveys, Assessment and Repair of Hull Structure</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Part 1</b>                                                                                                                                             | <b>Cargo hold region</b>                      | <b>Example No.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Area 3</b>                                                                                                                                             | <b>Cargo hold side structure</b>              | <b>1-a</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Detail of damage</b>                                                                                                                                   | Fractures in brackets at termination of frame |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Sketch of damage</b><br> <p><b>Separate bracket configuration</b></p> |                                               | <b>Sketch of repair</b><br> <p>S=Snipped end</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Notes on possible cause of damage</b><br>1. This type of damage is caused due to stress concentration.                                                 |                                               | <b>Notes on repairs</b><br>1. For small fractures, e. g. hairline fractures, the fracture can be veed-out, ground, examined by NDT for fractures, and rewelded.<br>2. For larger/significant fractures consideration is to be given to cropping and partly renewing/renewing the frame brackets. If renewing the brackets, end of frames can be snipped to soften them.<br>3. If felt prudent, soft toes are to be incorporated at the boundaries of the bracket to the hopper plating.<br>4. Attention to be given to the structure in wing tanks in way of the extended bracket arm. i.e. reinforcement provided in line with the bracket. |

**Figure 3 – Typical damage and repair example (reproduced from reference 3)**

## ANNEX 10

### REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENT AT AREAS OF SUBSTANTIAL CORROSION

#### Renewal survey of bulk carriers within the cargo area Shell plating

| Structural member                 | Extent of measurement                                                         | Pattern of measurement                                                |
|-----------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1 Bottom and side shell plating   | a Suspect plate, plus four adjacent plates                                    | a Five-point pattern for each panel between longitudinals             |
|                                   | b See other tables for particulars on gauging in way of tanks and cargo holds |                                                                       |
| 2 Bottom/side shell longitudinals | Minimum of three longitudinals in way of suspect areas                        | Three measurements in line across web<br>Three measurements on flange |

#### Transverse bulkheads in cargo holds

| Structural member     | Extent of measurement                                                                                                                    | Pattern of measurement                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 1 Lower stool         | a Transverse band within 25 mm of welded connection to inner bottom                                                                      | a Five-point pattern between stiffeners over 1 m length |
|                       | b Transverse band within 25 mm of welded connection to shelf plate                                                                       | b Ditto                                                 |
| 2 Transverse bulkhead | a Transverse band at approximately mid-height                                                                                            | a Five-point pattern over 1 m <sup>2</sup> of plating   |
|                       | b Transverse band at part of bulkhead adjacent to upper deck or below upper stool shelf plate (for those ships fitted with upper stools) | b Five-point pattern over 1 m <sup>2</sup> of plating   |

**Deck structure including cross strips, main cargo hatchways, hatch covers, coamings and topside tanks**

| <b>Structural member</b>      | <b>Extent of measurement</b>                                                             | <b>Pattern of measurement</b>                                    |
|-------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 1 Cross-deck strip plating    | Suspect cross-deck strip plating                                                         | Five-point pattern between under deck stiffeners over 1 m length |
| 2 Under-deck stiffeners       | a Transverse members                                                                     | a Five-point pattern at each end and mid-span                    |
|                               | b Longitudinal member                                                                    | b Five-point pattern on both web and flange                      |
| 3 Hatch covers                | a Side and end skirts, each three locations                                              | a Five-point pattern at each location                            |
|                               | b Three longitudinal bands, outboard strakes (2) and centerline strake (1)               | b Five-point measurement each band                               |
| 4 Hatch coamings              | Each side and end of coaming, one band lower third, one band upper two-thirds of coaming | Five-point measurement each band, i.e. end or side coaming       |
| 5 Topside water ballast tanks | a Watertight transverse bulkheads                                                        |                                                                  |
|                               | i lower third of bulkhead                                                                | i five-point pattern over 1 m <sup>2</sup> of plating            |
|                               | ii upper two-thirds of bulkhead                                                          | ii five-point pattern over 1 m <sup>2</sup> of plating           |
|                               | iii stiffeners                                                                           | iii five-point pattern over 1 m length                           |
|                               | b Two representative swash transverse bulkheads                                          |                                                                  |
|                               | i lower third of bulkhead                                                                | i five-point pattern over 1 m <sup>2</sup> of plating            |
|                               | ii upper two-thirds of bulkhead                                                          | ii five-point pattern over 1 m <sup>2</sup> of plating           |
|                               | iii stiffeners                                                                           | iii five-point pattern over 1 m length                           |
|                               | c Three representative bays of slope plating                                             |                                                                  |
|                               | i lower third of tank                                                                    | i five-point pattern over 1 m <sup>2</sup> of plating            |
|                               | ii upper two-thirds of tank                                                              | ii five-point pattern over 1 m <sup>2</sup> of plating           |
|                               | d Longitudinals, suspect and adjacent                                                    | d Five-point pattern both web and flange over 1 m length         |
| 6 Main deck plating           | Suspect plates and adjacent (4)                                                          | Five-point pattern over 1 m <sup>2</sup> of plating              |
| 7 Main deck longitudinals     | Minimum of three longitudinals where plating measured                                    | Five-point pattern on both web and flange over 1 m length        |
| 8 Web frames/transverses      | Suspect plates                                                                           | Five-point pattern over 1 m <sup>2</sup>                         |

### Double-bottom and hopper structure

| Structural member                           | Extent of measurement                                  | Pattern of measurement                                                  |
|---------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------|
| 1 Inner/double-bottom plating               | Suspect plate plus all adjacent plates                 | Five-point pattern for each panel between longitudinals over 1 m length |
| 2 Inner/double-bottom longitudinals         | Three longitudinals where plates measured              | Three measurements in line across web, and three measurements on flange |
| 3 Longitudinal girders or transverse floors | Suspect plates                                         | Five-point pattern over about 1 m <sup>2</sup>                          |
| 4 Watertight bulkheads (WT floors)          | a Lower third of tank                                  | a Five-point pattern over 1 m <sup>2</sup> of plating                   |
|                                             | b Upper two-thirds of tank                             | b Five-point pattern alternate plates over 1 m <sup>2</sup> of plating  |
| 5 Web frames                                | Suspect plate                                          | Five-point pattern over 1 m <sup>2</sup> of plating                     |
| 6 Bottom/side shell longitudinals           | Minimum of three longitudinals in way of suspect areas | Three measurements in line across web, and three measurements on flange |

### Cargo holds

| Structural member   | Extent of measurement           | Pattern of measurement                                                                     |
|---------------------|---------------------------------|--------------------------------------------------------------------------------------------|
| 1 Side shell frames | Suspect frame and each adjacent | a At each end and mid span: five-point pattern of both web and flange                      |
|                     |                                 | b Five-point pattern within 25 mm of welded attachment to both shell and lower slope plate |

## **ANNEX 11**

### **GUIDELINES FOR THE GAUGING OF THE VERTICALLY CORRUGATED TRANSVERSE WATERTIGHT BULKHEAD BETWEEN HOLDS NOS.1 AND 2**

**1** Gauging is necessary to determine the general condition of the structure and to define the extent of possible repairs and/or reinforcements of the vertically corrugated transverse watertight bulkhead for verification of the compliance with the bulk carrier bulkhead and double-bottom strength standards, defined in regulation XII/1.7 of the Convention.

**2** Taking into account the buckling model specified in the bulk carrier bulkhead and double-bottom strength standards, defined in regulation XII/1.7 of the Convention, in the evaluation of strength of the bulkhead, it is essential to determine the thickness diminution at the critical levels shown in figures 1 and 2 of this annex.

**3** The gauging should be carried out at the levels as described below. To adequately assess the scantlings of each individual vertical corrugation, each corrugation flange, web, shedder plate and gusset plate within each of the levels given below should be gauged.

Level (a) Ships without lower stool (see figure 1):

Locations:

- The mid-breadth of the corrugation flanges at approximately 200 mm above the line of shedder plates;
- The middle of gusset plates between corrugation flanges, where fitted;
- The middle of the shedder plates;
- The mid-breadth of the corrugation webs at approximately 200 mm above the line of shedder plates.

Level (b) Ships with lower stool (see figure 2):

Locations:

- The mid-breadth of the corrugation flanges at approximately 200 mm above the line of shedder plates;
- The middle of gusset plates between corrugation flanges, where fitted;
- The middle of the shedder plates;
- The mid-breadth of the corrugation webs at approximately 200 mm above the line of shedder plates.

Level (c) Ships with or without lower stool (see figures 1 and 2):

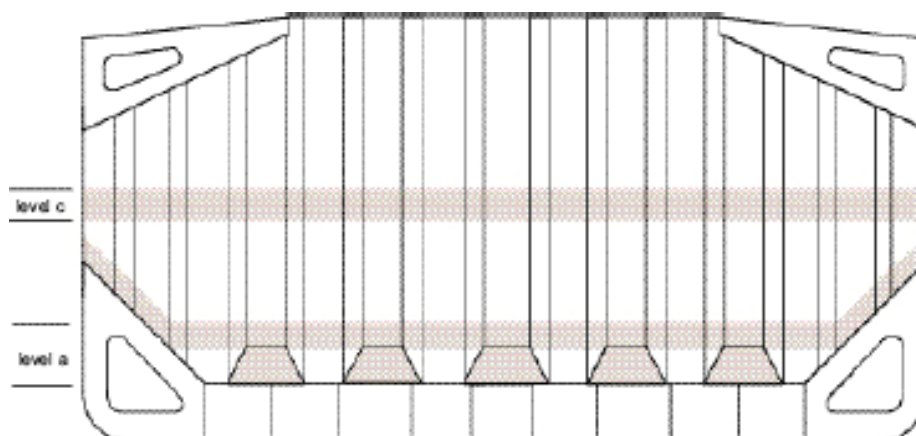
Locations:

- The mid-breadth of the corrugation flanges and webs at about the mid-height of the corrugation.

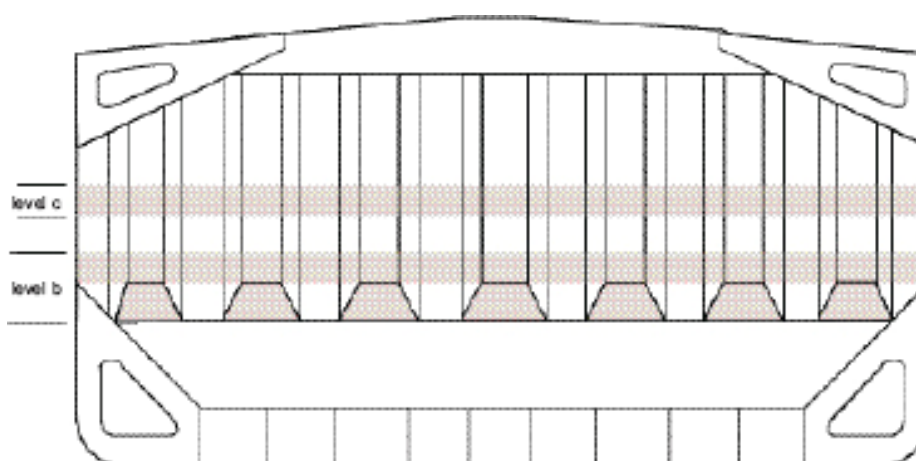
**4** Where the thickness changes within the horizontal levels, the thinner plate should be gauged.

**5** Steel renewal and/or reinforcement should comply with the bulk carrier bulkhead and double-bottom strength standards, defined in regulation XII/1.7 of the Convention.





**Figure 1** – Ships without lower stool



**Figure 2** – Ships with lower stool

## **ANNEX 12**

### **ADDITIONAL ANNUAL SURVEY REQUIREMENTS FOR THE FOREMOST CARGO HOLD OF SHIPS SUBJECT TO REGULATION XII/9 OF THE CONVENTION**

#### **1 General**

In the case of bulk carriers over 5 years of age, the annual survey should include, in addition to the requirements of the annual surveys prescribed in chapter 3 of the present Code, an examination of the following items.

#### **2 Extent of survey**

##### **2.1 *For bulk carriers of 5 to 15 years of age***

**2.1.1** An overall survey of the foremost cargo hold, including close-up survey of sufficient extent, minimum 25% of frames, should be carried out to establish the condition of:

- .1** shell frames including their upper and lower end attachments, adjacent shell plating, and transverse bulkheads; and
- .2** areas found to be suspect areas at the previous renewal survey.

**2.1.2** Where considered necessary by the surveyor as a result of the overall and close-up survey as described in 2.1.1 above, the survey should be extended to include a close-up survey of all of the shell frames and adjacent shell plating of the cargo hold.

##### **2.2 *For bulk carriers exceeding 15 years of age***

An overall survey of the foremost cargo hold, including close-up survey should be carried out to establish the condition of:

- .1** all shell frames including their upper and lower end attachments, adjacent shell plating, and transverse bulkheads; and
- .2** areas found to be suspect areas at the previous renewal survey.

#### **3 Extent of thickness measurement**

**3.1** Thickness measurement should be carried out to an extent sufficient to determine both general and local corrosion levels at areas subject to close-up survey, as described in 2.1 and 2.2 above. The minimum requirement for thickness measurements are areas found to be suspect areas at the previous renewal survey. Where substantial corrosion is found, the extent of thickness measurements should be increased with the requirements of annex 10.

**3.2** The thickness measurement may be dispensed with provided the surveyor is satisfied by the close-up survey, there is no structural diminution and the protective coating, where applied, remains effective.

#### **4 Special consideration**

Where the protective coating, as referred to in the explanatory note below, in the foremost cargo hold is found to be in GOOD condition, the extent of close-up surveys and thickness measurements may be specially considered.

***Explanatory note:***

For existing bulk carriers, where owners may elect to coat or recoat cargo holds as noted above, consideration may be given to the extent of the close-up and thickness measurement surveys. Prior to the coating of cargo holds of existing ships, scantlings should be ascertained in the presence of a surveyor.

## ANNEX 13

### STRENGTH OF CARGO HATCH COVER SECURING ARRANGEMENTS FOR BULK CARRIERS

#### 1 Securing devices

The strength of securing devices should comply with the following requirements:

- .1 Panel hatch covers should be secured by appropriate devices (bolts, wedges or similar) suitably spaced alongside the coamings and between cover elements. Arrangement and spacing should be determined with due attention to the effectiveness for weathertightness, depending upon the type and the size of the hatch cover, as well as on the stiffness of the cover edges between the securing devices.

- .2 The net sectional area of each securing device should not be less than:

$$A = 1.4 a / f \text{ (cm}^2\text{)}$$

where:

- $a$  = spacing between securing devices not to be taken less than 2 m  
 $f$  =  $(\sigma_Y / 235)^e$   
 $\sigma_Y$  = specified minimum upper yield stress in N/mm<sup>2</sup> of the steel used for fabrication, not to be taken greater than 70% of the ultimate tensile strength  
 $e$  = 0.75 for  $\sigma_Y > 235$   
1.0 for  $\sigma_Y \leq 235$ .

Rods or bolts should have a net diameter not less than 19 mm for hatchways exceeding 5 m<sup>2</sup> in area.

- .3 Between cover and coaming and at cross-joints, a packing line pressure sufficient to obtain weathertightness should be maintained by the securing devices. For packing line pressures exceeding 5 N/mm, the cross section area should be increased in direct proportion. The packing line pressure should be specified.

- .4 The cover edge stiffness should be sufficient to maintain adequate sealing pressure between securing devices. The moment of inertia,  $I$ , of edge elements should not be less than:

$$I = 6 p a^4 \text{ (cm}^4\text{)}$$

where:

- $p$  = packing line pressure in N/mm, minimum 5 N/mm  
 $a$  = spacing in metres of securing devices.

- .5 Securing devices should be of reliable construction and securely attached to the hatchway coamings, decks or covers. Individual securing devices on each cover are to have approximately the same stiffness characteristics.

- .6 Where rod cleats are fitted, resilient washers or cushions should be incorporated.
- .7 Where hydraulic cleating is adopted, a positive means should be provided to ensure that it remains mechanically locked in the closed position in the event of failure of the hydraulic system.

## **2 Stoppers**

**2.1** Nos.1 and 2 hatch covers should be effectively secured, by means of stoppers, against the transverse forces arising from a pressure of  $175 \text{ kN/m}^2$ .

**2.2** No.2 hatch covers should be effectively secured, by means of stoppers, against the longitudinal forces acting on the forward end arising from a pressure of  $175 \text{ kN/m}^2$ .

**2.3** No.1 hatch cover should be effectively secured, by means of stoppers, against the longitudinal forces acting on the forward end arising from a pressure of  $230 \text{ kN/m}^2$ . This pressure may be reduced to  $175 \text{ kN/m}^2$  if a forecastle is fitted.

**2.4** The equivalent stress in stoppers and their supporting structures and calculated in the throat of the stopper welds should not exceed the allowable value of  $0.8 \sigma_Y$ .

## **3 Materials and welding**

Where stoppers or securing devices are fitted to comply with this annex, they should be manufactured of materials, including welding electrodes, to the satisfaction of the Administration.

## **ANNEX 14**

### **PROCEDURAL REQUIREMENTS FOR THICKNESS MEASUREMENTS**

#### **1 General**

Thickness measurements required in the context of hull structural surveys, if not carried out by the society itself, should be witnessed by a surveyor. The attendance of the surveyor should be recorded. This also applies to thickness measurements taken during voyages.

#### **2 Survey meeting**

**2.1** Prior to commencement of the renewal or intermediate survey, a meeting should be held between the attending surveyor(s), the owner's representative(s) in attendance, the thickness measurement firm's representative(s) and the master of the ship or an appropriately qualified representative appointed by the master or Company so as to ensure the safe and efficient execution of the surveys and thickness measurements to be carried out on board.

**2.2** Communication with the thickness measurement operator(s) and owner's representative(s) should be agreed during the meeting, with respect to the following:

- .1** reporting of thickness measurements on regular basis; and
- .2** prompt notification to the surveyor in case of findings such as:
  - .1** excessive and/or extensive corrosion or pitting/grooving of any significance;
  - .2** structural defects like buckling, fractures and deformed structures;
  - .3** detached and/or holed structure; and
  - .4** corrosion of welds.

**2.3** The survey report should indicate where and when the meeting took place and who attended (the name of the surveyor(s), the owner's representative(s) and the thickness measurement firm's representative(s)).

#### **3 Monitoring of the thickness measurement process on board**

**3.1** The surveyor should decide final extent and location of thickness measurements after overall survey of representative spaces on board.

**3.2** In case the owner prefers to commence the thickness measurements prior to the overall survey, then the surveyor should advise that the planned extent and locations of thickness measurements are subject to confirmation during the overall survey. Based on findings, the surveyor may require additional thickness measurements to be taken.

**3.3** The surveyor should direct the gauging operation by selecting locations such that readings taken represent, on average, the condition of the structure for that area.

**3.4** Thickness measurements taken mainly to evaluate the extent of corrosion, which may affect the hull girder strength, should be carried out in a systematic manner such that all longitudinal structural members are gauged, as required.

**3.5** Where thickness measurements indicate substantial corrosion or wastage in excess of allowable diminution, the surveyor should direct locations for additional thickness measurements in order to delineate areas of substantial corrosion and to identify structural members for repairs/renewals.

**3.6** Thickness measurements of structures in areas where close-up surveys are required should be carried out simultaneously with close-up survey.

#### **4 Review and verification**

**4.1** Upon completion of the thickness measurements, the surveyor should confirm that no further gaugings are needed, or specify additional gaugings.

**4.2** Where these guidelines allow the extent of thickness measurements to be reduced after special considerations by the surveyor, these special considerations should be reported, where appropriate.

**4.3** In case thickness measurements are partly carried out, the extent of remaining thickness measurements should be reported for the use of the next surveyor.

## ANNEX 15

### GUIDELINES FOR THE THICKNESS MEASUREMENTS OF SIDE SHELL FRAMES AND BRACKETS IN SINGLE-SIDE SKIN BULK CARRIERS REQUIRED TO COMPLY WITH RESOLUTION MSC.168(79)

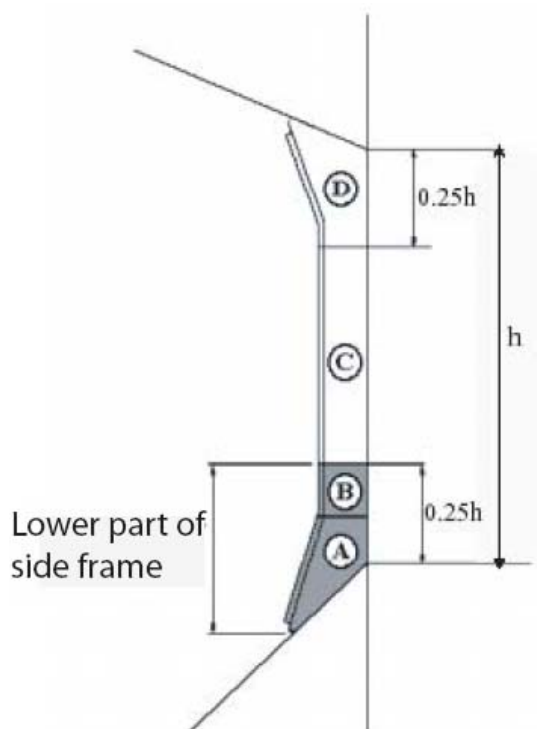
#### 1 General

Thickness measurements should be conducted to determine the general condition of the structure and to define the extent of possible steel renewals or other measures for the webs and flanges of side shell frames and brackets for verification of the compliance with resolution MSC.168(79).

#### 2 Zones of side shell frames and brackets

2.1 For the purpose of steel renewal, sand blasting and coating, four zones A, B, C and D are defined as shown in figure 1.

2.2 Zones A and B should be considered the most critical zones.



**Figure 1** – Zones of side shell frames and brackets

#### 3 Pitting and grooving

3.1 Pits can grow in a variety of shapes, some of which should be ground before assessment.

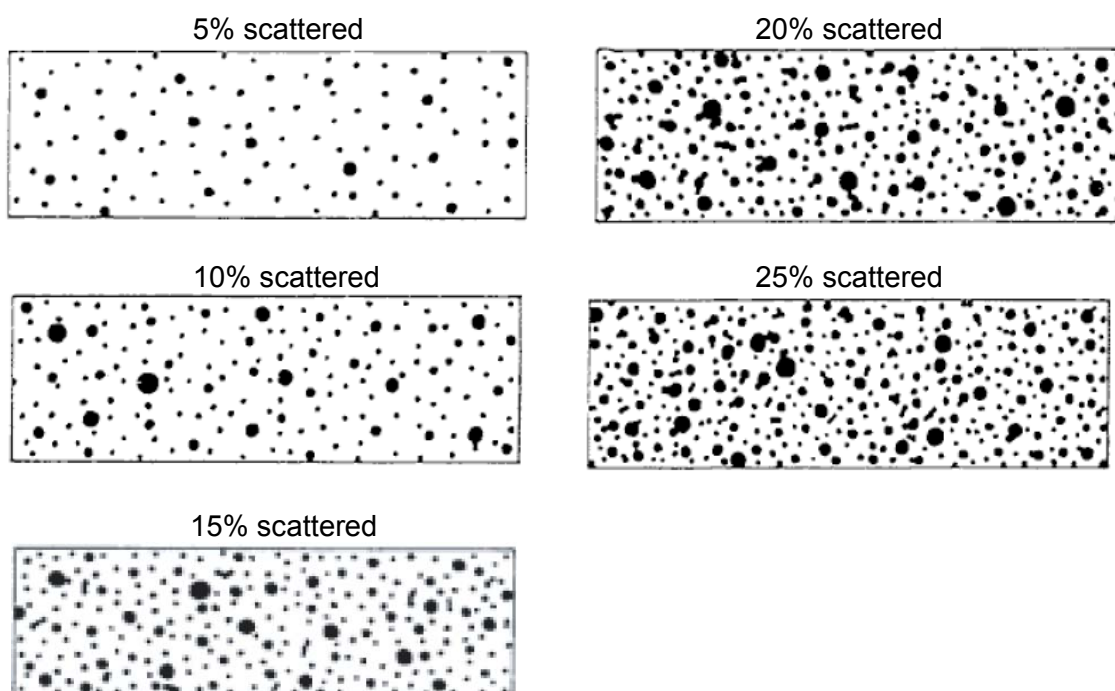
3.2 Pitting corrosion may be found under coating blisters, which should be removed before inspection.



3.3 To measure the remaining thickness of pits or grooving the normal ultrasonic transducer (generally 10 mm diameter) will not suffice. A miniature transducer (3 to 5 mm diameter) should be used. Alternatively the firm conducting the thickness measurement should use a pit gauge to measure the depth of the pits and grooving and calculate the remaining thickness.

**Assessment based upon area**

3.4 This is the method specified in paragraph 2.5 of annex 2 to resolution MSC.168(79) and is based upon the intensity determined from figure 2 below.



**Figure 2 – Pitting intensity diagrams (from 5% to 25% intensity)**

3.5 If pitting intensity is higher than 15% in an area (see figure 2), then thickness measurements should be taken to determine the extent of the pitting corrosion. The 15% is based upon pitting or grooving on only one side of the plate.

3.6 In cases where pitting is evident as defined above (exceeding 15%) then an area of 300 mm diameter or more (or, where this is impracticable on the frame flange or the side shell, hopper tank plating or topside tank plating attached to the side frame, an equivalent rectangular area), at the most pitted part, should be cleaned to bare metal, and the thickness measured in way of the five deepest pits within the cleaned area. The least thickness measured in way of any of these pits is to be taken as the thickness to be recorded.

3.7 The minimum acceptable remaining thickness in any pit or groove is equal to:

- .1 75% of the as built thickness, for pitting or grooving in the cargo hold side frame webs and flanges.
- .2 70% of the as built thickness, for pitting or grooving in the side shell, hopper tank and topside tank plating attached to the cargo hold side frame, over a width up to 30 mm from each side of it.

## 4 Thickness measurement methodology

4.1 Numbers of side frames to be measured should be equivalent to those of the renewal survey or intermediate survey corresponding to the ship's age. Representative thickness measurements should be taken for each zone as specified below.

4.2 Special consideration to the extent of the thickness measurements may be given by the Administration or organization recognized by the Administration, if the structural members show no thickness diminution with respect to the as built thicknesses and the coating is found in "as new" condition (i.e. without breakdown or rusting).

4.3 Where thickness measurement readings close to the criteria are found, the number of hold frames to be measured should be increased.

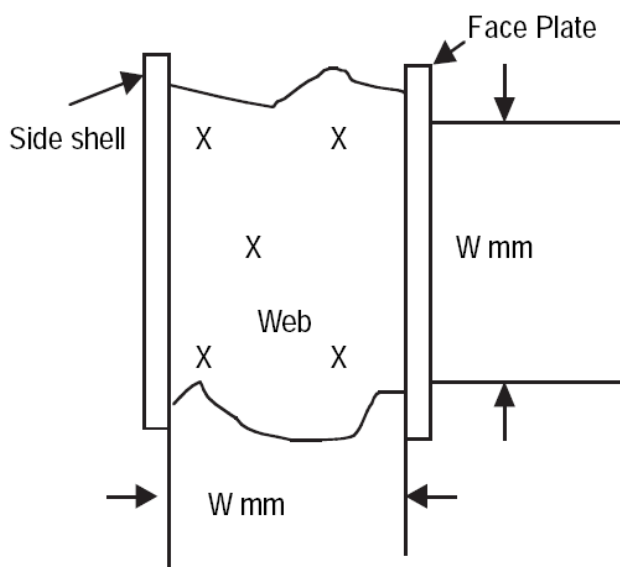
4.4 If renewal or other measures according to resolution MSC.168(79) should be applied on individual frames in a hold, then all frames in that hold should have thickness measurements taken.

4.5 There is a variety of construction methods used for side shell frames in bulk carriers. Some have faceplates (T sections) on the side shell frames, some have flanged plates and some have bulb plates. The use of faceplates and flanged sections is considered similar for thickness measurement purposes in that both the web and faceplate or web and flange plate should be measured. If bulb plate has been used, then web of the bulb plate should be measured in the normal manner and the sectional modulus should be specially considered if required.

*Gaugings for Zones A, B and D*

*Web plating*

4.6 The pattern for thickness measurements in Zones A, B and D should be a five-point pattern. See figure 3. The five-point pattern should be over the depth of the web and the same area vertically. The thickness measurement report should reflect the average reading.



**Figure 3** – Typical five-point pattern on the web plate

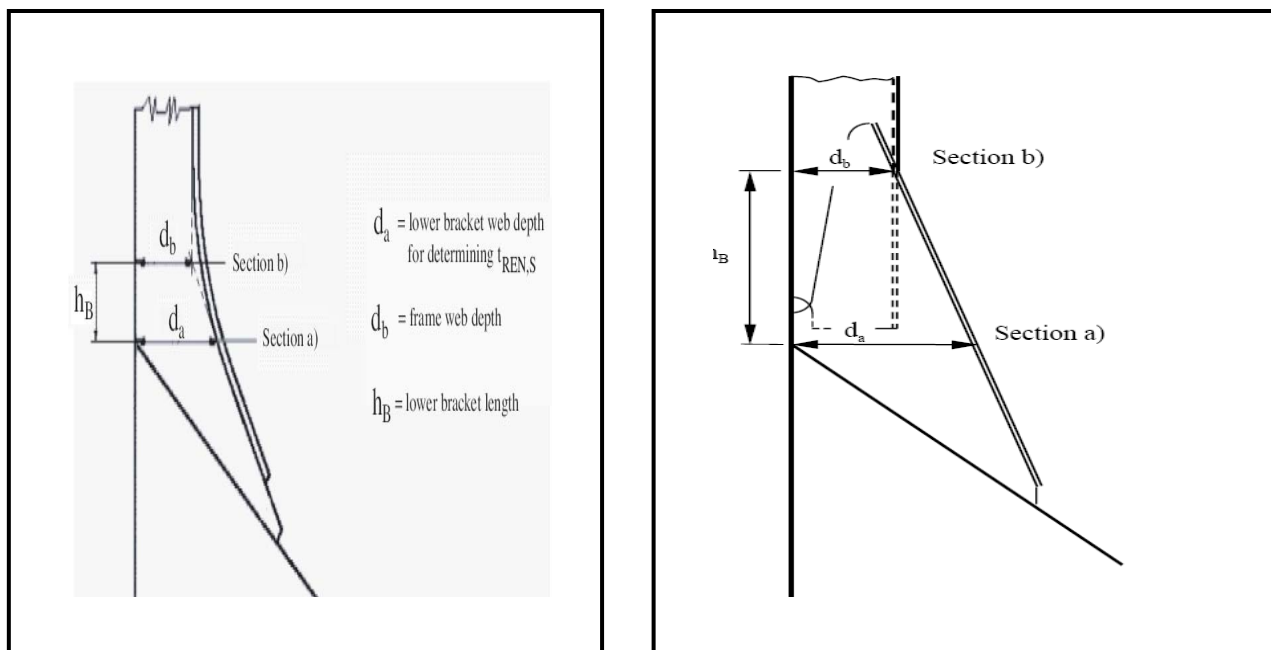
### Thickness measurements for Zone C

#### Web plating

4.7 Depending on the condition of the web in way of Zone C, the web may be measured by taking three readings over the length of Zone C and averaging them. The average reading should be compared with the allowable thickness. If the web plating has general corrosion then this pattern should be expanded to a five-point pattern as noted above.

#### Thickness measurements for section a) and b) (flanges and side shell plating)

4.8 Where the lower bracket length or depth does not meet the requirements in annex 1 to resolution MSC.168(79), thickness measurements should be taken at sections a) and b) to calculate the actual section modulus required in paragraph 3.4 of annex 2 to resolution MSC.168(79) (see figure 4). At least two readings on the flange/faceplate should be taken on each side of the frame (i.e. fore and aft) in way of section a) and section b).



**Figure 4 – Sections a) and b)**

## 5 Report on thickness measurement of cargo hold frames

See form TM7-BC(ii) as contained in appendix 2 to annex 8.

### References

IACS, Blue Books: Unified Requirement UR Z10.2 Hull Surveys of Bulk Carriers, Revision 27 2009, Annex V.

Resolution MSC.168(79) on Standards and criteria for side structures of bulk carriers of single-side skin construction.

## Part B

### CODE ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS HAVING DOUBLE-SIDE SKIN CONSTRUCTION

#### 1 General

##### 1.1 *Application*<sup>5</sup>

**1.1.1** The Code should apply to all self-propelled double-side skin bulk carriers of 500 gross tonnage and above.

**1.1.2** The Code should apply to surveys of hull structure and piping systems in way of cargo holds, cofferdams, pipe tunnels, void spaces, fuel oil tanks within the cargo length area and all ballast tanks.

**1.1.3** The Code contains the minimum extent of examination, thickness measurements and tank testing. The survey should be extended when substantial corrosion and/or structural defects are found and include additional close-up survey when necessary.

**1.1.4** For bulk carriers with hybrid cargo hold arrangements, e.g., with some cargo holds of single-side skin and others of double-side skin, the requirements of part A of annex A apply to cargo holds of single-side skin.

**1.1.5** The surveys should be carried out during the surveys prescribed by regulation I/10 of the Convention.

##### 1.2 *Definitions*

**1.2.1** *Double-side skin bulk carrier* is a ship which is constructed generally with single deck, topside tanks and hopper side tanks in cargo spaces, and is intended primarily to carry dry cargo in bulk including ore carriers and combination carriers, in which all cargo holds are bounded by a double-side skin (regardless of the width of the wing space).<sup>6</sup>

**1.2.2** *Ballast tank* is a tank which is used solely for salt water ballast or, where applicable, a space which is used for both cargo and salt water ballast will be treated as a ballast tank when substantial corrosion has been found in that space. A double-side tank should be considered, for survey purposes, as a separate tank even if it is in connection to either the topside tank or the hopper side tank.

**1.2.3** *Spaces* are separate compartments including holds, tanks, cofferdams and void spaces bounding cargo holds, decks and the outer hull.

**1.2.4** *Overall survey* is a survey intended to report on the overall condition of the hull structure and determine the extent of additional close-up surveys.

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<sup>5</sup> The intention of the Code is to ensure that an appropriate level of review of plans and documents is conducted and consistency in application is attained. Such evaluation of survey reports, survey programmes, planning documents, etc., should be carried out at the managerial level of the Administration or organization recognized by the Administration.

<sup>6</sup> For double-side skin combination carriers, additional requirements are specified in the Guidelines on the enhanced programme of inspections during surveys for oil tankers, set out in part A of annex B.

**1.2.5** *Close-up survey* is a survey where the details of structural components are within the close visual inspection range of the surveyor, i.e. normally within reach of hand.

**1.2.6** *Transverse section* is the cross section of the hull perpendicular to the ship's centerline and includes all longitudinal members such as plating, longitudinals and girders at the deck sides, bottom, inner bottom, hopper sides, inner sides, top wing inner sides and longitudinal bulkheads.

**1.2.7** *Representative spaces* are those which are expected to reflect the condition of other spaces of similar type and service and with similar corrosion prevention systems. When selecting representative spaces, account should be taken of the service and repair history on board and identifiable critical structural areas and/or suspect areas.

**1.2.8** *Suspect areas* are locations showing substantial corrosion and/or are considered by the surveyor to be prone to rapid wastage.

**1.2.9** *Substantial corrosion* is an extent of corrosion such that assessment of corrosion pattern indicates a wastage in excess of 75% of allowable margins, but within acceptable limits. For ships built under the IACS Common Structural Rules, substantial corrosion is an extent of corrosion such that the assessment of the corrosion pattern indicates a gauged (or measured) thickness between  $t_{\text{net}} + 0.5 \text{ mm}$  and  $t_{\text{net}}$ .

**1.2.10** A *corrosion prevention system* is normally considered a full hard protective coating. Hard protective coating should usually be epoxy coating or equivalent. Other coating systems which are neither soft nor semi-hard coatings, may be considered acceptable as alternatives provided that they are applied and maintained in compliance with the manufacturer's specifications.

**1.2.11** *Coating condition* is defined as follows:

GOOD condition with only minor spot rusting;

FAIR condition with local breakdown of coating at edges of stiffeners and weld connections and/or light rusting over 20% or more of areas under consideration, but less than as defined for POOR condition;

POOR condition with general breakdown of coating over 20% or more of areas or hard scale at 10% or more of areas under consideration.

**1.2.12** *Critical structural areas* are locations which have been identified from calculations to require monitoring or from the service history of the subject ship or from similar or sister ships (if applicable) to be sensitive to cracking, buckling or corrosion which would impair the structural integrity of the ship.

**1.2.13** *Cargo length area* is that part of the ship which includes all cargo holds and adjacent areas including fuel tanks, cofferdams, ballast tanks and void spaces.

**1.2.14** *Intermediate survey* is a survey carried out either at the second or third annual survey or between these surveys.

**1.2.15** A *prompt and thorough repair* is a permanent repair completed at the time of survey to the satisfaction of the surveyor, therein removing the need for the imposition of any associated condition of classification or recommendation.

**1.2.16** *Convention* means the International Convention for the Safety of Life at Sea, 1974, as amended.

**1.2.17** *Specially considered* (in connection with close-up surveys and thickness measurements) means sufficient close-up inspection and thickness measurements are taken to confirm the actual average condition of the structure under the coating.

### **1.3 Repairs**

**1.3.1** Any damage in association with wastage over the allowable limits (including buckling, grooving, detachment or fracture), or extensive areas of wastage over the allowable limits, which affects or, in the opinion of the Administration, will affect the ship's structural, watertight or weathertight integrity, should be promptly and thoroughly (see 1.2.15) repaired. Areas which should be considered include:

- .1 side structure and side plating;
- .2 deck structure and deck plating;
- .3 bottom structure and bottom plating;
- .4 inner bottom structure and inner bottom plating;
- .5 inner side structure and inner side plating;
- .6 watertight or oiltight bulkheads;
- .7 hatch covers or hatch coamings; and
- .8 items in 3.3.10.

For locations where adequate repair facilities are not available, the Administration may allow the ship to proceed directly to a repair facility. This may require discharging the cargo and/or temporary repairs for the intended voyage.

**1.3.2** Additionally, when a survey results in the identification of corrosion or structural defects, either of which, in the opinion of the Administration, will impair the ship's fitness for continued service, remedial measures should be implemented before the ship continues in service.

### **1.4 Surveyors**

For bulk carriers of 20,000 tons deadweight and above, two surveyors should jointly carry out the first scheduled renewal survey after the bulk carrier passes 10 years of age (i.e. third renewal survey), and all subsequent renewal surveys and intermediate surveys. If the surveys are carried out by a recognized organization, the surveyors should be exclusively employed by such recognized organizations.

### **1.5 Thickness measurements and close-up surveys**

In any kind of survey, i.e. renewal, intermediate, annual or other surveys having the scope of the foregoing ones, thickness measurements of structures in areas where close-up surveys are required should be carried out simultaneously with close-up surveys.

## **2 Renewal survey**

### **2.1 General**

**2.1.1** The renewal survey may be commenced at the fourth annual survey and be progressed during the succeeding year with a view to completion by the fifth anniversary date.

**2.1.2** As part of the preparation for the renewal survey, the survey programme should be dealt with in advance of the survey. The thickness measurement should not be held before the fourth annual survey.

**2.1.3** The survey should include, in addition to the requirements of the annual survey, examination, tests and checks of sufficient extent to ensure that the hull and related piping, as required in 2.1.5, is in satisfactory condition and is fit for its intended purpose for the new period of validity of the Cargo Ship Safety Construction Certificate, subject to proper maintenance and operation and to periodical surveys being carried out at the due dates.

**2.1.4** All cargo holds, ballast tanks, including double-bottom and double-side tanks, pipe tunnels, cofferdams and void spaces bounding cargo holds, decks and outer hull should be examined, and this examination should be supplemented by thickness measurement and testing, as required by 2.6 and 2.7, to ensure that the structural integrity remains effective. The examination should be sufficient to discover substantial corrosion, significant deformation, fractures, damages or other structural deterioration that may be present.

**2.1.5** All piping systems within the above spaces should be examined and operationally tested under working pressure to attending surveyor's satisfaction to ensure that tightness and condition remain satisfactory.

**2.1.6** The survey extent of ballast tanks converted to void spaces should be specially considered in relation to the requirements for ballast tanks.

**2.1.7** Concurrent crediting to both intermediate survey and renewal survey for surveys and thickness measurements of spaces should not be acceptable.

### **2.2 Dry dock survey**

**2.2.1** A survey in dry dock should be a part of the renewal survey. There should be a minimum of two inspections of the outside of the ship's bottom during the five-year period of the certificate. In all cases, the maximum interval between bottom inspections should not exceed 36 months.

**2.2.2** For ships of 15 years of age and over, inspection of the outside of the ship's bottom should be carried out with the ship in dry dock. For ships of less than 15 years of age, alternate inspections of the ship's bottom not conducted in conjunction with the renewal survey may be carried out with the ship afloat. Inspection of the ship afloat should only be carried out when the conditions are satisfactory and the proper equipment and suitably qualified staff is available.

**2.2.3** If a survey in dry-dock is not completed in conjunction with the enhanced survey during renewal survey or if the 36 month maximum interval referred to in 2.2.1 is not complied with, the Cargo Ship Safety Construction Certificate should cease to be valid until a survey in dry-dock is completed.

**2.2.4** The overall and close-up survey and thickness measurements, as applicable, of the lower portions of the cargo holds and ballast tanks should be carried out in accordance with the applicable requirements for renewal survey, if not already performed.

**Note:** Lower portions of the cargo holds and ballast tanks are considered to be the parts below light ballast water line.

## **2.3      *Space protection***

**2.3.1** Where provided, the condition of the corrosion prevention system of ballast tanks should be examined. For ballast tanks, excluding double-bottom tanks, where a hard protective coating is found in POOR condition as defined in 1.2.11, and it is not renewed, or where a soft or semi-hard coating has been applied, or where a hard protective coating has not been applied from the time of constructions, the tanks in question should be examined at annual intervals. Thickness measurements should be carried out as deemed necessary by the surveyor. When such breakdown of hard protective coating is found in water ballast double-bottom tanks and it is not renewed, or where a soft or semi-hard coating has been applied or where a hard protective coating has not been applied from the time of construction, the tanks in question may be examined at annual intervals. When considered necessary by the surveyor, or where extensive corrosion exists, thickness measurement should be carried out.

**2.3.2** Where a hard protective coating is provided in cargo holds and is found in good condition, the extent of close-up surveys and thickness measurements may be specially considered.

## **2.4      *Hatch covers and coamings***

The hatch covers and coamings should be surveyed as follows:

**2.4.1** A thorough inspection of the items listed in 3.3 should be carried out, in addition to all hatch covers and coamings.

**2.4.2** Checking of the satisfactory operation of all mechanically operated hatch covers should be made, including:

- .1          stowage and securing in open condition;
- .2          proper fit and efficiency of sealing in closed condition; and
- .3          operational testing of hydraulic and power components, wires, chains and link drives.

**2.4.3** The effectiveness of sealing arrangements of all hatch covers by hose testing or equivalent should be checked.

**2.4.4** Thickness measurement of the hatch cover and coaming plating and stiffeners should be carried out as given in annex 2.

## **2.5      *Extent of overall and close-up surveys***

**2.5.1** An overall survey of all tanks and spaces should be carried out at the renewal survey. Fuel oil tanks in the cargo length area should be surveyed as follows:



| Renewal survey<br>no.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Renewal survey<br>no.2   | Renewal survey<br>no.3    | Renewal survey<br>no.4 and subsequent |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------------------------|---------------------------------------|
| Age $\leq 5$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $5 < \text{Age} \leq 10$ | $10 < \text{Age} \leq 15$ | $15 < \text{Age}$                     |
| None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | One                      | Two                       | Half, minimum two                     |
| <b>Notes:</b> <ol style="list-style-type: none"> <li>1. These requirements apply to tanks of integral (structural) type.</li> <li>2. If a selection of tanks is accepted to be examined, then different tanks should be examined at each renewal survey, on a rotational basis.</li> <li>3. Peak tanks (all uses) should be examined internally at each renewal survey.</li> <li>4. At renewal survey no.3 and subsequent renewal surveys, one deep tank for fuel oil in the cargo area should be included, if fitted.</li> </ol> |                          |                           |                                       |

**2.5.2** The minimum requirements for close-up surveys at renewal surveys are given in annex 1, appendix 1 for double-side skin bulk carriers, excluding ore carriers, and in annex 1, appendix 2 for ore carriers respectively.

**2.5.3** The surveyor may extend the close-up survey as deemed necessary taking into account the maintenance of the spaces under survey, the condition of the corrosion prevention system and where spaces have structural arrangements or details which have suffered defects in similar spaces or on similar ships according to available information.

**2.5.4** For areas in spaces where hard protective coatings are found to be in a GOOD condition, the extent of close-up surveys according to annex 1 may be specially considered (refer also to 2.3.2).

## **2.6 Extent of thickness measurements**

**2.6.1** The minimum requirements for thickness measurements at the renewal survey are given in annex 2.

**2.6.2** Representative thickness measurements to determine both general and local levels of corrosion in the transverse web frames in all water ballast tanks should be carried out. Thickness measurements should also be carried out to determine the corrosion levels on the transverse bulkhead plating. The extent of thickness measurements may be specially considered provided the surveyor is satisfied by the close-up examination that there is no structural diminution, and the hard protective coating where applied remains efficient.

**2.6.3** Provisions for extended measurements for areas with substantial corrosion as defined in 1.2.9 are given in annex 10 and may be additionally specified in the survey programme as required by 5.1. These extended thickness measurements should be carried out before the survey is credited as completed. Suspect areas identified at previous surveys should be examined. Areas of substantial corrosion identified at previous surveys should have thickness measurements taken.

**2.6.4** The surveyor may further extend the thickness measurements as deemed necessary.

**2.6.5** For areas in tanks where hard protective coatings are found to be in GOOD condition as defined in 1.2.11, the extent of thickness measurements according to annex 2 may be specially considered by the Administration.

**2.6.6** Transverse sections should be chosen where the largest reductions are suspected to occur or are revealed from deck plating measurements.

## **2.7      *Extent of tank pressure testing***

**2.7.1** All boundaries of water ballast tanks, deep tanks and cargo holds used for water ballast within the cargo length area should be pressure tested. For fuel tanks, only representative tanks should be pressure tested.

**2.7.2** The surveyor may extend the tank testing as deemed necessary.

**2.7.3** Boundaries of ballast tanks should be tested with a head of liquid to the top of air pipes.

**2.7.4** Boundaries of ballast holds should be tested with a head of liquid to near to the top of hatches.

**2.7.5** Boundaries of fuel oil tanks should be tested with a head of liquid to the highest point that liquid will rise under service conditions. Tank testing of fuel oil tanks may be specially considered based on a satisfactory external examination of the tank boundaries and a confirmation from the master stating that the pressure testing has been carried out according to the requirements with satisfactory results.

**2.7.6** The testing of double-bottom tanks and other spaces not designated for the carriage of liquid may be omitted, provided a satisfactory internal examination together with an examination of the tank top is carried out.

## **2.8      *Additional renewal survey requirements after determining compliance with regulations XII/12 and XII/13 of the Convention***

**2.8.1** For ships complying with the requirements of regulation XII/12 of the Convention for hold, ballast and dry space water level detectors, the renewal survey should include an examination and a test of the water ingress detection system and of their alarms.

**2.8.2** For ships complying with the requirements of regulation XII/13 of the Convention for the availability of pumping systems, the renewal survey should include an examination and a test of the means for draining and pumping ballast tanks forward of the collision bulkhead and bilges of dry spaces any part of which extends forward of the foremost cargo hold, and of their controls.

## **3          *Annual survey***

### **3.1      *General***

The annual survey should consist of an examination for the purpose of ensuring, as far as practicable, that the hull weather decks, hatch covers, coamings and piping are maintained in a satisfactory condition and should take into account the service history, condition and extent of the corrosion prevention system of ballast tanks and areas identified in the survey report file.

### **3.2 Examination of the hull**

**3.2.1** Examination of the hull plating and its closing appliances should be carried out as far as can be seen.

**3.2.2** Examination of watertight penetrations should be carried out as far as practicable.

### **3.3 Examination of weather decks, hatch covers and coamings**

**3.3.1** It should be confirmed that no unapproved changes have been made to the hatch covers, hatch coamings and their securing and sealing devices since the last survey.

**3.3.2** A thorough survey of cargo hatch covers and coamings is only possible by examination in the open as well as closed positions and should include verification of proper opening and closing operation. As a result, the hatch cover sets within the forward 25% of the ship's length and at least one additional set, such that all sets on the ship are assessed at least once in every five-year period, should be surveyed open, closed and in operation to the full extent in each direction at each annual survey, including:

- .1** stowage and securing in open condition;
- .2** proper fit and efficiency of sealing in closed condition; and
- .3** operational testing of hydraulic and power components, wires, chains and link drives.

The closing of the covers should include the fastening of all peripheral, and cross joint cleats or other securing devices. Particular attention should be paid to the condition of hatch covers in the forward 25% of the ship's length, where sea loads are normally greatest.

**3.3.3** If there are indications of difficulty in operating and securing hatch covers, additional sets above those required by 3.3.2, at the discretion of the surveyor, should be tested in operation.

**3.3.4** Where the cargo hatch securing system does not function properly, repairs should be carried out under the supervision of the Administration. Where hatch covers or coamings undergo substantial repairs, the strength of securing devices should be upgraded to comply with annex 11.

**3.3.5** For each cargo hatch cover set, at each annual survey, the following items should be surveyed:

- .1** cover panels, including side plates, and stiffener attachments that may be accessible in the open position by close-up survey (for corrosion, cracks, deformation);
- .2** sealing arrangements of perimeter and cross joints (gaskets for condition and permanent deformation, flexible seals on combination carriers, gasket lips, compression bars, drainage channels and non-return valves);
- .3** clamping devices, retaining bars, cleating (for wastage, adjustment, and condition of rubber components);
- .4** closed cover locating devices (for distortion and attachment);

- .5 chain or rope pulleys;
- .6 guides;
- .7 guiderails and track wheels;
- .8 stoppers;
- .9 wires, chains, tensioners and gypsies;
- .10 hydraulic system, electrical safety devices and interlocks; and
- .11 end and interpanel hinges, pins and stools where fitted.

**3.3.6** At each hatchway, at each annual survey, the coamings, with plating, stiffeners and brackets should be checked for corrosion, cracks and deformation, especially of the coaming tops including close-up survey.

**3.3.7** Where considered necessary, the effectiveness of sealing arrangements may be proved by hose or chalk testing supplemented by dimensional measurements of seal compressing components.

**3.3.8** Where portable covers, wooden or steel pontoons are fitted, the satisfactory condition of the following should be confirmed:

- .1 wooden covers and portable beams, carriers or sockets for the portable beam, and their securing devices;
- .2 steel pontoons, including close-up survey of hatch cover plating;
- .3 tarpaulins;
- .4 cleats, battens and wedges;
- .5 hatch securing bars and their securing devices;
- .6 loading pads/bars and the side plate edge;
- .7 guide plates and chocks; and
- .8 compression bars, drainage channels and drain pipes (if any).

**3.3.9** The flame screens on vents to all bunker tanks should be examined.

**3.3.10** Bunker and vent piping systems, including ventilators, should be examined.

#### **3.4 Examination of cargo holds**

**3.4.1** For double-side skin bulk carriers of 10 to 15 years of age, the following should be carried out:

- .1 overall survey of two selected cargo holds;

- .2 when considered necessary by the surveyor, or where extensive corrosion exists, thickness measurement should be carried out. If the results of these thickness measurements indicate that substantial corrosion is found, the extent of thickness measurements should be increased in accordance with annex 10. These extended thickness measurements should be carried out before the survey is credited as complete. Suspect areas identified at previous surveys should be examined. Areas of substantial corrosion identified at previous surveys should have thickness measurements taken; and
- .3 all piping and penetrations in cargo holds, including overboard piping, should be examined.

**3.4.2** For double-side skin bulk carriers over 15 years of age, the following should be carried out:

- .1 overall survey of all cargo holds;
- .2 when considered necessary by the surveyor, or where extensive corrosion exists, thickness measurement should be carried out. If the results of these thickness measurements indicate that substantial corrosion is found, the extent of thickness measurements should be increased in accordance with annex 10. These extended thickness measurements should be carried out before the survey is credited as complete. Suspect areas identified at previous surveys should be examined. Areas of substantial corrosion identified at previous surveys should have thickness measurements taken; and
- .3 all piping and penetrations in cargo holds, including overboard piping, should be examined.

### **3.5 Examination of ballast tanks**

Examination of ballast tanks should be carried out when required as a consequence of the results of the renewal survey and intermediate survey. When considered necessary by the surveyor, or where extensive corrosion exists, thickness measurement should be carried out. If the results of these thickness measurements indicate that substantial corrosion is found, the extent of thickness measurements should be increased in accordance with annex 10. These extended thickness measurements should be carried out before the survey is credited as complete. Suspect areas identified at previous surveys should be examined. Areas of substantial corrosion identified at previous surveys should have thickness measurements taken.

### **3.6 Additional annual survey requirements after determining compliance with regulations XII/12 and XII/13 of the Convention**

**3.6.1** For ships complying with the requirements of regulation XII/12 of the Convention for hold, ballast and dry space water level detectors, the annual survey should include an examination and a test, at random, of the water ingress detection systems and of their alarms.

**3.6.2** For ships complying with the requirements of regulation XII/13 of the Convention for the availability of pumping systems, the annual survey should include an examination and a test, of the means for draining and pumping ballast tanks forward of the collision bulkhead and bilges of dry spaces any part of which extends forward of the foremost cargo hold, and of their controls.

## **4 Intermediate survey**

### **4.1 General**

**4.1.1** Items that are additional to the requirements of the annual survey may be surveyed either at the second or third annual survey or between these surveys.

**4.1.2** The extent of survey is dependent upon the age of the ship as specified in 4.2, 4.3 and 4.4.

**4.1.3** Concurrent crediting to both intermediate survey and renewal survey for surveys and thickness measurements of spaces should not be acceptable.

### **4.2 Double-side skin bulk carriers 5 to 10 years of age**

#### **4.2.1 Ballast tanks**

**4.2.1.1** For tanks used for water ballast, an overall survey of representative tanks selected by the surveyor should be carried out. The selection should include fore and aft peak tanks and a number of other tanks, taking into account the total number and type of ballast tanks. If such overall survey reveals no visible structural defects, the examination may be limited to verification that the corrosion prevention system remains efficient.

**4.2.1.2** Where POOR coating condition, corrosion or other defects are found in water ballast tanks or where a hard protective coating was not applied from the time of construction, the examination should be extended to other ballast tanks of the same type.

**4.2.1.3** In ballast tanks other than double-bottom tanks, where a hard protective coating is found in POOR condition and it is not renewed, or where soft or semi-hard coating has been applied, or where a hard protective coating was not applied from the time of construction, the tanks in question should be examined and thickness measurements carried out as considered necessary at annual intervals. When such breakdown of hard protective coating is found in ballast double-bottom tanks, where a soft or semi-hard coating has been applied, or where a hard protective coating has not been applied, the tanks in question may be examined at annual intervals. When considered necessary by the surveyor, or where extensive corrosion exists, thickness measurements should be carried out.

**4.2.1.4** In addition to the above requirements, areas found to be suspect areas at the previous renewal survey should be overall and close-up surveyed.

#### **4.2.2 Cargo holds**

**4.2.2.1** An overall survey of all cargo holds should be carried out.

**4.2.2.2** Where considered necessary by the surveyor as a result of the overall survey as described in 4.2.2.1, the survey should be extended to include a close-up survey those areas of the structure in the cargo holds selected by the surveyor.

#### **4.2.3 Extent of thickness measurements**

**4.2.3.1** Thickness measurements should be carried out to an extent sufficient to determine both general and local corrosion levels at areas subject to close-up survey where required as per 4.2.2.2 and as provided in 4.2.1.4.

**4.2.3.2** The extent of thickness measurements may be specially considered provided the surveyor is satisfied by the close-up survey that there is no structural diminution and the hard protective coatings are found to be in a GOOD condition.

**4.2.3.3** Where substantial corrosion is found, the extent of thickness measurements should be increased in accordance with the requirements of annex 10. These extended thickness measurements should be carried out before the survey is credited as completed. Suspect areas identified at previous surveys should be examined. Areas of substantial corrosion identified at previous surveys should have thickness measurements taken.

**4.2.3.4** Where the hard protective coating in cargo holds is found to be in GOOD condition, the extent of close-up surveys and thickness measurements may be specially considered.

#### **4.3      *Double-side skin bulk carriers 10 to 15 years of age***

**4.3.1** The requirements of the intermediate survey should be the same extent as the previous renewal survey as required in 2 and 5.1. However, internal examination of fuel oil tanks and pressure testing of all tanks are not required unless deemed necessary by the attending surveyor.

**4.3.2** In application of 4.3.1, the intermediate survey may be commenced at the second annual survey and be progressed during the succeeding year with a view to completion at the third annual survey in lieu of the application of 2.1.1.

**4.3.3** In application of 4.3.1, an underwater survey may be considered in lieu of the requirements of 2.2.

#### **4.4      *Double-side skin bulk carriers exceeding 15 years of age***

**4.4.1** The requirements of the intermediate survey should be to the same extent as the previous renewal survey required in 2 and 5.1. However, internal examination of fuel tanks and pressure testing of all tanks are not required unless deemed necessary by the attending surveyor.

**4.4.2** In application of 4.4.1, the intermediate survey may be commenced at the second annual survey and be progressed during the succeeding year with a view to completion at the third annual survey in lieu of the application of 2.1.1.

**4.4.3** In application of 4.4.1, a survey in dry dock should be part of the intermediate survey. The overall and close-up surveys and thickness measurements, as applicable, of the lower portions of the cargo holds and water ballast tanks should be carried out in accordance with the applicable requirements for intermediate surveys, if not already performed.

**Note:** Lower portions of the cargo holds and ballast tanks are considered to be the parts below light ballast water line.

### **5          Preparations for survey**

#### **5.1      *Survey programme***

**5.1.1** The owner in co-operation with the Administration should work out a specific survey programme prior to the commencement of any part of:

- .1 the renewal survey; and
- .2 the intermediate survey for double-side skin bulk carriers over 10 years of age.

The survey programme should be in a written format based on the information in annex 4A. The survey should not commence until the survey programme has been agreed.

**5.1.1.1** Prior to the development of the survey programme, the survey planning questionnaire should be completed by the owner based on the information set out in annex 4B, and forwarded to the Administration.

**5.1.1.2** The survey programme at intermediate survey may consist of the survey programme at the previous renewal survey supplemented by the condition evaluation report of that renewal survey and later relevant survey reports.

**5.1.1.3** The survey programme should be worked out taking into account any amendments to the survey requirements after the last renewal survey carried out.

**5.1.2** In developing the survey programme, the following documentation should be collected and consulted with a view to selecting tanks, holds, areas and structural elements to be examined:

- .1 survey status and basic ship information;
- .2 documentation on board, as described in 6.2 and 6.3;
- .3 main structural plans (scantlings drawings), including information regarding use of high-tensile steels (HTS);
- .4 relevant previous survey and inspection reports from both the classification society and the owner;
- .5 information regarding the use of ship's holds and tanks, typical cargoes and other relevant data;
- .6 information regarding corrosion prevention level on the new building; and
- .7 information regarding the relevant maintenance level during operation.

**5.1.3** The submitted survey programme should account for, and comply, as a minimum, with the provisions of annexes 1 and 2 and 2.7 for close-up survey, thickness measurement and tank testing, respectively, and should include relevant information, including at least:

- .1 basic ship information and particulars;
- .2 main structural plans (scantling drawings), including information regarding use of high-tensile steels (HTS);
- .3 plan of holds and tanks;
- .4 list of holds and tanks with information on use, protection and condition of coating;



- .5 conditions for survey (e.g., information regarding hold and tank cleaning, gas-freeing, ventilation, lighting, etc.);
- .6 provisions and methods for access to structures;
- .7 equipment for surveys;
- .8 nomination of holds and tanks and areas for close-up survey (see 2.5);
- .9 nomination of sections for thickness measurement (see 2.6);
- .10 nomination of tanks for testing (see 2.7); and
- .11 damage experience related to ship in question.

**5.1.4** The Administration should advise the owner of the maximum acceptable structural corrosion diminution levels applicable to the ship.

**5.1.5** Use may also be made of the Guidelines for technical assessment in conjunction with the planning of enhanced surveys for double-side skin bulk carriers, contained in annex 9. These Guidelines are a recommended tool which may be invoked at the discretion of the Administration, when considered necessary and appropriate, in conjunction with the preparation of the required survey programme.

## **5.2 Conditions for survey**

**5.2.1** The owner should provide the necessary facilities for a safe execution of the survey.

**5.2.2** In order to enable the attending surveyors to carry out the survey, provisions for proper and safe access, should be agreed between the owner and the Administration.

**5.2.3** Details of the means of access should be provided in the survey planning questionnaire.

**5.2.4** In cases where the provisions of safety and required access are judged by the attending surveyors not to be adequate, the survey of the spaces involved should not proceed.

**5.2.5** Cargo holds, tanks and spaces should be safe for access. Cargo holds, tanks and spaces should be gas free and properly ventilated. Prior to entering a tank, void or enclosed space, it should be verified that the atmosphere in the tank is free from hazardous gas and contains sufficient oxygen.

**5.2.6** In preparation for survey and thickness measurements and to allow for a thorough examination, all spaces should be cleaned including removal from surfaces of all loose accumulated corrosion scale. Spaces should be sufficiently clean and free from water, scale, dirt, oil residues, etc., to reveal corrosion, deformation, fractures, damages or other structural deterioration as well as the condition of the coating. However, those areas of structure whose renewal has already been decided by the owner need only be cleaned and descaled to the extent necessary to determine the limits of the areas to be renewed.

**5.2.7** Sufficient illumination should be provided to reveal corrosion, deformation, fractures, damages or other structural deterioration as well as the condition of the coating.

**5.2.8** Where soft or semi-hard coatings have been applied, safe access should be provided for the surveyor to verify the effectiveness of the coating and to carry out an assessment of the conditions of internal structures which may include spot removal of the coating. When safe access cannot be provided, the soft or semi-hard coating should be removed.

**5.2.9** The surveyor(s) should always be accompanied by, at least, one responsible person, assigned by the owner, experienced in tank and enclosed spaces inspection. In addition, a back-up team of at least two experienced persons should be stationed at the hatch opening of the tank or space that is being surveyed. The back-up team should continuously observe the work in the tank or space and should keep life-saving and evacuation equipment ready for use.

**5.2.10** A communication system should be arranged between the survey party in the cargo hold, tank or space being examined, the responsible officer on deck and, as the case may be, the navigation bridge. The communication arrangements should be maintained throughout the survey.

### **5.3 Access to structures<sup>7</sup>**

**5.3.1** For overall survey, means should be provided to enable the surveyor to examine the structure in a safe and practical way.

**5.3.2** For close-up survey, one or more of the following means for access, acceptable to the surveyor, should be provided:

- .1 permanent staging and passages through structures;
- .2 temporary staging and passages through structures;
- .3 lifts and moveable platforms;
- .4 portable ladders;
- .5 boats or rafts; and
- .6 other equivalent means.

### **5.4 Equipment for survey**

**5.4.1** Thickness measurements should normally be carried out by means of ultrasonic test equipment. The accuracy of the equipment should be proven to the surveyor as required.

**5.4.2** One or more of the following fracture detection procedures may be required if deemed necessary by the surveyor:

- .1 radiographic equipment;
- .2 ultrasonic equipment;
- .3 magnetic particle equipment; and
- .4 dye penetrant.

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<sup>7</sup> Refer to the Guidelines on the means of access to structures for inspection and maintenance of oil tankers and bulk carriers (MSC/Circ.686).

**5.4.3** Explosimeter, oxygen-meter, breathing apparatus, lifelines, riding belts with rope and hook and whistles together with instructions and guidance on their use should be made available during the survey. A safety checklist should be provided.

**5.4.4** Adequate and safe lighting should be provided for the safe and efficient conduct of the survey.

**5.4.5** Adequate protective clothing should be made available and used (e.g., safety helmet, gloves, safety shoes, etc.) during the survey.

## **5.5      *Surveys at sea or at anchorage***

**5.5.1** Surveys at sea or at anchorage may be accepted provided the surveyor is given the necessary assistance from the personnel on board. Necessary precautions and procedures for carrying out the survey should be in accordance with 5.1, 5.2, 5.3 and 5.4.

**5.5.2** A communication system should be arranged between the survey party in the spaces under examination and the responsible officer on deck. This system should also include the personnel in charge of ballast pump handling if boats or rafts are used.

**5.5.3** Surveys of tanks or applicable holds by means of boats or rafts may only be undertaken with the agreement of the surveyor, who should take into account the safety arrangements provided, including weather forecasting and ship response under foreseeable conditions and provided the expected rise of water within the tank does not exceed 0.25 m.

**5.5.4** When rafts or boats will be used for close-up survey, the following conditions should be observed:

- .1** only rough duty, inflatable rafts or boats, having satisfactory residual buoyancy and stability even if one chamber is ruptured, should be used;
- .2** the boat or raft should be tethered to the access ladder and an additional person should be stationed down the access ladder with a clear view of the boat or raft;
- .3** appropriate lifejackets should be available for all participants;
- .4** the surface of water in the tank or hold should be calm (under all foreseeable conditions the expected rise of water within the tank should not exceed 0.25 m) and the water level stationary. On no account should the level of the water be rising while the boat or raft is in use;
- .5** the tank, hold or space should contain clean ballast water only. Even a thin sheen of oil on the water is not acceptable; and
- .6** at no time should the water level be allowed to be within 1 m of the deepest under deck web face flat so that the survey team is not isolated from a direct escape route to the tank hatch. Filling to levels above the deck transverses should only be contemplated if a deck access manhole is fitted and open in the bay being examined, so that an escape route for the survey party is available at all times. Other effective means of escape to the deck may be considered.

**5.5.5** Rafts or boats alone may be allowed for inspection of the under deck areas for tanks or spaces, if the depth of the webs is 1.5 m or less.

**5.5.6** If the depth of the webs is more than 1.5 m, rafts or boats alone may be allowed only:

- .1 when the coating of the under deck structure is in GOOD condition and there is no evidence of wastage; or
- .2 if a permanent means of access is provided in each bay to allow safe entry and exit. This means:
  - .1 access direct from the deck via a vertical ladder and a small platform fitted approximately 2 m below the deck in each bay; or
  - .2 access to deck from a longitudinal permanent platform having ladders to deck in each end of the tank. The platform should, for the full length of the tank, be arranged in level with, or above, the maximum water level needed for rafting of under deck structure. For this purpose, the ullage corresponding to the maximum water level should be assumed not more than 3 m from the deck plate measured at the midspan of deck transverses and in the middle length of the tank.

If neither of the above conditions are met, then staging or other equivalent means should be provided for the survey of the under-deck areas.

**5.5.7** The use of rafts or boats alone in 5.5.5 and 5.5.6 does not preclude the use of boats or rafts to move about within a tank during a survey.

## **5.6** *Survey planning meeting*

**5.6.1** The establishment of proper preparation and the close co-operation between the attending surveyor(s) and the owner's representatives on board prior to and during the survey are an essential part in the safe and efficient conduct of the survey. During the survey on board safety meetings should be held regularly.

**5.6.2** Prior to commencement of any part of the renewal and intermediate survey, a survey planning meeting should be held between the attending surveyor(s), the owner's representative in attendance, the thickness measurement Company operator (as applicable) and the master of the ship or an appropriately qualified representative appointed by the master or Company for the purpose to ascertain that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the survey work to be carried out (see also 7.1.2).

**5.6.3** The following is an indicative list of items that should be addressed in the meeting:

- .1 schedule of the ship (i.e. the voyage, docking and undocking manoeuvres, periods alongside, cargo and ballast operations, etc.);
- .2 provisions and arrangements for thickness measurements (i.e. access, cleaning/descaling, illumination, ventilation, personal safety);
- .3 extent of the thickness measurements;

- .4 acceptance criteria (refer to the list of minimum thicknesses);
- .5 extent of close-up survey and thickness measurement considering the coating condition and suspect areas/areas of substantial corrosion;
- .6 execution of thickness measurements;
- .7 taking representative readings in general and where uneven corrosion/pitting is found;
- .8 mapping of areas of substantial corrosion; and
- .9 communication between attending surveyor(s) the thickness measurement company operator(s) and owner's representative(s) concerning findings.

## **6 Documentation on board**

### **6.1 General**

**6.1.1** The owner should obtain, supply and maintain on board documentation as specified in 6.2 and 6.3, which should be readily available for the surveyor. The condition evaluation report referred to in 6.2 should include a translation into English.

**6.1.2** The documentation should be kept on board for the lifetime of the ship.

### **6.2 Survey report file**

**6.2.1** A survey report file should be a part of the documentation on board consisting of:

- .1 reports of structural surveys (annex 6);
- .2 condition evaluation report (annex 7); and
- .3 thickness measurement reports (annex 8).

**6.2.2** The survey report file should be available also in the owner's and the Administration offices or in the office of the organization recognized by the Administration.

### **6.3 Supporting documents**

**6.3.1** The following additional documentation should be available on board:

- .1 survey programme, as required under 5.1 until such time as the renewal survey or intermediate survey, as applicable, has been completed;
- .2 main structural plans of cargo holds and ballast tanks;
- .3 previous repair history;
- .4 cargo and ballast history;

- .5 inspections by ship's personnel with reference to:
  - .1 structural deterioration in general;
  - .2 leakages in bulkheads and piping; and
  - .3 condition of coating or corrosion prevention system, if any. Guidance for reporting is shown in annex 3; and
- .6 any other information that would help to identify critical structural areas and/or suspect areas requiring inspection.

#### **6.4      *Review of documentation on board***

Prior to survey, the surveyor should examine the completeness of the documentation on board, and its contents as a basis for the survey.

### **7          *Procedures for thickness measurements***

#### **7.1      *General***

**7.1.1** The required thickness measurements, if not carried out by the recognized organization acting on behalf of the Administration itself, should be witnessed by a surveyor of the recognized organization. The surveyor should be on board to the extent necessary to control the process.

**7.1.2** The thickness measurement company should be part of the survey planning meeting to be held prior to commencing the survey.

**7.1.3** Thickness measurements of structures in areas where close-up surveys are required should be carried out simultaneously with close-up surveys.

**7.1.4** In all cases the extent of the thickness measurements should be sufficient as to represent the actual average condition.

**7.1.5** Procedural requirements for thickness measurements are set out in annex 12.

#### **7.2      *Certification of thickness measurement company***

The thickness measurements should be carried out by a qualified company certified by an organization recognized by the Administration according to principles stated in annex 5.

#### **7.3      *Reporting***

**7.3.1** A thickness measurement report should be prepared and submitted to the Administration. The report should give the location of measurements, the thickness measured as well as corresponding original thickness. Furthermore, the report should give the date when the measurements were carried out, type of measuring equipment, names of personnel and their qualifications and be signed by the operator. The thickness measurement report should follow the principles as specified in the recommended procedures for thickness measurements set out in annex 8.

**7.3.2** The surveyor should review the final thickness measurement reports and countersign the cover page.

## **8 Reporting and evaluation of survey**

### **8.1 *Evaluation of survey report***

**8.1.1** The data and information on the structural condition of the ship collected during the survey should be evaluated for acceptability and continued structural integrity of the ship.

**8.1.2** The analysis of data should be carried out and endorsed by the Administration or recognized organization authorized by the Administration and the conclusions of the analysis should form a part of the condition evaluation report.

### **8.2 *Reporting***

**8.2.1** Principles for survey reporting are shown in annex 6.

**8.2.2** When a survey is split between different survey stations, a report should be made for each portion of the survey. A list of items examined and/or tested (pressure testing, thickness measurements, etc.) and an indication of whether the item has been credited, should be made available to the next attending surveyor(s), prior to continuing or completing the survey.

**8.2.3** A condition evaluation report of the survey and results should be issued to the owner as shown in annex 7 and placed on board the ship for reference at future surveys. The condition evaluation report should be endorsed by the Administration or by the recognized organization on behalf of the Administration.

## ANNEX 1

### REQUIREMENTS FOR CLOSE-UP SURVEY AT RENEWAL SURVEYS

#### *Appendix 1 – Minimum requirements for close-up survey at renewal survey of double-side skin bulk carriers excluding ore carriers*

| <b>Age ≤ 5 years</b><br><b>Renewal Survey No.</b><br><b>1</b>                                                                                                                                                                                                                                                                                                                                                            | <b>5 &lt; Age ≤ 10 years</b><br><b>Renewal Survey No.</b><br><b>2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>10 &lt; Age ≤ 15 years</b><br><b>Renewal Survey No.</b><br><b>3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Age &gt; 15 years</b><br><b>Renewal Survey No.</b><br><b>4 and subsequent</b>                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| One transverse web with associated plating and longitudinals in two representative water ballast tanks of each type. This is to include the foremost topside and double-side water ballast tanks on either side. (A)<br>Two selected cargo hold transverse bulkheads, including internal structure of upper and lower stools, where fitted. (C)<br>All cargo hold hatch covers and coaming (plating and stiffeners). (D) | One transverse web with associated plating and longitudinals as applicable in each water ballast tank. (A)<br>Forward and aft transverse bulkheads including stiffening system in a transverse section including topside, hopper side and double-side ballast tanks. (A)<br>25% of ordinary transverse web frames in the foremost double-side tanks. (B)<br>One transverse bulkhead in each cargo hold, including internal structure of upper and lower stools, where fitted. (C)<br>All cargo hold hatch covers and coamings (plating and stiffeners). (D)<br>All deck plating and under-deck structure inside line of hatch openings between all cargo hold hatches. (E) | All transverse webs with associated plating and longitudinals as applicable in each water ballast tank. (A)<br>All transverse bulkheads including stiffening system in each water ballast tank. (A)<br>25% of ordinary transverse web frames in all double-side tanks. (B)<br>All cargo hold transverse bulkheads including internal structure of upper and lower stools, where fitted. (C)<br>All cargo hold hatch covers and coamings (plating and stiffeners). (D)<br>All deck plating and under-deck structure inside line of hatch openings between all cargo hold hatches. (E) | All transverse webs with associated plating and longitudinals as applicable in each water ballast tank. (A)<br>All transverse bulkheads including stiffening system in each water ballast tank. (A)<br>All ordinary transverse frames in all double-side tanks. (B)<br>Areas (C)–(E) as for column 3 |



(A), (B), (C), (D) and (E) are areas to be subjected to close-up surveys and thickness measurements (see sketches in appendix 3 to annex 8)

- (A) Transverse web frame or watertight transverse bulkhead in topside, hopper side and double-side ballast tanks. In fore and aft peak tanks transverse web frame means a complete transverse web frame ring including adjacent structural members.
- (B) Ordinary transverse frame in double-side tanks.
- (C) Cargo hold transverse bulkhead, plating, stiffeners and girders.
- (D) Cargo hold hatch covers and coamings.
- (E) Deck plating and under-deck structure inside line of hatch openings between cargo hold hatches.

**Note:** Close-up survey of transverse bulkheads should be carried out at four levels:

- Level (a) Immediately above the inner bottom and immediately above the line of gussets (if fitted) and shedders for ships without lower stool.
- Level (b) Immediately above and below the lower stool shelf plate (for those ships fitted with lower stools), and immediately above the line of the shedder plates.
- Level (c) About mid-height of the bulkhead.
- Level (d) Immediately below the upper deck plating and immediately adjacent to the upper wing tank, and immediately below the upper stool shelf plate for those ships fitted with upper stools, or immediately below the topside tanks.

## **Appendix 2**

### **Minimum requirements for close-up survey at renewal survey for ore carriers**

| <b>Age &lt; 5 years</b>                                                                                                                                                                                                           | <b>5 &lt; Age ≤ 10 years</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>age &gt; 10</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Renewal Survey No.1</b>                                                                                                                                                                                                        | <b>Renewal Survey No.2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Renewal Survey No.3 and subsequent</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>One web frame ring complete including adjacent structural members in a ballast wing tank. (A)</p> <p>One transverse bulkhead lower part – including girder system and adjacent structural members – in a ballast tank. (A)</p> | <p>All web frame rings complete including adjacent structural members in a ballast wing tank. (A)</p> <p>One deck transverse including adjacent deck structural members in each remaining ballast tank. (A)</p> <p>Forward and aft transverse bulkheads complete – including girder system and adjacent structural members – in a ballast wing tank.</p> <p>One transverse bulkhead lower part – including girder system and adjacent structural members – in each remaining ballast tank. (A)</p> | <p>All web frame rings complete including adjacent structural members in each ballast tank. (A)</p> <p>All transverse bulkheads complete – including girder system and adjacent structural members – in each ballast tank. (A)</p> <p>One web frame ring complete including adjacent structural members in each wing void space. (A)</p> <p>Additional web frame rings in void spaces as deemed necessary by the Administration or organization recognized by the Administration. (A)</p> |
| Two selected cargo hold transverse bulkheads, including internal structure of upper and lower stools, where fitted. (C)                                                                                                           | One transverse bulkhead in each cargo hold, including internal structure of upper and lower stools, where fitted. (C)                                                                                                                                                                                                                                                                                                                                                                              | All cargo hold transverse bulkheads, including internal structure of upper and lower stools, where fitted. (C)                                                                                                                                                                                                                                                                                                                                                                            |
| All cargo hold hatch covers and coamings (plating and stiffeners). (D)                                                                                                                                                            | All cargo hold hatch covers and coamings (plating and stiffeners). (D)                                                                                                                                                                                                                                                                                                                                                                                                                             | All cargo hold hatch covers and coamings (plating and stiffeners). (D)                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                                                                                                                                                                                                   | All deck plating and under deck structure inside line of hatch openings between all cargo hold hatches. (E)                                                                                                                                                                                                                                                                                                                                                                                        | All deck plating and under deck structure inside line of hatch openings between all cargo hold hatches. (E)                                                                                                                                                                                                                                                                                                                                                                               |

(A), (C), (D) and (E) are areas to be subjected to close-up surveys and thickness measurements (see sketches in appendix 3 to annex 8)

- (A) Transverse web frame or watertight transverse bulkhead in ballast wing tanks and void spaces. In fore and aft peak tanks transverse web frame means a complete transverse web frame ring including adjacent structural members.
- (C) Cargo hold transverse bulkhead plating, stiffeners and girders.
- (D) Cargo hold hatch covers and coamings.

- (E) Deck plating and under deck structure inside line of hatch openings between cargo hold hatches.

**Note:** Close-up survey of transverse bulkheads should be carried out at four levels:

- Level (a) Immediately above the inner bottom and immediately above the line of gussets (if fitted) and shedders for ships without lower stool.
- Level (b) Immediately above and below the lower stool shelf plate (for those ships fitted with lower stools), and immediately above the line of the shedder plates.
- Level (c) About mid-height of the bulkhead.
- Level (d) Immediately below the upper deck plating and immediately adjacent to the upper wing tank, and immediately below the upper stool shelf plate for those ships fitted with upper stools, or immediately below the topside tanks.

## ANNEX 2

### REQUIREMENTS FOR THICKNESS MEASUREMENTS AT RENEWAL SURVEYS

| Age $\leq$ 5 years<br>Renewal Survey No.<br>1 | 5 < Age $\leq$ 10 years<br>Renewal Survey No.<br>2                                                                                                                                                                                                                                                                                                                                                                                                           | 10 < Age $\leq$ 15 years<br>Renewal Survey No.<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Age > 15 years<br>Renewal Survey No.<br>4 and subsequent                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 Suspect areas                               | 1 Suspect areas<br>2 Within the cargo length area: two transverse sections of deck plating outside line of cargo hatch openings<br>3 Measurement, for general assessment and recording of corrosion pattern, of those structural members subject to close-up survey according to annex 1<br>4 Wind and water strakes in way of the two transverse sections considered under point 2 above<br>5 Selected wind and water strakes outside the cargo length area | 1 Suspect areas<br>2 Within the cargo length area:<br>.1 each deck plate outside line of cargo hatch openings<br>.2 two transverse sections, one of which should be in the amidship area, outside line of cargo hatch openings<br>3 Measurement, for general assessment and recording of corrosion pattern, of those structural members subject to close-up survey according to annex 1<br>4 All wind and water strakes within the cargo length area<br>5 Selected wind and water strakes outside the cargo length area | 1 Suspect areas<br>2 Within the cargo length area:<br>.1 each deck plate outside line of cargo hatch openings<br>.2 three transverse sections, one of which should be in the amidship area, outside line of cargo hatch openings<br>.3 each bottom plate<br>3 Point 3 referred to in column 3<br>4 All wind and water strakes, full length |

### ANNEX 3

## OWNER'S INSPECTION REPORT

### Structural condition

|                                                  |           |         |           |         |         |                     |       |
|--------------------------------------------------|-----------|---------|-----------|---------|---------|---------------------|-------|
| Ship's name: .....                               |           |         |           |         |         |                     |       |
| Owner's inspection report – Structural condition |           |         |           |         |         |                     |       |
| For tank/hold no: .....                          |           |         |           |         |         |                     |       |
| Grade of steel: deck: ..... side: .....          |           |         |           |         |         |                     |       |
| bottom: ..... longitudinal bulkhead: .....       |           |         |           |         |         |                     |       |
| Elements                                         | Cracks    | Buckles | Corrosion | Coating | Pitting | Modification/repair | Other |
|                                                  | condition |         |           |         |         |                     |       |
| Deck:                                            |           |         |           |         |         |                     |       |
| Bottom:                                          |           |         |           |         |         |                     |       |
| Side:                                            |           |         |           |         |         |                     |       |
| Side framing:                                    |           |         |           |         |         |                     |       |
| Longitudinal bulkheads:                          |           |         |           |         |         |                     |       |
| Transverse bulkheads:                            |           |         |           |         |         |                     |       |
| Repairs carried out due to:                      |           |         |           |         |         |                     |       |
| Thickness measurements carried out (dates):      |           |         |           |         |         |                     |       |
| Results in general:                              |           |         |           |         |         |                     |       |
| Overdue surveys:                                 |           |         |           |         |         |                     |       |
| Outstanding conditions of class:                 |           |         |           |         |         |                     |       |
| Comments:                                        |           |         |           |         |         |                     |       |
| Date of inspection: .....                        |           |         |           |         |         |                     |       |
| Inspected by: .....                              |           |         |           |         |         |                     |       |
| Signature: .....                                 |           |         |           |         |         |                     |       |

## ANNEX 4A

### SURVEY PROGRAMME

#### Basic information and particulars

|                                    |
|------------------------------------|
| Name of ship:                      |
| IMO number:                        |
| Flag State:                        |
| Port of registry:                  |
| Gross tonnage:                     |
| Deadweight (metric tonnes):        |
| Length between perpendiculars (m): |
| Shipbuilder:                       |
| Hull number:                       |
| Recognized organization (RO):      |
| RO ship identity:                  |
| Date of build of the ship:         |
| Owner:                             |
| Thickness measurement company:     |

#### **1 Preamble**

##### **1.1 Scope**

**1.1.1** The present survey programme covers the minimum extent of overall surveys, close-up surveys, thickness measurements and pressure testing within the cargo length area, cargo holds, ballast tanks, including fore and aft peak tanks, required by the Code.

**1.1.2** The arrangements and safety aspects of the survey should be acceptable to the attending surveyor(s).

##### **1.2 Documentation**

All documents used in the development of the survey programme should be available on board during the survey as required by section 6.

#### **2 Arrangement of cargo holds, tanks and spaces**

This section of the survey programme should provide information (either in the form of plans or text) on the arrangement of cargo holds, tanks and spaces that fall within the scope of the survey.

#### **3 List of cargo holds, tanks and spaces with information on their use, extent of coatings and corrosion prevention system**

This section of the survey programme should indicate any changes relating to (and should update) the information on the use of the holds and tanks of the ship, the extent of coatings and the corrosion prevention system provided in the survey planning questionnaire.

#### **4 Conditions for survey**

This section of the survey programme should provide information on the conditions for survey, e.g., information regarding cargo hold and tank cleaning, gas freeing, ventilation, lighting, etc.

#### **5 Provisions and method of access to structures**

This section of the survey programme should indicate any changes relating to (and should update) the information on the provisions and methods of access to structures provided in the survey planning questionnaire.

#### **6 List of equipment for survey**

This section of the survey programme should identify and list the equipment that will be made available for carrying out the survey and the required thickness measurements.

#### **7 Survey requirements**

##### **7.1 Overall survey**

This section of the survey programme should identify and list the spaces that should undergo an overall survey for this ship in accordance with 2.5.1.

##### **7.2 Close-up survey**

This section of the survey programme should identify and list the hull structures that should undergo a close-up survey for this ship in accordance with 2.5.2.

#### **8 Identification of tanks for tank testing**

This section of the survey programme should identify and list the cargo holds and tanks that should undergo tank testing for this ship in accordance with 2.7.

#### **9 Identification of areas and sections for thickness measurements**

This section of the survey programme should identify and list the areas and sections where thickness measurements should be taken in accordance with 2.6.1.

#### **10 Minimum thickness of hull structures**

This section of the survey programme should specify the minimum thickness for hull structures of this ship that are subject to survey, according to .1 or .2:

- .1** Determined from the attached wastage allowance table and the original thickness to the hull structure plans of the ship;
- .2** Given in the following table(s):

| Area or location                                             | Original as-built thickness (mm) | Minimum thickness (mm) | Substantial corrosion thickness (mm) |
|--------------------------------------------------------------|----------------------------------|------------------------|--------------------------------------|
| <b>Deck</b>                                                  |                                  |                        |                                      |
| Plating                                                      |                                  |                        |                                      |
| Longitudinals                                                |                                  |                        |                                      |
| Longitudinal girders                                         |                                  |                        |                                      |
| Cross deck plating                                           |                                  |                        |                                      |
| Cross deck stiffeners                                        |                                  |                        |                                      |
| <b>Bottom</b>                                                |                                  |                        |                                      |
| Plating                                                      |                                  |                        |                                      |
| Longitudinals                                                |                                  |                        |                                      |
| Longitudinal girders                                         |                                  |                        |                                      |
| <b>Inner bottom</b>                                          |                                  |                        |                                      |
| Plating                                                      |                                  |                        |                                      |
| Longitudinals                                                |                                  |                        |                                      |
| Longitudinal girders                                         |                                  |                        |                                      |
| Floors                                                       |                                  |                        |                                      |
| <b>Ship side in way of topside tanks</b>                     |                                  |                        |                                      |
| Plating                                                      |                                  |                        |                                      |
| Longitudinals                                                |                                  |                        |                                      |
| <b>Ship side in way of hopper side tanks</b>                 |                                  |                        |                                      |
| Plating                                                      |                                  |                        |                                      |
| Longitudinals                                                |                                  |                        |                                      |
| <b>Ship side in way of double-hull tanks (if applicable)</b> |                                  |                        |                                      |
| Plating                                                      |                                  |                        |                                      |
| Longitudinals or ordinary transverse frames                  |                                  |                        |                                      |
| Longitudinal stringers                                       |                                  |                        |                                      |
| <b>Longitudinal bulkhead inner side (if applicable)</b>      |                                  |                        |                                      |
| Plating                                                      |                                  |                        |                                      |
| Longitudinals if applicable)                                 |                                  |                        |                                      |
| Longitudinals or ordinary transverse frames                  |                                  |                        |                                      |
| Longitudinal girders (if applicable)                         |                                  |                        |                                      |
| <b>Transverse bulkheads</b>                                  |                                  |                        |                                      |
| Plating                                                      |                                  |                        |                                      |
| Stiffeners (if applicable)                                   |                                  |                        |                                      |
| Upper stool plating                                          |                                  |                        |                                      |
| Upper stool stiffeners                                       |                                  |                        |                                      |
| Lower stool plating                                          |                                  |                        |                                      |
| Lower stool stiffeners                                       |                                  |                        |                                      |
| <b>Transverse web frames in topside tanks</b>                |                                  |                        |                                      |
| Plating                                                      |                                  |                        |                                      |
| Flanges                                                      |                                  |                        |                                      |
| Stiffeners                                                   |                                  |                        |                                      |



| Area or location                                  | Original as-built thickness (mm) | Minimum thickness (mm) | Substantial corrosion thickness (mm) |
|---------------------------------------------------|----------------------------------|------------------------|--------------------------------------|
| <b>Transverse web frames in hopper tanks</b>      |                                  |                        |                                      |
| Plating                                           |                                  |                        |                                      |
| Flanges                                           |                                  |                        |                                      |
| Stiffeners                                        |                                  |                        |                                      |
| <b>Transverse web frames in double-side tanks</b> |                                  |                        |                                      |
| Plating                                           |                                  |                        |                                      |
| Flanges                                           |                                  |                        |                                      |
| Stiffeners                                        |                                  |                        |                                      |
| <b>Hatch covers</b>                               |                                  |                        |                                      |
| Plating                                           |                                  |                        |                                      |
| Stiffeners                                        |                                  |                        |                                      |
| <b>Hatch coamings</b>                             |                                  |                        |                                      |
| Plating                                           |                                  |                        |                                      |
| Stiffeners                                        |                                  |                        |                                      |

**Note:** The wastage allowance tables should be attached to the survey programme.

## 11 Thickness measurement company

This section of the survey programme should identify changes, if any, relating to the information on the thickness measurement company provided in the survey planning questionnaire.

## 12 Damage experience related to the ship

This section of the survey programme should, using the tables provided below, provide details of the hull damages for at least the last three years in way of the cargo holds, ballast tanks and void spaces within the cargo length area. These damages are subject to survey.

### Hull damages sorted by location for this ship

| Cargo hold, tank or space number or area | Possible cause, if known | Description of the damages | Location | Repair | Date of repair |
|------------------------------------------|--------------------------|----------------------------|----------|--------|----------------|
|                                          |                          |                            |          |        |                |
|                                          |                          |                            |          |        |                |
|                                          |                          |                            |          |        |                |
|                                          |                          |                            |          |        |                |
|                                          |                          |                            |          |        |                |
|                                          |                          |                            |          |        |                |

**Hull damages for sister or similar ships (if available) in the case of design-related damage**

| <b>Cargo hold, tank or space number or area</b> | <b>Possible cause, if known</b> | <b>Description of the damages</b> | <b>Location</b> | <b>Repair</b> | <b>Date of repair</b> |
|-------------------------------------------------|---------------------------------|-----------------------------------|-----------------|---------------|-----------------------|
|                                                 |                                 |                                   |                 |               |                       |
|                                                 |                                 |                                   |                 |               |                       |
|                                                 |                                 |                                   |                 |               |                       |
|                                                 |                                 |                                   |                 |               |                       |
|                                                 |                                 |                                   |                 |               |                       |
|                                                 |                                 |                                   |                 |               |                       |

**13 Areas identified with substantial corrosion from previous surveys**

This section of the survey programme should identify and list the areas of substantial corrosion from previous surveys.

**14 Critical structural areas and suspect areas**

This section of the survey programme should identify and list the critical structural areas and the suspect areas, when such information is available.

**15 Other relevant comments and information**

This section of the survey programme should provide any other comments and information relevant to the survey.

**Appendices**

**Appendix 1 – List of plans**

The provisions of 5.1.3.2 require that the main structural plans of cargo holds and ballast tanks (scantling drawings), including information regarding the use of high-tensile steel (HTS), should be available. This appendix of the survey programme should identify and list the main structural plans which form part of the survey programme.

**Appendix 2 – Survey planning questionnaire**

The survey planning questionnaire (annex 4B), which has been submitted by the owner, should be appended to the survey programme.

**Appendix 3 – Other documentation**

This part of the survey programme should identify and list any other documentation that forms part of the plan.

Prepared by the owner in co-operation with the Administration for compliance with 5.1.3.

Date:

*(name and signature of authorized owner's representative)*

Date:

*(name and signature of authorized representative of the Administration)*

## **ANNEX 4B**

### **SURVEY PLANNING QUESTIONNAIRE**

**1** The following information will enable the owner, in co-operation with the Administration, to develop a survey programme complying with the requirements of the Code. It is essential that the owner provides, when completing the present questionnaire, up-to-date information. The present questionnaire, when completed, should provide all information and material required by the Code.

#### **Particulars**

Ship's name:

IMO number:

Flag State:

Port of registry:

Owner:

Recognized organization:

RO Ship identity:

Gross tonnage:

Deadweight (metric tonnes):

Date of build:

#### **Information on access provision for close-up surveys and thickness measurement**

**2** The owner should indicate, in the table below, the means of access to the structures subject to close-up survey and thickness measurement. A close-up survey is an examination where the details of structural components are within the close visual inspection range of the attending surveyor, i.e. normally within reach of hand.

| Hold/Tank No.     | Structure                          | Temporary staging | Rafts | Ladders | Direct access | Other means (please specify) |
|-------------------|------------------------------------|-------------------|-------|---------|---------------|------------------------------|
| F.P.              | Fore peak                          |                   |       |         |               |                              |
| A.P.              | Aft peak                           |                   |       |         |               |                              |
| Cargo holds       | Hatch side coamings                |                   |       |         |               |                              |
|                   | Topside sloping plate              |                   |       |         |               |                              |
|                   | Upper stool plating                |                   |       |         |               |                              |
|                   | Cross deck                         |                   |       |         |               |                              |
|                   | Double-side tank plating           |                   |       |         |               |                              |
|                   | Transverse bulkhead                |                   |       |         |               |                              |
|                   | Hopper tank plating                |                   |       |         |               |                              |
|                   | Lower stool plating                |                   |       |         |               |                              |
|                   | Tank top                           |                   |       |         |               |                              |
| Topside tanks     | Under-deck structure               |                   |       |         |               |                              |
|                   | Side shell and structure           |                   |       |         |               |                              |
|                   | Sloping plate and structure        |                   |       |         |               |                              |
|                   | Webs and bulkheads                 |                   |       |         |               |                              |
| Hopper tanks      | Hopper sloping plate and structure |                   |       |         |               |                              |
|                   | Side shell and structure           |                   |       |         |               |                              |
|                   | Bottom structure                   |                   |       |         |               |                              |
|                   | Webs and bulkheads                 |                   |       |         |               |                              |
| Double-side tanks | Side shell and structure           |                   |       |         |               |                              |
|                   | Inner skin and structure           |                   |       |         |               |                              |
|                   | Webs and bulkheads                 |                   |       |         |               |                              |
|                   | Double-bottom structure            |                   |       |         |               |                              |
|                   | Upper stool internal structure     |                   |       |         |               |                              |
|                   | Lower stool internal structure     |                   |       |         |               |                              |

| Hold/Tank No.              | Structure                               | Temporary staging | Rafts | Ladders | Direct access | Other means (please specify) |
|----------------------------|-----------------------------------------|-------------------|-------|---------|---------------|------------------------------|
| Wing tanks of ore carriers | Underdeck and structure                 |                   |       |         |               |                              |
|                            | Side shell and structure                |                   |       |         |               |                              |
|                            | Side shell vertical web and structure   |                   |       |         |               |                              |
|                            | Longitudinal bulkhead and structure     |                   |       |         |               |                              |
|                            | Longitudinal bulkhead web and structure |                   |       |         |               |                              |
|                            | Bottom plating and structure            |                   |       |         |               |                              |
|                            | Cross ties/stringers                    |                   |       |         |               |                              |

| History of bulk cargoes of a corrosive nature (e.g., high sulphur content) |
|----------------------------------------------------------------------------|
|                                                                            |
|                                                                            |
|                                                                            |
|                                                                            |

### Owner's inspections

**3** Using a format similar to that of the table below (which is given as an example), the owner should provide details of the results of their inspections, for the last three years in accordance with the Code on all CARGO holds and BALLAST tanks and VOID spaces within the cargo area.

| Tank/Hold No.               | Corrosion protection (1) | Coating extent (2) | Coating condition (3) | Structural deterioration (4) | Hold and tank history (5) |
|-----------------------------|--------------------------|--------------------|-----------------------|------------------------------|---------------------------|
| Cargo holds                 |                          |                    |                       |                              |                           |
| Topside tanks               |                          |                    |                       |                              |                           |
| Hopper tanks                |                          |                    |                       |                              |                           |
| Double-side skin tanks      |                          |                    |                       |                              |                           |
| Double-bottom tanks         |                          |                    |                       |                              |                           |
| Upper stools                |                          |                    |                       |                              |                           |
| Lower stools                |                          |                    |                       |                              |                           |
| Wing tanks (ore tankers)    |                          |                    |                       |                              |                           |
| Fore peak                   |                          |                    |                       |                              |                           |
| Aft peak                    |                          |                    |                       |                              |                           |
| Miscellaneous other spaces: |                          |                    |                       |                              |                           |

### Note:

Indicate tanks which are used for oil/ballast.

- 1) HC = hard coating; SC = soft coating;  
SH=semi-hard coating; NP = no protection
- 2) U = upper part; M = middle part;  
L = lower part; C = complete
- 3) G = good; F = fair; P = poor;  
RC = recoated (during the last three years)
- 4) N = no findings recorded; Y = findings recorded,  
description of findings should be attached to this questionnaire
- 5) DR = damage & repair; L = leakages;  
CV = conversion (description to be attached to this questionnaire)

Name of owner's representative:

.....

Signature:.....

Date: .....

---

**Reports of port State control inspections**

|                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| List the reports of port State control inspections containing hull structural related deficiencies, and relevant information on rectification of the deficiencies: |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |

**Safety management system**

|                                                                                                 |
|-------------------------------------------------------------------------------------------------|
| List non-conformities related to hull maintenance, including the associated corrective actions: |
|-------------------------------------------------------------------------------------------------|

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |

**Name and address of the approved thickness measurement company:**

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |

## **ANNEX 5**

### **PROCEDURES FOR CERTIFICATION OF A COMPANY ENGAGED IN THICKNESS MEASUREMENT OF HULL STRUCTURES**

#### **1 Application**

This guidance applies for certification of the company which intends to engage in the thickness measurement of hull structures of ships.

#### **2 Procedures for certification**

##### ***Submission of documents***

**2.1** The following documents should be submitted to an organization recognized by the Administration for approval:

- .1** outline of the company, e.g., organization and management structure;
- .2** experience of the company on thickness measurement of hull structures of ships;
- .3** technicians' careers, i.e. experience of technicians as thickness measurement operators, technical knowledge and experience of hull structure, etc. Operators should be qualified according to a recognized industrial NDT Standard;
- .4** equipment used for thickness measurement such as ultrasonic testing machines and their maintenance/calibration procedures;
- .5** a guide for thickness measurement operators;
- .6** training programmes for technicians for thickness measurement;
- .7** measurement record format in accordance with recommended procedures for thickness measurements (see annex 8).

##### ***Auditing of the company***

**2.2** Upon reviewing the documents submitted with satisfactory results, the company should be audited in order to ascertain that the company is duly organized and managed in accordance with the documents submitted, and eventually is capable of conducting thickness measurement of the hull structure of ships.

**2.3** Certification is conditional upon an on board demonstration of thickness measurement as well as satisfactory reporting.

#### **3 Certification**

**3.1** Upon satisfactory results of both the audit of the company referred to in 2.2 and the demonstration tests referred to in 2.3, the Administration or organization recognized by the Administration should issue a certificate of approval as well as a notice to the effect that the thickness measurement operation system of the company has been certified.



**3.2** Renewal/endorsement of the certificate should be made at intervals not exceeding three years by verification that original conditions are maintained.

**4 Report of any alteration to the certified thickness measurement operation system**

In cases where any alteration to the certified thickness measurement operation system of the company is made, such an alteration should be immediately reported to the organization recognized by the Administration. Re-audit should be made where deemed necessary by the organization recognized by the Administration.

**5 Withdrawal of the certification**

The certification may be withdrawn in the following cases:

- .1** where the measurements were improperly carried out or the results were improperly reported;
- .2** where the surveyor found any deficiencies in the approved thickness measurement operation systems of the company; and
- .3** where the company failed to report any alteration referred to in 4 to the organization recognized by the Administration as required.

## **ANNEX 6**

### **SURVEY REPORTING PRINCIPLES**

As a principle, for bulk carriers subject to the Code, the surveyor should include the following contents in his report for survey of hull structure and piping systems, as relevant for the survey.

#### **1 General**

**1.1** A survey report should be generated in the following cases:

- .1** in connection with commencement, continuation and/or completion of periodical hull surveys, i.e. annual, intermediate and renewal surveys, as relevant;
- .2** when structural damages/defects have been found;
- .3** when repairs, renewals or modifications have been carried out; and
- .4** when condition of class (recommendation) has been imposed or has been deleted.

**1.2** The reporting should provide:

- .1** evidence that prescribed surveys have been carried out in accordance with applicable requirements;
- .2** documentation of surveys carried out with findings, repairs carried out and condition of class (recommendation) imposed or deleted;
- .3** survey records, including actions taken, which should form an auditable documentary trail. Survey reports should be kept in the survey report file required to be on board;
- .4** information for planning of future surveys; and
- .5** information which may be used as input for maintenance of classification rules and instructions.

**1.3** When a survey is split between different survey stations, a report should be made for each portion of the survey. A list of items surveyed, relevant findings and an indication of whether the item has been credited, should be made available to the next attending surveyor, prior to continuing or completing the survey. Thickness measurement and tank testing carried out should also be listed for the next surveyor.

#### **2 Extent of the survey**

**2.1** Identification of compartments where an overall survey has been carried out.

**2.2** Identification of locations, in each ballast tank and cargo hold including hatch covers and coamings, where a close-up survey has been carried out, together with information on the means of access used.

**2.3** Identification of locations, in each ballast tank and cargo hold including hatch covers and coamings, where thickness measurement has been carried out.

**Note:** As a minimum, the identification of location of close-up survey and thickness measurement should include a confirmation with description of individual structural members corresponding to the extent of requirements stipulated in part B of annex A based on type of periodical survey and the ship's age.

Where only partial survey is required, e.g., one transverse web, two selected cargo hold transverse bulkheads, the identification should include location within each ballast tank and cargo hold by reference to frame numbers.

**2.4** For areas in ballast tanks and cargo holds where protective coating is found to be in good condition and the extent of close-up survey and/or thickness measurement has been specially considered, structures subject to special consideration should be identified.

**2.5** Identification of tanks subject to tank testing.

**2.6** Identification of piping systems on deck and within cargo holds, ballast tanks, pipe tunnels, cofferdams and void spaces where:

- .1 examination including internal examination of piping with valves and fittings and thickness measurement, as relevant, has been carried out; and
- .2 operational test to working pressure has been carried out.

### **3 Result of the survey**

**3.1** Type, extent and condition of protective coating in each tank, as relevant (rated GOOD, FAIR or POOR).

**3.2** Structural condition of each compartment with information on the following, as relevant:

- .1 identification of findings, such as:
  - .1 corrosion with description of location, type and extent;
  - .2 areas with substantial corrosion;
  - .3 cracks/fractures with description of location and extent;
  - .4 buckling with description of location and extent; and
  - .5 indents with description of location and extent;
- .2 identification of compartments where no structural damages/defects are found. The report may be supplemented by sketches/photos; and
- .3 thickness measurement report should be verified and signed by the surveyor controlling the measurements on board.

#### **4 Actions taken with respect to findings**

**4.1** Whenever the attending surveyor is of the opinion that repairs are required, each item to be repaired should be identified in a numbered list. Whenever repairs are carried out, details of the repairs effected should be reported by making specific reference to relevant items in the numbered list.

**4.2** Repairs carried out should be reported with identification of:

- .1** compartment;
- .2** structural member;
- .3** repair method (i.e. renewal or modification), including:
  - .1** steel grades and scantlings (if different from the original); and
  - .2** sketches/photos, as appropriate;
- .4** repair extent; and
- .5** non-destructive test (NDT)/tests.

**4.3** For repairs not completed at the time of survey, condition of class/recommendation should be imposed with a specific time limit for the repairs. In order to provide correct and proper information to the surveyor attending for survey of the repairs, condition of class/recommendation should be sufficiently detailed with identification of each item to be repaired. For identification of extensive repairs, reference may be made to the survey report.

## ANNEX 7

### CONDITION EVALUATION REPORT

Issued upon completion of renewal survey

#### **General particulars**

|                           |                                                   |
|---------------------------|---------------------------------------------------|
| Ship's name:              | Class/Administration identity number:             |
|                           | Previous class/Administration identity number(s): |
|                           | IMO number:                                       |
| Port of registry:         | National flag:                                    |
|                           | Previous national flag(s):                        |
| Deadweight                | Gross tonnage:                                    |
| (metric tonnes):          | National:                                         |
|                           | ITC (1969):                                       |
| Date of build:            | Classification notation:                          |
| Date of major conversion: |                                                   |
| Type of conversion:       | Owner:                                            |
|                           | Previous owner(s):                                |

**1** The survey reports and documents listed below have been reviewed by the undersigned and found to be satisfactory.

**2** A summary of the survey is attached herewith on sheet 2.

**3** The renewal survey has been completed in accordance with the present Code on (date) .....

|                                          |                   |       |
|------------------------------------------|-------------------|-------|
| Condition evaluation report completed by | Name<br>Signature | Title |
| Office                                   | Date              |       |
| Condition evaluation report verified by  | Name<br>Signature | Title |
| Office                                   | Date              |       |

Attached reports and documents:

- 1)
- 2)
- 3)
- 4)
- 5)
- 6)

### ***Contents of condition evaluation report***

- |        |                                               |                                                                                      |
|--------|-----------------------------------------------|--------------------------------------------------------------------------------------|
| Part 1 | – General particulars:                        | – See front page                                                                     |
| Part 2 | – Report review:                              | – Where and how survey was done                                                      |
| Part 3 | – Close-up survey:                            | – Extent (which tanks/holds)                                                         |
| Part 4 | – Thickness measurements:                     | – Reference to thickness measurement report                                          |
|        |                                               | – Summary of where measured                                                          |
|        |                                               | – Separate form indicating the spaces with substantial corrosion, and corresponding: |
|        |                                               | – thickness diminution                                                               |
|        |                                               | – corrosion pattern                                                                  |
| Part 5 | – Tank/hold corrosion prevention system:      | – Separate form indicating:                                                          |
|        |                                               | – location of coating                                                                |
|        |                                               | – condition of coating (if applicable)                                               |
| Part 6 | – Repairs:                                    | – Identification of spaces/areas                                                     |
| Part 7 | – Condition of class/flag State requirements: |                                                                                      |
| Part 8 | – Memoranda:                                  | – Acceptable defects                                                                 |
|        |                                               | – Any points of attention for future surveys, e.g., for suspect areas                |
|        |                                               | – Extended annual/intermediate survey due to coating breakdown                       |
| Part 9 | – Conclusion:                                 | – Statement on evaluation/verification of survey report                              |

### ***Extract of thickness measurements***

Reference is made to the thickness measurement report:

| Position of substantially corroded tanks/areas <sup>1</sup> or areas with deep pitting <sup>3</sup> | Thickness diminution [%] | Corrosion pattern <sup>2</sup> | Remarks: (e.g., reference to attached sketches) |
|-----------------------------------------------------------------------------------------------------|--------------------------|--------------------------------|-------------------------------------------------|
|                                                                                                     |                          |                                |                                                 |

### ***Notes:***

- 1 Substantial corrosion, i.e. 75%–100% of acceptable margins wasted.
- 2 P = Pitting  
C = Corrosion in general
- 3 Any bottom plating with a pitting intensity of 20% or more, with wastage in the substantial corrosion range or having an average depth of pitting of  $\frac{1}{3}$  or more of actual plate thickness should be noted.

***Tank/hold corrosion prevention system***

| <b>Tank/hold nos.<sup>1</sup></b> | <b>Tank/hold corrosion prevention system<sup>2</sup></b> | <b>Coating condition<sup>3</sup></b> | <b>Remarks</b> |
|-----------------------------------|----------------------------------------------------------|--------------------------------------|----------------|
|                                   |                                                          |                                      |                |

**Notes:**

- 1 All ballast tanks and cargo holds should be listed.
- 2 C = Coating NP = No protection
- 3 Coating condition according to the following standard:

GOOD condition with only minor spot rusting.

FAIR condition with local breakdown of coating at edges of stiffeners and weld connections and/or light rusting over 20% or more of areas under consideration, but less than as defined for POOR condition.

POOR condition with general breakdown of coating over 20% or more of areas or hard scale at 10% or more of areas under consideration.

If coating condition POOR is given, extended annual surveys should be introduced. This should be noted in part 8 of the contents of condition evaluation report.

## **ANNEX 8**

### **RECOMMENDED PROCEDURES FOR THICKNESS MEASUREMENTS**

- 1** This annex should be used for recording thickness measurements as required by part B of annex A.
- 2** Thickness measurement sheet forms TM1-DSBC, TM2-DSBC(i), TM2-DSBC(ii), TM3-DSBC, TM4-DSBC, TM5-DSBC and TM6-DSBC (appendix 2) should be used, as appropriate, for recording thickness measurements. The maximum allowable diminution should be stated. The maximum allowable diminution could be stated in an attached document.
- 3** Appendix 3 contains guidance diagrams and notes relating to the reporting forms and the procedure for the thickness measurements.



## Appendix 1

### THICKNESS MEASUREMENT REPORT GENERAL PARTICULARS

Ship's name: .....  
IMO number: .....  
Class/Administration identity number: .....  
Port of registry: .....  
Gross tonnage: .....  
Deadweight: .....  
Date of build: .....  
Classification society: .....

Name of company performing thickness measurement: .....  
Thickness measurement company certified by: .....  
Certificate number: .....  
Certificate valid from: ..... to .....  
Place of measurement: .....  
First date of measurement: .....  
Last date of measurement: .....  
Renewal survey/intermediate survey\* due: .....  
Details of measurement equipment: .....  
Qualification of operator: .....

|                               |                              |
|-------------------------------|------------------------------|
| Report number: .....          | consisting of ..... pages    |
| Name of operator: .....       | Name of surveyor: .....      |
| Signature of operator: .....  | Signature of surveyor: ..... |
| Company official stamp: ..... | Administration: .....        |

Official stamp

---

\* Delete as appropriate.

## Appendix 2

### Report on thickness measurement of all deck plating, bottom shell plating and side shell plating\* (TM1-DSBC)

Ship's name..... Class Identity no..... Report no..... IMO number.....

| STRAKE POSITION |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
|-----------------|---------------|-----------------|-----------------|---|--------------|---|--------------|---|-------------|---|--------------|---|--------------|---|-------------------|---|----|
| PLATE POSITION  | No. or letter | Orig. thk. (mm) | Forward reading |   |              |   |              |   | Aft reading |   |              |   |              |   | Mean diminution % |   |    |
|                 |               |                 | Gauged          |   | Diminution P |   | Diminution S |   | Gauged      |   | Diminution P |   | Diminution S |   |                   |   |    |
|                 |               |                 | P               | S | mm           | % | mm           | % | P           | S | mm           | % | mm           | % | P                 | S | mm |
| 12th forward    |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 11th            |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 10th            |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 9th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 8th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 7th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 6th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 5th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 4th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 3rd             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 2nd             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 1st             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| Amidships       |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 1st aft         |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 2nd             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 3rd             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 4th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 5th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 6th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 7th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 8th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 9th             |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 10th            |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 11th            |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 12th            |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |

Operator's signature.....

Notes – see following page

\* Delete as appropriate.

**Notes to report TM1-DSBC:**

- 1** This report should be used for recording the thickness measurement of:
  - .1** all strength deck plating within cargo length area;
  - .2** all keel, bottom shell plating and bilge plating within the cargo length area;
  - .3** side shell plating including selected wind and water strakes outside cargo length area; and
  - .4** all wind and water strakes within cargo length area.
- 2** The strake position should be cleared as follows:
  - .1** for strength deck indicate the number of the strake of plating inboard from the stringer plate;
  - .2** for bottom plating indicate the number of the strake of plating outboard from the keel plate; and
  - .3** for side shell plating give number of the strake of plating below sheerstrake and letter as shown on shell expansion.
- 3** Only the deck plating strakes outside line of openings are to be recorded.
- 4** Measurements should be taken at the forward and aft areas of all plates and where plates cross ballast/cargo tank boundaries separate measurements for the area of plating in way of each type of tank should be recorded.
- 5** The single measurements recorded should represent the average of multiple measurements.
- 6** The maximum allowable diminution could be stated in an attached document.

**Report on thickness measurement of shell and deck plating at transverse sections (one, two or three transverse sections)  
(TM2-DSBC(i))**

Ship's name..... Class Identity no..... Report no..... IMO number.....

| STRENGTH DECK AND SHEERSTRAKE PLATING        |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
|----------------------------------------------|---------------|-----------|-------------------|--------|---|--------------|---|--------------|---|-----------------------------------------------|-----------|-------------------|--------|---|--------------|---|--------------|----------------------------------------------|---------------|-----------|-------------------|--------|---|--------------|---|--------------|---|
| FIRST TRANSVERSE SECTION AT FRAME NUMBER.... |               |           |                   |        |   |              |   |              |   | SECOND TRANSVERSE SECTION AT FRAME NUMBER.... |           |                   |        |   |              |   |              | THIRD TRANSVERSE SECTION AT FRAME NUMBER.... |               |           |                   |        |   |              |   |              |   |
| STRAKE POSITION                              | No. or letter | Orig. thk | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S |   | No. or letter                                 | Orig. thk | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S |                                              | No. or letter | Orig. thk | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S |   |
|                                              |               | (mm)      | (mm)              | P      | S | mm           | % | mm           | % |                                               | (mm)      | (mm)              | P      | S | mm           | % | mm           | %                                            |               | (mm)      | (mm)              | P      | S | mm           | % | mm           | % |
| Stringer plate                               |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 1st strake inboard                           |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 2nd                                          |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 3rd                                          |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 4th                                          |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 5th                                          |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 6th                                          |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 7th                                          |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 8th                                          |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 9th                                          |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 10th                                         |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 11th                                         |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 12th                                         |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 13th                                         |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 14th                                         |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| centre strake                                |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| sheerstrake                                  |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| TOPSIDE TOTAL                                |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |

Operator's signature.....

Notes – see following page

**Notes to report TM2-DSBC(i):**

- 1** This report should be used for recording the thickness measurement of:  
  
Strength deck plating and sheerstrake plating transverse sections:  
  
One, two or three sections within the cargo length area, comprising structural items (1), (2) and (3) as shown on the diagrams of typical transverse sections (appendix 3).
- 2** Only the deck plating strakes outside line of hatch openings should be recorded.
- 3** The top side area comprises deck plating, stringer plate and sheerstrake (including rounded gunwales).
- 4** The exact frame station of measurement should be stated.
- 5** The single measurements recorded should represent the average of multiple measurements.
- 6** The maximum allowable diminution could be stated in an attached document.

**Report on thickness measurement of shell and deck plating at transverse sections (one, two or three transverse sections)  
(TM2-DSBC(ii))**

Ship's name..... Class Identity no..... Report no..... IMO number.....

| SHELL PLATING                                |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
|----------------------------------------------|---------------|-----------|-------------------|--------|---|--------------|---|--------------|---|-----------------------------------------------|-----------|-------------------|--------|---|--------------|---|--------------|----------------------------------------------|---------------|-----------|-------------------|--------|---|--------------|---|--------------|---|
| FIRST TRANSVERSE SECTION AT FRAME NUMBER.... |               |           |                   |        |   |              |   |              |   | SECOND TRANSVERSE SECTION AT FRAME NUMBER.... |           |                   |        |   |              |   |              | THIRD TRANSVERSE SECTION AT FRAME NUMBER.... |               |           |                   |        |   |              |   |              |   |
| STRAKE POSITION                              | No. or letter | Orig. thk | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S |   | No. or letter                                 | Orig. thk | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S |                                              | No. or letter | Orig. thk | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S |   |
|                                              |               | (mm)      | (mm)              | P      | S | mm           | % | mm           | % |                                               | (mm)      | (mm)              | P      | S | mm           | % | mm           | %                                            |               | (mm)      | (mm)              | P      | S | mm           | % | mm           | % |
| 1st below sheerstrake                        |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 2nd                                          |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 3rd                                          |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 4th                                          |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 5th                                          |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 6th                                          |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 7th                                          |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 8th                                          |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 9th                                          |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 10th                                         |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 11th                                         |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 12th                                         |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 13th                                         |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 14th                                         |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 15th                                         |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 16th                                         |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 17th                                         |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 18th                                         |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 19th                                         |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| 20th                                         |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| keel strake                                  |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |
| BOTTOM TOTAL                                 |               |           |                   |        |   |              |   |              |   |                                               |           |                   |        |   |              |   |              |                                              |               |           |                   |        |   |              |   |              |   |

Operator's signature.....

Notes – see following page

**Notes to report TM2-DSBC(ii):**

- 1** This report should be used for recording the thickness measurement of:  
  
Shell plating at transverse sections:  
  
One, two or three sections within the cargo length area, comprising structural items (3), (4), (5) and (6) as shown on the diagrams of typical transverse sections in appendices 3 and 4.
- 2** The bottom area comprises keel, bottom and bilge plating.
- 3** The exact frame station of measurement should be stated.
- 4** The single measurements recorded should represent the average of multiple measurements.
- 5** The maximum allowable diminution could be stated in an attached document.

**Report on thickness measurement of longitudinal members at transverse sections (one, two or three transverse sections) (TM3-DSBC)**

Ship's name..... Class Identity no..... Report no..... IMO number.....

[illegible]

Operator's signature.....

Notes – see following page



**Notes to report TM3-DSBC:**

- 1** This report should be used for recording the thickness measurement of:

Longitudinal members at transverse sections:

Two, or three sections within the cargo length area comprising the appropriate structural items (10) to (25) as shown on diagrams of typical transverse sections in appendices 3 and 4.
- 2** The exact frame station of measurement should be stated.
- 3** The single measurements recorded should represent the average of multiple measurements.
- 4** The maximum allowable diminution could be stated in an attached document.

**Report on thickness measurement of transverse structural members in the double-bottom, hopper side and topside water ballast tanks (TM4-DSBC)**

Ship's name..... Class Identity no..... Report no..... IMO number.....

[illegible]

Operator's signature.....

Notes – see following page

**Notes to report TM4-DSBC:**

- 1**        This report should be used for recording the thickness measurement of:

Transverse structural members, comprising the appropriate structural items (30) to (34) as shown on diagrams of typical transverse sections illustrated in appendices 3 and 4.
- 2**        Guidance for areas of measurements is indicated in appendix 5.
- 3**        The single measurements recorded should represent the average of multiple measurements.
- 4**        The maximum allowable diminution could be stated in an attached document.

## Report on thickness measurement of watertight transverse bulkheads in cargo holds (TM5-DSBC)

Ship's name..... Class Identity no..... Report no..... IMO number.....

[illegible]

Operator's signature.....

*Notes – see following page*

***Notes to report TM5-DSBC:***

- 1**      This report should be used for recording the thickness measurement of:  
  
            Watertight transverse bulkheads in cargo holds.
- 2**      Guidance for areas of measurements is indicated in appendix 3.
- 3**      The single measurements recorded should represent the average of multiple measurements.
- 4**      The maximum allowable diminution could be stated in an attached document.

## Report on thickness measurement of miscellaneous structural members (TM6-DSBC)

Ship's name..... Class Identity no..... Report no..... IMO number.....

[illegible]

Operator's signature.....

Notes – see following page

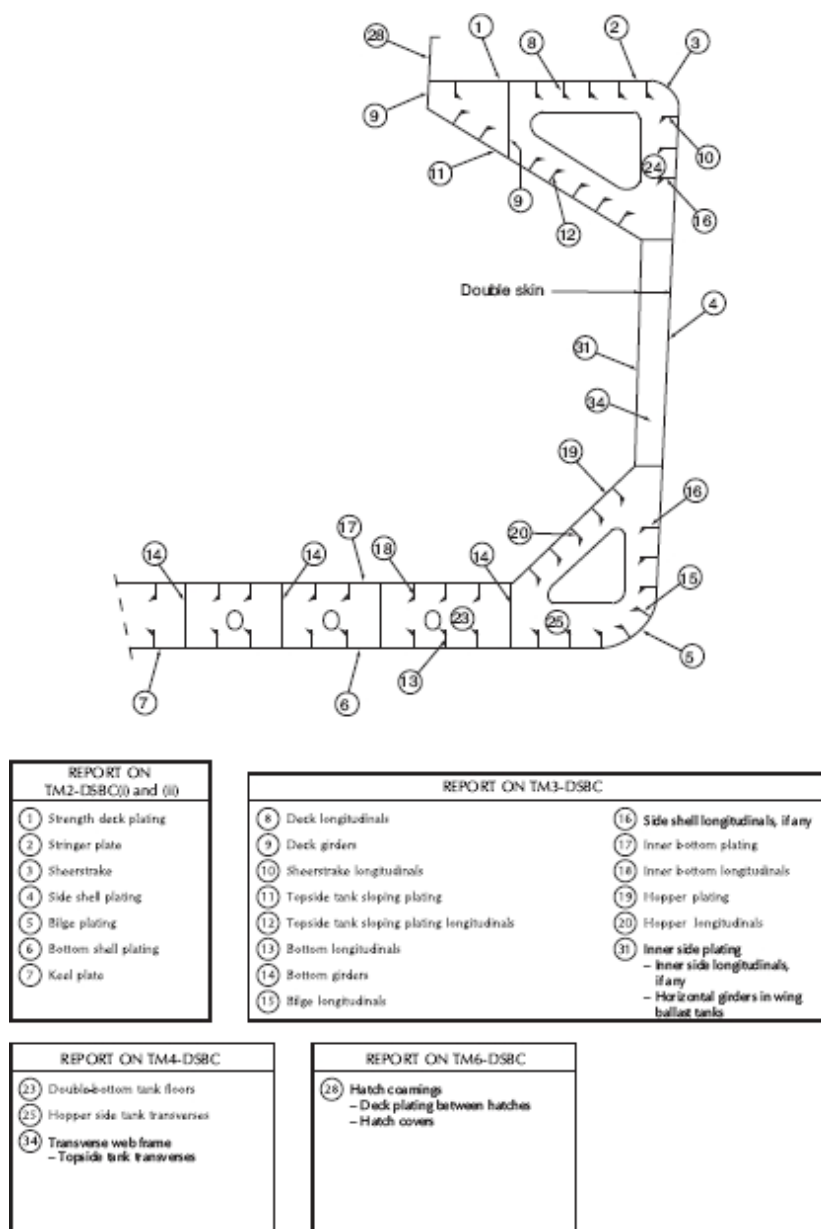
***Notes to report TM6-DSBC:***

- 1**      This report should be used for recording the thickness measurement of:  
  
Miscellaneous structural members including structural items (40), (41) and (42) as shown on diagrams of typical transverse sections illustrated in appendix 3.
- 2**      Guidance for areas of measurements is indicated in appendix 5.
- 3**      The single measurements recorded should represent the average of multiple measurements.
- 4**      The maximum allowable diminution could be stated in an attached document.

### Appendix 3

#### THICKNESS MEASUREMENT – DOUBLE-SIDE SKIN CONSTRUCTION SHIPS THICKNESS MEASUREMENT – DOUBLE-SIDE SKIN BULK CARRIER

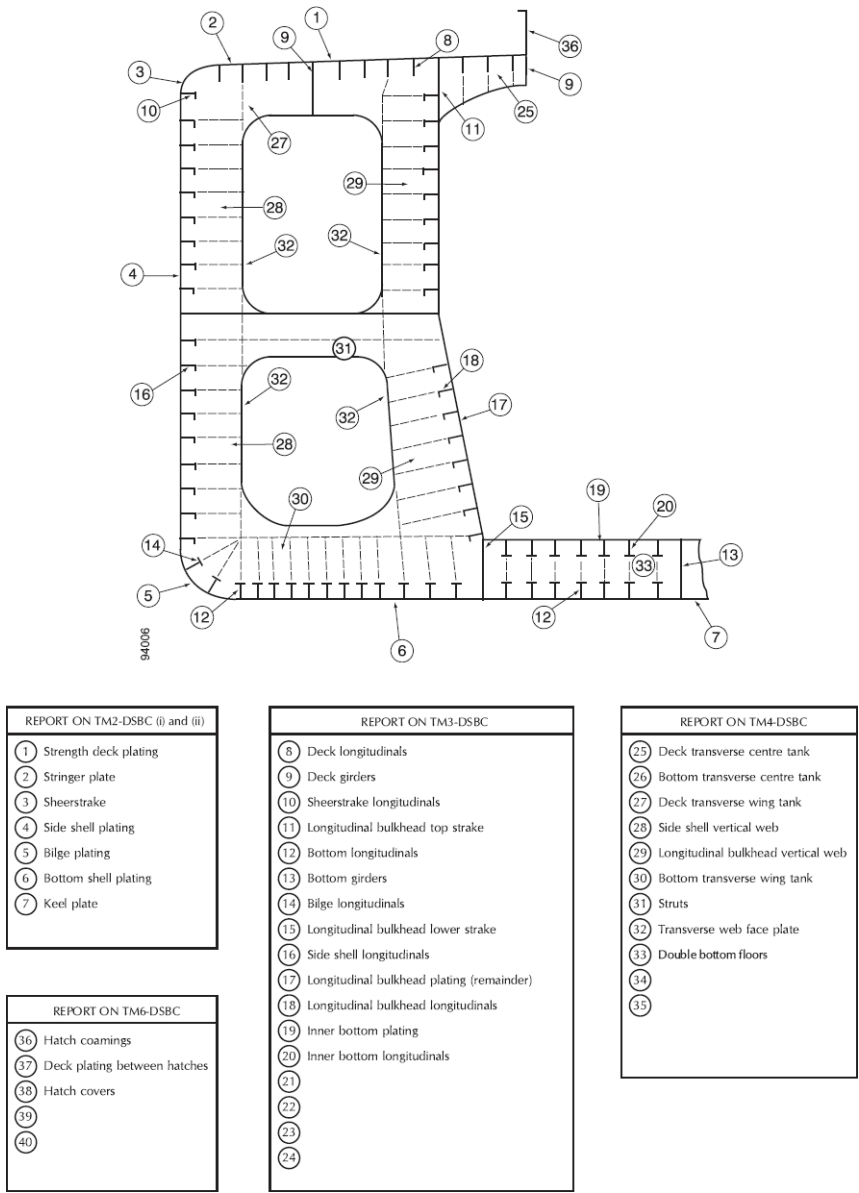
Typical transverse section of a double-side skin bulk carrier with indication of longitudinal and transverse members.





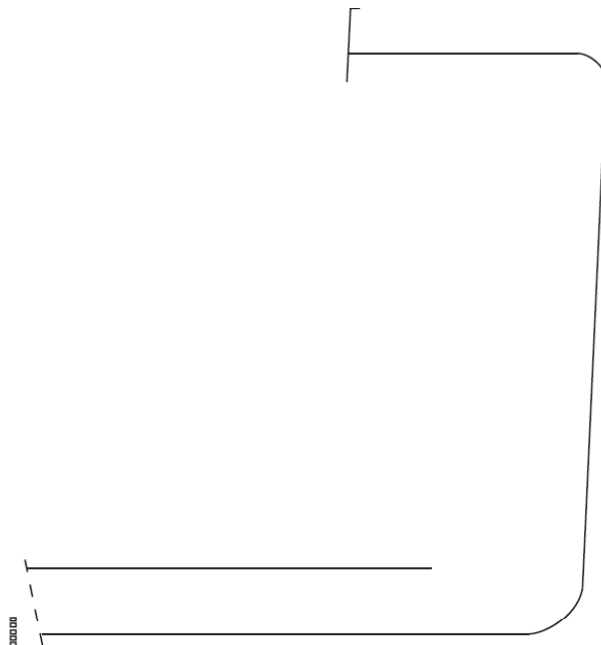
THICKNESS MEASUREMENT – ORE CARRIERS

Typical transverse section of an ore carrier with indication of longitudinal and transverse members.



## THICKNESS MEASUREMENT – DOUBLE-SIDE SKIN BULK CARRIER

Transverse section outline: the diagram may be used for those ships where the typical transverse sections are not suitable.



| REPORT ON<br>TM2-DSBC(i) and (ii) |                       |
|-----------------------------------|-----------------------|
| ①                                 | Strength deck plating |
| ②                                 | Stringer plate        |
| ③                                 | Sheerstrake           |
| ④                                 | Side shell plating    |
| ⑤                                 | Bilge plating         |
| ⑥                                 | Bottom shell plating  |
| ⑦                                 | Keel plate            |

| REPORT ON TM3-DSBC |                                                                                                        |
|--------------------|--------------------------------------------------------------------------------------------------------|
| ⑧                  | Deck longitudinals                                                                                     |
| ⑨                  | Deck girders                                                                                           |
| ⑩                  | Sheerstrake longitudinals                                                                              |
| ⑪                  | Topside tank sloping plating                                                                           |
| ⑫                  | Topside tank sloping plating longitudinals                                                             |
| ⑬                  | Bottom longitudinals                                                                                   |
| ⑭                  | Bottom girders                                                                                         |
| ⑮                  | Bilge longitudinals                                                                                    |
| ⑯                  | Side shell longitudinals, if any                                                                       |
| ⑰                  | Inner bottom plating                                                                                   |
| ⑱                  | Inner bottom longitudinals                                                                             |
| ⑲                  | Hopper plating                                                                                         |
| ⑳                  | Hopper longitudinals                                                                                   |
| ㉑                  | Inner side plating<br>– Inner side longitudinals, if any<br>– Horizontal girders in wing ballast tanks |

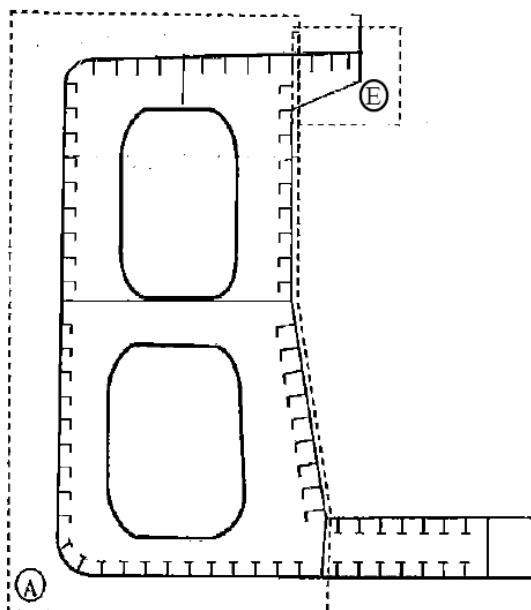
| REPORT ON TM4-DSBC |                                                    |
|--------------------|----------------------------------------------------|
| ㉒                  | Double-bottom tank floors                          |
| ㉓                  | Hopper side tank transverses                       |
| ㉔                  | Transverse web frame<br>– Topside tank transverses |

| REPORT ON TM6-DSBC |                                                                    |
|--------------------|--------------------------------------------------------------------|
| ㉕                  | Hatch coamings<br>– Deck plating between hatches<br>– Hatch covers |

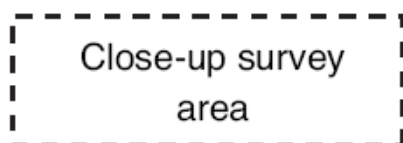
## THICKNESS MEASUREMENT – ORE CARRIERS

Areas subject to close-up survey and thickness measurements – areas (A), (C), (D) and (E) as defined in Annex 1 of Part B – Thickness to be reported on TM3-DSBC, TM4-DSBC, TM5-DSBC and TM6-DSBC as appropriate

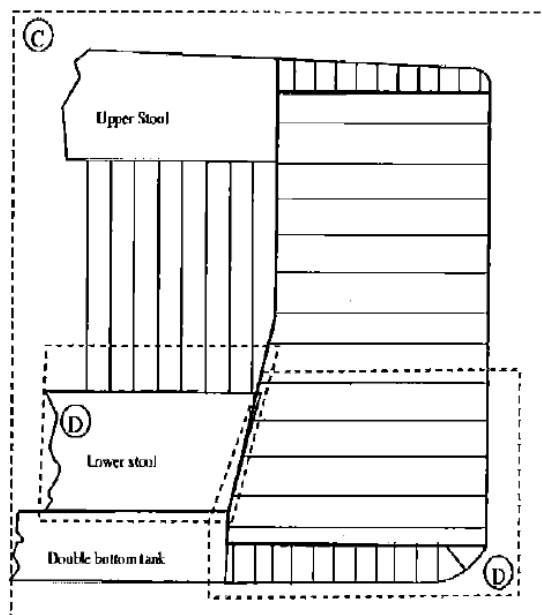
**Typical transverse section close-up survey**



Thickness to be reported on TM3-DSBC and TM4-DSBC as appropriate



**Typical transverse bulkhead**



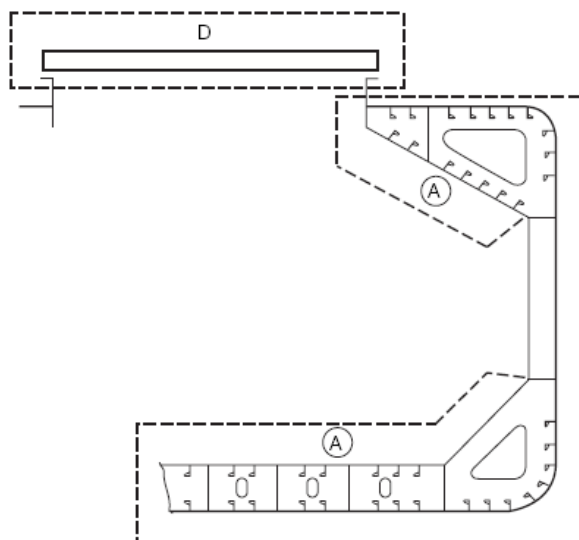
Thickness to be reported on TM5-DSBC

## THICKNESS MEASUREMENTS – DOUBLE-SIDE SKIN BULK CARRIER

Areas subject to close-up survey and thickness measurements – areas (A) to (E) as defined in Annex 1 of Part B – Thickness to be reported on TM3-DSBC, TM4-DSBC, TM5-DSBC and TM6-DSBC as appropriate.

Typical transverse section

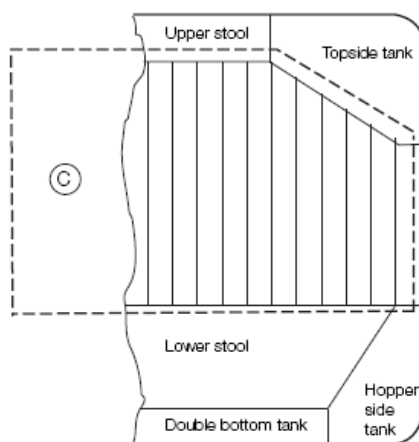
Areas (A) and (D)



Thickness to be reported on TM3-DSBC, TM4-DSBC, and TM6-DSBC, as appropriate

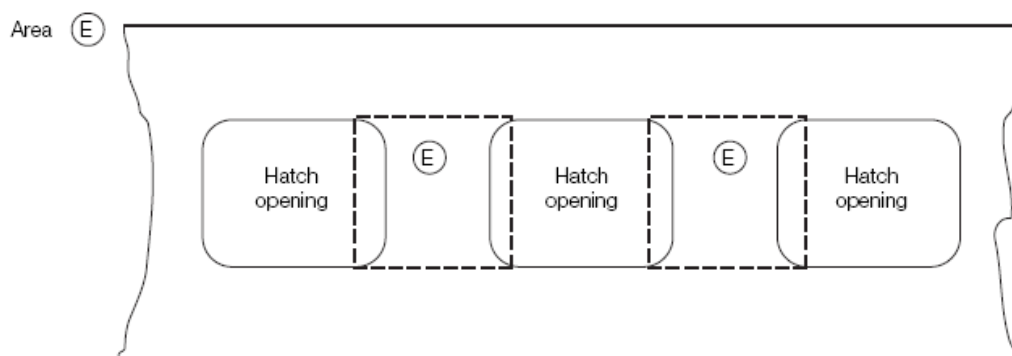
A cargo hold, transverse bulkhead

Area (C)



Thickness to be reported on TM5-DSBC

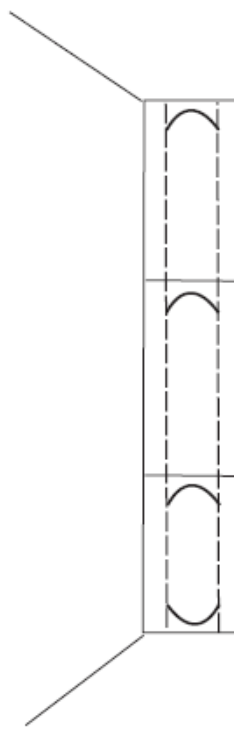
**Typical areas of deck plating and underdeck structure inside line  
of hatch openings between cargo hold hatches**



Thickness to be reported on TM1-DSBC

**Ordinary transverse frame in double-side skin tank**

Area (B)



Thickness to be reported on TM4-DSBC

## **ANNEX 9**

### **GUIDELINES FOR TECHNICAL ASSESSMENT IN CONJUNCTION WITH PLANNING FOR ENHANCED SURVEYS OF DOUBLE-SIDE SKIN BULK CARRIERS – RENEWAL SURVEY HULL**

#### **1 Introduction**

These guidelines contain information and suggestions concerning technical assessments, which may be of use in conjunction with the planning of enhanced renewal surveys of double-side skin bulk carriers. As indicated in 5.1.5, the guidelines are a recommended tool which may be invoked at the discretion of the Administration, when considered necessary and appropriate, in conjunction with the preparation of the required survey programme.

#### **2 Purpose and principles**

##### **2.1 Purpose**

**2.1.1** The purpose of the technical assessments described in these guidelines is to assist in identifying critical structural areas, nominating suspect areas and in focusing attention on structural elements or areas of structural elements which may be particularly susceptible to, or evidence a history of, wastage or damage. This information may be useful in nominating locations, areas holds and tanks for thickness measurement, close-up survey and tank testing.

**2.1.2** Critical structural areas are locations which have been identified from calculations to require monitoring or from the service history of the subject ship or from similar or sister ships (if available) to be sensitive to cracking, buckling or corrosion which would impair the structural integrity of the ship.

##### **2.2 Minimum requirements**

However, these guidelines may not be used to reduce the requirements pertaining to thickness measurement, close-up survey and tank testing contained in annexes 1 and 2 of part B and in 2.7, respectively, which, in all cases, should be complied with as a minimum.

##### **2.3 Timing**

As with other aspects of survey planning, the technical assessments described in these guidelines should be worked out by the owner or operator in cooperation with the Administration well in advance of the commencement of the renewal survey, i.e. prior to commencing the survey and normally at least 12 to 15 months before the survey's completion due date.

##### **2.4 Aspects to be considered**

**2.4.1** Technical assessments, which may include quantitative or qualitative evaluation of relative risks of possible deterioration, of the following aspects of a particular ship may be used as a basis for the nomination of holds, tanks and areas for survey:

- .1** design features such as stress levels on various structural elements, design details and extent of use of high-tensile steel;

- .2 former history with respect to corrosion, cracking, buckling, indents and repairs for the particular ship as well as similar vessels, where available; and
- .3 information with respect to types of cargo carried, use of different holds/tanks for cargo/ballast, protection of holds and tanks and condition of coating, if any.

**2.4.2** Technical assessments of the relative risks of susceptibility to damage or deterioration of various structural elements and areas should be judged and decided on the basis of recognized principles and practices, such as may be found in references 2, 3 and 4.

### **3 Technical assessment**

#### **3.1 General**

**3.1.1** There are three basic types of possible failure, which may be the subject of technical assessment in connection with planning of surveys; corrosion, cracks and buckling. Contact damages are not normally covered by the survey planning since indents are usually noted in memoranda and assumed to be dealt with as a normal routine by surveyors.

**3.1.2** Technical assessments performed in conjunction with the survey planning process should, in principle, be as shown schematically in figure 1.

The approach is basically an evaluation of the risk in the following aspects based on the knowledge and experience related to:

- .1 design; and
- .2 corrosion.

**3.1.3** The design should be considered with respect to structural details, which may be susceptible to buckling or cracking as a result of vibration, high stress levels or fatigue.

**3.1.4** Corrosion is related to the ageing process, and is closely connected with the quality of corrosion prevention systems fitted at new building, and subsequent maintenance during the service life. Corrosion may also lead to cracking and/or buckling.

#### **3.2 Methods**

##### **3.2.1 Design details**

**3.2.1.1** Damage experience related to the ship in question and sister and/or similar ships, where available, is the main source of information to be used in the process of planning. In addition, a selection of structural details from the design drawings should be included.

**3.2.1.2** Typical damage experience to be considered will consist of:

- .1 number, extent, location and frequency of cracks; and
- .2 location of buckles.

**3.2.1.3** This information may be found in the survey reports and/or the owner's files, including the results of the owner's own inspections. The defects should be analysed, noted and marked on sketches.

**3.2.1.4** In addition, general experience should be utilized. Also, reference should be made to reference 2, which contains a catalogue of typical damages and proposed repair methods for various structural details on single-skin bulk carriers. Reference should also be made to reference 3, which contains catalogues of typical damages and proposed repair methods for double-hull oil tanker structural details which may to some extent be similar to structural details in double-side skin bulk carriers. Such figures should be used together with a review of the main drawings, in order to compare with the actual structure and search for similar details that may be susceptible to damage. In particular, chapter 3 of reference 3 deals with various aspects specific to double-hull tankers, such as stress concentration locations, misalignment during construction, corrosion trends, fatigue considerations and areas requiring special attention, while chapter 4 of reference 3 addresses experience gained on structural defects in double-hulls (chemical tankers, OBO carriers, ore/oil carriers, gas carriers), which should also be considered in working out the survey planning.

**3.2.1.5** The review of the main structural drawings, in addition to using the above-mentioned figures, should include checking for typical design details where cracking has been experienced. The factors contributing to damage should be carefully considered.

**3.2.1.6** The use of high-tensile steel (HTS) is an important factor. Details showing good service experience where ordinary, mild steel has been used may be more susceptible to damage when HTS, and its higher associated stresses, are utilized. There is extensive and, in general, good experience, with the use of HTS for longitudinal material in deck and bottom structures. Experience in other locations, where the dynamic stresses may be higher, is less favourable, e.g., side structures.

**3.2.1.7** In this respect, stress calculations of typical and important components and details, in accordance with relevant methods, may prove useful and should be considered.

**3.2.1.8** The selected areas of the structure identified during this process should be recorded and marked on the structural drawings to be included in the survey programme.

## **3.2.2**     *Corrosion*

**3.2.2.1** In order to evaluate relative corrosion risks, the following information should generally be considered:

- .1        usage of tanks, holds and spaces;
- .2        condition of coatings;
- .3        cleaning procedures;
- .4        previous corrosion damage;
- .5        ballast use and time for cargo holds;
- .6        risk of corrosion in cargo holds and ballast tanks; and
- .7        location of ballast tanks adjacent to heated fuel oil tanks.



**3.2.2.2** Reference 4 gives definitive examples which can be used for judging and describing coating condition, using typical pictures of conditions.

**3.2.2.3** The evaluation of corrosion risks should be based on information in both reference 2 and reference 4, as far as applicable to double-side skin construction, together with relevant information on the anticipated condition of the ship as derived from the information collected in order to prepare the survey programme and the age of the ship. The various holds, tanks and spaces should be listed with the corrosion risks nominated accordingly.

**3.2.3** *Locations for close-up survey and thickness measurement*

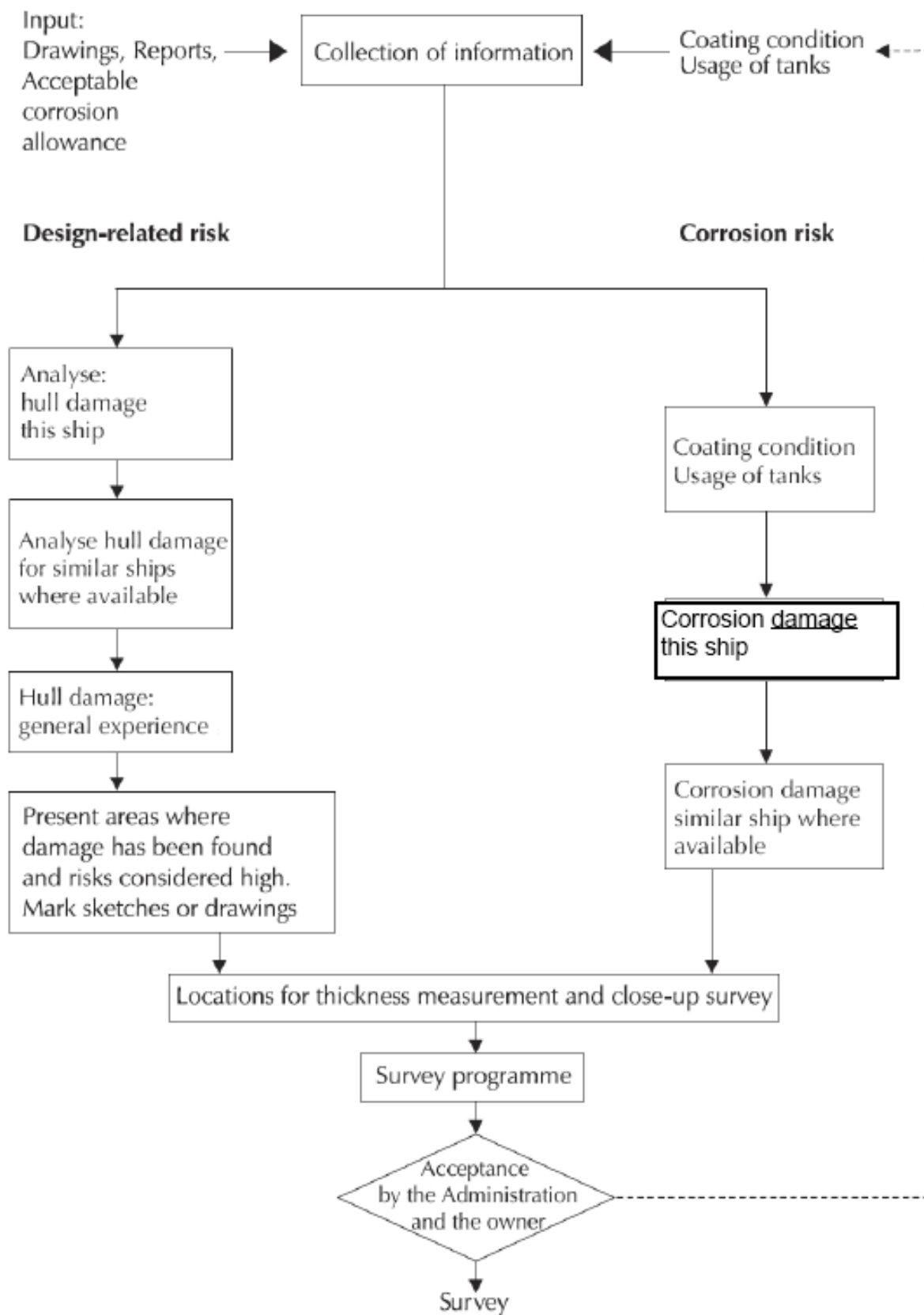
**3.2.3.1** On the basis of the table of corrosion risks and the evaluation of design experience, the locations for initial close-up survey and thickness measurement (areas and sections) may be nominated.

**3.2.3.2** The sections subject to thickness measurement should normally be nominated in tanks, holds and spaces where corrosion risk is judged to be the highest.

**3.2.3.3** The nomination of tanks, holds and spaces for close-up survey should initially be based on highest corrosion risk, and should always include ballast tanks. The principle for the selection should be that the extent is increased by age or where information is insufficient or unreliable.

## References

- 1** IACS, Unified Requirement Z10.5, Hull Surveys of Double Skin Bulk Carriers
- 2** IACS, Bulk Carriers: Guidelines for Surveys, Assessment and Repair of Hull Structures, January 2007
- 3** TSCF, Guidelines for the Inspection and Maintenance of Double Hull Tanker Structures, 1995
- 4** TSCF, Guidance Manual for Tanker Structures, 1997



**Figure 1** – Technical assessment and the survey planning process

## ANNEX 10

### REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT THOSE AREAS OF SUBSTANTIAL CORROSION OF BULK CARRIERS WITH DOUBLE-SIDE SKIN CONSTRUCTION WITHIN THE CARGO LENGTH AREA

**Table 1 – Bottom, inner bottom and hopper structure**

| Structural member                                                 | Extent of measurement                                                                                                       | Pattern of measurement                                                                                                                                                    |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bottom, inner bottom and hopper structure plating                 | Minimum of three bays across double-bottom tank, including aft bay<br>Measurements around and under all suction bell mouths | Five-point pattern for each panel between longitudinals and floors                                                                                                        |
| Bottom, inner bottom and hopper structure longitudinals           | Minimum of three longitudinals in each bay where bottom plating measured                                                    | Three measurements in line across flange and three measurements on the vertical web                                                                                       |
| Bottom girders, including the watertight ones                     | At fore and aft watertight floors and in centre of tanks                                                                    | Vertical line of single measurements on girder plating with one measurement between each panel stiffener, or a minimum of three measurements                              |
| Bottom floors, including the watertight ones                      | Three floors in the bays where bottom plating measured, with measurements at both ends and middle                           | Five-point pattern over 2 m <sup>2</sup> area                                                                                                                             |
| Hopper structure web frame ring                                   | Three floors in bays where bottom plating measured                                                                          | Five-point pattern over 1 m <sup>2</sup> of plating<br>Single measurements on flange                                                                                      |
| Hopper structure transverse watertight bulkhead or swash bulkhead | – lower ⅓ of bulkhead                                                                                                       | five-point pattern over 1 m <sup>2</sup> of plating                                                                                                                       |
|                                                                   | – upper ⅓ of bulkhead                                                                                                       | five-point pattern over 2 m <sup>2</sup> of plating                                                                                                                       |
|                                                                   | – stiffeners (minimum of three)                                                                                             | For web, five-point pattern over span (two measurements across web at each end and one at centre of span). For flange, single measurements at each end and centre of span |
| Panel stiffening                                                  | Where applicable                                                                                                            | Single measurements                                                                                                                                                       |

**Table 2 – Deck structure including cross strips, main cargo hatchways, hatch covers, coamings and topside tanks**

| Structural member        | Extent of measurement                                                                                                 | Pattern of measurement                                                                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Cross deck strip plating | Suspect cross deck strip plating                                                                                      | Five-point pattern between under deck stiffeners over 1 m length                                                                                 |
| Under deck stiffeners    | Transverse members<br>Longitudinal member                                                                             | Five-point pattern at each end and mid span<br>Five-point pattern on both web and flange                                                         |
| Hatch covers             | Side and end skirts, each three locations<br>Three longitudinal bands, outboard strakes (2) and centerline strake (1) | Five-point pattern at each location<br>Five-point measurement each band                                                                          |
| Hatch coamings           | Each side and end of coaming, one band lower ⅓, one band upper ⅓ of coaming                                           | Five-point measurement each band, i.e. end or side coaming                                                                                       |
| Topside ballast tanks    | a) watertight transverse bulkheads:<br>– Lower ⅓ of bulkhead<br>– Upper ⅓ of bulkhead<br>– Stiffeners                 | Five-point pattern over 1 m <sup>2</sup> of plating<br>Five-point pattern over 1 m <sup>2</sup> of plating<br>Five-point pattern over 1 m length |
| Topside ballast tanks    | b) two representative swash transverse bulkheads:<br>– Lower ⅓ of bulkhead<br>– Upper ⅓ of bulkhead<br>– Stiffeners   | Five-point pattern over 1 m <sup>2</sup> of plating<br>Five-point pattern over 1 m <sup>2</sup> of plating<br>Five-point pattern over 1 m length |
| Topside ballast tanks    | c) three representative bays of slope plating:<br>– Lower ⅓ of tank<br>– Upper ⅓ of tank                              | Five-point pattern over 1 m <sup>2</sup> of plating<br>Five point pattern over 1 m <sup>2</sup> of plating                                       |

| Structural member       | Extent of measurement                  | Pattern of measurement                                    |
|-------------------------|----------------------------------------|-----------------------------------------------------------|
| Topside ballast tanks   | d) Longitudinals, suspect and adjacent | Five point pattern on both web and flange over 1 m length |
| Main deck plating       | Suspect plates and adjacent (4)        | Five-point pattern over 1 m <sup>2</sup> of plating       |
| Main deck longitudinals | Suspect plates                         | Five point pattern on both web and flange over 1 m length |
| Web frames/transverses  | Suspect plates                         | Five-point pattern over 1 m <sup>2</sup> of plating       |

**Table 3 – Structure in double-side spaces of double-side skin bulk carriers including wing void spaces of ore carriers**

| Structural member                                                                                               | Extent of measurement                                                                                                                                                                | Pattern of measurement                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Side shell and inner plating:<br>– Upper strake and strakes in way of horizontal girders<br>– All other strakes | – Plating between each pair of transverse frames/longitudinals in a minimum of three bays (along the tank)<br>– Plating between every third pair of longitudinals in same three bays | – Single measurement<br>– Single measurement                                                                                   |
| Side shell and inner side transverse frames/ longitudinals on:<br>– upper strake<br>– all other strakes         | – Each transverse frame/ longitudinal in same three bays<br>– Every third transverse frame/longitudinal in same three bays                                                           | – Three measurements across web and one measurement on flange<br>– Three measurements across web and one measurement on flange |
| Transverse frames/ longitudinals:<br>– brackets                                                                 | Minimum of three at top, middle and bottom of tank in same three bays                                                                                                                | Five-point pattern over area of bracket                                                                                        |
| Vertical web and transverse bulkheads:<br>– strakes in a way of horizontal girders<br>– other strakes           | – Minimum of two webs and both transverse bulkheads<br>– Minimum of two webs and both transverse bulkheads                                                                           | – Five-point pattern over approximately 2 m <sup>2</sup> area<br>– Two measurements between each pair of vertical stiffeners   |
| Horizontal girders                                                                                              | Plating on each girder in a minimum of three bays                                                                                                                                    | Two measurements between each pair of longitudinal girder stiffeners                                                           |
| Panel stiffening                                                                                                | Where applicable                                                                                                                                                                     | Single measurements                                                                                                            |

**Table 4 – Transverse bulkheads in cargo holds**

| Structural member         | Extent of measurement                                                                                                                                                                     | Pattern of measurement                                                                                             |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Lower stool, where fitted | – Transverse band within 25 mm of welded connection to inner bottom<br>– Transverse bands within 25 mm of welded connection to shelf plate                                                | – Five-point pattern between stiffeners over 1 m length<br>– Five-point pattern between stiffeners over 1 m length |
| Transverse bulkheads      | – Transverse band at approximately mid height<br>– Transverse band at part of bulkhead adjacent to upper deck or below upper stool shelf plate (for those ships fitted with upper stools) | – Five-point pattern over 1 m <sup>2</sup> of plating<br>– Five-point pattern over 1 m <sup>2</sup> of plating     |

## ANNEX 11

### STRENGTH OF CARGO HATCH COVER SECURING ARRANGEMENTS FOR BULK CARRIERS

#### 1 Securing devices

The strength of securing devices should comply with the following requirements:

- .1 Panel hatch covers should be secured by appropriate devices (bolts, wedges or similar) suitably spaced alongside the coamings and between cover elements. Arrangement and spacing should be determined with due attention to the effectiveness for weathertightness, depending upon the type and the size of the hatch cover, as well as on the stiffness of the cover edges between the securing devices.

- .2 The net sectional area of each securing device should not be less than:

$$A = 1.4 a/f \text{ (cm}^2\text{)}$$

where:

- $a$  = spacing between securing devices not to be taken less than 2 m
- $f$  =  $(\sigma_Y / 235)^e$
- $\sigma_Y$  = specified minimum upper yield stress in N/mm<sup>2</sup> of the steel used for fabrication, not to be taken greater than 70% of the ultimate tensile strength
- $e$  = 0.75 for  $\sigma_Y > 235$   
= 1.0 for  $\sigma_Y \leq 235$

Rods or bolts should have a net diameter not less than 19 mm for hatchways exceeding 5 m<sup>2</sup> in area.

- .3 Between cover and coaming and at cross-joints, a packing line pressure sufficient to obtain weathertightness should be maintained by the securing devices. For packing line pressures exceeding 5 N/mm, the cross section area should be increased in direct proportion. The packing line pressure should be specified.

- .4 The cover edge stiffness should be sufficient to maintain adequate sealing pressure between securing devices. The moment of inertia,  $I$ , of edge elements should not be less than:

$$I = 6 p a^4 \text{ (cm}^4\text{)}$$

where:

- $p$  = packing line pressure in N/mm, minimum 5 N/mm
- $a$  = spacing in metres of securing devices.

- .5 Securing devices should be of reliable construction and securely attached to the hatchway coamings, decks or covers. Individual securing devices on each cover should have approximately the same stiffness characteristics.

- .6 Where rod cleats are fitted, resilient washers or cushions should be incorporated.
- .7 Where hydraulic cleating is adopted, a positive means should be provided to ensure that it remains mechanically locked in the closed position in the event of failure of the hydraulic system.

## **2 Stoppers**

- 2.1** Nos.1 and 2 hatch covers should be effectively secured, by means of stoppers, against the transverse forces arising from a pressure of 175 kN/m<sup>2</sup>.
- 2.2** No.2 hatch covers should be effectively secured, by means of stoppers, against the longitudinal forces acting on the forward end arising from a pressure of 175 kN/m<sup>2</sup>.
- 2.3** No.1 hatch cover should be effectively secured, by means of stoppers, against the longitudinal forces acting on the forward end arising from a pressure of 230 kN/m<sup>2</sup>. This pressure may be reduced to 175 kN/m<sup>2</sup> if a forecastle is fitted.
- 2.4** The equivalent stress in stoppers and their supporting structures and calculated in the throat of the stopper welds is not to exceed the allowable value of 0.8  $\sigma_Y$ .

## **3 Materials and welding**

Where stoppers or securing devices are fitted to comply with this annex, they should be manufactured of materials, including welding electrodes, to the satisfaction of the Administration.

## **ANNEX 12**

### **PROCEDURAL REQUIREMENTS FOR THICKNESS MEASUREMENTS**

#### **1 General**

Thickness measurements required in the context of hull structural surveys, if not carried out by the society itself should be witnessed by a surveyor. The attendance of the surveyor should be recorded. This also applies to thickness measurements taken during voyages.

#### **2 Survey meeting**

**2.1** Prior to commencement of the renewal or intermediate survey, a meeting should be held between the attending surveyor(s), the owner's representative(s) in attendance and the thickness measurement firm's representative(s) so as to ensure the safe and efficient execution of the surveys and thickness measurements to be carried out on board.

**2.2** Communication with the thickness measurement operator(s) and owner's representative(s) should be agreed during the meeting, with respect to the following:

- .1** reporting of thickness measurements on a regular basis;
- .2** prompt notification to the surveyor in case of findings such as:
  - .1** excessive and/or extensive corrosion or pitting/grooving of any significance;
  - .2** structural defects like buckling, fractures and deformed structures;
  - .3** detached and/or holed structure; and
  - .4** corrosion of welds.

**2.3** The survey report should indicate where and when the meeting took place and who attended (the name of the surveyor(s), the owner's representative(s) and the thickness measurement firm's representative(s)).

#### **3 Monitoring of the thickness measurement process on board**

**3.1** The surveyor should decide final extent and location of thickness measurements after overall survey of representative spaces on board.

**3.2** In case the owner prefers to commence the thickness measurements prior to the overall survey, then the surveyor should advise that the planned extent and locations of thickness measurements are subject to confirmation during the overall survey. Based on findings, the surveyor may require additional thickness measurements to be taken.

**3.3** The surveyor should direct the gauging operation by selecting locations such that readings taken represent, on average, the condition of the structure for that area.

**3.4** Thickness measurements taken mainly to evaluate the extent of corrosion, which may affect the hull girder strength, should be carried out in a systematic manner such that all longitudinal structural members are gauged, as required.

**3.5** Where thickness measurements indicate substantial corrosion or wastage in excess of allowable diminution, the surveyor should direct locations for additional thickness measurements in order to delineate areas of substantial corrosion and to identify structural members for repairs/renewals.

**3.6** Thickness measurements of structures in areas where close-up surveys are required should be carried out simultaneously with close-up survey.

#### **4 Review and verification**

**4.1** Upon completion of the thickness measurements, the surveyor should confirm that no further gaugings are needed, or specify additional gaugings.

**4.2** Where these guidelines allow the extent of thickness measurements to be reduced after special considerations by the surveyor, these special considerations should be reported, where appropriate.

**4.3** In case thickness measurements are partly carried out, the extent of remaining thickness measurements should be reported for the use of the next surveyor.



## **ANNEX B**

### **CODE ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF OIL TANKERS**

#### **Part A**

### **CODE ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF DOUBLE-HULL OIL TANKERS**

#### **1 General**

##### **1.1 *Application*<sup>8</sup>**

**1.1.1** The Code should apply to all self-propelled double-hull oil tankers of 500 gross tonnage and above.

**1.1.2** The Code should apply to surveys of hull structure and piping systems in way of cargo tanks, pump-rooms, cofferdams, pipe tunnels, void spaces within the cargo area and all ballast tanks.

**1.1.3** The Code contains the minimum extent of examination, thickness measurements and tank testing. The survey should be extended when substantial corrosion and/or structural defects are found and include additional close-up survey when necessary.

**1.1.4** The surveys should be carried out during the surveys prescribed by regulation I/10 of the Convention.

##### **1.2 *Definitions***

**1.2.1** *Double-hull oil tanker* is a ship which is constructed primarily for the carriage of oil<sup>9</sup> in bulk, which has the cargo tanks protected by a double-hull which extends for the entire length of the cargo area, consisting of double sides and double-bottom spaces for the carriage of water ballast or void spaces.

**1.2.2** *Ballast tank* is a tank which is used solely for the carriage of salt water ballast.

**1.2.3** *Combined cargo/ballast tank*, if referred to within the Code, is a tank which is used for the carriage of cargo or ballast water as a routine part of the vessel's operation and will be treated as a Ballast Tank. Cargo tanks in which water ballast might be carried only in exceptional cases per MARPOL regulation I/18.3 are to be treated as cargo tanks.

**1.2.4** *Overall survey* is a survey intended to report on the overall condition of the hull structure and determine the extent of additional close-up surveys.

**1.2.5** *Close-up survey* is a survey where the details of structural components are within the close visual inspection range of the surveyor, i.e. normally within reach of hand.

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<sup>8</sup> The intention of the Code is to ensure that an appropriate level of review of plans and documents is conducted and consistency in application is attained. Such evaluation of survey reports, survey programmes, planning documents, etc., should be carried out at the managerial level of the Administration or organization recognized by the Administration.

<sup>9</sup> MARPOL Annex I cargoes. The requirements of these Guidelines are also applicable to existing double-hull tankers not complying with MARPOL regulation I/19, but having a U-shaped midship section.

**1.2.6** *Transverse section* is the cross section of the hull perpendicular to the ship's centerline and includes all longitudinal members such as plating, longitudinals and girders at the deck, sides, bottom, inner bottom and longitudinal bulkheads.

**1.2.7** *Representative tanks* are those which are expected to reflect the condition of other tanks of similar type and service and with similar corrosion prevention systems. When selecting representative tanks, account should be taken of the service and repair history on board and identifiable critical and/or suspect areas.

**1.2.8** *Suspect areas* are locations showing substantial corrosion and/or are considered by the surveyor to be prone to rapid wastage.

**1.2.9** *Substantial corrosion* is an extent of corrosion such that assessment of corrosion pattern indicates a wastage in excess of 75% of allowable margins, but within acceptable limits. For ships built under the IACS Common Structural Rules, substantial corrosion is an extent of corrosion such that the assessment of the corrosion pattern indicates a gauged (or measured) thickness between  $t_{\text{net}} + 0.5\text{mm}$  and  $t_{\text{net}}$ .

**1.2.10** *Corrosion prevention system* is normally considered a full hard coating. Hard protective coating should usually be epoxy coating or equivalent. Other coating systems, which are neither soft nor semi-hard coatings, may be considered acceptable as alternatives provided that they are applied and maintained in compliance with the manufacturer's specification.

**1.2.11** *Coating condition* is defined as follows:

GOOD condition with only minor spot rusting;

FAIR condition with local breakdown of coating at edges of stiffeners and weld connections and/or light rusting over 20% or more of areas under consideration, but less than as defined for POOR condition; and

POOR condition with general breakdown of coating over 20% or more of areas or hard scale at 10% or more of areas under consideration.

**1.2.12** *Critical structural areas* are locations which have been identified from calculations to require monitoring or from the service history of the subject ship or from similar or sister ships to be sensitive to cracking, buckling or corrosion which would impair the structural integrity of the ship.

**1.2.13** *Cargo area* is that part of the ship which contains cargo tanks, slop tanks and cargo/ballast pump-rooms, cofferdams, ballast tanks and void spaces adjacent to cargo tanks and also deck areas throughout the entire length and breadth of the part of the ship over the above mentioned spaces.

**1.2.14** *Intermediate survey* is a survey carried out either at the second or the third annual survey.

**1.2.15** *Prompt and thorough repair* is a permanent repair completed at the time of survey to the satisfaction of the surveyor, therein removing the need for the imposition of any associated condition of classification or recommendation.

**1.2.16** *Specially considered* (in connection with close-up surveys and thickness measurements) means sufficient close-up inspection and thickness measurements are taken to confirm the actual average condition of the structure under coating.

### **1.3 Repairs**

**1.3.1** Any damage in association with wastage over the allowable limits (including buckling, grooving, detachment or fracture), or extensive areas of wastage over the allowable limits, which affects or, in the opinion of the Administration, will affect the ship's structural, watertight or weathertight integrity, should be promptly and thoroughly (see 1.2.15) repaired. Areas to be considered include:

- .1 bottom structure and bottom plating;
- .2 side structure and side plating;
- .3 deck structure and deck plating;
- .4 watertight or oiltight bulkheads; and
- .5 hatch covers and hatch coamings, where fitted.

For locations where adequate repair facilities are not available, the Administration may allow the ship to proceed directly to a repair facility. This may require discharging the cargo and/or temporary repairs for the intended voyage.

**1.3.2** Additionally, when a survey results in the identification of corrosion or structural defects, either of which, in the opinion of the Administration, will impair the ship's fitness for continued service, remedial measures should be implemented before the ship continues in service.

### **1.4 Surveyors**

For tankers of 20,000 tons deadweight and above, two surveyors should jointly carry out the first scheduled renewal survey after the tanker passes 10 years of age (i.e. third renewal survey), and all subsequent renewal surveys and intermediate surveys. If the surveys are carried out by a recognized organization, the surveyors should be exclusively employed by such recognized organizations.

### **1.5 Thickness measurements and close-up surveys**

In any kind of survey, i.e. renewal, intermediate, annual or other surveys having the scope of the foregoing ones, thickness measurements of structures in areas where close-up surveys are required should be carried out simultaneously with close-up surveys.

## **2 Renewal survey**

### **2.1 General**

**2.1.1** The renewal survey may be commenced at the fourth annual survey and be progressed during the succeeding year with a view to completion by the fifth anniversary date.

**2.1.2** As part of the preparation for the renewal survey the survey programme should be dealt with in advance of the renewal survey. The thickness measurement should not be carried out before the fourth annual survey.

**2.1.3** The survey should include, in addition to the requirements of the annual survey, examination, tests and checks of sufficient extent to ensure that the hull and related piping as required in 2.1.5 is in a satisfactory condition and is fit for its intended purpose for the new period of validity of the Cargo Ship Safety Construction Certificate, subject to proper maintenance and operation and to periodical surveys being carried out at the due dates.

**2.1.4** All cargo tanks, ballast tanks, and any other tanks in double-hull spaces, pump-rooms, pipe tunnels, cofferdams and void spaces bounding cargo tanks, decks and outer hull should be examined, and this examination should be supplemented by thickness measurement and testing as required in 2.5 and 2.6, to ensure that the structural integrity remains effective. The aim of the examination is to discover substantial corrosion, significant deformation, fractures, damages or other structural deterioration that may be present.

**2.1.5** Cargo piping on deck, including crude oil washing (COW) piping, and cargo and ballast piping within the above tanks and spaces should be examined and operationally tested to working pressure to attending surveyor's satisfaction to ensure that tightness and condition remain satisfactory. Special attention should be given to any ballast piping in cargo tanks and any cargo piping in ballast tanks and void spaces, and surveyors should be advised on all occasions when this piping, including valves and fittings, are open during repair periods and can be examined internally.

## **2.2** *Dry dock survey*

**2.2.1** A survey in dry dock should be a part of the renewal survey. There should be a minimum of two inspections of the outside of the ship's bottom during the five-year period of the Safety Construction Certificate. In all cases, the maximum interval between bottom inspections should not exceed 36 months.

**2.2.2** For ships of 15 years of age and over, inspection of the outside of the ship's bottom should be carried out with the ship in dry dock. For ships of less than 15 years of age, alternate inspections of the ship's bottom not conducted in conjunction with the renewal survey may be carried out with the ship afloat. Inspection of the ship afloat should only be carried out when the conditions are satisfactory and the proper equipment and suitably qualified staff is available.

**2.2.3** If a survey in dry dock is not completed in conjunction with the renewal survey or if the 36 month maximum interval referred to in 2.2.1 is not complied with, the Cargo Ship Safety Construction Certificate should cease to be valid until a survey in dry dock is completed.

**2.2.4** The overall and close-up surveys and thickness measurements, as applicable, of the lower portions of the cargo tanks and ballast tanks should be carried out in accordance with the applicable requirements for renewal surveys, if not already performed.

**Note:** Lower portions of the cargo and ballast tanks are considered to be the parts below light ballast water line.

### **2.3      *Tank corrosion prevention system***

Where provided, the condition of the corrosion prevention system of cargo tanks should be examined. A ballast tank should be examined at subsequent annual intervals where:

- .1      a hard protective coating has not been applied from the time of construction; or
- .2      a soft or semi-hard coating has been applied; or
- .3      substantial corrosion is found within the tank; or
- .4      the hard protective coating is found to be in less than GOOD condition and the hard protective coating is not repaired to the satisfaction of the surveyor.

Thickness measurement should be carried out as deemed necessary by the surveyor.

### **2.4      *Extent of overall and close-up surveys***

**2.4.1**    An overall survey of all tanks and spaces should be carried out at the renewal survey. Suspect areas identified at previous surveys should be examined.

**2.4.2**    The minimum requirements for close-up surveys at the renewal survey are given in annex 1.

**2.4.3**    The surveyor may extend the scope of the close-up survey as deemed necessary taking into account the maintenance of the tanks under survey, the condition of the corrosion prevention system and also in the following cases:

- .1      in particular, tanks having structural arrangements or details which have suffered defects in similar tanks or on similar ships according to available information; and
- .2      in tanks which have structures with reduced scantlings in association with a corrosion prevention system approved by the Administration.

**2.4.4**    For areas in tanks where hard protective coatings are found to be in GOOD condition as defined in 1.2.11, the extent of close-up surveys according to annex 1 may be specially considered by the Administration.

### **2.5      *Extent of thickness measurements***

**2.5.1**    The minimum requirements for thickness measurements at the renewal survey are given in annex 2.

**2.5.2**    Provisions for extended measurements for areas with substantial corrosion are given in annex 4, and may be additionally specified in the survey programme as required in 5.1. These extended thickness measurements should be carried out before the survey is credited as completed. Suspect areas identified at previous surveys should have thickness measurements taken.

**2.5.3**    The surveyor may further extend the thickness measurements as deemed necessary.

**2.5.4** For areas in tanks where hard protective coatings are found to be in GOOD condition as defined in 1.2.11, the extent of thickness measurements according to annex 2 may be specially considered by the Administration.

**2.5.5** Transverse sections should be chosen where the largest reductions are suspected to occur or are revealed from deck plating measurements.

**2.5.6** In cases where two or three sections are to be measured, at least one should include a ballast tank within 0.5L amidships. In case of oil tankers of 130 m in length and upwards (as defined in the International Convention on Load Lines in force) and more than 10 years of age, for the evaluation of the ship's longitudinal strength as required in 8.2, the sampling method of thickness measurements is given in annex 12.

## **2.6      *Extent of tank pressure testing***

**2.6.1** The minimum requirements for tank pressure testing at the renewal survey are given in annex 3.

**2.6.2** The surveyor may extend the tank pressure testing as deemed necessary.

**2.6.3** Boundaries of ballast tanks should be tested with a head of liquid to the top of air pipes.

**2.6.4** Boundaries of cargo tanks should be tested to the highest point that liquid will rise under service conditions.

**2.6.5** The testing of double-bottom tanks and other spaces not designed for the carriage of liquid may be omitted, provided a satisfactory internal examination together with an examination of the tanktop is carried out.

## **3          *Annual survey***

### **3.1      *General***

The annual survey should consist of an examination for the purpose of ensuring, as far as practicable, that the hull and piping are maintained in a satisfactory condition and should take into account the service history, condition and extent of the corrosion prevention system of ballast tanks and areas identified in the survey report file.

### **3.2      *Examination of the hull***

**3.2.1** Examination of the hull plating and its closing appliances should be carried out as far as can be seen.

**3.2.2** Examination of watertight penetrations should be carried out as far as practicable.

### **3.3      *Examination of weather decks***

**3.3.1** Examination of cargo tank openings including gaskets, covers, coamings and flame screens.

**3.3.2** Examination of cargo tank pressure/vacuum valves and flame screens.

**3.3.3** Examination of flame screens on vents to all bunker tanks.

**3.3.4** Examination of cargo, crude oil washing, bunker and vent piping systems, including vent masts and headers.

**3.4** ***Examination of cargo pump-rooms and pipe tunnels if fitted***

**3.4.1** Examination of all pump-room bulkheads for signs of oil leakage or fractures and, in particular, the sealing arrangements of all penetrations of pump-room bulkheads.

**3.4.2** Examination of the condition of all piping systems and pipe tunnels.

**3.5** ***Examination of ballast tanks***

**3.5.1** Examination of ballast tanks should be carried out when required as a consequence of the results of the renewal survey and intermediate survey. When considered necessary by the Administration or when extensive corrosion is found, thickness measurements should be carried out.

**3.5.2** Where substantial corrosion as defined in 1.2.9 is found, the extent of thickness measurements should be increased in accordance with the requirements in annex 4. These extended thickness measurements should be carried out before the survey is credited as completed. Suspect areas identified at previous surveys should be examined. Areas of substantial corrosion identified at previous surveys should have thickness measurements taken.

**4** ***Intermediate survey***

**4.1** ***General***

**4.1.1** Items that are additional to the requirements of the annual survey may be surveyed either at the second or third annual survey or between these surveys.

**4.1.2** The survey extent of cargo and ballast tanks dependent on the age of the ship is specified in 4.2, 4.3 and 4.4 and shown in annex 5.

**4.1.3** For weather decks, an examination as far as applicable of cargo, crude oil washing, bunker, ballast, steam and vent piping systems as well as vent masts and headers should be carried out. If upon examination there is any doubt as to the condition of the piping, the piping may be required to be pressure tested, thickness measured or both.

**4.1.4** Concurrent crediting to both intermediate survey and renewal survey for surveys and thickness measurements of spaces should not be acceptable.

**4.2** ***Oil tankers 5 to 10 years of age***

**4.2.1** The requirements of 4.1.3 apply.

**4.2.2** For tanks used for salt-water ballast, an overall survey of representative tanks selected by the surveyor should be carried out. If the overall survey of salt water ballast tanks reveals no visible structural defects, the examination may be limited to verification that the protective coatings remain in GOOD condition.

**4.2.3** A ballast tank should be examined at subsequent annual intervals where:

- .1 a hard protective coating has not been applied from the time of construction; or
- .2 a soft or semi-hard coating has been applied; or
- .3 substantial corrosion is found within the tank; or
- .4 the hard protective coating is found to be in less than GOOD condition and the hard protective coating is not repaired to the satisfaction of the surveyor.

#### **4.3 Oil tankers 10 to 15 years of age**

**4.3.1** The requirements of the intermediate survey should be to the same extent as the previous renewal survey as required in 2 and 5.1. However, pressure testing of cargo and ballast tanks and the requirements for longitudinal strength evaluation of hull girder as required in 8.1.2 are not required unless deemed necessary by the Administration.

**4.3.2** In application of 4.3.1, the intermediate survey may be commenced at the second annual survey and be progressed during the succeeding year with a view to completion at the third annual survey in lieu of application of 2.1.1.

**4.3.3** In application of 4.3.1., an underwater survey may be considered in lieu of the requirements of 2.2.

#### **4.4 Oil tankers exceeding 15 years of age**

**4.4.1** The requirements of the intermediate survey should be to the same extent as the previous renewal survey as required in 2 and 5.1. However, pressure testing of cargo and ballast tanks and the requirements for longitudinal strength evaluation of hull girder as required in 8.1.2 are not required unless deemed necessary by the Administration.

**4.4.2** In application of 4.4.1, the intermediate survey may be commenced at the second annual survey and be progressed during the succeeding year with a view to completion at the third annual survey in lieu of the application of 2.1.1.

**4.4.3** In application of 4.4.1, a survey in dry dock should be part of the intermediate survey. The overall and close-up surveys and thickness measurements, as applicable, of the lower portions of the cargo tanks and water ballast tanks should be carried out in accordance with the applicable requirements for intermediate surveys, if not already carried out.

**Note:** Lower portions of the cargo and ballast tanks are considered to be the parts below light ballast water line.

### **5 Preparations for survey**

#### **5.1 Survey programme**

**5.1.1** The owner in co-operation with the Administration or organization recognized by the Administration should work out a specific survey programme prior to the commencement of any part of:



- .1 the renewal survey; and
- .2 the intermediate survey for oil tanker over 10 years of age.

The survey programme at intermediate survey may consist of the survey programme at the previous renewal survey supplemented by the condition evaluation report of that renewal survey and later relevant survey reports. The survey programme should be worked out taking into account any amendments to the survey requirements implemented after the last renewal survey carried out. The survey programme should be in a written format based on the information in annex 6A. The survey should not commence until the survey programme has been agreed.

**5.1.1.1** Prior to the development of the survey programme, the survey planning questionnaire should be completed by the owner, based on the information set out in annex 6B, and forwarded to the Administration.

**5.1.2** In developing the survey programme, the following documentation should be collected and consulted with a view to selecting tanks, areas, and structural elements to be examined:

- .1 survey status and basic ship information;
- .2 documentation on board, as described in 6.2 and 6.3;
- .3 main structural plans of cargo and ballast tanks (scantlings drawings), including information regarding use of high-tensile steels (HTS);
- .4 Condition Evaluation Report, according to annex 9;
- .5 relevant previous damage and repair history;
- .6 relevant previous survey and inspection reports from both the recognized organization and the owner;
- .7 cargo and ballast history for the last three years, including carriage of cargo under heated conditions;
- .8 details of the inert gas plant and tank cleaning procedures;
- .9 information and other relevant data regarding conversion or modification of the ship's cargo and ballast tanks since the time of construction;
- .10 description and history of the coating and corrosion protection system (including anodes and previous class notations), if any;
- .11 inspections of the owner's personnel during the last three years with reference to structural deterioration in general, leakages in tank boundaries and piping and condition of the coating and corrosion protection system (including anodes) if any. Guidance for reporting is shown in annex 5;
- .12 information regarding the relevant maintenance level during operation including port State control reports of inspection containing hull related deficiencies, safety management system non-conformities relating to hull maintenance, including the associated corrective action(s); and

- .13 any other information that will help identify suspect areas and critical structural areas.

**5.1.3** The submitted survey programme should account for and comply, as a minimum, with the requirements of 2.6 and annexes 1, 2 and 3 for close-up survey, thickness measurement and tank testing, respectively, and should include relevant information including at least:

- .1 basic ship information and particulars;
- .2 main structural plans of cargo and ballast tanks (scantling drawings), including information regarding use of high tensile steels (HTS);
- .3 plan of tanks;
- .4 list of tanks with information on their use, corrosion prevention system and condition of coating;
- .5 conditions for survey (e.g., information regarding tank cleaning, gas freeing, ventilation, lighting, etc.);
- .6 provisions and methods for access to structures;
- .7 equipment for surveys;
- .8 identification of tanks and areas for close-up survey (see 2.4);
- .9 identification of areas and sections for thickness measurement (see 2.5);
- .10 identification of tanks for tank testing (see 2.6);
- .11 identification of the thickness measurement company;
- .12 damage experience related to the ship in question; and
- .13 critical structural areas and suspect areas, where relevant.

**5.1.4** The Administration will advise the owner of the maximum acceptable structural corrosion diminution levels applicable to the ship.

**5.1.5** Use may also be made of the Guidelines for technical assessment in conjunction with the planning of enhanced surveys for tankers, contained in annex 11. These Guidelines are a recommended tool which may be invoked at the discretion of the Administration, when considered necessary and appropriate, in conjunction with the preparation of the required survey programme.

## **5.2 Conditions for survey**

**5.2.1** The owner should provide the necessary facilities for a safe execution of the survey.

**5.2.1.1** In order to enable the attending surveyors to carry out the survey, provisions for proper and safe access should be agreed between the owner and the Administration.

**5.2.1.2** Details of the means of access should be provided in the survey planning questionnaire.

**5.2.1.3** In cases where the provisions of safety and required access are judged by the attending surveyors not to be adequate, the survey of the spaces involved should not proceed.

**5.2.2** Tanks and spaces should be safe for access. Tanks and spaces should be gas free and properly ventilated. Prior to entering a tank, void or enclosed space, it should be verified that the atmosphere in that space is free from hazardous gas and contains sufficient oxygen.

**5.2.3** In preparation for survey and thickness measurements and to allow for a thorough examination, all spaces should be cleaned including removal from surfaces of all loose accumulated corrosion scale. Spaces should be sufficiently clean and free from water, scale, dirt, oil residues, etc., to reveal corrosion, deformation, fractures, damages or other structural deterioration as well as the condition of the coating. However, those areas of structure whose renewal has already been decided by the owner need only be cleaned and descaled to the extent necessary to determine the limits of the areas to be renewed.

**5.2.4** Sufficient illumination should be provided to reveal corrosion, deformation, fractures, damages or other structural deterioration as well as the condition of the coating.

**5.2.5** Where soft or semi-hard coatings have been applied, safe access should be provided for the surveyor to verify the effectiveness of the coating and to carry out an assessment of the conditions of internal structures which may include spot removal of the coating. When safe access cannot be provided, the soft or semi-hard coating should be removed.

**5.2.6** The surveyor(s) should always be accompanied by at least one responsible person, assigned by the owner, experienced in tank and enclosed spaces inspection. In addition a backup team of at least two experienced persons should be stationed at the hatch opening of the tank or space that is being surveyed. The back-up team should continuously observe the work in the tank or space and should keep lifesaving and evacuation equipment ready for use.

**5.2.7** A communication system should be arranged between the survey party in the tank or space being examined, the responsible officer on deck and, as the case may be, the navigation bridge. The communication arrangements should be maintained throughout the survey.

### **5.3 Access to structures<sup>10</sup>**

**5.3.1** For overall surveys, means should be provided to enable the surveyor to examine the structure in a safe and practical way.

**5.3.2** For close-up surveys, one or more of the following means for access, acceptable to the surveyor, should be provided:

- .1 permanent staging and passages through structures;
- .2 temporary staging and passages through structures;

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<sup>10</sup> Refer to MSC/Circ.686, Guidelines on the means of access to structures for inspection and maintenance of oil tankers and bulk carriers

- .3 lifts and moveable platforms;
- .4 boats or rafts;
- .5 portable ladders; and
- .6 other equivalent means.

#### **5.4 *Equipment for survey***

**5.4.1** Thickness measurement should normally be carried out by means of ultrasonic test equipment. The accuracy of the equipment should be proven to the surveyor as required.

**5.4.2** One or more of the following fracture detection procedures may be required if deemed necessary by the surveyor:

- .1 radiographic equipment;
- .2 ultrasonic equipment;
- .3 magnetic particle equipment;
- .4 dye penetrant; and
- .5 other equivalent means.

**5.4.3** Explosimeter, oxygen-meter, breathing apparatus, lifelines, riding belts with rope and hook and whistles together with instructions and guidance on their use should be made available during the survey. A safety checklist should be provided.

**5.4.4** Adequate and safe lighting should be provided for the safe and efficient conduct of the survey.

**5.4.5** Adequate protective clothing should be made available and used during the survey (e.g., safety helmet, gloves, safety shoes, etc.).

#### **5.5 *Surveys at sea or at anchorage***

**5.5.1** Surveys at sea or at anchorage may be accepted provided the surveyor(s) is given the necessary assistance from the personnel on board. Necessary precautions and procedures for carrying out the survey should be in accordance with 5.1, 5.2, 5.3 and 5.4.

**5.5.2** A communication system should be arranged between the survey party in the tank and the responsible officer on deck. This system should also include the personnel in charge of ballast pump handling if boats or rafts are used.

**5.5.3** Surveys of tanks by means of boats or rafts may only be undertaken with the agreement of the surveyor, who should take into account the safety arrangements provided, including weather forecasting and ship response under foreseeable conditions and provided the expected rise of water within the tank does not exceed 0.25 m.

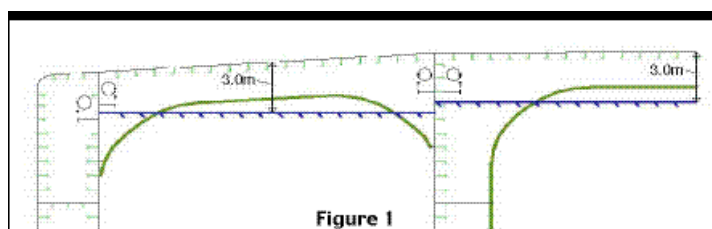
**5.5.4** When rafts or boats are used for close-up surveys, the following conditions should be observed:

- .1 only rough duty, inflatable rafts or boats, having satisfactory residual buoyancy and stability even if one chamber is ruptured, should be used;
- .2 the boat or raft should be tethered to the access ladder and an additional person should be stationed down the access ladder with a clear view of the boat or raft;
- .3 appropriate lifejackets should be available for all participants;
- .4 the surface of water in the tank should be calm (under all foreseeable conditions the expected rise of water within the tank should not exceed 0.25 m) and the water level stationary. On no account should the level of the water be rising while the boat or raft is in use;
- .5 the tank or space must contain clean ballast water only. Even a thin sheen of oil on the water is not acceptable;
- .6 at no time should the water level be allowed to be within 1 m of the deepest under-deck web face flat so that the survey team is not isolated from a direct escape route to the tank hatch. Filling to levels above the deck transverse should only be contemplated if a deck access manhole is fitted and open in the bay being examined, so that an escape route for the survey party is available at all times. Other effective means of escape to the deck may be considered; and
- .7 if the tanks (or spaces) are connected by a common venting system, or inert gas system, the tank in which the boat or raft should be used should be isolated to prevent a transfer of gas from other tanks (or spaces).

**5.5.5** Rafts or boats alone may be allowed for inspection of the under deck areas of tanks or spaces if the depth of the webs is 1.5 m or less.

**5.5.6** If the depth of the webs is more than 1.5 m, rafts or boats alone may be allowed only:

- .1 when the coating of the under-deck structure is in GOOD condition and there is no evidence of wastage; or
- .2 if a permanent means of access is provided in each bay to allow safe entry and exit. This means:
  - .1 access direct from the deck via a vertical ladder with a small platform fitted approximately 2 m below the deck in each bay; or
  - .2 access to deck from a longitudinal permanent platform having ladders to deck in each end of the tank. The platform should, for the full length of the tank, be arranged in level with, or above, the maximum water level needed for rafting of under deck structure. For this purpose, the ullage corresponding to the maximum water level should be assumed not more than 3 m from the deck plate measured at the midspan of deck transverse and in the middle length of the tank (see figure 1).



If neither of the above conditions are met, then staging or other equivalent means should be provided for the survey of the under-deck areas.

**5.5.7** The use of rafts or boats alone in 5.5.5 and 5.5.6 does not preclude the use of boats or rafts to move about within a tank during a survey.

## **5.6 Survey planning meeting**

**5.6.1** Proper preparation and close co-operation between the attending surveyor(s) and the owner's representatives on board prior to and during the survey are an essential part in the safe and efficient conduct of the survey. During the survey on board safety meetings should be held regularly.

**5.6.2** Prior to commencement of any part of the renewal and intermediate survey, a survey planning meeting should be held between the attending surveyor(s), the owner's representative in attendance, the thickness measurement company operator (as applicable) and the master of the ship or an appropriately qualified representative nominated by the master or company for the purpose of ascertaining that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the survey work to be carried out.

**5.6.3** The following is an indicative list of items that should be addressed in the meeting:

- .1 schedule of the vessel (i.e. the voyage, docking and undocking manoeuvres, periods alongside, cargo and ballast operations, etc.);
- .2 provisions and arrangements for thickness measurements (i.e. access, cleaning/descaling, illumination, ventilation, personal safety);
- .3 extent of the thickness measurements;
- .4 acceptance criteria (refer to the list of minimum thicknesses);
- .5 extent of close-up survey and thickness measurement considering the coating condition and suspect areas/areas of substantial corrosion;
- .6 execution of thickness measurements;
- .7 taking representative readings in general and where uneven corrosion/pitting is found;
- .8 mapping of areas of substantial corrosion; and
- .9 communication between attending surveyor(s) the thickness measurement company operator(s) and owner representative(s) concerning findings.

## **6 Documentation on board**

### **6.1 General**

**6.1.1** The owner should obtain, supply and maintain on board the ship, documentation as specified in 6.2 and 6.3 which should be readily available for the surveyor. The condition evaluation report referred to in 6.2 should include a translation into English.

**6.1.2** The documentation should be kept on board for the lifetime of the ship.

### **6.2 Survey report file**

**6.2.1** A survey report file should be a part of the documentation on board consisting of:

- .1 reports of structural surveys (annex 8);
- .2 condition evaluation report (annex 9); and
- .3 thickness measurement reports (annex 10).

**6.2.2** The survey report file should be available also in the owner's and the Administration's offices, or in the office of the organization recognized by the Administration.

### **6.3 Supporting documents**

The following additional documentation should be available on board:

- .1 survey programme as required by 5.1 until such time as the renewal survey or intermediate survey, as applicable, has been completed;
- .2 main structural plans of cargo and ballast tanks;
- .3 previous repair history;
- .4 cargo and ballast history;
- .5 extent of use of inert gas plant and tank cleaning procedures;
- .6 inspections by ship's personnel with reference to:
  - .1 structural deterioration in general;
  - .2 leakage in bulkheads and piping; and
  - .3 condition of coating or corrosion prevention system, if any; and
- .7 any other information that would help to identify critical structural areas and/or suspect areas requiring inspection.

#### **6.4      *Review of documentation on board***

Prior to survey, the surveyor should examine the completeness of the documentation on board and its contents as a basis for the survey.

### **7          *Procedures for thickness measurements***

#### **7.1      *General***

**7.1.1** The required thickness measurements, if not carried out by the recognized organization acting on behalf of the Administration, should be witnessed by a surveyor of the recognized organization. The surveyor should be on board to the extent necessary to control the process.

**7.1.2** The thickness measurement company should be part of the survey planning meeting to be held prior to commencing the survey.

**7.1.3** Thickness measurements of structures in areas where close-up surveys are required should be carried out simultaneously with close-up surveys.

**7.1.4** In all cases the extent of the thickness measurements should be sufficient as to represent the actual average condition.

#### **7.2      *Certification of thickness measurement company***

The thickness measurements should be carried out by a qualified company certified by an organization recognized by the Administration according to principles stated in annex 7.

#### **7.3      *Reporting***

**7.3.1** A thickness measurement report should be prepared and submitted to the Administration. The report should give the location of measurements, the thickness measured as well as corresponding original thickness. Furthermore, the report should give the date when the measurements were carried out, type of measuring equipment, names of personnel and their qualifications and be signed by the operator. The thickness measurement report should follow the principles as specified in the recommended procedures for thickness measurements set out in annex 10.

**7.3.2** The surveyor should review the final thickness measurement report and countersign the cover page.

### **8          *Reporting and evaluation of survey***

#### **8.1      *Evaluation of survey report***

**8.1.1** The data and information on the structural condition of the ship collected during the survey should be evaluated for acceptability and continued structural integrity of the ship.

**8.1.2** In case of oil tankers of 130 m in length and upwards (as defined in the International Convention on Load Lines in force), the ship's longitudinal strength should be evaluated by using the thickness of structural members measured, renewed and reinforced, as appropriate, during the renewal survey of safety construction carried out after the ship reached 10 years of age, in accordance with the criteria for longitudinal strength of the ship's hull girder for oil tankers specified in annex 12.



**8.1.3** The analysis of data should be carried out and endorsed by the Administration or recognized organization authorized by the Administration and the conclusions of the analysis should form a part of the condition evaluation report.

**8.1.4** The final result of the evaluation of the ship's longitudinal strength required in 8.1.2, after renewal or reinforcement work of structural members, if carried out as a result of initial evaluation, should be reported as a part of the condition evaluation report.

## **8.2**     ***Reporting***

**8.2.1** Principles for survey reporting are shown in annex 8.

**8.2.2** When a survey is split between different survey stations, a report should be made for each portion of the survey. A list of items examined and/or tested (pressure testing, thickness measurements, etc.) and an indication of whether the item has been credited, should be made available to the next attending surveyor(s), prior to continuing or completing the survey.

**8.2.3** A condition evaluation report of the survey and results should be issued to the owner as shown in annex 9 and placed on board the ship for reference at future surveys. The condition evaluation report should be endorsed by the Administration or recognized organization authorized by the Administration.

## ANNEX 1

### MINIMUM REQUIREMENTS FOR CLOSE-UP SURVEY AT RENEWAL SURVEY OF DOUBLE-HULL OIL TANKERS

| Age ≤ 5 years<br>Renewal Survey No.<br>1                                                                                     | 5 < Age ≤ 10 years<br>Renewal Survey No.<br>2                                                                                                                              | 10 < Age ≤ 15 years<br>Renewal Survey No.<br>3                                                                                                                                                                                                                                   | Age > 15 years<br>Renewal Survey No.<br>4 and subsequent                                                      |
|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| One web frame (1), in a complete ballast tank (see Note 1)                                                                   | All web frames (1), in a complete ballast tank (see Note 1)<br>The knuckle area and the upper part (5 m approximately) of one web frame in each remaining ballast tank (6) | All web frames (1), in all ballast tanks                                                                                                                                                                                                                                         | As for ships referred to in column 3<br>Additional transverse areas as deemed necessary by the Administration |
| One deck transverse, in a cargo oil tank (2)                                                                                 | One deck transverse, in two cargo oil tanks (2)                                                                                                                            | All web frames (7), including deck transverse and cross ties, if fitted, in a cargo oil tank<br>One web frame (7), including deck transverse and cross ties, if fitted, in each remaining cargo oil tank<br>All transverse bulkheads, in all cargo oil (3) and ballast (4) tanks |                                                                                                               |
| One transverse bulkhead (4), in a complete ballast tank (see Note 1)                                                         | One transverse bulkhead (4), in each complete ballast tank (see Note 1)                                                                                                    |                                                                                                                                                                                                                                                                                  |                                                                                                               |
| One transverse bulkhead (5) in a cargo oil centre tank<br>One transverse bulkhead (5), in a cargo oil wing tank (see Note 2) | One transverse bulkhead (5), in two cargo oil centre tanks<br>One transverse bulkhead (5), in a cargo oil wing tank (see Note 2)                                           |                                                                                                                                                                                                                                                                                  |                                                                                                               |

**Notes:**

**(1), (2), (3), (4), (5), (6) and (7)** are areas to be subjected to close-up surveys and thickness measurements (see appendix 3 of annex 10).

- (1)** Web frame in a ballast tank means vertical web in side tank, hopper web in hopper tank, floor in double-bottom tank and deck transverse in double-deck tank (where fitted), including adjacent structural members. In fore and aft peak tanks web frame means a complete transverse web frame ring including adjacent structural members.
- (2)** Deck transverse, including adjacent deck structural members (or external structure on deck in way of the tank, where applicable).
- (3)** Transverse bulkhead complete in cargo tanks, including girder system, adjacent structural members (such as longitudinal bulkheads) and internal structure of lower and upper stools, where fitted.
- (4)** Transverse bulkhead complete in ballast tanks, including girder system and adjacent structural members, such as longitudinal bulkheads, girders in double-bottom tanks, inner bottom plating, hopper side, connecting brackets.
- (5)** Transverse bulkhead lower part in cargo tank, including girder system, adjacent structural members (such as longitudinal bulkheads) and internal structure of lower stool, where fitted.
- (6)** The knuckle area and the upper part (5 m approximately), including adjacent structural members. Knuckle area is the area of the web frame around the connections of the slope hopper plating to the inner hull bulkhead and the inner bottom plating, up to 2 m from the corners both on the bulkhead and the double bottom.
- (7)** Web frame in a cargo oil tank means deck transverse, longitudinal bulkhead vertical girder and cross ties, where fitted, including adjacent structural members.

**Note 1:** Complete ballast tank: means double-bottom tank plus double-side tank plus double-deck tank, as applicable, even if these tanks are separate.

**Note 2:** Where no centre cargo tanks are fitted (as in the case of centre longitudinal bulkhead), transverse bulkheads in wing tanks should be surveyed.

## ANNEX 2

### MINIMUM REQUIREMENTS FOR THICKNESS MEASUREMENTS AT RENEWAL SURVEY OF DOUBLE-HULL OIL TANKERS

| Age ≤ 5 years<br>1                                                                                                                                   | 5 < Age ≤ 10 years<br>2                                                                                                                              | 10 < Age ≤ 15 years<br>3                                                                                                                             | Age > 15 years<br>4                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| One section of deck plating for the full beam of the ship within the cargo area                                                                      | Within the cargo area:<br>– each deck plate<br>– one transverse section                                                                              | Within the cargo area:<br>– each deck plate<br>– two transverse sections (1)<br>– all wind and water strakes                                         | Within the cargo area:<br>– each deck plate<br>– three transverse sections (1)<br>– each bottom plate                                                |
|                                                                                                                                                      | Selected wind and water strakes outside the cargo area                                                                                               | Selected wind and water strakes outside the cargo area                                                                                               | All wind and water strakes in full length                                                                                                            |
| Measurements, for general assessment and recording of corrosion pattern, of those structural members subject to close-up survey according to annex 1 | Measurements, for general assessment and recording of corrosion pattern, of those structural members subject to close-up survey according to annex 1 | Measurements, for general assessment and recording of corrosion pattern, of those structural members subject to close-up survey according to annex 1 | Measurements, for general assessment and recording of corrosion pattern, of those structural members subject to close-up survey according to annex 1 |
| Suspect areas                                                                                                                                        | Suspect areas                                                                                                                                        | Suspect areas                                                                                                                                        | Suspect areas                                                                                                                                        |
| (1): at least one section should be within 0.5L amidships.                                                                                           |                                                                                                                                                      |                                                                                                                                                      |                                                                                                                                                      |

### ANNEX 3

#### MINIMUM REQUIREMENTS FOR TANK TESTING AT RENEWAL SURVEY OF DOUBLE-HULL OIL TANKERS

| <b>Age ≤ 5 years</b>                                                                                                           | <b>Age &gt; 5 years</b>                                                                                                        | <b>Age &gt; 10 years</b>                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b>                                                                                                                       | <b>2</b>                                                                                                                       | <b>3</b>                                                                                                                       |
| All ballast tank boundaries                                                                                                    | All ballast tank boundaries                                                                                                    | All ballast tank boundaries                                                                                                    |
| Cargo tank boundaries facing ballast tanks, void spaces, pipe tunnels, representative fuel oil tanks, pump-rooms or cofferdams | Cargo tank boundaries facing ballast tanks, void spaces, pipe tunnels, representative fuel oil tanks, pump-rooms or cofferdams | Cargo tank boundaries facing ballast tanks, void spaces, pipe tunnels, representative fuel oil tanks, pump-rooms or cofferdams |
|                                                                                                                                | All cargo tank bulkheads which form the boundaries of segregated cargoes                                                       | All remaining cargo tank bulkheads                                                                                             |

**ANNEX 4/SHEET 1**

**REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT AREAS OF  
SUBSTANTIAL CORROSION OF DOUBLE-HULL OIL TANKERS**

**Renewal survey of double-hull oil tankers**

| <b>Bottom, inner bottom and hopper structure</b>                  |                                                                                                                             |                                                                                                                                                                           |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Structural member</b>                                          | <b>Extent of measurement</b>                                                                                                | <b>Pattern of measurement</b>                                                                                                                                             |
| Bottom, inner bottom and hopper structure plating                 | Minimum of three bays across double-bottom tank, including aft bay<br>Measurements around and under all suction bell mouths | Five-point pattern for each panel between longitudinals and floors                                                                                                        |
| Bottom, inner bottom and hopper structure longitudinals           | Minimum of three longitudinals in each bay where bottom plating measured                                                    | Three measurements in line across flange and three measurements on vertical web                                                                                           |
| Bottom girders, including the watertight ones                     | At fore and aft watertight floors and in centre of tanks                                                                    | Vertical line of single measurements on girder plating with one measurement between each panel stiffener, or a minimum of three measurements                              |
| Bottom floors, including the watertight ones                      | Three floors in bays where bottom plating measured, with measurements at both ends and middle                               | Five-point pattern over 2 m <sup>2</sup> area                                                                                                                             |
| Hopper structure web frame ring                                   | Three floors in bays where bottom plating measured                                                                          | Five-point pattern over 1 m <sup>2</sup> of plating. Single measurements on flange                                                                                        |
| Hopper structure transverse watertight bulkhead or swash bulkhead | – lower ⅓ of bulkhead                                                                                                       | Five-point pattern over 1 m <sup>2</sup> of plating                                                                                                                       |
|                                                                   | – upper ⅔ of bulkhead                                                                                                       | Five-point pattern over 2 m <sup>2</sup> of plating                                                                                                                       |
|                                                                   | – stiffeners (minimum of three)                                                                                             | For web, five-point pattern over span (two measurements across web at each end and one at centre of span). For flange, single measurements at each end and centre of span |
| Panel stiffening                                                  | Where applicable                                                                                                            | Single measurements                                                                                                                                                       |

**ANNEX 4/SHEET 2**

**REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT AREAS OF  
SUBSTANTIAL CORROSION OF DOUBLE-HULL OIL TANKERS**

**Renewal survey of double-hull oil tankers within the cargo area length**

| <b>Deck structure</b>                                                     |                                                                                  |                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Structural member</b>                                                  | <b>Extent of measurement</b>                                                     | <b>Pattern of measurement</b>                                                                                                                                                                                             |
| Deck plating                                                              | Two transverse bands across tank                                                 | Minimum of three measurements per plate per band                                                                                                                                                                          |
| Deck longitudinals                                                        | Every third longitudinal in each of two bands with a minimum of one longitudinal | Three measurements in line vertically on webs and two measurements on flange (if fitted)                                                                                                                                  |
| Deck girders and brackets (usually in cargo tanks only)                   | At fore and aft transverse bulkhead, bracket toes and in centre of tanks         | Vertical line of single measurements on web plating with one measurement between each panel stiffener, or a minimum of three measurements. Two measurements across flange. Five-point pattern on girder/bulkhead brackets |
| Deck transverse webs                                                      | Minimum of two webs, with measurements at both ends and middle of span           | Five-point pattern over 1 m <sup>2</sup> area. Single measurements on flange                                                                                                                                              |
| Vertical web and transverse bulkhead in wing ballast tank (2 m from deck) | Minimum of two webs, and both transverse bulkheads                               | Five-point pattern over 1 m <sup>2</sup> area                                                                                                                                                                             |
| Panel stiffening                                                          | Where applicable                                                                 | Single measurements                                                                                                                                                                                                       |

**ANNEX 4/SHEET 3**

**REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT AREAS OF  
SUBSTANTIAL CORROSION OF DOUBLE-HULL OIL TANKERS**

**Renewal survey of double-hull oil tankers within the cargo area length**

| <b>Structure in wing ballast tanks</b>                                                                                          |                                                                                                                                                                |                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Structural member</b>                                                                                                        | <b>Extent of measurement</b>                                                                                                                                   | <b>Pattern of measurement</b>                                                                                              |
| Side shell and longitudinal bulkhead plating:<br>– upper strake and strakes in way of horizontal girders<br>– all other strakes | Plating between each pair of longitudinals in a minimum of three bays (along the tank)<br>Plating between every third pair of longitudinals in same three bays | Single measurement<br>Single measurement                                                                                   |
| Side shell and longitudinal bulkhead longitudinals on:<br>– upper strake<br>– all other strakes                                 | Each longitudinal in same three bays<br>Every third longitudinal in same three bays                                                                            | Three measurements across web and one measurement on flange<br>Three measurements across web and one measurement on flange |
| Longitudinals brackets                                                                                                          | Minimum of three at top, middle and bottom of tank in same three bays                                                                                          | Five-point pattern over area of bracket                                                                                    |
| Vertical web and transverse bulkheads (excluding deckhead area):<br>– strakes in way of horizontal girders<br>– other strakes   | Minimum of two webs and both transverse bulkheads<br>Minimum of two webs and both transverse bulkheads                                                         | Five-point pattern over approximately 2 m <sup>2</sup> area<br>Two measurements between each pair of vertical stiffeners   |
| Horizontal girders                                                                                                              | Plating on each girder in a minimum of three bays                                                                                                              | Two measurements between each pair of longitudinal girder stiffeners                                                       |
| Panel stiffening                                                                                                                | Where applicable                                                                                                                                               | Single measurements                                                                                                        |



**ANNEX 4/SHEET 4**

**REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT AREAS OF  
SUBSTANTIAL CORROSION OF DOUBLE-HULL OIL TANKERS**

**Renewal survey of double-hull oil tankers within the cargo area length**

| <b>Longitudinal bulkheads in cargo tanks</b>                                                        |                                                                                                   |                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Structural member</b>                                                                            | <b>Extent of measurement</b>                                                                      | <b>Pattern of measurement</b>                                                                                                        |
| Deckhead and bottom strakes, and strakes in way of the horizontal stringers of transverse bulkheads | Plating between each pair of longitudinals in a minimum of three bays                             | Single measurement                                                                                                                   |
| All other strakes                                                                                   | Plating between every third pair of longitudinals in same three bays                              | Single measurement                                                                                                                   |
| Longitudinals on deckhead and bottom strakes                                                        | Each longitudinal in same three bays                                                              | Three measurements across web and one measurement on flange                                                                          |
| All other longitudinals                                                                             | Every third longitudinal in same three bays                                                       | Three measurements across web and one measurement on flange                                                                          |
| Longitudinals brackets                                                                              | Minimum of three at top, middle and bottom of tank in same three bays                             | Five-point pattern over area of bracket                                                                                              |
| Web frames and cross ties                                                                           | Three webs with minimum of three locations on each web, including in way of cross tie connections | Five-point pattern over approximately 2 m <sup>2</sup> area of webs, plus single measurements on flanges of web frame and cross ties |
| Lower end brackets (opposite side of web frame)                                                     | Minimum of three brackets                                                                         | Five-point pattern over approximately 2 m <sup>2</sup> area of brackets, plus single measurements on bracket flanges                 |

**ANNEX 4/SHEET 5**

**REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT AREAS OF  
SUBSTANTIAL CORROSION OF DOUBLE-HULL OIL TANKERS**

**Renewal survey of double-hull oil tankers within the cargo area length**

| <b>Transverse watertight and swash bulkheads in cargo tanks</b>         |                                                                                                                                                    |                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Structural member</b>                                                | <b>Extent of measurement</b>                                                                                                                       | <b>Pattern of measurement</b>                                                                                                                                                                                                     |
| Upper and lower stool, where fitted                                     | Transverse band within 25 mm of welded connection to inner bottom/deck plating<br>Transverse band within 25 mm of welded connection to shelf plate | Five-point pattern between stiffeners over 1 m length                                                                                                                                                                             |
| Deckhead and bottom strakes, and strakes in way of horizontal stringers | Plating between pair of stiffeners at three locations: approximately $\frac{1}{4}$ , $\frac{1}{2}$ and $\frac{3}{4}$ width of tank                 | Five-point pattern between stiffeners over 1 m length                                                                                                                                                                             |
| All other strakes                                                       | Plating between pair of stiffeners at middle location                                                                                              | Single measurement                                                                                                                                                                                                                |
| Strakes in corrugated bulkheads                                         | Plating for each change of scantling at centre of panel and at flange of fabricated connection                                                     | Five-point pattern over approximately 1 m <sup>2</sup> of plating                                                                                                                                                                 |
| Stiffeners                                                              | Minimum of three typical stiffeners                                                                                                                | For web, five-point pattern over span between bracket connections (two measurements across web at each bracket connection and one at centre of span)<br>For flange, single measurements at each bracket toe and at centre of span |
| Brackets                                                                | Minimum of three at top, middle and bottom of tank                                                                                                 | Five-point pattern over area of bracket                                                                                                                                                                                           |
| Horizontal stringers                                                    | All stringers with measurements at both ends and middle                                                                                            | Five-point pattern over 1 m <sup>2</sup> area, plus single measurements near bracket toes and on flanges                                                                                                                          |

## ANNEX 5

### OWNER'S INSPECTION REPORT

#### Structural condition

|                                             |        |         |           |         |         |                     |                 |
|---------------------------------------------|--------|---------|-----------|---------|---------|---------------------|-----------------|
| Ship's name: .....                          |        |         |           |         |         |                     |                 |
| For tank no: .....                          |        |         |           |         |         |                     |                 |
| Grade of steel: deck: ..... side: .....     |        |         |           |         |         |                     |                 |
| bottom: ..... longitudinal bulkhead: .....  |        |         |           |         |         |                     |                 |
| Elements                                    | Cracks | Buckles | Corrosion | Coating | Pitting | Modification/repair | Other condition |
| Deck:                                       |        |         |           |         |         |                     |                 |
| Bottom:                                     |        |         |           |         |         |                     |                 |
| Side:                                       |        |         |           |         |         |                     |                 |
| Longitudinal bulkhead:                      |        |         |           |         |         |                     |                 |
| Transverse bulkhead:                        |        |         |           |         |         |                     |                 |
| Repairs carried out due to:                 |        |         |           |         |         |                     |                 |
| Thickness measurements carried out (dates): |        |         |           |         |         |                     |                 |
| Results in general:                         |        |         |           |         |         |                     |                 |
| Overdue surveys:                            |        |         |           |         |         |                     |                 |
| Outstanding conditions of class:            |        |         |           |         |         |                     |                 |
| Comments:                                   |        |         |           |         |         |                     |                 |
| Date of inspection: .....                   |        |         |           |         |         |                     |                 |
| Inspected by: .....                         |        |         |           |         |         |                     |                 |
| Signature: .....                            |        |         |           |         |         |                     |                 |

## ANNEX 6A

### SURVEY PROGRAMME

#### Basic information and particulars

|                                    |
|------------------------------------|
| Name of ship:                      |
| IMO number:                        |
| Flag State:                        |
| Port of registry:                  |
| Gross tonnage:                     |
| Deadweight (metric tonnes):        |
| Length between perpendiculars (m): |
| Shipbuilder:                       |
| Hull number:                       |
| Recognized organization (RO):      |
| RO ship identity:                  |
| Date of delivery of the ship:      |
| Owner:                             |
| Thickness measurement company:     |

#### **1 Preamble**

##### **1.1 Scope**

**1.1.1** The present survey programme covers the minimum extent of overall surveys, close-up surveys, thickness measurements and pressure testing within the cargo area, ballast tanks, including fore and aft peak tanks, required by the Code.

**1.1.2** The arrangements and safety aspects of the survey should be acceptable to the attending surveyor(s).

##### **1.2 Documentation**

All documents used in the development of the survey programme should be available on board during the survey as required by section 6.

#### **2 Arrangement of tanks and spaces**

This section of the survey programme should provide information (either in the form of plans or text) on the arrangement of tanks and spaces that fall within the scope of the survey.

#### **3 List of tanks and spaces with information on their use, extent of coatings and corrosion prevention system**

This section of the survey programme should indicate any changes relating to (and should update) the information on the use of the tanks of the ship, the extent of coatings and the corrosion protective system provided in the survey planning questionnaire.

#### **4 Conditions for survey**

This section of the survey programme should provide information on the conditions for survey, e.g., information regarding cargo hold and tank cleaning, gas freeing, ventilation, lighting, etc.

#### **5 Provisions and method of access to structures**

This section of the survey programme should indicate any changes relating to (and should update) the information on the provisions and methods of access to structures provided in the survey planning questionnaire.

#### **6 List of equipment for survey**

This section of the survey programme should identify and list the equipment that will be made available for carrying out the survey and the required thickness measurements.

#### **7 Survey requirements**

##### **7.1 Overall survey**

This section of the survey programme should identify and list the spaces that should undergo an overall survey for the ship in accordance with 2.4.1.

##### **7.2 Close-up survey**

This section of the survey programme should identify and list the hull structures that should undergo a close-up survey for the ship in accordance with 2.4.2.

#### **8 Identification of tanks for tank testing**

This section of the survey programme should identify and list the tanks that should undergo tank testing for the ship in accordance with 2.6.

#### **9 Identification of areas and sections for thickness measurements**

This section of the survey programme should identify and list the areas and sections where thickness measurements should be taken in accordance with 2.5.1.

#### **10 Minimum thickness of hull structures**

This section of the survey programme should specify the minimum thickness for hull structures of the ship that are subject to the Code (indicate either (a) or preferably (b), if such information is available):

- (a) Determined from the attached wastage allowance table and the original thickness to the hull structure plans of the ship;
- (b) Given in the following table(s):

| Area or location                                   | Original as-built thickness (mm) | Minimum thickness (mm) | Substantial corrosion thickness (mm) |
|----------------------------------------------------|----------------------------------|------------------------|--------------------------------------|
| <b>Deck</b>                                        |                                  |                        |                                      |
| Plating                                            |                                  |                        |                                      |
| Longitudinals                                      |                                  |                        |                                      |
| Longitudinal girders                               |                                  |                        |                                      |
| <b>Bottom</b>                                      |                                  |                        |                                      |
| Plating                                            |                                  |                        |                                      |
| Longitudinals                                      |                                  |                        |                                      |
| Longitudinal girders                               |                                  |                        |                                      |
| <b>Ship side</b>                                   |                                  |                        |                                      |
| Plating                                            |                                  |                        |                                      |
| Longitudinals                                      |                                  |                        |                                      |
| Longitudinal girders                               |                                  |                        |                                      |
| <b>Longitudinal bulkhead</b>                       |                                  |                        |                                      |
| Plating                                            |                                  |                        |                                      |
| Longitudinals                                      |                                  |                        |                                      |
| Longitudinal girders                               |                                  |                        |                                      |
| <b>Inner bottom</b>                                |                                  |                        |                                      |
| Plating                                            |                                  |                        |                                      |
| Longitudinals                                      |                                  |                        |                                      |
| Longitudinal girders                               |                                  |                        |                                      |
| <b>Transverse bulkheads</b>                        |                                  |                        |                                      |
| Plating                                            |                                  |                        |                                      |
| Stiffeners                                         |                                  |                        |                                      |
| <b>Transverse web frames, floors and stringers</b> |                                  |                        |                                      |
| Plating                                            |                                  |                        |                                      |
| Flanges                                            |                                  |                        |                                      |
| Stiffeners                                         |                                  |                        |                                      |
| Cross ties                                         |                                  |                        |                                      |
| Flanges                                            |                                  |                        |                                      |
| Webs                                               |                                  |                        |                                      |

**Note:** The wastage allowance tables should be attached to the survey programme.

## 11 Thickness measurement company

This section of the survey programme should identify changes, if any, relating to the information on the thickness measurement company provided in the survey planning questionnaire.

## 12 Damage experience related to the ship

This section of the survey programme should, using the tables provided below, provide details of the hull damages for at least the last three years in way of the cargo and ballast tanks and void spaces within the cargo area. These damages are subject to survey.

**Hull damages sorted by location for this ship**

| <b>Tank or space number or area</b> | <b>Possible cause, if known</b> | <b>Description of the damages</b> | <b>Location</b> | <b>Repair</b> | <b>Date of repair</b> |
|-------------------------------------|---------------------------------|-----------------------------------|-----------------|---------------|-----------------------|
|                                     |                                 |                                   |                 |               |                       |
|                                     |                                 |                                   |                 |               |                       |
|                                     |                                 |                                   |                 |               |                       |
|                                     |                                 |                                   |                 |               |                       |
|                                     |                                 |                                   |                 |               |                       |
|                                     |                                 |                                   |                 |               |                       |
|                                     |                                 |                                   |                 |               |                       |
|                                     |                                 |                                   |                 |               |                       |

**Hull damages for sister or similar ships  
(if available) in the case of design related damage**

| <b>Tank or space number or area</b> | <b>Possible cause, if known</b> | <b>Description of the damages</b> | <b>Location</b> | <b>Repair</b> | <b>Date of repair</b> |
|-------------------------------------|---------------------------------|-----------------------------------|-----------------|---------------|-----------------------|
|                                     |                                 |                                   |                 |               |                       |
|                                     |                                 |                                   |                 |               |                       |
|                                     |                                 |                                   |                 |               |                       |
|                                     |                                 |                                   |                 |               |                       |
|                                     |                                 |                                   |                 |               |                       |
|                                     |                                 |                                   |                 |               |                       |
|                                     |                                 |                                   |                 |               |                       |
|                                     |                                 |                                   |                 |               |                       |

**13 Areas identified with substantial corrosion from previous surveys**

This section of the survey programme should identify and list the areas of substantial corrosion from previous surveys.

**14 Critical structural areas and suspect areas**

This section of the survey programme should identify and list the critical structural areas and the suspect areas, if such information is available.

**15 Other relevant comments and information**

This section of the survey programme should provide any other comments and information relevant to the survey.

## **Appendices**

### **Appendix 1 – List of plans**

Paragraph 5.1.3.2 requires that main structural plans of cargo and ballast tanks (scantling drawings), including information on regarding use of high tensile steel (HTS), to be available. This appendix of the survey programme should identify and list the main structural plans which form part of the survey programme.

### **Appendix 2 – Survey planning questionnaire**

The survey planning questionnaire (annex 6B), which has been submitted by the owner, should be appended to the survey programme.

### **Appendix 3 – Other documentation**

This part of the survey programme should identify and list any other documentation that forms part of the plan.

Prepared by the owner in co-operation with the Administration for compliance with 5.1.3.

Date:

*(name and signature of authorized owner's representative)*

Date:

*(name and signature of authorized representative of the Administration)*



## **ANNEX 6B**

### **SURVEY PLANNING QUESTIONNAIRE**

The following information will enable the owner in co-operation with the Administration to develop a survey programme complying with the requirements of the Code. It is essential that the owner provides, when completing the present questionnaire, up-to-date information. The present questionnaire, when completed, should provide all information and material required by the resolution.

#### **Particulars**

Ship's name:

IMO number:

Flag State:

Port of registry:

Owner:

Recognized organization (RO):

Gross tonnage:

Deadweight (metric tonnes):

Date of delivery:

#### **Information on access provision for close-up surveys and thickness measurement**

The owner should indicate, in the table below, the means of access to the structures subject to close-up survey and thickness measurement. A close-up survey is an examination where the details of structural components are within the close visual inspection range of the attending surveyor, i.e. normally within reach of hand.

| <b>Tank No.</b> | <b>Structure</b>  | <b>C (Cargo)/B (Ballast)</b> | <b>Temporary staging</b> | <b>Rafts</b> | <b>Ladders</b> | <b>Direct access</b> | <b>Other means (please specify)</b> |
|-----------------|-------------------|------------------------------|--------------------------|--------------|----------------|----------------------|-------------------------------------|
| F.P.            | Fore peak         |                              |                          |              |                |                      |                                     |
| A.P.            | Aft peak          |                              |                          |              |                |                      |                                     |
| Wing tanks      | Underdeck         |                              |                          |              |                |                      |                                     |
|                 | Side shell        |                              |                          |              |                |                      |                                     |
|                 | Bottom transverse |                              |                          |              |                |                      |                                     |
|                 | Longitudinal      |                              |                          |              |                |                      |                                     |
|                 | Transverse        |                              |                          |              |                |                      |                                     |
| Centre tanks    | Under deck        |                              |                          |              |                |                      |                                     |
|                 | Bottom transverse |                              |                          |              |                |                      |                                     |
|                 | Transverse        |                              |                          |              |                |                      |                                     |

**History of cargo with H<sub>2</sub>S content or heated cargo for the last three years together with indication as to whether cargo was heated and, where available, Material Safety Data Sheets (MSDS)<sup>11</sup>**

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |

### Owner's inspections

Using a format similar to that of the table below (which is given as an example), the owner should provide details of the results of their inspections for the last three years on all cargo and ballast tanks and void spaces within the cargo area, including peak tanks.

| Tank No.             | Corrosion protection (1) | Coating extent (2) | Coating condition (3) | Structural deterioration (4) | Tank damage history (5) |
|----------------------|--------------------------|--------------------|-----------------------|------------------------------|-------------------------|
| Cargo centre tanks   |                          |                    |                       |                              |                         |
| Cargo wing tanks     |                          |                    |                       |                              |                         |
| Slop                 |                          |                    |                       |                              |                         |
| Ballast tanks        |                          |                    |                       |                              |                         |
| Aft peak             |                          |                    |                       |                              |                         |
| Fore peak            |                          |                    |                       |                              |                         |
| Miscellaneous spaces |                          |                    |                       |                              |                         |

### Note:

Indicate tanks which are used for oil/ballast.

- 1) HC = hard coating; SC = soft coating;  
A = anodes; NP = no protection
- 2) U = upper part; M = middle part;  
L = lower part; C = complete
- 3) G = good; F = fair; P = poor;  
RC = recoated (during the last three years)
- 4) N = no findings recorded; Y = findings recorded,  
description of findings should be attached to this questionnaire
- 5) DR = Damage & Repair; L = Leakages;  
CV = Conversion (description to be attached to this questionnaire)

Name of owner's representative:

.....

Signature:.....

Date: .....

<sup>11</sup> Refer to resolution MSC.150(77) on Recommendation for material safety data sheets for MARPOL Annex I cargoes and marine fuel oils.

**Reports of port State control inspections**

|                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| List the reports of port State control inspections containing hull structural related deficiencies and relevant information on rectification of the deficiencies: |
|                                                                                                                                                                   |
|                                                                                                                                                                   |
|                                                                                                                                                                   |
|                                                                                                                                                                   |
|                                                                                                                                                                   |
|                                                                                                                                                                   |

**Safety management system**

|                                                                                                 |
|-------------------------------------------------------------------------------------------------|
| List non-conformities related to hull maintenance, including the associated corrective actions: |
|                                                                                                 |
|                                                                                                 |
|                                                                                                 |
|                                                                                                 |
|                                                                                                 |
|                                                                                                 |

**Name and address of the approved thickness measurement company:**

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |

## **ANNEX 7**

### **PROCEDURES FOR CERTIFICATION OF A COMPANY ENGAGED IN THICKNESS MEASUREMENT OF HULL STRUCTURES**

#### **1 Application**

This guidance applies for certification of the company which intends to engage in the thickness measurement of hull structures of ships.

#### **2 Procedures for certification**

##### ***Submission of documents***

**2.1** The following documents should be submitted to an organization recognized by the Administration for approval:

- .1** outline of the company, e.g., organization and management structure;
- .2** experience of the company in thickness measurement of hull structures of ships;
- .3** technicians' careers, i.e. experience of technicians as thickness measurement operators, technical knowledge and experience of hull structure, etc. Operators should be qualified according to a recognized industrial non-destructive test (NDT) standard;
- .4** equipment used for thickness measurement such as ultrasonic testing machines and their maintenance/calibration procedures;
- .5** a guide for thickness measurement operators;
- .6** training programmes for technicians for thickness measurement; and
- .7** measurement record format in accordance with recommended procedures for thickness measurements (see annex 10). Recommended procedures for thickness measurements of double-hull oil tankers are contained in annex 2.

##### ***Auditing of the company***

**2.2** Upon satisfactory review of the documents submitted, the company should be audited in order to ascertain that the company is duly organized and managed in accordance with the documents submitted and is capable of conducting thickness measurement of the hull structure of ships.

**2.3** Certification is conditional upon an on board demonstration of thickness measurement as well as satisfactory reporting.

### **3 Certification**

**3.1** Upon satisfactory results of both the audit of the company referred to in 2.2 and the demonstration tests referred to in 2.3, the Administration or organization recognized by the Administration should issue a certificate of approval as well as a notice to the effect that the thickness measurement operation system of the company has been certified.

**3.2** Renewal/endorsement of the certificate should be carried out at intervals not exceeding three years by verification that original conditions are maintained.

### **4 Report of any alteration to the certified thickness measurement operation system**

In case any alteration to the certified thickness measurement operation system of the company is made, such alteration should be immediately reported to the organization recognized by the Administration. A re-audit should be carried out where deemed necessary by the organization recognized by the Administration.

### **5 Withdrawal of certification**

The certification may be withdrawn in the following cases:

- .1** where the measurements were improperly carried out or the results were improperly reported;
- .2** where the surveyor found any deficiencies in the approved thickness measurement operation system of the company; and
- .3** where the company failed to report any alteration referred to in 4 to the organization recognized by the Administration as required.

## **ANNEX 8**

### **SURVEY REPORTING PRINCIPLES**

As a principle, for oil tankers subject to the Code, the surveyor should include the following contents in his report for survey of hull structure and piping systems, as relevant for the survey.

#### **1 General**

**1.1** A survey report should be generated in the following cases:

- .1** in connection with commencement, continuation and/or completion of periodical hull surveys, i.e. annual, intermediate and renewal surveys, as relevant;
- .2** when structural damages/defects have been found;
- .3** when repairs, renewals or modifications have been carried out; and
- .4** when condition of class (recommendation) has been imposed or deleted.

**1.2** The reporting should provide:

- .1** evidence that prescribed surveys have been carried out in accordance with applicable requirements;
- .2** documentation of surveys carried out with findings, repairs carried out and condition of class (recommendation) imposed or deleted;
- .3** survey records, including actions taken, which should form an auditable documentary trail. Survey reports should be kept in the survey report file required to be on board;
- .4** information for planning of future surveys; and
- .5** information which may be used as input for maintenance of classification rules and instructions.

**1.3** When a survey is split between different survey stations, a report should be made for each portion of the survey. A list of items surveyed, relevant findings and an indication of whether the item has been credited, should be made available to the next attending surveyor, prior to continuing or completing the survey. Thickness measurement and tank testing carried out should also be listed for the next surveyor.

#### **2 Extent of the survey**

**2.1** Identification of compartments where an overall survey has been carried out.

**2.2** Identification of locations, in each tank, where a close-up survey has been carried out, together with information of the means of access used.

**2.3** Identification of locations, in each tank, where thickness measurement has been carried out.

**Note:** As a minimum, the identification of location of close-up survey and thickness measurement should include a confirmation with description of individual structural members corresponding to the extent of requirements stipulated in this part of annex B based on type of periodical survey and the ship's age.

Where only partial survey is required, i.e. one web frame ring/one deck transverse, the identification should include location within each ballast tank and cargo hold by reference to frame numbers.

**2.4** For areas in tanks where protective coating is found to be in good condition and the extent of close-up survey and/or thickness measurement has been specially considered, structures subject to special consideration should be identified.

**2.5** Identification of tanks subject to tank testing.

**2.6** Identification of piping systems on deck, including crude oil washing (COW) piping, and ballast piping within cargo and ballast tanks, pipe tunnels, cofferdams and void spaces where:

- .1 examination including internal examination of piping with valves and fittings and thickness measurement, as relevant, has been carried out; and
- .2 operational test to working pressure has been carried out.

### **3 Result of survey**

**3.1** Type, extent and condition of protective coating in each tank, as relevant (rated GOOD, FAIR or POOR), including identification of tanks fitted with anodes.

**3.2** Structural condition of each compartment with information on the following, as relevant:

- .1 Identification of findings, such as:
  - .1 corrosion with description of location, type and extent;
  - .2 areas with substantial corrosion;
  - .3 cracks/fractures with description of location and extent;
  - .4 buckling with description of location and extent; and
  - .5 indents with description of location and extent.
- .2 Identification of compartments where no structural damage/defects are found. The report may be supplemented by sketches/photographs.
- .3 Thickness measurement report should be verified and signed by the surveyor controlling the measurements on board.



- .4 Evaluation result of longitudinal strength of the hull girder of oil tankers of 130 m in length and upwards and over 10 years of age. The following data should be included, as relevant:
  - .1 measured and as-built transverse sectional areas of deck and bottom flanges;
  - .2 diminution of transverse sectional areas of deck and bottom flanges; and
  - .3 details of renewals or reinforcements carried out, as relevant (see 4.2).

#### **4 Actions taken with respect to findings**

**4.1** Whenever the attending surveyor is of the opinion that repairs are required, each item to be repaired should be identified in a numbered list. Whenever repairs are carried out, details of the repairs effected should be reported by making specific reference to relevant items in the numbered list.

**4.2** Repairs carried out should be reported with identification of:

- .1 compartment;
- .2 structural member;
- .3 repair method (i.e. renewal or modification), including:
  - .1 steel grades and scantlings (if different from the original); and
  - .2 sketches/photographs, as appropriate;
- .4 repair extent; and
- .5 non-destructive tests (NDT).

**4.3** For repairs not completed at the time of survey, condition of class/recommendation should be imposed with a specific time limit for the repairs. In order to provide correct and proper information to the surveyor attending for survey of the repairs, condition of class/recommendation should be sufficiently detailed with identification of each item to be repaired. For identification of extensive repairs, reference may be given to the survey report.

## ANNEX 9

### CONDITION EVALUATION REPORT

#### Issued upon completion of renewal survey

##### *General particulars*

|                             |                                                   |
|-----------------------------|---------------------------------------------------|
| Ship's name:                | Class/Administration identity number:             |
|                             | Previous class/Administration identity number(s): |
|                             | IMO number:                                       |
| Port of registry:           | National flag:                                    |
|                             | Previous national flag(s):                        |
| Deadweight (metric tonnes): | Gross tonnage:                                    |
|                             | National:                                         |
|                             | ITC (1969):                                       |
| Date of build:              | Classification notation:                          |
| Date of major conversion:   |                                                   |
| Type of conversion:         | Owner:                                            |
|                             | Previous owner(s):                                |

**1** The survey reports and documents listed below have been reviewed by the undersigned and found to be satisfactory.

**2** A summary of the survey is attached herewith on sheet 2.

**3** The renewal survey has been completed in accordance with the present Code on (date) .....

|                                          |                   |       |
|------------------------------------------|-------------------|-------|
| Condition evaluation report completed by | Name<br>Signature | Title |
| Office                                   | Date              |       |
| Condition evaluation report verified by  | Name<br>Signature | Title |
| Office                                   | Date              |       |

Attached reports and documents:

- 1)
- 2)
- 3)
- 4)
- 5)
- 6)

### **Contents of condition evaluation report**

- |         |                                                                                                                                    |                                                                                                                                                                                                                     |
|---------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Part 1  | – General particulars:                                                                                                             | – See front page                                                                                                                                                                                                    |
| Part 2  | – Report review:                                                                                                                   | – Where and how survey was done                                                                                                                                                                                     |
| Part 3  | – Close-up survey:                                                                                                                 | – Extent (which tanks)                                                                                                                                                                                              |
| Part 4  | – Cargo and ballast piping system:                                                                                                 | – Examined<br>– Operationally tested                                                                                                                                                                                |
| Part 5  | – Thickness measurements:                                                                                                          | – Reference to thickness measurement report<br>– Summary of where measured<br>– Separate form indicating the spaces with substantial corrosion, and corresponding:<br>– thickness diminution<br>– corrosion pattern |
| Part 6  | – Tank corrosion prevention system:                                                                                                | – Separate form indicating:<br>– location of coating/anodes<br>– condition of coating (if applicable)                                                                                                               |
| Part 7  | – Repairs:                                                                                                                         | – Identification of tanks/areas                                                                                                                                                                                     |
| Part 8  | – Condition of class/flag State requirements:                                                                                      |                                                                                                                                                                                                                     |
| Part 9  | – Memoranda:                                                                                                                       | – Acceptable defects<br>– Any points of attention for future surveys, e.g., for suspect areas<br>– Extended annual/intermediate survey due to coating breakdown                                                     |
| Part 10 | – Evaluation results of the ship's longitudinal strength (for oil tankers of 130 m in length and upwards and over 10 years of age) |                                                                                                                                                                                                                     |
| Part 11 | – Conclusion:                                                                                                                      | – Statement on evaluation/ verification of survey report                                                                                                                                                            |

### **Extract of thickness measurements**

Reference is made to the thickness measurement report:

| <b>Position of substantially corroded tanks/areas<sup>1</sup> or areas with deep pitting<sup>3</sup></b> | <b>Thickness diminution [%]</b> | <b>Corrosion pattern<sup>2</sup></b> | <b>Remarks: (e.g., reference to attached sketches)</b> |
|----------------------------------------------------------------------------------------------------------|---------------------------------|--------------------------------------|--------------------------------------------------------|
|                                                                                                          |                                 |                                      |                                                        |

### **Notes:**

- 1 Substantial corrosion, i.e. 75%–100% of acceptable margins wasted.
- 2 P = Pitting  
C = Corrosion in general
- 3 Any bottom plating with a pitting intensity of 20% or more, with wastage in the substantial corrosion range or having an average depth of pitting of 1/3 or more of actual plate thickness should be noted.

**Tank corrosion prevention system**

| Tank nos. <sup>1</sup> | Tank corrosion prevention system <sup>2</sup> | Coating condition <sup>3</sup> | Remarks |
|------------------------|-----------------------------------------------|--------------------------------|---------|
|                        |                                               |                                |         |

**Notes:**

1 All segregated ballast tanks and combined cargo/ballast tanks should be listed.

2 C = Coating  
NP = No protection

3 Coating condition according to the following standard:

GOOD condition with only minor spot rusting.

FAIR condition with local breakdown of coating at edges of stiffeners and weld connections and/or light rusting over 20% or more of areas under consideration, but less than as defined for POOR condition.

POOR condition with general breakdown of coating over 20% or more of areas or hard scale at 10% or more of areas under consideration.

If coating condition less than GOOD is given, extended annual surveys should be introduced. This should be noted in part 9 of the Contents of condition evaluation report.

**Evaluation result of longitudinal strength of the hull girder of oil tankers of 130 m in length and upwards and of over 10 years of age**

*(of sections 1, 2 and 3 below, only one applicable section should be completed)*

1 This section applies to ships regardless of the date of construction: Transverse sectional areas of deck flange (deck plating and deck longitudinals) and bottom flange (bottom shell plating and bottom longitudinals) of the ship's hull girder have been calculated by using the thickness measured, renewed or reinforced, as appropriate, during the renewal survey of the Cargo Ship Safety Construction Certificate or the Cargo Ship Safety Certificate (SC renewal survey) most recently conducted after the ship reached 10 years of age, and found that the diminution of the transverse sectional area does not exceed 10% of the as-built area, as shown in the following table:

**Table 1 – Transverse sectional area of hull girder flange**

|                      |               | Measured        | As-built        | Diminution          |
|----------------------|---------------|-----------------|-----------------|---------------------|
| Transverse section 1 | Deck flange   | cm <sup>2</sup> | cm <sup>2</sup> | cm <sup>2</sup> (%) |
|                      | Bottom flange | cm <sup>2</sup> | cm <sup>2</sup> | cm <sup>2</sup> (%) |
| Transverse section 2 | Deck flange   | cm <sup>2</sup> | cm <sup>2</sup> | cm <sup>2</sup> (%) |
|                      | Bottom flange | cm <sup>2</sup> | cm <sup>2</sup> | cm <sup>2</sup> (%) |
| Transverse section 3 | Deck flange   | cm <sup>2</sup> | cm <sup>2</sup> | cm <sup>2</sup> (%) |
|                      | Bottom flange | cm <sup>2</sup> | cm <sup>2</sup> | cm <sup>2</sup> (%) |

**2** This section applies to ships constructed on or after 1 July 2002: Section moduli of transverse section of the ship's hull girder have been calculated by using the thickness of structural members measured, renewed or reinforced, as appropriate, during the SC renewal survey most recently conducted after the ship reached 10 years of age in accordance with the provisions of paragraph 2.2.1.1 of annex 12, and are found to be within their diminution limits determined by the Administration, taking into account the recommendations adopted by the Organization<sup>12</sup>, as shown in the following table:

**Table 2 – Transverse section modulus of hull girder**

|                      |            | $Z_{act} (cm^3)^1$ | $Z_{req} (cm^3)^2$ | Remarks |
|----------------------|------------|--------------------|--------------------|---------|
| Transverse section 1 | Upper deck |                    |                    |         |
|                      | Bottom     |                    |                    |         |
| Transverse section 2 | Upper deck |                    |                    |         |
|                      | Bottom     |                    |                    |         |
| Transverse section 3 | Upper deck |                    |                    |         |
|                      | Bottom     |                    |                    |         |

**Notes:**

- 1  $Z_{act}$  means the actual section moduli of the transverse section of the ship's hull girder calculated by using the thickness of structural members measured, renewed or reinforced, as appropriate, during the SC renewal survey, in accordance with the provisions of paragraph 2.2.1.1 of annex 12.
- 2  $Z_{req}$  means the diminution limit of the longitudinal bending strength of ships, as calculated in accordance with the provisions of paragraph 2.2.1.1 of annex 12.

The calculation sheets for  $Z_{act}$  should be attached to this report.

**3** This section applies to ships constructed before 1 July 2002: Section moduli of transverse sections of the ship's hull girder have been calculated by using the thickness of structural members measured, renewed or reinforced, as appropriate, during the SC renewal survey most recently conducted after the ship reached 10 years of age in accordance with the provisions of paragraph 2.2.1.2 of annex 12, and found to meet the criteria required by the Administration or the recognized classification society and that  $Z_{act}$  is not less than  $Z_{mc}$  (defined in note 2 below) as specified in appendix 2 to annex 12, as shown in the following table.

Describe the criteria for acceptance of the minimum section moduli of the ship's hull girder for ships in service required by the Administration or the recognized classification society.

<sup>12</sup> Refer to resolution MSC.108(73) on Recommendations on compliance with the requirements of paragraph 2.2.1.1 of annex 12 to annex B to resolution A.744(18).

**Table 3 – Transverse section modulus of hull girder**

|                      |            | $Z_{act} (cm^3)^1$ | $Z_{mc} (cm^3)^2$ | Remarks |
|----------------------|------------|--------------------|-------------------|---------|
| Transverse section 1 | Upper deck |                    |                   |         |
|                      | Bottom     |                    |                   |         |
| Transverse section 2 | Upper deck |                    |                   |         |
|                      | Bottom     |                    |                   |         |
| Transverse section 3 | Upper deck |                    |                   |         |
|                      | Bottom     |                    |                   |         |

**Notes:**

- 1 As defined in note 1 of table 2.
- 2  $Z_{mc}$  means the diminution limit of minimum section modulus calculated in accordance with the provisions of paragraph 2.2.1.2 of annex 12.

## **ANNEX 10**

### **RECOMMENDED PROCEDURES FOR THICKNESS MEASUREMENTS OF DOUBLE-HULL OIL TANKERS**

#### **General**

- 1** These procedures should be used for recording thickness measurements as required by annexes 2 and 4.
- 2** Reporting forms TM1-DHT, TM2-DHT(i), TM2-DHT(ii), TM3-DHT, TM4-DHT, TM5-DHT and TM6-DHT, set out in appendix 2, should be used for recording thickness measurements and the maximum allowable diminution should be stated. The maximum allowable diminution could be stated in an attached document.
- 3** Appendix 3 contains guidance diagrams and notes relating to the reporting forms and the requirements for thickness measurement.
- 4** The reporting forms should, where appropriate, be supplemented by data presented on structural sketches.

## Appendix 1

### GENERAL PARTICULARS

Ship's name: .....  
IMO number: .....  
Class/Administration identity number: .....  
Port of registry: .....  
Gross tonnage: .....  
Deadweight: .....  
Date of build: .....  
Classification society: .....

---

Name of company performing thickness measurement: .....  
.....  
Thickness measurement company certified by: .....  
Certificate number: .....  
Certificate valid from: ..... to .....  
Place of measurement: .....  
First date of measurement: .....  
Last date of measurement: .....  
Renewal survey/intermediate survey<sup>13</sup> due: .....  
Details of measurement equipment: .....  
Qualification of operator: .....  
Report number: ..... consisting of ..... pages  
Name of operator: ..... Name of surveyor: .....  
Signature of operator: ..... Signature of surveyor: .....  
Company official stamp: ..... Administration: .....  
Official stamp

---

<sup>13</sup> Delete as appropriate.



## Appendix 2

### Report on thickness measurement of all deck plating, all bottom shell plating or side shell plating\* (TM1-DHT)

Ship's name..... Class Identity no..... Report no..... IMO number.....

| STRAKE POSITION          |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
|--------------------------|---------------|---------------|-----------------|---|--------------|---|--------------|---|-------------|---|--------------|---|--------------|---|-------------------|---|----|
| PLATE POSITION           | No. or letter | Orig. thk. mm | Forward reading |   |              |   |              |   | Aft reading |   |              |   |              |   | Mean diminution % |   |    |
|                          |               |               | Gauged          |   | Diminution P |   | Diminution S |   | Gauged      |   | Diminution P |   | Diminution S |   |                   |   |    |
|                          |               |               | P               | S | mm           | % | mm           | % | P           | S | mm           | % | mm           | % | P                 | S | mm |
| 12 <sup>th</sup> forward |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 11th                     |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 10th                     |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 9th                      |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 8th                      |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 7th                      |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 6th                      |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 5th                      |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 4th                      |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 3rd                      |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 2nd                      |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 1st                      |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| Amidships                |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 1st aft                  |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 2nd                      |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 3rd                      |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 4th                      |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 5th                      |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 6th                      |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 7th                      |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 8th                      |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 9th                      |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 10th                     |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 11th                     |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 11th                     |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |
| 12th                     |               |               |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |    |

Operator's signature.....

Notes – see following page (\* – delete as appropriate)

**Notes to the report TM1-DHT:**

- 1** This report should be used for recording the thickness measurement of:
  - .1** All strength deck plating within the cargo area.
  - .2** All keel, bottom shell plating and bilge plating within the cargo area.
  - .3** Side shell plating including selected wind and water strakes outside cargo area.
  - .4** All wind and water strakes within cargo area.
- 2** The strake position should be clearly indicated as follows:
  - .1** For strength deck indicate the number of the strake of plating inboard from the stringer plate.
  - .2** For bottom plating indicate the number of the strake of plating outboard from the keel plate.
  - .3** For side shell plating give number of the strake of plating below sheerstrake and letter as shown on shell expansion.
- 3** Measurements should be taken at the forward and aft areas of all and where plates cross ballast/cargo tank boundaries separate measurements for the area of plating in way of each type of tank should be recorded.
- 4** The single measurements recorded should represent the average of multiple measurements.
- 5** The maximum allowable diminution could be stated in an attached document.

**Report on thickness measurement of shell and deck plating (one, two or three transverse sections) (TM2-DHT(i))**

Ship's name..... Class Identity no..... Report no..... IMO number.....

| STRENGTH DECK AND SHEERSTRAKE PLATING |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
|---------------------------------------|----------------------------------------------|------------|-------------------|--------|---|--------------|---|--------------|-----------------------------------------------|---------------|------------|-------------------|--------|---|--------------|---|----------------------------------------------|---|---------------|------------|-------------------|--------|---|--------------|---|--------------|---|
|                                       | FIRST TRANSVERSE SECTION AT FRAME NUMBER.... |            |                   |        |   |              |   |              | SECOND TRANSVERSE SECTION AT FRAME NUMBER.... |               |            |                   |        |   |              |   | THIRD TRANSVERSE SECTION AT FRAME NUMBER.... |   |               |            |                   |        |   |              |   |              |   |
| STRAKE POSITION                       | No. or letter                                | Orig. thk. | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S |                                               | No. or letter | Orig. thk. | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S                                 |   | No. or letter | Orig. thk. | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S |   |
|                                       |                                              | mm         | mm                | P      | S | mm           | % | mm           | %                                             |               | mm         | mm                | P      | S | mm           | % | mm                                           | % |               | mm         | mm                | P      | S | mm           | % | mm           | % |
| Stringer plate                        |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 1st strake inboard                    |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 2nd                                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 3rd                                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 4th                                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 5th                                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 6th                                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 7th                                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 8th                                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 9th                                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 10th                                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 11th                                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 12th                                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 13th                                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 14th                                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| centre strake                         |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| sheerstrake                           |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| TOPSIDE TOTAL                         |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |

Operator's signature.....

Notes – see following page

**Notes to the report TM2-DHT(i):**

- 1** This report form should be used for recording the thickness measurements of strength deck plating and sheerstrake plating transverse sections:

One, two or three sections within the cargo area comprising structural items (0), (1) and (2) as shown on the diagrams of typical transverse sections illustrated in appendix 3.
- 2** The topside area comprises deck plating, stringer plate and sheerstrake (including rounded gunwales).
- 3** The exact frame station of measurement should be stated.
- 4** The single measurements recorded should represent the average of multiple measurements.
- 5** The maximum allowable diminution could be stated in an attached document.

**Report on thickness measurement of shell and deck plating (one, two or three transverse sections) (TM2-DHT(ii))**

Ship's name..... Class Identity no..... Report no..... IMO number.....

| SHELL PLATING         |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
|-----------------------|----------------------------------------------|------------|-------------------|--------|---|--------------|---|--------------|-----------------------------------------------|---------------|------------|-------------------|--------|---|--------------|---|----------------------------------------------|---|---------------|------------|-------------------|--------|---|--------------|---|--------------|---|
|                       | FIRST TRANSVERSE SECTION AT FRAME NUMBER.... |            |                   |        |   |              |   |              | SECOND TRANSVERSE SECTION AT FRAME NUMBER.... |               |            |                   |        |   |              |   | THIRD TRANSVERSE SECTION AT FRAME NUMBER.... |   |               |            |                   |        |   |              |   |              |   |
| STRAKE POSITION       | No. or letter                                | Orig. thk. | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S |                                               | No. or letter | Orig. thk. | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S                                 |   | No. or letter | Orig. thk. | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S |   |
|                       |                                              | mm         | mm                | P      | S | mm           | % | mm           | %                                             |               | mm         | mm                | P      | S | mm           | % | mm                                           | % |               | mm         | mm                | P      | S | mm           | % | mm           | % |
| 1st below sheerstrake |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 2nd                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 3rd                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 4th                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 5th                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 6th                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 7th                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 8th                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 9th                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 10th                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 11th                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 12th                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 13th                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 14th                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 15th                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 16th                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 17th                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 18th                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 19th                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 20th                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| keel strake           |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| BOTTOM TOTAL          |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |

Operator's signature.....

Notes – see following page

**Notes to the report TM2-DHT(ii):**

- 1**      This report form should be used for recording the thickness measurements of shell plating transverse sections:

One, two or three sections within the cargo area comprising structural items (3), (4) and (5) and (6) as shown on the diagrams of typical transverse sections illustrated in appendix 3.
- 2**      The bottom area comprises keel, bottom and bilge plating.
- 3**      The exact frame station of measurement should be stated.
- 4**      The single measurements recorded should represent the average of multiple measurements.
- 5**      The maximum allowable diminution could be stated in an attached document.

**Report on thickness measurement of longitudinal members (one, two or three transverse sections) (TM3-DHT)**

Ship's name..... Class Identity no..... Report no..... IMO number.....

[illegible]

Operator's signature.....

Notes – see following page

***Notes to the report TM3-DHT:***

- 1**      This report should be used for recording the thickness measurements of longitudinal members at transverse sections:

One, two or three sections within the cargo area comprising appropriate structural items (10) to (29) as shown on the diagrams of typical transverse sections illustrated in appendix 3.
- 2**      The exact frame station of measurement should be stated.
- 3**      The single measurements recorded should represent the average of multiple measurements.
- 4**      The maximum allowable diminution could be stated in an attached document.



**Report on thickness measurement of transverse structural members (in the cargo oil and water ballast tanks within the cargo tank length) (TM4-DHT)**

Ship's name..... Class Identity no..... Report no..... IMO number.....

[illegible]

Operator's signature.....

Notes – see following page

***Notes to the report TM4-DHT:***

- 1**      This report should be used for recording the thickness measurements of transverse structural members, comprising appropriate structural items (30) to (36) as shown on diagrams of typical transverse sections illustrated in appendix 3.
- 2**      Guidance for areas of measurement is indicated in appendix 3. The single measurements recorded should represent the average of multiple measurements.
- 3**      The maximum allowable diminution could be stated in an attached document.

**Report on thickness measurement of W.T./O.T. transverse bulkheads (within the cargo tank or cargo hold spaces) (TM5-DHT)**

Ship's name..... Class Identity no..... Report no..... IMO number.....

[illegible]

Operator's signature.....

*Notes – see following page*

***Notes to the report TM5-DHT:***

- 1**      This report should be used for recording the thickness measurement of W.T./O.T. transverse bulkheads.
- 2**      Guidance for areas of measurement is indicated in appendix 3.
- 3**      The single measurements recorded should represent the average of multiple measurements.
- 4**      The maximum allowable diminution could be stated in an attached document.

## Report on thickness measurement of miscellaneous structural members (TM6-DHT)

Ship's name..... IMO number..... Class Identity no..... Report no.....

[illegible]

Operator's signature.....

Notes – see following page

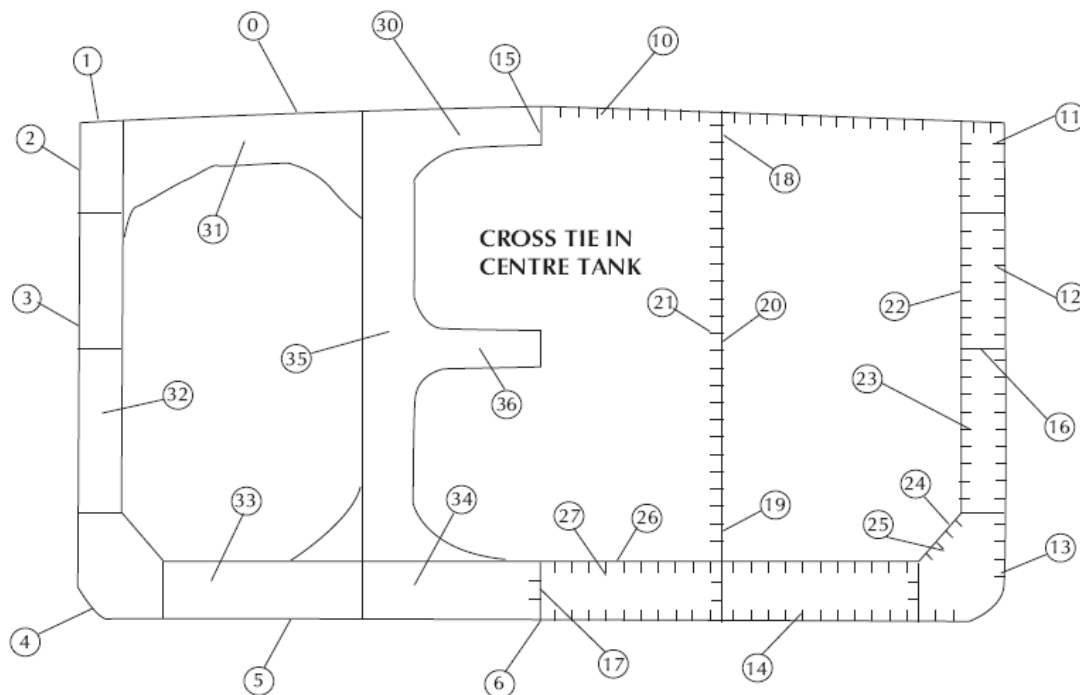
***Notes to the report TM6-DHT:***

- 1**      This report should be used for recording the thickness measurement of miscellaneous structural members.
- 2**      The single measurements recorded should represent the average of multiple measurements.
- 3**      The maximum allowable diminution could be stated in an attached document.

### Appendix 3

#### THICKNESS MEASUREMENT – DOUBLE-HULL OIL TANKERS

Typical transverse section of a double-hull oil tanker above 150,000 dwt with indication of longitudinal and transverse members.



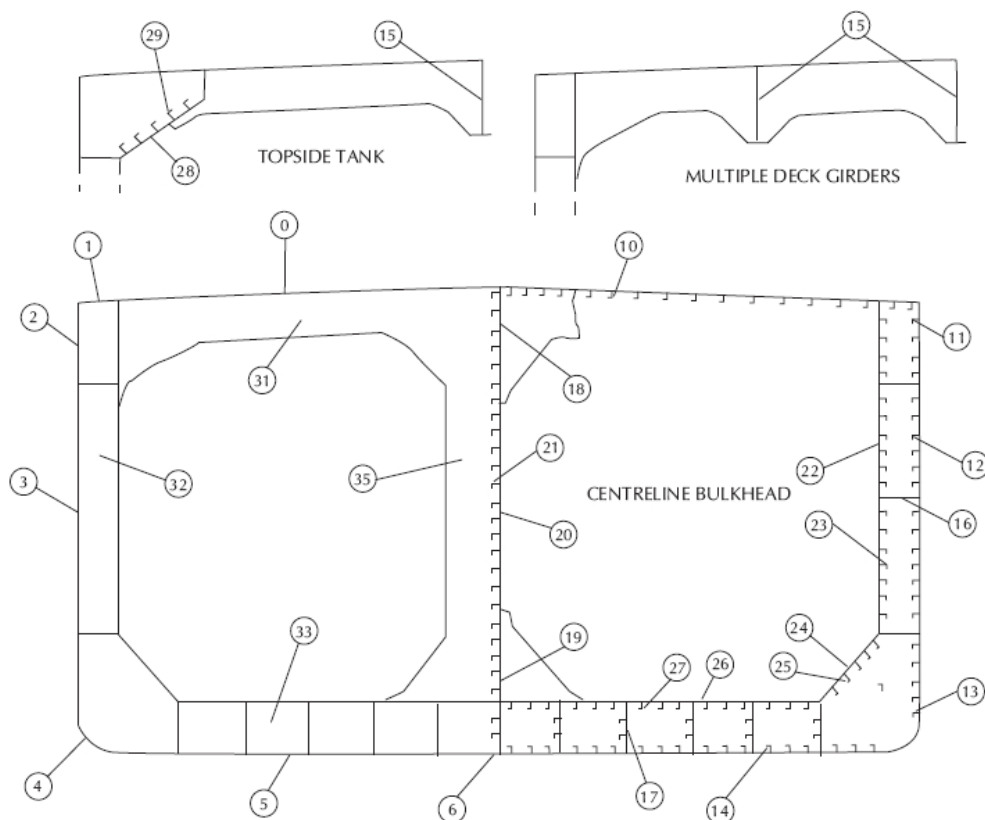
| REPORT ON<br>TM2-DHT(i) and (ii) |                       |
|----------------------------------|-----------------------|
| ①                                | Strength deck plating |
| ②                                | Stringer plate        |
| ③                                | Sheerstrake           |
| ④                                | Side shell plating    |
| ⑤                                | Bilge plating         |
| ⑥                                | Bottom shell plating  |
| ⑦                                | Keel plate            |

| REPORT ON TM3-DHT |                                           |
|-------------------|-------------------------------------------|
| ⑩                 | Deck longitudinals                        |
| ⑪                 | Sheerstrake longitudinals                 |
| ⑫                 | Side shell longitudinals                  |
| ⑬                 | Bilge longitudinals                       |
| ⑭                 | Bottom longitudinals                      |
| ⑮                 | Deck girders                              |
| ⑯                 | Horizontal girders in wing ballast tanks  |
| ⑰                 | Bottom girders                            |
| ⑱                 | Longitudinal bulkhead top strake          |
| ⑲                 | Longitudinal bulkhead bottom strake       |
| ⑳                 | Longitudinal bulkhead plating (remainder) |
| ㉑                 | Longitudinal bulkhead longitudinals       |
| ㉒                 | Inner side plating                        |
| ㉓                 | Inner side longitudinals                  |
| ㉔                 | Hopper plating                            |
| ㉕                 | Hopper longitudinals                      |
| ㉖                 | Inner bottom plating                      |
| ㉗                 | Inner bottom longitudinals                |
| ㉘                 | Topside tank plating                      |
| ㉙                 | Topside tank longitudinals                |

| REPORT ON TM4-DHT |                                    |
|-------------------|------------------------------------|
| ⑳                 | Deck transverse – centre tank      |
| ㉑                 | Deck transverse – wing tank        |
| ㉒                 | Vertical web in wing ballast tank  |
| ㉓                 | Double bottom floor – wing tank    |
| ㉔                 | Double bottom floor – centre tank  |
| ㉕                 | Longitudinal bulkhead vertical web |
| ㉖                 | Cross ties                         |

## THICKNESS MEASUREMENT – DOUBLE-HULL OIL TANKERS

Typical transverse section of a double-hull oil tanker up to 150,000 dwt with indication of longitudinal and transverse members.



| REPORT ON<br>TM2-DHT(i) and (ii) |                       |
|----------------------------------|-----------------------|
| 0                                | Strength deck plating |
| 1                                | Stringer plate        |
| 2                                | Sheerstrake           |
| 3                                | Side shell plating    |
| 4                                | Bilge plating         |
| 5                                | Bottom shell plating  |
| 6                                | Keel plate            |

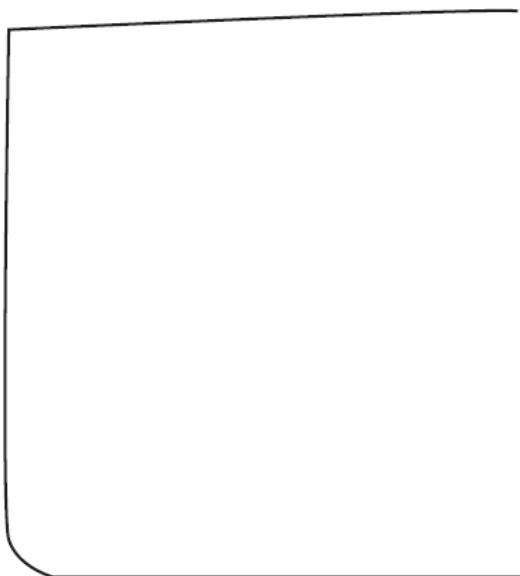
| REPORT ON TM3-DHT |                                           |
|-------------------|-------------------------------------------|
| 10                | Deck longitudinals                        |
| 11                | Sheerstrake longitudinals                 |
| 12                | Side shell longitudinals                  |
| 13                | Bilge longitudinals                       |
| 14                | Bottom longitudinals                      |
| 15                | Deck girders                              |
| 16                | Horizontal girders in wing ballast tanks  |
| 17                | Bottom girders                            |
| 18                | Longitudinal bulkhead top strake          |
| 19                | Longitudinal bulkhead bottom strake       |
| 20                | Longitudinal bulkhead plating (remainder) |
| 21                | Longitudinal bulkhead longitudinals       |
| 22                | Inner side plating                        |
| 23                | Inner side longitudinals                  |
| 24                | Hopper plating                            |
| 25                | Hopper longitudinals                      |
| 26                | Inner bottom plating                      |
| 27                | Inner bottom longitudinals                |
| 28                | Topside tank plating                      |
| 29                | Topside tank longitudinals                |

| REPORT ON TM4-DHT |                                    |
|-------------------|------------------------------------|
| 30                | Deck transverse – centre tank      |
| 31                | Deck transverse – wing tank        |
| 32                | Vertical web in wing ballast tank  |
| 33                | Double bottom floor – wing tank    |
| 34                | Double bottom floor – centre tank  |
| 35                | Longitudinal bulkhead vertical web |
| 36                | Cross ties                         |



## THICKNESS MEASUREMENT – DOUBLE-HULL OIL TANKERS

Transverse section outline. The diagram may be used for those ships where typical sections are not applicable



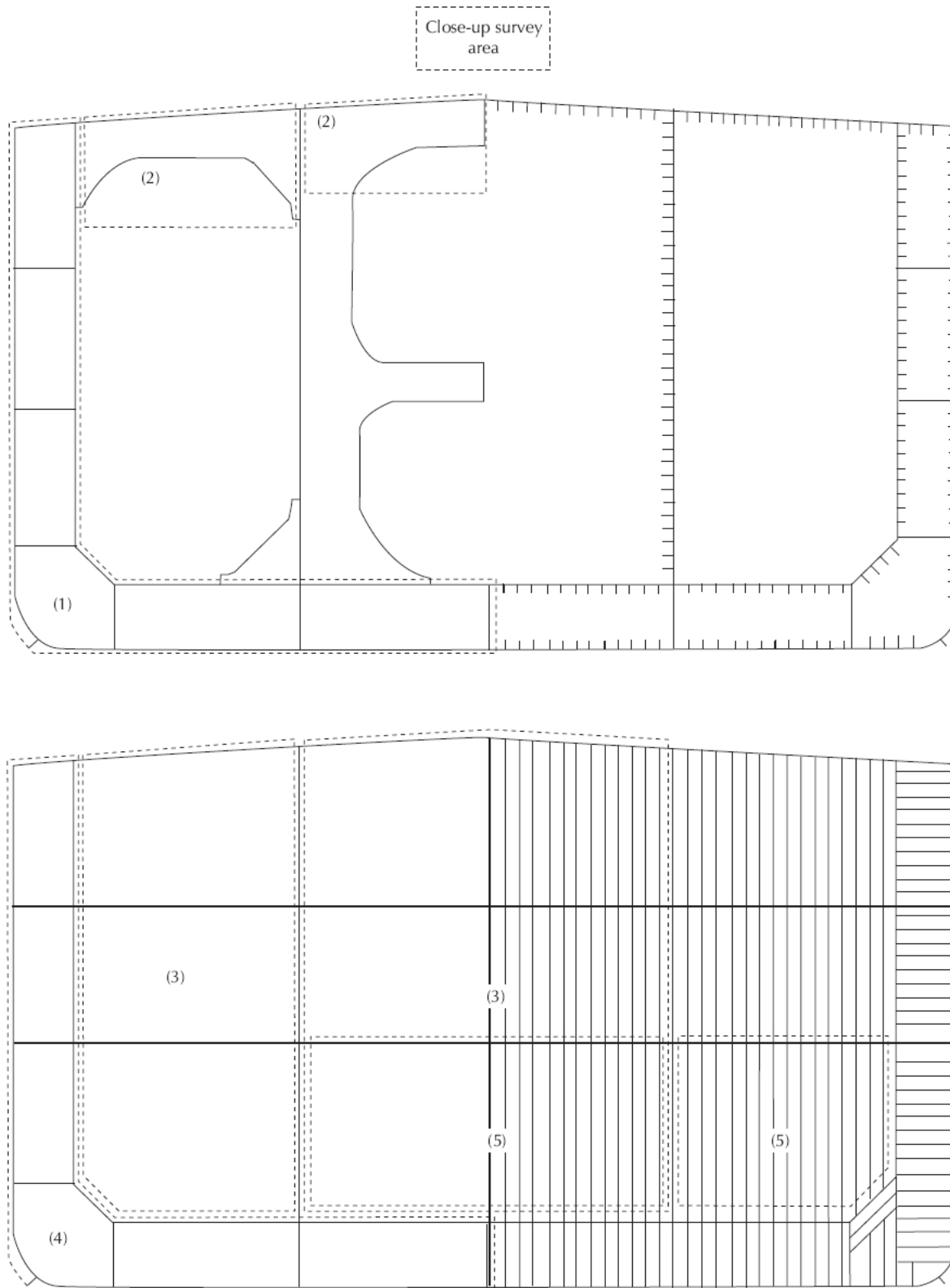
| REPORT ON<br>TM2-DHT(i) and (ii) |                       |
|----------------------------------|-----------------------|
| ①                                | Strength deck plating |
| ②                                | Stringer plate        |
| ③                                | Sheerstrake           |
| ④                                | Side shell plating    |
| ⑤                                | Bilge plating         |
| ⑥                                | Bottom shell plating  |
| ⑦                                | Keel plate            |

| REPORT ON TM3-DHT |                                           |
|-------------------|-------------------------------------------|
| ⑩                 | Deck longitudinals                        |
| ⑪                 | Sheerstrake longitudinals                 |
| ⑫                 | Side shell longitudinals                  |
| ⑬                 | Bilge longitudinals                       |
| ⑭                 | Bottom longitudinals                      |
| ⑮                 | Deck girders                              |
| ⑯                 | Horizontal girders in wing ballast tanks  |
| ⑰                 | Bottom girders                            |
| ⑱                 | Longitudinal bulkhead top strake          |
| ⑲                 | Longitudinal bulkhead bottom strake       |
| ⑳                 | Longitudinal bulkhead plating (remainder) |
| ㉑                 | Longitudinal bulkhead longitudinals       |
| ㉒                 | Inner side plating                        |
| ㉓                 | Inner side longitudinals                  |
| ㉔                 | Hopper plating                            |
| ㉕                 | Hopper longitudinals                      |
| ㉖                 | Inner bottom plating                      |
| ㉗                 | Inner bottom longitudinals                |
| ㉘                 | Topside tank plating                      |
| ㉙                 | Topside tank longitudinals                |

| REPORT ON TM4-DHT |                                    |
|-------------------|------------------------------------|
| ⑳                 | Deck transverse – centre tank      |
| ㉑                 | Deck transverse – wing tank        |
| ㉒                 | Vertical web in wing ballast tank  |
| ㉓                 | Double bottom floor – wing tank    |
| ㉔                 | Double bottom floor – centre tank  |
| ㉕                 | Longitudinal bulkhead vertical web |
| ㉖                 | Cross ties                         |

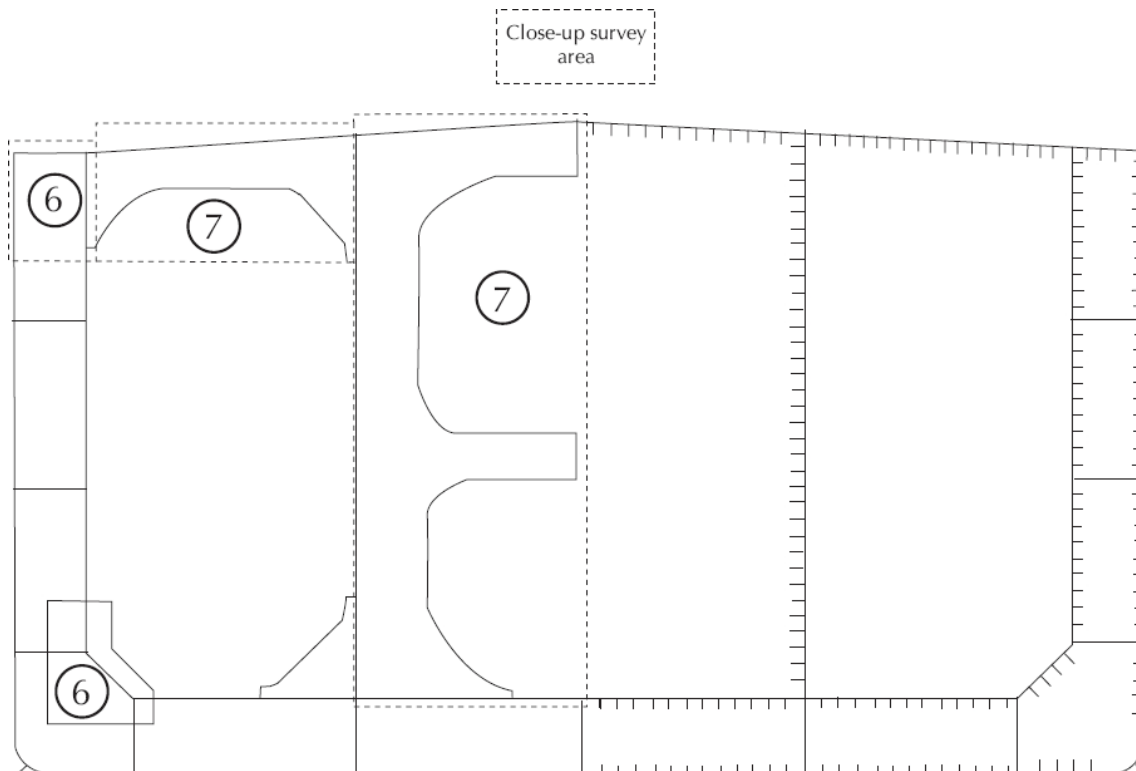
### THICKNESS MEASUREMENT – DOUBLE-HULL OIL TANKERS

Areas subject to close-up survey and thickness measurements – areas (1) to (5) as defined in annex 1 – Thickness to be reported on TM3-DHT, TM4-DHT and TM5-DHT as appropriate.



### THICKNESS MEASUREMENT – DOUBLE-HULL OIL TANKERS

Areas subject to close-up survey and thickness measurements – areas (6) to (7) as defined in annex 1 – Thickness to be reported on TM3-DHT and TM4-DHT as appropriate.



## **ANNEX 11**

### **GUIDELINES FOR TECHNICAL ASSESSMENT IN CONJUNCTION WITH THE PLANNING OF ENHANCED SURVEYS FOR OIL TANKERS**

#### **Renewal survey**

## **1 Introduction**

These Guidelines contain information and suggestions concerning technical assessments which may be of use in conjunction with the planning of renewal surveys of oil tankers. As indicated in 5.1.5, these Guidelines are a recommended tool which may be invoked at the discretion of an Administration, when considered necessary and appropriate, in conjunction with the preparation of the required survey programme.

## **2 Purpose and principles**

### **2.1 Purpose**

The technical assessments described in these Guidelines should assist in identifying critical structural areas, nominating suspect areas and in focusing attention on structural elements or areas of structural elements which may be particularly susceptible to, or evidence a history of, wastage or damage. This information may be useful in nominating locations, areas and tanks for thickness measurement, close-up survey and tank testing.

### **2.2 Minimum requirements**

These Guidelines may not be used to reduce the requirements of annexes 1, 2 and 3 for close-up survey, thickness measurement and tank testing, respectively, which are, in all cases, to be complied with as a minimum.

### **2.3 Timing**

As with other aspects of survey planning, the technical assessments described in these Guidelines should be carried out by the owner or operator in cooperation with the Administration well in advance of the commencement of the renewal survey, i.e. prior to commencing the survey and normally at least 12 to 15 months before the survey's completion due date.

### **2.4 Aspects to be considered**

Technical assessments, which may include quantitative or qualitative evaluation of relative risks of possible deterioration, of the following aspects of a particular ship may be used as a basis for the nomination of tanks and areas for survey of:

- .1** design features such as stress levels on various structural elements, design details and extent of use of high-tensile steel (HTS);
- .2** former history with respect to corrosion, cracking, buckling, indents and repairs for the particular ship as well as similar vessels, where available; and
- .3** information with respect to types of cargo carried, use of different tanks for cargo/ballast, protection of tanks and condition of coating, if any.

Technical assessments of the relative risks of susceptibility to damage or deterioration of various structural elements and areas should be judged and decided on the basis of recognized principles and practices, such as may be found in references 1 and 2.

### **3 Technical assessment**

#### **3.1 General**

**3.1.1** There are three basic types of possible failure which may be the subject of a technical assessment in connection with the planning of surveys: corrosion, cracks and buckling. Contact damages are not normally covered by the survey plan since indents are usually noted in memoranda and assumed to be dealt with as normal routine by surveyors.

**3.1.2** Technical assessments performed in conjunction with the survey planning process should, in principle, be as shown schematically in figure 1 which depicts how technical assessments can be carried out in conjunction with the survey planning process. The approach is basically an evaluation of the risk, based on the knowledge and experience related to design and corrosion.

**3.1.3** The design should be considered with respect to structural details which may be susceptible to buckling or cracking as a result of vibration, high stress levels or fatigue.

**3.1.4** Corrosion is related to the ageing process and is closely connected with the quality of corrosion protection at newbuilding and subsequent maintenance during the service life. Corrosion may also lead to cracking and/or buckling.

#### **3.2 Methods**

##### **3.2.1 Design details**

**3.2.1.1** Damage experience related to the ship in question and similar ships, where available, are the main source of information to be used in the process of planning. In addition, a selection of structural details from the design drawings should be included. Typical damage experience to be considered will consist of:

- .1 number, extent, location and frequency of cracks; and
- .2 location of buckles.

**3.2.1.2** This information may be found in the survey reports and/or the owner's files, including the results of the owner's own inspections. The defects should be analysed, noted and marked on sketches.

**3.2.1.3** In addition, general experience should be utilized. For example, reference should be made to reference 1, which contains a catalogue of typical damages and proposed repair methods for various tanker structural details.

**3.2.1.4** Such figures should be used together with a review of the main drawings, in order to compare with the actual structure and search for similar details which may be susceptible to damage. An example is shown in figure 2. In particular, chapter 3 of reference 1 deals with various aspects specific to double-hull tankers, such as stress concentration locations, misalignment during construction, corrosion trends, fatigue considerations and areas requiring special attention, which should be considered in working out the survey planning.

**3.2.1.5** The review of the main structural drawings, in addition to using the above-mentioned figures, should include checking for typical design details where cracking has been experienced. The factors contributing to damage should be carefully considered.

**3.2.1.6** The use of HTS is an important factor. Details showing good service experience where ordinary, mild steel has been used may be more susceptible to damage when HTS, and its higher associated stresses, are utilized. There is extensive and, in general, good experience, with the use of HTS for longitudinal material in deck and bottom structures. Experience in other locations, where the dynamic stresses may be higher, is less favourable, e.g., side structures.

**3.2.1.7** In this respect, stress calculations of typical and important components and details, in accordance with relevant methods, may prove useful and should be considered.

**3.2.1.8** The selected areas of the structure identified during this process should be recorded and marked on the structural drawings to be included in the survey programme.

## **3.2.2**     *Corrosion*

**3.2.2.1** In order to evaluate relative corrosion risks, the following information is generally to be considered:

- .1        usage of tanks and spaces;
- .2        condition of coatings;
- .3        condition of anodes;
- .4        cleaning procedures;
- .5        previous corrosion damage;
- .6        ballast use and time for cargo tanks;
- .7        corrosion risk scheme (see reference 2, table 2.1); and
- .8        location of heated tanks.

**3.2.2.2** Reference 2 gives definitive examples which can be used for judging and describing coating condition, using typical pictures of conditions.

**3.2.2.3** The evaluation of corrosion risks should be based on information in reference 2, together with the age of the ship and relevant information on the anticipated condition as derived from the information collected in order to prepare the survey programme.

**3.2.2.4** The various tanks and spaces should be listed with the corrosion risks nominated accordingly. Special attention should be given to the areas where the double-hull tanker is particularly exposed to corrosion. To this end, the specific aspects addressing corrosion in double-hull tankers indicated in 3.4 (Corrosion trends) of reference 1 should be taken into account.

### **3.2.3**    *Locations for close-up survey and thickness measurement*

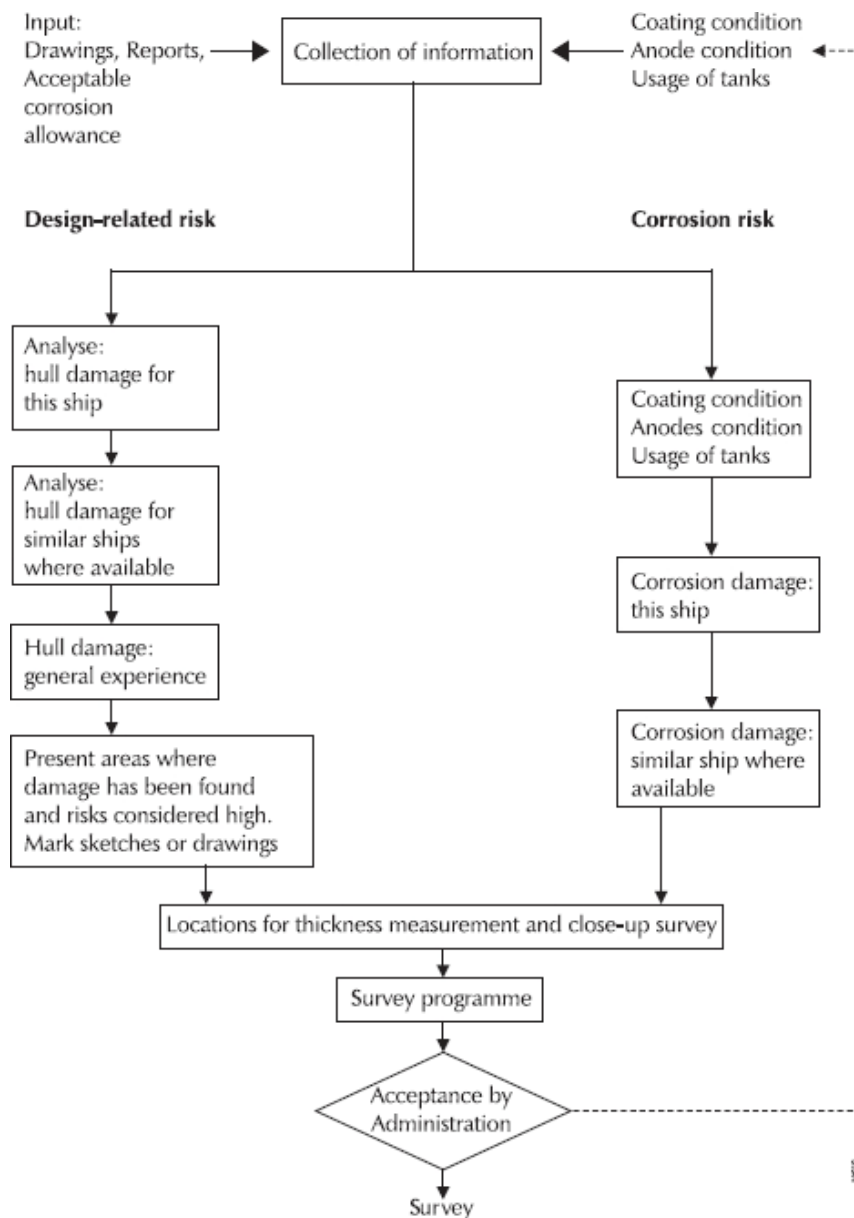
**3.2.3.1** On the basis of the table of corrosion risks and the evaluation of design experience, the locations for initial close-up survey and thickness measurement (sections) may be nominated.

**3.2.3.2** The sections subject to thickness measurement should normally be nominated in tanks and spaces where corrosion risk is judged to be the highest.

**3.2.3.3** The nomination of tanks and spaces for close-up survey should, initially, be based on highest corrosion risk and should always include ballast tanks. The principle for the selection should be that the extent is increased by age or where information is insufficient or unreliable.

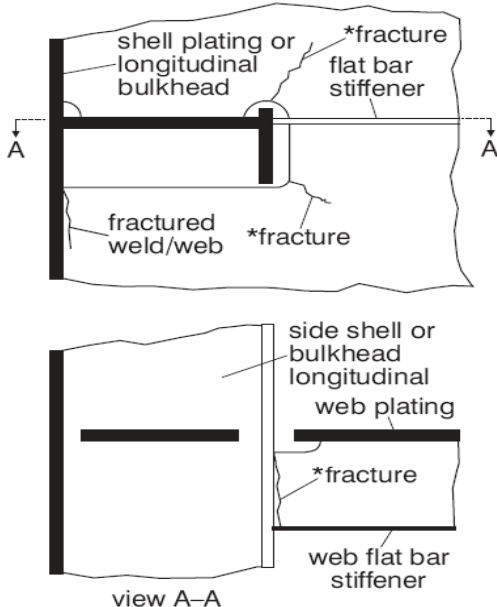
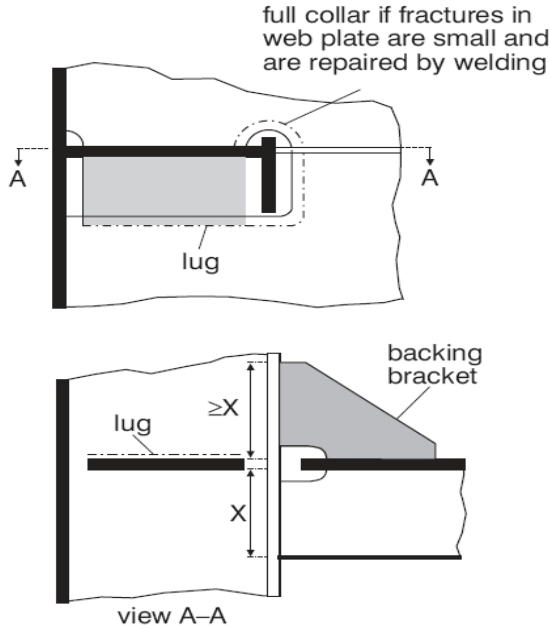
### **References**

- 1**        TSCF, Guidelines for the Inspection and Maintenance of Double Hull Tanker Structures, 1995.
- 2**        TSCF, Guidance Manual for Tanker Structures, 1997.



**Figure 1** – Technical assessment and the survey planning process



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                          |             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------|
| <b>LOCATION:</b> Connection of longitudinals to transverse webs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                          |             |
| <b>EXAMPLE NO. 1</b> Web and flat bar fractures at cut-outs for longitudinal stiffener connections                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                          |             |
| <p><b>TYPICAL DAMAGE</b></p>  <p><i>Note* one or more fractures may occur</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                          |             |
| <p><b>PROPOSED REPAIR</b></p>  <p><i>web and flat bar cropped and part renewed or alternatively welded</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                          |             |
| <p><b>FACTORS CONTRIBUTING TO DAMAGE</b></p> <ol style="list-style-type: none"> <li>1 Asymmetrical connection of flat bar stiffener resulting in high peak stresses at the heel of the stiffener under fatigue loading.</li> <li>2 Insufficient area of connection of longitudinal to web plate.</li> <li>3 Defective weld at return around the plate thickness.</li> <li>4 High localized corrosion at areas of stress concentration such as flat bar stiffener connections, corners of cut-out for the longitudinal and connection of web to shell at cut-outs.</li> <li>5 High shear stress in the web of the transverse.</li> <li>6 Dynamic seaway loads/ship motions.</li> </ol> |                                          |             |
| FIGURE<br>1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | TANKER STRUCTURE CO-OPERATIVE FORUM      | FIGURE<br>1 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SUBJECT: CATALOGUE OF STRUCTURAL DETAILS |             |

**Figure 2** – Typical damage and repair example (reproduced from reference 2)

## ANNEX 12

### CRITERIA FOR LONGITUDINAL STRENGTH OF HULL GIRDER FOR OIL TANKERS

#### 1 General

**1.1** These criteria should be used for the evaluation of the longitudinal strength of the ship's hull girder as required by 8.1.2.

**1.2** In order that the ship's longitudinal strength to be evaluated can be recognized as valid, fillet welding between longitudinal internal members and hull envelopes should be in sound condition so as to keep the integrity of longitudinal internal members with hull envelopes.

#### 2 Evaluation of longitudinal strength

On oil tankers of 130 m in length and upwards and over 10 years of age, the longitudinal strength of the ship's hull girder should be evaluated in compliance with the requirements of this annex on the basis of the thickness measured, renewed or reinforced, as appropriate, during the renewal survey of the Cargo Ship Safety Construction Certificate or Cargo Ship Safety Certificate (SC renewal survey). The condition of the hull girder for longitudinal strength evaluation should be determined in accordance with the methods specified in appendix 3.

##### **2.1 *Calculation of transverse sectional areas of deck and bottom flanges of hull girder***

**2.1.1** The transverse sectional areas of deck flange (deck plating and deck longitudinals) and bottom flange (bottom shell plating and bottom longitudinals) of the ship's hull girder should be calculated by using the thickness measured, renewed or reinforced, as appropriate, during the SC renewal survey.

**2.1.2** If the diminution of sectional areas of either deck or bottom flange exceeds 10% of their respective as-built area (i.e. original sectional area when the ship was built), either one of the following measures should be taken:

- .1** to renew or reinforce the deck or bottom flanges so that the actual sectional area is not less than 90% of the as-built area; or
- .2** to calculate the actual section of moduli ( $Z_{act}$ ) of transverse section of the ship's hull girder by applying the calculation method specified in appendix 1, by using the thickness measured, renewed or reinforced, as appropriate, during the SC renewal survey.

##### **2.2 *Requirements for transverse section modulus of hull girder***

**2.2.1** The actual section moduli of the transverse section of the ship's hull girder, calculated in accordance with 2.1.2.2, should satisfy either of the following provisions, as applicable:

- .1 for ships constructed on or after 1 July 2002, the actual section moduli ( $Z_{act}$ ) of the transverse section of the ship's hull girder calculated in accordance with the requirements of paragraph 2.1.2.2 should be not less than the diminution limits determined by the Administration, taking into account the recommendations adopted by the Organization;<sup>14</sup> or
- .2 for ships constructed before 1 July 2002, the actual section moduli ( $Z_{act}$ ) of the transverse section of the ship's hull girder calculated in accordance with the requirements of 2.1.2.2 should meet the criteria for minimum section modulus for ships in service required by the Administration or recognized classification society, provided that in no case  $Z_{act}$  should be less than the diminution limit of the minimum section modulus ( $Z_{mc}$ ) as specified in appendix 2.

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<sup>14</sup> Refer to resolution MSC.108(73) on Recommendation on compliance with the requirements of paragraph 2.2.1.1 of annex 12 to annex B to resolution A.744(18).

## Appendix 1

### CALCULATION CRITERIA OF SECTION MODULI OF MIDSHIP SECTION OF HULL GIRDER

- 1** When calculating the transverse section modulus of the ship's hull girder, the sectional area of all continuous longitudinal strength members should be taken into account.
- 2** Large openings, i.e. openings exceeding 2.5 m in length or 1.2 m in breadth, and scallops, where scallop welding is applied, are always to be deducted from the sectional areas used in the section modulus calculation.
- 3** Smaller openings (manholes, lightening holes, single scallops in way of seams, etc.) need not be deducted, provided that the sum of their breadths or shadow area breadths in one transverse section does not reduce the section modulus at deck or bottom by more than 3% and provided that the height of lightening holes, draining holes and single scallops in longitudinals or longitudinal girders does not exceed 25% of the web depth, for scallops maximum 75 mm.
- 4** A deduction-free sum of smaller opening breadths in one transverse section in the bottom or deck area of  $0.06(B - \Sigma b)$  (where  $B$  = breadth of ship,  $\Sigma b$  = total breadth of large openings) may be considered equivalent to the above reduction in sectional modulus.
- 5** The shadow area will be obtained by drawing two tangent lines with an opening angle of  $30^\circ$ .
- 6** The deck modulus is related to the moulded deck line at side.
- 7** The bottom modulus is related to the base line.
- 8** Continuous trunks and longitudinal hatch coamings should be included in the longitudinal sectional area provided they are effectively supported by longitudinal bulkheads or deep girders. The deck modulus is then to be calculated by dividing the moment of inertia by the following distance, provided this is greater than the distance to the deck line at side:

$$y_t = y \left( 0.9 + 0.2 \frac{x}{B} \right)$$

where:

$y$  = distance from neutral axis to top of continuous strength member;

$x$  = distance from top of continuous strength member to centerline of the ship;

$x$  and  $y$  to be measured to the point giving the largest value of  $y_t$ .

- 9** Longitudinal girders between multi-hatchways will be considered by special calculations.

## Appendix 2

### DIMINUTION LIMIT OF MINIMUM LONGITUDINAL STRENGTH OF SHIPS IN SERVICE

**1** The diminution limit of the minimum section modulus ( $Z_{mc}$ ) of oil tankers in service is given by the following formula:

$$Z_{mc} = cL^2B(C_b + 0.7)k \text{ (cm}^3\text{)}$$

where:

$L$  = Length of ships.  $L$  is the distance, in metres, on the summer load waterline from the fore-side of stem to the after side of the rudder post, or the centre of the rudder stock if there is no rudder post.  $L$  is not to be less than 96%, and need not be greater than 97%, of the extreme length on the summer load waterline. In ships with unusual stern and bow arrangement, the length  $L$  may be specially considered.

$B$  = Greatest moulded breadth in metres.

$C_b$  = Moulded block coefficient at draught  $d$  corresponding to summer load waterline, based on  $L$  and  $B$ .  $C_b$  is not to be taken less than 0.6.

$$C_b = \frac{\text{moulded displacement (m}^3\text{) at draught } d}{LBd}$$

$$c = 0.9c_n$$

$$c_n = 10.75 - \left(\frac{300-L}{100}\right)^{1.5} \quad \text{for } 130 \text{ m} \leq L \leq 300 \text{ m}$$

$$c_n = 10.75 \quad \text{for } 300 \text{ m} \leq L \leq 350 \text{ m}$$

$$c_n = 10.75 - \left(\frac{L-350}{150}\right)^{1.5} \quad \text{for } 350 \text{ m} \leq L \leq 500 \text{ m}$$

$k$  = material factor, e.g.,

$k$  = 1.0 for mild steel with yield stress of 235 N/mm<sup>2</sup> and over

$k$  = 0.78 for high-tensile steel with yield stress of 315 N/mm<sup>2</sup> and over

$k$  = 0.72 for high-tensile steel with yield stress of 355 N/mm<sup>2</sup> and over.

**2** Scantlings of all continuous longitudinal members of the ship's hull girder based on the section modulus requirement in 1 above are to be maintained within 0.4 $L$  amidships. However, in special cases, based on consideration of type of ship, hull form and loading conditions, the scantlings may be gradually reduced towards the end of 0.4 $L$  part, bearing in mind the desire not to inhibit the ship's loading flexibility.

**3** However, the above standard may not be applicable to ships of unusual type or design, e.g., for ships of unusual main proportions and/or weight distributions.

## **Appendix 3**

### **SAMPLING METHOD OF THICKNESS MEASUREMENTS FOR LONGITUDINAL STRENGTH EVALUATION AND REPAIR METHODS**

#### **1 Extent of longitudinal strength evaluation**

Longitudinal strength should be evaluated within  $0.4L$  amidships for the extent of the hull girder length that contains tanks therein and within  $0.5L$  amidships for adjacent tanks which may extend beyond  $0.4L$  amidships, where tanks means ballast tanks and cargo tanks.

#### **2 Sampling method of thickness measurement**

**2.1** Pursuant to the requirements of section 2.5, transverse sections should be chosen such that thickness measurements can be taken for as many different tanks in corrosive environments as possible, e.g., ballast tanks sharing a common plane boundary with cargo tanks fitted with heating coils, other ballast tanks, cargo tanks permitted to be filled with seawater and other cargo tanks. Ballast tanks sharing a common plane boundary with cargo tanks fitted with heating coils and cargo tanks permitted to be filled with seawater should be selected where present.

**2.2** The minimum number of transverse sections to be sampled should be in accordance with annex 2. The transverse sections should be located where the largest thickness reductions are suspected to occur or are revealed from deck and bottom plating measurements prescribed in 2.3 and should be clear of areas which have been locally renewed or reinforced.

**2.3** At least two points should be measured on each deck plate and/or bottom shell plate required to be measured within the cargo area in accordance with the requirements of annex 2.

**2.4** Within  $0.1D$  (where  $D$  is the ship's moulded depth) of the deck and bottom at each transverse section to be measured in accordance with the requirements of annex 2, every longitudinal and girder should be measured on the web and face plate, and every plate should be measured at one point between longitudinals.

**2.5** For longitudinal members other than those specified in 2.4 to be measured at each transverse section in accordance with the requirements of annex 2, every longitudinal and girder should be measured on the web and face plate, and every plate should be measured at least in one point per strake.

**2.6** The thickness of each component should be determined by averaging all of the measurements taken in way of the transverse section on each component.

#### **3 Additional measurements where the longitudinal strength is deficient**

**3.1** Where one or more of the transverse sections are found to be deficient in respect of the longitudinal strength requirements given in this annex, the number of transverse sections for thickness measurement should be increased such that each tank within the  $0.5L$  amidships region has been sampled. Tank spaces that are partially within, but extend beyond, the  $0.5L$  region, should be sampled.

**3.2** Additional thickness measurements should also be performed on one transverse section forward and one aft of each repaired area to the extent necessary to ensure that the areas bordering the repaired section also comply with the requirements of the Code.

#### **4 Effective repair methods**

**4.1** The extent of renewal or reinforcement carried out to comply with this annex should be in accordance with 4.2.

**4.2** The minimum continuous length of a renewed or reinforced structural member should be not less than twice the spacing of the primary members in way. In addition, the thickness diminution in way of the butt joint of each joining member forward and aft of the replaced member (plates, stiffeners, girder webs and flanges, etc.) should not be within the substantial corrosion range (75% of the allowable diminution associated with each particular member). Where differences in thickness at the butt joint exceed 15% of the lower thickness, a transition taper should be provided.

**4.3** Alternative repair methods involving the fitting of straps or structural member modification should be subject to special consideration. In considering the fitting of straps, it should be limited to the following conditions:

- .1** to restore and/or increase longitudinal strength;
- .2** the thickness diminution of the deck or bottom plating to be reinforced should not be within the substantial corrosion range (75% of the allowable diminution associated with the deck plating);
- .3** the alignment and arrangement, including the termination of the straps, is in accordance with a standard recognized by the Administration;
- .4** the straps are continuous over the entire 0.5L amidships length; and
- .5** continuous fillet welding and full penetration welds are used at butt welding and, depending on the width of the strap, slot welds. The welding procedures applied should be acceptable to the Administration.

**4.4** The existing structure adjacent to replacement areas and in conjunction with the fitted straps, etc., should be capable of withstanding the applied loads, taking into account the buckling resistance and the condition of welds between the longitudinal members and hull envelope plating.

## Part B

### CODE ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF OIL TANKERS OTHER THAN DOUBLE-HULL OIL TANKERS

#### 1 General

##### 1.1 *Application*<sup>15</sup>

**1.1.1** The Code should apply to self-propelled oil tankers of 500 gross tonnage and above other than double-hull oil tankers, as defined in 1.2.1 of part A of annex B.

**1.1.2** The Code should apply to surveys of hull structure and piping systems in way of cargo tanks, pump-rooms, cofferdams, pipe tunnels, void spaces within the cargo area and all ballast tanks.

**1.1.3** The Code contains the minimum extent of examination, thickness measurements and tank pressure testing. The survey should be extended when substantial corrosion and/or structural defects are found and include additional close-up survey when necessary.

**1.1.4** The surveys should be carried out during the surveys prescribed by regulation I/10 of the Convention.

##### 1.2 *Definitions*

**1.2.1** *Ballast tank* is a tank which is used solely for the carriage of salt water ballast.

**1.2.2** *Combined cargo/ballast tank*, if referred to within the Code, is a tank which is used for the carriage of cargo or ballast water as a routine part of the ship's operation and will be treated as a ballast tank. Cargo tanks in which water ballast might be carried only in exceptional cases per MARPOL regulation I/18.3 are to be treated as cargo tanks.

**1.2.3** *Overall survey* is a survey intended to report on the overall condition of the hull structure and determine the extent of additional close-up surveys.

**1.2.4** *Close-up survey* is a survey where the details of structural components are within the close visual inspection range of the surveyor, i.e. normally within reach of hand.

**1.2.5** *Transverse section* is the cross section of the hull perpendicular to the ship's centerline and includes all longitudinal members such as plating, longitudinals and girders at the deck, sides, bottom, inner bottom and longitudinal bulkheads.

**1.2.6** *Representative tanks* are those which are expected to reflect the condition of other tanks of similar type and service and with similar corrosion prevention systems. When selecting representative tanks, account should be taken of the service and repair history on board and identifiable critical and/or suspect areas.

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<sup>15</sup> The intention of the Code is to ensure that an appropriate level of review of plans and documents is conducted and consistency in application is attained. Such evaluation of survey reports, survey programmes, planning documents, etc., should be carried out at the managerial level of the Administration or organization recognized by the Administration.



**1.2.7** *Suspect areas* are locations showing substantial corrosion and/or are considered by the surveyor to be prone to rapid wastage.

**1.2.8** *Substantial corrosion* is an extent of corrosion such that assessment of corrosion pattern indicates a wastage in excess of 75% of allowable margins, but within acceptable limits.

**1.2.9** *Corrosion prevention system* is normally considered a full hard protective coating. Hard protective coating should usually be epoxy coating or equivalent. Other coating systems, which are neither soft nor semi-hard coatings, may be considered acceptable as alternatives provided that they are applied and maintained in compliance with the manufacturer's specifications.

**1.2.10** *Coating condition* is defined as follows:

GOOD condition with only minor spot rusting;

FAIR condition with local breakdown of coating at edges of stiffeners and weld connections and/or light rusting over 20% or more of areas under consideration, but less than as defined for POOR condition;

POOR condition with general breakdown of coating over 20% or more of areas or hard scale at 10% or more of areas under consideration.

**1.2.11** *Critical structural areas* are locations which have been identified from calculations to require monitoring or from the service history of the subject ship or from similar or sister ships to be sensitive to cracking, buckling or corrosion which would impair the structural integrity of the ship.

**1.2.12** *Cargo area* is that part of the ship which contains cargo tanks, slop tanks and cargo/ballast pump-rooms, cofferdams, ballast tanks and void spaces adjacent to cargo tanks and also deck areas throughout the entire length and breadth of the part of the ship over the above-mentioned spaces.

**1.2.13** *Intermediate survey* is a survey carried out either at the second or third annual survey.

**1.2.14** *Prompt and thorough repair* is a permanent repair completed at the time of survey to the satisfaction of the surveyor, therein removing the need for the imposition of any associated condition of classification or recommendation.

**1.2.15** *Specially considered* (in connection with close-up surveys and thickness measurements) means sufficient close-up inspection and thickness measurements are taken to confirm the actual average condition of the structure under coating.

### **1.3 Repairs**

**1.3.1** Any damage in association with wastage over the allowable limits (including buckling, grooving, detachment or fracture), or extensive areas of wastage over the allowable limits, which affects or, in the opinion of the Administration, will affect the ship's structural, watertight or weathertight integrity, should be promptly and thoroughly repaired. Areas to be considered include:

- .1 side shell frames, their end attachments or adjacent shell plating;
- .2 deck structure and deck plating;
- .3 bottom structure and bottom plating;
- .4 watertight or oiltight bulkheads; and
- .5 hatch covers and hatch coamings where fitted to combination carriers.

Where adequate repair facilities are not available, the Administration may allow the ship to proceed directly to a repair facility. This may require discharging the cargo and/or temporary repairs for the intended voyage.

**1.3.2** Additionally, when a survey results in the identification of corrosion or structural defects, either of which, in the opinion of the Administration, will impair the ship's fitness for continued service, remedial measures should be implemented before the ship continues in service.

#### **1.4 Surveyors**

For tankers of 20,000 tons deadweight and above, two surveyors should jointly carry out the first scheduled renewal survey after the tanker passes 10 years of age (i.e. third renewal survey), and all subsequent renewal surveys and intermediate surveys. If the surveys are carried out by a recognized organization, the surveyors should be exclusively employed by such recognized organizations.

#### **1.5 Thickness measurements and close-up surveys**

In any kind of survey, i.e. renewal, intermediate, annual or other surveys having the scope of the foregoing ones, thickness measurements of structures in areas where close-up surveys are required should be carried out simultaneously with close-up surveys.

### **2 Renewal survey**

#### **2.1 General**

**2.1.1** The renewal survey may be commenced at the fourth annual survey and be progressed during the succeeding year with a view to completion by the fifth anniversary date.

**2.1.2** As part of the preparation for the renewal survey, the survey programme should be dealt with, in advance of the renewal survey. The thickness measurement should not be held before the fourth annual survey.

**2.1.3** The survey should include, in addition to the requirements of the annual survey, examination, tests and checks of sufficient extent to ensure that the hull and related piping as required in 2.1.5 is in a satisfactory condition and is fit for its intended purpose for the new period of validity of the Cargo Ship Safety Construction Certificate, subject to proper maintenance and operation and to periodical surveys being carried out at the due dates.

**2.1.4** All cargo tanks, ballast tanks, and any other tanks within the cargo areas which are forming hull structures, pump-rooms, pipe tunnels, cofferdams and void spaces bounding cargo tanks, decks and outer hull should be examined, and this examination should be

supplemented by thickness measurement and testing as required in 2.5 and 2.6, to ensure that the structural integrity remains effective. The aim of the examination is to discover substantial corrosion, significant deformation, fractures, damages or other structural deterioration that may be present.

**2.1.5** Cargo piping on deck, including crude oil washing (COW) piping, and cargo and ballast piping within the above tanks and spaces should be examined and operationally tested to working pressure to attending surveyor's satisfaction to ensure that tightness and condition remain satisfactory. Special attention should be given to any ballast piping in cargo tanks and any cargo piping in any ballast tanks and void spaces, and surveyors should be advised on all occasions when this piping, including valves and fittings, are open during repair periods and can be examined internally.

**2.1.6** The survey extent of combined ballast/cargo tanks should be evaluated based on the records of ballast history and extent of the corrosion prevention system provided.

## **2.2      *Dry dock survey***

**2.2.1** A survey in dry dock should be a part of the renewal survey. There should be a minimum of two inspections of the outside of the ship's bottom during the five-year period of the Cargo Ship Safety Construction Certificate. In all cases, the maximum interval between bottom inspections should not exceed 36 months.

**2.2.2** For ships of 15 years of age and over, inspection of the outside of the ship's bottom should be carried out with the ship in dry dock. For ships of less than 15 years of age, alternate inspections of the ship's bottom not conducted in conjunction with the renewal survey may be carried out with the ship afloat. Inspection of the ship afloat should only be carried out when the conditions are satisfactory and the proper equipment and suitably qualified staff is available.

**2.2.3** If a survey in dry dock is not completed in conjunction with the renewal survey or if the 36 month maximum interval referred to in 2.2.1 is not complied with, the Cargo Ship Safety Construction Certificate should cease to be valid until a survey in dry dock is completed.

**2.2.4** The overall and close-up surveys and thickness measurements, as applicable, of the lower portions of the cargo tanks and ballast tanks should be carried out in accordance with the applicable requirements for renewal surveys, if not already performed.

**Note:** Lower portions of the cargo and ballast tanks are considered to be the parts below light ballast water line.

## **2.3      *Tank corrosion prevention system***

Where provided, the condition of the corrosion prevention system of cargo tanks should be examined. A ballast tank should be examined at annual intervals where:

- .1** a hard protective coating has not been applied from the time of construction; or
- .2** a soft or semi-hard coating has been applied; or
- .3** substantial corrosion is found within the tank; or

- .4 the hard protective coating is found to be in less than GOOD condition and the hard protective coating is not repaired to the satisfaction of the surveyor.

Thickness measurements should be carried out as deemed necessary by the surveyor.

## **2.4      *Extent of overall and close-up surveys***

**2.4.1** An overall survey of all tanks and spaces should be carried out at the renewal survey. Suspect areas identified at previous surveys should be examined.

**2.4.2** The minimum requirements for close-up surveys at the renewal survey are given in annex 1.

**2.4.3** The surveyor may extend the scope of the close-up survey as deemed necessary taking into account the maintenance of the tanks under survey, the condition of the corrosion prevention system and also in the following cases:

- .1 in particular, tanks having structural arrangements or details which have suffered defects in similar tanks or on similar ships according to available information; and
- .2 in tanks which have structures with reduced scantlings in association with a corrosion prevention system approved by the Administration.

**2.4.4** For areas in tanks where hard protective coatings are found to be in GOOD condition as defined in 1.2.10, the extent of close-up surveys according to annex 1 may be specially considered by the Administration.

## **2.5      *Extent of thickness measurements***

**2.5.1** The minimum requirements for thickness measurements at the renewal survey are given in annex 2.

**2.5.2** Provisions for extended measurements for areas with substantial corrosion are given in annex 4, and as may be additionally specified in the survey programme as required in 5.1. These extended thickness measurements should be carried out before the survey is credited as completed. Suspect areas identified at previous surveys should have thickness measurements taken.

**2.5.3** The surveyor may further extend the thickness measurements as deemed necessary.

**2.5.4** For areas in tanks where hard protective coatings are found to be in GOOD condition as defined in 1.2.10, the extent of thickness measurements according to annex 2 may be specially considered by the Administration.

**2.5.5** Transverse sections should be chosen where the largest reductions are suspected to occur or are revealed from deck plating measurements.

**2.5.6** In cases where two or three sections should be measured, at least one should include a ballast tank within 0.5L amidships. In case of oil tankers of 130 m in length and upwards (as defined in the International Convention on Load Lines in force) and more than 10 years of age, for the evaluation of the ship's longitudinal strength as required in 8.1.2, the sampling method of thickness measurements is given in annex 12.

## **2.6      *Extent of tank pressure testing***

**2.6.1**    The minimum requirements for tank pressure testing at the renewal survey are given in annex 3.

**2.6.2**    The surveyor may extend the tank pressure testing as deemed necessary.

**2.6.3**    Boundaries of ballast tanks should be tested with a head of liquid to the top of air pipes.

**2.6.4**    Boundaries of cargo tanks should be tested to the highest point that liquid will rise under service conditions.

## **3          *Annual survey***

### **3.1      *General***

The annual survey should consist of an examination for the purpose of ensuring, as far as practicable, that the hull and piping are maintained in a satisfactory condition and should take into account the service history, condition and extent of the corrosion prevention system of ballast tanks and areas identified in the survey report file.

### **3.2      *Examination of the hull***

**3.2.1**    Examination of the hull plating and its closing appliances should be carried out as far as can be seen.

**3.2.2**    Examination of watertight penetrations should be carried out as far as practicable.

### **3.3      *Examination of weather decks***

**3.3.1**    Examination of cargo tank openings including gaskets, covers, coamings and flame screens.

**3.3.2**    Examination of cargo tank pressure/vacuum valves and flame screens.

**3.3.3**    Examination of flame screens on vents to all bunker tanks.

**3.3.4**    Examination of cargo, crude oil washing, bunker and vent piping systems, including vent masts and headers.

### **3.4      *Examination of cargo pump-rooms and pipe tunnels if fitted***

**3.4.1**    Examination of all pump-room bulkheads for signs of oil leakage or fractures and, in particular, the sealing arrangements of all penetrations of pump-room bulkheads.

**3.4.2**    Examination of the condition of all piping systems and pipe tunnels.

### **3.5      *Examination of ballast tanks***

**3.5.1**    Examination of ballast tanks should be carried out when required as a consequence of the results of the renewal survey and intermediate survey. When considered necessary by the Administration or when extensive corrosion is found, thickness measurements should be carried out.

**3.5.2** Where substantial corrosion as defined in 1.2.8 is found, the extent of thickness measurements should be increased in accordance with the requirements in annex 4. These extended thickness measurements should be carried out before the survey is credited as completed. Suspect areas identified at previous surveys should be examined. Areas of substantial corrosion identified at previous surveys should have thickness measurements taken.

## **4 Intermediate survey**

### **4.1 General**

**4.1.1** Items that are additional to the requirements of the annual survey may be surveyed either at the second or third annual survey or between these surveys.

**4.1.2** The survey extent of cargo and ballast tanks dependent on the age of the ship is specified in 4.2, 4.3 and 4.4.

**4.1.3** For weather decks, an examination as far as applicable of cargo, crude oil washing, bunker, ballast, steam and vent piping systems as well as vent masts and headers should be carried out. If upon examination there is any doubt as to the condition of the piping, the piping may be required to be pressure tested, thickness measured or both.

**4.1.4** Concurrent crediting to both intermediate survey and renewal survey for surveys and thickness measurements of spaces should not be acceptable.

### **4.2 Oil tankers 5 to 10 years of age**

**4.2.1** The requirements of 4.1.3 apply.

**4.2.2** All ballast tanks should be examined. When considered necessary by the Administration, thickness measurement and testing should be carried out to ensure that the structural integrity remains effective.

**4.2.3** A ballast tank should be examined at subsequent annual intervals where:

- .1** a hard protective coating has not been applied from the time of construction; or
- .2** a soft or semi-hard coating has been applied; or
- .3** substantial corrosion is found within the tanks; or
- .4** the hard protective coating is found to be in less than GOOD condition and the hard protective coating is not repaired to the satisfaction of the surveyor.

**4.2.4** In addition to the requirements above, suspect areas identified at previous surveys should be examined.

### **4.3 Oil tankers 10 to 15 years of age**

**4.3.1** The requirements of the intermediate survey should be to the same extent as the previous renewal survey as required in 2 and 5.1. However, pressure testing of cargo and ballast tanks and the requirements for longitudinal strength evaluation of hull girder as required in 8.1.2 are not required unless deemed necessary by the Administration.

**4.3.2** In application of 4.3.1, the intermediate survey may be commenced at the second annual survey and be progressed during the succeeding year with a view to completion at the third annual survey in lieu of the application of 2.1.1.

**4.3.3** In application of 4.3.1, an underwater survey may be considered in lieu of the requirements of 2.2.

#### **4.4      *Oil tankers exceeding 15 years of age***

**4.4.1** The requirements of the intermediate survey should be to the same extent as the previous renewal survey as required in 2 and 5.1. However, pressure testing of cargo and ballast tanks and the requirements for longitudinal strength evaluation of hull girder as required in 8.1.2 are not required unless deemed necessary by the Administration.

**4.4.2** In application of 4.4.1, the intermediate survey may be commenced at the second annual survey and be progressed during the succeeding year with a view to completion at the third annual survey in lieu of the application of 2.1.1.

**4.4.3** In application of 4.4.1 a survey in dry dock should be part of the intermediate survey. The overall and close-up surveys and thickness measurements, as applicable, of the lower portions of the cargo tanks and water ballast tanks should be carried out in accordance with the applicable requirements for intermediate surveys, if not already performed.

**Note:** Lower portions of the cargo and ballast tanks are considered to be the parts below the light ballast water line.

### **5          Preparations for survey**

#### **5.1      *Survey programme***

**5.1.1** The owner in cooperation with the Administration or organization recognized by the Administration should work out a specific survey programme prior to the commencement of any part of:

- .1**          the renewal survey, and
- .2**          the intermediate survey for oil tanker over 10 years of age.

The survey programme at intermediate survey may consist of the survey programme at the previous renewal survey supplemented by the condition evaluation report of that renewal survey and later relevant survey reports. The survey programme should be worked out taking into account any amendments to the survey requirements implemented after the last renewal survey carried out. The survey programme should be in a written format based on the information in annex 6A. The survey should not commence until the survey programme has been agreed.

**5.1.1.1** Prior to the development of the survey programme, the survey planning questionnaire should be completed by the owner based on the information set out in annex 6B, and forwarded to the Administration.

**5.1.2** In developing the survey programme, the following documentation should be collected and consulted with a view to selecting tanks, areas, and structural elements to be examined:

- .1 survey status and basic ship information;
- .2 documentation on board, as described in 6.2 and 6.3;
- .3 main structural plans of cargo and ballast tanks (scantlings drawings), including information regarding use of high-tensile steels (HTS);
- .4 Condition Evaluation Report, according to annex 9;
- .5 relevant previous damage and repair history;
- .6 relevant previous survey and inspection reports from both the recognized organization and the owner;
- .7 cargo and ballast history for the last 3 years, including carriage of cargo under heated conditions;
- .8 details of the inert gas plant and tank cleaning procedures;
- .9 information and other relevant data regarding conversion or modification of the ship's cargo and ballast tanks since the time of construction;
- .10 description and history of the coating and corrosion protection system (including anodes and previous class notations), if any;
- .11 inspections of the owner's personnel during the last three years with reference to structural deterioration in general, leakages in tank boundaries and piping and condition of the coating and corrosion protection system (including anodes) if any. Guidance for reporting is shown in annex 5;
- .12 information regarding the relevant maintenance level during operation, including port State control reports of inspection containing hull related deficiencies, safety management system non-conformities relating to hull maintenance, including the associated corrective action(s); and
- .13 any other information that will help identify suspect areas and critical structural areas.

**5.1.3** The submitted survey programme should account for and comply, as a minimum, with the requirements of 2.6 and annexes 1, 2 and 3 for close-up survey, thickness measurement and tank testing, respectively, and should include relevant information including at least:

- .1 basic ship information and particulars;
- .2 main structural plans of cargo and ballast tanks (scantling drawings), including information regarding use of high tensile steels (HTS);
- .3 plan of tanks;
- .4 list of tanks with information on their use, corrosion prevention system and condition of coating;



- .5 conditions for survey (e.g., information regarding tank cleaning, gas freeing, ventilation, lighting, etc.);
- .6 provisions and methods for access to structures;
- .7 equipment for surveys;
- .8 identification of tanks and areas for close-up survey (see 2.4);
- .9 identification of areas and sections for thickness measurement (see 2.5);
- .10 identification of tanks for tank testing (see 2.6);
- .11 identification of the thickness measurement company;
- .12 damage experience related to the ship in question; and
- .13 critical structural areas and suspect areas, where relevant.

**5.1.4** The Administration will advise the owner of the maximum acceptable structural corrosion diminution levels applicable to the ship.

**5.1.5** Use may also be made of the Guidelines for technical assessment in conjunction with the planning of enhanced surveys for tankers, contained in annex 11. These Guidelines are a recommended tool which may be invoked at the discretion of the Administration, when considered necessary and appropriate, in conjunction with the preparation of the required survey programme.

## **5.2 Conditions for survey**

**5.2.1** The owner should provide the necessary facilities for a safe execution of the survey.<sup>16</sup>

**5.2.1.1** In order to enable the attending surveyors to carry out the survey, provisions for proper and safe access should be agreed between the owner and the Administration.

**5.2.1.2** Details of the means of access should be provided in the survey planning questionnaire.

**5.2.1.3** In cases where the provisions of safety and required access are judged by the attending surveyors not to be adequate, the survey of the spaces involved should not proceed.

**5.2.2** Tanks and spaces should be safe for access. Tanks and spaces should be gas-free and properly ventilated. Prior to entering tank, void or enclosed space, it should be verified that the atmosphere in that space is free from hazardous gas and contains sufficient oxygen.

**5.2.3** In preparation for survey and thickness measurements and to allow for a thorough examination, all spaces should be cleaned including removal from surfaces of all loose accumulated corrosion scale. Spaces should be sufficiently clean and free from water, scale, dirt, oil residues, etc., to reveal corrosion, deformation, fractures, damages or other structural

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<sup>16</sup> Refer to chapter 10 of the International Safety Guide for Oil Tankers and Terminals (ISGOTT), Entry into and working in enclosed spaces.

deterioration as well as the condition of the coating. However, those areas of structure whose renewal has already been decided by the owner need only be cleaned and descaled to the extent necessary to determine the limits of the areas to be renewed.

**5.2.4** Sufficient illumination should be provided to reveal corrosion, deformation, fractures, damages or other structural deterioration as well as the condition of the coating.

**5.2.5** Where soft or semi-hard coatings have been applied, safe access should be provided for the surveyor to verify the effectiveness of the coating and to carry out an assessment of the condition of internal structures which may include spot removal of the coating. When safe access cannot be provided, the soft or semi-hard coating should be removed.

**5.2.6** The surveyor(s) should always be accompanied by at least one responsible person, assigned by the owner, experienced in tank and enclosed spaces inspection. In addition a backup team of at least two experienced persons should be stationed at the hatch opening of the tank or space that is being surveyed. The back-up team should continuously observe the work in the tank or space and should keep life-saving and evacuation equipment ready for use.

**5.2.7** A communication system should be arranged between the survey party in the tank or space being examined, the responsible officer on deck and, as the case may be, the navigation bridge. The communication arrangements should be maintained throughout the survey.

### **5.3 Access to structures<sup>17</sup>**

**5.3.1** For overall survey, means should be provided to enable the surveyor to examine the structure in a safe and practical way.

**5.3.2** For close-up surveys, one or more of the following means for access, acceptable to the surveyor, should be provided:

- .1 permanent staging and passages through structures;
- .2 temporary staging and passages through structures;
- .3 lifts and moveable platforms;
- .4 boats or rafts;
- .5 portable ladders; and/or
- .6 other equivalent means.

### **5.4 Equipment for survey**

**5.4.1** Thickness measurement should normally be carried out by means of ultrasonic test equipment. The accuracy of the equipment should be proven to the surveyor as required.

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<sup>17</sup> Refer to MSC/Circ.686, Guidelines on the means of access to structures for inspection and maintenance of oil tankers and bulk carriers.

**5.4.2** One or more of the following fracture detection procedures may be required if deemed necessary by the surveyor:

- .1 radiographic equipment;
- .2 ultrasonic equipment;
- .3 magnetic particle equipment;
- .4 dye penetrant; and/or
- .5 other equivalent means.

**5.4.3** Explosimeter, oxygen-meter, breathing apparatus, lifelines, riding belts with rope and hook and whistles together with instructions and guidance on their use should be made available during the survey. A safety checklist should be provided.

**5.4.4** Adequate and safe lighting should be provided for the safe and efficient conduct of the survey.

**5.4.5** Adequate protective clothing should be made available and used (e.g., safety helmet, gloves, safety shoes, etc.) during the survey.

## **5.5 Surveys at sea or at anchorage**

**5.5.1** Surveys at sea or at anchorage may be accepted provided the surveyor(s) is given the necessary assistance from the personnel on board. Necessary precautions and procedures for carrying out the survey should be in accordance with 5.1, 5.2, 5.3 and 5.4.

**5.5.2** A communication system should be arranged between the survey party in the tank and the responsible officer on deck. This system should also include the personnel in charge of ballast pump handling if boats or rafts are used.

**5.5.3** Surveys of tanks by means of boats or rafts may only be undertaken with the agreement of the surveyor, who should take into account the safety arrangements provided, including weather forecasting and ship response under foreseeable conditions and provided the expected rise of water within the tank does not exceed 0.25 m.

**5.5.4** When rafts or boats are used for close-up survey the following conditions should be observed:

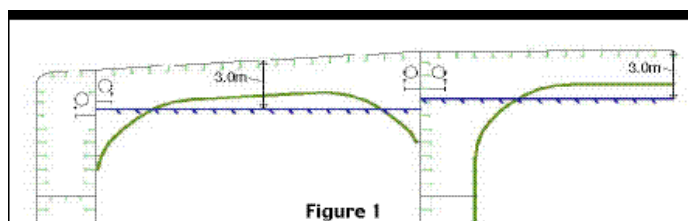
- .1 only rough duty, inflatable rafts or boats, having satisfactory residual buoyancy and stability even if one chamber is ruptured, should be used;
- .2 the boat or raft should be tethered to the access ladder and an additional person should be stationed down the access ladder with a clear view of the boat or raft;
- .3 appropriate lifejackets should be available for all participants;
- .4 the surface of water in the tank should be calm (under all foreseeable conditions the expected rise of water within the tank should not exceed 0.25 m) and the water level stationary. On no account should the level of the water be rising while the boat or raft is in use;

- .5 the tank or space must contain clean ballast water only. Even a thin sheen of oil on the water is not acceptable;
- .6 at no time should the water level be allowed to be within 1 m of the deepest under-deck web face flat so that the survey team is not isolated from a direct escape route to the tank hatch. Filling to levels above the deck transverses should only be contemplated if a deck access manhole is fitted and open in the bay being examined, so that an escape route for the survey party is available at all times. Other effective means of escape to the deck may be considered; and
- .7 if the tanks (or spaces) are connected by a common venting system, or inert gas system, the tank in which the boat or raft should be used should be isolated to prevent a transfer of gas from other tanks (or spaces).

**5.5.5** Rafts or boats alone may be allowed for inspection of the underdeck areas of tanks or spaces if the depth of the webs is 1.5 m or less.

**5.5.6** If the depth of the webs is more than 1.5 m, rafts or boats alone may be allowed only:

- .1 when the coating of the under-deck structure is in GOOD condition and there is no evidence of wastage; or
- .2 if a permanent means of access is provided in each bay to allow safe entry and exit. This means:
  - .1 access direct from the deck via a vertical ladder with a small platform fitted approximately 2 m below the deck; and
  - .2 access to deck from a longitudinal permanent platform having ladders to deck in each end of the tank. The platform should, for the full length of the tank, be arranged in level with, or above, the maximum water level needed for rafting of under deck structure. For this purpose, the ullage corresponding to the maximum water level should be assumed not more than 3 m from the deck plate measured at the midspan of deck transverses and in the middle length of the tanks (see figure 1).



If neither of the above conditions are met, then staging or other equivalent means should be provided for the survey of the under-deck areas.

**5.5.7** The use of rafts or boats alone in 5.5.5 and 5.5.6 does not preclude the use of boats or rafts to move about within a tank during a survey.

## **5.6      *Survey planning meeting***

**5.6.1** Proper preparation and close co-operation between the attending surveyor(s) and the owner's representatives on board prior to and during the survey are an essential part in the safe and efficient conduct of the survey. During the survey onboard safety meetings should be held regularly.

**5.6.2** Prior to commencement of any part of the renewal and intermediate survey, a survey planning meeting should be held between the attending surveyor(s), the owner's representative in attendance, the thickness measurement company operator (as applicable) and the master of the ship or an appropriately qualified representative nominated by the master or company for the purpose to ascertain that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the survey work to be carried out.

**5.6.3** The following is an indicative list of items that should be addressed in the meeting:

- .1      schedule of the ship (i.e. the voyage, docking and undocking manoeuvres, periods alongside, cargo and ballast operations, etc.);
- .2      provisions and arrangements for thickness measurements (i.e. access, cleaning/de-scaling, illumination, ventilation, personal safety);
- .3      extent of the thickness measurements;
- .4      acceptance criteria (refer to the list of minimum thicknesses);
- .5      extent of close-up survey and thickness measurement considering the coating condition and suspect areas/areas of substantial corrosion;
- .6      execution of thickness measurements;
- .7      taking representative readings in general and where uneven corrosion/pitting is found;
- .8      mapping of areas of substantial corrosion; and
- .9      communication between attending surveyor(s), the thickness measurement company operator(s), and owner representative(s) concerning findings.

## **6          *Documentation on board***

### **6.1      *General***

**6.1.1** The owner should obtain, supply and maintain on board the ship documentation as specified in 6.2 and 6.3, which should be readily available for the surveyor. The condition evaluation report referred to in 6.2 should include a translation into English.

**6.1.2** The documentation should be kept on board for the lifetime of the ship.

## **6.2      *Survey report file***

**6.2.1**    A survey report file should be a part of the documentation on board consisting of:

- .1          reports of structural surveys (annex 8);
- .2          condition evaluation report (annex 9); and
- .3          thickness measurement reports (annex 10).

**6.2.2**    The survey report file should be available also in the owner's and the Administration offices, or in the office of the organization recognized by the Administration.

## **6.3      *Supporting documents***

The following additional documentation should be available on board:

- .1          main structural plans of cargo and ballast tanks;
- .2          previous repair history;
- .3          cargo and ballast history;
- .4          extent of use of inert gas plant and tank cleaning procedures;
- .5          inspections by ship's personnel with reference to:
  - .1          structural deterioration in general;
  - .2          leakages in bulkheads and piping; and
  - .3          condition of coating or corrosion prevention system, if any. Guidance for reporting is shown in annex 5; and
- .6          survey programme as required by 5.1 until such time as the renewal survey has been completed; and
- .7          any other information that would help to identify critical structural areas and/or suspect areas requiring inspection.

## **6.4      *Review of documentation on board***

Prior to survey, the surveyor should examine the completeness of the documentation on board and its contents as a basis for the survey.

## **7          *Procedures for thickness measurements***

### **7.1      *General***

**7.1.1**    The required thickness measurements, if not carried out by the recognized organization acting on behalf of the Administration, should be witnessed by a surveyor of the recognized organization. The surveyor should be on board to the extent necessary to control the process.

**7.1.2** The thickness measurement company should be part of the survey planning meeting to be held prior to commencing the survey.

**7.1.3** Thickness measurements of structures in areas where close-up surveys are required should be carried out simultaneously with close-up surveys.

**7.1.4** In all cases the extent of the thickness measurements should be sufficient as to represent the actual average condition.

## **7.2      *Certification of thickness measurement company***

The thickness measurements should be carried out by a qualified company certified by an organization recognized by the Administration according to principles stated in annex 7.

## **7.3      *Reporting***

**7.3.1** A thickness measurement report should be prepared and submitted to the Administration. The report should give the location of measurements, the thickness measured as well as corresponding original thickness. Furthermore, the report should give the date when the measurements were carried out, type of measuring equipment, names of personnel and their qualifications and be signed by the operator. The thickness measurement report should follow the principles as specified in the recommended procedures for thickness measurements set out in annex 10.

**7.3.2** The surveyor should review the final thickness measurement report and countersign the cover page.

## **8          *Reporting and evaluation of survey***

### **8.1      *Evaluation of survey report***

**8.1.1** The data and information on the structural condition of the ship collected during the survey should be evaluated for acceptability and continued structural integrity of the ship.

**8.1.2** In case of oil tankers of 130 m in length and upwards (as defined in the International Convention on Load Lines in force), the ship's longitudinal strength should be evaluated by using the thickness of structural members measured, renewed and reinforced, as appropriate, during the renewal survey of safety construction carried out after the ship reached 10 years of age, in accordance with the criteria for longitudinal strength of the ship's hull girder for oil tankers specified in annex 12.

**8.1.3** The analysis of data should be carried out and endorsed by the Administration or recognized organization authorized by the Administration and the conclusions of the analysis should form a part of the condition evaluation report.

**8.1.4** The final result of the evaluation of the ship's longitudinal strength required in 8.1.2, after renewal or reinforcement work of structural members, if carried out as a result of initial evaluation, should be reported as a part of the condition evaluation report.

### **8.2      *Reporting***

**8.2.1** Principles for survey reporting are shown in annex 8.

**8.2.2** When a survey is split between different survey stations, a report should be made for each portion of the survey. A list of items examined and/or tested (pressure-testing, thickness measurements, etc.) and an indication of whether the item has been credited, should be made available to the next attending surveyor(s), prior to continuing or completing the survey.

**8.2.3** A condition evaluation report of the survey and results should be issued to the owner as shown in annex 9 and placed on board the ship for reference at future surveys. The condition evaluation report should be endorsed by the Administration or recognized organization authorized by the Administration.



# ANNEX 1

## REQUIREMENTS FOR CLOSE-UP SURVEY AT RENEWAL SURVEYS

| <b>Age ≤ 5 years</b><br><b>Renewal Survey No.</b><br><b>1</b>                                                                                                                                                                                                                                                                                       | <b>5 &lt; Age ≤ 10 years</b><br><b>Renewal Survey No.</b><br><b>2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>10 &lt; Age ≤ 15 years</b><br><b>Renewal Survey No.</b><br><b>3</b>                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Age &gt; 15 years</b><br><b>Renewal Survey No.</b><br><b>4 and subsequent</b>                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <p>(A) ONE WEB FRAME RING – in a ballast wing tank, if any, or a cargo wing tank used primarily for water ballast</p> <p>(B) ONE DECK TRANSVERSE – in a cargo tank</p> <p>(D) ONE TRANSVERSE BULKHEAD – in a ballast tank</p> <p>(D) ONE TRANSVERSE BULKHEAD – in a cargo wing tank</p> <p>(D) ONE TRANSVERSE BULKHEAD – in a cargo centre tank</p> | <p>(A) ALL WEB FRAME RINGS – in a ballast wing tank, if any, or a cargo wing tank used primarily for water ballast</p> <p>(B) ONE DECK TRANSVERSE – in each of the remaining ballast tanks, if any</p> <p>(B) ONE DECK TRANSVERSE – in a cargo wing tank</p> <p>(B) ONE DECK TRANSVERSE – in two cargo centre tanks</p> <p>(C) BOTH TRANSVERSE BULKHEADS in a wing ballast tank, if any, or a cargo wing tank used primarily for water ballast</p> <p>(D) ONE TRANSVERSE BULKHEAD – in each remaining ballast tank</p> <p>(D) ONE TRANSVERSE BULKHEAD – in a cargo wing tank</p> <p>(D) ONE TRANSVERSE BULKHEAD – in two cargo centre tanks</p> | <p>(A) ALL WEB FRAME RINGS – in all ballast tanks</p> <p>(A) ALL WEB FRAME RINGS – in a cargo wing tank</p> <p>(A) A minimum of 30% of all web frame rings in each remaining cargo wing tank (see note 1)</p> <p>(C) ALL TRANSVERSE BULKHEADS – in all cargo and ballast tanks</p> <p>(E) A minimum of 30% of deck and bottom transverses including adjacent structural members in each cargo centre tank (see note 1)</p> <p>(F) As considered necessary by the Administration</p> | <p>As for ships referred to in column 3</p> <p>Additional transverses included as deemed necessary by the Administration</p> |

**Note 1:**

The 30% should be rounded up to the next whole integer.

- (A) Complete transverse web frame ring including adjacent structural members
- (B) Deck transverse including adjacent deck structural members
- (C) Transverse bulkheads complete – including girder system and adjacent members
- (D) Transverse bulkhead lower part– including girder system and adjacent structural members
- (E) Deck and bottom transverse including adjacent structural members
- (F) Additional complete transverse web ring frame

## ANNEX 2

### REQUIREMENTS FOR THICKNESS MEASUREMENTS AT RENEWAL SURVEYS

| <b>Age ≤ 5 years</b><br><b>1</b>                                                                                                                                                                                                                                                                                                                            | <b>5 &lt; Age ≤ 10 years</b><br><b>2</b>                                                                                                                                                                                                                                                                                           | <b>10 &lt; Age ≤ 15 years</b><br><b>3</b>                                                                                                                                                                                                                                                                                                                                          | <b>Age &gt; 15 years</b><br><b>4</b>                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1 One section of deck plating for the full beam of the ship within the cargo area (in way of a ballast tank, if any, or a cargo tank used primarily for water ballast)</p> <p>2 Measurements of structural members subject to close-up survey according to annex 1, for general assessment and recording of corrosion pattern</p> <p>3 Suspect areas</p> | <p>1 Within the cargo area:<br/>.1 each deck plate<br/>.2 one transverse section</p> <p>2 Measurements of structural members subject to close-up survey according to annex 1, for general assessment and recording of corrosion pattern</p> <p>3 Suspect areas</p> <p>4 Selected wind and water strakes outside the cargo area</p> | <p>1 Within the cargo area:<br/>.1 each deck plate<br/>.2 two transverse sections<sup>18</sup><br/>.3 all wind and water strakes</p> <p>2 Measurements of structural members subject to close-up survey according to annex 1, for general assessment and recording of corrosion pattern</p> <p>3 Suspect areas</p> <p>4 Selected wind and water strakes outside the cargo area</p> | <p>1 Within the cargo area:<br/>.1 each deck plate<br/>.2 three transverse sections<br/>.3 each bottom plate</p> <p>2 Measurements of structural members subject to close-up survey according to annex 1, for general assessment and recording of corrosion pattern</p> <p>3 Suspect areas</p> <p>4 All wind and water strakes in full length</p> |

<sup>18</sup> At least one section should include a ballast tank within 0.5 L amidships.

**ANNEX 3**

**REQUIREMENTS FOR TANK PRESSURE TESTING AT RENEWAL SURVEYS**

| <b>Age ≤ 5 years</b> |                                                                                                                 | <b>Age &gt; 5 years</b>                                     |  |
|----------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|--|
| <b>1</b>             |                                                                                                                 | <b>2</b>                                                    |  |
| 1                    | Cargo tank boundaries facing ballast tanks, void spaces, pipe tunnels, fuel oil tanks, pump-rooms or cofferdams | 1 All ballast tank boundaries<br>2 All cargo tank bulkheads |  |
| 2                    | All ballast tank boundaries.                                                                                    |                                                             |  |

## ANNEX 4

### REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT AREAS OF SUBSTANTIAL CORROSION

#### Renewal survey within the cargo area

##### Bottom structure

| Structural member             | Extent of measurement                                                                               | Pattern of measurement                                                                                                                                                                                                       |
|-------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 Bottom plating              | Minimum of three bays across tank, including aft bay. Measurements around and under all bell mouths | Five-point pattern for each panel between longitudinals and webs                                                                                                                                                             |
| 2 Bottom longitudinals        | Minimum of three longitudinals in each bay where bottom plating measured                            | Three measurements in line across flange and three measurements on vertical web                                                                                                                                              |
| 3 Bottom girders and brackets | At fore and aft transverse bulkhead bracket toes and in centre of tanks                             | Vertical line of single measurements on web plating with one measurement between each panel stiffener, or a minimum of three measurements. Two measurements across face flat. Five-point pattern on girder/bulkhead brackets |
| 4 Bottom transverse webs      | Three webs in bays where bottom plating measured, with measurements at both ends and middle         | Five-point pattern over 2 m <sup>2</sup> area. Single measurements on face flat.                                                                                                                                             |
| 5 Panel stiffening            | Where fitted                                                                                        | Single measurements                                                                                                                                                                                                          |

##### Deck structure

| Structural member           | Extent of measurement                                                    | Pattern of measurement                                                                                                                                                                                                       |
|-----------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 Deck plating              | Two bands across tank                                                    | Minimum of three measurements per plate per band                                                                                                                                                                             |
| 2 Deck longitudinals        | Minimum of three longitudinals in each of two bays                       | Three measurements in line vertically on webs, and two measurements on flange (if fitted)                                                                                                                                    |
| 3 Deck girders and brackets | At fore and aft transverse bulkhead, bracket toes and in centre of tanks | Vertical line of single measurements on web plating with one measurement between each panel stiffener, or a minimum of three measurements. Two measurements across face flat. Five-point pattern on girder/bulkhead brackets |
| 4 Deck transverse webs      | Minimum of two webs with measurements at middle and both ends of span    | Five-point pattern over about 2 m <sup>2</sup> area. Single measurements on face flat                                                                                                                                        |
| 5 Panel stiffening          | Where available                                                          | Single measurements                                                                                                                                                                                                          |

### Shell and longitudinal bulkheads

| Structural member                                                       | Extent of measurement                                                                             | Pattern of measurement                                                                                              |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| 1 Deckhead and bottom strakes, and strakes in way of stringer platforms | Plating between each pair of longitudinals in a minimum of three bays                             | Single measurement                                                                                                  |
| 2 All other strakes                                                     | Plating between every third pair of longitudinals in same three bays                              | Single measurement                                                                                                  |
| 3 Longitudinals – deckhead and bottom strakes                           | Each longitudinal in same three bays                                                              | Three measurements across web and one measurement on flange                                                         |
| 4 Longitudinals – all others                                            | Every third longitudinal in same three bays                                                       | Three measurements across web and one measurement on flange                                                         |
| 5 Longitudinals – bracket                                               | Minimum of three at top, middle and bottom of tank in same three bays                             | Five-point pattern over area of bracket                                                                             |
| 6 Web frames and cross ties                                             | Three webs with minimum of three locations on each web, including in way of cross tie connections | Five-point pattern over about 2 m <sup>2</sup> area, plus single measurements on web frame and cross tie face flats |

### Transverse bulkheads and swash bulkheads

| Structural member                                                       | Extent of measurement                                                                                                | Pattern of measurement                                                                                                                                                                                                           |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 Deckhead and bottom strakes, and strakes in way of stringer platforms | Plating between pair of stiffeners at three locations – approximately quarter, half and three-quarters width of tank | Five-point pattern between stiffeners over 1 m length                                                                                                                                                                            |
| 2 All other strakes                                                     | Plating between pair of stiffeners at middle location                                                                | Single measurement                                                                                                                                                                                                               |
| 3 Strakes in corrugated bulkheads                                       | Plating for each change of scantling at centre of panel and at flange or fabricated connection                       | Five-point pattern over about 1 m <sup>2</sup> of plating                                                                                                                                                                        |
| 4 Stiffeners                                                            | Minimum of three typical stiffeners                                                                                  | For web, five-point pattern over span between bracket connections (two measurements across web at each bracket connection, and one at centre of span). For flange, single measurements at each bracket toe and at centre of span |
| 5 Brackets                                                              | Minimum of three at top, middle and bottom of tank                                                                   | Five-point pattern over area of bracket                                                                                                                                                                                          |
| 6 Deep webs and girders                                                 | Measurements at toe of bracket and centre of span                                                                    | For web, five-point pattern over about 1 m <sup>2</sup> area.<br>Three measurements across face flat                                                                                                                             |
| 7 Stringer platforms                                                    | All stringers with measurements at both ends and middle                                                              | Five-point pattern over 1 m <sup>2</sup> area plus single measurements near bracket toes and on face flats                                                                                                                       |

## ANNEX 5

### OWNER'S INSPECTION REPORT

|                                                   |        |         |           |         |         |                     |                 |
|---------------------------------------------------|--------|---------|-----------|---------|---------|---------------------|-----------------|
| Structural condition                              |        |         |           |         |         |                     |                 |
| Ship's name: .....                                |        |         |           |         |         |                     |                 |
| Owners's inspection report – Structural condition |        |         |           |         |         |                     |                 |
| For tank no: .....                                |        |         |           |         |         |                     |                 |
| Grade of steel: deck: ..... side: .....           |        |         |           |         |         |                     |                 |
| bottom: ..... longitudinal bulkhead: .....        |        |         |           |         |         |                     |                 |
| Elements                                          | Cracks | Buckles | Corrosion | Coating | Pitting | Modification/repair | Other condition |
| Deck:                                             |        |         |           |         |         |                     |                 |
| Bottom:                                           |        |         |           |         |         |                     |                 |
| Side:                                             |        |         |           |         |         |                     |                 |
| Longitudinal bulkheads:                           |        |         |           |         |         |                     |                 |
| Transverse bulkheads:                             |        |         |           |         |         |                     |                 |
| Repairs carried out due to:                       |        |         |           |         |         |                     |                 |
| Thickness measurements carried out (dates):       |        |         |           |         |         |                     |                 |
| Results in general:                               |        |         |           |         |         |                     |                 |
| Overdue surveys:                                  |        |         |           |         |         |                     |                 |
| Outstanding conditions of class:                  |        |         |           |         |         |                     |                 |
| Comments:                                         |        |         |           |         |         |                     |                 |
| Date of inspection: .....                         |        |         |           |         |         |                     |                 |
| Inspected by: .....                               |        |         |           |         |         |                     |                 |
| Signature: .....                                  |        |         |           |         |         |                     |                 |

## ANNEX 6A

### SURVEY PROGRAMME

#### Basic information and particulars

|                                    |
|------------------------------------|
| Name of ship:                      |
| IMO number:                        |
| Flag State:                        |
| Port of registry:                  |
| Gross tonnage:                     |
| Deadweight (metric tonnes):        |
| Length between perpendiculars (m): |
| Shipbuilder:                       |
| Hull number:                       |
| Recognized organization (RO):      |
| RO ship identity:                  |
| Date of delivery of the ship:      |
| Owner:                             |
| Thickness measurement company:     |

#### 1 Preamble

##### 1.1 Scope

**1.1.1** The present survey programme covers the minimum extent of overall surveys, close-up surveys, thickness measurements and pressure testing within the cargo area, ballast tanks, including fore and aft peak tanks, required by the Code.

**1.1.2** The arrangements and safety aspects of the survey should be acceptable to the attending surveyor(s).

##### 1.2 Documentation

All documents used in the development of the survey programme should be available on board during the survey as required by section 6.

#### 2 Arrangement of tanks and spaces

This section of the survey programme should provide information (either in the form of plans or text) on the arrangement of tanks and spaces that fall within the scope of the survey.

#### 3 List of tanks and spaces with information on their use, extent of coatings and corrosion prevention system

This section of the survey programme should indicate any changes relating to (and should update) the information on the use of the tanks of the ship, the extent of coatings and the corrosion protective system provided in the survey planning questionnaire.



#### **4 Conditions for survey**

This section of the survey programme should provide information on the conditions for survey, e.g., information regarding cargo hold and tank cleaning, gas freeing, ventilation, lighting, etc.

#### **5 Provisions and method of access to structures**

This section of the survey programme should indicate any changes relating to (and should update) the information on the provisions and methods of access to structures provided in the survey planning questionnaire.

#### **6 List of equipment for survey**

This section of the survey programme should identify and list the equipment that will be made available for carrying out the survey and the required thickness measurements.

#### **7 Survey requirements**

##### **7.1 Overall survey**

This section of the survey programme should identify and list the spaces that should undergo an overall survey for the ship in accordance with 2.4.1.

##### **7.2 Close-up survey**

This section of the survey programme should identify and list the hull structures that should undergo a close-up survey for the ship in accordance with 2.4.2.

#### **8 Identification of tanks for tank testing**

This section of the survey programme should identify and list the tanks that should undergo tank testing for the ship in accordance with 2.6.

#### **9 Identification of areas and sections for thickness measurements**

This section of the survey programme should identify and list the areas and sections where thickness measurements should be taken in accordance with 2.5.1.

#### **10 Minimum thickness of hull structures**

This section of the survey programme should specify the minimum thickness for hull structures of the ship that are subject to the Code (indicate either (a) or preferably (b), if such information is available):

- (a) Determined from the attached wastage allowance table and the original thickness to the hull structure plans of the ship;
- (b) Given in the following table(s):

| Area or location                                   | Original as-built thickness (mm) | Minimum thickness (mm) | Substantial corrosion thickness (mm) |
|----------------------------------------------------|----------------------------------|------------------------|--------------------------------------|
| <b>Deck</b>                                        |                                  |                        |                                      |
| Plating                                            |                                  |                        |                                      |
| Longitudinals                                      |                                  |                        |                                      |
| Longitudinal girders                               |                                  |                        |                                      |
| <b>Bottom</b>                                      |                                  |                        |                                      |
| Plating                                            |                                  |                        |                                      |
| Longitudinals                                      |                                  |                        |                                      |
| Longitudinal girders                               |                                  |                        |                                      |
| <b>Ship side</b>                                   |                                  |                        |                                      |
| Plating                                            |                                  |                        |                                      |
| Longitudinals                                      |                                  |                        |                                      |
| Longitudinal girders                               |                                  |                        |                                      |
| <b>Longitudinal bulkhead</b>                       |                                  |                        |                                      |
| Plating                                            |                                  |                        |                                      |
| Longitudinals                                      |                                  |                        |                                      |
| Longitudinal girders                               |                                  |                        |                                      |
| <b>Inner bottom</b>                                |                                  |                        |                                      |
| Plating                                            |                                  |                        |                                      |
| Longitudinals                                      |                                  |                        |                                      |
| Longitudinal girders                               |                                  |                        |                                      |
| <b>Transverse bulkheads</b>                        |                                  |                        |                                      |
| Plating                                            |                                  |                        |                                      |
| Stiffeners                                         |                                  |                        |                                      |
| <b>Transverse web frames, floors and stringers</b> |                                  |                        |                                      |
| Plating                                            |                                  |                        |                                      |
| Flanges                                            |                                  |                        |                                      |
| Stiffeners                                         |                                  |                        |                                      |
| Cross ties                                         |                                  |                        |                                      |
| Flanges                                            |                                  |                        |                                      |
| Webs                                               |                                  |                        |                                      |

**Note:** The wastage allowance tables should be attached to the survey programme.

## 11 Thickness measurement company

This section of the survey programme should identify changes, if any, relating to the information on the thickness measurement company provided in the survey planning questionnaire.

## 12 Damage experience related to the ship

This section of the survey programme should, using the tables provided below, provide details of the hull damages for at least the last three years in way of the cargo and ballast tanks and void spaces within the cargo area. These damages are subject to survey.

**Hull damages sorted by location for the ship**

| Tank or space number or area | Possible cause, if known | Description of the damages | Location | Repair | Date of repair |
|------------------------------|--------------------------|----------------------------|----------|--------|----------------|
|                              |                          |                            |          |        |                |
|                              |                          |                            |          |        |                |
|                              |                          |                            |          |        |                |
|                              |                          |                            |          |        |                |
|                              |                          |                            |          |        |                |
|                              |                          |                            |          |        |                |
|                              |                          |                            |          |        |                |
|                              |                          |                            |          |        |                |
|                              |                          |                            |          |        |                |

**Hull damages for sister or similar ships (if available) in the case of design related damage**

| Tank or space number or area | Possible cause, if known | Description of the damages | Location | Repair | Date of repair |
|------------------------------|--------------------------|----------------------------|----------|--------|----------------|
|                              |                          |                            |          |        |                |
|                              |                          |                            |          |        |                |
|                              |                          |                            |          |        |                |
|                              |                          |                            |          |        |                |
|                              |                          |                            |          |        |                |
|                              |                          |                            |          |        |                |
|                              |                          |                            |          |        |                |
|                              |                          |                            |          |        |                |
|                              |                          |                            |          |        |                |

**13 Areas identified with substantial corrosion from previous surveys**

This section of the survey programme should identify and list the areas of substantial corrosion from previous surveys.

**14 Critical structural areas and suspect areas**

This section of the survey programme should identify and list the critical structural areas and the suspect areas, if such information is available.

**15 Other relevant comments and information**

This section of the survey programme should provide any other comments and information relevant to the survey.

## **Appendices**

### **Appendix 1 – List of plans**

Paragraph 5.1.3.2 requires that main structural plans of cargo and ballast tanks (scantling drawings), including information on regarding use of high tensile steel (HTS), to be available. This appendix of the survey programme should identify and list the main structural plans which form part of the survey programme.

### **Appendix 2 – Survey planning questionnaire**

The survey planning questionnaire (annex 6B), which has been submitted by the owner, should be appended to the survey programme.

### **Appendix 3 – Other documentation**

This part of the survey programme should identify and list any other documentation that forms part of the plan.

Prepared by the owner in co-operation with the Administration for compliance with 5.1.3.

Date:

*(name and signature of authorized owner's representative)*

Date:

*(name and signature of authorized representative of the Administration)*

## **ANNEX 6B**

### **SURVEY PLANNING QUESTIONNAIRE**

The following information will enable the owner in co-operation with the Administration to develop a survey programme complying with the requirements of the Code. It is essential that the owner provides, when completing the present questionnaire, up-to-date information. The present questionnaire, when completed, should provide all information and material required by the resolution.

#### **Particulars**

Ship's name:  
IMO number:  
Flag State:  
Port of registry:  
Owner:  
Recognized organization (RO):  
Gross tonnage:  
Deadweight (metric tonnes):  
Date of delivery:

#### **Information on access provision for close-up surveys and thickness measurement**

The owner should indicate, in the table below, the means of access to the structures subject to close-up survey and thickness measurement. A close-up survey is an examination where the details of structural components are within the close visual inspection range of the attending surveyor, i.e. normally within reach of hand.

| <b>Tank No.</b> | <b>Structure</b>  | <b>C (Cargo)/B (Ballast)</b> | <b>Temporary staging</b> | <b>Rafts</b> | <b>Ladders</b> | <b>Direct access</b> | <b>Other means (please specify)</b> |
|-----------------|-------------------|------------------------------|--------------------------|--------------|----------------|----------------------|-------------------------------------|
| F.P.            | Fore peak         |                              |                          |              |                |                      |                                     |
| A.P.            | Aft peak          |                              |                          |              |                |                      |                                     |
| Wing tanks      | Underdeck         |                              |                          |              |                |                      |                                     |
|                 | Side shell        |                              |                          |              |                |                      |                                     |
|                 | Bottom transverse |                              |                          |              |                |                      |                                     |
|                 | Longitudinal      |                              |                          |              |                |                      |                                     |
|                 | Transverse        |                              |                          |              |                |                      |                                     |
| Centre tanks    | Under deck        |                              |                          |              |                |                      |                                     |
|                 | Bottom transverse |                              |                          |              |                |                      |                                     |
|                 | Transverse        |                              |                          |              |                |                      |                                     |

**History of cargo with H<sub>2</sub>S content or heated cargo for the last three years together with indication as to whether cargo was heated and, where available, Material Safety Data Sheets (MSDS)<sup>19</sup>**

### Owner's inspections

Using a format similar to that of the table below (which is given as an example), the owner should provide details of the results of their inspections for the last three years on all cargo and ballast tanks and void spaces within the cargo area, including peak tanks.

| Tank No.             | Corrosion protection (1) | Coating extent (2) | Coating condition (3) | Structural deterioration (4) | Tank damage history (5) |
|----------------------|--------------------------|--------------------|-----------------------|------------------------------|-------------------------|
| Cargo centre tanks   |                          |                    |                       |                              |                         |
| Cargo wing tanks     |                          |                    |                       |                              |                         |
| Slop                 |                          |                    |                       |                              |                         |
| Ballast tanks        |                          |                    |                       |                              |                         |
| Aft peak             |                          |                    |                       |                              |                         |
| Fore peak            |                          |                    |                       |                              |                         |
| Miscellaneous spaces |                          |                    |                       |                              |                         |

### Note:

Indicate tanks which are used for oil/ballast.

- 1) HC = hard coating; SC = soft coating;  
A = anodes; NP = no protection
- 2) U = upper part; M = middle part;  
L = lower part; C = complete
- 3) G = good; F = fair; P = poor;  
RC = recoated (during the last three years)
- 4) N = no findings recorded; Y = findings recorded,  
description of findings should be attached to this questionnaire
- 5) DR = Damage & Repair; L = Leakages;  
CV = Conversion (description to be attached to this questionnaire)

Name of owner's representative:

.....

Signature: .....

Date: .....

<sup>19</sup> Refer to resolution MSC.150(77) on Recommendation for material safety data sheets for MARPOL Annex I cargoes and marine fuel oils.

**Reports of port State control inspections**

|                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| List the reports of port State control inspections containing hull structural related deficiencies, relevant information on rectification of the deficiencies: |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |

**Safety management system**

|                                                                                                 |
|-------------------------------------------------------------------------------------------------|
| List non-conformities related to hull maintenance, including the associated corrective actions: |
|-------------------------------------------------------------------------------------------------|

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |

**Name and address of the approved thickness measurement company:**

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |



## **ANNEX 7**

### **PROCEDURES FOR CERTIFICATION OF A COMPANY ENGAGED IN THICKNESS MEASUREMENT OF HULL STRUCTURES**

#### **1 Application**

This guidance applies for certification of the company which intends to engage in the thickness measurement of hull structures of ships.

#### **2 Procedures for certification**

##### ***Submission of documents***

**2.1** The following documents should be submitted to an organization recognized by the Administration for approval:

- .1** outline of the company, e.g., organization and management structure;
- .2** experience of the company on thickness measurement of hull structures of ships;
- .3** technicians' careers, i.e. experience of technicians as thickness measurement operators, technical knowledge and experience of hull structure, etc. Operators should be qualified according to a recognized industrial NDT Standard;
- .4** equipment used for thickness measurement such as ultrasonic testing machines and their maintenance/calibration procedures;
- .5** a guide for thickness measurement operators;
- .6** training programmes for technicians for thickness measurement; and
- .7** measurement record format in accordance with recommended procedures for thickness measurements (see annex 10).

##### ***Auditing of the company***

**2.2** Upon reviewing the documents submitted with satisfactory results, the company should be audited in order to ascertain that the company is duly organized and managed in accordance with the documents submitted, and eventually is capable of conducting thickness measurement of the hull structure of ships.

**2.3** Certification is conditional upon an on board demonstration of thickness measurement as well as satisfactory reporting.

#### **3 Certification**

**3.1** Upon satisfactory results of both the audit of the company referred to in 2.2 and the demonstration tests referred to in 2.3, the Administration or organization recognized by the Administration should issue a certificate of approval as well as a notice to the effect that the thickness measurement operation system of the company has been certified.

**3.2** Renewal/endorsement of the certificate should be made at intervals not exceeding three years by verification that original conditions are maintained.

**4 Report of any alteration to the certified thickness measurement operation system**

In cases where any alteration to the certified thickness measurement operation system of the company is made, such an alteration should be immediately reported to the organization recognized by the Administration. Re-audit should be made where deemed necessary by the organization recognized by the Administration.

**5 Withdrawal of the certification**

The certification may be withdrawn in the following cases:

- .1** where the measurements were improperly carried out or the results were improperly reported;
- .2** where the surveyor found any deficiencies in the approved thickness measurement operation system of the company; and
- .3** where the company failed to report any alteration referred to in 4 to the organization recognized by the Administration as required.

## **ANNEX 8**

### **SURVEY REPORTING PRINCIPLES**

As a principle, for oil tankers subject to the Code, the surveyor should include the following contents in his report for survey of hull structure and piping systems, as relevant for the survey.

#### **1 General**

**1.1** A survey report should be generated in the following cases:

- .1** in connection with commencement, continuation and/or completion of periodical hull surveys, i.e. annual, intermediate and renewal surveys, as relevant;
- .2** when structural damages/defects have been found;
- .3** when repairs, renewals or modifications have been carried out; and
- .4** when condition of class (recommendation) has been imposed or deleted.

**1.2** The reporting should provide:

- .1** evidence that prescribed surveys have been carried out in accordance with applicable requirements;
- .2** documentation of surveys carried out with findings, repairs carried out and condition of class (recommendation) imposed or deleted;
- .3** survey records, including actions taken, which should form an auditable documentary trail. Survey reports should be kept in the survey report file required to be on board;
- .4** information for planning of future surveys; and
- .5** information which may be used as input for maintenance of classification rules and instructions.

**1.3** When a survey is split between different survey stations, a report should be made for each portion of the survey. A list of items surveyed, relevant findings and an indication of whether the item has been credited, should be made available to the next attending surveyor, prior to continuing or completing the survey. Thickness measurement and tank testing carried out should also be listed for the next surveyor.

#### **2 Extent of the survey**

**2.1** Identification of compartments where an overall survey has been carried out.

**2.2** Identification of locations, in each tank, where a close-up survey has been carried out, together with information of the means of access used.

**2.3** Identification of locations, in each tank, where thickness measurement has been carried out.

**Note:** As a minimum, the identification of location of close-up survey and thickness measurement should include a confirmation with description of individual structural members corresponding to the extent of requirements stipulated in this part of Annex B based on type of periodical survey and the ship's age.

Where only partial survey is required, i.e. one web frame ring/one deck transverse, the identification should include location within each ballast tank and cargo hold by reference to frame numbers.

**2.4** For areas in tanks where protective coating is found to be in good condition and the extent of close-up survey and/or thickness measurement has been specially considered, structures subject to special consideration should be identified.

**2.5** Identification of tanks subject to tank testing.

**2.6** Identification of piping systems on deck, including crude oil washing (COW) piping, and ballast piping within cargo and ballast tanks, pipe tunnels, cofferdams and void spaces where:

- .1 examination, including internal examination of piping with valves and fittings and thickness measurement, as relevant, has been carried out; and
- .2 operational test to working pressure has been carried out.

### **3 Result of survey**

**3.1** Type, extent and condition of protective coating in each tank, as relevant (rated GOOD, FAIR or POOR), including identification of tanks fitted with anodes.

**3.2** Structural condition of each compartment with information on the following, as relevant:

- .1 Identification of findings, such as:
  - .1 corrosion with description of location, type and extent;
  - .2 areas with substantial corrosion;
  - .3 cracks/fractures with description of location and extent;
  - .4 buckling with description of location and extent; and
  - .5 indents with description of location and extent.
- .2 Identification of compartments where no structural damages/defects are found. The report may be supplemented by sketches/photographs.
- .3 Thickness measurement report should be verified and signed by the surveyor controlling the measurements on board.

- .4 Evaluation result of longitudinal strength of the hull girder of oil tankers of 130 m in length and upwards and over 10 years of age. The following data should be included, as relevant:
  - .1 measured and as-built transverse sectional areas of deck and bottom flanges;
  - .2 diminution of transverse sectional areas of deck and bottom flanges; and
  - .3 details of renewals or reinforcements carried out, as relevant (see 4.2).

#### **4 Actions taken with respect to findings**

**4.1** Whenever the attending surveyor is of the opinion that repairs are required, each item to be repaired should be identified in a numbered list. Whenever repairs are carried out, details of the repairs effected should be reported by making specific reference to relevant items in the numbered list.

**4.2** Repairs carried out should be reported with identification of:

- .1 compartment;
- .2 structural member;
- .3 repair method (i.e. renewal or modification), including:
  - .3.1 steel grades and scantlings (if different from the original);
  - .3.2 sketches/photographs, as appropriate;
- .4 repair extent; and
- .5 non-destructive tests (NDT).

**4.3** For repairs not completed at the time of survey, condition of class/recommendation should be imposed with a specific time limit for the repairs. In order to provide correct and proper information to the surveyor attending for survey of the repairs, condition of class/recommendation should be sufficiently detailed with identification of each item to be repaired. For identification of extensive repairs, reference may be given to the survey report.

## ANNEX 9

### CONDITION EVALUATION REPORT

#### Issued upon completion of renewal survey

##### *General particulars*

|                             |                                                   |
|-----------------------------|---------------------------------------------------|
| Ship's name:                | Class/Administration identity number:             |
|                             | Previous class/Administration identity number(s): |
|                             | IMO number:                                       |
| Port of registry:           | National flag:                                    |
|                             | Previous national flag(s):                        |
| Deadweight (metric tonnes): | Gross tonnage:                                    |
|                             | National:                                         |
|                             | ITC (1969):                                       |
| Date of build:              | Classification notation:                          |
| Date of major conversion:   |                                                   |
| Type of conversion:         | Owner:                                            |
|                             | Previous owner(s):                                |

**1** The survey reports and documents listed below have been reviewed by the undersigned and found to be satisfactory.

**2** A summary of the survey is attached herewith on sheet 2

**3** The renewal survey has been completed in accordance with the present Code on (date) .....

|                                          |                   |       |
|------------------------------------------|-------------------|-------|
| Condition evaluation report completed by | Name<br>Signature | Title |
| Office                                   | Date              |       |
| Condition evaluation report verified by  | Name<br>Signature | Title |
| Office                                   | Date              |       |

Attached reports and documents:

- 1)
- 2)
- 3)
- 4)
- 5)
- 6)

### ***Contents of condition evaluation report***

- Part 1 – General particulars: – See front page
- Part 2 – Report review: – Where and how survey was done
- Part 3 – Close-up survey: – Extent (which tanks)
- Part 4 – Cargo and ballast piping system: – Examined
- Part 5 – Thickness measurements: – Operationally tested
- Reference to thickness measurement report
- Summary of where measured
- Separate form indicating the spaces with substantial corrosion, and corresponding:
  - thickness diminution
  - corrosion pattern
- Part 6 – Tank corrosion prevention system: – Separate form indicating:
  - location of coating/anodes
  - condition of coating (if applicable)
- Part 7 – Repairs: – Identification of spaces/areas
- Part 8 – Condition of class/flag State requirements:
- Part 9 – Memoranda:
  - Acceptable defects
  - Any points of attention for future surveys, e.g., for suspect areas
  - Extended annual/intermediate survey due to coating breakdown
- Part 10 – Evaluation results of the ship's longitudinal strength (for oil tankers of 130 m in length and upwards and over 10 years of age)
- Part 11 – Conclusion: – Statement on evaluation/verification of survey report

### ***Extract of thickness measurements***

Reference is made to the thickness measurement report:

| <b>Position of substantially corroded tanks/areas<sup>1</sup> or areas with deep pitting<sup>3</sup></b> | <b>Thickness diminution [%]</b> | <b>Corrosion pattern<sup>2</sup></b> | <b>Remarks:<br/>(e.g., reference to attached sketches)</b> |
|----------------------------------------------------------------------------------------------------------|---------------------------------|--------------------------------------|------------------------------------------------------------|
|                                                                                                          |                                 |                                      |                                                            |

**Notes:**

- 1 Substantial corrosion, i.e. 75%–100% of acceptable margins wasted.
- 2 P = Pitting  
C = Corrosion in general
- 3 Any bottom plating with a pitting intensity of 20% or more, with wastage in the substantial corrosion range or having an average depth of pitting of  $\frac{1}{3}$  or more of actual plate thickness should be noted.

***Tank corrosion prevention system***

| Tank nos. <sup>1</sup> | Tank corrosion prevention system <sup>2</sup> | Coating condition <sup>3</sup> | Remarks |
|------------------------|-----------------------------------------------|--------------------------------|---------|
|                        |                                               |                                |         |

**Notes:**

- 1 All segregated ballast tanks and combined cargo/ballast tanks should be listed.
- 2 C = Coating NP = No protection
- 3 Coating condition according to the following standard:  
  
GOOD condition with only minor spot rusting.  
  
FAIR condition with local breakdown of coating at edges of stiffeners and weld connections and/or light rusting over 20% or more of areas under consideration, but less than as defined for POOR condition.  
  
POOR condition with general breakdown of coating over 20% or more of areas or hard scale at 10% or more of areas under consideration.

If coating condition less than GOOD is given, extended annual surveys should be introduced. This should be noted in part 9 of the Contents of condition evaluation report.



**Evaluation result of longitudinal strength of the hull girder of oil tankers of 130 m in length and upwards and of over 10 years of age**  
(of sections 1, 2 and 3 below, only one applicable section should be completed)

**1** This section applies to ships regardless of the date of construction: Transverse sectional areas of deck flange (deck plating and deck longitudinals) and bottom flange (bottom shell plating and bottom longitudinals) of the ship's hull girder have been calculated by using the thickness measured, renewed or reinforced, as appropriate, during the renewal survey of the Cargo Ship Safety Construction Certificate or the Cargo Ship Safety Certificate (SC renewal survey) most recently conducted after the ship reached 10 years of age, and found that the diminution of the transverse sectional area does not exceed 10% of the as-built area, as shown in the following table:

**Table 1 – Transverse sectional area of hull girder flange**

|                      |               | Measured        | As-built        | Diminution          |
|----------------------|---------------|-----------------|-----------------|---------------------|
| Transverse section 1 | Deck flange   | cm <sup>2</sup> | cm <sup>2</sup> | cm <sup>2</sup> (%) |
|                      | Bottom flange | cm <sup>2</sup> | cm <sup>2</sup> | cm <sup>2</sup> (%) |
| Transverse section 2 | Deck flange   | cm <sup>2</sup> | cm <sup>2</sup> | cm <sup>2</sup> (%) |
|                      | Bottom flange | cm <sup>2</sup> | cm <sup>2</sup> | cm <sup>2</sup> (%) |
| Transverse section 3 | Deck flange   | cm <sup>2</sup> | cm <sup>2</sup> | cm <sup>2</sup> (%) |
|                      | Bottom flange | cm <sup>2</sup> | cm <sup>2</sup> | cm <sup>2</sup> (%) |

**2** This section applies to ships constructed on or after 1 July 2002: Section moduli of transverse section of the ship's hull girder have been calculated by using the thickness of structural members measured, renewed or reinforced, as appropriate, during the SC renewal survey most recently conducted after the ship reached 10 years of age in accordance with the provisions of paragraph 2.2.1.1 of annex 12, and are found to be within their diminution limits determined by the Administration, taking into account the recommendations adopted by the Organization,<sup>20</sup> as shown in the following table:

**Table 2 – Transverse section modulus of hull girder**

|                      |            | $Z_{act}$ (cm <sup>3</sup> ) <sup>1</sup> | $Z_{req}$ (cm <sup>3</sup> ) <sup>2</sup> | Remarks |
|----------------------|------------|-------------------------------------------|-------------------------------------------|---------|
| Transverse section 1 | Upper deck |                                           |                                           |         |
|                      | Bottom     |                                           |                                           |         |
| Transverse section 2 | Upper deck |                                           |                                           |         |
|                      | Bottom     |                                           |                                           |         |
| Transverse section 3 | Upper deck |                                           |                                           |         |
|                      | Bottom     |                                           |                                           |         |

**Notes:**

- 1  $Z_{act}$  means the actual section moduli of the transverse section of the ship's hull girder calculated by using the thickness of structural members measured, renewed or reinforced, as appropriate, during the SC renewal survey, in accordance with the provisions of paragraph 2.2.1.1 of annex 12.
- 2  $Z_{req}$  means the diminution limit of the longitudinal bending strength of ships, as calculated in accordance with the provisions of paragraph 2.2.1.1 of annex 12.

The calculation sheets for  $Z_{act}$  should be attached to this report.

<sup>20</sup> Refer to resolution MSC.108(73) on Recommendations on compliance with the requirements of paragraph 2.2.1.1 of annex 12 to annex B to resolution A.744(18).

**3** This section applies to ships constructed before 1 July 2002: Section moduli of transverse sections of the ship's hull girder have been calculated by using the thickness of structural members measured, renewed or reinforced, as appropriate, during the SC renewal survey most recently conducted after the ship reached 10 years of age in accordance with the provisions of paragraph 2.2.1.2 of annex 12, and found to meet the criteria required by the Administration or the recognized classification society and that  $Z_{act}$  is not less than  $Z_{mc}$  (defined in note 2 below) as specified in appendix 2 to annex 12, as shown in the following table.

Describe the criteria for acceptance of the minimum section moduli of the ship's hull girder for ships in service required by the Administration or the recognized classification society.

**Table 3 – Transverse section modulus of hull girder**

|                      |            | $Z_{act} (cm^3)^1$ | $Z_{mc} (cm^3)^2$ | Remarks |
|----------------------|------------|--------------------|-------------------|---------|
| Transverse section 1 | Upper deck |                    |                   |         |
|                      | Bottom     |                    |                   |         |
| Transverse section 2 | Upper deck |                    |                   |         |
|                      | Bottom     |                    |                   |         |
| Transverse section 3 | Upper deck |                    |                   |         |
|                      | Bottom     |                    |                   |         |

**Notes:**

- 1 As defined in note 1 of table 2.
- 2  $Z_{mc}$  means the diminution limit of minimum section modulus calculated in accordance with the provisions of paragraph 2.2.1.2 of annex 12.

## **ANNEX 10**

### **RECOMMENDED PROCEDURES FOR THICKNESS MEASUREMENTS**

#### **General**

- 1** These procedures should be used for recording thickness measurements as required by annexes 2 and 4.
- 2** Reporting forms TM1-T, TM2-T(i), TM2-T(ii), TM3-T, TM4-T, TM5-T and TM6-T, set out in appendix 2, should be used for recording thickness measurements and the maximum allowable diminution should be stated. The maximum allowable diminution could be stated in an attached document.
- 3** Appendix 3 contains guidance diagrams and notes relating to the reporting forms and the requirements for thickness measurement.
- 4** The reporting forms should, where appropriate, be supplemented by data presented on structural sketches.

## Appendix 1

### GENERAL PARTICULARS

Ship's name: .....  
IMO number: .....  
Class/Administration identity number: .....  
Port of registry: .....  
Gross tonnage: .....  
Deadweight: .....  
Date of build: .....  
Classification society: .....

---

Name of company performing thickness measurement: .....  
.....  
Thickness measurement company certified by: .....  
Certificate number: .....  
Certificate valid from: ..... to .....  
Place of measurement: .....  
First date of measurement: .....  
Last date of measurement: .....  
Periodical survey/intermediate survey<sup>21</sup> due: .....  
Details of measurement equipment: .....  
Qualification of operator: .....  
Report number: ..... consisting of ..... pages  
Name of operator: ..... Name of surveyor: .....  
Signature of operator: ..... Signature of surveyor: .....  
Company official stamp: ..... Administration: .....  
Official stamp

---

<sup>21</sup> Delete as appropriate.

## Appendix 2

### Report on thickness measurement of all deck plating, all bottom shell plating or side shell plating (TM1-T)

Ship's name..... Class Identity no..... Report no..... IMO number.....

| Ship's Name..... Class..... Report No..... IWS Number..... |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
|------------------------------------------------------------|---------------|-----------------|-----------------|---|--------------|---|--------------|---|-------------|---|--------------|---|--------------|---|-------------------|---|-------------------------|
| STRAKE POSITION                                            |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| PLATE POSITION                                             | No. or letter | Orig. thk. (mm) | Forward reading |   |              |   |              |   | Aft reading |   |              |   |              |   | Mean diminution % |   | Max. allow. dimin. (mm) |
|                                                            |               |                 | Gauged          |   | Diminution P |   | Diminution S |   | Gauged      |   | Diminution P |   | Diminution S |   | P                 | S |                         |
|                                                            |               |                 | P               | S | mm           | % | mm           | % | P           | S | mm           | % | mm           | % |                   |   |                         |
| 12th forward                                               |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 11th                                                       |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 10th                                                       |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 9th                                                        |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 8th                                                        |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 7th                                                        |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 6th                                                        |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 5th                                                        |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 4th                                                        |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 3rd                                                        |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 2nd                                                        |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 1st                                                        |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| Amidships                                                  |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 1st aft                                                    |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 2nd                                                        |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 3rd                                                        |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 4th                                                        |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 5th                                                        |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 6th                                                        |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 7th                                                        |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 8th                                                        |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 9th                                                        |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 10th                                                       |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 11th                                                       |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |
| 12th                                                       |               |                 |                 |   |              |   |              |   |             |   |              |   |              |   |                   |   |                         |

Operator's signature.....

Notes – see following page

**Notes to report TM1-T:**

- 1** This report should be used for recording the thickness measurement of:
  - .1** All strength deck plating within the cargo area.
  - .2** All keel, bottom shell plating and bilge plating within the cargo area.
  - .3** Side shell plating including selected wind and water strakes outside the cargo area.
  - .4** All wind and water strakes within the cargo area
- 2** The strake position should be clearly indicated as follows:
  - .1** For strength deck plating, indicate the number of the strake of plating inboard from the stringer plate.
  - .2** For bottom plating, indicate the number of the strake of plating outboard from the keel plate.
  - .3** For side shell plating, give number of the strake of plating below sheer strake and letter as shown on shell expansion.
- 3** For oil tankers, all deck plating strakes should be recorded, for ore/oil ships only the deck plating strakes outside line of openings should be recorded.
- 4** Measurements should be taken at the forward and aft areas of all plates and where plates cross ballast/cargo tank boundaries, separate measurements for the area of plating in way of each type of tank should be recorded.
- 5** The single measurements recorded should represent the average of multiple measurements.
- 6** The maximum allowable diminution could be stated in an attached document.

**Report on thickness measurement of shell and deck plating (one, two or three transverse sections) (TM2-T(i))**

Ship's name..... Class Identity no..... Report no..... IMO number.....

| STRENGTH DECK AND SHEERSTRAKE PLATING |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
|---------------------------------------|----------------------------------------------|------------|-------------------|--------|---|--------------|---|--------------|-----------------------------------------------|---------------|------------|-------------------|--------|---|--------------|---|----------------------------------------------|---|---------------|------------|-------------------|--------|---|--------------|---|--------------|---|
|                                       | FIRST TRANSVERSE SECTION AT FRAME NUMBER.... |            |                   |        |   |              |   |              | SECOND TRANSVERSE SECTION AT FRAME NUMBER.... |               |            |                   |        |   |              |   | THIRD TRANSVERSE SECTION AT FRAME NUMBER.... |   |               |            |                   |        |   |              |   |              |   |
| STRAKE POSITION                       | No. or letter                                | Orig. thk. | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S |                                               | No. or letter | Orig. thk. | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S                                 |   | No. or letter | Orig. thk. | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S |   |
|                                       |                                              | mm         | mm                | P      | S | mm           | % | mm           | %                                             |               | mm         | mm                | P      | S | mm           | % | mm                                           | % |               | mm         | mm                | P      | S | mm           | % | mm           | % |
| Stringer plate                        |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 1st stake inboard                     |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 2nd                                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 3rd                                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 4th                                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 5th                                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 6th                                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 7th                                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 8th                                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 9th                                   |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 10th                                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 11th                                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 12th                                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 13th                                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| 14th                                  |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| centre strake                         |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| sheerstrake                           |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |
| TOPSIDE TOTAL                         |                                              |            |                   |        |   |              |   |              |                                               |               |            |                   |        |   |              |   |                                              |   |               |            |                   |        |   |              |   |              |   |

Operator's signature.....

Notes – see following page

**Notes to report TM2-T(i):**

- 1** This report should be used for recording the thickness measurement of strength deck plating and sheerstrake plating transverse sections:

One, two or three sections within the cargo area comprising structural items (1), (2) and (3) as shown on the diagrams of typical transverse section indicating longitudinal and transverse members, in appendix 3.
- 2** For oil tankers all deck plating strakes should be recorded, for ore/oil ships only the deck plating strakes outside line of openings should be recorded.
- 3** The topside area comprises deck plating, stringer plate and sheerstrake (including rounded gunwales).
- 4** The exact frame station of measurement should be stated.
- 5** The single measurements recorded should represent the average of multiple measurements.
- 6** The maximum allowable diminution could be stated in an attached document.



**Report on thickness measurement of shell and deck plating (one, two or three transverse sections) (TM2-T(ii))**

Ship's name..... Class Identity no..... Report no..... IMO number.....

| SHELL PLATING         |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
|-----------------------|----------------------------------------------|------------|-------------------|--------|---|--------------|---|--------------|---|-----------------------------------------------|------------|-------------------|--------|---|--------------|---|--------------|---|----------------------------------------------|------------|-------------------|--------|---|--------------|---|--------------|---|
|                       | FIRST TRANSVERSE SECTION AT FRAME NUMBER.... |            |                   |        |   |              |   |              |   | SECOND TRANSVERSE SECTION AT FRAME NUMBER.... |            |                   |        |   |              |   |              |   | THIRD TRANSVERSE SECTION AT FRAME NUMBER.... |            |                   |        |   |              |   |              |   |
| STRAKE POSITION       | No. or letter                                | Orig. thk. | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S |   | No. or letter                                 | Orig. thk. | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S |   | No. or letter                                | Orig. thk. | Max allow. dimin. | Gauged |   | Diminution P |   | Diminution S |   |
|                       |                                              | mm         | mm                | P      | S | mm           | % | mm           | % |                                               | mm         | mm                | P      | S | mm           | % | mm           | % |                                              | mm         | mm                | P      | S | mm           | % | mm           | % |
| 1st below sheerstrake |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 2nd                   |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 3rd                   |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 4th                   |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 5th                   |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 6th                   |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 7th                   |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 8th                   |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 9th                   |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 10th                  |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 11th                  |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 12th                  |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 13th                  |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 14th                  |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 15th                  |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 16th                  |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 17th                  |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 18th                  |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 19th                  |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| 20th                  |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| keel strake           |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |
| BOTTOM TOTAL          |                                              |            |                   |        |   |              |   |              |   |                                               |            |                   |        |   |              |   |              |   |                                              |            |                   |        |   |              |   |              |   |

Operator's signature.....

Notes – see following page

**Notes to report TM2-T(ii):**

- 1** This report should be used for recording the thickness measurements of shell plating transverse sections:

One, two or three sections within the cargo area comprising structural items (4), (5), (6), and (7) as shown on the diagrams of typical transverse section indicating longitudinal and transverse members, in appendix 3.
- 2** The bottom area comprises keel, bottom and bilge plating.
- 3** The exact frame station of measurement should be stated.
- 4** The single measurements recorded should represent the average of multiple measurements.
- 5** The maximum allowable diminution could be stated in an attached document.

**Report on thickness measurement of longitudinal members (one, two or three transverse sections) (TM3-T)**

Ship's name..... Class Identity no..... Report no..... IMO number.....

[illegible]

Operator's signature.....

Notes – see following page

**Notes to report TM3-T:**

- 1** This report should be used for recording the thickness measurement of longitudinal members at transverse sections:

One, two or three sections within the cargo area comprising structural items (8) to (20) as shown on the diagrams of typical transverse section indicating longitudinal and transverse members, in appendix 3.
- 2** The exact frame station of measurement should be stated.
- 3** The single measurements recorded should represent the average of multiple measurements.
- 4** The maximum allowable diminution could be stated in an attached document.

**Report on thickness measurement of transverse structural members (in the cargo oil and water ballast tanks within the cargo tank length) (TM4-T)**

Ship's name..... Class Identity no..... Report no..... IMO number.....

[illegible]

Operator's signature.....

Notes – see following page

***Notes to report TM4-T:***

- 1**      This report should be used for recording the thickness measurement of transverse structural members, comprising appropriate structural items (25) to (32) as shown on the diagrams of typical transverse section indicating longitudinal and transverse members in appendix 3.
- 2**      Guidance for areas of measurement is indicated in appendix 3.
- 3**      The single measurements recorded should represent the average of multiple measurements.
- 4**      The maximum allowable diminution could be stated in an attached document.

**Report on thickness of W.T./O.T. transverse bulkheads (within the cargo tank or cargo hold spaces) (TM5-T)**

Ship's name..... Class Identity no..... Report no..... IMO number.....

[illegible]

Operator's signature.....

Notes – see following page

***Notes to report TM5-T:***

- 1**      This report should be used for recording the thickness measurement of W.T./O.T. transverse bulkheads.
- 2**      Guidance for areas of measurement is indicated in appendix 3.
- 3**      The single measurements recorded should represent the average of multiple measurements.
- 4**      The maximum allowable diminution could be stated in an attached document.



## Report on thickness measurement of miscellaneous structural members (TM6-T)

Ship's name..... Class Identity no..... Report no..... IMO number.....

[illegible]

Operator's signature.....

Notes – see following page

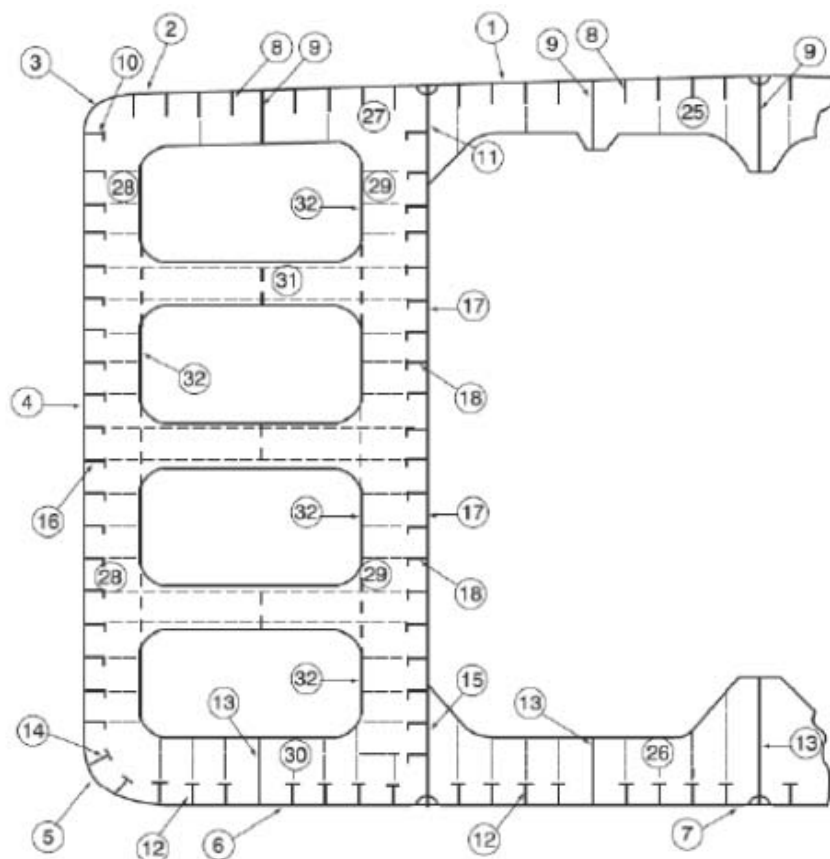
***Notes to report TM6-T:***

- 1**      This report should be used for recording the thickness measurement of miscellaneous structural members including structural items (36), (37) and (38) shown in appendix 3.
- 2**      The single measurements recorded should represent the average of multiple measurements.
- 3**      The maximum allowable diminution could be stated in an attached document.

### Appendix 3

#### GUIDANCE ON THICKNESS MEASUREMENT

Typical transverse section of oil tanker indicating longitudinal and transverse members



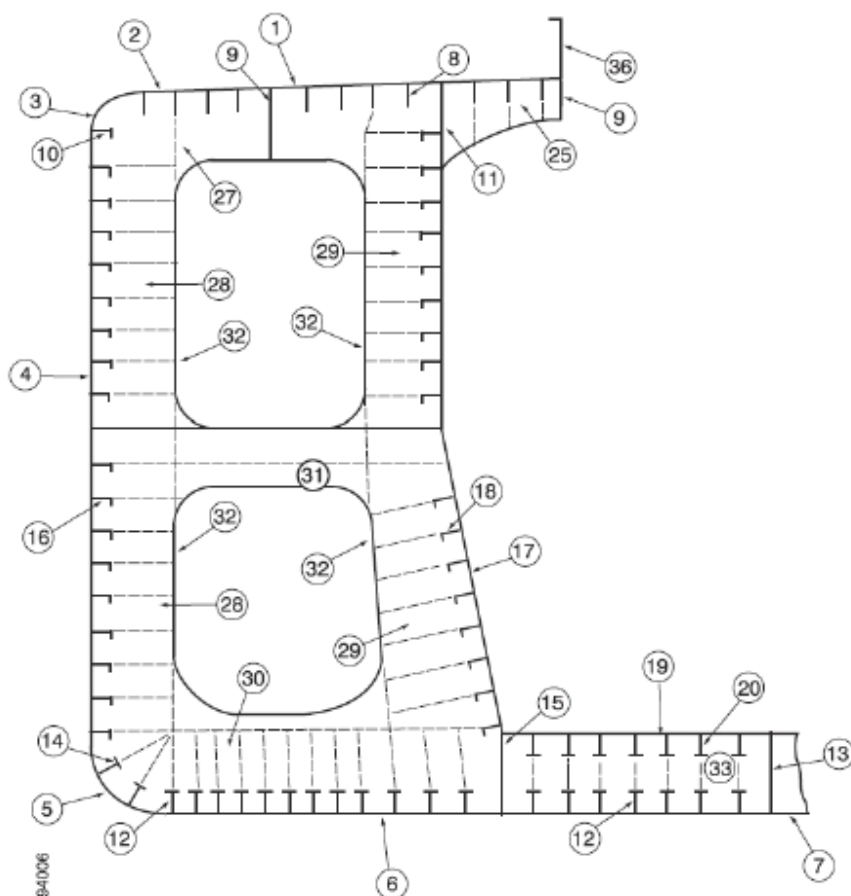
| REPORT ON TM2-T(i) & (ii) |                       |
|---------------------------|-----------------------|
| 1                         | Strength deck plating |
| 2                         | Stringer plate        |
| 3                         | Sheerstrake           |
| 4                         | Side shell plating    |
| 5                         | Bilge plating         |
| 6                         | Bottom shell plating  |
| 7                         | Keel plate            |

| REPORT ON TM6-T |                              |
|-----------------|------------------------------|
| 36              | Hatch coamings               |
| 37              | Deck plating between hatches |
| 38              | Hatch covers                 |
| 39              |                              |
| 40              |                              |

| REPORT ON TM3-T |                                           |
|-----------------|-------------------------------------------|
| 8               | Deck longitudinals                        |
| 9               | Deck girders                              |
| 10              | Sheerstrake longitudinals                 |
| 11              | Longitudinal bulkhead top strake          |
| 12              | Bottom longitudinals                      |
| 13              | Bottom girders                            |
| 14              | Bilge longitudinals                       |
| 15              | Longitudinal bulkhead lower strake        |
| 16              | Side shell longitudinals                  |
| 17              | Longitudinal bulkhead plating (remainder) |
| 18              | Longitudinal bulkhead longitudinals       |
| 19              | Inner bottom plating                      |
| 20              | Inner bottom longitudinals                |
| 21              |                                           |
| 22              |                                           |
| 23              |                                           |
| 24              |                                           |

| REPORT ON TM4-T |                                    |
|-----------------|------------------------------------|
| 25              | Deck transverse centre tank        |
| 26              | Bottom transverse centre tank      |
| 27              | Deck transverse wing tank          |
| 28              | Side shell vertical web            |
| 29              | Longitudinal bulkhead vertical web |
| 30              | Bottom transverse wing tank        |
| 31              | Struts                             |
| 32              | Transverse web face plate          |
| 33              | D.B. floors                        |
| 34              |                                    |
| 35              |                                    |

# Typical transverse section of ore/oil ship indicating longitudinal and transverse members



## REPORT ON TM2-T(i) & (ii)

- (1) Strength deck plating
- (2) Stringer plate
- (3) Sheerstrake
- (4) Side shell plating
- (5) Bilge plating
- (6) Bottom shell plating
- (7) Keel plate

## REPORT ON TM6-T

- (36) Hatch coamings
- (37) Deck plating between hatches
- (38) Hatch covers
- (39)
- (40)

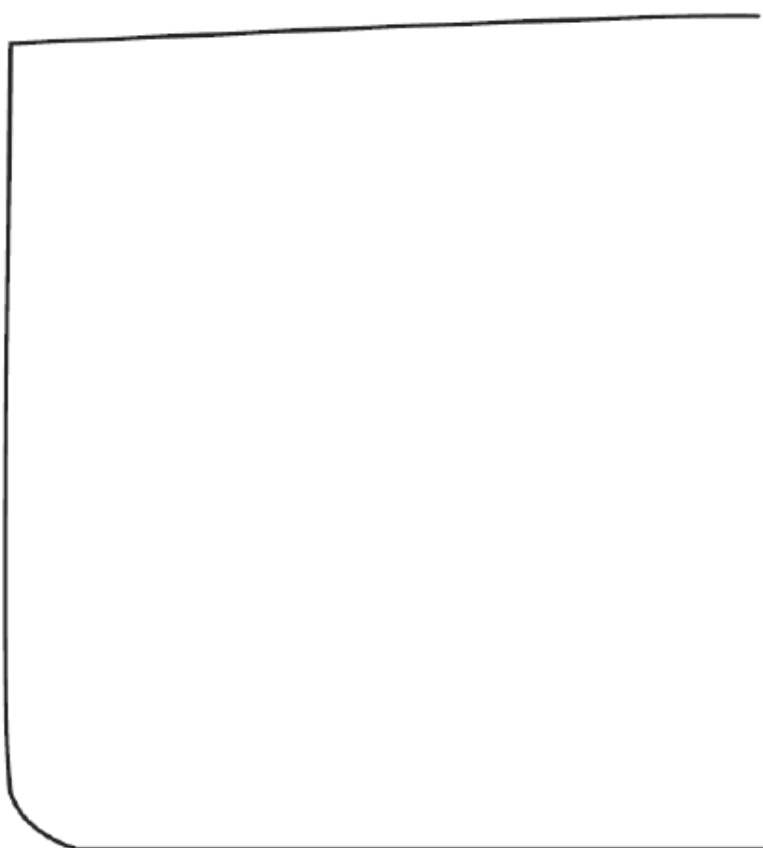
## REPORT ON TM3-T

- (8) Deck longitudinals
- (9) Deck girders
- (10) Sheerstrake longitudinals
- (11) Longitudinal bulkhead top strake
- (12) Bottom longitudinals
- (13) Bottom girders
- (14) Bilge longitudinals
- (15) Longitudinal bulkhead lower strake
- (16) Side shell longitudinals
- (17) Longitudinal bulkhead plating (remainder)
- (18) Longitudinal bulkhead longitudinals
- (19) Inner bottom plating
- (20) Inner bottom longitudinals
- (21)
- (22)
- (23)
- (24)

## REPORT ON TM4-T

- (25) Deck transverse centre tank
- (26) Bottom transverse centre tank
- (27) Deck transverse wing tank
- (28) Side shell vertical web
- (29) Longitudinal bulkhead vertical web
- (30) Bottom transverse wing tank
- (31) Struts
- (32) Transverse web face plate
- (33) D.B. floors
- (34)
- (35)

**Transverse section outline (to be used for longitudinal and transverse members where typical oil tanker or oil/ore ship sections are not applicable)**



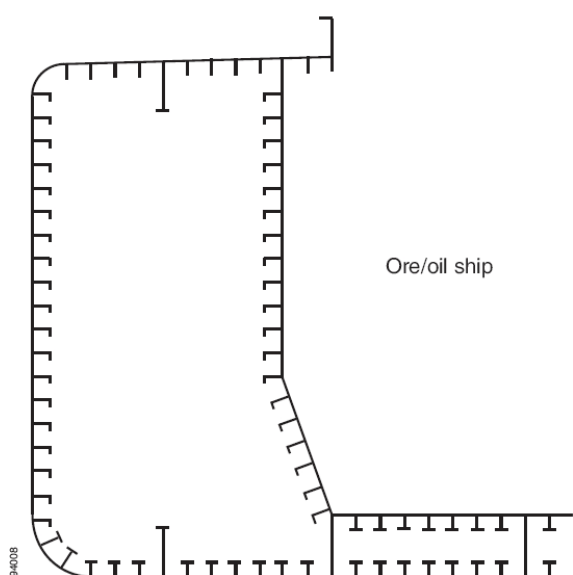
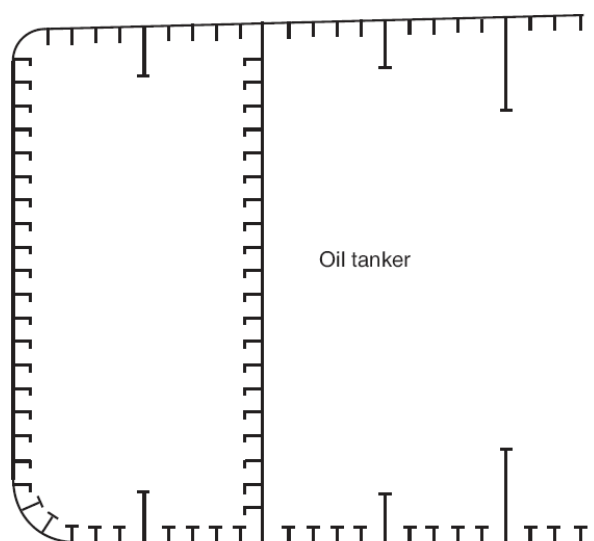
| REPORT ON TM2-T &  |                       |
|-------------------------------------------------------------------------------------------------------|-----------------------|
| ①                                                                                                     | Strength deck plating |
| ②                                                                                                     | Stringer plate        |
| ③                                                                                                     | Sheerstrake           |
| ④                                                                                                     | Side shell plating    |
| ⑤                                                                                                     | Bilge plating         |
| ⑥                                                                                                     | Bottom shell plating  |
| ⑦                                                                                                     | Keel plate            |

| REPORT ON TM6-T |                              |
|-----------------|------------------------------|
| ③⑥              | Hatch coamings               |
| ③⑦              | Deck plating between hatches |
| ③⑧              | Hatch covers                 |
| ③⑨              |                              |
| ④①              |                              |

| REPORT ON TM3-T |                                           |
|-----------------|-------------------------------------------|
| ⑧               | Deck longitudinals                        |
| ⑨               | Deck girders                              |
| ⑩               | Sheerstrake longitudinals                 |
| ⑪               | Longitudinal bulkhead top strake          |
| ⑫               | Bottom longitudinals                      |
| ⑬               | Bottom girders                            |
| ⑭               | Bilge longitudinals                       |
| ⑮               | Longitudinal bulkhead lower strake        |
| ⑯               | Side shell longitudinals                  |
| ⑰               | Longitudinal bulkhead plating (remainder) |
| ⑱               | Longitudinal bulkhead longitudinals       |
| ⑲               | Inner bottom plating                      |
| ⑳               | Inner bottom longitudinals                |
| ㉑               |                                           |
| ㉒               |                                           |
| ㉓               |                                           |
| ㉔               |                                           |

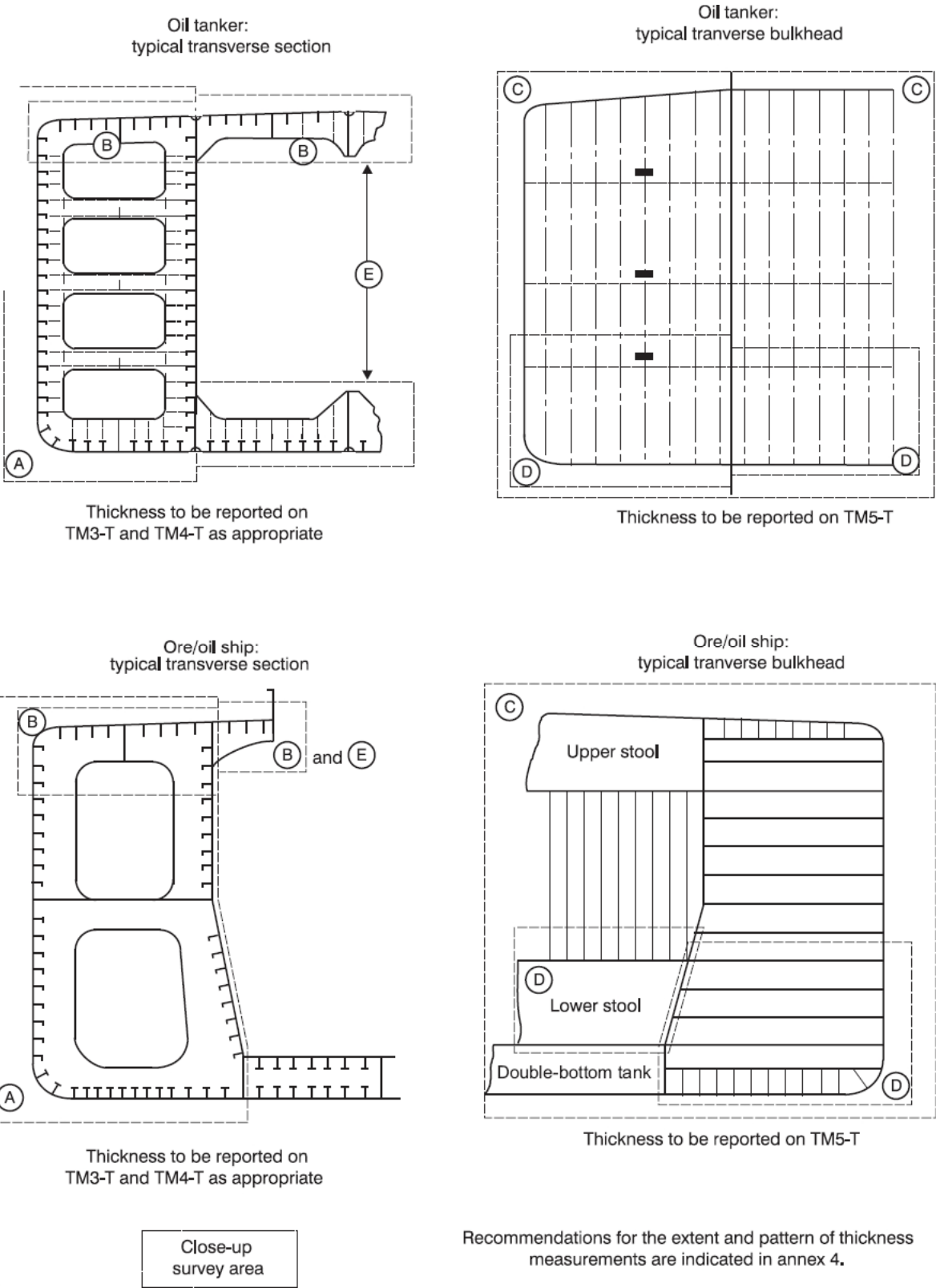
| REPORT ON TM4-T |                                    |
|-----------------|------------------------------------|
| ㉕               | Deck transverse centre tank        |
| ㉖               | Bottom transverse centre tank      |
| ㉗               | Deck transverse wing tank          |
| ㉘               | Side shell vertical web            |
| ㉙               | Longitudinal bulkhead vertical web |
| ㉚               | Bottom transverse wing tank        |
| ㉛               | Struts                             |
| ㉜               | Transverse web face plate          |
| ㉝               | D.B. floors                        |
| ㉞               |                                    |
| ㉟               |                                    |

**Typical transverse sections showing all longitudinal members to be reported on  
TM2-T(i) & (ii) and TM3-T**



Close-up survey requirements

(Transverse sections of oil tankers and ore/oil ships showing typical areas for thickness measurement in association with close-up survey requirements)



## **ANNEX 11**

### **GUIDELINES FOR TECHNICAL ASSESSMENT IN CONJUNCTION WITH THE PLANNING OF ENHANCED SURVEYS FOR OIL TANKERS**

#### **Renewal survey**

## **1 Introduction**

These Guidelines contain information and suggestions concerning technical assessments which may be of use in conjunction with the planning of enhanced special surveys of oil tankers. As indicated in 5.1.5, these Guidelines are a recommended tool which may be invoked at the discretion of an Administration, when considered necessary and appropriate, in conjunction with the preparation of the required survey programme.

## **2 Purpose and principles**

### **2.1 Purpose**

The purpose of the technical assessments described in these Guidelines is to assist in identifying critical structural areas, nominating suspect areas and in focusing attention on structural elements or areas of structural elements which may be particularly susceptible to, or evidence a history of, wastage or damage. This information may be useful in nominating locations, areas and tanks for thickness measurement, close-up survey and tank testing.

### **2.2 Minimum requirements**

These Guidelines may not be used to reduce the requirements of annexes 1, 2 and 3 for close-up survey, thickness measurement and tank testing, respectively, which are, in all cases, to be complied with as a minimum.

### **2.3 Timing**

As with other aspects of survey planning, the technical assessments described in these Guidelines should be carried out by the owner or operator in co-operation with the Administration well in advance of the commencement of the renewal survey, i.e. prior to commencing the survey and normally at least 12 to 15 months before the survey's completion due date.

### **2.4 Aspects to be considered**

**2.4.1** Technical assessments, which may include quantitative or qualitative evaluation of relative risks of possible deterioration, of the following aspects of a particular ship may be used as a basis for the nomination of tanks and areas for survey:

- .1** design features such as stress levels on various structural elements, design details and extent of use of high-tensile steel (HTS);
- .2** former history with respect to corrosion, cracking, buckling, indents and repairs for the particular ship as well as similar vessels, where available; and
- .3** information with respect to types of cargo carried, use of different tanks for cargo/ballast, protection of tanks and condition of coating, if any.



**2.4.2** Technical assessments of the relative risks of susceptibility to damage or deterioration of various structural elements and areas should be judged and decided on the basis of recognized principles and practices, such as may be found in references 1 and 2.

### **3 Technical assessment**

#### **3.1 General**

**3.1.1** There are three basic types of possible failure which may be the subject of technical assessment in connection with planning of surveys: corrosion, cracks and buckling. Contact damages are not normally covered by the survey plan since indents are usually noted in memoranda and assumed to be dealt with as a normal routine by surveyors.

**3.1.2** Technical assessments performed in conjunction with the survey planning process should, in principle, be as shown schematically in figure 1 which depicts, schematically, how technical assessments can be carried out in conjunction with the survey planning process. The approach is basically an evaluation of the risk based on the knowledge and experience related to design and corrosion.

**3.1.3** The design should be considered with respect to structural details which may be susceptible to buckling or cracking as a result of vibration, high stress levels or fatigue.

**3.1.4** Corrosion is related to the ageing process, and is closely connected with the quality of corrosion protection at newbuilding, and subsequent maintenance during the service life. Corrosion may also lead to cracking and/or buckling.

#### **3.2 Methods**

##### **3.2.1 Design details**

**3.2.1.1** Damage experience related to the ship in question and similar ships, where available, are the main source of information to be used in the process of planning. In addition, a selection of structural details from the design drawings should be included. Typical damage experience to be considered will consist of:

- .1 number, extent, location and frequency of cracks; and
- .2 location of buckles.

**3.2.1.2** This information may be found in the survey reports and/or the owner's files, including the results of the owner's own inspections. The defects should be analysed, noted and marked on sketches.

**3.2.1.3** In addition, general experience should be utilized. For example, reference should be made to reference 1, which contains a catalogue of typical damages and proposed repair methods for various tanker structural details.

**3.2.1.4** Such figures should be used together with a review of the main drawings, in order to compare with the actual structure and search for similar details which may be susceptible to damage. An example is shown in figure 2.

**3.2.1.5** The review of the main structural drawings, in addition to using the above-mentioned figures, should include checking for typical design details where cracking has been experienced. The factors contributing to damage should be carefully considered.

**3.2.1.6** The use of HTS is an important factor. Details showing good service experience where ordinary, mild steel has been used may be more susceptible to damage when HTS, and its higher associated stresses, are utilized. There is extensive and, in general, good experience, with the use of HTS for longitudinal material in deck and bottom structures. Experience in other locations, where the dynamic stresses may be higher, is less favourable, e.g., side structures.

**3.2.1.7** In this respect, stress calculations of typical and important components and details, in accordance with relevant methods, may prove useful and should be considered.

**3.2.1.8** The selected areas of the structure identified during this process should be recorded and marked on the structural drawings to be included in the survey programme.

### **3.2.2**     *Corrosion*

**3.2.2.1** In order to evaluate relative corrosion risks, the following information is generally to be considered:

- .1        usage of tanks and spaces;
- .2        condition of coatings;
- .3        condition of anodes;
- .4        cleaning procedures;
- .5        previous corrosion damage;
- .6        ballast use and time for cargo tanks;
- .7        corrosion risk scheme (see reference 2, table 3.1); and
- .8        location of heated tanks.

**3.2.2.2** Reference 2 gives definitive examples which can be used for judging and describing coating condition, using typical pictures of conditions.

**3.2.2.3** The evaluation of corrosion risks should be based on information in reference 2, together with the age of the ship and relevant information on the anticipated condition as derived from the information collected in order to prepare the survey programme.

**3.2.2.4** The various tanks and spaces should be listed with the corrosion risks nominated accordingly.

### **3.2.3**     *Locations for close-up survey and thickness measurement*

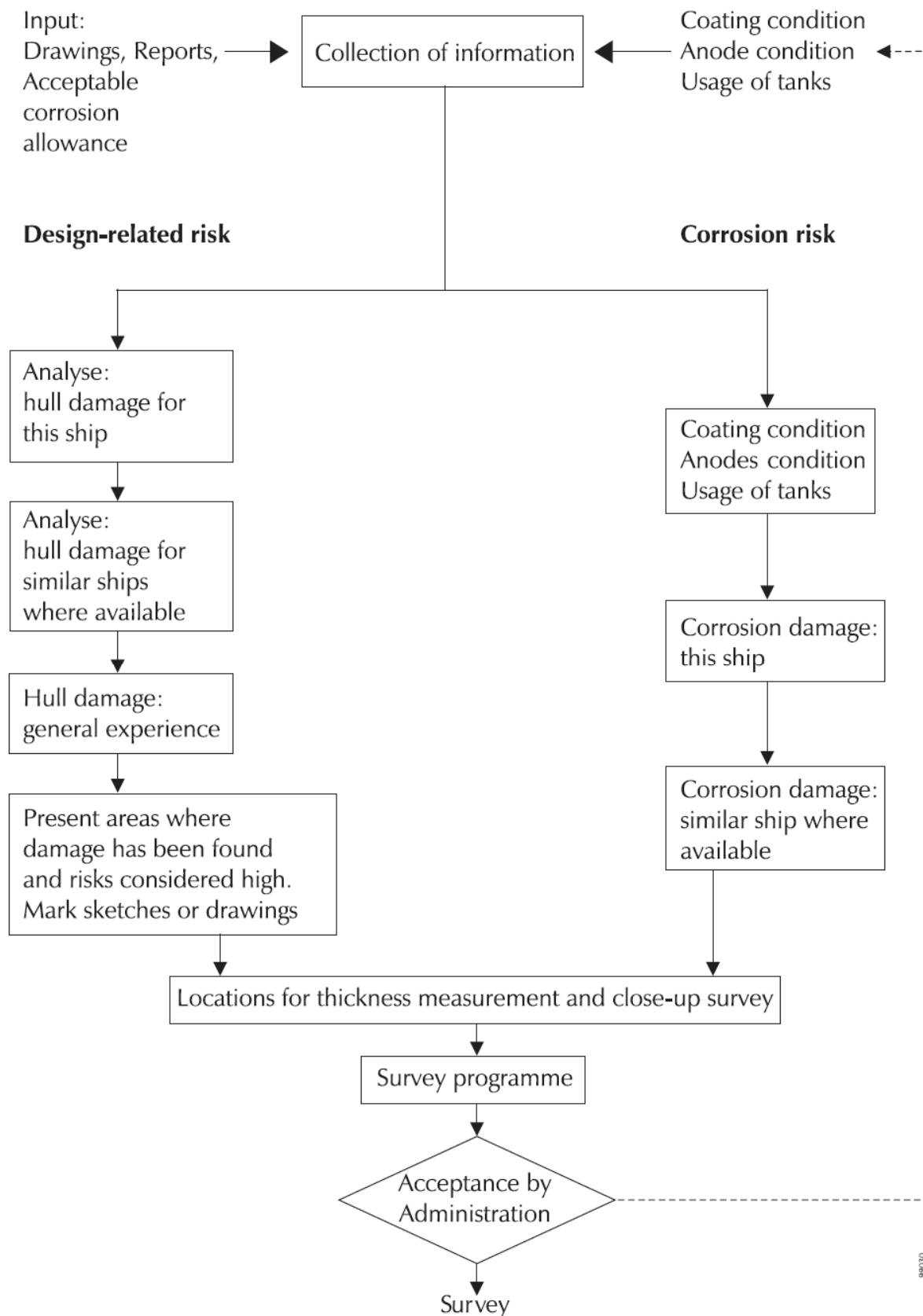
**3.2.3.1** On the basis of the table of corrosion risks and the evaluation of design experience, the locations for initial close-up survey and thickness measurement (sections) may be nominated.

**3.2.3.2** The sections subject to thickness measurement should normally be nominated in tanks and spaces where corrosion risk is judged to be the highest.

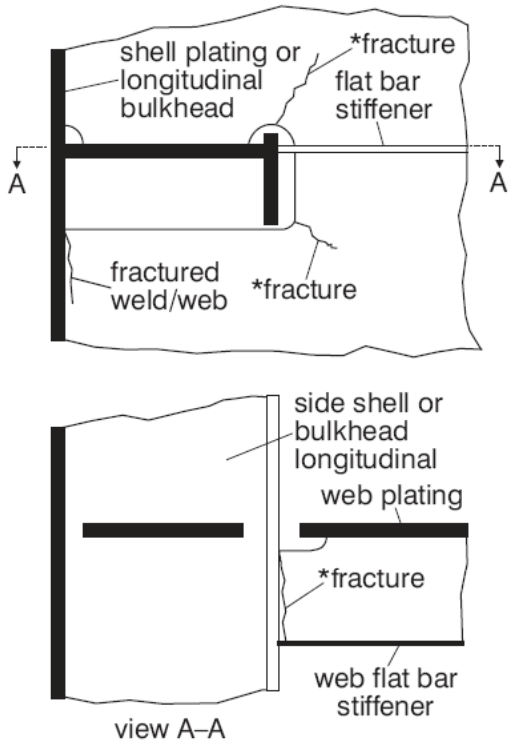
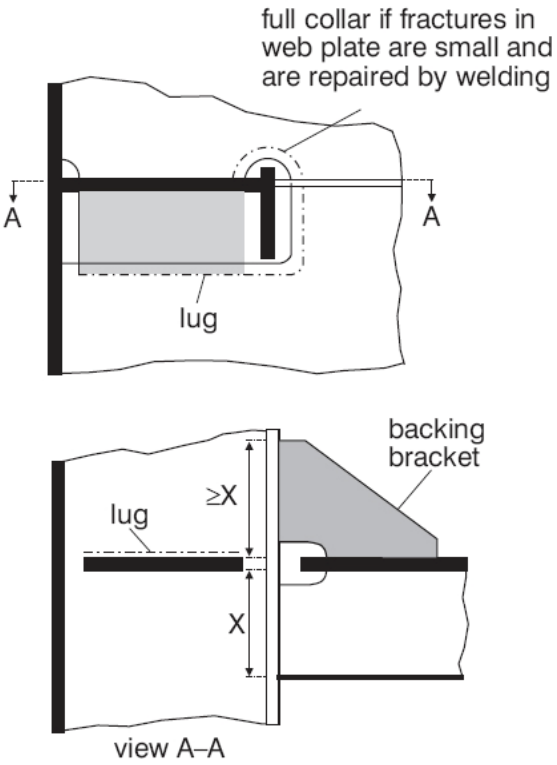
**3.2.3.3** The nomination of tanks and spaces for close-up survey should, initially, be based on highest corrosion risk, and should always include ballast tanks. The principle for the selection should be that the extent is increased by age or where information is insufficient or unreliable.

## References

- 1 TSCF, Guidance Manual for the Inspection and Condition Assessment of Tanker Structures, 1986.
- 2 TSCF, Condition Evaluation and Maintenance of Tanker Structures, 1992.



**Figure 1** – Planning process: technical assessment and the survey

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                          |             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------|
| <b>LOCATION:</b> Connection of longitudinals to transverse webs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                          |             |
| <b>EXAMPLE NO. 1</b> Web and flat bar fractures at cut-outs for longitudinal stiffener connections                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                          |             |
| <p><b>TYPICAL DAMAGE</b></p>  <p>view A-A</p> <p>Note* one or more fractures may occur</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                          |             |
| <p><b>PROPOSED REPAIR</b></p>  <p>view A-A</p> <p>web and flat bar cropped and part renewed or alternatively welded</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                          |             |
| <p><b>FACTORS CONTRIBUTING TO DAMAGE</b></p> <ol style="list-style-type: none"> <li>1 Asymmetrical connection of flat bar stiffener resulting in high peak stresses at the heel of the stiffener under fatigue loading.</li> <li>2 Insufficient area of connection of longitudinal to web plate.</li> <li>3 Defective weld at return around the plate thickness.</li> <li>4 High localized corrosion at areas of stress concentration such as flat bar stiffener connections, corners of cut-out for the longitudinal and connection of web to shell at cut-outs.</li> <li>5 High shear stress in the web of the transverse.</li> <li>6 Dynamic seaway loads/ship motions.</li> </ol> |                                          |             |
| FIGURE<br>1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | TANKER STRUCTURE CO-OPERATIVE FORUM      | FIGURE<br>1 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SUBJECT: CATALOGUE OF STRUCTURAL DETAILS |             |

**Figure 2 – Typical damage and repair example (reproduced from reference 1)**

## ANNEX 12

### CRITERIA FOR LONGITUDINAL STRENGTH OF HULL GIRDER FOR OIL TANKERS

#### 1 General

**1.1** These criteria should be used for the evaluation of the longitudinal strength of the ship's hull girder as required by 8.1.2.

**1.2** In order that the ship's longitudinal strength to be evaluated can be recognized as valid, fillet welding between longitudinal internal members and hull envelopes should be in sound condition so as to keep the integrity of longitudinal internal members with hull envelopes.

#### 2 Evaluation of longitudinal strength

On oil tankers of 130 m in length and upwards and over 10 years of age, the longitudinal strength of the ship's hull girder should be evaluated in compliance with the requirements of this annex on the basis of the thickness measured, renewed or reinforced, as appropriate, during the renewal survey of the Cargo Ship Safety Construction Certificate or Cargo Ship Safety Certificate (SC renewal survey). The condition of the hull girder for longitudinal strength evaluation should be determined in accordance with the methods specified in appendix 3.

##### **2.1** *Calculation of transverse sectional areas of deck and bottom flanges of hull girder*

**2.1.1** The transverse sectional areas of deck flange (deck plating and deck longitudinals) and bottom flange (bottom shell plating and bottom longitudinals) of the ship's hull girder should be calculated by using the thickness measured, renewed or reinforced, as appropriate, during the SC renewal survey.

**2.1.2** If the diminution of sectional areas of either deck or bottom flange exceeds 10% of their respective as-built area (i.e. original sectional area when the ship was built), either one of the following measures should be taken:

- .1** to renew or reinforce the deck or bottom flanges so that the actual sectional area is not less than 90% of the as-built area; or
- .2** to calculate the actual section of moduli ( $Z_{act}$ ) of transverse section of the ship's hull girder by applying the calculation method specified in appendix 1, by using the thickness measured, renewed or reinforced, as appropriate, during the SC renewal survey.

##### **2.2** *Requirements for transverse section modulus of hull girder*

**2.2.1** The actual section moduli of the transverse section of the ship's hull girder, calculated in accordance with paragraph 2.1.2.2, should satisfy either of the following provisions, as applicable:

- .1 for ships constructed on or after 1 July 2002, the actual section moduli ( $Z_{act}$ ) of the transverse section of the ship's hull girder calculated in accordance with the requirements of paragraph 2.1.2.2 should be not less than the diminution limits determined by the Administration, taking into account the recommendations adopted by the Organization;<sup>22</sup> or
- .2 for ships constructed before 1 July 2002, the actual section moduli ( $Z_{act}$ ) of the transverse section of the ship's hull girder calculated in accordance with the requirements of paragraph 2.1.2.2 should meet the criteria for minimum section modulus for ships in service required by the Administration or recognized classification society, provided that in no case  $Z_{act}$  should be less than the diminution limit of the minimum section modulus ( $Z_{mc}$ ) as specified in appendix 2.

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<sup>22</sup> Refer to resolution MSC.108(73), Recommendation on compliance with the requirements of paragraph 2.2.1.1 of annex 12 to annex B to resolution A.744(18).

## Appendix 1

### CALCULATION CRITERIA OF SECTION MODULI OF MIDSHIP SECTION OF HULL GIRDER

**1** When calculating the transverse section modulus of the ship's hull girder, the sectional area of all continuous longitudinal strength members should be taken into account.

**2** Large openings, i.e. openings exceeding 2.5 m in length or 1.2 m in breadth, and scallops, where scallop welding is applied, are always to be deducted from the sectional areas used in the section modulus calculation.

**3** Smaller openings (manholes, lightening holes, single scallops in way of seams, etc.) need not be deducted, provided that the sum of their breadths or shadow area breadths in one transverse section does not reduce the section modulus at deck or bottom by more than 3% and provided that the height of lightening holes, draining holes and single scallops in longitudinals or longitudinal girders does not exceed 25% of the web depth, for scallops of maximum 75 mm.

**4** A deduction-free sum of smaller opening breadths in one transverse section in the bottom or deck area of  $0.06 (B - \Sigma b)$  (where  $B$  = breadth of ship,  $\Sigma b$  = total breadth of large openings) may be considered equivalent to the above reduction in sectional modulus.

**5** The shadow area will be obtained by drawing two tangent lines with an opening angle of  $30^\circ$ .

**6** The deck modulus is related to the moulded deck line at side.

**7** The bottom modulus is related to the base line.

**8** Continuous trunks and longitudinal hatch coamings should be included in the longitudinal sectional area provided they are effectively supported by longitudinal bulkheads or deep girders. The deck modulus is then to be calculated by dividing the moment of inertia by the following distance, provided this is greater than the distance to the deck line at side:

$$y_t = y \left( 0.9 + 0.2 \frac{x}{B} \right)$$

where:

$y$  = distance from neutral axis to top of continuous strength member;

$x$  = distance from top of continuous strength member to centerline of the ship;

$x$  and  $y$  to be measured to the point giving the largest value of  $y_t$ .

**9** Longitudinal girders between multi-hatchways will be considered by special calculations.



## Appendix 2

### DIMINUTION LIMIT OF MINIMUM LONGITUDINAL STRENGTH OF SHIPS IN SERVICE

**1** The diminution limit of the minimum section modulus ( $Z_{mc}$ ) of oil tankers in service is given by the following formula:

$$Z_{mc} = cL^2 B (C_b + 0.7)k \quad (\text{cm}^3)$$

where:

$L$  = Length of ship.  $L$  is the distance, in metres, on the summer load waterline from the fore-side of stem to the after-side of the rudder post, or the centre of the rudder stock if there is no rudder post.  $L$  is not to be less than 96%, and need not be greater than 97%, of the extreme length on the summer load waterline. In ships with unusual stern and bow arrangement, the length  $L$  may be specially considered.

$B$  = Greatest moulded breadth in metres.

$C_b$  = Moulded block coefficient at draught  $d$  corresponding to summer load waterline, based on  $L$  and  $B$ .  $C_b$  is not to be taken less than 0.6.

$$C_b = \frac{\text{moulded displacement (m}^3\text{) at draught } d}{LBd}$$

$$c = 0.9c_n$$

$$c_n = 10.75 - \left(\frac{300-L}{100}\right)^{1.5} \quad \text{for } 130 \text{ m} \leq L \leq 300 \text{ m}$$

$$c_n = 10.75 \quad \text{for } 300 \text{ m} \leq L \leq 350 \text{ m}$$

$$c_n = 10.75 - \left(\frac{L-350}{150}\right)^{1.5} \quad \text{for } 350 \text{ m} \leq L \leq 500 \text{ m}$$

$k$  = material factor, e.g.,

$$k = 1.0 \text{ for mild steel with yield stress of } 235 \text{ N/mm}^2 \text{ and over}$$

$$k = 0.78 \text{ for high-tensile steel with yield stress of } 315 \text{ N/mm}^2 \text{ and over}$$

$$k = 0.72 \text{ for high-tensile steel with yield stress of } 355 \text{ N/mm}^2 \text{ and over.}$$

**2** Scantlings of all continuous longitudinal members of the ship's hull girder based on the section modulus requirement in 1 above should be maintained within  $0.4L$  amidships. However, in special cases, based on consideration of type of ship, hull form and loading conditions, the scantlings may be gradually reduced towards the end of  $0.4L$  part, bearing in mind the desire not to inhibit the ship's loading flexibility.

**3** However, the above standard may not be applicable to ships of unusual type or design, e.g., for ships of unusual main proportions and/or weight distributions.

## **Appendix 3**

### **SAMPLING METHOD OF THICKNESS MEASUREMENTS FOR LONGITUDINAL STRENGTH EVALUATION AND REPAIR METHODS**

#### **1 Extent of longitudinal strength evaluation**

Longitudinal strength should be evaluated within  $0.4L$  amidships for the extent of the hull girder length that contains tanks therein and within  $0.5L$  amidships for adjacent tanks which may extend beyond  $0.4L$  amidships, where tanks means ballast tanks and cargo tanks.

#### **2 Sampling method of thickness measurement**

**2.1** Pursuant to the requirements of section 2.5, transverse sections should be chosen such that thickness measurements can be taken for as many different tanks in corrosive environments as possible, e.g., ballast tanks sharing a common plane boundary with cargo tanks fitted with heating coils, other ballast tanks, cargo tanks permitted to be filled with sea water and other cargo tanks. Ballast tanks sharing a common plane boundary with cargo tanks fitted with heating coils and cargo tanks permitted to be filled with seawater should be selected where present.

**2.2** The minimum number of transverse sections to be sampled should be in accordance with annex 2. The transverse sections should be located where the largest thickness reductions are suspected to occur or are revealed from deck and bottom plating measurements prescribed in 2.3 and should be clear of areas which have been locally renewed or reinforced.

**2.3** At least two points should be measured on each deck plate and/or bottom shell plate required to be measured within the cargo area in accordance with the requirements of annex 2.

**2.4** Within  $0.1D$  (where  $D$  is the ship's moulded depth) of the deck and bottom at each transverse section to be measured in accordance with the requirements of annex 2, every longitudinal and girder should be measured on the web and face plate, and every plate should be measured at one point between longitudinals.

**2.5** For longitudinal members other than those specified in 2.4 to be measured at each transverse section in accordance with the requirements of annex 2, every longitudinal and girder should be measured on the web and face plate, and every plate should be measured at least in one point per strake.

**2.6** The thickness of each component should be determined by averaging all of the measurements taken in way of the transverse section on each component.

#### **3 Additional measurements where the longitudinal strength is deficient**

**3.1** Where one or more of the transverse sections are found to be deficient in respect of the longitudinal strength requirements given in this annex, the number of transverse sections for thickness measurement should be increased such that each tank within the  $0.5L$  amidships region has been sampled. Tank spaces that are partially within, but extend beyond, the  $0.5L$  region, should be sampled.

**3.2** Additional thickness measurements should also be performed on one transverse section forward and one aft of each repaired area to the extent necessary to ensure that the areas bordering the repaired section also comply with the requirements of the Code.

#### **4 Effective repair methods**

**4.1** The extent of renewal or reinforcement carried out to comply with this annex should be in accordance with 4.2.

**4.2** The minimum continuous length of a renewed or reinforced structural member should be not less than twice the spacing of the primary members in way. In addition, the thickness diminution in way of the butt joint of each joining member forward and aft of the replaced member (plates, stiffeners, girder webs and flanges, etc.) should not be within the substantial corrosion range (75% of the allowable diminution associated with each particular member). Where differences in thickness at the butt joint exceed 15% of the lower thickness, a transition taper should be provided.

**4.3** Alternative repair methods involving the fitting of straps or structural member modification should be subject to special consideration. In considering the fitting of straps, it should be limited to the following conditions:

- .1** to restore and/or increase longitudinal strength;
- .2** the thickness diminution of the deck or bottom plating to be reinforced should not be within the substantial corrosion range (75% of the allowable diminution associated with the deck plating);
- .3** the alignment and arrangement, including the termination of the straps, is in accordance with a standard recognized by the Administration;
- .4** the straps are continuous over the entire 0.5L amidships length; and
- .5** continuous fillet welding and full penetration welds are used at butt welding and, depending on the width of the strap, slot welds. The welding procedures applied should be acceptable to the Administration.

**4.4** The existing structure adjacent to replacement areas and in conjunction with the fitted straps, etc., should be capable of withstanding the applied loads, taking into account the buckling resistance and the condition of welds between the longitudinal members and hull envelope plating.

**ANNEX 9**

**DRAFT ASSEMBLY RESOLUTION**

**ADOPTION OF THE CODE OF SAFE PRACTICE FOR SHIPS  
CARRYING TIMBER DECK CARGOES, 2011 (2011 TDC CODE)**

THE ASSEMBLY,

RECALLING Article 15(j) of the Convention on the International Maritime Organization concerning the functions of the Assembly in relation to regulations and guidelines concerning maritime safety,

RECALLING further that, by resolution A.715(17), it had adopted the Code of Safe Practice for Ships Carrying Timber Deck Cargoes, 1991,

RECOGNIZING the need to improve the provisions contained in the Code in the light of experience gained,

HAVING CONSIDERED the recommendations made by the Maritime Safety Committee, at its eighty-ninth session,

1. ADOPTS the Code of Safe Practice for Ships Carrying Timber Deck Cargoes, 2011 (2011 TDC Code), set out in the Annex to the present resolution;
2. RECOMMENDS Governments to use the provisions of the 2011 TDC Code as a basis for relevant safety standards;
3. AUTHORIZES the Maritime Safety Committee to amend the Code as necessary in the light of further studies and experience gained from the implementation of the provisions contained therein;
4. REVOKES resolution A.715(17).

ANNEX

**DRAFT REVISED CODE OF SAFE PRACTICE FOR SHIPS CARRYING TIMBER DECK  
CARGOES, 2011 (2011 TDC CODE)**

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## **PREFACE**

The Code of Safe Practice for Ships Carrying Timber Deck Cargoes was first developed by the Organization in 1972 and subsequently amended in 1978.

The Code was revised by IMO resolution A.715(17) – Code of Safe Practice for Ships Carrying Timber Deck Cargoes, 1991, which was adopted on 6 November 1991.

This Code is based on the previous Code, which has been revised and amended in order to reflect the capability of today's ships and the equipment available on board and also taking expected future innovations in mind.

This Code is designed to assist:

- .1 shipowners, charterers, operating companies and ships' crew;
- .2 port industries, shippers and pre-packaging organizations, which are involved in preparation, loading, and stowing of timber deck cargoes; and
- .3 Administrations, manufacturers and designers of ships and equipment associated with the carriage of timber deck cargoes and those developing cargo securing manuals,

in the carriage of timber deck cargoes.

This Code is directed primarily at providing recommendations for the safe carriage of timber deck cargoes.

### **Status of references**

The references given in this consolidated text do not form part of the Code but are inserted for ease of reference.

## CHAPTER 1 – GENERAL

### 1.1 Purpose

1.1.1 The purpose of the Code is to ensure that timber deck cargoes are loaded, stowed and secured to prevent, as far as practicable, throughout the voyage, damage or hazard to the ship and persons on board as well as loss of cargo overboard<sup>(1)</sup>.

1.1.2 The Code provides:

- .1 practices for safe transportation;
- .2 methodologies for safe stowage and securing;
- .3 design principles for securing systems;
- .4 guidance for developing procedures and instructions to be included in ships' cargo securing manuals on safe stowage and securing; and
- .5 sample checklists for safe stowage and securing.

### 1.2 Application

1.2.1 The provisions of this Code apply to all ships of 24 metres or more in length, carrying a timber deck cargo. This Code will be effective from [*to be decided*].

1.2.2 Cargo securing of timber deck cargoes should be in accordance with the requirements in the ship's Cargo Securing Manual (CSM), based on the principles in chapter 5 or chapter 6 of Part B of this Code.

1.2.3 The Master should note that national requirements may exist which may restrict the application of either chapter 5 or chapter 6, and these may also require third party inspections to ensure that the cargo has been properly secured according to the ship's cargo securing manual.

1.2.4 Cargo securing manuals for timber deck cargoes, approved following the implementation date of this Code, should meet the contents of this Code. Existing cargo securing manuals approved under the previous Timber Deck Cargo Code (resolution A.715(17)) may remain valid.

### 1.3 Definitions

1.3.1 The following *definitions* apply to this Code:

#### ***General expressions***

- .1 *Administration* means the Government of the State whose flag the ship is entitled to fly.
- .2 *Company* means the Owner of the ship or any other organization or person such as the Manager, or the Bareboat Charterer, who has assumed the responsibility for operation of the ship from the Ship owner and who, on assuming such responsibility, has agreed to take over all duties and responsibilities imposed by SOLAS<sup>(2)</sup>.



- .3 *Load Lines Convention* means the International Convention on Load Lines, 1966, or the 1988 Protocol relating thereto, as applicable.
- .4 *Organization* means the International Maritime Organization (IMO).
- .5 *Port industries* means the port facilities and/or stevedoring companies serving ships engaged in the stowage of timber deck cargoes.
- .6 *Shipper* means any person, organization or Government which prepares or provides a consignment for transport<sup>(3)</sup>.
- .7 *SOLAS* means the International Convention for the Safety of Life at Sea, 1974, as amended.
- .8 *2008 IS Code* means the International Code on Intact Stability, 2008.

***Cargo related expressions***

- .9 *Cant* means a log which is "slab-cut", i.e. ripped lengthwise so that the resulting thick pieces have two opposing, parallel flat sides and in some cases a third side which is sawn flat.
- .10 *Non-rigid cargo* means sawn wood or lumber, cants, logs, poles, pulpwood and all other types of loose timber or timber in packaged forms not fulfilling specified strength requirement, as defined in section 4.7.
- .11 *Rigid cargo package* means sawn wood or lumber, cants, logs, poles, pulpwood and all other types of timber in packaged forms, fulfilling specified strength requirement, as defined in section 4.7.
- .12 *Round wood* means parts of trees that have not been sawn on more than one long side. The term includes, among others, logs, poles and pulpwood in loose or packed form.
- .13 *Sawn wood* means parts of trees that have been sawn so that they have at least two parallel flat long sides. The term includes, among others, lumber and cants in loose or packed form.
- .14 *Timber* is used as a collective expression used for all types of wooden material covered by this Code, including both round and sawn wood but excluding wood pulp and similar cargo.

***Technically related expressions***

- .15 *Blocking device* means physical measures to prevent sliding and/or tipping of cargoes and/or collapse of stow.
- .16 *Lashing plan* means a sketch or drawing showing the required number and strength of securing items for the timber deck cargo to obtain safe stowage and securing of timber deck cargoes.
- .17 *Timber deck cargo* means a cargo of timber carried on an uncovered part of a freeboard or superstructure deck.

- .18 *Timber load line* means a special load line assigned to ships complying with certain conditions set out in the International Convention on Load Lines.
- .19 *Stowage Factor (SF)* means the volume occupied by one tonne of a cargo when stowed and separated in the accepted manner.
- .20 *Weather deck* means the uppermost complete deck exposed to weather and sea.
- .21 *Reeving* means the process where a rope, chain or any other type of lashing can freely move through a sheave or over a fulcrum such as a rounded angle piece, in such a manner so as to minimize the frictional effect of such movement.
- .22 *Height of cargo* means the distance from the base of the deck cargo stow to the highest part of the cargo.

## **PART A – OPERATIONAL REQUIREMENTS**

### **CHAPTER 2 – GENERAL RECOMMENDATIONS ON STOWAGE AND SECURING OF TIMBER DECK CARGOES**

#### **2.1 Goals**

2.1.1 The stowage and cargo securing arrangements for timber deck cargoes should enable a safe yet rational securing of the cargo so that it is satisfactorily prevented from shifting by collapsing, sliding or tipping in any direction, taking into account the acceleration forces the cargo may be subjected to throughout the voyage in the worst sea and weather conditions which may be expected.

2.1.2 This chapter lists measures and factors that should be taken under consideration in order to achieve such level of cargo securing.

2.1.3 Procedures should be established for the preparation of plans and instructions, including checklists as appropriate, for key shipboard operations<sup>(5)</sup>. Guidance is provided in Annex A to assist the development of such checklists.

#### **2.2 Pre-loading operation**

2.2.1 Prior to loading the vessel, relevant cargo information,<sup>(4)</sup> as defined in chapter 4 of this Code, should be provided by the shipper, according to the custom of the trade.

2.2.2 The master of the vessel should study the relevant cargo information and take the precautions necessary for proper stowage, securing and safe carriage of the cargo as defined in this Code and as prescribed in the vessel's Cargo Securing Manual.

2.2.3 Prior to loading, the stevedoring company should be made aware of specific requirements according to the ship's Cargo Securing Manual regarding stowage and securing of timber deck cargoes.

2.2.4 During loading of deck cargo the master should ensure that all tanks are maintained in such a condition that free surface effects are minimized. Ballast tanks should as far as practicable be either full or empty and ballast movement during loading operations should be avoided.

2.2.5 Before timber deck cargo is loaded on any area of the weather deck:

- .1 hatch covers and other openings to spaces below that area should be securely closed and battened down;
- .2 air pipes and ventilators should be effectively protected and check-valves or similar devices should be examined to ascertain their effectiveness against the entry of water;
- .3 objects which might obstruct cargo stowage on deck should be removed and safely secured in places appropriate for storage;
- .4 the condition of friction enhancing arrangements, where fitted, should be checked;

- .5 accumulations of ice and snow on such area should be removed;
- .6 it is normally preferable to have all deck lashings, uprights, etc., readily available before loading on that specific area. This will be necessary should a preloading examination of securing equipment be required in the loading port; and
- .7 all sounding pipes on the deck should be reviewed and arrangements made that access to these remain as far as practicable.

2.2.6 Further aspects to be considered during pre-loading operations are given in Annex A, chapter A.1.

## **2.3 Permitted loading weights on decks and hatch covers**

2.3.1 The hatch cover securing and support arrangements, chocks, etc., as well as coamings should be designed and reinforced as necessary for carriage of timber deck cargoes. Potential weight increase of timber deck cargoes due to water absorption, icing, etc., should be taken under consideration.

2.3.2 Care should be taken not to exceed the designed maximum permissible loads on weather deck and hatch covers during any stage of the voyage<sup>(6)</sup>.

## **2.4 Stability**

2.4.1 The master should ensure that the ship condition complies with its stability booklet at all times.

2.4.2 A ship carrying timber deck cargo must comply with applicable parts of the damage stability requirements as well as the 2008 IS Code<sup>(11)</sup>, particularly the timber deck cargo requirements. Since excessive GM values induce large accelerations, GM should preferably not exceed 3% of the breadth of the vessel, as indicated in paragraph 3.7.5 of the 2008 IS Code.

2.4.3 Ballast water exchange operations should be carried out in accordance with instructions in the Ballast Water Management Plan, if available<sup>(12)</sup>. The ballast water exchange operation, if required, should be considered when planning the amount of cargo to be loaded on deck.

2.4.4 According to the 2008 IS Code<sup>(11)</sup>, account may be taken of the buoyancy of timber deck cargo when calculating stability curves, assuming that such cargo has a permeability up to 25%. Permeability is defined as the percentage of empty space of the volume occupied by the deck cargo. Additional curves of stability may be required if the Administration considers it necessary to investigate the influence of different permeabilities and/or assumed effective height of the deck cargo. 25% permeability corresponds to sawn wood cargo and 40%-60% permeability corresponds to round wood cargo with increasing permeability with increasing log diameters.

## **2.5 Load line**

Ships assigned and making use of their timber load line should follow relevant regulations of the applicable Load Lines Convention<sup>(13)</sup> for stowage and securing of timber as prescribed in the ship's Cargo Securing Manual.

## **2.6 Timber freeboard**

2.6.1 The timber freeboard, if applicable, will be found in the ship's Load Line Certificate.

2.6.2 Instructions on computation of the timber freeboard are given in the applicable Load Lines Convention<sup>(14)</sup>.

## **2.7 Visibility**

2.7.1 Timber deck cargo should be loaded in such a manner as to ensure that the ship complies with the visibility requirements contained in SOLAS chapter V. National deviations may exist and should be taken into consideration as required dependent on the intended voyage.

2.7.2 The SOLAS requirements on visibility as well as instructions on how to calculate the visibility range are given in chapter 3.

## **2.8 Work safety and work environment aspects**

2.8.1 The Company should establish procedures by which the ship's personnel receive relevant information on the Safety Management System<sup>(16)</sup> in a working language or languages understood by them.

2.8.2 When deck cargo is being lashed and secured, special measures may be needed to ensure safe access to the top of, and across, the cargo so that the risk of falling is minimized. Safety helmets, proper footwear and non-obstructive high visibility garments should be worn during work on deck.

2.8.3 The risk of slipping should especially be considered during winter time when loading timber packages covered by plastic wrapping or tarpaulins. Plastic wrapping on packages with lumber of uneven length should be avoided otherwise clearly identified.

2.8.4 Lighting during loading and discharge operations should be reasonably constant and arranged to minimize glare and dazzle, the formation of deep shadows and sharp contrasts in the level of illumination between one area and another.

2.8.5 Any obstruction such as lashings or securing points in the access way of escape routes and spaces essential to operation of the vessel, such as machinery spaces and crew's quarters, as well as obstructions to safety equipment, fire-fighting equipment and sounding pipes, should be clearly marked. In no case should an obstruction prevent safe access or egress of escape arrangements and spaces referred to above.

2.8.6 During the course of the voyage, if there is no convenient passage for the crew on or below the deck of the ship<sup>(18)</sup> giving safe means of access from the accommodation to all parts used in the necessary working of the ship, guard lines or rails, not more than 330 mm apart vertically, should be provided on each side of the deck cargo to a height of at least 1 m above the cargo. In addition, a lifeline, preferably wire rope, set up taut with a tightening device should be provided as near as practicable to the centreline of the ship. The stanchion supports to all guardrails or lifelines should be spaced so as to prevent undue sagging. Where the cargo is uneven, a safe walking surface of not less than 600 mm in width should be fitted over the cargo and effectively secured beneath, or adjacent to, the lifeline.

2.8.7 Fencing or means of closing should be provided for all openings in the stow such as at masts, winches, etc.

2.8.8 Where uprights are not fitted or where alternative to the provisions of 2.8.6 are permitted, a walkway of substantial construction should be provided having an even walking surface and consisting of two fore and aft sets of guardlines or rails about 1 m apart, each having a minimum of three courses of guardlines or rails to a height of not less than 1 m above the walking surface. Such guardlines or rails should be supported by rigid stanchions spaced not more than 3 m apart and lines should be set up taut by tightening devices.

2.8.9 As an alternative to 2.8.6, 2.8.7 and 2.8.8, a lifeline, preferably wire rope, may be erected above the timber deck cargo such that a crew member equipped with a fall protection system can hook on to it and work about the timber deck cargo. The lifeline should be:

- .1 erected about 2 m above the timber deck cargo as near as practicable to the centreline of the ship;
- .2 stretched sufficiently taut with a tightening device to support a fallen crew member without collapse or failure.

2.8.10 Properly constructed ladders, steps or ramps fitted with guard lines or handrails should be provided from the top of the cargo to the deck, and in other cases where the cargo is stepped, in order to provide reasonable access.

2.8.11 Personnel safety equipment referred to in this chapter should be kept in an easily accessible place.

2.8.12 When lashings need to be checked and/or retightened during voyage, the Master should take appropriate actions to reduce the motion of the vessel during such operation.

2.8.13 Additional guidance regarding work safety and work environment aspects can be found in the relevant International Labour Organization (ILO) Conventions<sup>(17)</sup>.

## **2.9 Stowage**

2.9.1 The basic principle for the safe carriage of timber deck cargo is to make the stow as solid, compact and stable as practicable. The purpose of this is to:

- .1 prevent movement in the stow which could cause the lashings to slacken;
- .2 produce a binding effect within the stow; and
- .3 reduce to a minimum the permeability of the stow.

2.9.2 Openings in the deck exposed to weather over which cargo is stowed should be securely closed and battened down. The ventilators and air pipes should be effectively protected<sup>(19)</sup>.

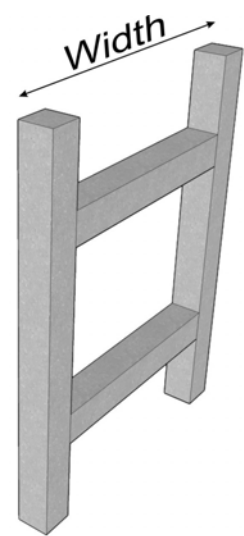
2.9.3 Deck cargo should be stowed so that access is provided to and from designated escape routes and spaces essential to operation of the vessel, such as machinery spaces and crew's quarters, as well as to safety equipment, fire-fighting equipment and sounding pipes<sup>(18)</sup>. It should not interfere in any way with the navigation and necessary work of the ship<sup>(19)</sup>.

2.9.4 When cargo is loaded voids may occur in the stow between packages as well as between bulwarks or gantry crane rails, etc., and other fixed constructions such as the hatch coaming.

2.9.5 Care should be taken to avoid the creation of voids or open spaces when loading cargo. Voids, where created, should be filled with loose timber or blocked by vertical H-frames with required strength to avoid cargo from shifting. The MSL for double H-frames of different widths and dimensions are given in the table below. The values apply to H-frames made of sound softwood timber without knots.

**Table 2.1.** MSL (maximum secure load) of H-frames for different dimensions

| Dimensions of battens<br>mm | MSL in kN of double H-frames with different widths |       |       |       |
|-----------------------------|----------------------------------------------------|-------|-------|-------|
|                             | 0.5 m                                              | 1.0 m | 1.5 m | 2.0 m |
| 50 x 50                     | 75                                                 | 53    | 30    | 17    |
| 50 x 75                     | 113                                                | 79    | 46    | 26    |
| 50 x 100                    | 151                                                | 106   | 61    | 34    |
| 50 x 150                    | 226                                                | 159   | 91    | 51    |
| 75 x 75                     | 186                                                | 153   | 119   | 85    |
| 75 x 100                    | 248                                                | 203   | 159   | 114   |
| 75 x 150                    |                                                    | 305   | 238   | 171   |
| 75 x 200                    |                                                    |       | 317   | 227   |
| 100 x 100                   |                                                    | 301   | 256   | 212   |



2.9.6 Timber deck cargo which substantially overhangs (one-third of the package length) hatch coamings or other structures in the longitudinal direction, should be supported at the outer end by other cargo stowed on deck or railing or equivalent structure of sufficient strength to support it.

2.9.7 For ships assigned and making use of a timber load line, additional practices apply in accordance with the applicable Load Lines Convention.<sup>(19)</sup>

## 2.10 Securing

2.10.1 One or more of the following principal methods may be used to secure timber deck cargoes, by themselves or in combination with each other:

- .1 different types of lashing arrangements; and
- .2 bottom blocking of the base tier in combination with lashing arrangements;
- .3 blocking over the full height of the cargo by, e.g., uprights alternatively complemented by lashing arrangements;
- .4 frictional securing, taking into account scientific research and appropriate weather and voyage criteria; and
- .5 other practical securing enhancement, (taking into account appropriate weather and voyage criteria), such as:

- .1 non slip paints on hatch covers;
- .2 liberal use of dunnage in the stow to shore and bridge gaps;
- .3 double lashing in exposed areas; and
- .4 consideration given to the use of locking tiers.

2.10.2 Securing arrangements used should be designed in accordance with Part B and documented in accordance with section 2.13 of this Code.

### ***Lashings***

2.10.3 Different lashing arrangements are described in Part B of this Code.

2.10.4 The following three types of lashing equipment with different strength and elongation characteristics are most frequently used for securing timber deck cargoes. Individual suitability should be determined by such factors as ship type, size and area of operation, and as described in this Code and as prescribed in the cargo securing manual:

- .1 chain lashings;
- .2 wire lashings; and
- .3 fabricated web lashings.



*Chain lashing*



*Wire lashing*



*Fabricated web lashing*



*Chain lashing*



*Wire lashing*



*Fabricated web lashing*

**Figure 2.1** *Examples of different types of lashing equipment*



Open hooks, which may loosen if the lashing becomes slack, should not be used in securing arrangements for timber deck cargoes. Web lashing should not be used in combination with chain or wire lashings.

2.10.5 The appropriate safety factors for the different types of equipment are described in Annex 13 to the Code of Safe Practice for Cargo Stowage and Securing (CSS Code).

2.10.6 All lashing equipment should be visually examined according to the instruction in the cargo securing manual before use and only equipment fit for purpose should be used for securing of timber deck cargoes.

2.10.7 The necessary pre-tension in the lashings used should be maintained throughout the voyage. It is of paramount importance that all lashings be carefully examined and tightened at the beginning of the voyage as the vibration and working of the ship will cause the cargo to settle and compact. They should be further examined at regular intervals during the voyage and tightened as necessary.

2.10.8 Entries of all examinations and adjustments to lashings should be made in the ship's log-book.

2.10.9 Slip hooks or other appropriate methods may be used for quick and safe adjustment of lashings. Pelican hooks, when used, should be moused.

2.10.10 Corner protectors should be used to prevent lashings from cutting into the cargo and to protect lashings from sharp corners. The latter especially applies to fabricated web lashings.

2.10.11 Every lashing should be provided with a tightening device or system so placed that it can safely and efficiently operate when required.

### ***Uprights***

2.10.12 Uprights should be fitted when required by this Code and as prescribed in the ship's cargo securing manual in accordance with the nature, height or character of the timber deck cargo. They should be designed in accordance with the criteria in chapter 7 of this Code and fitted in accordance with the ship's cargo securing manual. If there is an operational limit of the uprights (in terms of wave heights) this should be indicated in the ship's cargo securing manual.

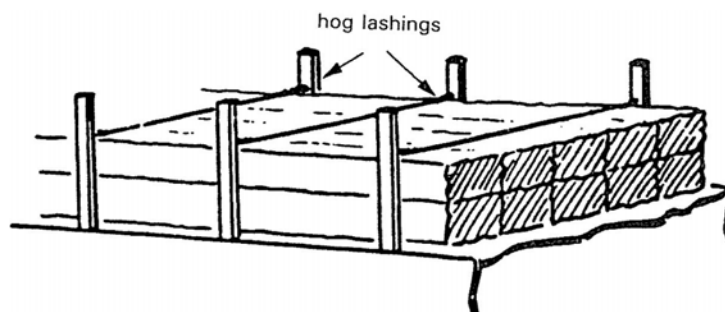
2.10.13 The uprights should be well fastened to the deck, hatches or coamings of the vessel (where adequate strength exists) and restrained from falling inwards during loading and discharging operations.

### ***Lashing arrangements***

2.10.14 In order to achieve a more secure stowage of logs when stowed on deck hog wires may be utilized. Such hog wire should be installed in the following manner:

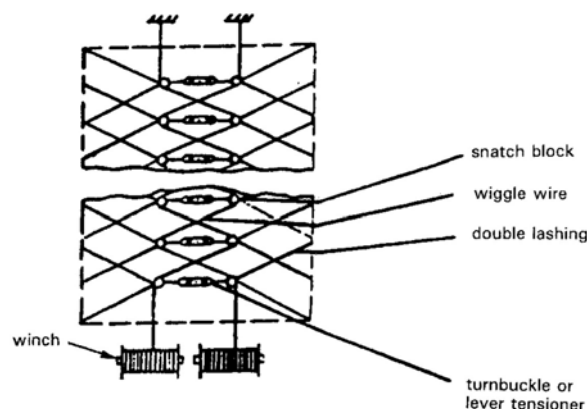
- .1 At approximately three quarters of the height of the stow, the hog wire should be rove through a padeye attached to the uprights at this level so as to run transversely, connecting the respective port and starboard uprights. The hog lashing wire should not be too tight when laid so that it becomes taut when overstowed with other logs.

- .2 A second hog wire may be applied in a similar manner if the height of the hatch cover is less than 2 m. Such second hog wire should be installed approximately 1 m above the hatch covers.
- .3 The aim of having the hog wires applied in this manner is to assist in obtaining as even a tension as possible throughout, thus producing an inboard pull on the respective uprights.

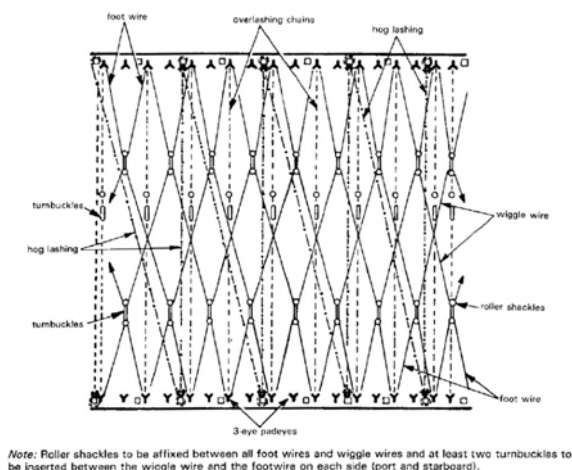


**Figure 2.2** Example of hog lashings

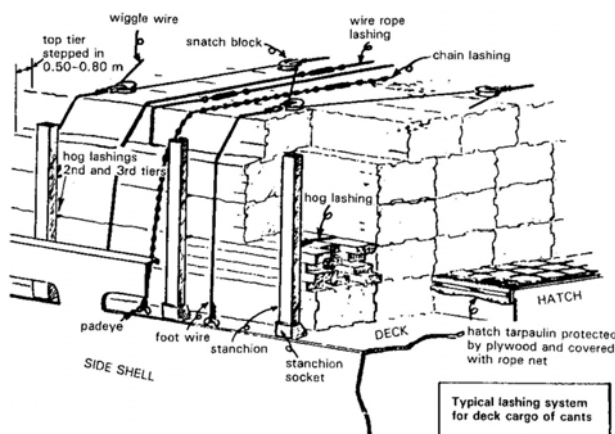
2.10.15 In addition to uprights and hog lashings, an arrangement with top-over and continuous wiggle lashings (wiggle wires), as shown in the following figures, may be utilized at each hatch meeting the specifications of chapter 5.



**Figure 2.3.** Example of wiggle lashings



**Figure 2.4.** Arrangement of wiggle and top-over lashings



**Figure 2.5.** Example of an arrangement with hog, top-over and wiggle lashings

2.10.16 If a wiggle wire is not fitted, then extra chain or chain/wire combination overlashings should be fitted instead, as described in 5.4.1.

## **2.11 Post-loading operation**

The Company should establish procedures for the preparation of plans and instructions, including checklists as appropriate, for key post loading operations.<sup>[5]</sup>

## **2.12 Voyage planning**

2.12.1 Prior to proceeding to sea, the master should ensure that the intended voyage has been planned using the appropriate nautical charts and nautical publications for the area concerned, taking into account the guidelines and recommendations developed by the Organization<sup>[23]</sup>.

2.12.2 In order to reduce excessive accelerations, the master should plan the voyage so as to avoid potential severe weather and sea conditions. To this effect, weather reports, weather facsimiles or where available weather routing may be consulted and the latest available weather information should always be used.<sup>[24]</sup>

2.12.3 If deviation from the intended voyage plan is considered during the voyage, the same procedure as described in 2.12.1 and 2.12.2 should be followed.

2.12.4 In cases where severe weather and sea conditions are unavoidable, the Master should be conscious of the need to reduce speed and/or alter course at an early stage in order to minimize the forces imposed on the cargo, structure and lashings. The lashings are not designed to provide a means of securing against imprudent ship handling in severe weather and sea conditions. There can be no substitute for good seamanship. The following precautions should be observed:

- .1 in the case of marked roll resonance with amplitudes above 30° to either side, the cargo securing arrangements could be overstressed. Effective measures should be taken to avoid this condition;
- .2 in the case of heading into the seas at high speed with marked slamming shocks, excessive longitudinal and vertical acceleration may occur. An appropriate reduction of speed should be considered; and
- .3 in the case of running before large stern or quartering seas with a stability which does not amply exceed the accepted minimum requirements, large roll amplitudes should be expected with great transverse accelerations as a result. An appropriate change of heading should be considered.

### ***Foreseeable risks***

2.12.5 During voyage planning, all foreseeable risks, which could lead to either excessive accelerations causing cargo to shift or conditions leading to water absorption and ice aggregation, should be considered. The following list comprises the most significant situations that should be taken under consideration to that effect:

- .1 extreme weather conditions predicted by weather forecasts;
- .2 severe wave conditions that have been known to appear in certain navigational areas;
- .3 unfavourable directions of encountered waves<sup>(25)</sup>; and
- .4 swell caused by recent weather phenomena in the vicinity of the area of the intended voyage.

## **2.13 Cargo Securing Manual**

2.13.1 Timber deck cargoes should be loaded, stowed and secured, throughout the voyage, in accordance with the Cargo Securing Manual as required by SOLAS chapter VI.

2.13.2 The Cargo Securing Manual should be based on the guidelines in this Code and drawn up to a standard at least equivalent to the guidelines developed by the Organization <sup>(26), (27)</sup> and approved by the Administration.<sup>(26)</sup>

2.13.3 Each cargo securing arrangement for timber deck cargoes should be documented in the ship's Cargo Securing Manual in accordance with the instructions in MSC/Circ.745.

2.13.4 According to the CSS Code and MSC/Circ.745, among others, the following parameters should be taken into account at the design stage of cargo securing systems:

- .1 duration of the voyage;
- .2 geographical area of the voyage;
- .3 sea conditions which may be expected;
- .4 dimensions, design and characteristics of the ship;
- .5 expected static and dynamic forces during the voyage;
- .6 type and packaging of cargo units;
- .7 intended stowage pattern of the cargo units; and
- .8 mass and dimensions of the cargo units.

2.13.5 In the Cargo Securing Manual, each stowage and securing arrangements should additionally be documented by a Lashing Plan showing at least the following:

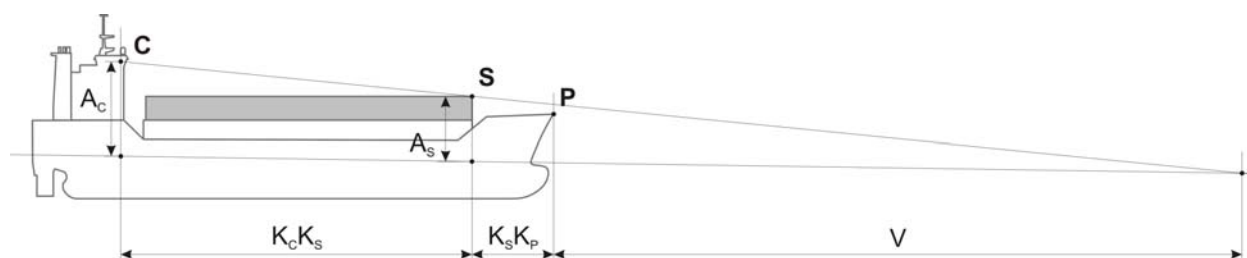
- .1 maximum cargo weight for which the arrangement is designed;
- .2 maximum stowage height;
- .3 required number and strength of blocking devices and lashings as applicable;
- .4 required pretension in lashings;
- .5 other cargo properties of importance for the securing arrangement such as friction, rigidity of timber packages, etc.;
- .6 illustrations of all securing items that might be used; and
- .7 any restriction regarding maximum accelerations, weather criteria, for non-winter conditions only, restricted sea areas, etc.

## CHAPTER 3 – VISIBILITY

3.1 According to SOLAS chapter V, the view of the sea surface from the conning position should not be obscured by more than two ship lengths, or 500 m, whichever is the less, forward of the bow to 10° on either side under all conditions of draught, trim and deck cargo. National deviations may exist and should be taken into consideration as required dependent on the intended voyage.

3.2 No blind sector, caused by cargo, cargo gear or other obstructions outside of the wheelhouse forward of the beam which obstructs the view of the sea surface as seen from the conning position, should exceed 10°. The total arc of blind sectors should not exceed 20°. The clear sectors between blind sectors should be at least 5°. However, in the view described in 3.1, each individual blind sector should not exceed 5°.

3.3 The following formula can be used for calculating the bridge visibility:



**Figure 3.1.** Distances used for calculating the bridge visibility

$$V = \frac{K_C K_S \cdot A_S}{A_C - A_S} - K_S K_P$$

Where:

|           |                                                           |
|-----------|-----------------------------------------------------------|
| $K_C K_S$ | Horizontal distance from conning position to position 'S' |
| $K_S K_P$ | Horizontal distance from position 'S' to position 'P'     |
| $A_C$     | Airdraft of conning position                              |
| $A_S$     | Airdraft of position 'S'                                  |

## CHAPTER 4 – PHYSICAL PROPERTIES OF TIMBER CARGOES

### 4.1 Stowage factors

4.1.1 Typical values for density and stowage factors are given in the table below for different types of timber deck cargoes.

*Table 4.1. Typical values for density and stowage factors*

| Type of timber cargo                   | Density<br>[ton / m <sup>3</sup> ] | Volume factor<br>[m <sup>3</sup> hold space<br>/ m <sup>3</sup> cargo] | Stowage factor<br>[m <sup>3</sup> hold space<br>/ ton of cargo] |
|----------------------------------------|------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>Sawn wood</b>                       |                                    |                                                                        |                                                                 |
| Packages of sawn wood with even ends   | 0.5 – 0.8                          | 1.4 - 1.7                                                              | 1.8 – 3.4                                                       |
| Packages of sawn wood with uneven ends | 0.5 – 0.8                          | 1.6 – 1.9                                                              | 2.0 - 3.8                                                       |
| Packages of planed wood with even ends | 0.5                                | 1.2 – 1.4                                                              | 2.4 - 2.8                                                       |
| <b>Round wood</b>                      |                                    |                                                                        |                                                                 |
| Coniferous round wood, fresh (bark on) | 0.9 – 1.1                          | 1.5 - 2.0                                                              | 1.4 - 2.2                                                       |
| Broad-leaf round wood, fresh (bark on) | 0.9 – 1.5                          | 2.0 - 2.5                                                              | 1.3 - 2.8                                                       |
| Round wood, dried (bark on)            | 0.65                               | 1.5 - 2.0                                                              | 2.3 - 3.1                                                       |
| Debarked coniferous round wood, fresh  | 0.85 – 1.2                         | 1.5 – 2.0                                                              | 1.2 – 2.4                                                       |
| Debarked broad-leaf round wood, fresh  | 0.9 – 1.0                          | 1.5 – 2.5                                                              | 1.5 – 2.8                                                       |
| Debarked round wood, dried             | 0.6 – 0.75                         | 1.2 – 2.0                                                              | 1.6 – 3.3                                                       |

4.1.2 The densities and stowage factors in the table above are presented for information purpose only to aid preplanning operations. The corresponding values for actual loads may vary significantly from those presented in the table depending on the timber type and condition. During actual loading more accurate values of the cargo weight are obtained by repeated checks of the vessel's displacement. The weights of sawn wooden packages are normally more accurate.

4.1.3 The weight of uncovered timber cargo may change during a voyage due to loss or absorption of water (but wrapped bundled cargoes do not). Timber cargo stowed under deck may lose weight whereas timber stowed on deck may gain weight by absorption of water, see special instruction in Annex C. Particular attention should be given to the impact that these and other changing conditions have on stability throughout a voyage.

### 4.2 Friction factors

4.2.1 Cargo at rest is prevented from sliding by static friction. When movement has been initiated the resistance of the material contact is reduced and sliding is counteracted by dynamic friction, see 4.2.6, instead.

4.2.2 The static friction may be determined by an inclination test. The angle  $\rho$  is measured when the timber cargo starts to slide. The static friction is calculated as:

$$\mu = \tan (\rho).$$

4.2.3 Five inclination tests should be performed with the same combination of materials. The highest and the lowest values should be disregarded and the friction factor is taken as the average of the three middle values. This average figure should be rounded down to the nearest fraction of 0.05.

4.2.4 If the values are intended to be used for non-winter conditions, the coefficient of friction for both dry and wet contact surfaces should be measured in separate series of tests and the lower of the two values are to be the used when designing cargo securing arrangements.

4.2.5 If the values are intended to be used for winter conditions when exposed surfaces are covered by snow and ice, the lowest coefficient of friction found for either dry, wet or snowy and icy contact surfaces should be used when designing cargo securing arrangements.

4.2.6 If not specially measured the dynamic friction factor may be taken as 70% of the static values.

4.2.7 The following values of static friction for the mentioned conditions may be used when designing securing arrangements for timber deck cargoes unless the actual coefficient of friction is measured and documented as described above.

**Table 4.2.** *Typical values of static friction for different material combinations*

| Contact surface                                              | Non winter conditions<br><i>Dry or wet</i> | Winter conditions |
|--------------------------------------------------------------|--------------------------------------------|-------------------|
| <b>Sawn wooden package</b>                                   |                                            |                   |
| <i>against</i> painted steel                                 | 0.45                                       | 0.05              |
| <i>against</i> sawn wood                                     | 0.50                                       | 0.30              |
| <i>against</i> plastic cover or webbing slings               | 0.30                                       | 0.25              |
| <b>Round wood</b>                                            |                                            |                   |
| coniferous round wood (bark on) <i>against</i> painted steel | 0.35                                       |                   |
| coniferous round wood (bark on) <i>between layers</i>        | 0.75                                       |                   |

4.2.8 Static friction may be used for tight block stowage arrangements as well as for the design of frictional lashing systems such as top-over lashing systems.

4.2.9 Dynamic friction should be used for non rigid lashing systems, which due to elasticity of securing equipment allow for minor dislocation of the cargo before full capacity of the securing arrangement is reached.

### **4.3 Plastic covers**

4.3.1 Plastic sheeting is often used on packages of sawn wood to protect the cargo. High friction coatings (friction coefficient 0.5 and above) can be incorporated into plastic sheeting as an important means of improving the safe transport of these cargoes.

4.3.2 Special precautions should be taken to prevent slippery plastic hoods with low friction coefficients, from being used as a sawn wood package cargo covering on deck.

#### 4.4 Package marking

All sawn wooden packages should be clearly marked with the volume of the package. The marking should be clearly visible on the top of the package as well as both long sides. The approximate weight should also be shown<sup>(29)</sup>.

#### 4.5 Water absorption

Sea spray may increase the weight of the timber deck cargo and thus influence the stability. The weight increase of the timber varies with time, exposure and type of timber. The value of increased weight of timber deck cargo due to water absorption should be considered in accordance with the 2008 IS Code and special instructions in Annex C.

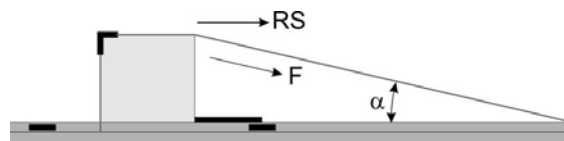
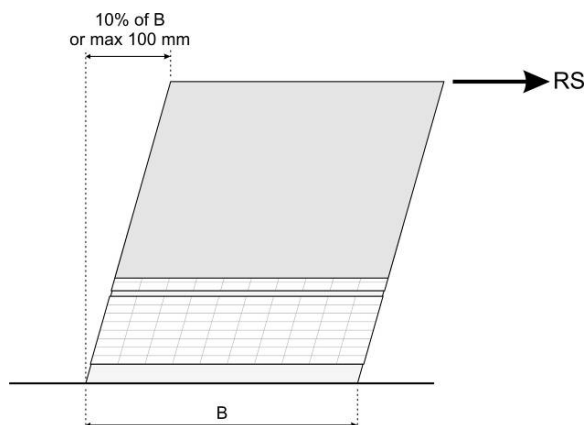
#### 4.6 Weight of ice

During cold weather conditions ice may form from sea spray and the stability may be affected as the ice can add weight rapidly. Increasing of the weight due to icing should be considered in accordance with the 2008 IS Code.

#### 4.7 Rigidity of sawn wood packages

4.7.1 The Racking Strength, RS, of a sawn wood package is defined as the horizontal force that a package can withstand per metre package length without collapsing or deforming more than 10% of its width, B, or a maximum of 100 mm as shown in figure 4.1.

4.7.2 The racking strength of timber packages can be measured by a test setup as shown in figure 4.2. The angle  $\alpha$  should not be greater than 30°.



**Figure 4.1.** Racking strength of timber packages      **Figure 4.2.** Test setup for racking strength

4.7.3 The Racking Strength, RS, is taken as the applied force  $F \cdot \cos \alpha$  (see figure above) when the package collapses or when the deflection in the top is 10% of the package width, B, or maximum 100 mm.

4.7.4 Racking strength measurements will have to be carried out by the shipper and the information should be provided to the master as part of the required cargo information mentioned in SOLAS chapter VI.



## PART B – DESIGN OF CARGO SECURING ARRANGEMENTS

To accommodate proven designs and practices but to also embrace advances in technology and materials, part B has been split into two chapters, each providing different design principles. Chapter 5: (Design Principles) incorporates **prescriptive** requirements. Chapter 6: (Alternative Design Principles) provides for alternative designs and equipment to be developed and includes **functional** requirements.

### CHAPTER 5 – DESIGN PRINCIPLES

This chapter applies primarily, but is not limited to, ships of 24 metres in beam and above engaged in international deep-sea trade and incorporates experience based prescriptive requirements on the securing of timber deck cargoes. It primarily applies the use of steel components for lashings but is not limited to their sole use. Consideration may be given to allowing chapter 5 ships to make use of proven alternative technologies in cargo securing design, which provide at least the level of safety as specified in this chapter.

#### 5.1 General

5.1.1 Every lashing should pass over the timber deck cargo and be secured to suitable eyeplates, lashing bollards or other devices adequate for the intended purpose which are efficiently attached to the deck stringer plate or other strengthened points. They should be installed in such a manner as to be, as far as practicable, in contact with the timber deck cargo throughout its full height.

5.1.2 All lashings and components used for securing should:

- .1 possess a breaking strength of not less than 133 kN;
- .2 after initial stressing, show an elongation of not more than 5% at 80% of their breaking strength; and
- .3 show no permanent deformation after having been subjected to a proof load of not less than 40% of their original breaking strength.

5.1.3 Every lashing should be provided with a tightening device or system so placed that it can safely and efficiently operate when required. The load to be produced by the tightening device or system should not be less than:

- .1 27 kN in the horizontal part; and
- .2 16 kN in the vertical part.

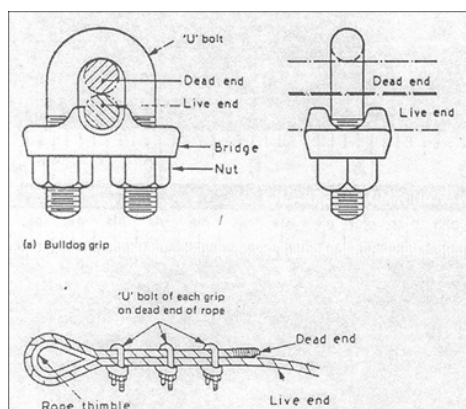
5.1.4 Upon completion and after the initial securing, the tightening device or system should be left with not less than half the threaded length of screw or of tightening capacity available for future use.

5.1.5 Every lashing should be provided with a device or an installation to permit the length of the lashing to be adjusted.

5.1.6 The spacing of the lashings should be such that the two lashings at each end of each length of continuous deck stow are positioned as close as practicable to the extreme end of the timber deck cargo.

5.1.7 If wire rope clips are used to make a joint in a wire lashing, the following conditions should be observed to avoid a significant reduction in strength:

- .1 the number and size of rope clips utilized should be in proportion to the diameter of the wire rope and should not be less than three, each spaced at intervals of not less than 150 mm;
- .2 the saddle portion of the clip should be applied to the live load segment and the U-bolt to the dead or shortened end segment; and
- .3 rope clips should be initially tightened so that they visibly compress the wire rope and subsequently be re-tightened after the lashing has been stressed.



**Figure 5.1. Wire rope clips**

5.1.8 Greasing the threads of grips, clips, shackles and turnbuckles increases their holding capacity and prevents corrosion.

5.1.9 Bulldog grips are only suitable for a standard wire rope of right-hand lay having six strands. Left-hand lay or different construction should not be used with such grips.

## **5.2 Uprights**

5.2.1 Uprights, designed in accordance with chapter 7, should be used when required by the nature, height or character of the timber deck cargo as outlined in this code.

5.2.2 When uprights are used, they should:

- .1 be made of material of adequate strength, taking into account relevant parameters such as; the breadth of the deck cargo, the weight and height of the cargo, the type of timber cargo, friction factors, additional lashings, etc.;
- .2 be spaced at intervals between the centrelines of two uprights not exceeding 3 m so that preferably all sections of the stow are supported by at least two uprights; and
- .3 be fixed to the deck and/or hatch cover by angles, sockets or equally efficient means and be secured in position as required by the CSM.

### **5.3 Loose or packaged sawn wood**

5.3.1 Uprights should be used for loose sawn wood. Uprights or stoppers (low uprights) should also be used to prevent packaged sawn wood loaded on top of the hatch covers only from sliding. The timber deck cargo should in addition be secured throughout its length by independent lashings.

5.3.2 Subject to 5.3.3, the maximum spacing of the lashings referred to above should be determined by the maximum height of the timber deck cargo in the vicinity of the lashings:

- .1 for a height of 2.5 m and below, the maximum spacing should be 3 m;
- .2 for heights of above 2.5 m, the maximum spacing should be 1.5 m; and
- .3 on the foremost and aft most sections of the deck cargo the distance between the lashings according to above should be halved.

5.3.3 As far as practicable, long and sturdy packages should be stowed in the outer rows of the stow and the packages stowed at the upper outboard edge should be secured by at least two lashings each.

5.3.4 When the outboard packages of the timber deck cargo are in lengths of less than 3.6 m, the spacing of the lashings should be reduced as necessary or other suitable provisions made to suit the length of timber.

5.3.5 Rounded angle pieces of suitable material and design should be used along the upper outboard edge of the stow to bear the stress and permit free reeving of the lashings.

5.3.6 Timber packages may alternatively be secured by a chain or wire loop lashing system, based on the design principles contained in chapter 6.

### **5.4 Logs, poles, cants or similar cargo**

5.4.1 The round wood deck cargo should be supported by uprights and secured throughout its length by independent lashings spaced not more than 1.5 m apart.

5.4.2 If the round wood deck cargo is stowed over the hatches and higher, it should, in addition to being secured by the lashings recommended in 5.4.1, be further secured by a system of athwartship lashings (hog lashings as described in section 2.10.14) joining each port and starboard pair of uprights.

5.4.3 If winches or other adequate tensioning systems are available on board, every other of the lashings mentioned in 5.4.1 may be connected to a wiggle wire system as described in section 2.10.15.

5.4.4 The recommendation of 5.3.5 should apply to a timber deck cargo of cants.

### **5.5 Testing, marking, examination and certification**

All lashings and components used for the securing of the timber deck cargo should be tested, marked, examined and certified, as per the guidelines in MSC/Circ.745<sup>(27)</sup>, and be specific to the requirements for lashing and components outlined in 5.1.2 and 5.1.3.

## **5.6 Lashing plans**

One or more generic lashing plans complying with the recommendations of this Code should be provided and maintained on board a ship carrying timber deck cargo. Lashing plans should be incorporated in the cargo securing manual and the most relevant lashing plan should be consulted when stowing and securing timber deck cargoes.

## CHAPTER 6 – ALTERNATIVE DESIGN PRINCIPLES

This chapter permits the development (and use) of new designs and securing arrangements, by providing functional based requirements on the securing of timber deck cargoes, which may be used as an alternative to the requirements in chapter 5 for ships of less than 24 metres in beam and for designers considering alternative technologies in cargo securing. When chapter 6 is applied, operational procedures based on design risk assessment should be included within the ship's safety management system.

### 6.1 General requirements

6.1.1 The construction of deck, bulwarks, uprights, hatches and coamings should be of a design that allows a load of timber deck cargo to be carried in a satisfactory manner.

6.1.2 The goal is to prevent cargo shifting as far as practicable and the securing system should be designed according to the principles laid down in this chapter.

6.1.3 Loose sawn or round wood should as a general rule be longitudinally stowed and supported on the sides by uprights to the full height of the stow.

6.1.4 Packaged sawn wood deck cargoes may be secured without uprights if the racking strength of the packages has been tested and found sufficient and sliding is prevented by bottom blocking, friction or lashing.

6.1.5 If the friction is sufficient and the expected transverse accelerations are limited, unpackaged sawn wood cargo may be transversely stowed.

6.1.6 All denotations used in the formulae in this chapter are listed in section 6.7 of this Code.

### 6.2 Accelerations and forces acting on the cargo

6.2.1 The cargo securing arrangement should in the transverse direction be designed for accelerations generated as well as forces by wind and sea according to the CSS Code, Annex 13.

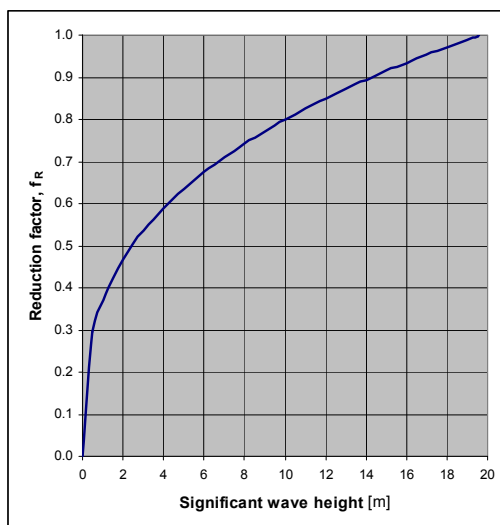
6.2.2 Special securing of timber deck cargoes in the longitudinal direction may be dispensed with only if great care is taken to avoid excessive acceleration forces in heavy head seas.

6.2.3 To take account of the factors mentioned in 2.13.4, the acceleration data calculated according to Annex 13 of the CSS Code may be multiplied by a reduction factor ranging from 0 to 1, depending on expected maximum significant wave height during the intended voyage. The reduction factor is obtained by the following formula:

$$f_R = \sqrt[3]{\frac{H_M}{19.6}}$$

Where the variable  $H_M$  means the maximum expected significant wave height in metres.

(The value 19.6 is the assumed twenty year wave that will occur in the Northern Atlantic Ocean. Relevant significant wave heights for different sea areas and seasons can be obtained from "Ocean Wave Statistics".)



**Figure 6.1.** Plot of the reduction factor as a function of the expected significant wave height

6.2.4 Reduced acceleration may be used for the design of securing arrangements for timber deck cargoes in any of the following ways:

- .1 Required securing arrangements are designed for different wave heights and the securing arrangement is selected according to the maximum expected wave height for each voyage.
- .2 The maximum wave height that a particular securing arrangement can withstand is calculated and the vessel is limited to operate in wave heights up to the maximum calculated. Examples on such arrangements are unsecured transversely stowed timber deck cargoes in restricted sea areas.
- .3 The required securing arrangement is calculated for the maximum expected twenty year wave in a particular restricted area and the cargo is always secured according to the designed arrangement when operating in that area.

6.2.5 If one of the two first mentioned methods in 6.2.4 are used for decision on securing arrangements, it is important that procedures for forecasting the maximum expected wave height on intended voyages is developed and followed and documented in the ship's approved Cargo Securing Manual.

### 6.3 Physical properties of timber deck cargoes

6.3.1 Prior to loading of timber deck cargoes, all relevant cargo information, as described in this section and in chapter 4, should be provided to the master of the vessel.

#### **Friction**

6.3.2 Friction is one of the most important factors preventing cargo from shifting. Deck cargo may shift due to a lack of internal friction. Snow, ice, frost, rain, and other slippery

surface conditions drastically affect friction. Special consideration should be given to package materials, contact surfaces, and weather conditions.

6.3.3 Static friction may be used for tight block stowage arrangements as well as for the design of frictional lashing systems such as top-over lashing systems.

6.3.4 Dynamic friction should be used for non rigid lashing systems, e.g., loop lashings, which due to elasticity of securing equipment allow for minor dislocation, see 6.5.16, of the cargo before full capacity of the securing arrangement is reached.

6.3.5 Test procedures for determining coefficients of friction as well as generic friction values for material contacts common for timber deck cargoes are given in chapter 4.

### ***Rigidity of timber packages***

6.3.6 The rigidity of timber packages is of great importance for the stability of the deck cargo and the racking strength of the timber packages should be taken into consideration when securing systems are designed.



**Figure 6.2.** *Example of poor rigidity*

6.3.7 The definition of the rigidity of timber packages for the purpose of this Code as well as methods for determining it are presented in chapter 4. The racking strength should not be less than 3.5 kN/m of package length.

## **6.4 Safety factors**

6.4.1 Safety factors are to be used when:

- .1 calculating the Maximum Securing Load (MSL) of the lashings from the Minimum Breaking Load (MBL); and
- .2 calculating the maximum allowed Calculated Strength (CS) in the lashings as function of MSL.

6.4.2 MSL as function of the MBL should be taken according to Annex 13 of the CSS Code, provided inspection and maintenance of the equipment have been carried out in accordance with the ship's cargo securing manual.

6.4.3 The maximum allowed Calculated Strength (CS) in lashings and uprights used in the calculations should be taken from the following formula:

$$CS \leq \frac{MSL}{1.35}$$

## 6.5 Design criteria for different securing arrangements

6.5.1 Securing arrangements for timber deck cargoes should be based on accelerations, physical properties and safety factors as described in 6.4 above.

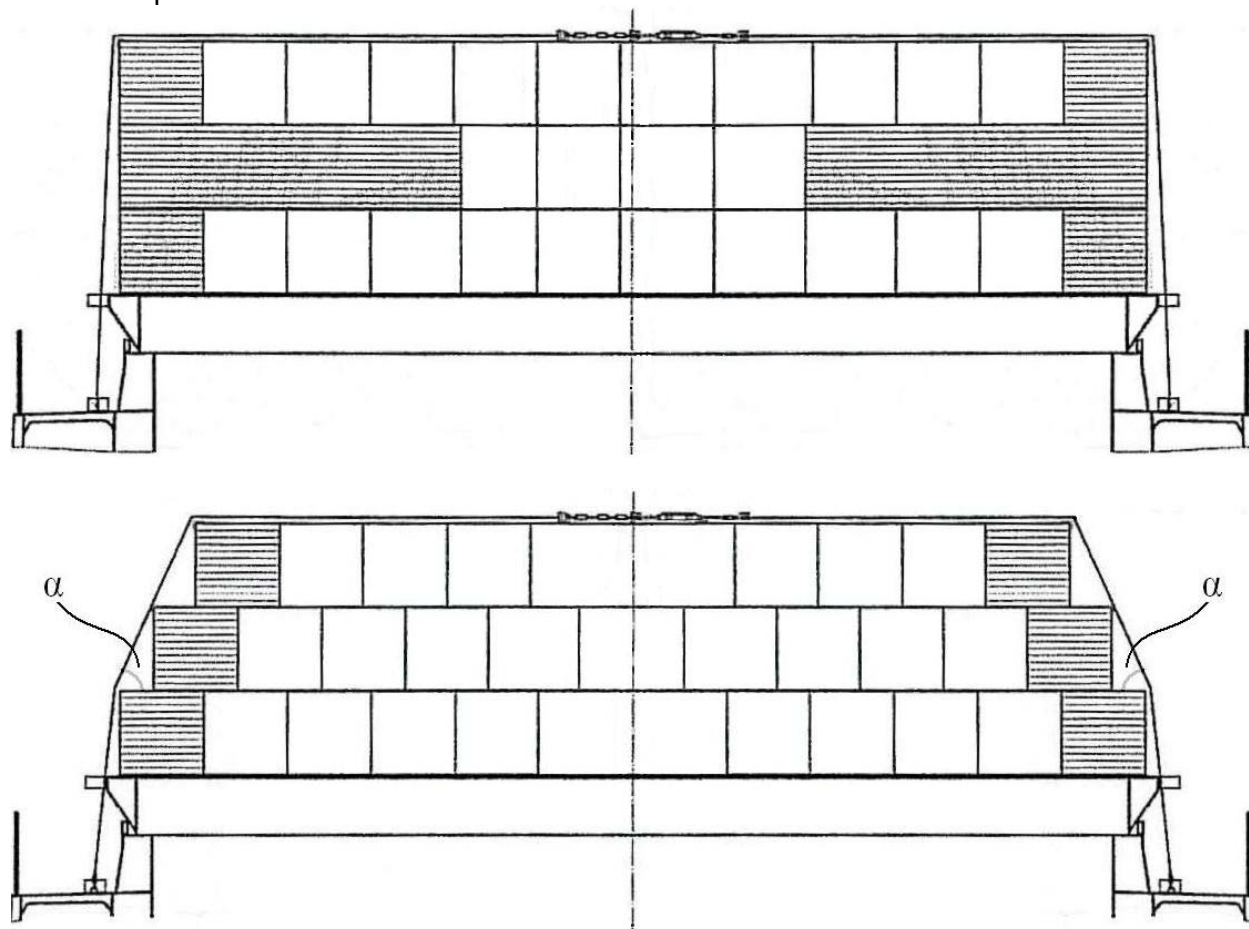
6.5.2 Design criteria for some different securing arrangements are given below. Other securing arrangements may also be used as long as the system is designed according to the principles given in this code.

6.5.3 In Annex B detailed descriptions and example design calculations are given for some stowage and securing arrangements.

6.5.4 The denotations used in the formulas in this chapter are listed in chapter 8.

### *Top-over lashed longitudinally stowed timber packages*

6.5.5 Top-over lashing alone is a frictional lashing method and the effect of the lashing is to apply vertical pressure increasing the friction force between the outer stows of deck cargo and the ship's deck/hatch cover.



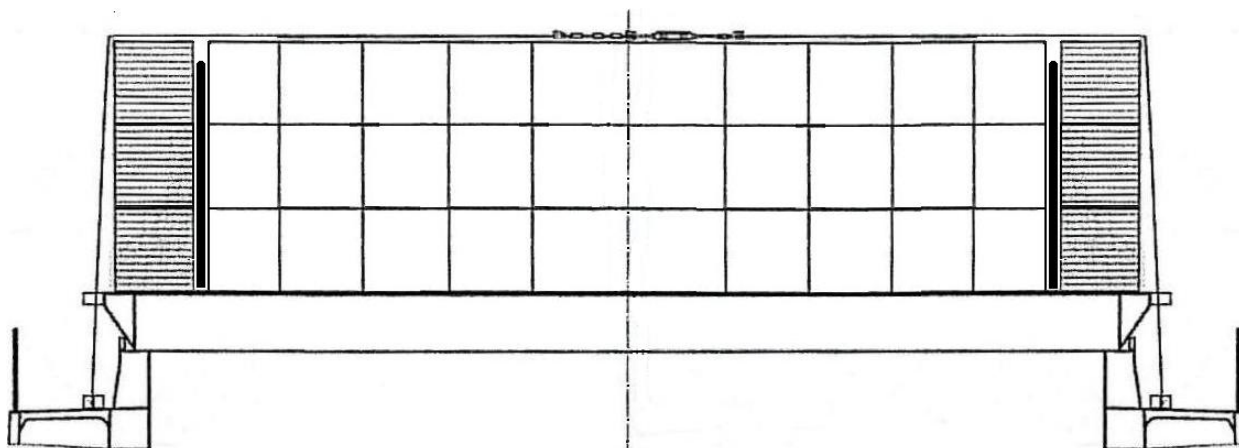
**Figure 6.3.** Principles for top-over lashing



6.5.6 For pure top-over lashing arrangements the friction alone will have to counteract the transverse forces so that the following equilibrium of forces is satisfied:

$$(m \cdot g_0 + 2 \cdot n \cdot PT_V \cdot \sin \alpha) \cdot \mu_{static} \geq m \cdot a_t + PW + PS$$

6.5.7 In practice, sliding between the layers is often prevented due to slightly different heights of the timber packages. Alternatively it may be prevented by inserting vertical sturdy battens of proper dimensions between the columns.



**Figure 6.4.** Sliding of upper layer prevented by vertical sturdy battens

6.5.8 If sliding between layers is not prevented, sliding between each individual layer should be considered by the following equilibrium of forces:

$$(m_a \cdot g_0 + 2 \cdot n \cdot PT_V \cdot \sin \alpha) \cdot \mu_{static\ a} \geq m_a \cdot a_t + PW_a + PS_a$$

Units denoted with <sub>a</sub> consider cargo units above the sliding level only.

6.5.9 To prevent the packages in the bottom layer from collapsing due to racking, the weight of the cargo stowed on top of the bottom layer should be limited so that the following equilibrium of forces is satisfied:

$$n_p \cdot L \cdot RS \geq m_a \cdot (a_t - 0.5g_0) + PW_a + PS_a$$

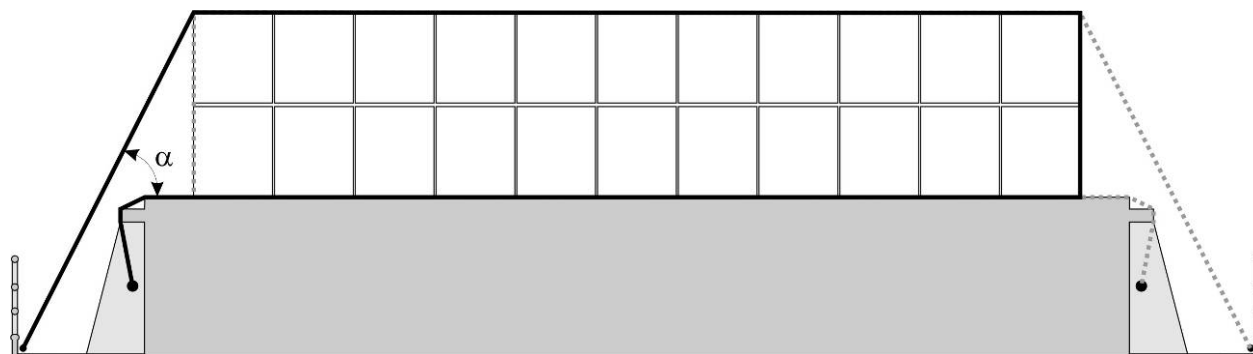
Units denoted with <sub>a</sub> consider cargo units above the bottom layer only.

6.5.10 Lashings used should comply with 6.5.20 and 6.5.21. It is extremely important to keep the lashings tight when a top-over lashing arrangement is used as the arrangement is based on the vertical pressure from the lashings.

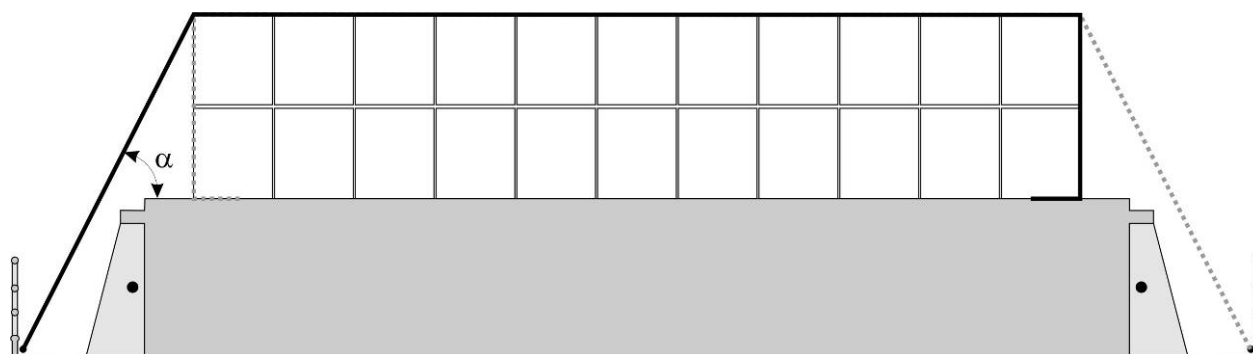
6.5.11 When top-over lashings are used as the only means of securing longitudinally stowed packages of sawn wood, adequate friction against the hatch covers should be sought and/or the transverse accelerations should if possible be limited.

### Loop lashed longitudinally stowed timber packages

6.5.12 Loop lashings are always applied in pairs as shown in the figure below. The lashings are drawn from one side of the cargo, under the cargo to the other side, up over the cargo and back to the same side. Alternatively, the lower part of the lashing may be fastened to a securing point on top of the hatch cover underneath the cargo.



**Figure 6.5.** Principles of loop lashing alternative 1 (be aware of chafing where lashings are lead around ship's structure as shown in the above figure, see section 2.10.10)



**Figure 6.6.** Principles for loop lashing alternative 2. The shorter length of the lashing compared to alternative 1 reduces the movement of the cargo due to elongation of the lashing

6.5.13 The number and strength of the lashings are to be chosen so that the following equilibrium is satisfied:

$$(m \cdot g_0 + n \cdot CS \cdot \sin \alpha) \cdot \mu_{dynamic} + n \cdot CS + n \cdot CS \cdot \cos \alpha \geq m \cdot a_t + PW + PS$$

6.5.14 Sliding between the layers should be prevented (see 6.5.7).

6.5.15 To prevent the packages in the bottom layer from racking, the weight of the cargo stowed on top of the bottom layer should be limited so that the following equilibrium is satisfied:

$$n_p \cdot L \cdot RS + n \cdot CS \cdot \cos \alpha \geq m_a \cdot (a_t - 0.5g_0) + PW_a + PS_a$$

Units denoted with <sub>a</sub> consider cargo units above the bottom layer only.

6.5.16 The transverse movement of the deck cargo due to elongation of the lashings is calculated according to the following formula:

$$\delta = L_L \cdot \frac{(CS - PT_V)}{MSL} \cdot \varepsilon$$

The elongation factor  $\varepsilon$  should be taken as 2% for chain and wire lashings and 7% for web lashings unless otherwise specified by certificate from the manufacturer.

The maximum heeling angle of the vessel due to a small transverse movement of the cargo should in no case be more than 5°, based on the full timber deck load condition of the vessel calculated according to the following formula:

$$HA = \arctan\left(\frac{HM}{G'M \cdot \Delta}\right)$$

Where:

|          |   |                                                                          |
|----------|---|--------------------------------------------------------------------------|
| HA       | = | Heeling angle in degrees                                                 |
| HM       | = | Heeling moment due to transverse movement of the deck cargo in tonmetres |
| G'M      | = | Metacentric height corrected for free surface moments in metres          |
| $\Delta$ | = | Ship's actual displacement in tons                                       |

#### ***Bottom blocked and top-over lashed longitudinally stowed timber packages***

6.5.17 Blocking means that the cargo is stowed against a blocking structure or fixture on the ship. If the cargo consists of packages with large racking capacity, bottom blocking should be sufficient in combination with top-over lashings.



**Figure 6.7.** Example of uprights for bottom blocking

6.5.18 The required strength, MSL, of the bottom blocking devices is calculated by satisfying the following equilibrium:

$$(m \cdot g_0 + 2 \cdot n \cdot PT_V \cdot \sin \alpha) \cdot \mu_{static} + n_b \cdot \frac{MSL}{1.35} \geq m \cdot a_t + PW + PS$$

6.5.19 The spacing between top-over lashings in a longitudinal direction should be maximum 3 m for stowage heights below 2.5 m and maximum 1.5 m for stowage heights above 2.5 m.

6.5.20 The pretension  $PT_V$  in the vertical part of the lashings should be not less than 16 kN and the pretension  $PT_H$  in the horizontal part of the lashing should not be less than 27 kN.

6.5.21 All lashings and components used for securing in combination with bottom blocking should:

- .1 possess a breaking strength MBL of not less than 133 kN;
- .2 after initial stressing, show an elongation of not more than 5% at 80% of their breaking strength; and
- .3 show no permanent deformation after having been subjected to a proof load of not less than 40% of their original breaking strength.

6.5.22 The bottom blocking devices are to be placed on both sides of the deck cargo equally spaced. Two blocking device per side should be used per cargo section and the height should extend to a height of at least 200 mm.

6.5.23 Sliding between the layers should be prevented (see 6.5.7). If no such measures are taken, sliding between layers should be checked by the calculation for equilibrium of forces in 6.5.8.

6.5.24 To prevent the packages in the bottom layer from racking, the weight of the cargo stowed on top of the bottom layer should be limited so that the following equilibrium of forces is satisfied:

$$n_p \cdot L \cdot RS \geq m_a \cdot (a_t - 0.5g_0) + PW_a + PS_a$$

Units denoted with  $_a$  consider cargo units above the bottom layer only.

***Uprights blocked and top-over lashed longitudinally stowed sawn wood packages and round wood***

6.5.25 Longitudinally stowed sawn wood packages, loose sawn wood or round wood may be supported by uprights in combination depending on trading pattern with or without top-over lashings or hog-wires.

6.5.26 The uprights should be designed in accordance with chapter 7.

6.5.27 The uprights should be placed on both sides of the cargo, equally spaced. Each cargo block of the stow should be supported by at least two uprights per side.

6.5.28 The spacing of top-over lashings should for packaged sawn wood be a maximum of 3 m for stowage heights below 2.5 m and maximum 1.5 m for stowage heights above 2.5 m for round wood the spacing should be 1.5 m irrespective of the height.

6.5.29 The pretension  $PT_V$  in the vertical part of the lashings should be not less than 16 kN and the pretension  $PT_H$  in the horizontal part of the lashing should not be less than 27 kN.

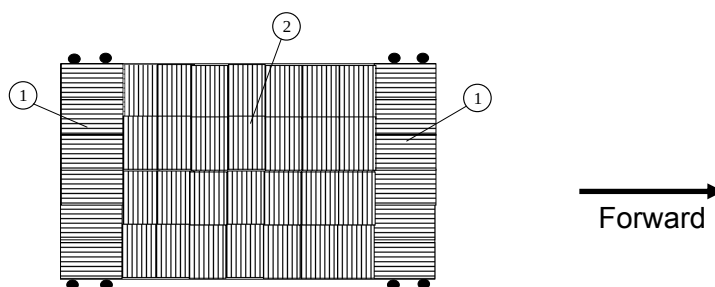
6.5.30 All lashings and components used for securing in combination with bottom blocking should:

- .1 possess a breaking strength MBL of not less than 133 kN;

- .2 after initial stressing, show an elongation of not more than 5% at 80% of their breaking strength; and
- .3 show no permanent deformation after having been subjected to a proof load of not less than 40% of their original breaking strength.

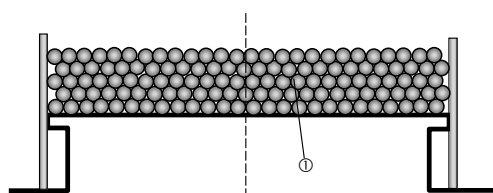
### Frictional securing

6.5.31 In restricted sea areas, round wood may be transversely stowed and secured by bottom blocking and/or friction between tiers only. This may be done only if the friction between layers is sufficient and the expected transverse accelerations are limited. When the friction is sufficient between bottom layers and deck/hatch, then the bottom blocking may not be required.

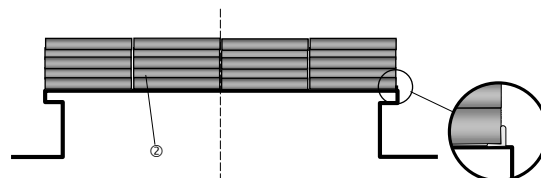


*Example of round wood stowage pattern for restricted sea areas.*

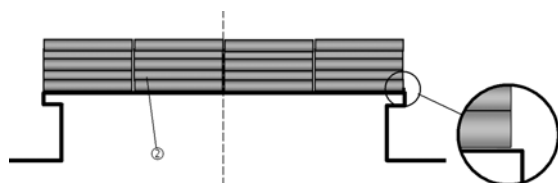
*Sections marked 1 are longitudinally stowed round wood secured by uprights. Section marked 2 are transversely stowed round wood secured by friction in combination with or without bottom blocking.*



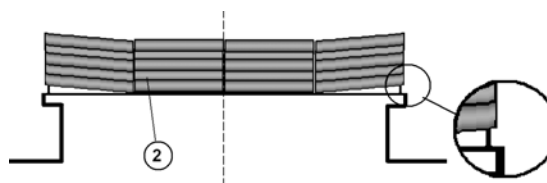
*Section with longitudinally stowed round wood secured by uprights.*



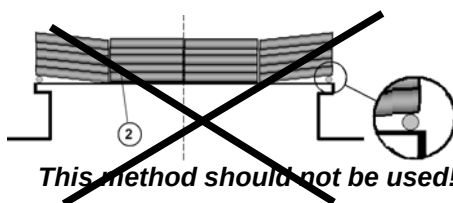
*Section with transversely stowed timber cargo secured by friction in combination with bottom blocking.*



*Section with transversely stowed round wood secured by friction only (Alternative 1). Non-slip paint on hatch cover or non-slip material between hatch cover and round wood should be used.*



*Section with transversely stowed round wood secured by friction only (Alternative 2). Non-slip paint on hatch cover or non-slip material between hatch cover and round wood should be used.*



***This method should not be used!***

*Section with transversely stowed round wood secured by friction only (Alternative 3).*  
**Figure 6.8. Principles for friction securing of round wood in restricted sea areas**

6.5.32 The required strength, MSL, of the bottom blocking devices is calculated by satisfying the following equilibrium:

$$m \cdot g_0 \cdot \mu_{static} + n_b \cdot \frac{MSL}{1.35} \geq m \cdot a_t + PW + PS$$

6.5.33 The required friction between the layers can be calculated by satisfying the following equilibrium:

$$m \cdot g_0 \cdot \mu_{static} \geq m \cdot a_t + PW + PS$$

## CHAPTER 7 – UPRIGHTS

7.1 Longitudinally stowed round wood, loose sawn wood and sawn wood packages with limited racking strength should be supported by uprights at least as high as the stow.

7.2 Uprights should be designed for the forces they have to take up according to the formulas in this section. Especially the design of high uprights should be such that the deflection is limited. Uprights may be complemented by different lashing arrangements.



**Figure 6.9.** Uprights for blocking over the entire height of the stow

7.3 For vessels carrying loose sawn wood and round timber, the design bending moment per upright is calculated as the greater of the two moments given by the following formulas:

$$CM_{bending1} = 0.1 \cdot \frac{H^2}{k \cdot B \cdot N} \cdot m \cdot g_0$$

$$CM_{bending2} = \frac{H}{3 \cdot k \cdot N} \cdot (m \cdot (a_t - 0.6 \cdot \mu_{static} \cdot g_0) + PW + PS)$$

$$M_{bending} \geq 1.35 \cdot \max(CM_{bending1}, CM_{bending2})$$

In the table below the required bending resistance for uprights supporting loose sawn wood or round wood have been calculated based on the formulae above and by using typical cargo properties and configurations.

| Height<br>[m] | Transverse Acceleration [m/s <sup>2</sup> ] |      |       |       |       |       |       |       |
|---------------|---------------------------------------------|------|-------|-------|-------|-------|-------|-------|
|               | 3.0                                         | 3.5  | 4.0   | 4.5   | 5.0   | 5.5   | 6.0   | 6.5   |
| 2             | 107                                         | 150  | 193   | 235   | 278   | 321   | 363   | 406   |
| 3             | 330                                         | 474  | 618   | 762   | 906   | 1050  | 1194  | 1338  |
| 4             | 756                                         | 1097 | 1438  | 1780  | 2121  | 2462  | 2803  | 3144  |
| 5             | 1452                                        | 2118 | 2785  | 3451  | 4118  | 4784  | 5451  | 6117  |
| 6             | 2486                                        | 3638 | 4790  | 5941  | 7093  | 8245  | 9396  | 10548 |
| 7             | 3926                                        | 5755 | 7584  | 9413  | 11242 | 13070 | 14899 |       |
| 8             | 5840                                        | 8570 | 11300 | 14030 | 16759 |       |       |       |

**Table 7.1.** Required bending resistance in cm<sup>3</sup> on uprights supporting round wood.

If **top-over lashings** are applied in accordance with sections 5.4 or 6.5.28 – 6.5.30, the bending moment of the uprights may be reduced by 12%.

7.4 The design bending moment per upright supporting timber packages is to be taken as the greatest of the three moments given by the following formulas:

$$CM_{bending1} = \frac{m}{n_p \cdot k \cdot N} \cdot \left( a_t \cdot \frac{H}{2} - g_0 \cdot \frac{b}{2} \right) \cdot \frac{1 - (1 - f_i)^n}{f_i} \quad (\text{Moment required to prevent tipping})$$

$$\text{where:} \quad f_i = \mu_{\text{internal}} \cdot \frac{2b}{H} \quad (f_i = \text{Factor for considering internal moment})$$

$$CM_{bending2} = \frac{H}{2 \cdot k \cdot N} \cdot m \cdot (a_t - \mu_{\text{internal}} \cdot g_0) \cdot \frac{q-1}{2q} \quad (\text{Moment required to prevent sliding})$$

$$CM_{bending3} = \frac{H}{k \cdot N} \cdot (m \cdot a_t - (n_p - 4)(q - 2) \cdot L \cdot RS) \cdot \frac{q-1}{2q} \quad (\text{Moment required to prevent racking})$$

$$M_{bending} \geq 1.35 \cdot \max(CM_{bending1}, CM_{bending2}, CM_{bending3})$$

In the tables below the required bending resistance for uprights supporting timber packages have been calculated based on the formulae above and by using typical cargo properties and configurations for **sturdy** timber packages with a racking strength of 7 kN/m and for weaker packages with a racking strength of 3.5 kN/m.

| Height<br>[m] | Transverse Acceleration [m/s <sup>2</sup> ] |      |      |      |       |       |       |  |
|---------------|---------------------------------------------|------|------|------|-------|-------|-------|--|
|               | 3.0                                         | 3.5  | 4.0  | 4.5  | 5.0   | 5.5   | 6.0   |  |
| 2             |                                             |      |      |      | 26    | 70    | 115   |  |
| 3             |                                             | 22   | 70   | 118  | 165   | 213   | 378   |  |
| 4             | 124                                         | 237  | 350  | 463  | 576   | 689   | 953   |  |
| 5             | 458                                         | 679  | 900  | 1120 | 1341  | 1562  | 1927  |  |
| 6             | 1040                                        | 1421 | 1803 | 2184 | 2565  | 2946  | 3405  |  |
| 7             | 1934                                        | 2539 | 3144 | 3748 | 4353  | 4958  | 5563  |  |
| 8             | 3202                                        | 4104 | 5007 | 5909 | 6812  | 7714  | 8617  |  |
| 9             | 4907                                        | 6192 | 7477 | 8761 | 10046 | 11331 | 12615 |  |

**Table 7.2.** Required bending resistance in cm<sup>3</sup> on uprights supporting **sturdy** packages of sawn wood.



| Height<br>[m] | Transverse Acceleration [m/s <sup>2</sup> ] |      |      |      |      |       |       |
|---------------|---------------------------------------------|------|------|------|------|-------|-------|
|               | 3.0                                         | 3.5  | 4.0  | 4.5  | 5.0  | 5.5   | 6.0   |
| 2             | 3                                           | 32   | 61   | 90   | 118  | 147   | 176   |
| 3             | 524                                         | 660  | 797  | 934  | 1071 | 1207  | 1344  |
| 4             | 724                                         | 1095 | 1466 | 1837 | 2208 | 2579  | 2950  |
| 5             | 725                                         | 1304 | 2084 | 2864 | 3644 | 4423  | 5203  |
| 6             | 1645                                        | 2248 | 2982 | 4393 | 5804 | 7215  | 8626  |
| 7             | 3055                                        | 4011 | 4966 | 7200 | 9512 | 11824 | 14136 |

**Table 7.3.** Required bending resistance in cm<sup>3</sup>  
on uprights supporting **weaker** packages of sawn wood.

7.5 If hog lashings are used, the required MSL of each hog lashing is calculated by the following formula:

$$MSL \geq \frac{M_{bending}}{2 \cdot h}$$

7.6 The design bending moment should not produce greater stress than 50% of the ultimate stress for the material in any part of the uprights.

## CHAPTER 8 – DENOTATIONS USED

The denotations used in the formulas in the design criteria of this code are listed below:

|                  |                                                                                                                                                       |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| $a_t$            | = Largest transverse acceleration at the centre of gravity of the deck cargo in the forward or aft end of the stow in $m/s^2$                         |
| $B$              | = Width of deck cargo in metres                                                                                                                       |
| $b$              | = Width of each individual stack of packages                                                                                                          |
| $CS$             | = Calculated strength of lashing in kN, see section 6.4                                                                                               |
| $f_R$            | = Reduction factor for accelerations due to expected sea state                                                                                        |
| $g_0$            | = Gravity acceleration $9.81 m/s^2$                                                                                                                   |
| $H$              | = Height of deck cargo in metres                                                                                                                      |
| $H_M$            | = Maximum significant wave height                                                                                                                     |
| $h$              | = Height above deck at which hoglashings are attached to the uprights in metres                                                                       |
| $k$              | = Factor for considering hog lashings:<br>$k = 1$ if no hog lashings are used<br>$k = 1.8$ if hog lashings are used                                   |
| $L$              | = Length of the deck cargo or section to be secured in metres                                                                                         |
| $L_L$            | = Length of each lashing in metres                                                                                                                    |
| $M_{bending}$    | = Design bending moment on uprights in kNm                                                                                                            |
| $MSL$            | = Maximum Securing Load in kN of cargo securing devices                                                                                               |
| $m$              | = Mass of the deck cargo or section to be secured in tonnes, including absorbed water and possible icing                                              |
| $N$              | = Number of uprights supporting the considered section on each side                                                                                   |
| $n$              | = Number of lashings                                                                                                                                  |
| $n_b$            | = Number of bottom blocking devices per side of the deck cargo                                                                                        |
| $n_p$            | = Number of stacks of packages abreast in each row                                                                                                    |
| $PS$             | = Pressure from unavoidable sea sloshing in kN based on $1 kN$ per $m^2$ exposed area, see CSS Code, Annex 13                                         |
| $PT_V$           | = Pretension in the vertical part of the lashings in kN                                                                                               |
| $PT_H$           | = Pretension in the horizontal part of the lashings in kN                                                                                             |
| $PW$             | = Wind pressure in kN based on $1 kN$ per $m^2$ wind exposed area, see CSS Code, Annex 13                                                             |
| $q$              | = Number of layers of timber packages                                                                                                                 |
| $RS$             | = Racking Strength per metre of timber package in $kN/m$ , see section 4.7                                                                            |
| $\alpha$         | = Angle between the hatch cover top plating and the lashings in degrees                                                                               |
| $\delta$         | = Small transverse movement of deck cargo in metres due to elasticity of lashing arrangement                                                          |
| $\varepsilon$    | = Elasticity factor for lashing equipment, taken as fraction of elongation experienced at the load of MSL for the lashing                             |
| $\mu_{dynamic}$  | = Dynamic coefficient of friction between the timber deck cargo and the ship's deck/hatch cover and considered to be 70% of the static friction value |
| $\mu_{internal}$ | = Coefficient of dynamic friction found internally between the packages of sawn wood                                                                  |
| $\mu_{static}$   | = Static coefficient of friction between the timber deck cargo and the ship's deck/hatch cover                                                        |

## **ANNEX A – GUIDANCE IN DEVELOPING PROCEDURES AND CHECKLISTS**

Items in A.1 to A.5 should be taken into account when developing the checklists for timber deck cargo operations.

### **A.1 Preparations before loading of timber deck cargoes**

#### ***General preparations***

A.1.1 The following information as applicable for each parcel of cargo should be provided by the shipper and collected by the master or his representative:

- .1 total amount of cargo intended as deck cargo;
- .2 typical dimensions of the cargo;
- .3 number of bundles;
- .4 density of the cargo;
- .5 stowage factor of the cargo;
- .6 racking strength for packaged cargo;
- .7 type of cover of packages and whether non-slip type; and
- .8 relevant coefficients of friction including covers of sawn wooden packages if applicable.

A.1.2 A confirmation on when the deck cargo will be ready for loading should be received.

A.1.3 A pre-loading plan according to the ship's Trim and Stability Book should be done and the following should be calculated and checked:

- .1 stowage height;
- .2 weight per m<sup>2</sup>;
- .3 required amount of water ballast; and
- .4 displacement, draught, trim and stability at departure and arrival.

A.1.4 The stability should be within required limits during the entire voyage.

A.1.5 When undertaking stability calculations, variation in displacement, centre of gravity and free surface moments due to the following factors should be considered:

- .1 absorption of water in timber carried as timber deck cargo according to special instruction, see annex c;
- .2 ice accretion, if applicable;
- .3 variations in consumables; and
- .4 ballast water exchange operations, in accordance with approved procedures.

A.1.6 Proper instructions for ballast water exchange operations, if applicable for the intended voyage, should be available in the Ballast Water Management Plan.

A.1.7 A lashing plan according to the ship's Cargo Securing Manual (CSM) should be prepared and the following calculated:

- .1 weight and height of stows per hatch;
- .2 number of sections in longitudinal direction per hatch;
- .3 required number of pieces of lashing equipment; and
- .4 required number of uprights, if applicable.

A.1.8 The certificates for the lashing equipment should be available in the ship's Cargo Securing Manual.

A.1.9 When the initial stability calculations and lashing plan have been satisfactorily completed, the maximum cargo intake should be confirmed.

A.1.10 Pre-load, loading and pre-lashing plans should be distributed to all involved parties (i.e. supercargo, stevedores, agent, etc.).

A.1.11 Weather report for loading period and forecasted weather for the sea voyage should be checked.

A.1.12 It should be confirmed that the stevedoring company is aware of the ship's specific requirements regarding stowage and securing of timber deck cargoes.

### ***Ship readiness***

A.1.13 All ballast tanks required for the voyage and included in the stability calculations should be filled before the commencement of loading on deck and it should be ensured that free surfaces are eliminated in all tanks intended to be completely full or empty.

A.1.14 Hatch covers and other openings to spaces below deck should be closed, secured and battened down.

A.1.15 Air pipes, ventilators, etc., should be protected and examined to ascertain their effectiveness against entry of water.

A.1.16 Objects which might obstruct cargo stowage on deck should be removed and secured safely in places appropriate for storage.

A.1.17 Accumulation of ice and snow on areas to be loaded and on packaged timber should be removed.

A.1.18 All sounding pipes on the deck should be reviewed and necessary precautions should be taken that safe access to these remains.

A.1.19 Cargo securing equipment should be examined in preparation for use in securing of timber deck cargoes and any defective equipment found should be removed from service, tagged for repair and replaced.

A.1.20 It should be confirmed that uprights utilized are in compliance with the requirements in the ship's Cargo Securing Manual.

A.1.21 A firm and level stowage surface should be prepared. Dunnage, where used, should be of rough lumber and placed in the direction which will spread the load across the ship's hatches or main deck structure and assist in draining.

A.1.22 Extra lashing points, if required, should be approved by the Administration.

A.1.23 It should be ensured that dunnage is readily available and in good condition.

A.1.24 Friction enhancing arrangements, where fitted, should be checked for their condition.

A.1.25 Cranes with wires, brakes, micro switches and signals (if they are to be used) should be controlled.

A.1.26 It should be verified that illumination on deck is working and ready for use.

### ***Ship to shore communication***

A.1.27 Radio channels to be used during cargo operations should be assigned and tested.

A.1.28 It should be confirmed that crane drivers and loading stevedores/crew understands signals to be used during cargo operations.

A.1.29 A plan should be worked out to halt loading or unloading operations due to any unforeseen circumstances that may jeopardize safety of ship and/or anyone on board.

## **A.2 Safety during loading and securing of timber deck cargoes**

### ***Lashing equipment***

A.2.1 If applicable, uprights should be mounted before loading on deck is commenced.

A.2.2 It should be checked that all lashing equipment is in place.

### ***Ship's safety***

A.2.3 All loading operations should be planned to immediately cease if a list develops for which there is no satisfactory explanation.

A.2.4 In the event that the vessel takes up an unexplained list, then no further work should be undertaken until all ship's tanks are sounded and assessment made of the ship's stability condition.

A.2.5 If deemed necessary, samples of the timber cargo should be weighed during loading and their actual weight should be compared to the weight stated by the shipper, in order to correctly assess the ship's stability.

A.2.6 Draught checks should be regularly carried out during the course of loading and the ship's displacement should be calculated to ensure the ship's stability and draft in the final condition are within prescribed limits.

A.2.7 Permitted loading weights on deck and hatches should not be exceeded.

A.2.8 The stability of the ship should at all times be positive and in compliance with the ship's intact stability requirements.

A.2.9 Emergency escape routes should be free and ready for use.

A.2.10 There should be free access to ventilation ducts and valves if required.

A.2.11 Any obstruction such as lashings or securing points in the access way of escape routes or operational spaces and to safety equipment, fire fighting equipment or sounding pipes should be kept to a minimum and in any case they should be clearly marked.

A.2.12 An approximation of the metacentric height should be obtained (when safe to do so) from the rolling period or static list at a late stage of loading. Rolling or static list may be initiated by quick or slow (as appropriate) shifting of cargo with the deck cranes or lowering cargo bundles onto other deck cargo at one side of the ship.

### ***Stowage***

A.2.13 The stow of the deck cargo should be as solid, compact and stable as practicable. Slack in the stow should be prevented as such could cause lashings to slacken and/or water to accumulate.

A.2.14 A binding effect should, as far as practicable, be obtained within the stow to enhance the stability of stack structure and to minimize the risk of cargo shifting during the sea voyage.

A.2.15 Stowage of damaged timber packages should not be allowed. Timber packages that have deformed or are found with broken bands should be returned to shore for rectification.

A.2.16 Cargo should not be stowed overhanging the ship's side.

A.2.17 Timber deck cargo which overhangs the outer side of hatch coamings or other structures, should be supported at the outer end by other cargo stowed on deck or railing or equivalent structure of sufficient strength to support it (refer to 2.9.6).

### ***Avoid the risk of sliding in the stow***

A.2.18 Ice and snow accretions should be cleared from the hatches and deck cargo before placing further cargo layers in order to obtain a high coefficient of friction in the stow.

A.2.19 Sliding between the layers should if possible be prevented by stowing timber packages of different heights in the same layer or by inserting vertical, sturdy battens between the layers. Transverse tipping of wooden packages could be prevented by overlapping packages in successive tiers so as to create a binding stow (refer to 6.5.7).

### ***Work safety***

A.2.20 Personnel involved in the loading process should be equipped with protective clothing, i.e. hardhats, proper footwear, gloves, etc., according to ship's and harbour requirements.

A.2.21 Personnel working on cargo stowed at heights 2 m and above, within 1 m of an unguarded edge, should if deemed necessary be protected from falls with fall restraint equipment such as a safety harness or other fall restraining devices approved by the Administration.

A.2.22 While working on the cargo there should be provisions to attach a safety harness.

A.2.23 Safe access should be available to the top of, and across the cargo stow.

A.2.24 Personnel should exercise caution when working or moving on timber packages covered by plastic wrapping or tarpaulins.

### **A.3 Securing of timber deck cargoes**

#### ***Basic requirements on the securing***

A.3.1 The stevedoring company and the crew should be informed about the requirements on the securing arrangements.

A.3.2 Uprights, when used, should be well fastened and protected from falling inwards during loading and discharging operations.

A.3.3 If required by this Code and as prescribed in the cargo securing manual, uprights should be connected by hog lashings, running between each pair of uprights on opposing sides of the stow.

#### ***Repair or replacement of damaged securing equipment***

A.3.4 Only undamaged cargo securing equipment should be used for securing timber deck cargo.

A.3.5 Damaged equipment that is beyond repair should be marked as unserviceable and removed from the vessel.

A.3.6 If any damage is noted on any of the uprights or their support on deck, coamings or hatches, this should immediately be repaired.

A.3.7 If any damage is noted on the fixed lashing equipment this should immediately be repaired.

A.3.8 If any damage is noted on the portable lashing equipment this should immediately be repaired or the equipment should be exchanged by new certified equipment.

#### ***Tightening of lashings***

A.3.9 Threads on turnbuckles should be greased to increase pre-tension in the lashings.

A.3.10 All lashings should be thoroughly tightened and all bolts and screws on shackles and turnbuckles should be tightly fastened.

A.3.11 Turnbuckles should have sufficient threads remaining to permit lashings to be tightened during the voyage as needed.

A.3.12 Lashings should be tensioned as specified in this Code and as prescribed in the cargo securing manual.

A.3.13 Edge protectors should be used when required according to this code and as prescribed in the ship's Cargo Securing Manual to obtain good pretension in both vertical and horizontal parts of the lashings.

### ***Provision of catwalk***

A.3.14 If there is no convenient passage on or below the deck of the ship, a sturdy catwalk with strong railings should be provided above the deck cargo (refer to 2.8.6).

### ***Securing according to the ship's Cargo Securing Manual***

A.3.15 The timber deck cargo should be stowed and secured according to this code and as prescribed in the ship's Cargo Securing Manual.

A.3.16 Number and strength of uprights and lashing equipment used for the securing of the timber deck cargo should be in accordance with this code and as prescribed in the ship's Cargo Securing Manual.

## **A.4 Actions to be taken during the voyage**

### ***Voyage planning***

A.4.1 During voyage planning, all foreseeable risks which could lead to either excessive accelerations causing cargo to shift or sloshing sea causing water absorption and ice aggregation, should be taken under consideration.

A.4.2 Before the ship proceeds to sea, the following should be verified:

- .1 The ship is upright;
- .2 The ship has an adequate metacentric height;
- .3 The ship meets the required stability criteria; and
- .4 The cargo is properly secured.

A.4.3 Soundings of tanks should be regularly carried out throughout the voyage.

A.4.4 The rolling period of the ship should be regularly checked in order to establish that the metacentric height is still within the acceptable range.

A.4.5 In cases where severe weather and sea conditions are unavoidable, the Master should be conscious of the need to reduce speed and/or alter course at an early stage in order to minimize the forces imposed on the cargo, structure and lashings.

A.4.6 If deviation from the intended voyage plan is considered during the voyage, a new plan should be made.

### ***Cargo safety inspections during sea voyages***

A.4.7 Cargo safety inspections, in accordance with the items below, should be frequently conducted throughout the voyage.

A.4.8 Prior to any inspections being commenced on deck, the Master should take appropriate actions to reduce the motions of the ship during such operations.

A.4.9 Close attention should be given to any movement of the cargo which could compromise the safety of the ship.



A.4.10 When safety permits fixed and portable lashing equipment should be visually examined for any abnormal wear and tear or other damages.

A.4.11 Since vibrations and working of the ship will cause the cargo to settle and compact, lashing equipment should be retightened to produce the necessary pre-tension, as needed.

A.4.12 Uprights should be checked for any damage or deformation.

A.4.13 Supports for upright should be undamaged.

A.4.14 Corner protections should still be in place.

A.4.15 All examinations and adjustments to cargo securing equipment during the voyage should be entered in the ship's log-book.

***List during voyage***

A.4.16 If a list occurs that cannot be attributed to normal use of consumables the matter should be immediately investigated. This should consider that the cause may be due to one or more of the following:

- .1 cargo shift;
- .2 water ingresses; and
- .3 an angle of loll (inadequate GM).

A.4.17 Even if no major shift of the deck cargo is apparent, it should be examined whether the deck cargo has shifted slightly or if there has been a shift of cargo below deck. However, prior to entering any closed hold that contains timber the atmosphere should be checked to make sure that the hold atmosphere has not been oxygen depleted by the timber.

A.4.18 It should be considered whether the weather conditions are such that sending the crew to release or tighten the lashings on a moving or shifted cargo present a greater hazard than retaining an overhanging load.

A.4.19 The possibility of water ingress should be determined by sounding throughout the vessel. In the event that unexplained water is detected, all available pumps, as appropriate, should be used to bring the situation under control.

A.4.20 An approximation of the current metacentric height should be determined by timing the rolling period.

A.4.21 If the list is corrected by ballasting and deballasting operations, the order in which tanks are filled and emptied should be decided with consideration to the following factors:

- .1 when the draft of the vessel increases, water ingress may occur through openings and ventilation pipes;
- .2 if ballast has been shifted to counteract a cargo shift or water ingress, a far greater list may rapidly develop to the opposite side;
- .3 if the list is due to the ship lolling, and if empty divided double bottom space is available, the tank on the lower side should be ballasted first in order to immediately provide additional metacentric height – after which the tank on the high side should also be ballasted; and

- .4 free surface moments should be kept at a minimum by operating only one tank at a time.

A.4.22 As a final resort when all other options have been exhausted if the list is to be corrected by jettisoning deck cargo, the following aspects should be noted:

- .1 jettisoning is unlikely to improve the situation entirely as the whole stack would probably not fall at once;
- .2 severe damage may be sustained by the propeller if it is still turning when the timber is jettisoned;
- .3 it will be inherently dangerous to anyone involved in the actual jettison procedure; and
- .4 the position of the jettisoning procedure and estimated navigational hazard must be immediately reported to coastal authorities.

A.4.23 If the whole or partial timber deck load is either jettisoned or accidentally lost overboard, the information on a direct danger to navigation<sup>(28)</sup> should be communicated by the master by all means at his disposal to the following parties:

- .1 ships in the vicinity; and
- .2 competent authorities at the first point on the coast with which he can communicate directly.

Such information is to include the following:

- .3 the kind of danger;
- .4 the position of the danger when last observed; and
- .5 the time and date (coordinated universal time) when the danger was last observed.

## **A.5 Safety during discharge of timber deck cargoes**

### ***Cargo securing equipment***

A.5.1 The cargo securing equipment should be collected and examined and damaged equipment should be either repaired or scrapped.

A.5.2 Uprights, when used, should be well fastened to the deck, hatches or coamings of the vessel and protected from falling inwards during discharging operations.

### ***Ship's safety***

A.5.3 All discharge operations should be planned to immediately cease if a list develops for which there is no satisfactory explanation and it would be imprudent to continue loading.

A.5.4 The stability of the ship should, at all times, be positive and in compliance with the vessels intact stability requirements.

A.5.5 Emergency escape routes should be free and ready for use.

**Work safety**

A.5.6 Personnel involved in the discharge process should be dressed with protective clothing, i.e. hardhats, proper footwear, gloves, etc., according to ship's and harbour requirements.

A.5.7 While working on the cargo there should be provisions to attach a safety harness.

A.5.8 Correct signals should be agreed and used with crane operator(s).

A.5.9 Safe access should be available to the top of, and across the cargo stow.

A.5.10 All possible actions should be taken to minimize the risk of slipping on the cargo (i.e. when plastic wrapping or tarpaulins are used as covers).

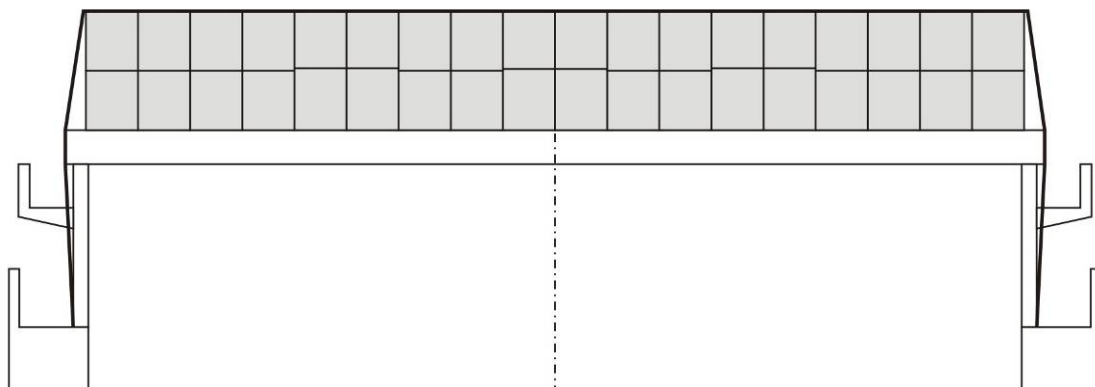
A.5.11 Illumination should be used when required during the cargo operation.

## ANNEX B – SAMPLES OF STOWAGE AND SECURING ARRANGEMENTS

### B.1 Example calculation – Top-over lashings

In the examples below, the number of lashings required to secure packages of sawn wood on deck as well as the required racking strength in the packages in the bottom layer are calculated for a 16,600 DWT ship.

#### Example B.1.1 – Top-over lashings on a 16,600 DWT ship



**Figure B.1.** Midship section of 16,600 DWT ship with packages of sawn wood in two layers secured with top-over lashings

#### Ship particulars

|                                     |             |
|-------------------------------------|-------------|
| Length between perpendiculars, LPP: | 134 metres  |
| Moulded breadth, BM:                | 22 metres   |
| Service speed:                      | 14.5 knots  |
| Metacentric height, GM:             | 0.70 metres |

The deck cargo has the dimensions  $L \times B \times H = 80 \times 19.7 \times 2.4$  metres. The total weight of the deck cargo is taken as 1,600 tons. Sliding between the layers is prevented by packages of different heights in the bottom layer.

#### Dimensioning transverse acceleration

With ship particulars as above and considering a stowage position on deck low, Annex 13 of the CSS Code gives a transverse acceleration of  $a_t = 5.3 \text{ m/s}^2$ , using the following basic acceleration and correction factors:

|                       |   |      |                |   |                                        |
|-----------------------|---|------|----------------|---|----------------------------------------|
| $a_{t \text{ basic}}$ | = | 6.5  | $\text{m/s}^2$ | = | Basic transverse acceleration          |
| $f_{R1}$              | = | 0.81 |                | = | Correction factor for length and speed |
| $f_{R2}$              | = | 1.00 |                | = | Correction factor for $B_M/GM$         |

$$a_t = a_{t \text{ basic}} \cdot f_{R1} \cdot f_{R2} = 6.5 \cdot 0.81 \cdot 1.00 = 5.3 \text{ m/s}^2$$

### Cargo properties

|                |   |           |   |                                                                                                                    |
|----------------|---|-----------|---|--------------------------------------------------------------------------------------------------------------------|
| $m$            | = | 1,600 ton | = | Mass of the section to be secured in tons, including absorbed water and possible icing                             |
| $\mu_{static}$ | = | 0.45      | = | Coefficient of static friction between the timber deck cargo and the ship's deck/hatch cover                       |
| $H$            | = | 2.4 m     | = | Height of deck cargo in metres                                                                                     |
| $B$            | = | 19.7 m    | = | Width of deck cargo in metres                                                                                      |
| $L$            | = | 80 m      | = | Length of the deck cargo or section to be secured in metres                                                        |
| $PW$           | = | 192 kN    | = | Wind pressure in kN based on 1 kN per m <sup>2</sup> wind exposed area, see CSS Code, Annex 13                     |
| $PS$           | = | 160 kN    | = | Pressure from unavoidable sea sloshing in kN based on 1 kN per m <sup>2</sup> exposed area, see CSS Code, Annex 13 |
| $PT_V$         | = | 16 kN     | = | Pretension in the vertical part of the lashings in kN                                                              |
| $\alpha$       | = | 85°       | = | Angle between the horizontal plane and the lashings in degrees                                                     |
| $n_p$          | = | 18 pcs    | = | Number of stacks of packages abreast in each row                                                                   |

### Number of required top-over lashings

For pure top-over lashing arrangements with no bottom blocking, the friction alone will have to counteract the transverse forces so that the following equilibrium of forces is satisfied:

$$(m \cdot g_0 + 2 \cdot n \cdot PT_V \cdot \sin \alpha) \cdot \mu_{static} \geq m \cdot a_t + PW + PS$$

Units denoted with <sub>a</sub> consider cargo units above the bottom layer only.

Thus the required number of top-over lashings can be calculated as:

$$n \geq \frac{\frac{m \cdot a_t + PW + PS}{\mu_{static}} - m \cdot g_0}{2 \cdot PT_V \cdot \sin \alpha} = \frac{\frac{1600 \cdot 5.3 + 192 + 160}{0.45} - 1600 \cdot 9.81}{2 \cdot 16 \cdot \sin 85} = 123 \text{ pcs}$$

### Racking strength

To prevent the packages in the bottom layer from collapsing due to racking, the weight of the cargo stowed on top of the bottom layer should be limited so that the following equilibrium of forces is satisfied:

$$n_p \cdot L \cdot RS \geq m_a \cdot (a_t - 0.5g_0) + PW_a + PS_a$$

Units denoted with <sub>a</sub> consider cargo units above the bottom layer only.

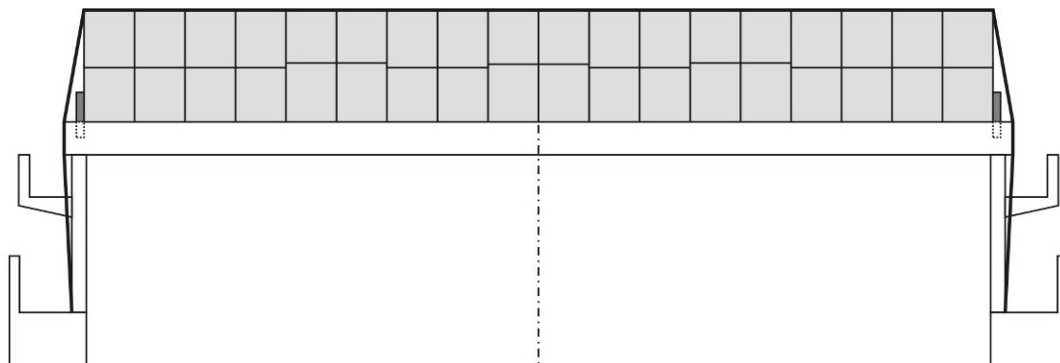
Thus the required racking strength can be calculated to 0.33 kN/metre:

$$RS \geq \frac{m_a \cdot (a_t - 0.5 \cdot g_0) + PW_a + PS_a}{n_p \cdot L} = \frac{800 \cdot (5.3 - 0.5 \cdot 9.81) + 96 + 64}{18 \cdot 80} = 0.33 \text{ kN / m} = 0.034 \text{ ton / m}$$

## B.2 Example calculation – Bottom blocking and top-over lashings

In the example below, the required strength of the bottom blocking devices are calculated for a deck load of packages of sawn wood. The number of lashings used and the pretension of the lashings have been taken in accordance with sections 6.5.19 and 6.5.20 of this Code.

### Example B.2.1 – Bottom blocking and top-over lashings on a 16,600 DWT ship



**Figure B.2.** Midship section of 16,600 DWT ship with packages of sawn wood in two layers secured with bottom blocking devices and top-over lashings

#### Ship particulars

|                                     |             |
|-------------------------------------|-------------|
| Length between perpendiculars, LPP: | 134 metres  |
| Moulded breadth, BM:                | 22 metres   |
| Service speed:                      | 14.5 knots  |
| Metacentric height, GM:             | 0.70 metres |

The deck cargo has the dimensions  $L \times B \times H = 80 \times 19.7 \times 2.4$  metres. The total weight of the deck cargo is taken as 1,600 tons. Sliding between the layers is prevented by packages of different heights in the bottom layer.

#### Dimensioning transverse acceleration

With ship particulars as above and considering a stowage position on deck low, Annex 13 of the CSS Code gives a transverse acceleration of  $a_t = 5.3 \text{ m/s}^2$ , using the following basic acceleration and correction factors:

|                       |   |                    |   |                                        |
|-----------------------|---|--------------------|---|----------------------------------------|
| $a_{t \text{ basic}}$ | = | 6.5 $\text{m/s}^2$ | = | Basic transverse acceleration          |
| $f_{R1}$              | = | 0.81               | = | Correction factor for length and speed |
| $f_{R2}$              | = | 1.00               | = | Correction factor for $B_M/GM$         |

$$a_t = a_{t \text{ basic}} \cdot f_{R1} \cdot f_{R2} = 6.5 \cdot 0.81 \cdot 1.00 = 5.3 \text{ m/s}^2$$

#### Cargo properties

|                       |   |           |   |                                                                                              |
|-----------------------|---|-----------|---|----------------------------------------------------------------------------------------------|
| m                     | = | 1,600 ton | = | Mass of the section to be secured in tons, including absorbed water and possible icing       |
| $\mu_{\text{static}}$ | = | 0.45      | = | Coefficient of static friction between the timber deck cargo and the ship's deck/hatch cover |
| H                     | = | 2.4 m     | = | Height of deck cargo in metres                                                               |
| B                     | = | 19.7 m    | = | Width of deck cargo in metres                                                                |

|                 |   |        |   |                                                                                                                    |
|-----------------|---|--------|---|--------------------------------------------------------------------------------------------------------------------|
| L               | = | 80 m   | = | Length of the deck cargo or section to be secured in metres                                                        |
| PW              | = | 192 kN | = | Wind pressure in kN based on 1 kN per m <sup>2</sup> wind exposed area, see CSS Code, Annex 13                     |
| PS              | = | 160 kN | = | Pressure from unavoidable sea sloshing in kN based on 1 kN per m <sup>2</sup> exposed area, see CSS Code, Annex 13 |
| n               | = | 26 pcs | = | Number of top-over lashings                                                                                        |
| PT <sub>V</sub> | = | 16 kN  | = | Pretension in the vertical part of the lashings in kN                                                              |
| α               | = | 85°    | = | Angle between the horizontal plane and the lashings in degrees                                                     |
| n <sub>p</sub>  | = | 18 pcs | = | Number of stacks of packages abreast in each row                                                                   |
| n <sub>b</sub>  | = | 26 pcs | = | Number of bottom blocking devices per side of the deck cargo                                                       |

### Required strength of the bottom blocking

The required strength, MSL, of the bottom blocking devices is given by the following equilibrium:

$$(m \cdot g_0 + 2 \cdot n \cdot PT_V \cdot \sin \alpha) \cdot \mu_{static} + n_b \frac{MSL}{1.35} \geq m \cdot a_t + PW + PS$$

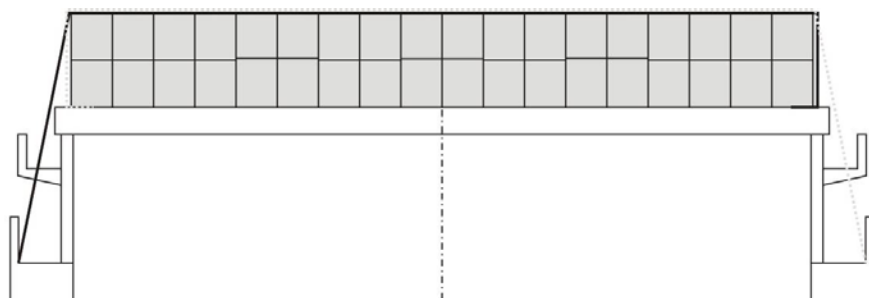
$$MSL \geq \frac{1.35}{n_b} (m \cdot a_t + PW + PS - (m \cdot g_0 + 2 \cdot n \cdot PT_V \cdot \sin \alpha) \cdot \mu_{static})$$

$$MSL \geq \frac{1.35}{26} (2000 \cdot 5.3 + 192 + 160 - (2000 \cdot 9.81 + 2 \cdot 26 \cdot 16 \cdot \sin 85) \cdot 0.45) = 91kN$$

### B.3 Example calculation – Loop lashings

In the example below, the required strength in loop lashings used for secure packages of sawn wood on deck is calculated.

#### Example B.3.1 – Loop lashings on a 16,600 DWT ship



**Figure B.3.** Midship section of 16,600 DWT ship with packages of sawn wood secured with loop lashings

#### Ship particulars

|                                     |             |
|-------------------------------------|-------------|
| Length between perpendiculars, LPP: | 134 metres  |
| Moulded breadth, BM:                | 22 metres   |
| Service speed:                      | 14.5 knots  |
| Metacentric height, GM:             | 0.70 metres |

The deck cargo has the dimensions  $L \times B \times H = 80 \times 19.7 \times 2.4$  metres. The total weight of the deck cargo is taken as 1,600 tons. Sliding between the layers is prevented by packages of different heights in the bottom layer.

### Dimensioning transverse acceleration

With vessel particulars as above and considering a stowage position on deck low, Annex 13 of the CSS Code gives a transverse acceleration of  $a_t = 5.3 \text{ m/s}^2$ , using the following basic acceleration and correction factors:

$$\begin{aligned} a_{t \text{ basic}} &= 6.5 \text{ m/s}^2 &= \text{Basic transverse acceleration} \\ f_{R1} &= 0.81 &= \text{Correction factor for length and speed} \\ f_{R2} &= 1.00 &= \text{Correction factor for } B_M/GM \end{aligned}$$

$$a_t = a_{t \text{ basic}} \cdot f_{R1} \cdot f_{R2} = 6.5 \cdot 0.81 \cdot 1.00 = 5.3 \text{ m/s}^2$$

### Cargo properties

|                 |   |           |   |                                                                                                                    |
|-----------------|---|-----------|---|--------------------------------------------------------------------------------------------------------------------|
| $m$             | = | 1,600 ton | = | Mass of the section to be secured in tons, including absorbed water and possible icing                             |
| $\mu_{dynamic}$ | = | 0.32      | = | Coefficient of dynamic friction between the timber deck cargo and the ship's deck/hatch cover                      |
| $H$             | = | 2.4 m     | = | Height of deck cargo in metres                                                                                     |
| $B$             | = | 19.7 m    | = | Width of deck cargo in metres                                                                                      |
| $L$             | = | 80 m      | = | Length of the deck cargo or section to be secured in metres                                                        |
| $PW$            | = | 192 kN    | = | Wind pressure in kN based on 1 kN per m <sup>2</sup> wind exposed area, see CSS Code, Annex 13                     |
| $PS$            | = | 160 kN    | = | Pressure from unavoidable sea sloshing in kN based on 1 kN per m <sup>2</sup> exposed area, see CSS Code, Annex 13 |
| $\alpha$        | = | 70°       | = | Angle between the horizontal plane and the lashings in degrees                                                     |
| $n$             | = | 36 pcs    | = | Number of loop lashings pairs                                                                                      |
| $L_L$           | = | 25 m      | = | Length of each lashing in metres                                                                                   |
| $PT_V$          | = | 16 kN     | = | Pretension in the vertical part of the lashings in kN                                                              |
| $n_p$           | = | 13 pcs    | = | Number of stacks of packages abreast in each row                                                                   |

### Number of required loop lashings

The number and strength of the lashings are to be chosen so that the following equilibrium is satisfied:

$$(m \cdot g_0 + n \cdot CS \cdot \sin \alpha) \cdot \mu_{dynamic} + n \cdot CS + n \cdot CS \cdot \cos \alpha \geq m \cdot a_t + PW + PS$$

If the number of loop lashings pairs is 36 then the required strength in the lashings can be calculated as:

$$CS \geq \frac{m \cdot (a_t - g_0 \cdot \mu_{dynamic}) + PW + PS}{n \cdot (\sin \alpha \cdot \mu_{dynamic} + 1 + \cos \alpha)} = \frac{1600 \cdot (5.3 - 9.81 \cdot 0.32) + 192 + 160}{36 \cdot (\sin 70 \cdot 0.32 + 1 + \cos 70)} = 64 \text{ kN}$$



The required MSL in the lashings is calculated as:

$$MSL = CS \cdot 1.35 = 64 \cdot 1.35 = 86 \text{ kN} = 8.8 \text{ ton}$$

### ***Transverse movement of cargo due to elongation in lashings***

The transverse movement of the deck cargo due to elongation of the lashings is calculated according to the formula below. If chains are used the elongation factor is set to  $\varepsilon = 0.02$ , and the transverse movement is calculated as:

$$\delta = L_L \cdot \frac{(CS - PT_V)}{MSL} \cdot \varepsilon = 25 \cdot \frac{(64 - 16)}{86} \cdot 0.02 = 0.28 \text{ m}$$

If web lashings are used the elongation factor is set to  $\varepsilon = 0.07$ , and the transverse movement is calculated as:

$$\delta = L_L \cdot \frac{(CS - PT_V)}{MSL} \cdot \varepsilon = 25 \cdot \frac{(64 - 16)}{86} \cdot 0.07 = 0.98 \text{ m}$$

In accordance with 6.5.16 the transverse movement of the cargo should not generate a greater heeling angle than 5 degrees. In order to comply with this requirement significantly more and/or stronger lashings than described above have to be used.

### ***Racking strength***

To prevent the packages in the bottom layer from collapsing due to racking, the weight of the cargo stowed on top of the bottom layer should be limited so that the following equilibrium of forces is satisfied:

$$n_p \cdot L \cdot RS + n \cdot CS \cdot \cos \alpha \geq m_a \cdot (a_t - 0.5g_0) + PW_a + PS_a$$

Units denoted with <sub>a</sub> consider cargo units above the bottom layer only.

Thus the required racking strength can be calculated as:

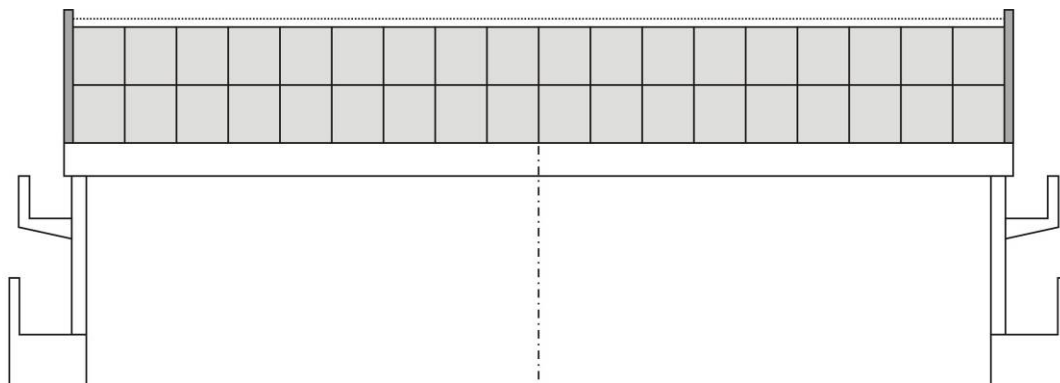
$$\begin{aligned} RS &\geq \frac{m_a \cdot (a_t - 0.5 \cdot g_0) + PW_a + PS_a - n \cdot CS \cdot \cos \alpha}{n_p \cdot L} = \\ &= \frac{800 \cdot (5.3 - 0.5 \cdot 9.81) + 96 + 64 - 46 \cdot 62 \cdot \cos 70}{13 \cdot 80} < 0 \text{ kN / m} \end{aligned}$$

There is no requirement on the racking strength of the packages, since the calculated value is less than zero.

## **B.4 Example Calculation – Uprights for packages of sawn wood**

In the example below, the dimensioning moment for uprights supporting packages of sawn wood on deck is calculated for a 16,600 DWT ship.

### Example B.4.1 – Uprights on a 16,600 DWT Vessel



**Figure B.4.** Midship section of ship with timber packages secured with uprights

#### Ship particulars

|                                           |            |
|-------------------------------------------|------------|
| Length between perpendiculars, $L_{PP}$ : | 134 metres |
| Moulded breadth, $B_M$ :                  | 22 metres  |
| Service speed:                            | 14.5 knots |
| Metacentric height, GM:                   | 0.7 metres |

The deck cargo has the dimensions  $L \times B \times H = 80 \times 19.7 \times 2.4$  metres. The total weight of the deck cargo is taken as 1,600 tons.

With ship particulars as above and considering a stowage position on deck low, Annex 13 of the CSS Code gives a transverse acceleration of  $a_t = 5.3 \text{ m/s}^2$ , using the following basic acceleration and correction factors:

|                       |   |                    |   |                                        |
|-----------------------|---|--------------------|---|----------------------------------------|
| $a_{t \text{ basic}}$ | = | 6.5 $\text{m/s}^2$ | = | Basic transverse acceleration          |
| $f_{R1}$              | = | 0.80               | = | Correction factor for length and speed |
| $f_{R2}$              | = | 1.00               | = | Correction factor for $B_M/GM$         |

$$a_t = a_{t \text{ basic}} \cdot f_{R1} \cdot f_{R2} = 6.5 \cdot 0.81 \cdot 1.00 = 5.3 \text{ m/s}^2$$

#### Cargo properties

|                         |   |           |   |                                                                                        |
|-------------------------|---|-----------|---|----------------------------------------------------------------------------------------|
| m                       | = | 1,600 ton | = | Mass of the section to be secured in tons, including absorbed water and possible icing |
| $\mu_{\text{internal}}$ | = | 0.30      | = | Coefficient of internal friction between the timber packages                           |
| H                       | = | 2.4 m     | = | Height of deck cargo in metres                                                         |
| b                       | = | 1.1 m     | = | Width of each individual stack of packages                                             |
| $n_p$                   | = | 18 pcs    | = | Number of stacks of timber packages abreast in each row                                |
| q                       | = | 2 pcs     | = | Number of layers of timber packages                                                    |
| RS                      | = | 3.5 kN    | = | Racking Strength per timber package in kN                                              |
| N                       | = | 36 pcs    | = | Number of uprights supporting the considered section on each side                      |
| H                       | = | 2.4 m     | = | Height above deck at which hoglashings are attached to the uprights in metres          |

K = 1.8 = Factor for considering hog lashings  
k = 1 if no hog lashings are used  
k = 1.8 if hog lashings are used

### **Bending moment in uprights**

The design bending moment per upright supporting timber packages is to be taken as the greatest of the three moments given by the following formulas:

$$CM_{bending1} = \frac{m}{n_p \cdot k \cdot N} \cdot \left( a_t \cdot \frac{H}{2} - g_0 \cdot \frac{b}{2} \right) \cdot \frac{1 - (1 - f_i)^{n_p}}{f_i} \quad (\text{Moment required to prevent tipping})$$

$$\text{where} \quad f_i = \mu_{internal} \cdot \frac{2b}{H} \quad (f_i = \text{Factor for considering internal moment})$$

$$CM_{bending2} = \frac{H}{2 \cdot k \cdot N} \cdot m \cdot (a_t - \mu_{internal} \cdot g_0) \cdot \frac{q-1}{2q} \quad (\text{Moment required to prevent sliding})$$

$$CM_{bending3} = \frac{H}{k \cdot N} \cdot (m \cdot a_t - (n_p - 4)(q-2) \cdot L \cdot RS) \cdot \frac{(q-1)}{2q} \quad (\text{Moment required to prevent racking})$$

With cargo properties and acceleration as given above, the following bending moments are calculated:

$$f_i = 0.3 \cdot \frac{2 \cdot 1.1}{2.4} = 0.275$$

$$CM_{bending1} = \frac{1600}{18 \cdot 1.8 \cdot 36} \cdot \left( 5.3 \cdot \frac{2.4}{2} - 9.81 \cdot \frac{1.1}{2} \right) \cdot \frac{1 - (1 - 0.275)^{18}}{0.275} = 4.8 \text{ kNm}$$

$$CM_{bending2} = \frac{2.4}{2 \cdot 1.8 \cdot 36} \cdot 1600 \cdot (5.3 - 0.30 \cdot 9.81) \cdot \frac{2-1}{2 \cdot 2} = 17.5 \text{ kNm}$$

$$CM_{bending3} = \frac{2.4}{1.8 \cdot 36} \cdot (1600 \cdot 5.3 - (18 - 4)(2 - 2) \cdot 80 \cdot 3.5) \cdot \frac{(2-1)}{2 \cdot 2} = 78.5 \text{ kNm}$$

The design bending moment, taken as the maximum bending moment calculated by the three formulae above multiplied with the safety factor of 1.35, thus becomes 106 kNm:

$$M_{bending} \geq 1.35 \cdot \max(CM_{bending1}, CM_{bending2}, CM_{bending3}) = 1.35 \cdot 78.5 = 106 \text{ kNm}$$

### **Suitable dimensions for uprights**

With MSL taken as 50% of the MBL for steel with the ultimate strength 360 MPa (N/mm<sup>2</sup>), the required bending resistance, W, can be calculated as:

$$W = \frac{M_{bending}}{50\% \text{ of } 360 \text{ MPa}} = \frac{106 \cdot 10^6}{180} = 589 \cdot 10^3 \text{ mm}^3 = 589 \text{ cm}^3$$

Thus, uprights made from either HE220A profiles or a cylindrical profile with an outer diameter of 324 mm and a wall thickness of 10.3 mm are suitable (see section B.7).

### **Strength in hoglashings**

The required MSL of each hog lashing is calculated by the following formula:

$$MSL \geq \frac{M_{bending}}{2 \cdot h}$$

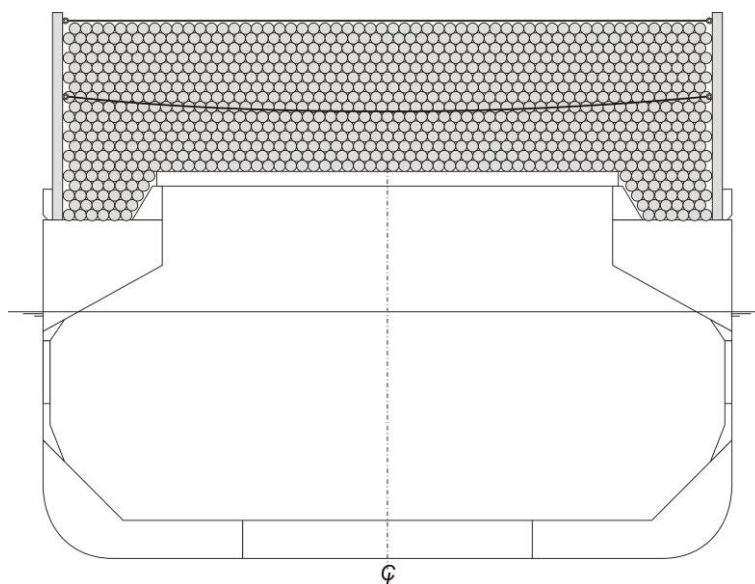
In this case, the hoglashings are attached at a height of  $h = 3.5$  m and the required strength is calculated as:

$$MSL \geq \frac{M_{bending}}{2 \cdot h} = \frac{106}{2 \cdot 3.5} = 15 \text{ kN} \approx 1.5 \text{ ton}$$

## **B.5 Example Calculation – Uprights for round wood**

In the examples below, the dimensioning moments for uprights supporting round wood on deck are calculated for three different ships of varying sizes.

### **Example B.5.1 – Uprights for round wood on a 28,400 DWT ship**



**Figure B.5.** Midship section of 28,400 DWT ship with round wood secured with uprights

### **Ship particulars**

|                                     |             |
|-------------------------------------|-------------|
| Length between perpendiculars, LPP: | 160 metres  |
| Moulded breadth, BM:                | 27 metres   |
| Service speed:                      | 14 knots    |
| Metacentric height, GM:             | 0.80 metres |

The deck cargo has the dimensions  $L \times B \times H = 110 \times 25.6 \times 7$  metres and is supported by 42 uprights on each side. The total weight is taken as 10,500 tons.

In addition to the uprights and hog-lashings, the cargo has been secured with top-over lashings applied in accordance with sections 5.4 and 6.5.28 – 6.5.30 .

With ship particulars as above and considering a stowage position on deck low, Annex 13 of the CSS Code gives a transverse acceleration of  $a_t = 4.6 \text{ m/s}^2$ , using the following basic acceleration and correction factors:

|                       |   |      |                |   |                                        |
|-----------------------|---|------|----------------|---|----------------------------------------|
| $a_{t \text{ basic}}$ | = | 6.5  | $\text{m/s}^2$ | = | Basic transverse acceleration          |
| $f_{R1}$              | = | 0.71 |                | = | Correction factor for length and speed |
| $f_{R2}$              | = | 1.00 |                | = | Correction factor for $B_M/GM$         |

$$a_t = a_{t \text{ basic}} \cdot k_1 \cdot k_2 = 6.5 \cdot 0.71 \cdot 1.00 = 4.6 \text{ m/s}^2$$

### Cargo properties

|                       |   |           |     |   |                                                                                                                  |
|-----------------------|---|-----------|-----|---|------------------------------------------------------------------------------------------------------------------|
| $M$                   | = | 10,500    | ton | = | Mass of the section to be secured in tons, including absorbed water and possible icing                           |
| $\mu_{\text{static}}$ | = | 0.5       |     | = | Coefficient of static friction between the timber deck cargo and the ship's deck/hatch cover                     |
| $H$                   | = | 7         | m   | = | Height of deck cargo in metres                                                                                   |
| $B$                   | = | 25.6      | m   | = | Width of deck cargo in metres                                                                                    |
| $L$                   | = | 110       | m   | = | Length of the deck cargo or section to be secured in metres                                                      |
| $PW$                  | = | 770       | kN  | = | Wind pressure in kN based on 1 kN per $\text{m}^2$ wind exposed area, see CSS Code, Annex 13                     |
| $PS$                  | = | 220       | kN  | = | Pressure from unavoidable sea sloshing in kN based on 1 kN per $\text{m}^2$ exposed area, see CSS Code, Annex 13 |
| $N$                   | = | 42        | pcs | = | Number of uprights supporting the considered section on each side                                                |
| $h$                   | = | 3.7 / 6.7 | m   | = | Height above deck at which hog lashings are attached to the uprights in metres                                   |
| $n_{\text{hog}}$      | = | 2         | pcs | = | Number of hog lashings for each upright                                                                          |
| $k$                   | = | 1.8       |     | = | Factor for considering hog lashings;<br>k = 1 if no hog lashings are used<br>k = 1.8 if hog lashings are used    |

### Bending moment in uprights

For ships carrying loose sawn wood and round wood, the design bending moment per upright is calculated as the greater of the two moments given by the following formulas:

$$CM_{\text{bending1}} = 0.1 \cdot \frac{H^2}{k \cdot B \cdot N} \cdot m \cdot g_0$$

$$CM_{\text{bending2}} = \frac{H}{3 \cdot k \cdot N} \cdot (m \cdot (a_t - 0.6 \cdot \mu_{\text{static}} \cdot g_0) + PW + PS)$$

With cargo properties and acceleration as given above, the following bending moments are calculated:

$$CM_{bending1} = 0.1 \cdot \frac{7^2}{1.8 \cdot 25.6 \cdot 42} \cdot 10500 \cdot 9.81 = 260 \text{ kNm}$$

$$CM_{bending2} = \frac{7}{3 \cdot 1.8 \cdot 42} \cdot (10500 \cdot (4.6 - 0.6 \cdot 0.5 \cdot 9.81) + 770 + 220) = 568 \text{ kNm}$$

The design bending moment, taken as the maximum bending moment calculated by the formulae above multiplied with a safety factor of 1.35 and considering the 12% reduction allowed for by the use of properly applied top-over lashings, thus becomes:

$$M_{bending} \geq 88\% \cdot 1.35 \cdot \max(CM_{bending1}, CM_{bending2}) = 0.88 \cdot 1.35 \cdot 568 = 675 \text{ kNm}$$

### **Suitable dimensions for uprights**

With MSL taken as 50% of the MBL for steel with the ultimate strength 360 MPa (N/mm<sup>2</sup>), the required bending resistance, W, can be calculated as:

$$W = \frac{M_{bending}}{50\% \text{ of } 360 \text{ MPa}} = \frac{675 \cdot 10^6}{180} = 3749 \cdot 10^3 \text{ mm}^3 = 3749 \text{ cm}^3$$

Thus, uprights made from either HE500B profiles or a cylindrical profile with an outer diameter of 508 mm and a wall thickness of 26.2 mm are suitable (see section B.7).

### **Strength in hog lashings**

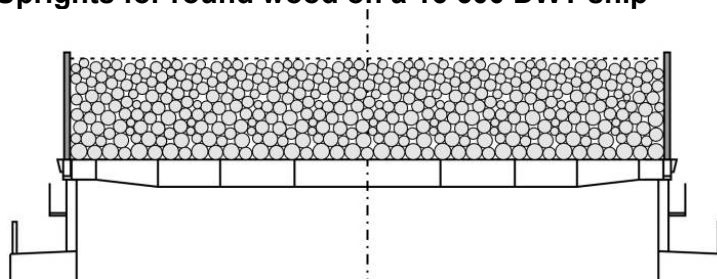
The required MSL of each hog lashing is calculated by the following formula:

$$MSL \geq \frac{M_{bending}}{2 \cdot h \cdot n_{hog}}$$

In this case, the hog lashings are attached at the heights 3.7 and 6.7 metres (mean height=5.2) and the required strength is calculated as:

$$MSL \geq \frac{M_{bending}}{2 \cdot h \cdot n_{hog}} = \frac{528}{2 \cdot 5.2 \cdot 2} = 25 \text{ kN} \approx 2.5 \text{ ton}$$

### **Example B.5.2 – Uprights for round wood on a 16 600 DWT ship**



**Figure B.6.** Midship section of 16 600 DWT ship with round wood secured with uprights

### Ship particulars

|                                     |             |
|-------------------------------------|-------------|
| Length between perpendiculars, LPP: | 134 metres  |
| Moulded breadth, BM:                | 22 metres   |
| Service speed:                      | 14.5 knots  |
| Metacentric height, GM:             | 0.70 metres |

The deck cargo has the dimensions  $L \times B \times H = 80 \times 19.7 \times 3.7$  metres and is supported by 30 uprights on each side. The weight of the cargo is taken as 3,000 tons.

With ship particulars as above and considering a stowage position on deck low, Annex 13 of the CSS Code gives a transverse acceleration of  $a_t = 5.3 \text{ m/s}^2$ , using the following basic acceleration and correction factors:

|                       |   |                    |   |                                        |
|-----------------------|---|--------------------|---|----------------------------------------|
| $a_{t \text{ basic}}$ | = | 6.5 $\text{m/s}^2$ | = | Basic transverse acceleration          |
| $f_{R1}$              | = | 0.81               | = | Correction factor for length and speed |
| $f_{R2}$              | = | 1.00               | = | Correction factor for $B_M/GM$         |

$$a_t = a_{t \text{ basic}} \cdot k_1 \cdot k_2 = 6.5 \cdot 0.81 \cdot 1.00 = 5.3 \text{ m/s}^2$$

### Cargo properties

|                       |   |           |   |                                                                                                                  |
|-----------------------|---|-----------|---|------------------------------------------------------------------------------------------------------------------|
| $M$                   | = | 3,000 ton | = | Mass of the section to be secured in tons, including absorbed water and possible icing                           |
| $\mu_{\text{static}}$ | = | 0.35      | = | Coefficient of static friction between the timber deck cargo and the ship's deck / hatch cover                   |
| $H$                   | = | 3.7 m     | = | Height of deck cargo in metres                                                                                   |
| $B$                   | = | 19.7 m    | = | Width of deck cargo in metres                                                                                    |
| $L$                   | = | 80 m      | = | Length of the deck cargo or section to be secured in metres                                                      |
| $PW$                  | = | 296 kN    | = | Wind pressure in kN based on 1 kN per $\text{m}^2$ wind exposed area, see CSS Code, Annex 13                     |
| $PS$                  | = | 160 kN    | = | Pressure from unavoidable sea sloshing in kN based on 1 kN per $\text{m}^2$ exposed area, see CSS Code, Annex 13 |
| $N$                   | = | 30 pcs    | = | Number of uprights supporting on each side                                                                       |
| $h$                   | = | 3.7 m     | = | Height above deck at which hog lashings are attached to the uprights in metres                                   |
| $n_{\text{hog}}$      | = | 1 pcs     | = | Number of hog lashings for each uprights                                                                         |
| $k$                   | = | 1.8       | = | Factor for considering hog lashings;<br>k = 1 if no hog lashings are used<br>k = 1.8 if hog lashings are used    |

### Bending moment in uprights

For ships carrying loose sawn wood and round timber, the design bending moment per upright is calculated as the greater of the two moments given by the following formulas:

$$CM_{\text{bending1}} = 0.1 \cdot \frac{H^2}{k \cdot B \cdot N} \cdot m \cdot g_0$$

$$CM_{\text{bending2}} = \frac{H}{3 \cdot k \cdot N} \cdot (m \cdot (a_t - 0.6 \cdot \mu_{\text{static}} \cdot g_0) + PW + PS)$$

With cargo properties and acceleration as given above, the following bending moments are calculated:

$$CM_{bending1} = 0.1 \cdot \frac{3.7^2}{19.7 \cdot 30} \cdot 3000 \cdot 9.81 = 68 \text{ kNm}$$

$$CM_{bending2} = \frac{3.7}{3 \cdot 2 \cdot 30} \cdot (3000 \cdot (5.3 - 0.6 \cdot 0.35 \cdot 9.81) + 296 + 160) = 209 \text{ kNm}$$

The design bending moment, taken as the maximum bending moment calculated by the formulae above multiplied with a safety factor of 1.35, thus becomes 282 kNm:

$$M_{bending} \geq 1.35 \cdot \max(CM_{bending1}, CM_{bending2}) = 1.35 \cdot 209 = 282 \text{ kNm}$$

### ***Suitable dimensions for uprights***

With MSL taken as 50% of the MBL for steel with the ultimate strength 360 MPa (N/mm<sup>2</sup>), the required bending resistance, W, can be calculated as:

$$W = \frac{M_{bending}}{50\% \text{ of } 360 \text{ MPa}} = \frac{282 \cdot 10^6}{180} = 1568 \cdot 10^3 \text{ mm}^3 = 1568 \text{ cm}^3$$

Thus, uprights made from either HE320B profiles or a cylindrical profile with an outer diameter of 406 mm and a wall thickness of 16.7 mm are suitable (see section B.7).

### ***Strength in hog lashings***

The required MSL of each hog lashing is calculated by the following formula:

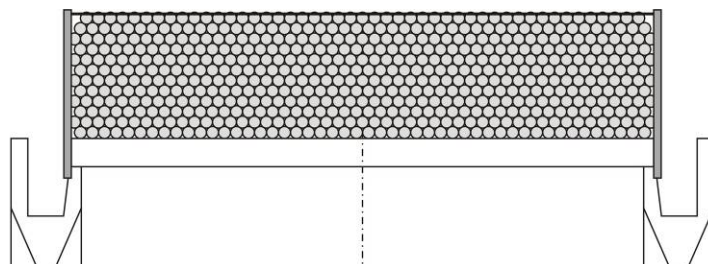
$$MSL \geq \frac{M_{bending}}{2 \cdot h \cdot n_{hog}}$$

In this case, the hog lashings are attached at a height of 3.7 metres and the required strength is calculated as:

$$MSL \geq \frac{M_{bending}}{2 \cdot h \cdot n_{hog}} = \frac{282}{2 \cdot 3.7 \cdot 1} = 38 \text{ kN} \approx 3.9 \text{ ton}$$



### Example B.5.3 – Uprights for round wood on a 6,000 DWT ship on the Baltic Sea



**Figure B.7.** Midship section of 6,000 DWT ship with round wood secured with uprights

#### Ship particulars

|                                     |             |
|-------------------------------------|-------------|
| Length between perpendiculars, LPP: | 101 metres  |
| Moulded breadth, BM:                | 17.5 metres |
| Service speed:                      | 13 knots    |
| Metacentric height, GM:             | 0.50 metres |

The deck cargo has the dimensions  $L \times B \times H = 65 \times 14.5 \times 3.1$  metres and is supported by 25 uprights on each side. The weight of the cargo is taken as 1,500 tons.

With ship particulars as above and considering a stowage position on deck low, Annex 13 of the CSS Code gives the following basic transverse acceleration and correction factors:

|                       |   |      |         |   |                                        |
|-----------------------|---|------|---------|---|----------------------------------------|
| $a_{t \text{ basic}}$ | = | 6.5  | $m/s^2$ | = | Basic transverse acceleration          |
| $f_{R1}$              | = | 0.93 |         | = | Correction factor for length and speed |
| $f_{R2}$              | = | 1.00 |         | = | Correction factor for $B_M/GM$         |

The ship is trading in the Baltic Sea where the maximum expected significant wave height on a 20-year basis can be taken as 8.5 metres. Thus, the reduction factor for operation in restricted waters is taken as:

$$f_R = \sqrt[3]{\frac{H_M}{19.6}} = \sqrt[3]{\frac{8.5}{19.6}} = 0.76$$

$$a_t = a_{t \text{ basic}} \cdot f_{R1} \cdot f_{R2} \cdot f_R = 6.5 \cdot 0.93 \cdot 1.00 \cdot 0.76 = 4.6 \text{ m/s}^2$$

#### Cargo properties

|                       |   |           |   |                                                                                              |
|-----------------------|---|-----------|---|----------------------------------------------------------------------------------------------|
| M                     | = | 1,500 ton | = | Mass of the section to be secured in tons, including absorbed water and possible icing       |
| $\mu_{\text{static}}$ | = | 0.35      | = | Coefficient of static friction between the timber deck cargo and the ship's deck/hatch cover |
| H                     | = | 3.1 m     | = | Height of deck cargo in metres                                                               |
| B                     | = | 14.5 m    | = | Width of deck cargo in metres                                                                |
| L                     | = | 65 m      | = | Length of the deck cargo or section to be secured in metres                                  |
| PW                    | = | 202 kN    | = | Wind pressure in kN based on 1 kN per $m^2$ wind exposed area, see CSS Code, Annex 13        |

|                  |   |     |     |   |                                                                                                                    |
|------------------|---|-----|-----|---|--------------------------------------------------------------------------------------------------------------------|
| PS               | = | 130 | kN  | = | Pressure from unavoidable sea sloshing in kN based on 1 kN per m <sup>2</sup> exposed area, see CSS Code, Annex 13 |
| N                | = | 25  | pcs | = | Number of uprights supporting the considered section on each side                                                  |
| h                | = | 3.1 | m   | = | Height above deck at which hog lashings are attached to the uprights in metres                                     |
| n <sub>hog</sub> | = | 1   | pcs | = | Number of hog lashings for each uprights                                                                           |
| k                | = | 1.8 |     | = | Factor for considering hog lashings;<br>k = 1 if no hog lashings are used<br>k = 1.8 if hog lashings are used      |

### ***Bending moment in uprights***

For ships carrying loose sawn wood and round timber, the design bending moment per upright is calculated as the greater of the two moments given by the following formulas:

$$CM_{bending1} = 0.1 \cdot \frac{H^2}{k \cdot B \cdot N} \cdot m \cdot g_0$$

$$CM_{bending2} = \frac{H}{3 \cdot k \cdot N} \cdot (m \cdot (a_t - 0.6 \cdot \mu_{static} \cdot g_0) + PW + PS)$$

With cargo properties and acceleration as given above, the following bending moments are calculated:

$$CM_{bending1} = 0.1 \cdot \frac{3.1^2}{14.5 \cdot 25} \cdot 1500 \cdot 9.81 = 39 \text{ kNm}$$

$$CM_{bending2} = \frac{3.1}{3 \cdot 1.8 \cdot 25} \cdot (1500 \cdot (4.6 - 0.6 \cdot 0.35 \cdot 9.81) + 202 + 130) = 95 \text{ kNm}$$

The design bending moment, taken as the maximum bending moment calculated by the formulae above multiplied with a safety factor of 1.35, thus becomes 128 kNm:

$$M_{bending} \geq 1.35 \cdot \max(CM_{bending1}, CM_{bending2}) = 1.35 \cdot 95 = 128 \text{ kNm}$$

### ***Suitable dimensions for uprights***

With MSL taken as 50% of the MBL for steel with the ultimate strength 360 MPa (N/mm<sup>2</sup>), the required bending resistance, W, can be calculated as:

$$W = \frac{M_{bending}}{50\% \text{ of } 360 \text{ MPa}} = \frac{128 \cdot 10^6}{180} = 713 \cdot 10^3 \text{ mm}^3 = 713 \text{ cm}^3$$

Thus, uprights made from either HE220 B profiles or a cylindrical profile with an outer diameter of 324 mm and a wall thickness of 10 mm are suitable (see section B.7).

### Strength in hog lashings

The required MSL of each hog lashing is calculated by the following formula:

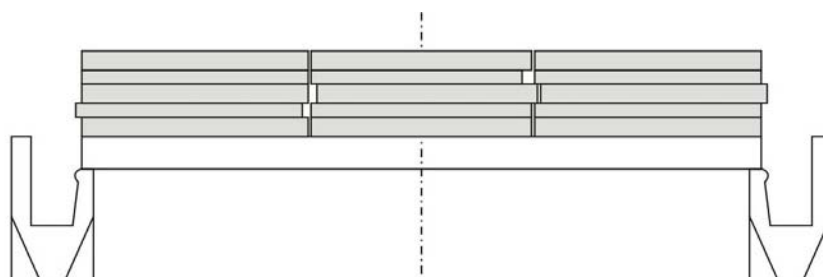
$$MSL \geq \frac{M_{bending}}{2 \cdot h \cdot n_{hog}}$$

In this case, the hog lashings are attached at a height of 3.7 m and the required strength is calculated as:

$$MSL \geq \frac{M_{bending}}{2 \cdot h \cdot n_{hog}} = \frac{128}{2 \cdot 3.1 \cdot 1} = 20.6 \text{ kN} \approx 2.1 \text{ ton}$$

## B.6 Example calculation – frictional securing of transversely stowed round wood

### Example B.6.1 – Frictional securing of round wood on a 6,000 DWT ship



**Figure B.8.** Midship section of 6,000 DWT ship frictional secured wood secured

#### Ship particulars

|                                     |             |
|-------------------------------------|-------------|
| Length between perpendiculars, LPP: | 101 metres  |
| Moulded breadth, BM:                | 17.5 metres |
| Service speed:                      | 13 knots    |
| Metacentric height, GM:             | 0.50 metres |

The deck cargo has the dimensions  $L \times B \times H = 65 \times 14.5 \times 3.1$  metres. The weight of the cargo is taken as 1,500 tons.

#### Cargo properties

|                |   |           |   |                                                                                                                    |
|----------------|---|-----------|---|--------------------------------------------------------------------------------------------------------------------|
| M              | = | 1,500 ton | = | Mass of the section to be secured in tons, including absorbed water and possible icing                             |
| $\mu_{static}$ | = | 0.35      | = | Coefficient of static friction between the timber deck cargo and the ship's deck/hatch cover                       |
| H              | = | 3.1 m     | = | Height of deck cargo in metres                                                                                     |
| B              | = | 14.5 m    | = | Width of deck cargo in metres                                                                                      |
| L              | = | 65 m      | = | Length of the deck cargo or section to be considered in metres                                                     |
| PW             | = | 202 kN    | = | Wind pressure in kN based on 1 kN per m <sup>2</sup> wind exposed area, see CSS Code, Annex 13                     |
| PS             | = | 130 kN    | = | Pressure from unavoidable sea sloshing in kN based on 1 kN per m <sup>2</sup> exposed area, see CSS Code, Annex 13 |

### Transverse acceleration

With a static friction of 0.35 between the layers of wood and between the wood and the hatch cover the maximum acceptable transverse acceleration can be calculated by satisfying the following equilibrium:

$$m \cdot g_0 \cdot \mu_{static} \geq m \cdot a_t + PW + PS$$

In this case transverse acceleration can not exceed 3.2 m/s<sup>2</sup> as shown below:

$$a_t \leq \frac{m \cdot g_0 \cdot \mu_{static} - PW - PS}{m}$$

$$a_t \leq \frac{1500 \cdot 9.81 \cdot 0.35 - 202 - 130}{1500} = 3.2 \text{ m/s}^2$$

With vessel particulars as above and considering a stowage position on deck low, Annex 13 of the CSS Code gives the following basic acceleration and correction factors:

|                       |   |                      |   |                                          |
|-----------------------|---|----------------------|---|------------------------------------------|
| $a_{t \text{ basic}}$ | = | 6.5 m/s <sup>2</sup> | = | Basic transverse acceleration            |
| $f_{R1}$              | = | 0.93                 | = | Correction factor for length and speed   |
| $f_{R2}$              | = | 1.00                 | = | Correction factor for B <sub>M</sub> /GM |

The maximum allowed significant wave height with this stowage arrangement is calculated to 2.9 m according to the following:

$$a_t = a_{t \text{ basic}} \cdot f_{R1} \cdot f_{R2} \cdot f_R$$

$$f_R = \frac{a_t}{a_{t \text{ basic}} \cdot f_{R1} \cdot f_{R2}} = \frac{3.2}{6.5 \cdot 0.93 \cdot 1.00} = 0.53 \text{ m/s}^2$$

$$f_R = \sqrt[3]{\frac{H_M}{19.6}}$$

$$H_M = 19.6 \cdot f_R^3 = 19.6 \cdot 0.53^3 = 2.9 \text{ m}$$

## B.7 Maximum bending resistance in common profiles for uprights

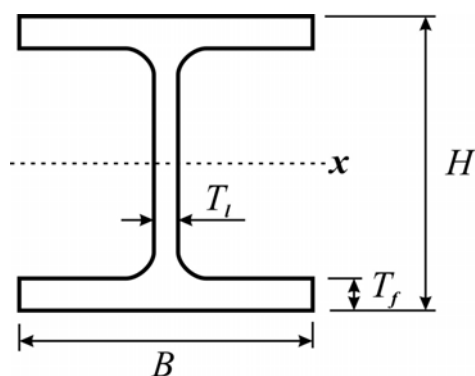
### HE-A beams

| Size     | H<br>[mm] | B<br>[mm] | T <sub>i</sub><br>[mm] | T <sub>f</sub><br>[mm] | Maximum bending resistance<br>W <sub>x</sub> [cm <sup>3</sup> ] |
|----------|-----------|-----------|------------------------|------------------------|-----------------------------------------------------------------|
| HE 220 A | 210       | 220       | 7                      | 11                     | 515                                                             |
| HE 240 A | 230       | 240       | 7.5                    | 12                     | 675                                                             |
| HE 260 A | 250       | 260       | 7.5                    | 12.5                   | 836                                                             |
| HE 280 A | 270       | 280       | 8                      | 13                     | 1010                                                            |
| HE 300 A | 290       | 300       | 8.5                    | 14                     | 1260                                                            |
| HE 320 A | 310       | 300       | 9                      | 15.5                   | 1480                                                            |
| HE 340 A | 330       | 300       | 9.5                    | 16.5                   | 1680                                                            |
| HE 360 A | 350       | 300       | 10                     | 17.5                   | 1890                                                            |
| HE 400 A | 390       | 300       | 11                     | 19                     | 2310                                                            |
| HE 450 A | 440       | 300       | 11.5                   | 21                     | 2900                                                            |
| HE 500 A | 490       | 300       | 12                     | 23                     | 3550                                                            |

| Size     | H<br>[mm] | B<br>[mm] | T <sub>l</sub><br>[mm] | T <sub>f</sub><br>[mm] | Maximum bending resistance<br>W <sub>x</sub> [cm <sup>3</sup> ] |
|----------|-----------|-----------|------------------------|------------------------|-----------------------------------------------------------------|
| HE 550 A | 540       | 300       | 12.5                   | 24                     | 4150                                                            |
| HE 600 A | 590       | 300       | 13                     | 25                     | 4790                                                            |
| HE 650 A | 640       | 300       | 13.5                   | 27                     | 5470                                                            |

#### HE-B beams

| Size     | H<br>[mm] | B<br>[mm] | T <sub>l</sub><br>[mm] | T <sub>f</sub><br>[mm] | Maximum bending resistance,<br>W <sub>x</sub> [cm <sup>3</sup> ] |
|----------|-----------|-----------|------------------------|------------------------|------------------------------------------------------------------|
| HE 220 B | 210       | 220       | 9.5                    | 16                     | 736                                                              |
| HE 240 B | 230       | 240       | 10                     | 17                     | 938                                                              |
| HE 260 B | 250       | 260       | 10                     | 17.5                   | 1150                                                             |
| HE 280 B | 270       | 280       | 10.5                   | 18                     | 1380                                                             |
| HE 300 B | 290       | 300       | 11                     | 19                     | 1680                                                             |
| HE 320 B | 310       | 300       | 11.5                   | 20.5                   | 1930                                                             |
| HE 340 B | 330       | 300       | 12                     | 21.5                   | 2160                                                             |
| HE 360 B | 350       | 300       | 12.5                   | 22.5                   | 2400                                                             |
| HE 400 B | 390       | 300       | 13.5                   | 24                     | 2880                                                             |
| HE 450 B | 440       | 300       | 14                     | 26                     | 3550                                                             |
| HE 500 B | 490       | 300       | 14.5                   | 28                     | 4290                                                             |
| HE 550 B | 540       | 300       | 15                     | 29                     | 4970                                                             |
| HE 600 B | 590       | 300       | 15.5                   | 30                     | 5700                                                             |
| HE 650 B | 640       | 300       | 16                     | 31                     | 6480                                                             |



#### Pipes

| Size | Schedule | Outer diameter<br>[mm] | Wall thickness<br>[mm] | Bending resistance, W<br>[cm <sup>3</sup> ] |
|------|----------|------------------------|------------------------|---------------------------------------------|
| 8"   | 40       | 219.1                  | 8.2                    | 276                                         |
|      | 60       | 219.1                  | 10.3                   | 337                                         |
|      | 80       | 219.1                  | 12.7                   | 402                                         |
| 12"  | 40       | 323.9                  | 10.3                   | 772                                         |
|      | 60       | 323.9                  | 14.3                   | 1029                                        |
|      | 80       | 323.9                  | 17.5                   | 1223                                        |
| 16"  | 40       | 406.4                  | 12.7                   | 1499                                        |
|      | 60       | 406.4                  | 16.7                   | 1910                                        |
|      | 80       | 406.4                  | 21.4                   | 2371                                        |
| 18"  | 40       | 457.2                  | 14.3                   | 2132                                        |
|      | 60       | 457.2                  | 19.1                   | 2758                                        |
|      | 80       | 457.2                  | 23.8                   | 3342                                        |
| 20"  | 40       | 508.0                  | 15.1                   | 2797                                        |

| Size | Schedule | Outer diameter<br>[mm] | Wall thickness<br>[mm] | Bending resistance, W<br>[cm <sup>3</sup> ] |
|------|----------|------------------------|------------------------|---------------------------------------------|
|      | 60       | 508.0                  | 20.6                   | 3697                                        |
|      | 80       | 508.0                  | 26.2                   | 4542                                        |
|      | 100      | 508.0                  | 32.5                   | 5433                                        |
| 24"  | 40       | 610.0                  | 17.5                   | 4686                                        |
|      | 60       | 610.0                  | 24.6                   | 6368                                        |
|      | 80       | 610.0                  | 31.0                   | 7761                                        |

## ANNEX C

### INSTRUCTION TO A MASTER ON CALCULATION OF MASS CHANGE OF A TIMBER DECK CARGO DUE TO WATER ABSORPTION

C.1 Mass increase due to water absorption for a timber deck cargo in protective packaging or covered by a protective awning or timber that has been immersed in water until loaded on board should not be taken into account in the ship's stability calculation for arrival at the port of destination.

C.2 Calculation of mass change  $P$  of a timber deck cargo should be done by the formula:

$$\delta P, \% = T_{pl} \cdot \delta P_{day}, \%$$

where:

- $T_{pl}$  – planned duration of the voyage, days;
- $\delta P_{day}, \%$  – wood mass change per day, to be chosen from table C.1

C.3 Corresponding line in table C.1 should be chosen by means of comparison of the forthcoming voyage with the timber cargo transportation lines specified in the leftmost column "Line".

C.4 With calculation value being  $\delta P \leq 2\%$ , water absorption of a timber deck cargo should not be taken into account in the ship's stability calculations as it is commensurable with initial calculation data determination errors.

C.5 With calculation value being  $\delta P \geq 10\%$ , water absorption of a timber deck cargo  $\delta P = 10\%$  should be taken into account.

**Table C.1.** Daily wood mass change

| Line                                  | Deck cargo mass change per day, $\delta P_{day}, \%$ |                  |
|---------------------------------------|------------------------------------------------------|------------------|
|                                       | Sawn wood                                            | Round wood cargo |
| Vladivostok – ports of Japan          | 1.00                                                 | 0.14             |
| Ports of Malaysia – ports of Japan    | 0.73                                                 | 0.10             |
| Ports of Canada, USA – ports of Japan | 1.00                                                 | 0.14             |
| Saint-Petersburg – London             | 0.83                                                 | 0.11             |
| Arkhangelsk – Manchester              | 1.16                                                 | 0.15             |
| Australasia – North Asia              | -                                                    | -0.10            |

## ANNEX D

### REFERENCES

- (1) **SOLAS** – Chapter VI Regulation 5 Section 1 – Carriage of cargoes – Stowage and securing
- (2) **ISM Code** – Part A – Implementation – Paragraph 1.1.2
- (3) **IMDG Code** – Part 1 – Chapter 1.2.1 – Definitions
- (4) **SOLAS** – Chapter VI Regulation 2 – Cargo information
- (5) **ISM Code** – Paragraph 7 – Development of plans for shipboard operations
- (6) **Load Lines, 1966** – Annex I – Chapter II – Regulation 16 – Regulations for determining load lines – Conditions of assignment of freeboard
- (7) **SOLAS** – Chapter II-1 – Part B-1 – Regulation 25-8 – Stability information
- (8) **2008 IS Code** – Part A – Chapter 3.3 Cargo ships carrying timber deck cargoes
- (9) **2008 IS Code** – Part B – Chapter 3.6.3 Stability booklet (for ships carrying timber deck cargoes)
- (10) **2008 IS Code** – Part B – Chapter 3.7 Operational measures for ships carrying timber deck cargoes
- (11) **2008 IS Code**
- (12) **MEPC.127(53)** – Development of Ballast Water Management Plans
- (13) **Load Lines Convention, 1966** – Annex I – Regulations for determining load lines – Chapter IV – Special requirements for ships assigned timber freeboards – Regulation 44 – Stowage, uprights, lashings, stability, protection of crew, access to machinery spaces, etc.
- (14) **Load Lines Convention, 1966** – Annex I – Regulations for determining load lines – Chapter IV – Special requirements for ships assigned timber freeboards – Regulation 45 – Computation for freeboard
- (15) **SOLAS** – Chapter V – Regulation 22 – Navigational bridge visibility
- (16) **ISM Code** – Paragraph 6.6 – Resources and personnel
- (17) **ILO Convention No. 152** – Convention Concerning Occupational Safety and Health in Dock Work
- (18) **Load Lines Convention, 1966** – Annex I – Regulations for determining load lines – Chapter II – Conditions of assignment of freeboard – Regulation 25 – Protection of the crew



- (19) **Load Lines Convention, 1966** – Annex I – Regulations for determining load lines – Chapter IV – Special requirements for ships assigned timber freeboards – Regulation 44 – Stowage, uprights, lashings, stability, protection of crew, access to machinery spaces, etc.
- (20) **CSS Code** – Annex 13 – Chapter 4 – Strength of securing equipment.
- (21) **ISM Code** – Paragraph 7 – Development of plans for shipboard operations
- (22) **STCW Code** – Section A – VIII/2 – Watchkeeping arrangements and principles to be observed – Part 2 – Voyage planning
- (23) **SOLAS** – Chapter V – Regulation 34 – Safe navigation
- (24) **CSS Code** – Chapter 6 – Actions which may be taken in heavy weather – 6.3
- (25) **MCS/Circ.1228** – Revised guidance to the master for avoiding dangerous situations in adverse weather and sea conditions
- (26) **SOLAS** – Chapter VI – Regulation 5 – Section 2 – Carriage of cargoes – Stowage and securing
- (27) **MSC/Circ.745** – Guidelines for the preparation of the Cargo Securing Manual
- (28) **SOLAS** – Chapter V – Regulation 31– Safety of navigation – Danger messages
- (29) **ILO Convention No. 27** – Marking of weight (packages transported by vessels) Convention, 1929.

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**ANNEX 28**

**DRAFT ASSEMBLY RESOLUTION**

**ADOPTION OF THE IMO/WMO WORLDWIDE MET-OCEAN INFORMATION AND  
WARNING SERVICE GUIDANCE DOCUMENT**

THE ASSEMBLY,

RECALLING Article 15(j) of the Convention on the International Maritime Organization concerning the functions of the Assembly in relation to regulations and guidelines concerning maritime safety,

NOTING the International Convention for the Safety of Life at Sea (SOLAS), 1974, in particular Chapter V (Safety of Navigation), Regulation 5 (Meteorological services and warnings),

BEARING IN MIND the decisions of the 62nd Council of the World Meteorological Organization (WMO),

RECOGNIZING that the existing WMO GMDSS Marine Broadcast System, as amended, is an integral part of the IMO/WMO Worldwide Met-Ocean Information and Warning Service (WWMIWS),

FURTHER RECOGNIZING that the WMO GMDSS Marine Broadcast System needs to be fully in harmony with the IMO/IHO World-Wide Navigational Warning Services, and to respond to requirements for maritime safety services expressed by IMO,

NOTING the provisions made for the promulgation of maritime safety information by the 1988 amendments to the 1974 SOLAS Convention concerning radio communications for the Global Maritime Distress and Safety System (GMDSS),

HAVING CONSIDERED the recommendations made by the Maritime Safety Committee at its eighty-ninth session,

1. ADOPTS the IMO/WMO Worldwide Met-Ocean Information and Warning Service – Guidance Document, as set out in the Annex to the present resolution;
2. RECOMMENDS Governments to implement the IMO/WMO Worldwide Met-Ocean Information and Warning Service;
3. AUTHORIZES the Maritime Safety Committee to keep the annexed Guidance Document under review and update it as necessary in light of experience gained in its application, in accordance with the procedure set out in section 7 of the annex to the present resolution.

## ANNEX

### IMO/WMO WORLDWIDE MET-OCEAN INFORMATION AND WARNING SERVICE GUIDANCE DOCUMENT

#### 1 INTRODUCTION

1.1 The International Convention for the Safety of Life at Sea (SOLAS), 1974 (1974 SOLAS Convention), Chapter V (Safety of Navigation), as amended, Regulation 5 (Meteorological services and warnings), states:

"2 In particular, Contracting Governments undertake to co-operate in carrying out, as far as practicable, the following meteorological arrangements:

.10 To endeavour to obtain a uniform procedure in regard to the international meteorological services already specified, and, as far as is practicable, to conform to the Technical Regulations and recommendations made by the World Meteorological Organization, to which the Contracting Governments may refer for study and advice any meteorological question which may arise in carrying out the present Convention".

1.2 Resolution A.705(17), as amended, on the promulgation of maritime safety information, sets out the organization, standards and methods which should be used for the promulgation and reception of maritime safety information, including navigational and meteorological warnings, meteorological forecasts and other urgent safety-related messages broadcast to ships, as documented in the 1974 SOLAS Convention. The WMO Executive Council, at its sixty-first session (June 2009), requested WMO to establish and develop, in collaboration with IMO, terms of reference for the development of an IMO/WMO Worldwide Met-Ocean Information and Warning Service Guidance document (WWMIWS), to complement the existing IMO/IHO World-wide Navigational Warning Services Guidance document (WWNWS), provided in resolution A.706(17), as amended. In this context, this document is intended to provide specific guidance for the promulgation of internationally co-ordinated meteorological information, forecast and warnings services, which does not apply to purely national services.

1.3 The regulatory framework for the provision of marine meteorological services within the new WMO GMDSS Marine Broadcast System was developed from Recommendation 3 (CMM-XI) in 1993, endorsed by the WMO Executive Council at its forty-fourth session. This new system reflects the evolution since the advent of the Global Maritime Distress and Safety System (GMDSS), as adopted by the Conference of Contracting Governments to the 1974 SOLAS Convention on the Global Maritime Distress and Safety System in November 1988, effective on 1 February 1992. The WMO GMDSS Marine Broadcast System is an integral part of the WWMIWS.

1.4 Future amendments to this guidance document will be considered formally and approved by both WMO and IMO in accordance with the procedure set out in section 7. Proposed amendments should be evaluated by the Joint WMO-IOC Technical Commission for Oceanography and Marine Meteorology (JCOMM) Expert Team on Maritime Safety Services, which includes an *ex-officio* representative of the IMO Secretariat, prior to any extensive WMO and IMO consideration.

## 2 DEFINITIONS

For the purposes of meteorological information, the following definitions apply:

- .1 *Coastal and offshore area* applies to areas for which Member States issue weather and sea bulletins, governed by the procedures in WMO-No.558 – Manual on Marine Meteorological Services;
- .2 *Global Maritime Distress and Safety System (GMDSS)* means the global communications service based upon automated systems, both satellite and terrestrial, to provide distress alerting and promulgation of maritime safety information for mariners;
- .3 *HF NBDP* means High Frequency narrow-band direct-printing, using radio telegraphy as defined in Recommendation ITU-R M.688;
- .4 *International NAVTEX service* means the co-ordinated broadcast and automatic reception on 518 kHz of maritime safety information by means of narrow-band direct-printing telegraphy using the English language<sup>1</sup>;
- .5 *International SafetyNET service* means the co-ordinated broadcasting and automated reception of maritime safety information via the Inmarsat Enhanced Group Call (EGC) system, using the English language, in accordance with the provisions of the 1974 SOLAS Convention;
- .6 *Issuing service* means a National Meteorological Service which has accepted responsibility for ensuring that meteorological warnings and forecasts for shipping are disseminated through the Inmarsat SafetyNET service to the designated area (METAREA) for which the Service has accepted responsibility under the broadcast requirements of the GMDSS<sup>2</sup>;
- .7 *Maritime safety information (MSI)*<sup>3</sup> means navigational and meteorological warnings, meteorological forecasts and other urgent safety-related messages broadcast to ships;
- .8 *METAREA* means a geographical sea area<sup>4</sup> established for the purpose of co-ordinating the broadcast of marine meteorological information. The term METAREA followed by a roman numeral may be used to identify a particular sea area. The delimitation of such areas is not related to and should not prejudice the delimitation of any boundaries between States;
- .9 *METAREA Co-ordinator* means the authority charged with co-ordinating Marine Meteorological Information broadcasts by one or more National Meteorological Services acting as Preparation or Issuing Services within the METAREA;

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<sup>1</sup> As set out in the IMO NAVTEX Manual.

<sup>2</sup> As defined in WMO-No 558.

<sup>3</sup> As defined in Regulation IV/2 of the 1974 SOLAS Convention, as amended.

<sup>4</sup> Which may include inland seas, lakes and waterways navigable by sea-going ships.

- .10 *National NAVTEX service* means the broadcast and automatic reception of maritime safety information by means of narrow-band direct-printing telegraphy using frequencies other than 518 kHz and languages as decided by the Administration concerned;
- .11 *National SafetyNET service* means the broadcasting and automated reception of maritime safety information via the Inmarsat EGC system, using languages as decided by the Administration concerned;
- .12 *NAVAREA* means a geographical sea area<sup>5</sup> established for the purpose of co-ordinating the broadcast of navigational warnings. The term NAVAREA followed by a roman numeral may be used to identify a particular sea area. The delimitation of such areas is not related to and should not prejudice the delimitation of any boundaries between States;
- .13 *NAVTEX* means the system for the broadcast and automatic reception of maritime safety information by means of narrow band direct-printing telegraphy;
- .14 *NAVTEX co-ordinator* means the authority charged with operating and managing one or more NAVTEX stations broadcasting maritime safety information as part of the International NAVTEX service;
- .15 *Preparation service* means a National Meteorological Service which has accepted responsibility for the preparation of forecasts and warnings for parts of or an entire designated area (METAREA) in the WMO system for the dissemination of meteorological forecasts and warning to shipping under the GMDSS and for their transfer to the relevant Issuing Service for broadcast;
- .16 *SafetyNET* means the international service for the broadcasting and automatic reception of maritime safety information via the Inmarsat EGC system. SafetyNET receiving capability is part of the mandatory equipment which is required to be carried by certain ships in accordance with the provisions of the 1974 SOLAS Convention;
- .17 *Sub-Area* means a sub-division of a METAREA in which a number of countries have established a co-ordinated system for the promulgation of marine meteorological information. The delimitation of such areas is not related to and should not prejudice the delimitation of any boundaries between States; and
- .18 In the operating procedures *co-ordination* means that the allocation of the time for data broadcast is centralized, the format and criteria of data transmissions are compliant as described in the Joint IMO/IHO/WMO Manual on Maritime Safety Information and that all services are managed as set out in resolution A.705(17) on Promulgation of Maritime Safety Information, as amended.

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<sup>5</sup> Which may include inland seas, lakes and waterways navigable by sea-going ships.

### **3 METEOROLOGICAL INFORMATION BROADCASTS**

#### **3.1 Methods**

3.1.1 Two principal methods are used for broadcasting marine meteorological information as part of MSI in accordance with the provisions of the 1974 SOLAS Convention, in the areas covered by these methods, as follows:

- .1 NAVTEX: broadcasts to coastal and offshore areas; and
- .2 *SafetyNET*: broadcasts which cover all the waters of the globe except for sea area A4, as defined by resolution A.801(19) on Provision of radio services for the GMDSS, Annex 3, paragraph 4, as amended.

3.1.2 Information has to be provided for unique and precisely defined sea areas, each being served only by the most appropriate of the above systems. Although there will be some duplication to allow a ship to change from one system to another, the majority of messages will only be broadcast on one system.

3.1.3 NAVTEX broadcasts have to be made in accordance with the standards and procedures set out in the NAVTEX Manual.

3.1.4 *SafetyNET* broadcasts have to be made in accordance with the standards and procedures set out in the International *SafetyNET* Manual.

3.1.5 HF NBDP may be used to promulgate marine meteorological information in areas outside Inmarsat coverage (SOLAS regulation IV/7.1.5).

3.1.6 In addition, Administrations may also provide marine meteorological information by other means.

#### **3.2 Scheduling**

##### **3.2.1 Automated methods (NAVTEX/*SafetyNET*)**

3.2.1.1 Meteorological warnings have to be broadcasted as soon as possible or as dictated by the nature and timing of the event. Normally, the initial broadcast should be made as follows:

- .1 for NAVTEX, at the next scheduled broadcast, unless circumstances indicate the use of procedures for VITAL or IMPORTANT warnings; and
- .2 for *SafetyNET*, within 30 min of receipt of original information, or at the next scheduled broadcast.

3.2.1.2 Meteorological warnings have to be repeated in scheduled broadcasts in accordance with the guidelines promulgated in the NAVTEX Manual and International *SafetyNET* Manual as appropriate.

3.2.1.3 At least two scheduled daily broadcast times are necessary to provide adequate promulgation of meteorological information.

### **3.2.2 Schedule changes**

3.2.2.1 Broadcast times for NAVTEX are defined by the B1 character of the station, allocated by the IMO NAVTEX Co-ordinating Panel.

3.2.2.2 Times of scheduled broadcasts under the international SafetyNET service are co-ordinated through the IMO SafetyNET Co-ordinating Panel.

3.2.2.3 Information on broadcast schedules and the content of bulletins are contained in WMO-No. 9 (Weather Reports), Volume D (Information for shipping).

### **3.3 Guidance**

Guidance for handling and formatting meteorological information is given in the Joint IMO/IHO/WMO Manual on Maritime Safety Information, as approved under MSC.1/Circ.1310, the NAVTEX Manual, the International SafetyNET Manual and the Manual on Marine Meteorological Services (WMO-No. 558).

### **3.4 Language**

3.4.1 All meteorological information have to be broadcasted in English only in the International NAVTEX and SafetyNET services.

3.4.2 In addition to the required broadcasts in English, meteorological information may be broadcast in a national language using national NAVTEX and SafetyNET services and/or other means.

## **4 METEOROLOGICAL INFORMATION**

### **4.1 General**

4.1.1 Marine meteorological services are provided to satisfy the requirements for information on marine environmental conditions and phenomena, established by national practices and international conventions in relation to marine operations.

4.1.2 Marine meteorological services are designed for the safety of marine operations and to promote, where possible, the efficiency and economy of marine activities.

4.1.3 There are three types of marine meteorological information: forecasts and warnings for the High Seas, forecasts and warnings for coastal and offshore areas and services for ports and harbour areas. The Marine Meteorological Information guidance and co-ordination are involved with only two of them:

- .1 services for the High Seas, which will comprise:
  - .1 warnings of gales and storms;
  - .2 weather and sea bulletins, which have to include, in the order given hereafter:

Part I – Storm warnings;

Part II – Synopsis of major features of the surface weather chart and, to the extent possible, significant characteristics of corresponding sea-surface conditions; and

Part III –Forecasts.

- .2 services for coastal and offshore areas, which will comprise warnings, synopses and forecasts.

4.1.4 Operational guidance for handling and formatting meteorological information is given in detail in the Annex IV of the WMO Technical Regulations (Manual on Marine Meteorological Services – WMO-No. 558). It is summarized in the following sections 4.2 and 4.3.

## **4.2 Services for the High Seas**

Services for the High Seas should consist of:

### **4.2.1 Warnings**

4.2.1.1 Warnings have to be given for gales (Beaufort force 8 or 9) and storms (Beaufort force 10 or over), and for tropical cyclones (hurricanes in the North Atlantic and eastern North Pacific, typhoons in the Western Pacific, cyclones in the Indian Ocean and cyclones of similar nature in other regions). Warnings should include:

- .1 type of warning;
- .2 date and time of reference in UTC;
- .3 location of disturbance in terms of latitude and longitude or with reference to well-known landmarks;
- .4 extent of affected area; and
- .5 wind speed or force and direction in the affected areas.

4.2.1.2 Warnings for other severe conditions such as poor visibility, severe sea states (swell, risk of abnormal waves), ice accretion, etc., have also to be issued, as necessary.

4.2.1.3 When no warnings for gales, storms or tropical cyclones are to be issued, that fact has to be positively stated in Part I of each weather and sea bulletin.

### **4.2.2 Synopses**

4.2.2.1 Synopses will be broadcast as part of routine meteorological information, within Part II of weather and sea bulletins, and should have the following content and order of items:

- .1 date and time of reference in UTC;
- .2 synopsis of major features of the surface weather chart; and



- .3 direction and speed of movement of significant pressure systems and tropical disturbances.

#### **4.2.3 Forecasts**

4.2.3.1 The forecasts given in Part III of weather and sea bulletins should have the following content and order of items:

- .1 the valid period of forecast;
- .2 name or designation of forecast area(s) within the main MSI area; and
- .3 a description of:
  - .1 wind speed or force and direction;
  - .2 sea state;
  - .3 visibility when forecast is less than five nautical miles; and
  - .4 ice accretion, where applicable.

4.2.3.2 The forecasts should include expected significant changes during the forecast period, significant meteors such as freezing precipitation, snowfall or rainfall, and an outlook for a period beyond 24 hours. In addition, phenomena such as breaking seas, cross seas and abnormal or rogue waves should also be included, if feasible.

#### **4.3 Services for the coastal and offshore areas**

Services for the coastal and offshore areas should consist of:

##### **4.3.1 Warnings**

4.3.1.1 When included, warnings have to be placed at the beginning of the bulletin.

4.3.1.2 Warnings have to be given for:

- .1 tropical cyclones (hurricanes in the North Atlantic and eastern North Pacific, typhoons in the Western Pacific, cyclones in the Indian Ocean and cyclones of similar nature in other regions);
- .2 gales (Beaufort force 8 or 9) and storms (Beaufort force 10 or over); and
- .3 ice accretion.

##### **4.3.2 Synopses and forecasts**

4.3.2.1 Synopses and forecasts should have the following content:

- .1 a synopsis of major features of the surface weather chart;
- .2 the valid period of forecast;
- .3 name or designation of forecast area(s); and

- .4 a description of:
  - .1 wind speed or force and direction;
  - .2 visibility when forecast is less than five nautical miles;
  - .3 ice accretion, where applicable; and
  - .4 sea and swell.

## **5 ISSUING AND PREPARATION SERVICES**

### **5.1 Issuing service**

An issuing service is a National Meteorological Service which has accepted responsibility for ensuring that meteorological forecasts and warnings for shipping are disseminated through the Inmarsat SafetyNET and NAVTEX services to the designated area for which the Service has accepted responsibility under the broadcast requirements of the GMDSS. The forecasts and warnings for broadcasts may have been prepared solely by the issuing service, or by another preparation service, or a combination of both, on the basis of negotiations between the services concerned, or otherwise, as appropriate. The issuing service is responsible for composing a complete broadcast bulletin on the basis of information input from the relevant preparation services and for broadcasting this in accordance with the guidelines contained within the International SafetyNET Manual and the International NAVTEX Manual. The issuing service is also responsible for monitoring the broadcasts of SafetyNET information to its designated area of responsibility.

#### **NOTES:**

- (1) For some METAREAS there may be only one preparation service, which will be the same National Meteorological Service as the issuing service (e.g., United Kingdom for area I, Argentina for area VI and Australia for area X).
- (2) An appropriate format for the attribution of the origins of the forecast and warning information contained in a broadcast bulletin may be developed on the basis of negotiations among the services concerned.
- (3) In situations where appropriate information, data or advice from other designated preparation services for a given area of responsibility is not available, it is the responsibility of the issuing service for that area to ensure that complete broadcast coverage for the area is maintained.

### **5.2 Preparation service**

A preparation service is a National Meteorological Service which has accepted responsibility for the preparation of forecasts and warnings for parts of, or an entire, designated area (METAREA) in the WMO system for the dissemination of meteorological forecasts and warnings to shipping under the GMDSS and for their transfer to the relevant issuing service for broadcast.

## **6 METAREA CO-ORDINATOR RESOURCES AND RESPONSIBILITIES**

### **6.1 METAREA Co-ordinator resources**

The METAREA co-ordinator should have:

- .1 the expertise and information sources of National Meteorological Services; and
- .2 effective communications, e.g., telephone, e-mail, facsimile, internet, telex, etc., with National Meteorological Services in the METAREA, with other METAREA co-ordinators, and with other data providers.

### **6.2 METAREA Co-ordinator responsibilities**

6.2.1 The METAREA co-ordinator has to:

- .1 act as the central point of contact on matters relating to meteorological information and warnings within the METAREA;
- .2 promote and oversee the use of established international standards and practices in the promulgation of meteorological information and warnings throughout the METAREA;
- .3 co-ordinate preliminary discussions between neighbouring Members, seeking to establish and operate NAVTEX services, prior to formal application; and
- .4 contribute to the development of international standards and practices through attendance and participation in the JCOMM Expert Team on Maritime Safety Services meetings, and also attend and participate in relevant IMO, IHO and WMO meetings as appropriate and required.

6.2.2 The METAREA co-ordinator has to also ensure that within its METAREA, National Meteorological Services which act as Issuing Services have the capability to:

- .1 select meteorological information and warnings for broadcast in accordance with the guidance given in paragraphs 4 and 5 above; and
- .2 monitor the SafetyNET transmission of their bulletins, broadcast by the Issuing Service.

6.2.3 The METAREA co-ordinator has to further ensure that within its METAREA, National Meteorological Services which act as Preparation Services have the capability to:

- .1 endeavour to be informed of all meteorological events that could significantly affect the safety of navigation within their area of responsibility;
- .2 assess all meteorological information immediately upon receipt in the light of expert knowledge for relevance to navigation within their area of responsibility;

- .3 forward marine meteorological information that may require wider promulgation directly to adjacent METAREA co-ordinators and/or others as appropriate, using the quickest possible means;
- .4 ensure that information concerning all meteorological warning subject areas listed in paragraph 4 that may not require a METAREA warning within their own area of responsibility is forwarded immediately to the appropriate National Meteorological Services and METAREA co-ordinators affected by the meteorological event; and
- .5 maintain records of source data relating to meteorological information and warning messages within their area of responsibility.

## **7 PROCEDURE FOR AMENDING THE WORLDWIDE MET-OCEAN INFORMATION AND WARNING SERVICE GUIDANCE DOCUMENT**

7.1 Proposed amendments to the Worldwide Met-Ocean Information and Warning Service should be submitted to the Maritime Safety Committee for evaluation.

7.2 Amendments to the service should normally come into force at intervals of approximately two years or at such longer periods as determined by the Maritime Safety Committee at the time of adoption. Amendments adopted by the Maritime Safety Committee will be notified to all concerned, will provide at least 12 months' notification and will come into force on 1 January of the following year.

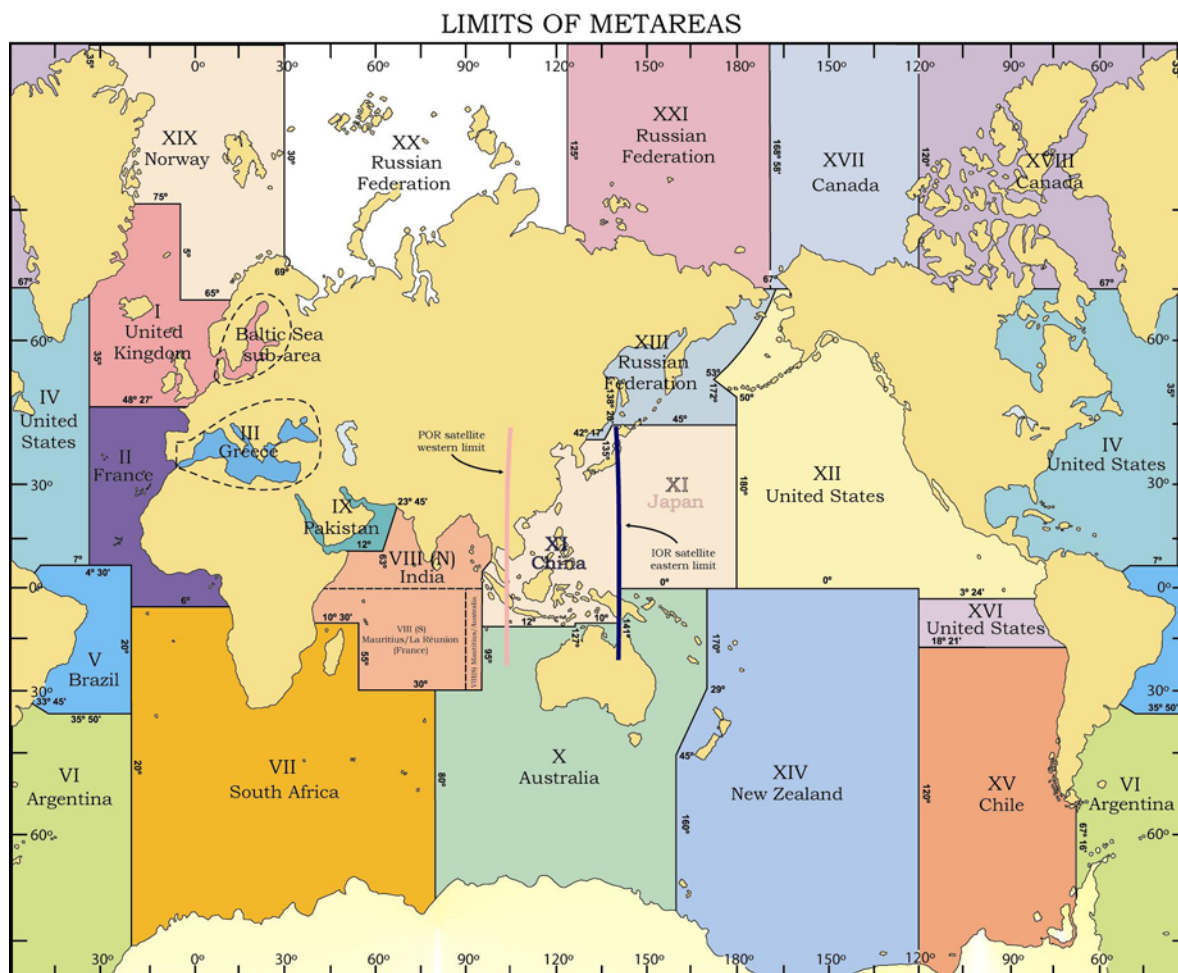
7.3 The agreement of the World Meteorological Organization and the active participation of other bodies should be sought according to the nature of the proposed amendments.

7.4 When the proposals for amendment have been examined in substance, the Maritime Safety Committee will entrust the Sub-Committee on Radiocommunications and Search and Rescue with the ensuing editorial tasks.

7.5 The METAREA schedule of broadcast times and frequencies, not being an integral part of the service and being subject to frequent changes, will not be subject to the amendment procedures.

## Appendix

### METAREAS for co-ordinating and promulgating meteorological warnings and forecasts



**The delimitation of such areas is not related to and should not prejudice  
the delimitation of any boundaries between States**

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**ANNEX 24**

**DRAFT ASSEMBLY RESOLUTION**

**PROCEDURES FOR PORT STATE CONTROL, 2011**

THE ASSEMBLY,

RECALLING Article 15(j) of the Convention on the International Maritime Organization concerning the functions of the Assembly in relation to regulations and guidelines concerning maritime safety and the prevention and control of marine pollution from ships,

RECALLING ALSO resolution A.787(19), by which it adopted Procedures for port State control and resolution A.882(21), by which it adopted amendments to the Procedures for port State control as adopted by resolution A.787(19),

RECALLING FURTHER that, at its twenty-first session, when adopting resolution A.882(21), it requested the Maritime Safety Committee and the Marine Environment Protection Committee to keep the Procedures, as amended, under review, on the basis of experience gained from their implementation,

RECOGNIZING that efforts by port States have greatly contributed to enhanced maritime safety and security, and prevention of marine pollution,

RECOGNIZING FURTHER the need to update the Procedures, as amended, to take account of the amendments to IMO instruments which have entered into force or have become effective since the adoption of resolutions A.787(19) and A.882(21),

HAVING CONSIDERED the recommendations made by the Maritime Safety Committee at its eighty-ninth session and the Marine Environment Protection Committee at its [sixty-second] session,

1. ADOPTS the Procedures for port State control, 2011, as set out in the annex to the present resolution;
2. INVITES Governments, when exercising port State control, to implement the aforementioned Procedures;
3. REQUESTS the Maritime Safety Committee and the Marine Environment Protection Committee to keep the Procedures under review and to amend them as necessary;
4. REVOKES resolutions A.787(19) and A.882(21).

## ANNEX

### PROCEDURES FOR PORT STATE CONTROL 2011

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## **CHAPTER 1 – GENERAL**

### **1.1 PURPOSE**

This document is intended to provide basic guidance on the conduct of port State control inspections and afford consistency in the conduct of these inspections, the recognition of deficiencies of a ship, its equipment, or its crew, and the application of control procedures.

### **1.2 APPLICATION**

1.2.1 These Procedures apply to ships falling under the provisions of:

- .1 the International Convention for the Safety of Life at Sea, 1974, as amended (SOLAS);
- .2 the Protocol of 1988 relating to the International Convention for the Safety of Life at Sea, 1974 (SOLAS Protocol 1988);
- .3 the International Convention on Load Lines, 1966 (Load Lines);
- .4 the Protocol of 1988 relating to the International Convention on Load Lines, 1966 (Load Lines Protocol, 1988);
- .5 the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocols of 1978 and 1997 relating thereto, as amended (MARPOL);
- .6 the International Convention on Standards of Training, Certification and Watchkeeping for Seafarers, 1978, as amended (STCW);
- .7 the International Convention on Tonnage Measurement of Ships, 1969 (Tonnage); and
- .8 the International Convention on the Control of Harmful Anti-Fouling Systems on Ships (AFS),

hereafter referred to as the applicable conventions.

1.2.2 Ships of non-Parties or below convention size should be given no more favourable treatment (see sections 1.5 and 1.6).

1.2.3 In exercising port State control, Parties should only apply those provisions of the conventions which are in force and which they have accepted.

1.2.4 If a port Convention State exercises port State control based on International Labour Organization (ILO) Convention No. 147, "Merchant Shipping (Minimum Standards) Convention, 1976", guidance on the conduct of such control inspections is given in the ILO publication "Inspection of Labour Conditions on board Ship: Guidelines for Procedure".

### **1.3 INTRODUCTION**

1.3.1 Under the provisions of the applicable conventions set out in section 1.2 above, the Administration (i.e. the Government of the flag State) is responsible for promulgating laws and regulations and for taking all other steps which may be necessary to give the applicable

conventions full and complete effect so as to ensure that, from the point of view of safety of life and pollution prevention, a ship is fit for the service for which it is intended and seafarers are qualified and fit for their duties.

1.3.2 In some cases it may be difficult for the Administration to exercise full and continuous control over some ships entitled to fly the flag of its State, for instance those ships which do not regularly call at a port of the flag State. The problem can be, and has been, partly overcome by appointing inspectors at foreign ports and/or authorizing recognized organizations to act on behalf of the flag State Administration.

1.3.3 The following control procedures should be regarded as complementary to national measures taken by Administrations of flag States in their countries and abroad and are intended to provide assistance to flag State Administrations in securing compliance with convention provisions in safeguarding the safety of crew, passengers and ships, and ensuring the prevention of pollution.

## **1.4 PROVISION FOR PORT STATE CONTROL**

Regulation 19 of chapter I, regulation 6.2 of chapter IX, regulation 4 of chapter XI-1 and regulation 9 of chapter XI-2 of SOLAS, as modified by the SOLAS Protocol 1988; article 21 of Load Lines, as modified by the Load Lines Protocol 1988; articles 5 and 6, regulation 11 of Annex I, regulation 16.9 of Annex II, regulation 8 of Annex III, regulation 13 of Annex IV, regulation 8 of Annex V and regulation 10 of Annex VI of MARPOL; article X of STCW; article 12 of Tonnage and article 11 of AFS provide for control procedures to be followed by a Party to a relevant convention with regard to foreign ships visiting their ports. The authorities of port States should make effective use of these provisions for the purposes of identifying deficiencies, if any, in such ships which may render them substandard (see section 3.1), and ensuring that remedial measures are taken.

## **1.5 SHIPS OF NON-PARTIES**

1.5.1 Article I(3) of the SOLAS Protocol 1988, article 5(4) of MARPOL and article X(5) of STCW provide that no more favourable treatment is to be given to the ships of countries which are not Party to the relevant Convention. All Parties should, as a matter of principle, apply these Procedures to ships of non-Parties in order to ensure that equivalent surveys and inspections are conducted and an equivalent level of safety and protection of the marine environment is ensured.

1.5.2 As ships of non-Parties are not provided with SOLAS, Load Line or MARPOL certificates, as applicable, or the crew members may not hold STCW certificates, the Port State Control Officer (PSCO), taking into account the principles established in these Procedures, should be satisfied that the ship and crew do not present a danger to those on board or an unreasonable threat of harm to the marine environment. If the ship or crew has some form of certification other than that required by a convention, the PSCO may take the form and content of this documentation into account in the evaluation of that ship. The conditions of and on such a ship and its equipment and the certification of the crew and the flag State's minimum manning standard should be compatible with the aims of the provisions of the conventions; otherwise, the ship should be subject to such restrictions as are necessary to obtain a comparable level of safety and protection of the marine environment.

## 1.6 SHIPS BELOW CONVENTION SIZE

1.6.1 In the exercise of their functions, the PSCOs should be guided by any certificates and other documents issued by or on behalf of the flag State Administration. In such cases, the PSCOs should limit the scope of inspection to the verification of compliance with those certificates and documents.

1.6.2 To the extent a relevant instrument is not applicable to a ship below convention size, the PSCO's task should be to assess whether the ship is of an acceptable standard in regard to safety and the environment. In making that assessment, the PSCO should take due account of such factors as the length and nature of the intended voyage or service, the size and type of the ship, the equipment provided and the nature of the cargo.

## 1.7 DEFINITIONS

1.7.1 **Bulk carrier:** whilst noting the definitions in SOLAS regulations IX/1.6 and XII/1.1 and resolution MSC.277(85), for the purposes of port State control, PSCOs should be guided by the ship's type indicated in the ship's certificates in determining whether a ship is a bulk carrier and recognize that a ship which is not designated as a bulk carrier as the ship type on the ship certificate may carry certain bulk cargo as provided for in the above instruments.

1.7.2 **Clear grounds:** Evidence that the ship, its equipment, or its crew does not correspond substantially with the requirements of the relevant conventions or that the master or crew members are not familiar with essential shipboard procedures relating to the safety of ships or the prevention of pollution. Examples of clear grounds are included in section 2.4.

1.7.3 **Deficiency:** A condition found not to be in compliance with the requirements of the relevant convention.

1.7.4 **Detention:** Intervention action taken by the port State when the condition of the ship or its crew does not correspond substantially with the applicable conventions to ensure that the ship will not sail until it can proceed to sea without presenting a danger to the ship or persons on board, or without presenting an unreasonable threat of harm to the marine environment, whether or not such action will affect the normal schedule of the departure of the ship.

1.7.5 **Inspection:** A visit on board a ship to check both the validity of the relevant certificates and other documents, and the overall condition of the ship, its equipment and its crew.

1.7.6 **More detailed inspection:** An inspection conducted when there are clear grounds for believing that the condition of the ship, its equipment or its crew does not correspond substantially to the particulars of the certificates.

1.7.7 **Port State Control Officer (PSCO):** A person duly authorized by the competent authority of a Party to a relevant convention to carry out port State control inspections, and responsible exclusively to that Party.

1.7.8 **Recognized organization:** An organization which meets the relevant conditions set forth by resolution A.739(18), as amended by resolution MSC.208(81), and has been authorized by the flag State Administration to provide the necessary statutory services and certification to ships entitled to fly its flag.

**1.7.9 Stoppage of an operation:** Formal prohibition against a ship to continue an operation due to an identified deficiency(ies) which, singly or together, render the continuation of such operation hazardous.

**1.7.10 Substandard ship:** A ship whose hull, machinery, equipment or operational safety is substantially below the standards required by the relevant convention or whose crew is not in conformance with the safe manning document.

**1.7.11 Valid certificates:** A certificate that has been issued directly by a Party to a relevant convention or on its behalf by a recognized organization and contains accurate and effective dates meets the provisions of the relevant convention and to which the particulars of the ship, its crew and its equipment correspond.

## **1.8 PROFESSIONAL PROFILE OF PSCOs**

**1.8.1** Port State control should be carried out only by qualified PSCOs who fulfil the qualifications and training specified in section 1.9.

**1.8.2** When the required professional expertise cannot be provided by the PSCO, the PSCO may be assisted by any person with the required expertise, as acceptable to the port State.

**1.8.3** The PSCOs and the persons assisting them should have no commercial interest, either in the port of inspection, or in the ships inspected, nor should PSCOs be employed by, or undertake work on behalf, of recognized organizations.

**1.8.4** A PSCO should carry a personal document in the form of an identity card issued by the port State and indicating that the PSCO is authorized to carry out the control.

## **1.9 QUALIFICATION AND TRAINING REQUIREMENTS OF PSCOs**

**1.9.1** The PSCO should be an experienced officer qualified as flag State surveyor.

**1.9.2** The PSCO should be able to communicate in English with the key crew.

**1.9.3** Training should be provided for PSCOs to give the necessary knowledge of the provisions of the applicable conventions which are relevant to the conduct of port State control, taking into account the latest IMO Model Courses for port State control.

**1.9.4** In specifying the qualifications and training requirements for PSCOs, the Administration should take into account, as appropriate, which of the internationally agreed instruments are relevant for the control by the port State and the variety of types of ships which may enter its ports.

**1.9.5** PSCOs carrying out inspections of operational requirements should be qualified as a master or chief engineer and have appropriate seagoing experience, or have qualifications from an institution recognized by the Administration in a maritime related field and have specialized training to ensure adequate competence and skill, or be a qualified officer of the Administration with an equivalent level of experience and training, for performing inspections of the relevant operational requirements.

1.9.6 Periodical seminars for PSCOs should be held in order to update their knowledge with respect to instruments related to port State control.

## **CHAPTER 2 – PORT STATE INSPECTIONS**

### **2.1 GENERAL**

2.1.1 In accordance with the provisions of the applicable conventions, Parties may conduct inspections by PSCOs of foreign ships in their ports.

2.1.2 Such inspections may be undertaken on the basis of:

- .1 the initiative of the Party;
- .2 the request of, or on the basis of, information regarding a ship provided by another Party; or
- .3 information regarding a ship provided by a member of the crew, a professional body, an association, a trade union or any other individual with an interest in the safety of the ship, its crew and passengers, or the protection of the marine environment.

2.1.3 Whereas Parties may entrust surveys and inspections of ships entitled to fly their own flag either to inspectors nominated for this purpose or to recognized organizations, they should be aware that, under the applicable conventions, foreign ships are subject to port State control, including boarding, inspection, remedial action and possible detention, only by officers duly authorized by the port State. This authorization of PSCOs may be a general grant of authority or may be specific on a case-by-case basis.

2.1.4 All possible efforts should be made to avoid a ship being unduly detained or delayed. If a ship is unduly detained or delayed, it should be entitled to compensation for any loss or damage suffered.

### **2.2 INITIAL INSPECTIONS**

2.2.1 In the pursuance of control procedures under the applicable conventions, which, for instance, may arise from information given to a port State regarding a ship, a PSCO may proceed to the ship and, before boarding, gain, from its appearance in the water, an impression of its standard of maintenance from such items as the condition of its paintwork, corrosion or pitting or unrepaired damage.

2.2.2 At the earliest possible opportunity, the PSCO should ascertain the type of ship, year of build and size of the ship for the purpose of determining which provisions of the conventions are applicable.

2.2.3 On boarding and introduction to the master or the responsible ship's officer, the PSCO should examine the ship's relevant certificates and documents, as listed in appendix 12. When examining 1969 International Tonnage Certificates, the PSCO should be guided by appendix 10.

2.2.4 If the certificates are valid and the PSCO's general impression and visual observations on board confirm a good standard of maintenance, the PSCO should generally confine the inspection to reported or observed deficiencies, if any.

2.2.5 In conducting an initial inspection, the PSCO should check both the validity of the relevant certificates and other documents and the overall condition of the ship, including its equipment, navigational bridge, decks including forecastle, cargo holds/areas, engine-room and pilot transfer arrangement.

2.2.6 In pursuance of control procedures under chapter IX of SOLAS in relation to the International Management Code for the Safe Operation of Ships and for Pollution Prevention (ISM Code), the PSCO should utilize the guidelines in appendix 8.

2.2.7 If, however, the PSCO from general impression or observations on board has clear grounds for believing that the ship, its equipment or its crew do not substantially meet the requirements, the PSCO should proceed to a more detailed inspection, taking into consideration sections 2.4 and 2.5. In forming such an impression, the PSCO should utilize the guidelines in relevant appendices.

### **2.3 GENERAL PROCEDURAL GUIDELINES FOR PSCOs**

2.3.1 The PSCO should observe the Code of Good Practice for port State control officers (MSC-MEPC.4/Circ.2), as shown in appendix 1, use professional judgement in carrying out all duties and consider consulting others as deemed appropriate.

2.3.2 When boarding a ship, the PSCO should present to the master or to the representative of the owner, if requested to do so, the PSCO identity card. This card should be accepted as documented evidence that the PSCO in question is duly authorized by the Administration to carry out port State control inspections.

2.3.3 If the PSCO has clear grounds for carrying out a more detailed inspection, the master should be immediately informed of these grounds and advised that, if so desired, the master may contact the Administration or, as appropriate, the recognized organization responsible for issuing the certificate and invite their presence on board.

2.3.4 In the case that an inspection is initiated based on a report or complaint, especially if it is from a crew member, the source of the information should not be disclosed.

2.3.5 When exercising control, all possible efforts should be made to avoid a ship being unduly detained or delayed. It should be borne in mind that the main purpose of port State control is to prevent a ship proceeding to sea if it is unsafe or presents an unreasonable threat of harm to the marine environment. The PSCO should exercise professional judgement to determine whether to detain a ship until the deficiencies are corrected or to allow it to sail with certain deficiencies, having regard to the particular circumstances of the intended voyage.

2.3.6 It should be recognized that all equipment is subject to failure and spares or replacement parts may not be readily available. In such cases, undue delay should not be caused if, in the opinion of the PSCO, safe alternative arrangements have been made.

2.3.7 Where the grounds for detention are the result of accidental damage suffered on the ship's voyage to a port, no detention order should be issued, provided that:

- .1 due account has been given to the convention requirements regarding notification to the flag State Administration, the nominated surveyor or the recognized organization responsible for issuing the relevant certificate;

- .2 prior to entering a port, the master or company has submitted to the port State Authority details of the circumstances of the accident and the damage suffered and information about the required notification of the flag State Administration;
- .3 appropriate remedial action, to the satisfaction of the port State Authority, is being taken by the ship; and
- .4 the port State Authority has ensured, having been notified of the completion of the remedial action, that deficiencies which were clearly hazardous to safety, health or environment have been rectified.

2.3.8 Since detention of a ship is a serious matter involving many issues, it may be in the best interest of the PSCO to act together with other interested parties (see paragraph 4.1.3). For example, the officer may request the owner's representatives to provide proposals for correcting the situation. The PSCO should also consider co-operating with the flag State Administration's representatives or the recognized organization responsible for issuing the relevant certificates, and consulting them regarding their acceptance of the owner's proposals and their possible additional requirements. Without limiting the PSCO's discretion in any way, the involvement of other parties could result in a safer ship, avoid subsequent arguments relating to the circumstances of the detention and prove advantageous in the case of litigation involving "undue delay".

2.3.9 Where deficiencies cannot be remedied at the port of inspection, the PSCO may allow the ship to proceed to another port, subject to any appropriate conditions determined. In such circumstances, the PSCO should ensure that the competent authority of the next port of call and the flag State are notified.

2.3.10 Detention reports to the flag State should be in sufficient detail for an assessment to be made of the severity of the deficiencies giving rise to the detention.

2.3.11 The company or its representative have a right of appeal against a detention taken by the Authority of a port State. The appeal should not cause the detention to be suspended. The PSCO should properly inform the master of the right of appeal.

2.3.12 To ensure consistent enforcement of port State control requirements, PSCOs should carry an extract of section 2.3 (General procedural guidelines for PSCOs) for ready reference when carrying out any port State control inspections.

2.3.13 PSCOs should also be familiar with the detailed guidelines given in the appendices to these Procedures.

## **2.4 CLEAR GROUNDS**

2.4.1 When a PSCO inspects a foreign ship which is required to hold a convention certificate, and which is in a port or an offshore terminal under the jurisdiction of the port State, any such inspection should be limited to verifying that there are on board valid certificates and other relevant documentation and the PSCO forming an impression of the overall condition of the ship, its equipment and its crew, unless there are "clear grounds" for believing that the condition of the ship or its equipment does not correspond substantially with the particulars of the certificates.

2.4.2 "Clear grounds" to conduct a more detailed inspection include:

- .1 the absence of principal equipment or arrangements required by the applicable conventions;
- .2 evidence from a review of the ship's certificates that a certificate or certificates are clearly invalid;
- .3 evidence that documentation required by the applicable conventions and listed in appendix 12 is not on board, incomplete, not maintained or falsely maintained;
- .4 evidence from the PSCO's general impressions and observations that serious hull or structural deterioration or deficiencies exist that may place at risk the structural, watertight or weathertight integrity of the ship;
- .5 evidence from the PSCO's general impressions or observations that serious deficiencies exist in the safety, pollution prevention or navigational equipment;
- .6 information or evidence that the master or crew is not familiar with essential shipboard operations relating to the safety of ships or the prevention of pollution, or that such operations have not been carried out;
- .7 indications that key crew members may not be able to communicate with each other or with other persons on board;
- .8 the emission of false distress alerts not followed by proper cancellation procedures; and
- .9 receipt of a report or complaint containing information that a ship appears to be substandard.

## **2.5 MORE DETAILED INSPECTIONS**

2.5.1 If the ship does not carry valid certificates, or if the PSCO from general impressions or observations on board has clear grounds for believing that the condition of the ship or its equipment does not correspond substantially with the particulars of the certificates or that the master or crew is not familiar with essential shipboard procedures, a more detailed inspection as described in this chapter should be carried out, utilizing relevant appendices.



2.5.2 It is not envisaged that all of the equipment and procedures outlined in this chapter would be checked during a single port State control inspection, unless the condition of the ship or the familiarity of the master or crew with essential shipboard procedures necessitates such a detailed inspection. In addition, these Procedures are not intended to impose the seafarer certification programme of the port State on a ship entitled to fly the flag of another Party to STCW or to impose control procedures on foreign ships in excess of those imposed on ships of the port State.

## **CHAPTER 3 – CONTRAVENTION AND DETENTION**

### **3.1 IDENTIFICATION OF A SUBSTANDARD SHIP**

3.1.1 In general, a ship is regarded as substandard if the hull, machinery, equipment or operational safety, is substantially below the standards required by the applicable conventions or if the crew is not in conformance with the safe manning document, owing to, *inter alia*:

- .1 the absence of principal equipment or arrangement required by the conventions;
- .2 non-compliance of equipment or arrangement with relevant specifications of the conventions;
- .3 substantial deterioration of the ship or its equipment, for example, because of poor maintenance;
- .4 insufficiency of operational proficiency, or unfamiliarity of essential operational procedures by the crew; and
- .5 insufficiency of manning or insufficiency of certification of seafarers.

3.1.2 If these evident factors as a whole or individually make the ship unseaworthy and put at risk the ship or the life of persons on board or present an unreasonable threat of harm to the marine environment if it were allowed to proceed to sea, it should be regarded as a substandard ship. The PSCO should also take into account the guidelines in appendix 2.

### **3.2 SUBMISSION OF INFORMATION CONCERNING DEFICIENCIES**

3.2.1 Information that a ship appears to be substandard could be submitted to the appropriate authorities of the port State (see section 3.3 below) by a member of the crew, a professional body, an association, a trade union or any other individual with an interest in the safety of the ship, its crew and passengers, or the protection of the marine environment.

3.2.2 This information should be submitted in writing to permit proper documentation of the case and of the alleged deficiencies. If the information is passed verbally, the filing of a written report should be required, identifying, for the purposes of the port State's records, the individual or body providing the information. The attending PSCO may collect this information and submit it as part of the PSCO's report if the originator is unable to do so.

3.2.3 Information which may cause an investigation should be submitted as early as possible after the arrival of the ship, giving adequate time to the authorities to act as necessary.

3.2.4 Each Party to the relevant convention should determine which authorities should receive information on substandard ships and initiate action. Measures should be taken to ensure that information submitted to the wrong department should be promptly passed on by such department to the appropriate authority for action.

### **3.3 PORT STATE ACTION IN RESPONSE TO ALLEGED SUBSTANDARD SHIPS**

3.3.1 On receipt of information about an alleged substandard ship or alleged pollution risk, the authorities should immediately investigate the matter and take the action required by the circumstances in accordance with the preceding sections.

3.3.2 Authorities which receive information about a substandard ship that could give rise to detention should forthwith notify any maritime, consular and/or diplomatic representatives of the flag State in the area of the ship and request them to initiate or cooperate with investigations. Likewise, the recognized organization which has issued the relevant certificates on behalf of the flag State should be notified. These provisions will not, however, relieve the authorities of the port State, being a Party to a relevant convention, from the responsibility for taking appropriate action in accordance with its powers under the relevant conventions.

3.3.3 If the port State receiving information is unable to take action because there is insufficient time or no PSCOs can be made available before the ship sails, the information should be passed to the authorities of the country of the next appropriate port of call, to the flag State and also to the recognized organization in that port, where appropriate.

### **3.4 RESPONSIBILITIES OF PORT STATE TO TAKE REMEDIAL ACTION**

If a PSCO determines that a ship can be regarded as substandard as specified in section 3.1 and appendix 2, the port State should immediately ensure that corrective action is taken to safeguard the safety of the ship and passengers and/or crew and eliminate any threat of harm to the marine environment before permitting the ship to sail.

### **3.5 GUIDANCE FOR THE DETENTION OF SHIPS**

Notwithstanding the fact that it is impracticable to define a ship as substandard solely by reference to a list of qualifying defects, guidance for the detention of ships is given in appendix 2.

### **3.6 SUSPENSION OF INSPECTION**

3.6.1 In exceptional circumstances where, as a result of a more detailed inspection, the overall condition of a ship and its equipment, also taking into account the crew conditions, are found to be obviously substandard, the PSCO may suspend an inspection.

3.6.2 Prior to suspending an inspection, the PSCO should have recorded detainable deficiencies in the areas set out in appendix 2, as appropriate.

3.6.3 The suspension of the inspection may continue until the responsible parties have taken the steps necessary to ensure that the ship complies with the requirements of the relevant instruments.

3.6.4 In cases where the ship is detained and an inspection is suspended, the port State Authority should notify the responsible parties without delay. The notification should include information about the detention, and state that the inspection is suspended until that authority has been informed that the ship complies with all relevant requirements.

### **3.7 PROCEDURES FOR RECTIFICATION OF DEFICIENCIES AND RELEASE**

3.7.1 The PSCO should endeavour to secure the rectification of all deficiencies detected.

3.7.2 In the case of deficiencies which are clearly hazardous to safety or the environment, the PSCO should, except as provided in paragraph 3.7.3, ensure that the hazard is removed before the ship is allowed to proceed to sea. For this purpose, appropriate action should be taken, which may include detention or a formal prohibition of a ship to continue an operation due to established deficiencies which, individually or together, would render the continued operation hazardous.

3.7.3 Where deficiencies which caused a detention, as referred to in paragraph 3.7.2, cannot be remedied in the port of inspection, the port State Authority may allow the ship concerned to proceed to the nearest appropriate repair yard available, as chosen by the master and agreed to by that authority, provided that the conditions agreed between the port State Authority and the flag State are complied with. Such conditions will ensure that the ship should not sail until it can proceed without risk to the safety of the passengers or crew, or risk to other ships, or without presenting an unreasonable threat of harm to the marine environment. Such conditions may include confirmation from the flag State that remedial action has been taken on the ship in question. In such circumstances the port State Authority should notify the authority of the ship's next port of call, the parties mentioned in paragraph 4.1.4 and any other authority as appropriate. Notification to authorities should be made in the form shown in appendix 14. The authority receiving such notification should inform the notifying authority of action taken and may use the form shown in appendix 15.

3.7.4 On the condition that all possible efforts have been made to rectify all other deficiencies, except those referred to in paragraphs 3.7.2 and 3.7.3, the ship may be allowed to proceed to a port where any such deficiencies can be rectified.

3.7.5 If a ship referred to in paragraph 3.7.3 proceeds to sea without complying with the conditions agreed to by the Authority of the port of inspection that port State Authority should immediately alert the next port, if known, the flag State and all other authorities it considers appropriate.

3.7.6 If a ship referred to in paragraph 3.7.3 does not call at the nominated repair port, the port State Authority of the repair port should immediately alert the flag State and detaining port State, which may take appropriate action, and notify any other authority it considers appropriate.

## **CHAPTER 4 – REPORTING REQUIREMENTS**

### **4.1 PORT STATE REPORTING**

4.1.1 Port State authorities should ensure that, at the conclusion of an inspection, the master of the ship is provided with a document showing the results of the inspection, details of any action taken by the PSCO, and a list of any corrective action to be initiated by the master and/or company. Such reports should be made in accordance with the format in appendix 13.

4.1.2 Where, in the exercise of port State control, a Party denies a foreign ship entry to the ports or offshore terminals under its jurisdiction, whether or not as a result of information about a substandard ship, it should forthwith provide the master and flag State with reasons for the denial of entry.

4.1.3 In the case of a detention, at least an initial notification should be made to the flag State Administration as soon as practicable (see paragraph 2.3.8). If such notification is made verbally, it should be subsequently confirmed in writing. As a minimum, the notification should include details of the ship's name, the IMO number, copies of Forms A and B as set out in appendix 13, time of detention and copies of any detention order. Likewise, the recognized organizations which have issued the relevant certificates on behalf of the flag State should be notified, where appropriate. The parties above should also be notified in writing of the release of detention. As a minimum, this information should include the ship's name, the IMO number, the date and time of release and a copy of Form B as set out in appendix 13.

4.1.4 If the ship has been allowed to sail with known deficiencies, the authorities of the port State should communicate all the facts to the authorities of the country of the next appropriate port of call, to the flag State, and to the recognized organization, where appropriate.

4.1.5 Parties to a relevant convention, when they have exercised control giving rise to detention, should submit to the Organization reports in accordance with SOLAS regulation I/19, article 11 of MARPOL, article 21 of Load Lines, or article X(3) of STCW. Such deficiency reports should be made in accordance with the form given in appendix 13 or 16, as appropriate, or may be submitted electronically by the port State or a regional PSC regime.

4.1.6 Copies of such deficiency reports should, in addition to being forwarded to the Organization, be sent by the port State without delay to the authorities of the flag State and, where appropriate, to the recognized organization which had issued the relevant certificate. Deficiencies found which are not related to the applicable conventions, or which involve ships of non-convention countries or below convention size, should be submitted to flag States and/or to appropriate organizations but not to IMO.

4.1.7 Relevant telephone numbers and addresses of flag States Headquarters to which reports should be sent as outlined above, as well as addresses of flag State offices which provide inspection services should be provided to the Organization\*.

## **4.2 FLAG STATE REPORTING**

4.2.1 On receiving a report on detention, the flag State and, where appropriate, the recognized organization through the flag State Administration, should, as soon as possible, inform the Organization of remedial action taken in respect of the detention which may be submitted electronically by the flag State to GISIS or in a format shown in appendix 17.

4.2.2 Relevant telephone numbers and addresses of port State control offices, Headquarters and those who provide inspection services should be provided to the Organization.

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Such addresses are available in MSC-MEPC.6/Circ.9 (National contact points for safety and pollution prevention and response), as amended, the IMO Internet Home Page and the GISIS module on contact points (<http://gisis.imo.org/Public>).

### **4.3 REPORTING OF ALLEGATIONS UNDER MARPOL**

4.3.1 A report on alleged deficiencies or on alleged contravention of the discharge provisions relating to the provisions of MARPOL should be forwarded to the flag State as soon as possible, preferably no later than sixty days after the observation of the deficiencies or contravention. Such reports may be made in accordance with the format in appendix 13 or 16, as appropriate. If a contravention of the discharge provisions is suspected, then the information should be supplemented by evidence of violations which, as a minimum, should include the information specified in parts 2 and 3 of appendices 3 and 4 of these Procedures.

4.3.2 On receiving a report on alleged deficiencies or alleged contravention of the discharge provisions, the flag State and, where appropriate, the recognized organization through the flag State Administration, should, as soon as possible, inform the Party submitting the report of immediate action taken in respect of the alleged deficiencies or contravention. That Party and the Organization should, upon completion of such action, be informed of the outcome and details, where appropriate, be included in the mandatory annual report to the Organization.

## **CHAPTER 5 – REVIEW PROCEDURES**

### **5.1 REPORT OF COMMENTS**

5.1.1 In the interest of making information regarding deficiencies and remedial measures generally available, a summary of such reports should be made by the Organization in a timely manner in order that the information can be disseminated in accordance with the Organization's procedures to all Parties to the applicable conventions. In the summary of deficiency reports, an indication should be given of flag State action or whether a comment by the flag State concerned is outstanding.

5.1.2 The appropriate Committee should periodically evaluate the summary of the deficiency reports in order to identify measures that may be necessary to ensure more consistent and effective application of IMO instruments, paying close attention to the difficulties reported by Parties to the relevant conventions, particularly in respect to developing countries in their capacity as port States.

5.1.3 Recommendations to address such difficulties, when recognized by the appropriate Committee, should, where appropriate, be incorporated into the applicable IMO instrument and any modifications relating to the procedures and obligations should be made in the port State documentation.

## APPENDIX 1

### **CODE OF GOOD PRACTICE FOR PORT STATE CONTROL OFFICERS CONDUCTING INSPECTIONS WITHIN THE FRAMEWORK OF THE REGIONAL MEMORANDA OF UNDERSTANDING AND AGREEMENT ON PORT STATE CONTROL (MSC-MEPC.4/Circ.2)**

#### **Introduction**

1 This document provides guidelines regarding the standards of integrity, professionalism and transparency that regional port State control (PSC) regimes expect of all port State control officers (PSCOs) who are involved in or associated with port State control inspections.

#### **Objective**

2 The object of this Code is to assist PSCOs in conducting their inspections to the highest professional level. PSCOs are central to achieving the aims of the regional PSC regime. They are the daily contact with the shipping world. They are expected to act within the law, within the rules of their Government and in a fair, open, impartial and consistent manner.

#### **Fundamental principles of the Code**

3 The Code of good practice encompasses three fundamental principles against which all actions of PSCOs are judged: integrity, professionalism and transparency. These are defined as follows:

- .1 integrity is the state of moral soundness, honesty and freedom from corrupting influences or motives;
- .2 professionalism is applying accepted professional standards of conduct and technical knowledge. For PSCOs standards of behaviour are established by the maritime Authority and the general consent of the port State members; and
- .3 transparency implies openness and accountability.

4 The list of the actions and behaviour expected of PSCOs in applying these principles are set out in the annex to this appendix.

5 Adhering to professional standards provides greater credibility to PSCOs and places more significance on their findings.

6 Nothing in the Code shall absolve the PSCOs from complying with the specific requirements of the PSC instruments and applicable national laws.

## ANNEX

### CODE OF GOOD PRACTICE FOR PORT STATE CONTROL OFFICERS

#### Actions and behaviour of PSCOs

The PSCOs should:

- 1 use their professional judgement in carrying out their duties;

#### Respect

- 2 remember that a ship is a home as well as a workplace for the ship's personnel and not unduly disturb their rest or privacy;
- 3 comply with any ship housekeeping rules such as removing dirty shoes or work clothes;
- 4 not be prejudiced by the race, gender, religion or nationality of the crew when making decisions and treat all personnel on board with respect;
- 5 respect the authority of the master or his deputy;
- 6 be polite but professional and firm as required;
- 7 never become threatening, abrasive or dictatorial or use language that may cause offence;
- 8 expect to be treated with courtesy and respect;

#### Conduct of inspections

- 9 comply with all health and safety requirements of the ship and their administration, e.g., wearing of personal protective clothing, and not take any action or cause any action to be taken which could compromise the safety of the PSCO or the ship's crew;
- 10 comply with all security requirements of the ship and wait to be escorted around the ship by a responsible person;
- 11 present their identity cards to the master or the representative of the owner at the start of the inspection;
- 12 explain the reason for the inspections. However, where the inspection is triggered by a report or complaint they must not reveal the identity of the person making the complaint;
- 13 apply the procedures of PSC and the convention requirements in a consistent and professional way and interpret them pragmatically when necessary;
- 14 not try to mislead the crew, for example by asking them to do things that are contrary to the Conventions;

- 15 request the crew to demonstrate the functioning of equipment and operational activities, such as drills and not make tests themselves;
- 16 seek advice when they are unsure of a requirement or of their findings rather than making an uninformed decision, for example by consulting colleagues, publications, the flag Administration, the recognized organization;
- 17 where it is safe to do so accommodate the operational needs of the port and the ship;
- 18 explain clearly to the master the findings of the inspection and the corrective action required and ensure that the report of inspection is clearly understood;
- 19 issue to the master a legible and comprehensible report of inspection before leaving the ship;

### **Disagreements**

- 20 deal with any disagreement over the conduct or findings of the inspection calmly and patiently;
- 21 advise the master of the complaints procedure in place if the disagreement cannot be resolved within a reasonable time;
- 22 advise the master of the right of appeal and relevant procedures in the case of detention;

### **Integrity**

- 23 be independent and not have any commercial interest in their ports and the ships they inspect or companies providing services in their ports. For example, the PSCOs should not be employed from time to time by companies which operate ships in their ports or the PSCOs should not have an interest in the repair companies in their ports;
- 24 be free to make decisions based on the findings of their inspections and not on any commercial considerations of the port;
- 25 always follow the rules of their administrations regarding the acceptance of gifts and favours, e.g., meals on board;
- 26 firmly refuse any attempts of bribery and report any blatant cases to the maritime Authority;
- 27 not misuse their authority for benefit, financial or otherwise; and

### **Updating knowledge**

- 28 update their technical knowledge regularly.



## APPENDIX 2

### **GUIDELINES FOR THE DETENTION OF SHIPS**

#### **1 Introduction**

1.1 When deciding whether the deficiencies found in a ship are sufficiently serious to merit detention, the PSCO should assess whether:

- .1 the ship has relevant, valid documentation; and
- .2 the ship has the crew required in the minimum Safe Manning Document.

1.2 During inspection, the PSCO should further assess whether the ship and/or crew, throughout its forthcoming voyage, is able to:

- .1 navigate safely;
- .2 safely handle, carry and monitor the condition of the cargo;
- .3 operate the engine-room safely;
- .4 maintain proper propulsion and steering;
- .5 fight fires effectively in any part of the ship if necessary;
- .6 abandon ship speedily and safely and effect rescue if necessary;
- .7 prevent pollution of the environment;
- .8 maintain adequate stability;
- .9 maintain adequate watertight integrity;
- .10 communicate in distress situations if necessary; and
- .11 provide safe and healthy conditions on board.

1.3 If the result of any of these assessments is negative, taking into account all deficiencies found, the ship should be strongly considered for detention. A combination of deficiencies of a less serious nature may also warrant the detention of the ship. Ships which are unsafe to proceed to sea should be detained upon the first inspection, irrespective of the time the ship will stay in port.

#### **2 General**

The lack of valid certificates as required by the relevant instruments may warrant the detention of ships. However, ships flying the flag of States not a Party to a convention or not having implemented another relevant instrument, are not entitled to carry the certificates provided for by the convention or other relevant instrument. Therefore, absence of the required certificates should not by itself constitute a reason to detain these ships; however, in applying the "no more favourable treatment" clause, substantial compliance with the provisions and criteria specified in these Procedures must be required before the ship sails.

### **3 Detainable deficiencies**

To assist the PSCO in the use of these Guidelines, there follows a list of deficiencies, grouped under relevant conventions and/or codes, which are considered to be of such a serious nature that they may warrant the detention of the ship involved. This list is not considered exhaustive, but is intended to give examples of relevant items.

#### **Areas under the SOLAS Convention**

- 1 Failure of proper operation of propulsion and other essential machinery, as well as electrical installations.
- 2 Insufficient cleanliness of engine-room, excess amount of oily-water mixture in bilges, insulation of piping including exhaust pipes in engine-room contaminated by oil, and improper operation of bilge pumping arrangements.
- 3 Failure of the proper operation of emergency generator, lighting, batteries and switches.
- 4 Failure of proper operation of the main and auxiliary steering gear.
- 5 Absence, insufficient capacity or serious deterioration of personal life-saving appliances, survival craft and launching and recovery arrangements.
- 6 Absence, non-compliance or substantial deterioration to the extent that it can not comply with its intended use of fire detection system, fire alarms, fire-fighting equipment, fixed fire-extinguishing installation, ventilation valves, fire dampers, and quick-closing devices.
- 7 Absence, substantial deterioration or failure of proper operation of the cargo deck area fire protection on tankers.
- 8 Absence, non-compliance or serious deterioration of lights, shapes or sound signals.
- 9 Absence or failure of the proper operation of the radio equipment for distress and safety communication.
- 10 Absence or failure of the proper operation of navigation equipment, taking the relevant provisions of SOLAS regulation V/16.2 into account.
- 11 Absence of corrected navigational charts, and/or all other relevant nautical publications necessary for the intended voyage, taking into account that electronic charts may be used as a substitute for the charts.
- 12 Absence of non-sparking exhaust ventilation for cargo pump-rooms.
- 13 Serious deficiency in the operational requirements listed in appendix 7.
- 14 Number, composition or certification of crew not corresponding with safe manning document.
- 15 Non-implementation or failure to carry out the enhanced survey programme in accordance with SOLAS regulation XI-1/2 and resolution A.744(18), as amended.

- 16 Absence or failure of a voyage data recorder (VDR), when its use is compulsory.

**Areas under the IBC Code**

- 1 Transport of a substance not mentioned in the Certificate of Fitness or missing cargo information.
- 2 Missing or damaged high pressure safety devices.
- 3 Electrical installations not intrinsically safe or not corresponding to the Code requirements.
- 4 Sources of ignition in hazardous locations.
- 5 Contravention of special requirements.
- 6 Exceeding of maximum allowable cargo quantity per tank.
- 7 Insufficient heat protection for sensitive products.
- 8 Pressure alarms for cargo tanks not operable.
- 9 Transport of substances to be inhibited without valid inhibitor certificate.

**Areas under the IGC Code**

- 1 Transport of a substance not mentioned in the Certificate of Fitness or missing cargo information.
- 2 Missing closing devices for accommodations or service spaces.
- 3 Bulkhead not gastight.
- 4 Defective air locks.
- 5 Missing or defective quick-closing valves.
- 6 Missing or defective safety valves.
- 7 Electrical installations not intrinsically safe or not corresponding to the Code requirements.
- 8 Ventilators in cargo area not operable.
- 9 Pressure alarms for cargo tanks not operable.
- 10 Gas detection plant and/or toxic gas detection plant defective.
- 11 Transport of substances to be inhibited without valid inhibitor certificate.

**Areas under the Load Lines Convention**

- 1      Significant areas of damage or corrosion, or pitting of plating and associated stiffening in decks and hull effecting seaworthiness or strength to take local loads, unless properly authorized temporary repairs for a voyage to a port for permanent repairs have been carried out.
- 2      A recognized case of insufficient stability.

- 3 The absence of sufficient and reliable information, in an approved form, which by rapid and simple means, enables the master to arrange for the loading and ballasting of the ship in such a way that a safe margin of stability is maintained at all stages and at varying conditions of the voyage, and that the creation of any unacceptable stresses in the ship's structure are avoided.
- 4 Absence, substantial deterioration or defective closing devices, hatch closing arrangements and watertight/weathertight doors.
- 5 Overloading.
- 6 Absence of, or impossibility to read, draught marks and/or Load Line marks.

#### **Areas under the MARPOL Convention, Annex I**

- 1 Absence, serious deterioration or failure of proper operation of the oily-water filtering equipment, the oil discharge monitoring and control system or the 15 ppm alarm arrangements.
- 2 Remaining capacity of slop and/or sludge tank insufficient for the intended voyage.
- 3 Oil Record Book not available.
- 4 Unauthorized discharge bypass fitted.
- 5 Failure to meet the requirements of regulation 20.4 or alternative requirements specified in regulation 20.7.

#### **Areas under the MARPOL Convention, Annex II**

- 1 Absence of P and A Manual.
- 2 Cargo is not categorized.
- 3 No Cargo Record Book available.
- 4 Unauthorized discharge bypass fitted.

#### **Areas under the MARPOL Convention, Annex IV**

To be developed.

#### **Areas under the MARPOL Convention, Annex V**

- 1 Absence of the garbage management plan.
- 2 No garbage record book available.
- 3 Ship's personnel not familiar with disposal/discharge requirements of garbage management plan.

#### **Areas under the MARPOL Convention, Annex VI**

- 1 Absence of valid IAPP Certificate and where relevant EIAPP Certificates and Technical Files.
- 2 A diesel engine, with a power output of more than 130 kW, which is installed on board a ship constructed on or after 1 January 2000, or a diesel engine having undergone a major conversion on or after 1 January 2000, which does not comply with the No<sub>x</sub> Technical Code.
- 3 The sulphur content of any fuel oil used on board ships exceeds 4.5% m/m.
- 4 The sulphur content of any fuel used on board exceeds a maximum of 1.5% m/m while operating within a SO<sub>x</sub> emission control area.
- 5 An incinerator installed on board the ship on or after 1 January 2000 does not comply with requirements contained in appendix IV to the Annex, or the standard specifications for shipboard incinerators developed by the Organization (resolutions MEPC.76(40) and MEPC.93(45)).
- 6 The master or crew are not familiar with essential procedures regarding the operation of air pollution prevention equipment.

#### **Areas under the STCW Convention**

- 1 Failure of seafarers to hold a certificate, to have an appropriate certificate, to have a valid dispensation or to provide documentary proof that an application for an endorsement has been submitted to the Administration.
- 2 Failure to comply with the applicable safe manning requirements of the Administration.
- 3 Failure of navigational or engineering watch arrangements to conform to the requirements specified for the ship by the Administration.
- 4 Absence in a watch of a person qualified to operate equipment essential to safe navigation, safety radiocommunications or the prevention of marine pollution.
- 5 Inability to provide for the first watch at the commencement of a voyage and for subsequent relieving watches persons who are sufficiently rested and otherwise fit for duty.
- 6 Failure to provide proof of professional proficiency for the duties assigned to seafarers for the safety of the ship and the prevention of pollution.

#### **Areas which may not warrant a detention, but where, e.g., cargo operations have to be suspended**

Failure of the proper operation (or maintenance) of inert gas systems, cargo related gear or machinery should be considered sufficient grounds to stop cargo operation.

## APPENDIX 3

### GUIDELINES FOR INVESTIGATIONS AND INSPECTIONS CARRIED OUT UNDER ANNEX I OF MARPOL

#### PART 1

#### INSPECTION OF IOPP CERTIFICATE, SHIP AND EQUIPMENT

##### 1 Ships required to carry an IOPP Certificate

1.1 On boarding and introduction to the master or responsible ship's officer, the PSCO should examine the IOPP Certificate, including the attached Record of Construction and Equipment, and the Oil Record Book.

1.2 The certificate carries the information on the type of ship and the dates of surveys and inspections. As a preliminary check it should be confirmed that the dates of surveys and inspections are still valid. Furthermore it should be established if the ship carries an oil cargo and whether the carriage of such oil cargo is in conformity with the certificate (see also paragraph 1.11 of the Record of Construction and Equipment for Oil Tankers).

1.3 Through examining the Record of Construction and Equipment, the PSCO may establish how the ship is equipped for the prevention of marine pollution.

1.4 If the certificate is valid and the general impression and visual observations on board confirm a good standard of maintenance, the PSCO should generally confine the inspection to reported deficiencies, if any.

1.5 If, however, the PSCO from general impressions or observations on board has clear grounds for believing that the condition of the ship or its equipment does not correspond substantially with the particulars of the certificate, a more detailed inspection should be initiated.

1.6 The inspection of the engine-room should begin with forming a general impression of the state of the engine-room, the presence of traces of oil in the engine-room bilges and the ship's routine for disposing of oil contaminated water from the engine-room spaces.

1.7 Next a closer examination of the ship's equipment as listed in the IOPP Certificate may take place. This examination should also confirm that no unapproved modifications have been made to the ship and its equipment.

1.8 Should any doubt arise as to the maintenance or the condition of the ship or its equipment, then further examination and testing may be conducted as considered necessary. In this respect reference is made to annex 3 to the Survey Guidelines under the Harmonized System of Survey and Certification (HSSC), 2011 (resolution A.[...](27)).

1.9 The PSCO should bear in mind that a ship may be equipped over and above the requirements of Annex I of MARPOL. If such equipment is malfunctioning the flag State should be informed. This alone however should not cause a ship to be detained unless the discrepancy presents an unreasonable threat of harm to the marine environment.

1.10 In cases of oil tankers, the inspection should include the cargo tank and pump-room area of the ship and should begin with forming a general impression of the layout of the tanks, the cargoes carried, and the routine of cargo slops disposal.

## **2 Ships of non-Parties to the Convention and other ships not required to carry an IOPP Certificate**

2.1 As this category of ships is not provided with an IOPP Certificate, the PSCO should be satisfied with regard to the construction and equipment standards relevant to the ship on the basis of the requirements set out in Annex I of MARPOL.

2.2 In all other respects the PSCO should be guided by the procedures for ships referred to in section 1 above.

2.3 If the ship has some form of certification other than the IOPP Certificate, the PSCO may take the form and content of this documentation into account in the evaluation of that ship.

## **3 Control**

In exercising the control functions, the PSCO should use professional judgement to determine whether to detain the ship until any noted deficiencies are corrected or to allow it to sail with certain deficiencies which do not pose an unreasonable threat of harm to the marine environment. In doing this, the PSCO should be guided by the principle that the requirements contained in Annex I of MARPOL, in respect of construction and equipment and the operation of ships, are essential for the protection of the marine environment and that departure from these requirements could constitute an unreasonable threat of harm to the marine environment.

## **PART 2**

### **CONTRAVENTION OF DISCHARGE PROVISIONS**

1 Experience has shown that information furnished to the flag State as envisaged in appendix 5 of these Procedures is often inadequate to enable the flag State to cause proceedings to be brought in respect of the alleged violation of the discharge requirements. This appendix is intended to identify information which is often needed by a flag State for the prosecution of such possible violations.

2 It is recommended that, in preparing a port State report on deficiencies, where contravention of the discharge requirements is involved, the authorities of the coastal or port State be guided by the itemized list of possible evidence as shown in part 3 of this appendix. It should be borne in mind in this connection that:

- .1 the report aims to provide the optimal collation of obtainable data; however, even if all the information cannot be provided, as much information as possible should be submitted;
- .2 it is important for all the information included in the report to be supported by facts which, when considered as a whole, would lead the port or coastal State to believe a contravention had occurred.



3 In addition to the port State report on deficiencies, a report should be completed by a port or coastal State, on the basis of the itemized list of possible evidence. It is important that these reports are supplemented by documents such as:

- .1 a statement by the observer of the pollution. In addition to the information required under section 1 of part 3 of this appendix the statement should include considerations which lead the observer to conclude that none of any other possible pollution sources is in fact the source;
- .2 statements concerning the sampling procedures both of the slick and on board. These should include location of and time when samples were taken, identity of person(s) taking the samples and receipts identifying the persons having custody and receiving transfer of the samples;
- .3 reports of analyses of samples taken of the slick and on board; the reports should include the results of the analyses, a description of the method employed, reference to or copies of scientific documentation attesting to the accuracy and validity of the method employed and names of persons performing the analyses and their experience;
- .4 a statement by the PSCO on board together with the PSCO's rank and organization;
- .5 statements by persons being questioned;
- .6 statements by witnesses. All observations, photographs and documentation should be supported by a signed verification of their authenticity. All certifications, authentications or verifications shall be executed in accordance with the laws of the State which prepares them. All statements should be signed and dated by the person making the statement and, if possible, by a witness to the signing. The names of the persons signing statements should be printed in legible script above or below the signature;
- .7 photographs of the oil slick; and
- .8 copies of relevant recordings, etc., pages of Oil Record Books, log-books, discharge.

4 The report referred to in paragraphs 2 and 3 should be sent to the flag State. If the coastal State observing the pollution and the port State carrying out the investigation on board are not the same, the State carrying out the latter investigation should also send a copy of its findings to the State observing the pollution and requesting the investigation.

## **PART 3**

### **ITEMIZED LIST OF POSSIBLE EVIDENCE ON ALLEGED CONTRAVENTION OF THE MARPOL ANNEX I DISCHARGE PROVISIONS**

#### **1 Action on sighting oil pollution**

##### **1.1 Particulars of ship or ships suspected of contravention**

- .1 Name of ship
- .2 Reasons for suspecting the ship
- .3 Date and time (UTC) of observation or identification
- .4 Position of ship
- .5 Flag and port of registry
- .6 Type (e.g., tanker, cargo ship, passenger ship, fishing vessel), size (estimated tonnage) and other descriptive data (e.g., superstructure colour and funnel mark)
- .7 Draught condition (loaded or in ballast)
- .8 Approximate course and speed
- .9 Position of slick in relation to ship (e.g., astern, port, starboard)
- .10 Part of the ship from which side discharge was seen emanating
- .11 Whether discharge ceased when ship was observed or contacted by radio

##### **1.2 Particulars of slick**

- .1 Date and time (UTC) of observation if different from paragraph 1.1.3
- .2 Position of oil slick in longitude and latitude if different from paragraph 1.1.4
- .3 Approximate distance in nautical miles from the nearest landmark
- .4 Approximate overall dimension of oil slick (length, width and percentage thereof covered by oil)
- .5 Physical description of oil slick (direction and form, e.g., continuous, in patches or in windrows)
- .6 Appearance of oil slick (indicate categories)
  - Category A: Barely visible under most favourable light condition
  - Category B: Visible as silvery sheen on water surface
  - Category C: First trace of colour may be observed
  - Category D: Bright band of colour
  - Category E: Colours begin to turn dull
  - Category F: Colours are much darker
- .7 Sky conditions (bright sunshine, overcast, etc.), lightfall and visibility (kilometres) at the time of observation
- .8 Sea state
- .9 Direction and speed of surface wind
- .10 Direction and speed of current

##### **1.3 Identification of the observer(s)**

- .1 Name of the observer
- .2 Organization with which observer is affiliated (if any)
- .3 Observer's status within the organization
- .4 Observation made from aircraft/ship/shore/otherwise
- .5 Name or identity of ship or aircraft from which the observation was made
- .6 Specific location of ship, aircraft, place on shore or otherwise from which observation was made

- .7 Activity engaged in by observer when observation was made, for example: patrol, voyage, flight (en route from ... to ...), etc.

#### **1.4 Method of observation and documentation**

- .1 Visual
- .2 Conventional photographs
- .3 Remote sensing records and/or remote sensing photographs
- .4 Samples taken from slick
- .5 Any other form of observation (specify)

**Note:** A photograph of the discharge should preferably be in colour. Photographs can provide the following information: that a material on the sea surface is oil; that the quantity of oil discharged does constitute a violation of the Convention; that the oil is being, or has been discharged from a particular ship; and the identity of the ship.

Experience has shown that the aforementioned can be obtained with the following three photographs:

- details of the slick taken almost vertically down from an altitude of less than 300 m with the sun behind the photographer;
- an overall view of the ship and "slick" showing oil emanating from a particular ship; and
- details of the ship for the purposes of identification.

#### **1.5 Other information if radio contact can be established**

- .1 Master informed of pollution
- .2 Explanation of master
- .3 Ship's last port of call
- .4 Ship's next port of call
- .5 Name of ship's master and owner
- .6 Ship's call sign

### **2 Investigation on board**

#### **2.1 Inspection of IOPP Certificate**

- .1 Name of ship
- .2 Distinctive number or letters
- .3 Port of registry
- .4 Type of ship
- .5 Date and place of issue
- .6 Date and place of endorsement.

**Note:** If the ship is not issued an IOPP Certificate, as much as possible of the requested information should be given.

#### **2.2 Inspection of supplement of the IOPP Certificate**

- .1 Applicable paragraphs of sections 2, 3, 4, 5 and 6 of the supplement (non-oil tankers)

- .2 Applicable paragraphs of sections 2, 3, 4, 5, 6, 7, 8, 9 and 10 of the supplement (oil tankers)

**Note:** If the ship does not have an IOPP Certificate, a description should be given of the equipment and arrangements on board, designed to prevent marine pollution.

### **2.3 Inspection of Oil Record Book (O.R.B.)**

- .1 Copy sufficient pages of the O.R.B. – part I to cover a period of 30 days prior to the reported incident.
- .2 Copy sufficient pages of the O.R.B. – part II (if on board) to cover a full loading/unloading/ballasting and tank cleaning cycle of the ship. Also copy the tank diagram.

### **2.4 Inspection of log-book**

- .1 Last port, date of departure, draught forward and aft
- .2 Current port, date of arrival, draught forward and aft
- .3 Ship's position at or near the time the incident was reported
- .4 Spot check if positions mentioned in the log-book agree with positions noted in the O.R.B.

### **2.5 Inspection of other documentation on board**

Other documentation relevant for evidence (if necessary make copies) such as:

- recent ullage sheets
- records of monitoring and control equipment.

### **2.6 Inspection of ship**

- .1 Ship's equipment in accordance with the supplement of the IOPP Certificate
- .2 Samples taken. State location on board
- .3 Traces of oil in vicinity of overboard discharge outlets
- .4 Condition of engine-room and contents of bilges
- .5 Condition of oily water separator, filtering equipment and alarm, stopping or monitoring arrangements
- .6 Contents of sludge and/or holding tanks
- .7 Sources of considerable leakage on oil tankers.

The following additional evidence may be pertinent:

- .8 Oil on surface of segregated or dedicated clean ballast
- .9 Condition of pump-room bilges
- .10 Condition of COW system
- .11 Condition of IG system
- .12 Condition of monitoring and control system
- .13 Slop tank contents (estimate quantity of water and of oil).

## **2.7 Statements of persons concerned**

If the O.R.B. – part I has not been properly completed, information on the following questions may be pertinent:

- .1 Was there a discharge (accidental or intentional) at the time indicated on the incident report?
- .2 Is the bilge discharge controlled automatically?
- .3 If so, at what time was this system last put into operation and at what time was this system last put on manual mode?
- .4 If not, what were date and time of the last bilge discharge?
- .5 What was the date of the last disposal of residue and how was disposal effected?
- .6 Is it usual to effect discharge of bilge water directly to the sea, or to store bilge water first in a collecting tank? Identify the collecting tank
- .7 Have oil fuel tanks recently been used as ballast tanks?

If the O.R.B. – part II has not been properly completed, information on the following questions may be pertinent:

- .8 What was the cargo/ballast distribution in the ship on departure from the last port?
- .9 What was the cargo/ballast distribution in the ship on arrival in the current port?
- .10 When and where was the last loading effected?
- .11 When and where was the last unloading effected?
- .12 When and where was the last discharge of dirty ballast?
- .13 When and where was the last cleaning of cargo tanks?
- .14 When and where was the last COW operation and which tanks were washed?
- .15 When and where was the last decanting of slop tanks?
- .16 What is the ullage in the slop tanks and the corresponding height of interface?
- .17 Which tanks contained the dirty ballast during the ballast voyage (if ship arrived in ballast)?
- .18 Which tanks contained the clean ballast during the ballast voyage (if ship arrived in ballast)?

In addition the following information may be pertinent:

- .19 Details of the present voyage of the ship (previous ports, next ports, trade)
- .20 Contents of oil fuel and ballast tanks
- .21 Previous and next bunkering, type of oil fuel
- .22 Availability or non-availability of reception facilities for oily wastes during the present voyage
- .23 Internal transfer of oil fuel during the present voyage.

In the case of oil tankers the following additional information may be pertinent:

- .24 The trade the ship is engaged in, such as short/long distance, crude or product or alternating crude/product, lightering service, oil/dry bulk
- .25 Which tanks clean and dirty
- .26 Repairs carried out or envisaged in cargo tanks.

Miscellaneous information:

- .27 Comments in respect of condition of ship's equipment
- .28 Comments in respect of pollution report
- .29 Other comments.

### **3 Investigation ashore**

#### **3.1 Analyses of oil samples**

Indicate method and results of the samples' analyses.

#### **3.2 Further information**

Additional information on the ship, obtained from oil terminal staff, tank cleaning contractors or shore reception facilities may be pertinent.

**Note:** Any information under this heading is, if practicable, to be corroborated by documentation such as signed statements, invoices, receipts, etc.

### **4 Information not covered by the foregoing**

### **5 Conclusion**

- .1 Summing up of the investigator's technical conclusions
- .2 Indication of applicable provisions of Annex I of MARPOL which the ship is suspected of having contravened.
- .3 Did the results of the investigation warrant the filing of a deficiency report?

## **PART 4**

### **GUIDELINES FOR IN-PORT INSPECTION OF CRUDE OIL WASHING PROCEDURES**

#### **1 Preamble**

1.1 Guidelines for the in-port inspection of crude oil washing (COW) procedures, as called for by resolution 7 of the International Conference on Tanker Safety and Pollution Prevention, 1978, are required to provide a uniform and effective control of crude oil washing to ensure compliance of ships at all times with the provisions of MARPOL.

1.2 The design of the crude oil washing installation is subject to the approval of the flag Administration. However, although the operational aspect of crude oil washing is also subject to the approval of the same Administration, it might be necessary for a port State Authority to see to it that continuing compliance with agreed procedures and parameters is ensured.

1.3 The COW Operations and Equipment Manual has been so specified that it contains all the necessary information relating to the operation of crude oil washing on a particular tanker. The objectives of the inspection would then be to ensure that the provisions of the Manual dealing with safety procedures and with pollution prevention are being strictly adhered to.

1.4 The method of the inspection is at the discretion of the port State Authority and may cover the entire operation or only those parts of the operation which occur when the PSCO is on board.

1.5 Inspection will be governed by articles 5 and 6 of MARPOL.

## **2 Inspections**

2.1 A port State should make the appropriate arrangements so as to ensure compliance with requirements governing the crude oil washing of oil tankers. This is not, however, to be construed as relieving terminal operators and ship owners of their obligations to ensure that the operation is undertaken in accordance with the regulations.

2.2 The inspection may cover the entire operation of crude oil washing or only certain aspects of it. It is thus in the interest of all concerned that the ship's records with regard to the COW operations are maintained at all times so that a PSCO may verify those operations undertaken prior to the inspection.

## **3 Ship's personnel**

3.1 The person in charge and the other nominated persons who have responsibility in respect of the crude oil washing operation should be identified. They must, if required, be able to show that their qualifications meet the requirements, as appropriate, of paragraphs 5.2 and 5.3 of the revised Specifications for the Design, Operation and Control of Crude Oil Washing Systems (resolution A.446(XI), as amended).

3.2 The verification may be accomplished by reference to the individual's discharge papers, testimonials issued by the ship's operator or by certificates issued by a training centre approved by an Administration. The numbers of such personnel should be at least as stated in the Manual.

## **4 Documentation**

4.1 The following documents should be available for inspection:

- .1 the IOPP Certificate and the Record of Construction and Equipment, to determine:
  - .1 whether the ship is fitted with a crude oil washing system as required in regulation 33 of MARPOL Annex I;
  - .2 whether the crude oil washing system is according to and complying with the requirements of regulations 33 and 35 of MARPOL Annex I;
  - .3 the validity and date of the Operations and Equipment Manual; and
  - .4 the validity of the Certificate;
- .2 the approved Manual;
- .3 the Oil Record Book; and

- .4 the Cargo Ship Safety Equipment Certificate to confirm that the inert gas system conforms to regulations contained in chapter II-2 of SOLAS, as amended.

## **5 Inert gas system**

5.1 Inert gas system regulations require that instrumentation shall be fitted for continuously indicating and permanently recording at all times when inert gas is being supplied, the pressure and the oxygen content of the gas in the inert gas supply main. Reference to the permanent recorder would indicate if the system had been operating before and during the cargo discharge in a satisfactory manner.

5.2 If conditions specified in the Manual are not being met then the washing must be stopped until satisfactory conditions are restored.

5.3 As a further precautionary measure, the oxygen level in each tank to be washed is to be determined at the tank. The meters used should be calibrated and inspected to ensure that they are in good working order. Readings from tanks already washed in port prior to inspection should be available for checking. Spot checks on readings may be instituted.

## **6 Electrostatic generation**

It should be confirmed either from the cargo log or by questioning the person in charge that presence of water in the crude oil is being minimized as required by paragraph 6.7 of the revised Specifications (resolution A.446(XI) as amended).

## **7 Communication**

It should be established that effective means of communication exist between the person in charge and the other persons concerned with the COW operation.

## **8 Leakage on deck**

PSCOs should ensure that the COW piping system has been operationally tested for leakage before cargo discharge and that the test has been noted in the ship's Oil Record Book.

## **9 Exclusion of oil from engine-room**

It should be ascertained that the method of excluding cargo oil from the machinery space is being maintained by inspecting the isolating arrangements of the tank washing heater (if fitted) or of any part of the tank washing system which enters the machinery space.

## **10 Suitability of the crude oil**

In judging the suitability of the oil for crude oil washing, the guidance and criteria contained in section 9 of the COW Operations and Equipment Manual should be taken into account.

## **11 Checklist**

It should be determined from the ship's records that the pre-crude oil wash operational checklist was carried out and all instruments functioned correctly. Spot checks on certain items may be instituted.



## **12 Wash programmes**

12.1 Where the tanker is engaged in a multiple port discharge, the Oil Record Book would indicate if tanks were crude oil washed at previous discharge ports or at sea. It should be determined that all tanks which will, or may be, used to contain ballast on the forthcoming voyage will be crude oil washed before the ship departs from the port. There is no obligation to wash any tank other than ballast tanks at a discharge port except that each of these other tanks must be washed at least in accordance with paragraph 6.1 of the revised Specifications (resolution A.446(XI) as amended). The Oil Record Book should be inspected to check that this is being complied with.

12.2 All crude oil washing must be completed before a ship leaves its final port of discharge.

12.3 If tanks are not being washed in one of the preferred orders given in the Manual the PSCO should determine that the reason for this, and the proposed order of tank washing, are acceptable.

12.4 For each tank being washed it should be ensured that the operation is in accordance with the Manual in that:

- .1 the deck mounted machines and the submerged machines are operating either by reference to indicators, the sound patterns or other approved methods;
- .2 the deck mounted machines, where applicable, are programmed as stated;
- .3 the duration of the wash is as required; and
- .4 the number of tank washing machines being used simultaneously does not exceed that specified.

## **13 Stripping of tanks**

13.1 The minimum trim conditions and the parameters of the stripping operations are to be stated in the Manual.

13.2 All tanks which have been crude oil washed are to be stripped. The adequacy of the stripping is to be checked by hand dipping at least in the after most hand dipping location in each tank or by such other means provided and described in the Manual. It should be ascertained that the adequacy of stripping has been checked or will be checked before the ship leaves its final port of discharge.

## **14 Ballasting**

14.1 Tanks that were crude oil washed at sea will be recorded in the Oil Record Book. These tanks must be left empty between discharge ports for inspection at the next discharge port. Where these tanks are the designated departure ballast tanks they may be required to be ballasted at a very early stage of the discharge. This is for operational reasons and also because they must be ballasted during cargo discharge if hydrocarbon emission is to be contained on the ship. If these tanks are to be inspected when empty, then this must be done shortly after the tanker berths. If a PSCO arrives after the tanks have begun accepting ballast, then the sounding of the tank bottom would not be available. However, an examination of the surface of the ballast water is then possible. The thickness of the oil film

should not be greater than that specified in paragraph 4.2.10(b) of the revised Specifications (resolution A.446(XI) as amended).

14.2 The tanks that are designated ballast tanks will be listed in the Manual. It is, however, left to the discretion of the master or responsible officer to decide which tanks may be used for ballast on the forthcoming voyage. It should be determined from the Oil Record Book that all such tanks have been washed before the tanker leaves its last discharge port. It should be noted that where a tanker back-loads a cargo of crude oil at an intermediate port into tanks designated for ballast, then it should not be required to wash those tanks at that particular port but at a subsequent port.

14.3 It should be determined from the Oil Record Book that additional ballast water has not been put into tanks which had not been crude oil washed during previous voyages.

14.4 It should be verified that the departure ballast tanks are stripped as completely as possible. Where departure ballast is filled through cargo lines and pumps these must be stripped either into another cargo tank, or ashore by the special small diameter line provided for this purpose.

14.5 The methods to avoid vapour emission where locally required will be provided in the Manual and they must be adhered to. The PSCO should ensure that this is being complied with.

14.6 The typical procedures for ballasting listed in the Manual must be observed. The PSCO should ensure this is being complied with.

14.7 When departure ballast is to be shifted, the discharge into the sea must be in compliance with regulations 15 and 34 of Annex I of MARPOL. The Oil Record Book should be inspected to ensure that the ship is complying with this.

## APPENDIX 4

### **GUIDELINES FOR INVESTIGATIONS AND INSPECTIONS CARRIED OUT UNDER ANNEX II OF MARPOL**

#### **PART 1**

#### **INSPECTION OF CERTIFICATE (COF OR NLS CERTIFICATE), SHIP AND EQUIPMENT**

##### **1 Ships required to hold a Certificate**

1.1 On boarding and after introducing oneself to the master or responsible ship's officer, the PSCO should examine the Certificate of Fitness or NLS Certificate and Cargo Record Book.

1.2 The Certificate includes information on the type of ship, the dates of surveys and a list of the products which the ship is certified to carry.

1.3 As a preliminary check, the Certificate's validity should be confirmed by verifying that the Certificate is properly completed and signed and that required surveys have been performed. In reviewing the Certificate particular attention should be given to verifying that only those noxious liquid substances which are listed on the Certificate are carried and that these substances are in tanks approved for their carriage.

1.4 The Cargo Record Book should be inspected to ensure that the records are up to date. The PSCO should check whether the ship left the previous port(s) with residues of noxious liquid substances on board which could not be discharged into the sea. The book could also have relevant entries from the appropriate authorities in the previous ports. If the examination reveals that the ship was permitted to sail from its last unloading port under certain conditions, the PSCO should ascertain that such conditions have been or will be adhered to. If the PSCO discovers an operational violation in this respect, the flag State should be informed by means of a deficiency report.

1.5 If the Certificate is valid and the PSCO's general impressions and visual observations on board confirm a good standard of maintenance, the PSCO should, provided that the Cargo Record Book entries do not show any operational violations, confine the inspection to reported deficiencies, if any.

1.6 If, however, the PSCO's general impressions or observations on board show clear grounds for believing that the condition of the ship, its equipment, or its cargo and slops handling operations do not correspond substantially with the particulars of the Certificate, the PSCO should proceed to a more detailed inspection:

- .1 initially this requires an examination of the ship's approved Procedures and Arrangements Manual (P and A Manual);
- .2 the more detailed inspection should include the cargo and pump-room areas of the ship and should begin with forming a general impression of the layout of the tanks, the cargoes carried, pumping and stripping conditions and cargo;

- .3 next a closer examination of the ship's equipment as shown in the P and A Manual may take place. This examination should also confirm that no unapproved modifications have been made to the ship and its equipment; and
- .4 should any doubt arise as to the maintenance or the condition of the ship or its equipment then further examination and testing may be conducted as may be necessary. In this respect reference is made to the Guidelines for Surveys under the Harmonized System of Survey and Certification (resolution A.[...](27)), as appropriate.

1.7 The PSCO should bear in mind that a ship may be equipped over and above the requirements of Annex II of MARPOL. If such equipment is malfunctioning the flag State should be informed. This alone, however, should not cause a ship to be detained unless the malfunction presents an unreasonable threat of harm to the marine environment.

## **2 Ships of non-Parties to the Convention**

2.1 As this category of ship is not provided with a COF or NLS Certificate as required by Annex II of MARPOL, the PSCO should be satisfied with regard to the construction and equipment standards relevant to the ship on the basis of the requirements set out in Annex II of MARPOL and the Standards for Procedures and Arrangements.

2.2 In all other respects the PSCO should be guided by the procedures for ships referred to in section 1 above (i.e., ships required to hold a Certificate).

2.3 If the ship has some form of certification other than the required Certificate, the PSCO may take the form and content of this document into account in the evaluation of that ship. Such a form of certification, however, is only of value to the PSCO if the ship has been provided with a P and A Manual.

## **3 Control**

In exercising the control functions, the PSCO should use professional judgement to determine whether to detain the ship until any noted deficiencies are rectified or to allow it to sail with certain deficiencies which do not pose an unreasonable threat of harm to the marine environment. In doing this, the PSCO should be guided by the principle that the requirements contained in Annex II of MARPOL, in respect of construction and equipment and the operation of ships, are essential for the protection of the marine environment and that departure from these requirements could constitute an unreasonable threat of harm to the marine environment.

## **PART 2**

### **CONTRAVENTION OF DISCHARGE PROVISIONS**

1 With illegal discharges, past experience has shown that information furnished to the flag State is often inadequate to enable the flag State to cause proceedings to be brought in respect of the alleged violation of the discharge requirements. This appendix is intended to identify information which will be needed by a flag State for the prosecution of violations of the discharge provisions under Annex II of MARPOL.

2 It is recommended that in preparing a port State report on deficiencies, where contravention of the discharge requirements is involved, the authorities of a coastal or port State should be guided by the itemized list of possible evidence as shown in part 3 of this appendix. It should be borne in mind in this connection that:

- .1 the report aims to provide the optimal collation of obtainable data; however, even if all the information cannot be provided, as much information as possible should be submitted;
- .2 it is important for all the information included in the report to be supported by facts which, when considered as a whole, would lead the port or coastal State to believe a contravention has occurred; and
- .3 the discharge may have been oil, in which case part 2 to Appendix 3 of this resolution applies (Guidelines for Investigation and Inspections carried out under Annex I of MARPOL).

3 In addition to the port State report on deficiencies, a report should be completed by a port or coastal State, on the basis of the itemized list of possible evidence. It is important that these reports are supplemented by documents such as:

- .1 a statement by the observer of the pollution. In addition to the information required under section 1 of part 3 of this appendix, the statement should include considerations which have led the observer to conclude that none of any other possible pollution sources is in fact the source;
- .2 statements concerning the sampling procedures both of the slick and on board. These include location of and time when samples were taken, identity of person(s) taking the samples and receipts identifying the persons having custody and receiving transfer of the samples;
- .3 reports of analyses of samples taken of the slick and on board; the reports should include the results of the analyses, a description of the method employed, reference to or copies of scientific documentation attesting to the accuracy and validity of the method employed and names of persons performing the analyses and their experience;
- .4 a statement by the PSCO on board together with the PSCO's rank and organization;
- .5 statements by persons being questioned;
- .6 statements by witnesses;
- .7 photographs of the slick; and
- .8 copies of relevant pages of the Cargo Record Book, log-books, discharge recordings, etc.

4 All observations, photographs and documentation should be supported by a signed verification of their authenticity. All certifications, authentications or verifications shall be executed in accordance with the laws of the State which prepares them. All statements should be signed and dated by the person making the statement and, if possible, by a witness

to the signing. The names of the persons signing statements should be printed in legible script above or below the signature.

5 The report referred to in paragraphs 2 and 3 should be sent to the flag State. If the coastal State observing the pollution and the port State carrying out the investigation on board are not the same, the State carrying out the latter investigation should also send a copy of its findings to the State observing the pollution and requesting the investigation.

### **PART 3**

#### **ITEMIZED LIST OF POSSIBLE EVIDENCE ON ALLEGED CONTRAVENTION OF THE MARPOL ANNEX II DISCHARGE PROVISIONS**

##### **1 Action on sighting pollution**

###### **1.1 Particulars of ship or ships suspected of contravention**

- .1 Name of ship and IMO Number
- .2 Reasons for suspecting the ship
- .3 Date and time (UTC) of observation or identification
- .4 Position of ship
- .5 Flag and port of registry
- .6 Type, size (estimated tonnage) and other descriptive data (e.g., superstructure, colour and funnel mark)
- .7 Draught condition (loaded or in ballast)
- .8 Approximate course and speed
- .9 Position of slick in relating to ship (e.g., astern, port, starboard)
- .10 Part of the ship from which discharge was seen emanating
- .11 Whether discharge ceased when ship was observed or contacted by radio

###### **1.2 Particulars of slick**

- .1 Date and time (UTC) of observation if different from item 1.1.3
- .2 Position of slick in longitude and latitude if different from item 1.1.4
- .3 Approximate distance in nautical miles from the nearest land
- .4 Depth of water according to sea chart
- .5 Approximate overall dimension of slick (length, width and percentage thereof covered)
- .6 Physical description of slick (direction and form, e.g., continuous, in patches or in windrows)
- .7 Colour of slick
- .8 Sky conditions (bright sunshine, overcast, etc.), lightfall and visibility (kms) at the time of observation
- .9 Sea state
- .10 Direction and speed of surface wind
- .11 Direction and speed of current

###### **1.3 Identification of the observer(s)**

- .1 Name of the observer
- .2 Organization with which observer is affiliated (if any)
- .3 Observer's status within the organization
- .4 Observation made from aircraft (ship) (shore) or otherwise

- .5 Name or identity of ship or aircraft from which the observation was made
- .6 Specific location of ship, aircraft, place on shore or otherwise from which observation was made
- .7 Activity engaged in by observer when observation was made, for example: patrol, voyage, flight (en route from ... to ...), etc.

#### **1.4 Method of observation and documentation**

- .1 Visual
- .2 Conventional photographs
- .3 Remote sensing records and/or remote sensing photographs
- .4 Samples taken from slick
- .5 Any other form of observation (specify)

**Note:** A photograph of the discharge should preferably be in colour. The best results may be obtained with the following three photographs:

- details of the slick taken almost vertically down from an altitude of less than 300 metres with the sun behind the photographer;
- an overall view of the ship and "slick" showing a substance emanating from the particular ship; and
- details of the ship for the purposes of identification

#### **1.5 Other information if radio contact can be established**

- .1 Master informed of pollution
- .2 Explanation of master
- .3 Ship's last port of call
- .4 Ship's next port of call
- .5 Name of ship's master and owner
- .6 Ship's call sign

### **2 Investigation on board**

#### **2.1 Inspection of the Certificate (COF or NLS Certificate)**

- .1 Name of ship and IMO Number
- .2 Distinctive number or letters
- .3 Port of registry
- .4 Type of ship
- .5 Date and place of issue
- .6 Date and place of endorsement
- .7 List of Annex II substances the ship is certified to carry
- .8 Limitation as to tanks in which these substances may be carried

#### **2.2 Inspection of P and A Manual**

- .1 Ship equipped with an efficient stripping system
- .2 Residue quantities established at survey

## **2.3 Inspection of Cargo Record Book (CRB)**

Copy sufficient pages of the CRB to cover a full loading/unloading/ballasting and tank cleaning cycle of the ship. Also copy the tank diagram

## **2.4 Inspection of log-book**

- .1 Last port, date of departure, draught forward and aft
- .2 Current port, date of arrival, draught forward and aft
- .3 Ship's position at or near the time the incident was reported
- .4 Spot check if times entered in the Cargo Record Book in respect of discharges correspond with sufficient distance from the nearest land, the required ship's speed and with sufficient water depth

## **2.5 Inspection of other documentation on board**

Other documentation relevant for evidence (if necessary make copies) such as:

- cargo documents of cargo presently or recently carried, together with relevant information on required unloading temperature, viscosity and/or melting point
- records of temperature of substances during unloading
- records of monitoring equipment if fitted

## **2.6 Inspection of ship**

- .1 Ship's equipment in accordance with the P and A Manual
- .2 Samples taken; state location on board
- .3 Sources of considerable leakage
- .4 Cargo residues on surface of segregated or dedicated clean ballast
- .5 Condition of pump-room bilges
- .6 Condition of monitoring system
- .7 Slop tank contents (estimate quantity of water and residues)

## **2.7 Statements of persons concerned if the CRB has not been properly completed, information on the following questions may be pertinent:**

- .1 Was there a discharge (accidental or intentional) at the time indicated on the incident report?
- .2 Which tanks are going to be loaded in the port?
- .3 Which tanks needed cleaning at sea? Had the tanks been prewashed?
- .4 When and where were these cleaned?
- .5 Residues of which substances were involved?
- .6 What was done with the tank washing slops?
- .7 Was the slop tank, or cargo tank used as a slop tank, discharged at sea?
- .8 When and where was the discharge effected?
- .9 What are the contents of the slop tank or cargo tank used as slop tank?
- .10 Which tanks contained the dirty ballast during the ballast voyage (if ship arrived in ballast)?
- .11 Which tanks contained the clean ballast during the ballast voyage (if ship arrived in ballast)?
- .12 Details of the present voyage of the ship (previous ports, next ports, trade)
- .13 Difficulties experienced with discharge to shore reception facilities



- .14 Difficulties experienced with efficient stripping operations
- .15 Which tanks are clean or dirty on arrival?
- .16 Repairs carried out or envisaged in cargo tanks

Miscellaneous information

- .17 Comments in respect of condition of ship's equipment
- .18 Comments in respect of pollution report
- .19 Other comments

### **3 Investigation ashore**

#### **3.1 Analyses of samples**

Indicate method and results of the samples' analyses

#### **3.2 Further information**

Additional information on the ship, obtained from terminal staff, tank cleaning contractors or shore reception facilities may be pertinent

**Note:** Any information under this heading is, if practicable, to be corroborated by documentation such as signed statements, invoices, receipts, etc.

#### **3.3 Information from previous unloading port terminal**

- .1 Confirmation that the ship unloaded, stripped or prewashed in accordance with its P and A Manual
- .2 The nature of difficulties if any
- .3 Restrictions by authorities under which the ship was permitted to sail
- .4 Restrictions in respect of shore reception facilities

### **4 Information not covered by the foregoing**

### **5 Conclusion**

- .1 Summing up of the investigator's conclusions
- .2 Indication of applicable provisions of Annex II of MARPOL which the ship is suspected of having contravened
- .3 Did the results of the investigation warrant the filing of a deficiency report?

## **PART 4**

### **PROCEDURES FOR INSPECTION OF UNLOADING, STRIPPING AND PREWASHING OPERATIONS (MAINLY IN UNLOADING PORTS)**

#### **1 Introduction**

The PSCO or the surveyor authorized by the Administration exercising control in accordance with regulation 16 of MARPOL Annex II should be thoroughly acquainted with Annex II of MARPOL and the custom of the port as of relevance to cargo handling, tank washing, cleaning berths, prohibition of lighters alongside, etc.

#### **2 Documentation**

2.1 The documentation required for the inspection referred to in this appendix consists of:

- .1 COF or NLS Certificate;
- .2 cargo plan and shipping document;
- .3 Procedures and Arrangements (P and A) Manual; and
- .4 Cargo Record Book.

#### **3 Information by ship's staff**

3.1 Of relevance to the PSCO or the surveyor appointed or authorized by the Administration is the following:

- .1 the intended loading and unloading programme of the ship;
- .2 whether unloading and stripping operations can be effected in accordance with the P and A Manual and if not the reason why it cannot be done;
- .3 the constraints, if any, under which the efficient stripping system operates (i.e. back pressure, ambient air temperature, malfunctioning, etc.); and
- .4 whether the ship requests an exemption from the prewashing and the discharge of residues in the unloading port.

3.2 When tank washing is required without the use of water the PSCO or the surveyor appointed or authorized by the Administration is to be informed about the tank washing procedure and disposal of residues.

3.3 When the Cargo Record Book is not up to date, any information on prewash and residue disposal operations outstanding should be supplied.

#### **4 Information from terminal staff**

Terminal staff should supply information on limitations imposed upon the ship in respect of back pressure and/or reception facilities.

#### **5 Control**

5.1 On boarding and introduction to the master or responsible ship officers, the PSCO or the surveyor appointed or authorized by the Administration should examine the necessary documentation.

5.2 The documentation may be used to establish the following:

- .1 noxious liquid substances to be unloaded, their categories and stowage (cargo plan, P and A Manual);
- .2 details of efficient stripping system, if fitted (P and A Manual);
- .3 tanks which require prewashing with disposal of tank washings to reception facilities (shipping document and cargo temperature);
- .4 tanks which require prewashing with disposal of tank washings either to reception facilities or into the sea (P and A Manual, shipping document and cargo temperature);
- .5 prewash operations and/or residue disposal operations outstanding (Cargo Record Book); and
- .6 tanks which may not be washed with water due to the nature of substances involved (P and A Manual).

5.3 In respect of the prewash operations referred to under paragraph 5.2, the following information is of relevance (P and A Manual):

- .1 pressure required for tank washing machines;
- .2 duration of one cycle of the tank washing machine and quantity of water used;
- .3 washing programmes for the substances involved;
- .4 required temperature of washing water; and
- .5 special procedures.

5.4 The PSCO or the surveyor authorized by the Administration, in accordance with regulation 16 of MARPOL Annex II, should ascertain that unloading, stripping and/or prewash operations are carried out in conformance with the information obtained in accordance with paragraph 2 (Documentation) of this Part. If this cannot be achieved, alternative measures should be taken to ensure that the ship does not proceed to sea with more than the quantities of residue specified in regulation 12 of MARPOL Annex II, as applicable. If the residue quantities cannot be reduced by alternative measures the PSCO or the surveyor appointed or authorized by the Administration should inform the port State Administration.

5.5 Care should be taken to ensure that cargo hoses and piping systems of the terminal are not drained back to the ship.

5.6 If a ship is exempted from certain pumping efficiency requirements under regulation 4.4 of MARPOL Annex II or requests an exemption from certain stripping or prewashing procedures under regulation 13.4 of MARPOL Annex II the conditions for such exemption set out in the said regulations should be observed. These concern:

- .1 regulations 4.2 and 4.3: the ship is constructed before 1 July 1986 and is exempted from the requirement for reducing its residue quantities to specified limits of regulation 12 (i.e. category X or Y substances 300 litres and category Z substances 900 litres). This is subject to the conditions of regulation 4.3 that whenever a cargo tank is to be washed or ballasted, a prewash is required with disposal of prewash slops to shore reception facilities. The COF or NLS Certificate should have been endorsed to the effect that the ship is solely engaged in restricted voyages;

- .2 regulation 4.4: the ship is never required to ballast its cargo tanks and tank washing is only required for repair or dry-docking. The COF or NLS Certificate should indicate the particulars of the exemption. Each cargo tank should be certified for the carriage of only one named substance;
- .3 regulation 13.4.1: cargo tanks will not be washed or ballasted prior to the next loading;
- .4 regulation 13.4.2: cargo tanks will be washed and prewash slops will be discharged to reception facilities in another port. It should be confirmed in writing that an adequate reception facility is available at that port for such purpose; and
- .5 regulation 13.4.3: the cargo residues can be removed by ventilation.

5.7 The PSCO or the surveyor appointed or authorized by the Administration must endorse the Cargo Record Book under section J whenever an exemption under regulation 13.4 referred to under paragraph 5.6 above has been granted, or whenever a tank having unloaded category X substances has been prewashed in accordance with the P and A Manual.

5.8 Alternatively, for category X substances, regulation 13.6.1.1 of MARPOL Annex II, residual concentration should be measured by the procedures which each port State authorizes. In this case the PSCO or the surveyor authorized by the Administration must endorse in the Cargo Record Book under section K whenever the required residual concentration has been achieved.

5.9 In addition to paragraph 5.7 above, the PSCO or the surveyor authorized by the Administration shall endorse the Cargo Record Book whenever the unloading, stripping or prewash of category Y and Z substances, in accordance with the P and A Manual, has actually been witnessed.

## APPENDIX 5

### GUIDELINES FOR DISCHARGE REQUIREMENTS UNDER ANNEXES I AND II OF MARPOL

#### 1 Introduction

1.1 Regulations 15 and 34 of MARPOL Annex I prohibit the discharge into the sea of oil and regulation 13 of Annex I prohibits the discharge into the sea of noxious liquid substances except under precisely defined conditions. A record of these operations shall be completed, where appropriate, in the form of an Oil or Cargo Record Book as applicable and shall be kept in such a place as to be readily available for inspection at all reasonable times.

1.2 The regulations referred to above provide that whenever visible traces of oil are observed on or below the surface of the water in the immediate vicinity of a ship or of its wake, a Party should, to the extent that it is reasonably able to do so, promptly investigate the facts bearing on the issue of whether or not there has been a violation of the discharge provisions.

1.3 The conditions under which noxious liquid substances are permitted to be discharged into the seas include quantity, quality, and position limitations, which depend on category of substance and sea area.

1.4 An investigation into an alleged contravention should therefore aim to establish whether a noxious liquid substance has been discharged and whether the operations leading to that discharge were in accordance with the ship's Procedures and Arrangements Manual (P and A Manual).

1.5 Recognizing the likelihood that many of the violations of the discharge provisions will take place outside the immediate control and knowledge of the flag State, Article 6 of MARPOL provides that Parties shall cooperate in the detection of violations and the enforcement of the provisions using all appropriate and practicable measures of detection and environmental monitoring, adequate procedures for reporting and gathering evidence. MARPOL also contains a number of more specific provisions designed to facilitate that cooperation.

1.6 Several sources of information about possible violations of the discharge provisions can be indicated. These include:

- .1 Reports by masters: Article 8 and Protocol I of MARPOL require, *inter alia*, a ship's master to report certain incidents involving the discharge or the probability of a discharge of oil or oily mixtures, or noxious liquid substances or mixtures containing such substances;
- .2 Reports by official bodies: Article 8 of MARPOL requires furthermore that a Party issue instructions to its maritime inspection vessels and aircraft and to other appropriate services to report to its authorities incidents involving the discharge or the probability of a discharge of oil or oily mixtures, or noxious liquid substances or mixtures containing such substances;
- .3 Reports by other Parties: Article 6 of MARPOL provides that a Party may request another Party to inspect a ship. The Party making the request shall supply sufficient evidence that the ship has discharged oil or oily mixtures,

noxious liquid substances or mixtures containing such substances, or that the ship has departed from the unloading port with residues of noxious liquid substances in excess of those permitted to be discharged into the sea; and

- .4 Reports by others: It is not possible to list exhaustively all sources of information concerning alleged contravention of the discharge provisions. Parties should take all circumstances into account when deciding upon investigating such reports.

1.7 Action which can be taken by States other than the flag or port States that have information on discharge violations (hereinafter referred to as coastal States):

- .1 Coastal States, Parties to MARPOL, upon receiving a report of pollution by oil or noxious liquid substances allegedly caused by a ship, may investigate the matter and collect such evidence as can be collected. For details of the desired evidence reference is made to appendices 3 and 4;
- .2 If the investigation referred to under subparagraph .1 above discloses that the next port of call of the ship in question lies within its jurisdiction, the coastal State should also take port State action as set out in paragraphs 2.1 to 2.6 below;
- .3 If the investigation referred to in subparagraph .1 above discloses that the next port of call of the ship in question lies within the jurisdiction of another Party, then the coastal State should in appropriate cases furnish the evidence to that other Party and request that Party to take port State action in accordance with paragraphs 2.1 to 2.6 below; and
- .4 In either case referred to in subparagraphs .2 and .3 above and if the next port of call of the ship in question cannot be ascertained, the coastal State shall inform the flag State of the incident and of the evidence obtained.

## **2 Port State action**

2.1 Parties shall appoint or authorize officers to carry out investigations for the purpose of verifying whether a ship has discharged oil or noxious liquid substances in violation of the provisions of MARPOL.

2.2 Parties may undertake such investigations on the basis of reports received from sources indicated in paragraph 1.6 above.

2.3 These investigations should be directed toward the gathering of sufficient evidence to establish whether the ship has violated the discharge requirements. Guidelines for the optimal collation of evidence are given in appendices 3 and 4.

2.4 If the investigations provide evidence that a violation of the discharge requirements took place within the jurisdiction of the port State, that port State shall either cause proceedings to be taken in accordance with its law, or furnish to the flag State all information and evidence in its possession about the alleged violation. When the port State causes proceedings to be taken, it shall inform the flag State.

2.5 Details of the report to be submitted to the flag State are set out in appendix 16.

2.6 The investigation might provide evidence that pollution was caused through damage to the ship or its equipment. This might indicate that a ship is not guilty of a violation of the discharge requirements of Annex I or Annex II of MARPOL provided that:

- .1 all reasonable precautions have been taken after the occurrence of the damage or discovery of the discharge for the purpose of preventing or minimizing the discharge; and
- .2 the owner or the master did not act either with intent to cause damage or recklessly and with knowledge that damage would probably result.

2.7 However, action by the port State as set out in chapter 3 of these Procedures may be called for.

### **3 Inspection of crude oil washing (COW) operations**

3.1 Regulations 18, 33 and 35 of MARPOL Annex I, *inter alia*, require that crude oil washing of cargo tanks be performed on certain categories of crude carriers. A sufficient number of tanks shall be washed in order that ballast water is put only in cargo tanks which have been crude oil washed. The remaining cargo tanks shall be washed on a rotational basis for sludge control.

3.2 Port State Authorities may carry out inspections to ensure that crude oil washing is performed by all crude carriers either required to have a COW system or where the owner or operator chooses to install a COW system in order to comply with regulation 18 of MARPOL Annex I. In addition compliance should be ensured with the operational requirements set out in the revised Specifications for the Design, Operation and Control of Crude Oil Washing Systems (resolution A.446(XI) as amended). This can best be done in the ports where the cargo is unloaded.

3.3 Parties should be aware that the inspection referred to in paragraph 3.2 may also lead to the identification of a pollution risk, necessitating additional action by the port State as set out in chapter 3 of these Procedures.

3.4 Detailed guidelines for in-port inspections of crude oil washing procedures have been approved and published by IMO (Crude Oil Washing Systems, revised edition, 1983) and are set out in part 4 to appendix 3.

### **4 Inspection of unloading, stripping and prewash operations**

4.1 Regulation 16 of MARPOL Annex II requires Parties to MARPOL to appoint or authorize surveyors for the purpose of implementing the regulation.

4.2 The provisions of regulation 16 are aimed at ensuring in principle that a ship having unloaded, to the maximum possible extent, noxious liquid substances of category X, Y or Z, proceeds to sea only if residues of such substances have been reduced to such quantities as may be discharged into the sea.

4.3 Compliance with these provisions is in principle ensured in the case of categories X, Y and Z substances through the application of a prewash in the unloading port and the discharge of prewash residue water mixtures to reception facilities, except that in the case of non-solidifying and low viscosity categories Y and Z substances, requirements for the efficient stripping of a tank to negligible quantities apply in lieu of the application of a

prewash. Alternatively for a number of substances ventilation procedures may be employed for removing cargo residues from a tank.

4.4 Regulation 16.6 permits the Government of the receiving Party to exempt a ship proceeding to a port or terminal under the jurisdiction of another Party from the requirement to prewash cargo tanks and discharge residue/water mixtures to a reception facility.

4.5 Existing chemical tankers engaged on restricted voyages may by virtue of regulation 4.3 of MARPOL Annex II be exempted from the quantity limitation requirements of regulations 12.1 to 12.3. If a cargo tank is to be ballasted or washed, a prewash is required after unloading category Y or Z substances and prewash residue water mixtures must be discharged to shore reception facilities. The exemption should be indicated on the certificate.

4.6 A ship whose constructional and operational features are such that ballasting of cargo tanks is not required and cargo tank washing is only required for repairs or dry-docking may by virtue of regulation 4.4 be exempted from the provisions of regulation 12 of MARPOL Annex II provided that all conditions mentioned in regulation 4.4 are complied with. Consequentially, the certificate of the ship should indicate that each cargo tank is only certified for the carriage of one named substance. It should also indicate the particulars of the exemption granted by the Administration in respect of pumping, piping and discharge arrangements.

4.7 Detailed instructions on efficient stripping and prewash procedures are included in a ship's Procedures and Arrangements Manual. The Manual also contains alternative procedures to be followed in case of equipment failure.

4.8 Parties should be aware that the inspection referred to in paragraphs 1.3 and 1.4 above may lead to the identification of a pollution risk or of a contravention of the discharge provisions, necessitating port State action as set out in chapter 3 of these Procedures.

4.9 For details in respect of inspections under this section reference is made to appendix 4.



## APPENDIX 6

### **GUIDELINES FOR MORE DETAILED INSPECTIONS OF SHIP STRUCTURAL AND EQUIPMENT REQUIREMENTS**

#### **1 Introduction**

If the PSCO from general impressions or observations on board has clear grounds for believing that the ship might be substandard, the PSCO should proceed to a more detailed inspection, taking the following considerations into account.

#### **2 Structure**

2.1 The PSCO's impression of hull maintenance and the general state on deck, the condition of such items as ladderways, guard rails, pipe coverings and areas of corrosion or pitting should influence the PSCO's decision as to whether it is necessary to make the fullest possible examination of the structure with the ship afloat. Significant areas of damage or corrosion, or pitting of plating and associated stiffening in decks and hull affecting seaworthiness or strength to take local loads, may justify detention. It may be necessary for the underwater portion of the ship to be checked. In reaching a decision, the PSCO should have regard to the seaworthiness and not the age of the ship, making an allowance for fair wear and tear over the minimum acceptable scantlings. Damage not affecting seaworthiness will not constitute grounds for judging that a ship should be detained, nor will damage that has been temporarily but effectively repaired for a voyage to a port for permanent repairs. However, in this assessment of the effect of damage, the PSCO should have regard to the location of crew accommodation and whether the damage substantially affects its habitability.

2.2 The PSCO should pay particular attention to the structural integrity and seaworthiness of bulk carriers and oil tankers and note that these ships must undergo the enhanced programme of inspection during surveys under the provision of regulation XI-1/2 of SOLAS.

2.3 The PSCO's assessment of the safety of the structure of those ships should be based on the Survey Report File carried on board. This file should contain reports of structural surveys, condition evaluation reports (translated into English and endorsed by or on behalf of the Administration), thickness measurement reports and a survey planning document. The PSCO should note that there may be a short delay in the update of the Survey Report File following survey. Where there is doubt that the required survey has taken place, the PSCO should seek confirmation from the recognized organization.

2.4 If the Survey Report File necessitates a more detailed inspection of the structure of the ship or if no such report is carried, special attention should be given by the PSCO, as appropriate, to hull structure, piping systems in way of cargo tanks or holds, pump-rooms, cofferdams, pipe tunnels, void spaces within the cargo area, and ballast tanks.

2.5 For bulk carriers, PSCOs should inspect holds' main structure for any obviously unauthorized repairs. For bulk carriers the port State control officer should verify that the bulk carrier booklet has been endorsed, the water level alarms in cargo holds are fitted, and where applicable, that any restrictions imposed on the carriage of solid bulk cargoes have been recorded in the booklet and the bulk carrier loading triangle is permanently marked.

### **3 Machinery spaces**

3.1 The PSCO should assess the condition of the machinery and of the electrical installations such that they are capable of providing sufficient continuous power for propulsion and for auxiliary services.

3.2 During inspection of the machinery spaces, the PSCO should form an impression of the standard of maintenance. Frayed, disconnected or inoperative quick-closing valve wires, disconnected or inoperative extended control rods or machinery trip mechanisms, missing valve hand wheels, evidence of chronic steam, water and oil leaks, dirty tank tops and bilges or extensive corrosion of machinery foundations are pointers to an unsatisfactory organization of the systems' maintenance. A large number of temporary repairs, including pipe clips or cement boxes, will indicate reluctance to make permanent repairs.

3.3 While it is not possible to determine the condition of the machinery without performance trials, general deficiencies, such as leaking pump glands, dirty water gauge glasses, inoperable pressure gauges, rusted relief valves, inoperative or disconnected safety or control devices, evidence of repeated operation of diesel engine scavenge belt or crankcase relief valves, malfunctioning or inoperative automatic equipment and alarm systems, and leaking boiler casings or uptakes, would warrant inspection of the engine-room log-book and investigation into the record of machinery failures and accidents and a request for running tests of machinery.

3.4 If one electrical generator is out of commission, the PSCO should investigate whether power is available to maintain essential and emergency services and should conduct tests.

3.5 If evidence of neglect becomes evident, the PSCO should extend the scope of an investigation to include, for example, tests on the main and auxiliary steering gear arrangements, overspeed trips, circuit breakers, etc.

3.6 It must be stressed that while detection of one or more of the above deficiencies would afford guidance to a substandard condition, the actual combination is a matter for professional judgement in each case.

### **4 Conditions of assignment of load lines**

It may be that the PSCO has concluded that a hull inspection is unnecessary but, if dissatisfied on the basis of observations on deck, with items such as defective hatch closing arrangements, corroded air pipes and vent coamings, the PSCO should examine closely the conditions of assignment of load lines, paying particular attention to closing appliances, means of freeing water from the deck and arrangements concerned with the protection of the crew.

### **5 Life-saving appliances**

5.1 The effectiveness of life-saving appliances depends heavily on good maintenance by the crew and their use in regular drills. The lapse of time since the last survey for a Safety Equipment Certificate can be a significant factor in the degree of deterioration of equipment if it has not been subject to regular inspection by the crew. Apart from failure to carry equipment required by a convention or obvious defects such as holed lifeboats, the PSCO should look for signs of disuse of, obstructions to, or defects with survival craft launching and recovery equipment which may include paint accumulation, seizing of pivot points, absence of greasing, condition of blocks and falls, condition of lifeboat lifting hook attachment to the lifeboat hull and improper lashing or stowing of deck cargo.

5.2 Should such signs be evident, the PSCO would be justified in making a detailed inspection of all life-saving appliances. Such an examination might include the lowering of survival craft, a check on the servicing of liferafts, the number and condition of lifejackets and lifebuoys and ensuring that the pyrotechnics are still within their period of validity. It would not normally be as detailed as that for a renewal of the Safety Equipment Certificate and would concentrate on essentials for safe abandonment of the ship, but in an extreme case could progress to a full Safety Equipment Certificate inspection. The provision and functioning of effective overside lighting, means of alerting the crew and passengers and provision of illuminated routes to assembly points and embarkation positions should be given importance in the inspection.

## **6 Fire safety**

6.1 Ships in general: The poor condition of fire and wash deck lines and hydrants and the possible absence of fire hoses and extinguishers in accommodation spaces might be a guide to a need for a close inspection of all fire safety equipment. In addition to compliance with convention requirements, the PSCO should look for evidence of a higher than normal fire risk; this might be brought about by a poor standard of cleanliness in the machinery space, which together with significant deficiencies of fixed or portable fire-extinguishing equipment could lead to a judgement of the ship being substandard.

6.2 Passenger ships: The PSCO should initially form an opinion of the need for inspection of the fire safety arrangements on the basis of consideration of the ship under the previous headings and, in particular, that dealing with fire safety equipment. If the PSCO considers that a more detailed inspection of fire safety arrangements is necessary, the PSCO should examine the fire control plan on board in order to obtain a general picture of the fire safety measures provided in the ship and consider their compliance with convention requirements for the year of build. Queries on the method of structural protection should be addressed to the flag Administration and the PSCO should generally confine the inspection to the effectiveness of the arrangements provided.

6.3 The spread of fire could be accelerated if fire doors are not readily operable. The PSCO should inspect for the operability and securing arrangements of those doors in the main zone bulkheads and stairway enclosures and in boundaries of high fire risk spaces, such as main machinery rooms and galleys, giving particular attention to those retained in the open position. Attention should also be given to main vertical zones which may have been compromised through new construction. An additional hazard in the event of fire is the spread of smoke through ventilation systems. Spot checks might be made on dampers and smoke flaps to ascertain the standard of operability. The PSCO should also ensure that ventilation fans can be stopped from the master controls and that means are available for closing main inlets and outlets of ventilation systems.

6.4 Attention should be given to the effectiveness of escape routes by ensuring that vital doors are not maintained locked and that alleyways and stairways are not obstructed.

## **7 Regulations for preventing collisions at sea**

A vital aspect of ensuring safety of life at sea is full compliance with the collision regulations. Based on observations on deck, the PSCO should consider the need for close inspection of lanterns and their screening and means of making sound and distress signals.

## **8 Cargo Ship Safety Construction Certificate**

The general condition of the ship may lead the PSCO to consider matters other than those concerned with safety equipment and assignment of load lines, but nevertheless associated with the safety of the vessel, such as the effectiveness of items associated with the Cargo Ship Safety Construction Certificate, which can include pumping arrangements, means for shutting off air and oil supplies in the event of fire, alarm systems and emergency power supplies.

## **9 Cargo Ship Safety Radio Certificates**

The validity of the Cargo Ship Safety Radio Certificates and associated Record of Equipment (Form R) may be accepted as proof of the provision and effectiveness of its associated equipment, but the PSCO should ensure that appropriate certificated personnel are carried for its operation and for listening periods. Requirements for maintenance of radio equipment are contained in SOLAS regulation IV/15. The radio log or radio records should be examined. Where considered necessary, operational checks may be carried out.

## **10 Means of access to ship**

10.1 Prior to boarding a ship, the PSCO should assess the means of embarkation on and disembarkation from the ship. The PSCO should be guided by SOLAS regulation II-1/3-9 noting its application for ships constructed on or after 1 January 2010 but also noting that paragraph 3 of this regulation applies to all ships and requires that:

- .1 the means of embarkation and disembarkation shall be inspected and maintained in suitable condition for their intended purpose, taking into account any restrictions related to safe loading; and
- .2 all wires used to support the means of embarkation and disembarkation shall be maintained as specified in SOLAS regulation III/20.4.

10.2 In regard to the maintenance of the means of embarkation and disembarkation, the PSCO should refer to the Guidelines for construction, installation, maintenance and inspection/survey of means of embarkation and disembarkation (MSC.1/Circ.1331).

10.3 During the inspection, the PSCO should also ensure that the pilot transfer arrangements comply with SOLAS regulation V/23 and the Unified interpretation of SOLAS regulation V/23 (MSC.1/Circ.1375).

## **11 Equipment in excess of convention or flag State requirements**

Equipment on board which is expected to be relied on in situations affecting safety or pollution prevention must be in operating condition. If such equipment is inoperative and is in excess of the equipment required by an appropriate convention and/or the flag State, it should be repaired, removed or, if removal is not practicable, clearly marked as inoperative and secured.

## APPENDIX 7

### GUIDELINES FOR CONTROL OF OPERATIONAL REQUIREMENTS

#### 1 Introduction

1.1 When, during a port State control inspection, the PSCO has clear grounds according to section 2.4 of the present Procedures, the following onboard operational procedures may be checked in accordance with this resolution. However, in exercising controls recommended in these guidelines, the PSCO should not include any operational tests or impose physical demands which, in the judgement of the master, could jeopardize the safety of the ship, crew, passengers, control officers or cargo.

1.2 When carrying out operational control, the PSCO should ensure, as far as possible, no interference with normal shipboard operations, such as loading and unloading of cargo and ballasting, which is carried out under the responsibility of the master, nor should the PSCO require demonstration of operational aspects which would unnecessarily delay the ship.

1.3 Having assessed the extent to which operational requirements are complied with, the PSCO then has to exercise professional judgement to determine whether the operational proficiency of the crew as a whole is of a sufficient level to allow the ship to sail without danger to the ship or persons on board, or presenting an unreasonable threat of harm to the marine environment.

1.4 When assessing the crew's ability to conduct an operational drill, the mandatory minimum requirements for familiarization and basic safety training for seafarers, as stated in the Convention STCW, shall be used as a benchmark.

#### 2 Muster list

2.1 The PSCO may determine if the crew members are aware of their duties indicated in the muster list.

2.2 The PSCO may ensure that muster lists are exhibited in conspicuous places throughout the ship, including the navigational bridge, the engine-room and the crew accommodation spaces. When determining if the muster list is in accordance with the regulations, the PSCO may verify whether:

- .1 the muster list shows the duties assigned to the different members of the crew;
- .2 the muster list specifies which officers are assigned to ensure that life-saving and fire appliances are maintained in good condition and are ready for immediate use;
- .3 the muster list specifies the substitutes for key persons who may become disabled, taking into account that different emergencies may call for different actions;
- .4 the muster list shows the duties assigned to crew members in relation to passengers in case of emergency; and

- .5 the format of the muster list used on passenger ships is approved and is drawn up in the language or languages required by the ship's flag State and in the English language.

2.3 To determine whether the muster list is up to date, the PSCO may require an up-to-date crew list, if available, to verify this.

2.4 The PSCO may determine whether the duties assigned to crew members manning the survival craft (lifeboats or liferafts) are in accordance with the regulations and verify that a deck officer or certificated person is placed in charge of each survival craft to be used. However, the Administration (of the flag State), having due regard to the nature of the voyage, the number of persons on board and the characteristics of the ship, may permit persons practised in the handling and operation of liferafts to be placed in charge of liferafts in lieu of persons qualified as above. A second-in-command shall also be nominated in the case of lifeboats.

2.5 The PSCO may determine whether the crew members are familiar with the duties assigned to them in the muster list and are aware of the locations where they should perform their duties.

### **3 Communication**

3.1 The PSCO may determine if the key crew members are able to communicate with each other, and with passengers as appropriate, in such a way that the safe operation of the ship is not impaired, especially in emergency situations.

3.2 The PSCO may ask the master which languages are used as the working languages and may verify whether the language has been recorded in the log-book.

3.3 The PSCO may ensure that the key crew members are able to understand each other during the inspection or drills. The crew members assigned to assist passengers should be able to give the necessary information to the passengers in case of an emergency.

### **4 Search and Rescue Plan**

For passenger ships, the PSCO may verify that there is on board an approved plan for cooperation with appropriate search and rescue services in event of an emergency.

### **5 Fire and abandon ship drills**

5.1 The PSCO witnessing a fire and abandon ship drill should ensure that the crew members are familiar with their duties and the proper use of the ship's installations and equipment.

5.2 When setting a drill scenario, witnessing the drill and finally assessing the standard of the drill, it is important to emphasize that the PSCO is not looking for an exceptional drill, particularly on cargo ships. The main points for the PSCO to be satisfied are:

- .1 In the event of a shipboard emergency can the crew organize themselves into an effective team to tackle the emergency?
- .2 Can the crew communicate effectively?

- .3 Is the master in control and is information flowing to/from the command centre? and
- .4 In the event of the situation getting out of hand can the crew safely abandon the ship?

5.3 It is important that when setting the scenario the PSCO clearly explains to the master exactly what is required and expected during the drill, bearing in mind there may be language difficulties. PSCOs should not be intimidating, not interfere during the drill nor offer advice. The PSCO should stand back and observe only, making appropriate notes. It is important to emphasize that the PSCO's role is not to teach or train but to witness.

5.4 Drills should be carried out at a safe speed. PSCOs should not expect to see operational drills conducted in real time. During drills, care should be taken to ensure that everybody familiarizes themselves with their duties and with the equipment. If necessary, drills should be stopped if the PSCO considers that the crew are carrying out unsafe practices or if there is a real emergency.

5.5 Language difficulty between the PSCO and non-English speaking crews can make it difficult to put across the intentions for the conduct of the exercise. Care needs to be exercised when an unsatisfactory drill takes place, this is to ensure differentiation between the crew possibly failing to understand the attending PSCO's intention and failure through lack of crew competence.

## **6 Fire drills**

6.1 The PSCO may witness a fire drill carried out by the crew assigned to these duties on the muster list. After consultation with the master of the vessel, one or more specific locations of the ship may be selected for a simulated fire. A crew member may be sent to the location(s) and activate a fire alarm system or use other means to give alarm.

6.2 At the location the PSCO can describe the fire indication to the crew member and observe how the report of fire is relayed to the bridge or damage control centre. At this point most ships will sound the crew alarm to summon the fire-fighting parties to their stations. The PSCO should observe the fire-fighting party arriving on the scene, breaking out their equipment and fighting the simulated fire. Team leaders should be giving orders as appropriate to their crews and passing the word back to the bridge or damage control centre on the conditions. The fire-fighting crews should be observed for proper donning and the use of their equipment. The PSCO should make sure that all the gear is complete. Merely mustering the crew with their gear is not acceptable. Crew response to personnel injuries can be checked by selecting a crew member as a simulated casualty. The PSCO should observe how the word is passed and the response of stretcher and medical teams. Handling a stretcher properly through narrow passageways, doors and stairways is difficult and takes practice.

6.3 The drill should, as far as practicable, be conducted as if there were an actual emergency.

6.4 Those crew members assigned to other duties related to a fire drill, such as the manning of the emergency generators, the CO<sub>2</sub> room, the sprinkler and emergency fire pumps, should also be involved in the drill. The PSCO may ask these crew members to explain their duties and if possible to demonstrate their familiarity.

6.5 On passenger ships, special attention should be paid to the duties of those crew members assigned to the closing of manually operated doors and fire dampers. These closing devices should be operated by the responsible persons in the areas of the simulated fire(s) during the drill. Crew members not assigned to the fire-fighting teams are generally assigned to locations throughout the passenger accommodations to assist in passenger evacuation. These crew members should be asked to explain their duties and the meaning of the various emergency signals and asked to point out the two means of escape from the area, and where the passengers are to report. Crew members assigned to assist passengers should be able to communicate at least enough information to direct a passenger to the proper muster and embarkation stations.

## **7 Abandon ship drills**

7.1 After consultation with the master, the PSCO may require an abandon ship drill for one or more survival craft. The essence of this drill is that the survival craft are manned and operated by the crew members assigned to them on the muster list. If possible the PSCO should include the rescue boat(s) in this drill. SOLAS chapter III gives specific requirements on abandon ship training and drills, of which the following principles are particularly relevant.

7.2 The drill should, as far as practicable, be conducted as if there was an actual emergency.

7.3 The abandon ship drill should include:

- .1 summoning of (passengers and) crew to the muster station(s) with the required alarm and ensuring that they are aware of the order to abandon ship as specified in the muster list;
- .2 reporting to the stations and preparing for the duties described in the muster list;
- .3 checking that (passengers and) crew are suitably dressed;
- .4 checking that lifejackets are correctly donned;
- .5 lowering of at least one lifeboat after the necessary preparation for launching;
- .6 starting and operating the lifeboat engine;
- .7 operation of the davits used for launching liferafts;
- .8 a mock search and rescue of passenger trapped in their staterooms (if applicable);
- .9 instructions in the use of radio life-saving appliances;
- .10 testing of emergency lighting for mustering and abandonment; and
- .11 if the ship is fitted with marine evacuation systems, exercising of the procedures required for the deployment of such systems up to the point immediately preceding actual deployment.



7.4 If the lifeboat lowered during the drill is not the rescue boat, the rescue boat should be lowered as well, taking into account that it is boarded and launched in the shortest possible time. The PSCO should ensure that crew members are familiar with the duties assigned to them during abandon ship operations and that the crew member in charge of the survival craft has complete knowledge of the operation and equipment of the survival craft. Care needs to be taken when requiring a ship to lower lifeboats. The number of persons inside the lifeboats during launching for the purpose of a drill should be at the master's discretion noting that SOLAS does not require persons in the lifeboat during lowering and recovery. The purpose of this is to reduce the risk of accidents during launching and recovery, however this must be balanced out with the risk of embarking/disembarking the boat whilst it is in the water, if the boat is to be taken away and run.

7.5 Each survival craft should be stowed in a state of continuous readiness so that two crew members can carry out preparations for embarking and launching in less than five minutes.

7.6 On passenger ships, it is required that lifeboats and davit-launched liferafts are capable of being launched within a period of 30 min after all persons have been assembled with lifejackets donned.

7.7 On cargo ships, it is required that lifeboats and davit launched liferafts are capable of being launched within a period of 10 min.

## **8 Assessment of drills**

8.1 When witnessing a drill, the PSCO should seek:

- .1 confirmation that the crew follow what is required of them by the muster list;
- .2 confirmation that there are sufficient personnel assigned to the various parties to cope with the duties given to them;
- .3 confirmation that there is an effective means of communication between the party, the party leader and the bridge and that relevant information is being passed bi-directionally;
- .4 confirmation of the efficiency of the crew working as a team. This would be based on questioning of personnel and observation of their actions. The response times should be noted of the various parties in assembling at their stations. The reaction of the parties to unplanned events should also be noted;
- .5 confirmation that key members of the crew are able to understand each other;
- .6 confirmation of the efficiency of the equipment used, for example:
  - .1 that the fire alarms are audible and efficient;
  - .2 that the fire doors close as required; and
  - .3 that items of personal fire-fighting equipment appear well maintained; and

- .7 confirmation that the response time was considered fast enough (taking into account safety of the drill as indicated in paragraph 5.4 of this appendix), considering the size of the ship and the locations of fire, personnel and fire-fighting equipment.

8.2 If the PSCO determines that the crew are unfamiliar with their duties or incapable of safely operating the life-saving and fire-fighting equipment, the PSCO should halt the drill, notify the master that the drill was unsuccessful and use their professional judgement to establish the next steps, noting the likelihood that this will establish "clear grounds" for a more detailed inspection.

## **9 Damage control plan and Shipboard Oil Pollution Emergency Plan (SOPEP) or Shipboard Marine Pollution Emergency Plans (SMPEP)**

9.1 The PSCO may determine if a damage control plan is provided on a passenger ship and whether the crew members are familiar with their duties and the proper use of the ship's installations and equipment for damage control purposes. The same applies with regard to SOPEP on all ships and SMPEP where applicable.

9.2 The PSCO may determine if the officers of the ship are aware of the contents of the damage control booklet which should be available to them, or of the damage control plan.

9.3 The officers may be asked to explain the action to be taken in various damage conditions.

9.4 The officers may also be asked to explain about the boundaries of the watertight compartments, the openings therein with the means of closure and position of any controls thereof and the arrangements for the correction of any list due to flooding.

9.5 The officers should have a sound knowledge of the effect of trim and stability of their ship in the event of damage to and consequent flooding of a compartment and counter-measures to be taken.

## **10 Fire control plan**

10.1 The PSCO may determine if a fire control plan or booklet is provided and whether the crew members are familiar with the information given in the fire control plan or booklet.

10.2 The PSCO may verify that fire control plans are permanently exhibited for the guidance of the ship's officers. Alternatively, booklets containing the information of the fire control plan may be supplied to each officer, and one copy should at all times be available on board in an accessible position. Plans and booklets should be kept up to date, any alterations being recorded thereon as soon as possible.

10.3 The PSCO may determine that the responsible officers, especially those who are assigned to related duties on the muster list, are aware of the information provided by the fire control plan or booklet and how to act in case of a fire.

10.4 The PSCO may ensure that the officers in charge of the ship are familiar with the principal structural members which form part of the various fire sections and the means of access to the different compartments.

## **11 Bridge operation**

11.1 The PSCO may determine if officers in charge of a navigational watch are familiar with bridge control and navigational equipment, changing the steering mode from automatic to manual and vice versa, and the ship's manoeuvring characteristics.

11.2 The officer in charge of a navigational watch should have knowledge of the location and operation of all safety and navigational equipment. Moreover, this officer should be familiar with procedures which apply to the navigation of the ship in all circumstances and should be aware of all information available.

11.3 The PSCO may also verify the familiarity of the officers on all the information available to them such as manoeuvring characteristics of the ship, life-saving signals, up-to-date nautical publications, checklists concerning bridge procedures, instructions, manuals, etc.

11.4 The Permit to Operate High-Speed Craft includes limitations of the maximum significant wave height (and wind force for hovercraft) within which the craft may operate. When carrying out inspections of HSC, PSCOs may verify by the log-book and the weather records whether these limitations have been respected. PSCOs may find that a voyage had to be completed when worse weather conditions than permitted were encountered, but a new voyage should not commence in such conditions.

11.5 The PSCO may verify the familiarity of the officers with procedures such as periodical tests and checks of equipment, preparations for arrival and departure, change over of steering modes, signalling, communications, manoeuvring, emergencies and log-book entries.

## **12 Cargo operation**

12.1 The PSCO may determine if ship's personnel assigned specific duties related to the cargo and cargo equipment are familiar with those duties, any dangers posed by the cargo and with the measures to be taken in such a context.

12.2 With respect to the carriage of solid bulk cargoes, the PSCO should verify, as appropriate, that cargo loading is performed in accordance with a ship's loading plan and unloading in accordance with a ship's unloading plan agreed by the ship and the terminal, taking into account the information provided by the loading instrument, where fitted.

12.3 The PSCO, when appropriate, may determine whether the responsible crew members are familiar with the relevant provisions of the International Maritime Solid Bulk Cargoes (IMSBC) Code, particularly those concerning moisture limits and trimming of the cargo, the Code of Safe Practice for Ships Carrying Timber Deck Cargoes (2011 TDC code) and the Code of Safe Practice for Cargo Stowage and Securing.

12.4 Some solid materials transported in bulk can present a hazard during transport because of their chemical nature or physical properties. Section 2 of the IMSBC Code gives general precautions. Section 4 of the IMSBC Code contains the obligation imposed on the shipper to provide all necessary information to ensure a safe transport of the cargo. The PSCO may determine whether all relevant details, including all relevant certificates of tests, have been provided to the master from the shipper.

12.5 For some cargoes, such as cargoes which are subject to liquefaction, special precautions are given (see section 7 of the IMSBC Code). The PSCO may determine

whether all precautions are met with special attention for the stability of those ships engaged in the transport of cargoes subject to liquefaction and solid hazardous waste in bulk.

12.6 Officers responsible for cargo handling and operation and key crew members of oil tankers, chemical tankers and liquefied gas carriers should be familiar with the cargo and cargo equipment and with the safety measures as stipulated in the relevant sections of the IBC and IGC Codes.

12.7 For the carriage of grain in bulk, reference is made to part C, chapter VI of SOLAS and the International Code for the Safe Carriage of Grain in Bulk (resolution MSC.23(59)).

12.8 The PSCO may determine whether the operations and loading manuals include all the relevant information for safe loading and unloading operations in port as well as in transit conditions.

### **13 Operation of the machinery**

13.1 The PSCO may determine if responsible ship's personnel are familiar with their duties related to operating essential machinery, such as:

- .1 emergency and stand-by sources of electrical power;
- .2 auxiliary steering gear;
- .3 bilge and fire pumps; and
- .4 any other equipment essential in emergency situations.

13.2 The PSCO may verify whether the responsible ship's personnel are familiar with, *inter alia*:

- .1 emergency generator:
  - .1 actions which are necessary before the engine can be started;
  - .2 different possibilities to start the engine in combination with the source of starting energy; and
  - .3 procedures when the first attempts to start the engine fail.
- .2 stand-by generator engine:
  - .1 possibilities to start the stand-by engine, automatic or by hand;
  - .2 blackout procedures; and
  - .3 load-sharing system.

13.3 The PSCO may verify whether the responsible ship's personnel are familiar with, *inter alia*:

- .1 which type of auxiliary steering gear system applies to the ship;

- .2 how it is indicated which steering gear unit is in operation; and
- .3 what action is needed to bring the auxiliary steering gear into operation.

13.4 The PSCO may verify whether the responsible ship's personnel are familiar with, *inter alia*:

- .1 bilge pumps:
  - .1 number and location of bilge pumps installed on board the ship (including emergency bilge pumps);
  - .2 starting procedures for all these bilge pumps;
  - .3 appropriate valves to operate; and
  - .4 most likely causes of failure of bilge pump operation and their possible remedies.
- .2 fire pumps:
  - .1 number and location of fire pumps installed on board the ship (including the emergency fire pump);
  - .2 starting procedures for all these pumps; and
  - .3 appropriate valves to operate.

13.5 The PSCO may verify whether the responsible ship's personnel are familiar with, *inter alia*:

- .1 starting and maintenance of lifeboat engine and/or rescue boat engine;
- .2 local control procedures for those systems which are normally controlled from the navigating bridge;
- .3 use of the emergency and fully independent sources of electrical power of radio installations;
- .4 maintenance procedures for batteries;
- .5 emergency stops, fire detection system and alarm system operation of watertight and fire doors (stored energy systems); and
- .6 change of control from automatic to manual for cooling water and lube oil systems for main and auxiliary engines.

## **14 Manuals, instructions, etc.**

14.1 The PSCO may determine if the appropriate crew members are able to understand the information given in manuals, instructions, etc., relevant to the safe condition and operation of the ship and its equipment and that they are aware of the requirements for maintenance, periodical testing, training, drills and recording of log-book entries.

14.2 The following information should, *inter alia*, be provided on board and PSCOs may determine whether it is in a language or languages understood by the crew and whether crew members concerned are aware of the contents and are able to respond accordingly:

- .1 instructions concerning the maintenance and operation of all the equipment and installations on board for the fighting and containment of fire should be kept under one cover, readily available in an accessible position;
- .2 clear instructions to be followed in the event of an emergency should be provided for every person on board;
- .3 illustrations and instructions in appropriate languages should be posted in passenger cabins and be conspicuously displayed at muster stations and other passenger spaces to inform passengers of their muster station, the essential action they must take in an emergency and the method of donning lifejackets;
- .4 posters and signs should be provided on or in the vicinity of survival craft and their launching controls and shall illustrate the purpose of controls and the procedures for operating the appliance and give relevant instructions or warnings;
- .5 instructions for onboard maintenance of life-saving appliances;
- .6 training manuals should be provided in each crew mess room and recreation room or in each crew cabin. The training manual, which may comprise several volumes, should contain instructions and information, in easily understood terms illustrated wherever possible, on the life-saving appliances provided in the ship and on the best method of survival;
- .7 Shipboard Oil Pollution Emergency Plan in accordance with regulation 37 of MARPOL Annex I, or Shipboard Marine Pollution Emergency Plan in accordance with regulation 17 of MARPOL Annex II, where applicable; and
- .8 stability booklet, associated stability plans and stability information.

## **15 Oil and oily mixtures from machinery spaces**

15.1 The PSCO may determine if all operational requirements of Annex I of MARPOL have been met, taking into account:

- .1 the quantity of oil residues generated;
- .2 the capacity of sludge and bilge water holding tank; and
- .3 the capacity of the oily water separator.

15.2 An inspection of the Oil Record Book should be made. The PSCO may determine if reception facilities have been used and note any alleged inadequacy of such facilities.

15.3 The PSCO may determine whether the responsible officer is familiar with the handling of sludge and bilge water. The relevant items from the guidelines for systems for handling oily wastes in machinery spaces of ships may be used as guidance. Taking into account the above, the PSCO may determine if the ullage of the sludge tank is sufficient for

the expected generated sludge during the next intended voyage. The PSCO may verify that, in respect of ships for which the Administration has waived the requirements of regulations 14(1) and (2) of MARPOL Annex I, all oily bilge water is retained on board for subsequent discharge to a reception facility.

15.4 When reception facilities in other ports have not been used because of inadequacy, the PSCO should advise the master to report the inadequacy of the reception facility to the ship's flag State, in conformity with the Revised consolidated format for reporting alleged inadequacy of port reception facilities (MEPC/Circ.349 of November 1998).

## **16 Loading, unloading and cleaning procedures for cargo spaces of tankers**

16.1 The PSCO may determine if all operational requirements of Annexes I or II of MARPOL have been met taking into account the type of tanker and the type of cargo carried, including the inspection of the Oil Record Book and/or Cargo Record Book. The PSCO may determine if the reception facilities have been used and note any alleged inadequacy of such facilities.

16.2 For the control on loading, unloading and cleaning procedures for tankers carrying oil, reference is made to paragraphs 3.1 to 3.4 in appendix 5 where guidance is given for the inspection of crude oil washing (COW) operations. In appendix 3, the PSCO may find detailed guidelines for in-port inspection of crude oil washing procedures.

16.3 For the control on loading, unloading and cleaning procedures for tankers carrying noxious liquid substances, reference is made to paragraphs 4.1 to 4.9 in appendix 5 where guidance is given for the inspection of unloading, stripping and prewash operations. In appendix 4 more detailed guidelines for these inspections are given.

16.4 When reception facilities in other ports have not been used because of inadequacy, the PSCO should advise the master to report the inadequacy of the reception facility to the ship's flag State, in conformity with MEPC/Circ.349 (November 1998).

16.5 When a vessel is permitted to proceed to the next port with residues of noxious liquid substances on board in excess of those permitted to be discharged into the sea during the ship's passage, it should be ascertained that the residues can be received by that port. At the same time that port should be informed if practicable.

## **17 Dangerous goods and harmful substances in packaged form**

17.1 The PSCO may determine if the required shipping documents for the carriage of dangerous goods and harmful substances carried in packaged form are provided on board and whether the dangerous goods and harmful substances are properly stowed and segregated and the crew members are familiar with the essential action to be taken in an emergency involving such packaged cargo (see SOLAS regulation VII/3).

17.2 Ships types and cargo spaces of ships over 500 gross tonnage built on, or after, 1 September 1984 and ships types and cargo spaces of ships less than 500 gross tonnage built on, or after, 1 February 1992 are to fully comply with the requirements of SOLAS chapter II-2. Administrations may reduce the requirements for ships of less than 500 gross tonnage but such reductions shall be recorded in the document of compliance. A document of compliance is not required for ships which only carry class 6.2, class 7 or dangerous goods in limited quantities.

17.3 Annex III of MARPOL contains requirements for the carriage of harmful substances in packaged form which are identified in the IMDG Code as marine pollutants. Cargoes which are determined to be marine pollutants should be labelled and stowed in accordance with Annex III of MARPOL.

17.4 The PSCO may determine whether a Document of Compliance is on board and whether the ship's personnel are familiar with this document provided by the Administration as evidence of compliance of construction and equipment with the requirements. Additional control may consist of:

- .1 whether the dangerous goods have been stowed on board in conformity with the Document of Compliance, using the dangerous goods manifest or the stowage plan, required by SOLAS chapter VII. This manifest or stowage plan may be combined with the one required under Annex III of MARPOL;
- .2 whether inadvertent pumping of leaking flammable or toxic liquids is not possible in case these substances are carried in under-deck cargo spaces; or
- .3 determining whether the ship's personnel are familiar with the relevant provisions of the Medical First Aid Guide and Emergency Procedures for Ships Carrying Dangerous Goods.

## **18 Garbage**

18.1 The PSCO may determine if all operational requirements of Annex V of MARPOL have been met. The PSCO may determine if the reception facilities have been used and note any alleged inadequacy of such facilities.

18.2 Guidelines for the implementation of Annex V of MARPOL were approved by the MEPC at its twenty-ninth session and have been amended on numerous occasions. The Guidelines can be found within the consolidated text of MARPOL Annex V. One of the objectives of these guidelines is to assist ship operators complying with the requirements set forth in Annex V and domestic laws.

18.3 The PSCO may determine whether:

- .1 ship's personnel are aware of these Guidelines, in particular section 3 "Minimizing the amount of potential garbage" and section 4 "Shipboard garbage handling and storage procedures"; and
- .2 ship's personnel are familiar with the disposal and discharge requirements under Annex V of MARPOL inside and outside a special area and are aware of the areas determined as special areas under Annex V of MARPOL.

18.4 When reception facilities in other ports have not been used because of inadequacy, the PSCO should advise the master to report the inadequacy of the reception facility to the ship's flag State, in conformity with MEPC/Circ.349 (November 1998).



## **19 Sewage**

19.1 The PSCO may determine:

- .1 if all operational requirements of Annex IV of MARPOL have been met. The PSCO may determine if the sewage treatment system, comminuting and disinfecting system or holding tank has been used and note any alleged inadequacy of the system or holding tank; and
- .2 that appropriate ship's personnel are familiar with the correct operation of the sewage treatment system, comminuting and disinfecting system or holding tank.

19.2 The PSCO may determine whether appropriate ship's personnel are familiar with the discharge requirements under regulation 11 of MARPOL Annex IV.

19.3 When reception facilities in other ports have not been used because of inadequacy, the PSCO should advise the master to report the inadequacy of the reception facility to the ship's flag State, in conformity with the Waste reception facility reporting requirements (MEPC/Circ.470 of 27 July 2005).

## **20 Air pollution prevention**

The PSCO may determine whether:

- .1 the master or crew is familiar with the procedures to prevent emissions of ozone-depleting substances;
- .2 the master or crew is familiar with the proper operation and maintenance of diesel engines, in accordance with their Technical Files;
- .3 the master or crew has undertaken the necessary fuel changeover procedures or equivalent, associated with demonstrating compliance within a SO<sub>x</sub> emission control area;
- .4 the master or crew is familiar with the garbage screening procedure to ensure that prohibited garbage is not incinerated;
- .5 the master or crew is familiar with the operation of the shipboard incinerator, as required by regulation 16(2) of MARPOL Annex VI, within the limit provided in appendix IV to the Annex, in accordance with the operational manual;
- .6 the master or crew recognizes the regulation of emissions of volatile organic compounds (VOCs), when the ship is in ports or terminals under the jurisdiction of a Party to the 1997 Protocol to MARPOL in which VOCs emissions are to be regulated, and is familiar with the proper operation of a vapour collection system approved by the Administration (in case the ship is a tanker as defined in regulation 2(12) of MARPOL Annex VI); and
- .7 the master or crew is familiar with bunker delivery procedures in respect of bunker delivery notes and retained samples as required by regulation 18 of MARPOL Annex VI.

## APPENDIX 8

### GUIDELINES FOR PORT STATE CONTROL RELATED TO THE ISM CODE

1 To the extent applicable, the PSCO should examine the copy of the Document of Compliance (DOC), issued to the company, and the Safety Management Certificate (SMC), issued to the ship. An SMC is not valid unless the company holds a valid DOC for that ship type. The PSCO should in particular verify that the type of ship is included in the DOC and that the company's particulars are the same on both the DOC and the SMC.

2 During the examination of on board documents and certificates, PSCOs should recognize:

- .1 the common practice of issuing, after successfully completing an audit, SMCs and DOCs valid for a period not exceeding 5 months, to cover the period between completion of the audit and issuance of the full term certificate by either the Administration or the recognized organization; and
- .2 that the current valid DOC with proper annual endorsements is normally only available in the company to which it has been issued and that the copy on board may not reflect the annual endorsements that exist on the valid DOC held by the company.

3 If a ship has on board Interim Certificates (DOC and/or SMC), the PSCO should check whether they have been issued in accordance with the provisions of ISM Code paragraphs 14.1 and 14.2. The PSCO should take into consideration the planned arrangements for the implementation of the Safety Management System as referred to in the ISM Code, paragraph 14.4, and should recognize that the full and effective functioning of the SMS has not been audited under an Interim SMC as per the ISM Code.

4 A more detailed inspection of the Safety Management System (SMS) should be carried out if clear grounds are established. Clear grounds may include absent or inaccurate ISM Code certification or detainable (or many non-detainable) deficiencies in other areas.

5 When carrying out a more detailed inspection, the PSCO may utilize, but not be limited to, the following questions to ascertain the extent of compliance with the ISM Code (references to the relevant paragraphs of the ISM Code are given in *italic print in brackets*).

- .1 Is there a company safety and environmental protection policy and is the appropriate ship's personnel familiar with it? (*paragraph 2.2*)
- .2 Is safety management documentation (e.g., manual) readily available on board? (*paragraph 11.3*)
- .3 Is relevant documentation on the SMS in a working language or language understood by the ship's personnel? (*paragraph 6.6*)
- .4 Can senior ship officers identify the company responsible for the operation of the ship and does this correspond with the entity specified on the ISM Code certificates? (*paragraph 3*)
- .5 Can senior ship officers identify the "designated person"? (*paragraph 4*)

- .6 Are procedures in place for establishing and maintaining contact with shore management in an emergency? (*paragraph 8.3*)
- .7 Are programmes for drills and exercises to prepare for emergency actions available on board? (*paragraph 8.2*)
- .8 How have new crew members been made familiar with their duties if they have recently joined the ship and are instructions which are essential prior to sailing available? (*paragraph 6.3*)
- .9 Can the master provide documented proof of his responsibilities and authority, which must include his overriding authority? (*paragraph 5*)
- .10 Have non-conformities been reported to the company and has corrective action been taken by the company? PSCOs should not normally scrutinize the contents of any Non Conformity Note (NCN) resulting from internal audits. (*paragraphs 9.1 and 9.2*)
- .11 Does the ship have a maintenance routine and are records available? (*paragraph 10.2*)

6 Deficiencies in the Safety Management System should be recorded in the PSCO's inspection report. The port State Authority should, if necessary, inform the flag State of deficiencies found in the SMS. Those deficiencies identified in the SMS, which are defined as major non-conformities in the Revised guidelines on implementation of the ISM Code by Administrations (resolution A.913(22)), have to be rectified by removing the immediate threat or hazard before sailing. Whenever the deficiencies identified during the inspection are indicative of the existence of a major non-conformity resulting in the vessel's detention, an additional audit shall be carried out by the flag State or the recognized organization acting on its behalf to determine compliance or non-compliance in accordance with the procedures for safety management audits. The procedures set out in chapter 3 of those Procedures are applicable.

## APPENDIX 9

### GUIDELINES FOR PORT STATE CONTROL RELATED TO LRIT

#### 1 PURPOSE

These Guidelines are intended to provide basic guidance to PSCOs to verify compliance with the requirements of SOLAS for Long Range Identification and Tracking (LRIT).

#### 2 APPLICATION

2.1 LRIT equipment is required by SOLAS regulation V/19-1, and resolution MSC.263(84) requires all passenger ships, cargo ships (including high-speed craft) over 300 tons and Mobile Offshore Drilling Units (MODU) to send LRIT position information at least every 6 hours. Ships fitted with Automatic Identification System (AIS) and operated exclusively within sea area A1 are not required to comply with LRIT. Sea area A1 is defined by SOLAS regulation IV/2.1.12 as "an area within the radiotelephone coverage of at least one VHF coast station in which continuous DSC alerting is available, as may be defined by a Contracting Government".

2.2 SOLAS Contracting Governments are expected to maintain a LRIT Data Centre, either on a national basis, or on a regional or cooperative basis with other flag States, and notify the IMO of it. In turn, these LRIT Data Centres will forward, upon request, LRIT information from ships entitled to fly their flag, to other SOLAS Contracting Governments through the International LRIT Data Exchange. Port States are entitled to request the LRIT information from foreign ships that have indicated their intention to enter a port, port facility or place under its jurisdiction.

2.3 In most cases a stand-alone Inmarsat C or Inmarsat Mini-C terminal used for GMDSS or Ship Security Alert System will function as the LRIT terminal, but other equipment may be employed for the LRIT function (example Inmarsat D+ or Iridium).

#### 3 INSPECTION OF SHIPS REQUIRED TO CARRY LRIT EQUIPMENT

##### 3.1 Initial inspection

3.1.1 The PSCO should first establish the sea area the ship is certified to operate in. This verification should ensure that the ship is subject to the LRIT regulation in relation to its ship type and tonnage. After the certificate check, the PSCO should verify that:

- .1 the Record of Equipment (Form E, P or C) indicates LRIT as required, if applicable\*;
- .2 a Statement of Conformity/Conformance Test Report (see MSC.1/Circ.1307) is on board; and
- .3 the equipment identified by the ship's representative as the designated LRIT terminal is switched on.\*\*

---

\* Noting that a Record of Equipment is required for cargo ships greater than 500 gross tonnage and passenger ships.

\*\* Note: In exceptional circumstances and for shortest duration possible LRIT is capable of being switched off or may transmit less frequently (SOLAS regulation V/19-1.7.2) and resolution MSC.263(84), paragraph 4.4.1.

3.1.2 In case of recent transfer of flag, the PSCO may further ensure that:

- .1 a conformance test report has been re-issued if the new flag State does not recognize the issuing body of the existing conformance test report; or
- .2 a new conformance test has been carried out by the Application Service Provider (ASP) on behalf of the Administration before issuance of a new test report and certificate.

### **3.2 Clear grounds**

Conditions which may warrant a more detailed inspection of equipment used for LRIT may comprise the following:

- .1 defective main or emergency source of energy;
- .2 information or indication that LRIT equipment is not functioning properly;
- .3 ship does not hold conformance test report; and
- .4 the "record of navigational activities" indicates that the LRIT installation has been switched off and that this has not been reported to the flag Administration as required by SOLAS regulation V/19-1.7.2.

### **3.3 More detailed inspection**

3.3.1 In case of doubt or reports of malfunctioning of the LRIT installation, the flag Administration may be contacted to determine if the ship's LRIT information has been reliably relayed to the LRIT Data Centre.

3.3.2 If any issues are identified at the initial inspection, a more detailed inspection of equipment used for LRIT may comprise the following:

- .1 verification of the power supply which should be connected to the main source of energy and the emergency source of energy, there is no requirement for an uninterrupted power source. If the LRIT is part of the GMDSS radio-installation, the power supply should conform to GMDSS regulations;
- .2 inspection of the "record of navigational activities" log to establish if and when the installation has been switched off and if this has been reported to the flag Administration (SOLAS regulation V/19-1.7.2 and resolution MSC.263(84), paragraph 4.4.1); and
- .3 ensuring that any conformance test report is issued on behalf of the flag State, even by itself or by an authorized Application Service Provider (see MSC.1/Circ.1377/Rev.2, as may be updated), available for a ship that has installation of LRIT.

#### **4 Deficiencies warranting detention**

4.1 A PSCO should use professional judgment to determine whether to detain the ship until any noted deficiencies are corrected or to permit a vessel to sail with deficiencies\*.

4.2 In order to assist the PSCO in the use of these Guidelines, the following deficiencies should be considered to be of such nature that they may warrant the detention of a ship:

- .1 absence of a valid LRIT Conformance test report; and
- .2 the master or the responsible officer are not familiar with essential shipboard operational procedures relating to LRIT.

4.3 Taking into account the guidance found in the Guidance on the implementation of the LRIT system (MSC.1/Circ.1298), PSCOs are also advised that ships should not be detained if the LRIT installation on board works, but the shore-side installation or organization is not able to receive, relay or process the information.

4.4 PSCOs are advised that a flag State may issue a short-term certificate. This could happen if, following a successful inspection for the issuance of a Conformity Test report, the ASP has not been able to issue a document yet, or if the ASP is not able to perform a conformance test in due time upon request of the shipowner.

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\* SOLAS regulation V/16: "Whilst all reasonable steps shall be taken to maintain the equipment required by this chapter in efficient working order, malfunctions of that equipment shall not be considered as making the ship unseaworthy or as a reason for delaying the ship in ports where repair facilities are not readily available, provided suitable arrangements are made by the master to take the inoperative equipment or unavailable information into account in planning and executing a safe voyage to a port where repairs can take place."

## APPENDIX 10

### **GUIDELINES FOR PORT STATE CONTROL UNDER THE 1969 TONNAGE CONVENTION**

1 The International Convention on Tonnage Measurement of Ships, 1969, which came into force on 18 July 1982, applies to:

- .1 new ships, i.e. ships the keels of which were laid on or after 18 July 1982; and
- .2 existing ships, i.e. ships the keels of which were laid before 18 July 1982, as from 18 July 1994,

except that for the purpose of application of the SOLAS, MARPOL and STCW Conventions, the following interim schemes indicated in paragraph 2 may apply.

2 In accordance with the interim schemes adopted by the Organization\*, the Administration may, at the request of the shipowner, use the gross tonnage determined in accordance with national rules prior to the coming into force of the 1969 Tonnage Convention, for the following ships:

- .1 for the purpose of SOLAS:
  - .1 ships the keels of which were laid before 1 January 1986;
  - .2 in respect of SOLAS regulation IV/3, ships the keels of which were laid on or after 1 January 1986 but before 18 July 1994; and
  - .3 cargo ships of less than 1,600 tons gross tonnage (as determined under the national tonnage rules) the keels of which were laid on or after 1 January 1986 but before 18 July 1994;
- .2 for the purpose of STCW, ships falling under the categories of paragraphs 2.1.1 and 2.1.3 above, except that for the purpose of 1995 amendments to STCW the interim scheme does not apply (see regulation I/15.3 of the 1995 STCW amendments); and
- .3 for the purpose of MARPOL ships of less than 400 tons gross tonnage (as determined under the national tonnage rules) the keel of which were laid before 18 July 1994.

3 For ships to which the above interim schemes apply, a statement to the effect that the gross tonnage has been measured in accordance with the national tonnage rules should be included in the "REMARKS" column of the International Tonnage Certificate (1969) and in the footnote to the figure of the gross tonnage in the relevant SOLAS, MARPOL and STCW certificates.

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\* Resolutions A.494(XII) in respect to SOLAS, A.540(13) in respect to STCW 78, and A.541(13) in respect to MARPOL.

4 The PSCO should take the following actions as appropriate when deficiencies are found in relation to the 1969 Tonnage Convention:

- .1 if a ship does not hold a valid International Tonnage Certificate (1969), the ship loses all privileges of the 1969 Tonnage Convention, and the flag State should be informed without delay;
- .2 if the required remarks and footnote are not included in the relevant certificates on ships to which the interim schemes apply, this deficiency should be notified to the master; and
- .3 if the main characteristics of the ship differ from those entered on the International Tonnage Certificate (1969), so as to lead to an increase in the gross tonnage or net tonnage, the flag State should be informed without delay.

5 The control provisions of article 12 of the 1969 Tonnage Convention do not include the provision for detention of a ship holding a valid International Tonnage Certificate (1969).



## APPENDIX 11

### MINIMUM MANNING STANDARDS AND CERTIFICATION

#### 1 Introduction

The guiding principles for port State control of the manning of a foreign ship should be to establish conformity with:

- .1 the flag State's safe manning requirements. Where this is in doubt the flag State should be consulted; and
- .2 the international provisions as laid down in SOLAS, STCW and in the Principles of minimum safe manning (resolution A.[...](27)).

#### 2 Manning control

2.1 If a ship is manned in accordance with a safe manning document or equivalent document issued by the flag State, the PSCO should accept that the ship is safely manned unless the document has clearly been issued without regard to the principles contained in the relevant instruments in which case the PSCO should act according to the procedures defined in paragraph 2.3.

2.2 If the actual crew number or composition does not conform to the manning document, the port State should request the flag State for advice as to whether or not the ship should be allowed to sail with the actual number of crew and its composition. Such a request and response should be by expedient means and either Party may request this communication in writing. If the actual crew number or composition is not brought into accordance with the safe manning document or the flag State does not advise that the ship could sail, the ship may be considered for detention after the criteria set out in section 4 have been taken into proper account.

2.3 If the ship does not carry a safe manning document or equivalent, the port State should request the flag State to specify the required number of crew and its composition and to issue a document as quickly as possible.

2.4 In case the actual number or composition of the crew does not conform to the specifications received from the flag State the procedure as contained in paragraph 2.2 applies.

2.5 If the flag State does not respond to the request this should be considered as clear grounds for a more detailed inspection to ensure that the number and composition of the crew is in accordance with the principles laid down in section 1 above. The ship should only be allowed to proceed to sea if it is safe to do so, taking into account the criteria for detention indicated in section 4. In any such case the minimum standards to be applied should be no more stringent than those applied to ships flying the flag of the port State.

### **3 Control under the provisions of STCW**

Control exercised by the PSCO should be limited to the following:

- .1 verification that all seafarers serving on board, who are required to be certificated, hold an appropriate certificate or a valid dispensation, or provide documentary proof that an application for an endorsement has been submitted to the Administration;
- .2 verification that the numbers and certificates of the seafarers serving on board are in conformity with the applicable safe manning requirements of the Administration; and
- .3 assessment of the ability of the seafarers of the ship to maintain watchkeeping standards as required by the Convention if there are clear grounds for believing that such standards are not being maintained because any of the following have occurred:
  - .1 the ship has been involved in a collision, grounding or stranding, or
  - .2 there has been a discharge of substances from the ship when underway, at anchor or at berth which is illegal under any international convention, or
  - .3 the ship has been manoeuvred in an erratic or unsafe manner whereby routing measures adopted by the Organization or safe navigation practices and procedures have not been followed, or
  - .4 the ship is otherwise being operated in such a manner as to pose a danger to persons, property or the environment.

### **4 Detention related to minimum manning standards and certification**

Before detaining a ship, the following should be considered:

- .1 length and nature of the intended voyage or service;
- .2 whether or not the deficiency poses a danger to ships, persons on board or the environment;
- .3 whether or not appropriate rest periods of the crew can be observed;
- .4 size and type of ship and equipment provided; and
- .5 nature of cargo.

## APPENDIX 12

### LIST OF CERTIFICATES AND DOCUMENTS

List of certificates and documents which to the extent applicable should be checked during the inspection referred to in paragraph 2.2.3 (as appropriate):

- 1 International Tonnage Certificate (1969);
- 2 Reports of previous port State control inspections;
- 3 Passenger Ship Safety Certificate (SOLAS reg.I/12);
- 4 Cargo Ship Safety Construction Certificate (SOLAS reg.I/12);
- 5 Cargo Ship Safety Equipment Certificate (SOLAS reg.I/12);
- 6 Cargo Ship Safety Radio Certificate (SOLAS reg.I/12);
- 7 Cargo Ship Safety Certificate (SOLAS reg.I/12);
- 8 Special Purpose Ship Safety Certificate (SOLAS reg.I/12, SPS Code reg.1.7.12 and res.A.791(19));
- 9 For ro-ro passenger ships, information on the A/A-max ratio (SOLAS reg.II-1/8-1\*);
- 10 Damage control plans and booklets (SOLAS reg.II-1/19);
- 11 Stability information (SOLAS reg.II-1/5-1 and LLC66/88 reg.10);
- 12 Manoeuvring Booklet and information (SOLAS reg.II-1/28.2);
- 13 Unattended machinery spaces (UMS) evidence (SOLAS reg.II-I/46.3);
- 14 Exemption Certificate and any list of cargoes (SOLAS reg.II-2/10.7.1.4);
- 15 Fire control plan (SOLAS reg.II-2/15.2.4);
- 16 Fire safety operational booklet (SOLAS reg.II-2/16.3.1);
- 17 Dangerous goods special list or manifest, or detailed stowage plan (SOLAS reg.II-2/19 and ILO Convention No. 134 art.4.3(h));
- 18 Document of compliance Dangerous Goods (SOLAS reg.II-2/19.4);
- 19 Ship's log-book with respect to the records of drills, including security drills, and the log for records of inspection and maintenance of life-saving appliances and arrangements and fire-fighting appliances and arrangements (SOLAS regs.III/19.5 and 20.6);
- 20 Minimum Safe Manning Document (SOLAS reg.V/14.2);

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\* Refer to Resolution 1 of the 1995 SOLAS Conference.

- 21 SAR coordination plan for passenger ships trading on fixed routes (SOLAS reg.V/7.3);
- 22 LRIT Conformance Test Report
- 23 Copy of the Document of compliance issued by the testing facility, stating the date of compliance and the applicable performance standards of VDR (voyage data recorder) (SOLAS reg.V/18.8);
- 24 For passenger ships, List of operational limitations (SOLAS reg.V/30.2);
- 25 Cargo Securing Manual (SOLAS reg.VI/5.6);
- 26 Bulk Carrier Booklet (SOLAS reg.VI/7.2);
- 27 Loading/Unloading Plan for bulk carriers (SOLAS reg.VI/7.3);
- 28 Document of authorization for the carriage of grain (SOLAS reg.VI/9);
- 29 INF (International Code for the Safe Carriage of Packaged Irradiated Nuclear Fuel, Plutonium and High-Level Radioactive Wastes on Board Ships) Certificate of Fitness (SOLAS reg.VII/16 and INF Code reg.1.3);
- 30 Copy of Document of Compliance issued in accordance with the International Management Code for the Safe Operation of Ships and for Pollution Prevention (DoC) ISM Code (SOLAS reg.IX/4.2);
- 31 Safety Management Certificate issued in accordance with the International Management Code for the Safe Operation of Ships and for Pollution Prevention (SMC) (SOLAS reg.IX/4.3);
- 32 High-Speed Craft Safety Certificate and Permit to Operate High-Speed Craft (SOLAS reg.X/3.2 and HSC Code 94/00 reg.1.8.1);
- 33 Continuous Synopsis Record (SOLAS reg.XI-1/5)
- 34 International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk, or the Certificate of Fitness for the Carriage of Liquefied Gases in Bulk, whichever is appropriate (IGC Code reg.1.5.4);
- 35 International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk, or the Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk, whichever is appropriate (IBC Code reg.1.5.4 and BCH Code reg.1.6.3);
- 36 International Oil Pollution Prevention Certificate (MARPOL Annex I reg.7.1);
- 37 Survey Report Files (in case of bulk carriers or oil tankers) (SOLAS reg.XI/2);
- 38 Oil Record Book, parts I and II (MARPOL Annex I regs.17 and 36);
- 39 Shipboard Marine pollution emergency plan for Noxious Liquid Substances (MARPOL Annex II reg.17);

- 40 (Interim) Statement of compliance Condition Assessment Scheme (CAS) (MARPOL Annex I regs.20.6 and 21.6.1);
- 41 For oil tankers, the record of oil discharge monitoring and control system for the last ballast voyage (MARPOL Annex I reg.31.2);
- 42 Shipboard Oil Pollution Emergency Plan (MARPOL Annex I reg.37.1);
- 43 International Pollution Prevention Certificate for the Carriage of Noxious Liquid Substances in Bulk (NLS) (MARPOL Annex II reg.9.1);
- 44 Cargo Record Book (MARPOL Annex II reg.15);
- 45 Procedures and Arrangements Manual (chemical tankers) (MARPOL Annex II reg.14.1);
- 46 International Sewage Pollution Prevention Certificate (ISPPC) (MARPOL Annex IV reg.5.1);
- 47 Garbage Management Plan (MARPOL Annex V reg.9.2);
- 48 Garbage Record Book (MARPOL Annex V reg.9.3);
- 49 International Air Pollution Prevention Certificate (IAPPC) (MARPOL Annex VI reg.6.1);
- 50 Log-book for fuel oil change-over (MARPOL Annex VI reg.14.6);
- 51 Type approval certificate of incinerator (MARPOL Annex VI reg.16.6);
- 52 Bunker delivery notes (MARPOL Annex VI reg.18.3);
- 53 Engine International Air Pollution Prevention Certificate (EIAPPC) (Nox Technical Code 2008 reg.2.1.1.1);
- 54 Technical files (Nox Technical Code 2008 reg.2.3.6);
- 55 Record book of engine parameters (Nox Technical Code reg.6.2.2.7.1);
- 56 International Load Line Certificate (1966) (LLC66/88 art.16.1);
- 57 International Load Line Exemption Certificate (LLC66/88 art.16.2);
- 58 Certificates issued in accordance with STCW Convention (STCW art.VI, reg.I/2 and STCW Code Section A-I/2);
- 59 Table of shipboard working arrangements (STCW Code Section A-VIII/1.5 and ILO Convention No. 180 art. 5.7);
- 60 Mobile Offshore Drilling Unit Safety Certificate (MODU Code 2009 chapter I section 6);

- 61 Certificate of insurance or any other financial security in respect of civil liability for oil pollution damage (CLC69/92 art.VII.2);
- 62 Certificate of insurance or any other financial security in respect of civil liability for Bunker oil pollution damage (BUNKERS 2001 art.7.2);
- 63 International Ship Security Certificate (ISSC) (ISPS Code part A/19.2);
- 64 Record of AFS (AFS 2001 Annex 4 reg.2);
- 65 International Anti-Fouling System Certificate (IAFS Certificate) (AFS 2001 Annex 4 reg.2); and
- 66 Declaration on AFS (AFS 2001 Annex 4 reg.5).

For reference:

- 1 Certificate of Registry or other document of nationality (UNCLOS art.9.1.2);
- 2 Certificates as to the ship's hull strength and machinery installations issued by the classification society in question (only to be required if the ship maintains its class with a classification society);
- 3 Cargo Gear Record Book (ILO Convention No. 32 art.9.2(4) and ILO Convention No. 152 art.25);
- 4 Certificates loading and unloading equipment (ILO Convention No. 134 art.4.3(e) and ILO Convention No. 32 art.9(4));
- 5 Medical certificates (ILO Convention No. 73); and
- 6 Records of hours of work or rest of seafarers (ILO Convention No.180 part II art. 8.1).

APPENDIX 13

**REPORT OF INSPECTION IN ACCORDANCE WITH IMO PORT STATE CONTROL  
PROCEDURES\***

**FORM A**

(Reporting authority)  
(Address)  
(Telephone)  
(Telefax)

Copy to: Master  
Head office  
PSCO

If ship is detained, copy to:  
Flag State  
IMO  
Recognized organization, if applicable

- |                                                                              |                                              |
|------------------------------------------------------------------------------|----------------------------------------------|
| <b>1</b> Name of reporting authority .....                                   | <b>2</b> Name of ship .....                  |
| <b>3</b> Flag of ship .....                                                  | <b>4</b> Type of ship .....                  |
| <b>5</b> Call sign .....                                                     | <b>6</b> IMO number .....                    |
| <b>7</b> Gross tonnage .....                                                 | <b>8</b> Deadweight (where applicable) ..... |
| <b>9</b> Year of build .....                                                 | <b>10</b> Date of inspection .....           |
| <b>11</b> Place of inspection .....                                          | <b>12</b> Classification society .....       |
| <b>13</b> Date of release from detention** .....                             |                                              |
| <b>14</b> Particulars of ISM company (details or IMO Company Number)** ..... |                                              |

**15** Relevant certificate(s)\*\*

|           | a) Title | b) Issuing authority | c) Dates of issue and expiry |
|-----------|----------|----------------------|------------------------------|
| <b>1</b>  | .....    | .....                | .....                        |
| <b>2</b>  | .....    | .....                | .....                        |
| <b>3</b>  | .....    | .....                | .....                        |
| <b>4</b>  | .....    | .....                | .....                        |
| <b>5</b>  | .....    | .....                | .....                        |
| <b>6</b>  | .....    | .....                | .....                        |
| <b>7</b>  | .....    | .....                | .....                        |
| <b>8</b>  | .....    | .....                | .....                        |
| <b>9</b>  | .....    | .....                | .....                        |
| <b>10</b> | .....    | .....                | .....                        |
| <b>11</b> | .....    | .....                | .....                        |
| <b>12</b> | .....    | .....                | .....                        |

d) Information on last intermediate or annual survey\*\*

|    | Date  | Surveying authority | Place |
|----|-------|---------------------|-------|
| 1  | ..... | .....               | ..... |
| 2  | ..... | .....               | ..... |
| 3  | ..... | .....               | ..... |
| 4  | ..... | .....               | ..... |
| 5  | ..... | .....               | ..... |
| 6  | ..... | .....               | ..... |
| 7  | ..... | .....               | ..... |
| 8  | ..... | .....               | ..... |
| 9  | ..... | .....               | ..... |
| 10 | ..... | .....               | ..... |
| 11 | ..... | .....               | ..... |
| 12 | ..... | .....               | ..... |

16 Deficiencies                      No              Yes (see attached FORM B)

17 Penalty imposed              No              Yes              Amount:

18 Ship detained              No              Yes \*\*\*

19 Supporting documentation              No              Yes (see annex)

Issuing office ..... Name .....  
(duly authorized PSCO of reporting authority)

Telephone .....

Telefax ..... Signature .....

**This report must be retained on board for a period of two years and must be available for consultation by port State control officers at all times.**

\* This inspection report has been issued solely for the purposes of informing the master and other port States that an inspection by the port State, mentioned in the heading, has taken place. This inspection report cannot be construed as a seaworthiness certificate in excess of the certificate the ship is required to carry.

\*\* To be completed in the event of a detention.

\*\*\* Masters, shipowners and/or operators are advised that detailed information on a detention may be subject to future publication.



**REPORT OF INSPECTION IN ACCORDANCE WITH IMO PORT STATE CONTROL  
PROCEDURES**

**FORM B**

(Reporting authority)  
(Address)  
(Telephone)  
(Telefax)

Copy to: Master  
Head office  
PSCO

If ship is detained, copy to:  
Flag State  
IMO  
Recognized organization, if applicable

**2** Name of ship ..... **6** IMO number .....

**10** Date of inspection ..... **4** Place of inspection .....

**21** Nature of deficiency<sup>1)</sup>

Convention<sup>2)</sup>

**22** Action taken<sup>3)</sup>

|       |       |       |
|-------|-------|-------|
| ..... | ..... | ..... |
| ..... | ..... | ..... |
| ..... | ..... | ..... |
| ..... | ..... | ..... |
| ..... | ..... | ..... |
| ..... | ..... | ..... |
| ..... | ..... | ..... |
| ..... | ..... | ..... |
| ..... | ..... | ..... |
| ..... | ..... | ..... |
| ..... | ..... | ..... |

Name .....  
**(duly authorized PSCO of reporting  
authority)**

Signature .....

---

<sup>1</sup> This inspection was not a full survey and deficiencies listed may not be exhaustive. In the event of a detention, it is recommended that full survey is carried out and all deficiencies are rectified before an application for re-inspection is made.

<sup>2</sup> To be completed in the event of a detention.

<sup>3</sup> Actions taken include i.e.: ship detained/released, flag State informed, classification society informed, next port informed.

APPENDIX 14

**REPORT OF DEFICIENCIES  
NOT FULLY RECTIFIED OR ONLY PROVISIONALLY REPAIRED**

**In accordance with the provision of paragraph 3.7.3 of IMO Port State Control Procedures  
(resolution A.[...](27))**

**(Copy to maritime Authority of next port of call, flag Administration, or other certifying  
authority as appropriate)**

|                                                       |                                         |
|-------------------------------------------------------|-----------------------------------------|
| 1 From (Country/region) .....                         | 2 Port .....                            |
| 3 To (Country/region) .....                           | 4 Port .....                            |
| 5 Name of ship .....                                  | 6 Date departed .....                   |
| 7 Estimated place and time of arrival .....           |                                         |
| 8 IMO number .....                                    | 9 Flag of ship and POR .....            |
| 10 Type of ship .....                                 | 11 Call sign .....                      |
| 12 Gross tonnage .....                                | 13 Year of build .....                  |
| 14 Issuing authority of relevant certificate(s) ..... |                                         |
| 15 Nature of deficiencies to be rectified             | 16 Suggested action                     |
| .....                                                 | (including action at next port of call) |
| .....                                                 | .....                                   |
| .....                                                 | .....                                   |
| .....                                                 | .....                                   |
| .....                                                 | .....                                   |
| .....                                                 | .....                                   |
| .....                                                 | .....                                   |

**17 Action taken**

.....  
.....  
.....  
.....  
.....  
.....  
.....

|                                               |               |
|-----------------------------------------------|---------------|
| Reporting Authority .....                     | Office .....  |
| Name .....                                    | Telefax ..... |
| (duly authorized PSCO of reporting authority) |               |
| Signature .....                               | Date .....    |

APPENDIX 15

**REPORT OF ACTION TAKEN TO THE NOTIFYING AUTHORITY**

**In accordance with the provision of paragraph 3.7.3 of IMO Port State Control Procedures  
(resolution A.[....](27))**

**(by Telefax and/or Mail)**

- 1** To: (Name) .....  
(Position) .....  
(Authority) .....  
Telephone ..... Telefax .....  
Date: .....
- 2** From: (Name) .....  
(Position) .....  
(Authority) .....  
Telephone ..... Telefax .....
- 3** Name of ship .....
- 4** Call sign ..... **5** IMO Number .....
- 6** Port of inspection .....
- 7** Date of inspection .....
- 8** Action taken
- | a) Deficiencies | b) Action taken |
|-----------------|-----------------|
| .....           | .....           |
| .....           | .....           |
| .....           | .....           |
| .....           | .....           |
| .....           | .....           |
| .....           | .....           |
| .....           | .....           |
| .....           | .....           |
| .....           | .....           |
| .....           | .....           |
- 9** Next port ..... (Date) .....
- 10** Supporting documentation    No    Yes    (See attached)

Signature .....

APPENDIX 16

**FORMAT FOR THE REPORT OF CONTRAVENTION OF MARPOL (article 6)**

**IMO PORT STATE CONTROL PROCEDURES**

**(resolution A.[...](27))**

(Issuing authority)  
(Address)  
(Telephone)  
(Telefax)

Copy to: Master

- |           |                             |           |                                         |
|-----------|-----------------------------|-----------|-----------------------------------------|
| <b>1</b>  | Reporting country .....     |           |                                         |
| <b>2</b>  | Name of ship .....          |           |                                         |
| <b>3</b>  | Flag of ship .....          |           |                                         |
| <b>4</b>  | Type of ship .....          |           |                                         |
| <b>5</b>  | Call sign .....             | <b>6</b>  | IMO number .....                        |
| <b>7</b>  | Gross tonnage .....         | <b>8</b>  | Deadweight .....<br>(where appropriate) |
| <b>9</b>  | Year of build .....         | <b>10</b> | Classification society .....            |
| <b>11</b> | Date of incident .....      | <b>12</b> | Place of incident .....                 |
| <b>13</b> | Date of investigation ..... |           |                                         |

**14** In case of contravention of discharge provisions, a report may be completed in addition to port State report on deficiencies. This report should be in accordance with parts 2 and 3 of appendix 3 and/or parts 2 and 3 of appendix 4, as applicable, and should be supplemented by documents, such as:

- .1 a statement by the observer of the pollution;
- .2 the appropriate information listed under section 1 of part 3 of appendices 3 and 4 to the Procedures, the statement should include considerations which lead the observer to conclude that none of any other possible pollution sources is in fact the source;
- .3 statements concerning the sampling procedures both of the slick and on board. These should include location of and time when samples were taken, identity of person(s) taking the samples and receipts identifying the persons having custody and receiving transfer of the samples;

- .4 reports of analyses of samples taken of the slick and on board; the reports should include the results of the analyses, a description of the method employed, reference to or copies of scientific documentation attesting to the accuracy and validity of the method employed and names of persons performing the analyses and their experience;
- .5 if applicable, a statement by the PSCO on board together with the PSCO's rank and organization;
- .6 statements by persons being questioned;
- .7 statements by witnesses;
- .8 photographs of the slick; and
- .9 copies of relevant pages of Oil/Cargo Record Books, log-books, discharge recordings, etc.

Name and Title (duly authorized contravention investigation official)

.....  
.....  
.....  
.....

Signature

APPENDIX 17

**COMMENTS BY FLAG STATE ON DETENTION REPORT**

Name of ship: .....

IMO number/call sign: .....

Flag State: .....

Gross tonnage: .....

Deadweight (where appropriate): .....

Date of report: .....

Report by: .....

Classification Society

Recognized Organization involved: .....

.....

☐ Did you receive the notification of detention? (tick the box if the answer is 'yes')

Action taken

a) Deficiencies

b) Cause

c) Action taken

|       |       |       |
|-------|-------|-------|
| ..... | ..... | ..... |
| ..... | ..... | ..... |
| ..... | ..... | ..... |
| ..... | ..... | ..... |
| ..... | ..... | ..... |
| ..... | ..... | ..... |
| ..... | ..... | ..... |
| ..... | ..... | ..... |
| ..... | ..... | ..... |
| ..... | ..... | ..... |

Additional Information: .....

APPENDIX 18

**LIST OF INSTRUMENTS RELEVANT TO PORT STATE CONTROL PROCEDURES**

| <b>Instrument</b>             | <b>Name</b>                                                                                                                                                | <b>IMO body</b> | <b>Remark</b> |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------|
| A.797(19)                     | Safety of ships carrying solid bulk cargoes                                                                                                                | DSC             |               |
| A.[...](27)                   | Principles of minimum safe manning                                                                                                                         | MSC/STW         |               |
| MSC.159(78)                   | Interim guidance on control and compliance measures to enhance maritime security                                                                           | MSC/FSI         |               |
| MSC.286(86)                   | Recommendations for material safety data sheets (MSDS) for MARPOL Annex I                                                                                  | BLG             |               |
| MSC/Circ.447                  | Control under SOLAS regulation I/19 – Recommendation on radar reflectors for liferafts and on training manuals                                             | DE              |               |
| MSC/Circ.592                  | Carriage of dangerous goods                                                                                                                                | DSC             |               |
| MSC/Circ.606                  | Port State concurrence with SOLAS exemptions                                                                                                               | FSI             |               |
| MSC/Circ.635                  | Tonnage measurement of certain ships relevant to the International Convention on Standards of Training, Certification and Watchkeeping for Seafarers, 1978 | SLF             |               |
| MSC/Circ.656                  | Safety of ships carrying solid bulk cargoes                                                                                                                | DSC             |               |
| MSC/Circ.811                  | Identification of float-free arrangements for liferafts                                                                                                    | DE              |               |
| MSC/Circ.887                  | Interpretation of the term "other strategic points" in SOLAS regulation III/50 and LSA Code section VII/7.2                                                | DE              |               |
| MSC/Circ.890<br>MEPC/Circ.354 | Interim guidelines for Port State Control related to the ISM Code                                                                                          | FSI             |               |
| MSC/Circ.907                  | Application of SOLAS regulation III/28.2 concerning helicopter landing areas on non ro-ro passenger ships                                                  | DE              |               |
| MSC/Circ.918                  | Guidance for port State control officers in respect of certificates of competency issued under the provision of the STCW Convention                        | FSI/STW         |               |

| Instrument                      | Name                                                                                                                                                                                                                        | IMO body | Remark |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| MSC/Circ.955                    | Servicing of life-saving appliances and radiocommunication equipment under the harmonized system of survey and certification (HSSC)                                                                                         | FSI      |        |
| MSC/Circ.1011,<br>MEPC/Circ.383 | Measures to improve port State control procedures                                                                                                                                                                           | FSI      |        |
| MSC/Circ.1012                   | Endorsement of certificates with the date of completion of the survey on which they are based                                                                                                                               | FSI      |        |
| MSC/Circ.1016                   | Application of SOLAS regulation III/26 concerning fast rescue boats and means of rescue systems on ro-ro passenger ships                                                                                                    | DE       |        |
| MSC/Circ.1030                   | Guidance for port State control officers on issues related to certificates of competency                                                                                                                                    | FSI/STW  |        |
| MSC/Circ.1032                   | Guidance for port State control officers on references to STCW 95 in certificates, endorsements and documentary evidence                                                                                                    | FSI/STW  |        |
| MSC/Circ.1089                   | Guidance on recommended anti-fraud measures and forgery prevention measures for seafarers' certificate                                                                                                                      | FSI/STW  |        |
| MSC/Circ.1097                   | Guidance relating to the implementation of SOLAS chapter XI-2 and the ISPS Code                                                                                                                                             | MSC      |        |
| MSC/Circ.1107                   | Application of SOLAS regulation II-1/3-6 on access to and within spaces in, and forward of, the cargo area of oil tankers and bulk carriers and application of the technical provisions for means of access for inspections | DE       |        |
| MSC/Circ.1111                   | Guidance relating to the implementation of SOLAS chapter XI-2 and the ISPS Code                                                                                                                                             | MSC      |        |
| MSC/Circ.1112                   | Shore leave and access to ships under the ISPS Code                                                                                                                                                                         | MSC      |        |
| MSC/Circ.1113                   | Guidance to port State control officers on the non-security related elements of the 2002 SOLAS amendment                                                                                                                    | MSC      |        |
| MSC/Circ.1117                   | Guidance for checking the structure of bulk carriers                                                                                                                                                                        | DE       |        |
| MSC/Circ.1133                   | Reminder of the obligation to notify flag States when exercising control and compliance measures                                                                                                                            | FSI      |        |
| MSC/Circ.1059<br>MEPC/Circ.401  | Procedures concerning observed ISM Code major non-conformities                                                                                                                                                              | FSI      |        |



| Instrument                                      | Name                                                                                                                                                                                                                                                                                                                                                | IMO body | Remark |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| MSC/Circ.1151<br>MEPC/Circ.426<br>FAL.2/Circ.87 | Revised list of certificates and documents required to be carried on board ships                                                                                                                                                                                                                                                                    | FAL      |        |
| MSC/Circ.1156                                   | Guidance on the access of public authorities, emergency response services and pilots on board ships to which SOLAS chapter XI-2 and the ISPS Code apply                                                                                                                                                                                             | MSC      |        |
| MSC/Circ.1176                                   | Unified interpretations to SOLAS chapters II-1 and XII and to the technical provisions for means of access for inspections                                                                                                                                                                                                                          | DE       |        |
| MSC/Circ.1191                                   | Further reminder of the obligation to notify flag States when exercising control and compliance measures                                                                                                                                                                                                                                            | MSC/FSI  |        |
| MSC.1/Circ.1196                                 | Means of embarkation on and disembarkation from ships                                                                                                                                                                                                                                                                                               | DE       |        |
| MSC.1/Circ.1197                                 | Amendments to the unified interpretations to SOLAS chapters II-1 and XII approved by MSC/Circ.1176                                                                                                                                                                                                                                                  | DE       |        |
| MSC.1/Circ.1198                                 | Application of SOLAS regulation XII/6.3 on corrosion prevention of dedicated seawater ballast tanks in all types of ships and double-sided skin spaces of bulk carriers and application of the performance standard for protective coatings for dedicated seawater ballast tanks in all types of ships and double-side skin spaces of bulk carriers | DE       |        |
| MSC.1/Circ.1199                                 | Interim guidance on compliance of ships carrying dry cargoes in bulk with requirements of SOLAS chapters II-1, III, IX, XI-1 and XII                                                                                                                                                                                                                | DE       |        |
| MSC.1/Circ.1208                                 | Promoting and verifying continued familiarization of GMDSS operators on board ships                                                                                                                                                                                                                                                                 | STW      |        |
| MSC.1/Circ.1221                                 | Validity of type approval certification for marine products                                                                                                                                                                                                                                                                                         | FSI      |        |
| MSC.1/Circ.1235                                 | Guidelines on security-related training and familiarization for shipboard personnel                                                                                                                                                                                                                                                                 | STW      |        |
| MSC.1/Circ.1326                                 | Clarification of SOLAS regulation III/19                                                                                                                                                                                                                                                                                                            | DE       |        |
| MSC.1/Circ.1331                                 | Guidelines for construction, installation, maintenance and inspection/survey of means of embarkation and disembarkation                                                                                                                                                                                                                             | DE       |        |
| MSC.1/Circ.1402                                 | Safety of pilot transfer arrangements                                                                                                                                                                                                                                                                                                               |          |        |

| Instrument                    | Name                                                                                                                                                                                      | IMO body | Remark |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| MEPC.104(49)                  | Guidelines for brief sampling of anti-fouling systems on ships                                                                                                                            | FSI      |        |
| MEPC.[...](62)                | 2011 Guidelines for inspection of anti-fouling systems on ships                                                                                                                           | FSI      |        |
| MEPC.129(53)<br>MEPC/Circ.472 | Guidelines for port State control under MARPOL Annex VI                                                                                                                                   | FSI      |        |
| MEPC.173(58)                  | Guidelines for ballast water sampling (G2)                                                                                                                                                | MEPC/BLG |        |
| MEPC.174(58)                  | Guidelines for approval of ballast water management systems (G8)                                                                                                                          | MEPC     |        |
| MEPC.181(59)                  | 2009 Guidelines for port State control under the revised MARPOL Annex VI                                                                                                                  | MEPC/BLG |        |
| MEPC.184(59)                  | 2009 Guidelines for exhaust gas cleaning system                                                                                                                                           | MEPC/BLG |        |
| MEPC/Circ.411                 | Guidance for port State control officers on issues related to the Form of the Oil Record Book Part I                                                                                      | MEPC     |        |
| MEPC/Circ.479                 | Guidelines for port State control officers whilst checking compliance with the Condition Assessment Scheme(CAS)                                                                           | MEPC/FSI |        |
| MEPC/Circ.479/<br>Corr.1      | Guidelines for port State control officers whilst checking compliance with the Condition Assessment Scheme(CAS)                                                                           | MEPC/FSI |        |
| MEPC/Circ.508                 | Bunker delivery note and fuel oil sampling                                                                                                                                                | MEPC/FSI |        |
| MEPC/Circ.513                 | Validity of the IOPP Certificate and Supplements issued under the current MARPOL Annex I after 1 January 2007                                                                             | MEPC     |        |
| MEPC.1/Circ.516               | Public access to the condition assessment scheme (CAS) database                                                                                                                           | MEPC     |        |
| MEPC.1/Circ.637               | Fuel oil availability and quality                                                                                                                                                         | MEPC     |        |
| MEPC.1/Circ.640               | Interim guidance on the use of the oil record book concerning voluntary declaration of quantities retained on board in oily bilge water holding tanks and heating of oil residue (sludge) | DE       |        |

| <b>Instrument</b> | <b>Name</b>                                                                                                                                                                    | <b>IMO body</b> | <b>Remark</b>     |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------|
| MEPC.1/Circ.675   | Discharge of cargo hold washing water in the Gulfs area and Mediterranean Sea area under MARPOL Annex V                                                                        | MEPC            |                   |
| MSC-MEPC.2/Circ.2 | IMO requirements on carriage of publication on board ships                                                                                                                     | FSI/NAV         |                   |
| MSC-MEPC.4/Circ.1 | Retention of original records/documents on board ships                                                                                                                         | FSI             |                   |
| MSC-MEPC.4/Circ.2 | Code of good practice for port State control officers                                                                                                                          | MSC/MEPC        |                   |
| MSC-MEPC.4/Circ.3 | Blanking of bilge discharge piping systems in port                                                                                                                             | MSC/MEPC        |                   |
| MSC-MEPC.5/Circ.6 | Guidance on the timing of replacement of existing certificates by the certificates issued after the entry into force of amendments to certificates in IMO instruments          | FSI             |                   |
| STCW.7/Circ.12    | Advice for port State control officers and recognized organizations on action to be taken in cases where not all seafarers carry certificates and endorsements meeting STCW 95 | STW             |                   |
|                   | Guidelines on port State control under the 2004 BWM Convention                                                                                                                 | MEPC/FSI        | Under development |

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**ANNEX 25**

**DRAFT ASSEMBLY RESOLUTION**

**SURVEY GUIDELINES UNDER THE HARMONIZED SYSTEM OF SURVEY AND  
CERTIFICATION (HSSC), 2011**

THE ASSEMBLY,

RECALLING Article 15(j) of the Convention on the International Maritime Organization concerning the functions of the Assembly in relation to regulations and guidelines concerning maritime safety and the prevention and control of marine pollution from ships,

RECALLING ALSO the adoption by:

- (a) the International Conference on the Harmonized System of Survey and Certification, 1988, of the Protocol of 1988 relating to the International Convention for the Safety of Life at Sea, 1974, and the Protocol of 1988 relating to the International Convention on Load Lines, 1966, which, *inter alia*, introduced the harmonized system of survey and certification under the International Convention for the Safety of Life at Sea, 1974 and the International Convention on Load Lines, 1966, respectively;
- (b) resolution MEPC.39(29), of amendments to introduce the harmonized system of survey and certification into the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the 1978 Protocol relating thereto (MARPOL 73/78);
- (c) resolution MEPC.132(53), of amendments to introduce the harmonized system of survey and certification to the MARPOL Annex VI; and
- (d) the resolutions given below, of amendments to introduce the harmonized system of survey and certification into:
  - (i) the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk (IBC Code) (resolutions MEPC.40(29) and MSC.16(58));
  - (ii) the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk (IGC Code) (resolution MSC.17(58)); and
  - (iii) the Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk (BCH Code) (resolutions MEPC.41(29) and MSC.18(58)),

RECALLING FURTHER that, by resolution A.997(25), it adopted the Survey Guidelines under the Harmonized System of Survey and Certification, 2007, with a view to assisting Governments in the implementation of the requirements of the aforementioned instruments, and, by resolution A.1020(26), it adopted amendments to the Survey Guidelines,

RECOGNIZING the need for the Survey Guidelines to be further revised to take into account the amendments to the IMO instruments referred to above, which have entered into force or become effective since the adoption of resolution A.1020(26),

HAVING CONSIDERED the recommendations made by the Maritime Safety Committee, at its eighty-ninth session, and the Marine Environment Protection Committee, [at its sixty-second] session,

1. ADOPTS the Survey Guidelines under the Harmonized System of Survey and Certification, 2011, set out in the annex to the present resolution;
2. INVITES Governments carrying out surveys required by the relevant IMO instruments to follow the provisions of the annexed Survey Guidelines;
3. REQUESTS the Maritime Safety Committee and the Marine Environment Protection Committee to keep the Survey Guidelines under review and amend them as necessary;
4. SUPERCEDES resolutions A.997(25), A.1020(26) and MEPC.180(59).

\* \* \*

ANNEX

**SURVEY GUIDELINES UNDER THE HARMONIZED  
SYSTEM OF SURVEY AND CERTIFICATION (HSSC), 2011**

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ANNEX 1

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- |       |     |                                                                                     |
|-------|-----|-------------------------------------------------------------------------------------|
| (E)   | 1   | GUIDELINES FOR SURVEYS FOR THE CARGO SHIP SAFETY<br>EQUIPMENT CERTIFICATE           |
| (EI)  | 1.1 | Initial surveys                                                                     |
| (EA)  | 1.2 | Annual surveys                                                                      |
| (EP)  | 1.3 | Periodical surveys                                                                  |
| (ER)  | 1.4 | Renewal surveys                                                                     |
| (C)   | 2   | GUIDELINES FOR SURVEYS FOR THE CARGO SHIP SAFETY<br>CONSTRUCTION CERTIFICATE        |
| (CI)  | 2.1 | Initial surveys                                                                     |
| (CA)  | 2.2 | Annual surveys                                                                      |
| (CIn) | 2.3 | Intermediate surveys                                                                |
| (CR)  | 2.4 | Renewal surveys                                                                     |
| (B)   | 3   | GUIDELINES FOR THE INSPECTION OF THE OUTSIDE OF THE<br>SHIP'S BOTTOM OF CARGO SHIPS |
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## ANNEX 2

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## GENERAL

### 1 INTRODUCTION

1.1 These Guidelines supersede the guidelines adopted by resolution A.997(25), as amended by resolution A.1020(26), and resolution MEPC.180(59), and take account of the Harmonized System of Survey and Certification in the following instruments:

- .1 International Convention for the Safety of Life at Sea, 1974 (SOLAS 1974), as modified by its 1988 Protocol and as amended (SOLAS 74/88);
- .2 International Convention on Load Lines, 1966 (LLC 1966) and as modified by its 1988 Protocol, as amended (LLC 66/88);
- .3 International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto, as amended (MARPOL 73/78);
- .4 Protocol of 1997 to amend the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto, as amended (MARPOL PROT 1997);
- .5 International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk, as amended (IBC Code);
- .6 International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk, as amended (IGC Code);
- .7 Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk, as amended (BCH Code).

1.2 These Guidelines take into account amendments to statutory instruments which have entered into force up to and including 31 December 2011 (see appendix 1) and contain the following:

- .1 Survey Guidelines under the 1974 SOLAS Convention, as modified by the 1988 Protocol relating thereto (annex 1);
- .2 Survey Guidelines under the 1966 Load Line Convention, as modified by the 1988 Protocol relating thereto (annex 2);
- .3 Survey Guidelines under the MARPOL Convention (annex 3); and
- .4 Survey Guidelines under mandatory Codes (annex 4).

1.3 The harmonized system, a diagrammatic arrangement of which is given in appendix 2, provides for:

- .1 a one-year standard interval between surveys, based on initial, annual, intermediate, periodical and renewal surveys, as appropriate, except for MARPOL Annex IV which is based on initial and renewal surveys;

- .2 a scheme providing the necessary flexibility to execute each survey, with provision for:
    - .1 completion of the renewal survey within 3 months before the expiry date of the existing certificate with no loss of its period of validity; and
    - .2 a "time window" of 6 months – from 3 months before to 3 months after the anniversary date of the certificate for annual, intermediate and periodical surveys;
  - .3 a maximum period of validity of five years for all cargo ship certificates;
  - .4 a maximum period of validity of 12 months for the Passenger Ship Safety Certificate;
  - .5 a system for the extension of certificates limited to three months, enabling a ship to complete its voyage, or one month for ships engaged on short voyages;
  - .6 when an extension has been granted, the period of validity of the new certificate starting from the expiry date of the existing certificate before its extension;
  - .7 a flexible system for inspection of the outside of the ship's bottom on the following conditions:
    - .1 a minimum of two inspections during any five-year period of validity of the Cargo Ship Safety Construction Certificate or the Cargo Ship Safety Certificate; and
    - .2 the interval between any two such inspections should not exceed 36 months;
  - .8 a Cargo Ship Safety Certificate under SOLAS 74/88, as an alternative to separate Cargo Ship Safety Construction, Cargo Ship Safety Equipment and Cargo Ship Safety Radio Certificates;
  - .9 a flexible system concerning the frequency and the period of validity of certificates, subject to the minimum pattern of surveys being maintained.
- 1.4 In implementing the harmonized system, the following principal changes made to the survey and certification requirements of SOLAS 74/88 have been taken into account:
- .1 unscheduled inspections are no longer included and annual surveys are mandatory for cargo ships;
  - .2 intervals between the periodical surveys of equipment covered by the Cargo Ship Safety Equipment Certificate are alternately two and three years instead of two years;
  - .3 intermediate surveys are required for all ships under the Cargo Ship Safety Construction Certificate;

- .4 inspection of the outside of the ship's bottom is required for all cargo ships;
- .5 intermediate surveys for the Cargo Ship Safety Construction Certificate are held within three months of either the second or third anniversary date;
- .6 all cargo ship certificates may be issued for any period of validity up to and including five years;
- .7 there is provision for a Cargo Ship Safety Certificate;
- .8 the extension provisions have been reduced from five months to three months to enable a ship to complete its voyage and the extension for one month for a period of grace is limited to ships engaged on short voyages.

1.5 With regard to LLC 66/88, the principal changes to the requirements for survey and certification are the introduction of similar extension provisions (see 1.4.8) and linking of the period of validity of the new certificate to the expiry date of the previous certificate (see 1.3.6).

1.6 With regard to MARPOL and IBC Code, IGC Code and BCH Code, the main changes are the linking of the period of validity of the new certificate to the expiry date of the previous certificate (see 1.3.6), the holding of the intermediate survey within three months of the second or third anniversary date and the introduction of the same extension provisions (see 1.4.8).

## 2 TYPES OF SURVEY

The types of survey used in the harmonized system are as follows:

- (I) 2.1 *An initial survey* is a complete inspection before a ship is put into service of all the items relating to a particular certificate, to ensure that the relevant requirements are complied with and that these items are satisfactory for the service for which the ship is intended.
- (P) 2.2 *A periodical survey* is an inspection of the items relating to the particular certificate to ensure that they are in a satisfactory condition and fit for the service for which the ship is intended.
- (R) 2.3 *A renewal survey* is the same as a periodical survey but also leads to the issue of a new certificate.
- (In) 2.4 *An intermediate survey* is an inspection of specified items relevant to the particular certificate to ensure that they are in a satisfactory condition and fit for the service for which the ship is intended.
- (A) 2.5 *An annual survey* is a general inspection of the items relating to the particular certificate to ensure that they have been maintained and remain satisfactory for the service for which the ship is intended.
- (B) 2.6 *An inspection of the outside of the ship's bottom* is an inspection of the underwater part of the ship and related items to ensure that they are in a satisfactory condition and fit for the service for which the ship is intended.

- (Ad) 2.7 *An additional survey* is an inspection, either general or partial according to the circumstances, to be made after a repair resulting from investigations or whenever any important repairs or renewals are made.
- 2.8 List of types of survey in Conventions and Codes
- (I) 2.8.1 *Initial surveys*
- SOLAS 74/88 chapter I regulation 7(a)(i)  
chapter I regulation 8(a)(i)  
chapter I regulation 9(a)(i)  
chapter I regulation 10(a)(i)  
LLC 66/88 article 14(1)(a)  
MARPOL Annex I regulation 6.1.1  
MARPOL Annex II regulation 8.1.1  
MARPOL Annex IV regulation 4.1.1  
MARPOL Annex VI regulation 5.1.1  
IBC Code regulation 1.5.2.1.1  
IGC Code regulation 1.5.2.1.1  
BCH Code regulation 1.6.2.1.1
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chapter I regulation 9(a)(iii)
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chapter I regulation 9(a)(ii)  
chapter I regulation 10(a)(ii)  
LLC 66/88 article 14(1)(b)  
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IGC Code regulation 1.5.2.1.2  
BCH Code regulation 1.6.2.1.2
- (In) 2.8.4 *Intermediate surveys*
- SOLAS 74/88 chapter I regulation 10(a)(iii)  
MARPOL Annex I regulation 6.1.3  
MARPOL Annex II regulation 8.1.3  
MARPOL Annex VI regulation 5.1.3  
IBC Code regulation 1.5.2.1.3  
IGC Code regulation 1.5.2.1.3  
BCH Code regulation 1.6.2.1.3
- (A) 2.8.5 *Annual surveys*
- SOLAS 74/88 chapter I regulations 8(a)(iv) and 10(a)(iv)  
LLC 66/88 article 14(1)(c)  
MARPOL Annex I regulation 6.1.4  
MARPOL Annex II regulation 8.1.4  
MARPOL Annex VI regulation 5.1.4

- IBC Code regulation 1.5.2.1.4
- IGC Code regulation 1.5.2.1.4
- BCH Code regulation 1.6.2.1.4
- (B) 2.8.6 *Inspection of the outside of the ship's bottom*
  - SOLAS 74/88 chapter I regulation 10(a)(v)
- (Ad) 2.8.7 *Additional surveys*
  - SOLAS 74/88 chapter I regulation 7(a)(iii)
  - chapter I regulation 7(b)(iii)
  - chapter I regulation 8(a)(iv)
  - chapter I regulation 10(a)(iv)
  - LLC 66/88 article 14(1)(c)
  - MARPOL Annex I regulation 6.1.5
  - MARPOL Annex II regulation 8.1.5
  - MARPOL Annex IV regulation 4.1.3
  - MARPOL Annex VI regulation 5.1.5
  - IBC Code regulation 1.5.2.1.4
  - IGC Code regulation 1.5.2.1.4
  - BCH Code regulation 1.6.2.1.4

### **3 APPLICATION AND ARRANGEMENT OF THE GUIDELINES**

3.1 The Guidelines provide a general framework upon which Administrations will be able to base their arrangements for carrying out surveys. It is recognized that survey provisions contained in the Guidelines are not necessarily applicable to all types and sizes of ship.

3.2 Whilst the Guidelines are intended to cover instruments listed in 1.1, they should be applied, as appropriate, to drilling rigs and other platforms covered by MARPOL Annex I regulation 39 and Annex VI regulation 5.

3.3 A description of the various types of survey is given in section 4 and, as shown on the contents page, this is followed by the detailed requirements for the various surveys for each of the certificates.

3.4 When appropriate, the detailed requirements for the various surveys contain a section that is applicable to all cargo ships followed by a section that only applies to oil tankers.

3.5 Whilst the Convention or Code references are included, where possible, it should be noted that, in general, it has not been possible to indicate where there are differing requirements dependent upon the ship's year of build. Consequently, care should be taken in applying specific requirements, particularly where there have been amendments that are only applicable to ships built after a certain date.

3.6 Although also part of the requirements for the Cargo Ship Safety Construction Certificate, a separate section is provided for inspection of the outside of the ship's bottom.

3.7 SOLAS 74/88 regulation I/12(v) provides for a Cargo Ship Safety Certificate to be issued as an alternative to the Cargo Ship Safety Equipment Certificate, the Cargo Ship Safety Construction Certificate and the Cargo Ship Safety Radio Certificate. Consequently, the surveys for the issue and renewal of the Cargo Ship Safety Certificate should be in accordance with the certificates it replaces and, similarly, the annual and intermediate

surveys should be the same as those required for the replaced certificates and the appropriate sections of the Cargo Ship Safety Certificate, endorsed accordingly.

3.8 On the left-hand side of each item to be surveyed may be found two letters in brackets, the first indicating the certificate to which the survey relates, as follows:

- (E) for the Cargo Ship Safety *Equipment* Certificate;
- (C) for the Cargo Ship Safety *Construction* Certificate;
- (R) for the Cargo Ship Safety *Radio* Certificate;
- (L) for the International *Load Line* Certificate;
- (O) for the International *Oil Pollution* Prevention Certificate;
- (N) for the International Pollution Prevention Certificate for Carriage of *Noxious Liquid Substances* in Bulk;
- (S) for the International Sewage Pollution Prevention Certificate;
- (A) for the International Air Pollution Prevention Certificate;
- (D) for the International Certificate of Fitness for the Carriage of *Dangerous Chemicals* in Bulk or the Certificate of Fitness for the Carriage of *Dangerous Chemicals* in Bulk;
- (G) for the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk;
- (P) for the *Passenger Ship* Safety Certificate;

and the second for the type of survey, as follows:

- (I) for the *initial* survey;
- (A) for the *annual* survey;
- (In) for the *intermediate* survey;
- (P) for the *periodical* survey;
- (R) for the *renewal* survey;
- (B) for inspection of the outside of the *ship's bottom*;
- (Ad) for *additional* survey.

Consequently, for example, "(EI)", "(OIn)" and "(PR)" indicate the initial survey for the Cargo Ship Safety Equipment Certificate, the intermediate survey for the International Oil Pollution Prevention Certificate and the renewal survey for the Passenger Ship Safety Certificate respectively.

3.9 The amplification of various terms and conditions are given in section 5.

## **4 DESCRIPTION OF THE VARIOUS TYPES OF SURVEYS**

### **(I) 4.1 Initial surveys**

#### **4.1.1 Frequency**

The initial survey, as required by the relevant regulations (see 2.8.1), should be held before the ship is put in service, or when a new instrument applies to an existing ship, and the appropriate certificate is issued for the first time.

#### **4.1.2 General**

4.1.2.1 The initial survey should include a complete inspection, with tests when necessary, of the structure, machinery and equipment to ensure that the requirements relevant to the particular certificate are complied with and that the structure, machinery and equipment are fit for the service for which the ship is intended.

4.1.2.2 The initial survey should consist of:

- .1 an examination of the plans, diagrams, specifications, calculations and other technical documentation to verify that the structure, machinery and equipment comply with the requirements relevant to the particular certificate;
- .2 an inspection of the structure, machinery and equipment to ensure that the materials, scantlings, construction and arrangements, as appropriate, are in accordance with the approved plans, diagrams, specifications, calculations and other technical documentation and that the workmanship and installation are in all respects satisfactory; and
- .3 a check that all the certificates, record books, operating manuals and other instructions and documentation specified in the requirements relevant to the particular certificate have been placed on board the ship.

#### **4.1.3 Examination of plans and designs**

An application for an initial survey should be accompanied by plans and designs referred to in sections 1, 2, 4 and 5 of Annex 1 and in Annexes 2, 3 and 4, as appropriate, together with:

- .1 the particulars of the ship;
- .2 any exemptions sought; and
- .3 any special conditions.

### **(A) 4.2 Annual surveys**

#### **4.2.1 Frequency**

The annual survey, as required by the relevant regulations (see 2.8.5) and as shown diagrammatically in appendix 2, should be held within three months before or after each anniversary date of the certificate.



#### 4.2.2 *General*

4.2.2.1 An annual survey should enable the Administration to verify that the condition of the ship, its machinery and equipment is being maintained in accordance with the relevant requirements.

4.2.2.2 In general, the scope of the annual survey should be as follows:

- .1 it should consist of a certificate examination, a visual examination of a sufficient extent of the ship and its equipment, and certain tests to confirm that their condition is being properly maintained;
- .2 it should also include a visual examination to confirm that no unapproved modifications have been made to the ship and its equipment;
- .3 the content of each annual survey is given in the respective guidelines. The thoroughness and stringency of the survey should depend upon the condition of the ship and its equipment; and
- .4 should any doubt arise as to the maintenance of the condition of the ship or its equipment, further examination and testing should be conducted as considered necessary.

4.2.3 Where an annual survey has not been carried out within the due dates, reference should be made to 5.6.

### (In) **4.3 Intermediate surveys**

#### 4.3.1 *Frequency*

The intermediate survey, as required by the relevant regulations (see 2.8.4) and as shown diagrammatically in appendix 2, should be held within three months before or after the second anniversary date or within three months before or after the third anniversary date of the appropriate certificate and should take the place of one of the annual surveys.

#### 4.3.2 *General*

4.3.2.1 The intermediate survey should be an inspection of items relevant to the particular certificate to ensure that they are in a satisfactory condition and are fit for the service for which the ship is intended.

4.3.2.2 When specifying items of hull and machinery for detailed examination, due account should be taken of any continuous survey schemes that may be applied by classification societies.

4.3.2.3 Where an intermediate survey has not been carried out within the due dates, reference should be made to 5.6.

**(P) 4.4 Periodical surveys**

**4.4.1 Frequency**

The periodical survey, as required by the relevant regulations (see 2.8.2) and as shown diagrammatically in appendix 2, should be held within three months before or after the second anniversary date or within three months before or after the third anniversary date in the case of the cargo ship safety equipment certificate and should take the place of one of the annual surveys and within three months before or after each anniversary date in the case of the cargo ship safety radio certificate.

**4.4.2 General**

4.4.2.1 The periodical survey should consist of an inspection, with tests when necessary, of the equipment to ensure that requirements relevant to the particular certificate are complied with and that they are in a satisfactory condition and are fit for the service for which the ship is intended.

4.4.2.2 The periodical survey should also consist of a check that all the certificates, record books, operating manuals and other instructions and documentation specified in the requirements relevant to the particular certificate are on board the ship.

4.4.2.3 Where a periodical survey has not been carried out within the due dates, reference should be made to 5.6.

**(R) 4.5 Renewal surveys**

**4.5.1 Frequency**

The renewal survey, as required by the relevant regulations (see 2.8.3) and as shown diagrammatically in the appendix 2, should be held before the appropriate certificate is renewed.

**4.5.2 General**

4.5.2.1 The renewal survey should consist of an inspection, with tests when necessary, of the structure, machinery and equipment to ensure that the requirements relevant to the particular certificate are complied with and that they are in a satisfactory condition and are fit for the service for which the ship is intended.

4.5.2.2 The renewal survey should also consist of a check that all the certificates, record books, operating manuals and other instructions and documentation specified in the requirements relevant to the particular certificate are on board the ship.

**(B) 4.6 Inspections of the outside of the ship's bottom of cargo ships**

**4.6.1 Frequency**

There should be a minimum of two inspections of the outside of the ship's bottom during any five year period (see 5.7), except where SOLAS 74/88 regulation I/14(e) or (f) is applicable. One such inspection should be carried out on or after the fourth annual survey in conjunction with the renewal of the Cargo Ship Safety Construction Certificate or the Cargo Ship Safety Certificate. Where the Cargo Ship Safety Construction Certificate or the Cargo Ship Safety Certificate has been extended under SOLAS 74/88 regulation I/14(e) or (f), this five-year period may be extended to coincide with the validity of the certificate. In all cases the interval between any two such inspections should not exceed 36 months.

#### 4.6.2 *General*

4.6.2.1 The inspection of the outside of the ship's bottom and the survey of related items (see 5.1) should include an inspection to ensure that they are in a satisfactory condition and fit for the service for which the ship is intended<sup>1</sup>.

4.6.2.2 Inspections of the outside of the ship's bottom should normally be carried out with the ship in a dry dock. However, consideration may be given to alternate inspections being carried out with the ship afloat. Special consideration should be given before ships of 15 years of age and over other than bulk carriers and oil tankers are permitted to have such surveys afloat. Inspection of the outside of the ship's bottom of bulk carriers and oil tankers of 15 years of age and over should be carried out with the ship in dry dock. Inspections with the ship afloat should only be carried out when the conditions are satisfactory and the proper equipment and suitably trained staff are available. For ships subject to enhanced survey, the provisions of paragraph 2.2.2<sup>2</sup> of Annexes A or B, as applicable, of the International Code on the enhanced programme of inspections during surveys of bulk carriers and oil tankers, 2011 (2011 ESP Code) (resolution A...(27)) should apply.

4.6.3 Where an inspection of the ships bottom has not been carried out before the due dates reference should be made to 5.6.

#### (Ad) **4.7 Additional surveys**

Whenever an accident occurs to a ship or a defect is discovered which affects the safety or integrity of the ship or the efficiency or completeness of its equipment, the master or owner should make a report at the earliest opportunity to the Administration, the nominated surveyor or recognized organization responsible for issuing the relevant certificate. The Administration, the nominated surveyor or recognized organization responsible for issuing the relevant certificate should then initiate an investigation to determine whether a survey, as required by the regulations applicable to the particular certificate, is necessary. This additional survey, which may be general or partial according to the circumstances, should be such as to ensure that the repairs and any renewals have been effectively made and that the ship and its equipment continue to be fit for the service for which the ship is intended.

#### **4.8 Completion of surveys**

4.8.1 If a survey shows that the condition of the ship or its equipment is unsatisfactory, the officer of the Administration, nominated surveyor or recognized organization should be guided by the requirements of SOLAS 74/88 regulation I/6(c), MARPOL Annex I regulation 3.4, MARPOL Annex II regulation 8.2.5, MARPOL Annex IV regulation 4.5, MARPOL Annex VI regulation 5.3.3, IBC Code regulation 1.5.1.3, IGC Code regulation 1.5.1.3 and BCH Code regulation 1.6.1.3. These instruments require that corrective action be taken immediately and the Administration notified in due course. In cases where the corrective action has not been undertaken the relevant certificate should be withdrawn and the Administration notified immediately. If the ship is in the port of another Party, the appropriate authorities of the port State should also be notified immediately.

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<sup>1</sup> Refer to the Guidelines for pre-planning of surveys in dry dock of ships which are not subject to the enhanced programme of inspections (MSC.1/Circ.1223).

<sup>2</sup> "2.2.2 For ships 15 years of age and over, inspection of the outside of the ship's bottom should be carried out with the ship in dry dock. For ships less than 15 years of age, alternate inspections of the ship's bottom not conducted in conjunction with the renewal survey may be carried out with the ship afloat. Inspections of the ship afloat should only be carried out when the conditions are satisfactory and the proper equipment and suitably qualified staff is available."

4.8.2 Although LLC 66/88 does not contain specific requirements, if a load line survey shows the condition of the ship or its equipment is unsatisfactory, the officer of the Administration, nominated surveyor or recognized organization should, nevertheless, be guided by 4.8.1.

## **5 AMPLIFICATION OF TERMS AND CONDITIONS**

### **5.1 Definition of related items**

Reference: SOLAS 74/88 regulation I/10(b)(v).

Related items mean those items which may only be inspected when the ship is in dry dock or undergoing an in-water examination of the outside of its bottom. For oil tankers, chemical tankers and gas carriers, this may mean that the ship has to be specially prepared by, for example, being cleaned and gas-freed. Then the survey of items such as the internal examination of cargo tanks, as referred to in (CIn) 2.3.2 and (CIn) 2.3.3 in Annex 1 may be undertaken at the same time.

### **5.2 Extending to five years a certificate issued for less than five years**

References: SOLAS 74/88 regulation I/14(c), LLC 66/88 article 19(3), MARPOL Annex I regulation 10.3, MARPOL 90/04 Annex II regulation 10.3, MARPOL Annex IV regulation 8.3, MARPOL Annex VI regulation 9.3, IBC Code regulation 1.5.6.3, IGC Code regulation 1.5.6.3, BCH Code regulation 1.6.6.3.

Where a certificate has been issued for a period of less than five years, it is permissible under these regulations or article to extend the certificate so that its maximum period of validity is five years provided that the pattern of surveys for a certificate with a five-year period of validity is maintained (see appendix 2). This means that, for example, if a request is made to extend a two-year Cargo Ship Safety Equipment Certificate to five years, then a periodical and two further annual surveys, as detailed in SOLAS 74/88 regulation I/8, would be required. Also, for example, if it was intended to extend a four-year Cargo Ship Safety Construction Certificate to five years, then an additional annual survey would be required, as detailed in SOLAS 74/88 regulation I/10. Where a certificate has been so extended, it is still permissible to also extend the certificate under SOLAS 74/88 regulations I/14(e) and (f), LLC 66/88 articles 19(5) and (6), MARPOL Annex I regulations 10.5 and .6, MARPOL Annex II regulations 10.5 and 6, MARPOL Annex IV regulations 8.5 and 8.6, MARPOL Annex VI regulations 9.5 and 9.6, IBC Code regulations 1.5.6.5 and 1.5.6.6, IGC Code, regulations 1.5.6.5 and 1.5.6.6, BCH Code regulations 1.6.6.5 and 1.6.6.6, when no additional surveys would be required but, of course, the new certificate issued after the renewal survey would date from the five-year expiry of the existing certificate, in accordance with SOLAS 74/88 regulation I/14(b)(ii), LLC 66/88 article 19(2)(b), MARPOL Annex I regulation 10.2.2, MARPOL Annex II regulation 10.2.2, MARPOL Annex IV regulation 8.2.2, MARPOL Annex VI regulation 9.2.2, IBC Code regulation 1.5.6.2.2, IGC Code regulation 1.5.6.2.2 and BCH Code regulation 1.6.6.2.2.

### **5.3 Extending the period between inspections of the outside of the ship's bottom**

Reference: SOLAS 74/88 regulation I/10(a)(v).

This permits the period of five years in which two inspections of the ship's bottom are to be carried out to be extended when the Cargo Ship Safety Construction Certificate is extended under regulation I/14(e) and (f). However, no extension should be permitted on the period of 36 months between any two such inspections. If the first ship's bottom inspection is carried out between 24 and 27 months then the thirty-sixth-month limitation may prevent the certificate being extended by the periods permitted in regulation I/14(e) and (f).

#### **5.4 Definition of "short voyage"**

References: SOLAS 74/88 regulation I/14(f), LLC 66/88 article 19(6), MARPOL Annex I regulation 10.6, MARPOL Annex II regulation 10.6, MARPOL Annex IV regulation 8.6, MARPOL Annex VI regulation 9.6, IBC Code regulation 1.5.6.6, IGC Code regulation 1.5.6.6, BCH Code regulation 1.6.6.6.

For the purpose of these regulations or article, a "short voyage" means a voyage where neither the distance from the port in which the voyage begins and the final port of destination nor the return voyage exceeds 1,000 miles.

#### **5.5 Application of "special circumstances"**

References: SOLAS 74/88 regulation I/14(g), LLC 66/88 article 19(7), MARPOL Annex I, Regulation 10.7, MARPOL Annex II regulation 10.7, MARPOL Annex IV regulation 8.7, MARPOL Annex VI regulation 9.7, IBC Code regulation 1.5.6.7, IGC Code regulation 1.5.6.7 and the BCH Code regulation 1.6.6.7.

The purpose of these regulations or article is to permit Administrations to waive the requirement that a certificate issued following a renewal survey that is completed after the expiry of the existing certificate should be dated from the expiry date of the existing certificate. The special circumstances when this could be permitted are where the ship has been laid-up or has been out of service for a considerable period because of a major repair or modification. Whilst the renewal survey would be as extensive as if the ship had continued in service, the Administration should consider whether additional surveys or examinations are required depending on how long the ship was out of service and the measures taken to protect the hull and machinery during this period. Where this regulation is invoked, it is reasonable to expect an examination of the outside of the ship's bottom to be held at the same time as the renewal survey when it would not be necessary to include any special requirements for cargo ships for the continued application of SOLAS 74/88 regulation I/10(a)(v).

#### **5.6 Revalidation of certificates**

References: SOLAS 74/88 regulation I/14(i)(i), LLC 66/88 article 19(9)(c), MARPOL Annex I Regulation 10.9.1, MARPOL Annex II regulation 10.9.1, MARPOL Annex IV regulation 8.8.1, MARPOL Annex VI regulation 9.9.1, IBC Code regulation 1.5.6.9.1, IGC Code regulation 1.5.6.9.1 and BCH Code regulation 1.6.6.9.1.

A certificate ceases to be valid if the periodical, intermediate or annual survey, as appropriate, or the inspection of the outside of the ship's bottom is not completed within the periods specified in the relevant regulation or article. The validity of the certificate should be restored by carrying out the appropriate survey which, in such circumstances, should consist of the requirements of the survey that was not carried out, but its thoroughness and stringency should have regard to the time this survey was allowed to lapse. The Administration concerned should then ascertain why the survey was allowed to lapse and consider further action.

#### **5.7 Meaning of "any five-year period"**

Reference: SOLAS 74/88 regulation I/10(a)(v).

*Any five-year period* is the five-year period of validity of the Cargo Ship Safety Construction Certificate or the Cargo Ship Safety Certificate.

## **5.8 Surveys required after transfer of the ship to the flag of another State**

The certificates cease to be valid when a ship transfers to the flag of another State and it is required that the Government of the State to which the ship transfers should not issue new certificates until it is fully satisfied that the ship is being properly maintained and that there have been no unauthorized changes made to the structure, machinery and equipment. When so requested, the Government of the State whose flag the ship was formerly entitled to fly is obliged to forward, as soon as possible, to the new Administration copies of certificates carried by the ship before the transfer and, if available, copies of the relevant survey reports and records, such as record of safety equipment and conditions of assignment for load line. When fully satisfied by an inspection that the ship is being properly maintained and that there have been no unauthorized changes, in order to maintain the harmonization of the surveys the new Administration may give due recognition to initial and subsequent surveys carried out by, or on behalf of, the former Administration and issue new certificates having the same expiry date as the certificates that ceased to be valid because of the change of flag.

## **5.9 Recommended conditions for extending the period of validity of a certificate**

5.9.1 In SOLAS and other mandatory IMO instruments the following provision applies: "If a ship at the time when a certificate expires is not in a port in which it is to be surveyed, the Administration may extend the period of validity of a certificate but this extension should be granted only for the purpose of allowing the ship to complete its voyage to the port in which it is to be surveyed, and then only in cases *where it appears proper and reasonable to do so*. No certificate should be extended for a period longer than three months, and a ship to which an extension is granted should not, on its arrival in the port in which is to be surveyed, be entitled by virtue of such extension to leave that port without having a new certificate."

5.9.2 If a ship is in a port where the required survey cannot be completed, and where the Convention allows the Administration to extend the certificate when it is proper and reasonable to do so, the Administration should be guided by the following:

- .1 an additional survey, equivalent to at least the same scope of an annual survey required by the relevant certificate(s) should be carried out;
- .2 the renewal survey should be carried out to the maximum extent possible;
- .3 in cases where a dry docking is required, but cannot be carried out, an underwater inspection of the ship's bottom should be carried out;
- .4 in cases where an underwater inspection is not possible (e.g., poor water visibility, draft restrictions, excessive current, refusal by the port Authority), an internal inspection of the ship's bottom structure, to the maximum extent practicable, should be carried out;
- .5 the ship should be allowed to sail directly to a named final agreed cargo discharge port and then directly to a named agreed port to complete the survey and/or dry docking;
- .6 the extension period should be for the minimum amount of time needed to complete the survey and/or dry docking under the relevant certificate(s);
- .7 the condition of the ship found by the surveys indicated above should be considered in determining the duration, distance and operational restrictions, if any, of the voyage needed to complete the survey and/or dry docking; and

- .8 the extension period of the relevant statutory certificate(s) should not exceed the period of validity of the certificate which may be issued to document compliance with the structural, mechanical and electrical requirements of the recognized classification society.

## **5.10 Inspection of the outside of a passenger ship's bottom**

5.10.1 A minimum of two of the inspections of the outside of the ship's bottom during any five-year period should be conducted in dry-dock. In all cases, the maximum interval between any two dry-dock bottom inspections should not exceed 36 months.

5.10.2 Where acceptable to the Administration, the minimum number of inspections in dry-dock of the outside of the bottom of a passenger ship (which is not a ro-ro passenger ship) in any five-year period may be reduced from two to one<sup>3</sup>. In such cases the interval between consecutive inspections in dry-dock should not exceed 60 months.

**Note:** The definition of "any five-year period" is the five-year period of validity of the International Load Line Certificate.

5.10.3 Inspections of the ship's bottom required for the renewal survey that are not conducted in dry-dock may be carried out with the ship afloat. The bottom inspection, regardless of method, should be carried out within the allowable time window for the Passenger Ship Safety Certificate renewal survey (i.e. within the 3 months time window before the expiry date of the certificate). Additionally, inspections of the outside of the ship's bottom conducted afloat should only be carried out when the conditions are satisfactory and the proper equipment and suitably qualified staff is available. Rudder bearing clearances specified in (PR) 5.2.2.1 need not be taken at the afloat inspections.

5.10.4 Special consideration should be given to ships 15 years of age or over before being permitted to credit inspections afloat.

5.10.5 If a survey in dry-dock is not completed within the maximum intervals referred to above, the Passenger Ship Safety Certificate should cease to be valid until the survey in dry-dock is completed.

## **5.11 Survey of radio installations**

The survey of the radio installations, including those used in life-saving appliances, should always be carried out by a qualified radio surveyor who has necessary knowledge of the requirements of SOLAS 1974, the International Telecommunication Union's Radio Regulations and the associated performance standards for radio equipment. The radio survey should be carried out using suitable test equipment capable of performing all the relevant measurements required by these guidelines. On satisfactory completion of the survey, the radio surveyor should forward a report of the survey, which should also state the organization he represents, to the authorities responsible for the issue of the ship's Cargo Ship Safety Radio Certificate or Passenger Ship Safety Certificate.

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<sup>3</sup> Refer to the Guidelines for the assessment of technical provisions for the performance of an in-water survey in lieu of bottom inspection in dry-dock to permit one dry-dock examination in any five-year period for passenger ships other than ro-ro passenger ships (MSC.1/Circ.1348).

#### **5.12 Survey of the automatic identification system (AIS)**

The survey of the automatic identification system should always be carried out by a qualified radio surveyor who has necessary knowledge of the requirements of SOLAS 1974, the International Telecommunication Union's Radio Regulations and the associated performance standards for radio equipment. The survey of the automatic identification system should be carried out using suitable test equipment capable of performing all the relevant measurements required by and in accordance with the Guidelines on annual testing of the automatic Identification System (AIS) (MSC.1/Circ.1252).



ANNEX 1

**SURVEY GUIDELINES UNDER THE 1974 SOLAS CONVENTION,  
AS MODIFIED BY THE 1988 PROTOCOL RELATING THERETO**

- (E)     **1**               **GUIDELINES FOR SURVEYS FOR THE CARGO SHIP SAFETY EQUIPMENT CERTIFICATE**
- (EI)    **1.1**            **Initial surveys** – see part "General" section 4.1.
- (EI)    1.1.1           For the life-saving appliances and the other equipment of cargo ships the examination of plans and designs should consist of:
- (EI)    1.1.1.1       examining the plans for the fire pumps, fire mains, hydrants, hoses and nozzles and the international shore connection (SOLAS 74/00 regs.II-2/10.2 and 10.4.4 and FSSC chs.2 and 12);
- (EI)    1.1.1.2       checking the provision, specification and arrangements of the fire extinguishers (SOLAS 74/00 reg.II-2/10.3) (SOLAS 74/88 reg.II-2/6);
- (EI)    1.1.1.3       checking the provision, specification and arrangements of the fire fighters' outfits and emergency escape breathing devices (EEBDs) (SOLAS 74/00 regs.II-2/10.10, 13.3.4 and 13.4.3; FSSC ch.3) (SOLAS 74/88 reg.II-2/17) (BCH Code ch.III Part E);
- (EI)    1.1.1.4       examining the plans for the fire-extinguishing arrangements in the machinery spaces (SOLAS 74/00 regs.II-2/10.4 and 10.5 (except 10.5.5); FSSC chs.5, 6 and 7) (SOLAS 74/88 reg.II-2/7);
- (EI)    1.1.1.5       examining the plans for the special arrangements in the machinery spaces (SOLAS 74/00 regs.II-2/5.2, 8.3 and 9.5) (SOLAS 74/88 reg.II-2/11);
- (EI)    1.1.1.6       checking the provision of a fixed fire detection and fire alarm system for machinery spaces including periodically unattended machinery spaces (SOLAS 74/00 regs.II-2/7.2, 7.3 and 7.4) (SOLAS 74/88 regs.II-2/13 and 14);
- (EI)    1.1.1.7       checking the provision of a fixed fire detection and fire alarm system and/or a sprinkler, fire detection and fire alarm system in accommodation and service spaces and control stations (SOLAS 74/00 regs.II-2/7.2, 7.3, 7.5.5, 7.7 and 10.6.2; FSSC chs.8 and 9) (SOLAS 74/88 reg.II-2/52);
- (EI)    1.1.1.8       checking the provision of a fire-extinguishing system for spaces containing paint and/or flammable liquids and deep-fat cooking equipment in accommodation and service spaces (SOLAS 74/00 regs.II-2/10.6.3 and 10.6.4; FSSC chs.5 and 7) (SOLAS 74/88 regs.II-2/18.7) (BCH Code ch.III Part E);
- (EI)    1.1.1.9       examining the arrangements for remote closing of valves for oil fuel, lubricating oil and other flammable oils (SOLAS 74/00 reg.II-2/4.2.2.3.4) (SOLAS 74/88 reg.II-2/15.2.5);

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- |      |          |                                                                                                                                                                                                                                                                                                                                                                          |
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| (EI) | 1.1.1.10 | examining the plans for the fire protection arrangements in cargo spaces for general cargo and dangerous goods (SOLAS 74/00 regs.II-2/10.7 and 19) (SOLAS 74/88 regs.II-2/53 and 54);                                                                                                                                                                                    |
| (EI) | 1.1.1.11 | examining the plans for the fire protection arrangements in vehicle, special category and ro-ro spaces (SOLAS 74/00 reg.II-2/20 (except 20.2.2 and 20.5); FSSC chs.5, 6, 7, 9 and 10) (SOLAS 74/88 regs.II-2/37, 38 and 53);                                                                                                                                             |
| (EI) | 1.1.1.12 | checking navigation bridge visibility (SOLAS 74/00 reg.V/22);                                                                                                                                                                                                                                                                                                            |
| (EI) | 1.1.1.13 | examining the plans for the helicopter facilities (SOLAS 74/00 reg.II-2/18) (SOLAS 74/88 regs.II-2/18.8);                                                                                                                                                                                                                                                                |
| (EI) | 1.1.1.14 | examining the plans for the special arrangements for the carriage of dangerous goods, when appropriate, including water supplies, electrical equipment and wiring, fire detection, ventilation, bilge pumping, personnel protection and any water spray system (SOLAS 74/00 reg.II-2/19 (except 19.3.8, 19.3.10 and 19.4); FSSC chs.9 and 10) (SOLAS 74/88 reg.II-2/54); |
| (EI) | 1.1.1.15 | examining the provision and disposition of the survival craft and rescue boats and, where applicable, marine evacuation systems (SOLAS 74/88 regs.III/11 to 16, 31 and 33);                                                                                                                                                                                              |
| (EI) | 1.1.1.16 | examining, where applicable, the approved documentation for the alternative design and arrangements (SOLAS 06 reg.III/38);                                                                                                                                                                                                                                               |
| (EI) | 1.1.1.17 | examining the design of the survival craft, including their equipment, launching and recovery appliances and embarkation and launching arrangements (SOLAS 74/96/06 regs.III/4,16, 31, 32 to 33; LSAC sections. 3.2, 4.1 to 4.9, 6.1 and 6.2);                                                                                                                           |
| (EI) | 1.1.1.18 | checking that the life-saving appliances are to be of international or vivid reddish orange, or a comparably highly visible colour on all parts where this will assist detection at sea (LSAC section 1.2.2.6);                                                                                                                                                          |
| (EI) | 1.1.1.19 | examining the design of the rescue boats, including their equipment and launching and recovery appliances and arrangements (SOLAS 74/00 regs.III/17 and 31; LSAC sections 5.1 and 6.1);                                                                                                                                                                                  |
| (EI) | 1.1.1.20 | examining the provision, specification and stowage of two-way VHF radiotelephone apparatus and search and rescue locating devices (SOLAS 74/88/08 reg.III/6);                                                                                                                                                                                                            |
| (EI) | 1.1.1.21 | examining the provision, specification and stowage of the distress flares and the line-throwing appliance and the provision of onboard communications equipment and the general alarm system (SOLAS 74/00 reg.II-2/12.1 and 12.2, and regs.III/6 and 18; and LSAC sections 3.1, 7.1 and 7.2);                                                                            |
| (EI) | 1.1.1.22 | examining the provision, specification and stowage of the lifebuoys, including those fitted with self-igniting lights, self-activating smoke                                                                                                                                                                                                                             |

- signals and buoyant lines, lifejackets<sup>4</sup>, immersion suits and anti-exposure suits (SOLAS 74/00/06 regs.III/7 and 32; LSAC sections 2.1 to 2.5 and 3.1 to 3.3);
- (EI) 1.1.1.23 checking that immersion suits designed to be worn in conjunction with a lifejacket are suitably marked (LSAC section 2.3.1);
- (EI) 1.1.1.24 examining the plans for the lighting of the muster and embarkation stations and the alleyways, stairways and exits giving access to the muster and embarkation stations, including the supply from the emergency source of power (SOLAS 74/88 regs.II-1/43 and III/11);
- (EI) 1.1.1.25 examining the plans for the positioning of, and the specification for, the navigation lights, shapes and sound signalling equipment (International Regulations for Preventing Collisions at Sea (COLREG) in force, rules 20 to 24, 27 to 30 and 33);
- (EI) 1.1.1.26 examining the plans relating to the bridge design and arrangement of navigational systems and equipment and bridge procedures (SOLAS 74/00 regs.V/15 and 19);
- (EI) 1.1.1.27 checking the provision and specification of the following navigation equipment as appropriate: daylight signalling lamp, magnetic compass, transmitting heading device, gyro compass, gyro compass repeaters, radar installation(s), automatic identification system, electronic plotting aid, automatic tracking aid(s) or automatic radar plotting aid(s), echo-sounding device, speed and distance measuring device(s), rudder angle indicator, propeller rate of revolution indicator, variable-pitch propeller pitch and operational mode indicator, rate-of-turn indicator, heading or track control system, GNSS receiver, terrestrial radio navigation system and sound reception system, means of communication with emergency steering position, a pelorus or compass bearing device, means for correcting heading and bearings, a Bridge Navigational Watch Alarm System (BNWAS) as applicable and ECDIS including back-up arrangements as applicable (SOLAS 74/00/09 reg.V/19);
- (EI) 1.1.1.28 checking the provision and specification of voyage data recorder (SOLAS 74/00 reg.V/20);
- (EI) 1.1.1.29 checking the provision and specification of the long-range identification and tracking system (SOLAS 04 reg.V/19-1);
- (EI) 1.1.1.30 checking the provision and specification of the pilot ladders and hoists/pilot transfer arrangements (SOLAS 74/88 reg.V/23);
- (EI) 1.1.1.31 checking the provision of means of embarkation and disembarkation from ships for use in port and in port-related operations, such as gangways and accommodation ladders (SOLAS 08 reg.II-1/3-9).
- (EI) 1.1.2 For the examination of plans and designs of the life-saving appliances and the other equipment of cargo ships the additional requirements for oil tankers should consist of:

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<sup>4</sup> Regulation III/7.2.1.5 should be considered.

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| (EI) | 1.1.2.1 | examining the plans for the cargo tank protection (SOLAS 74/00 regs.II-2/4.5.3, 4.5.5, 4.5.6, 4.5.7 and 10.8; FSSC chs.14 and 15) (SOLAS 74/88 regs.II-2/60 and 62); and                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| (EI) | 1.1.2.2 | examining the plans for protection of the cargo pump rooms (SOLAS 78/00 regs.II-2/4.5.10 and 10.9) (SOLAS 74/88 reg.II-2/63).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| (EI) | 1.1.3   | For the life-saving appliances and the other equipment of cargo ships the survey during construction and after installation should consist of:                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| (EI) | 1.1.3.1 | examining the fire pumps and fire main and the disposition of the hydrants, hoses and nozzles and the international shore connection and checking that each fire pump, including the emergency fire pump, can be operated separately so that two jets of water are produced simultaneously from different hydrants at any part of the ship whilst the required pressure is maintained in the fire main (SOLAS 74/00 reg.II-2/10.2; FSSC chs.2 and 12) (SOLAS 74/88 regs.II-2/4 and 19);                                                                                                                      |
| (EI) | 1.1.3.2 | examining the provision and disposition of the fire extinguishers (SOLAS 74/00 reg.II-2/10.3; FSSC ch.4) (SOLAS 74/88 reg.II-2/17);                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| (EI) | 1.1.3.3 | examining the fire fighters' outfits and emergency escape breathing devices (EEBDs) (SOLAS 74/00 regs.II-2/10.10, 13.3.4 and 13.4.3; FSSC ch.3) (SOLAS 74/88 reg.II-2/17) (BCH Code ch.III Part E);                                                                                                                                                                                                                                                                                                                                                                                                          |
| (EI) | 1.1.3.4 | checking the operational readiness and maintenance of fire-fighting systems (SOLAS 74/00 reg.II-2/14.1) (SOLAS 74/88 reg.II-2/21);                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| (EI) | 1.1.3.5 | examining the fixed fire-fighting system for the machinery, cargo, vehicle, special category and ro-ro spaces, as appropriate, and confirming that the installation tests have been satisfactorily completed and that its means of operation are clearly marked (SOLAS 74/00/08 regs.II-2/10.4, 10.5, 10.7 and 20.6.1; FSSC chs.5 to 7) (SOLAS 74/88 regs.II-2/7 and 53);                                                                                                                                                                                                                                    |
| (EI) | 1.1.3.6 | checking that fixed carbon dioxide fire-extinguishing systems for the protection of machinery spaces and cargo pump-rooms, where applicable, are provided with two separate controls, one for opening of the gas piping and one for discharging the gas from the storage container, each of them located in a release box clearly identified for the particular space (SOLAS 08 reg.II-2/10.4.1.5);                                                                                                                                                                                                          |
| (EI) | 1.1.3.7 | examining the fire-extinguishing and special arrangements in the machinery spaces and confirming, as far as practicable and as appropriate, the operation of the remote means of control provided for the opening and closing of the skylights, the release of smoke, the closure of the funnel and ventilation openings, the closure of power-operated and other doors, the stopping of ventilation and boiler forced and induced draft fans and the stopping of oil fuel and other pumps that discharge flammable liquids (SOLAS 74/00 regs.II-2/5.2, 8.3, 9.5 and 10.5) (SOLAS 74/88 regs.II-2/7 and 11); |
| (EI) | 1.1.3.8 | examining any fire detection and alarm system and any automatic sprinkler, fire detection and fire alarm system and confirming that installation tests have been satisfactorily completed (SOLAS 74/00                                                                                                                                                                                                                                                                                                                                                                                                       |

- regs.II-2/7.2, 7.3, 7.4, 7.5.1, 7.5.5, 19.3.3 and 20.4; FSSC ch.9) (SOLAS 74/88 regs.II-2/11, 13, 14, 53 and 54);
- (EI) 1.1.3.9 examining the fire-extinguishing system for spaces containing paint and/or flammable liquids and deep-fat cooking equipment in accommodation and service spaces and confirming that installation tests have been satisfactorily completed and that its means of operation are clearly marked (SOLAS 74/00 regs.II-2/10.6.3 and 10.6.4; FSSC chs.4 to 7) (SOLAS 74/88 reg.II-2/18.7) (BCH Code ch.III Part E);
- (EI) 1.1.3.10 examining the arrangements for remote closing of valves for oil fuel, lubricating oil and other flammable oils and confirming, as far as practicable and as appropriate, the operation of the remote means of closing the valves on the tanks that contain oil fuel, lubricating oil and other flammable oils (SOLAS 74/00 reg.II-2/4.2.2.3.4) (SOLAS 74/88 reg.II-2/15.2.5);
- (EI) 1.1.3.11 examining the fire protection arrangements in cargo vehicle and ro-ro spaces and confirming, as far as practicable and as appropriate, the operation of the means for closing the various openings (SOLAS 74/00 regs.II-2/10.7, 20.2.1, 20.3 and 20.6.2) (SOLAS 74/88 reg.II-2/53);
- (EI) 1.1.3.12 examining, when appropriate, the special arrangements for carrying dangerous goods, including checking the electrical equipment and wiring, the ventilation, the provision of protective clothing and portable appliances and the testing of the water supply, bilge pumping and any water spray system (SOLAS 74/00/08 reg.II-2/19 (except 19.3.8, 19.3.10 and 19.4); FSSC chs.9 and 10) (SOLAS 74/88 reg.II-2/54);
- (EI) 1.1.3.13 checking that the life-saving appliances are of international or vivid reddish orange, or a comparably highly visible colour on all parts where this will assist detection at sea (LSAC section 1.2.2.6);
- (EI) 1.1.3.14 checking the provision and disposition of the survival craft, where applicable, marine evacuation systems and rescue boats (SOLAS 74/88 regs.III/11 to 16 and 31; LSAC section 6.2);
- (EI) 1.1.3.15 deployment of 50% of the MES after installation (LSAC paragraph 6.2.2.2);
- (EI) 1.1.3.16 examining each survival craft, including its equipment. For liferafts provided for easy side to side transfer, verifying that they are less than 185 kg (SOLAS 74/88 reg.III/31; LSAC sections 2.5, 3.1 to 3.3 and 4.1 to 4.9) (SOLAS 74/00 reg.III/31.1);
- (EI) 1.1.3.17 examining the embarkation arrangements for each survival craft and the testing of each launching appliance, including overload tests, tests to establish the lowering speed and the lowering of each survival craft to the water with the ship at its lightest sea-going draught, and, where applicable, launching underway at 5 knots, checking the recovery of each lifeboat (SOLAS 74/00 regs.III/11, 12, 13, 16, 31 and 33; LSAC section 6.1);

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| (EI) | 1.1.3.18 | examining the embarkation arrangements for each marine evacuation device, where applicable, and the launching arrangements, including inspection for lack of side shell opening between the embarkation station and waterline, review of distance to the propeller and other life-saving appliances and ensuring that the stowed position is protected from heavy weather damage, as much as practicable (SOLAS 74/00 reg.III/15; LSAC section 6.2); |
| (EI) | 1.1.3.19 | examining each rescue boat, including its equipment. For inflatable rescue boats, confirming that they are stowed in a fully inflated condition (SOLAS 74/88 regs.III/14 and 31; LSAC sections 2.5, 5.1 and 6.1);                                                                                                                                                                                                                                    |
| (EI) | 1.1.3.20 | examining the embarkation and recovery arrangements for each rescue boat and testing each launching and recovery appliance, including overload tests, tests to establish the lowering and recovery speeds and ensuring that each rescue boat can be lowered to the water and recovered with the ship at its lightest sea-going draught, launching underway at 5 knots (SOLAS 74/88 regs.III/14, 17 and 31; LSAC section 6.1);                        |
| (EI) | 1.1.3.21 | testing that the engine of the rescue boat(s) and of each lifeboat, when so fitted, start satisfactorily and operate both ahead and astern (SOLAS 74/00 reg.III/19);                                                                                                                                                                                                                                                                                 |
| (EI) | 1.1.3.22 | confirming that there are posters or signs in the vicinity of survival craft and their launching stations and containers, brackets, racks and other similar stowage locations for life-saving equipment (SOLAS 74/88 regs.III/9 and 20);                                                                                                                                                                                                             |
| (EI) | 1.1.3.23 | examining the provision and stowage and checking the operation of portable on board communications equipment, if provided, and two-way VHF radiotelephone apparatus and search and rescue locating devices (SOLAS 74/88/08 regs.II-2/12.2 and III/6);                                                                                                                                                                                                |
| (EI) | 1.1.3.24 | examining the provision and stowage of the distress flares and the line-throwing appliance, checking the provision and operation of fixed on board communications equipment, if provided, and testing the means of operation of the general alarm system (SOLAS 74/00 regs.III/6 and 18; LSAC sections 3.1, 7.1 and 7.2);                                                                                                                            |
| (EI) | 1.1.3.25 | examining the provision, disposition and stowage of the lifebuoys, including those fitted with self-igniting lights, self-activating smoke signals and buoyant lines, lifejackets <sup>5</sup> , immersion suits and anti-exposure suits (SOLAS 74/00/06 regs.III/7 and 32; LSAC sections 2.1 to 2.5 and 3.1 to 3.3);                                                                                                                                |
| (EI) | 1.1.3.26 | checking the lighting of the muster and embarkation stations and the alleyways, stairways and exits giving access to the muster and embarkation stations, including when supplied from the emergency source of power (SOLAS 74/88 regs.II-1/43 and III/11);                                                                                                                                                                                          |

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<sup>5</sup> Regulation III/7.2.1.5 should be considered.

- (EI) 1.1.3.27 examining the provision and positioning and checking the operation of, as appropriate, the navigation lights, shapes and sound signalling equipment (International Regulations for Preventing Collisions at Sea (COLREG) in force, rules 20 to 24, 27 to 30 and 33);
- (EI) 1.1.3.28 checking that the minimum safe distances from the steering and standard magnetic compasses for all electrical equipment are complied with (SOLAS 74/00 regs.V/17 and 19);
- (EI) 1.1.3.29 checking the electromagnetic compatibility of electrical and electronic equipment on or in the vicinity of the bridge (SOLAS 74/00 reg.V/17);
- (EI) 1.1.3.30 checking, as appropriate, the provision and operation of the following shipborne navigational systems equipment (SOLAS 74/00 reg.V/19):
  - (EI) 1.1.3.30.1 the magnetic compass, including examining the siting, movement, illumination and a pelorus or compass bearing device (SOLAS 74/00 reg.V/19);
  - (EI) 1.1.3.30.2 nautical charts and nautical publications necessary for the intended voyage are available and have been updated, and, where an electronic chart display and information systems (ECDIS) is used, the electronic charts have been updated and the required back-up system is provided and updated (SOLAS 74/00/09 reg.V/19);
  - (EI) 1.1.3.30.3 global navigation satellite system receiver or terrestrial radionavigation system;
  - (EI) 1.1.3.30.4 sound-reception system, when bridge is totally enclosed;
  - (EI) 1.1.3.30.5 means of communication to emergency steering position, where provided;
  - (EI) 1.1.3.30.6 spare magnetic compass;
  - (EI) 1.1.3.30.7 daylight signalling lamp;
  - (EI) 1.1.3.30.8 echo sounding device;
  - (EI) 1.1.3.30.9 radar(s), including examining the waveguide and cable runs for routing and protection and the display unit confirming lighting, correct operation of all controls, and functions;
  - (EI) 1.1.3.30.10 electronic plotting aid, automatic tracking aid or automatic radar plotting aid as appropriate, using the appropriate test facilities;
  - (EI) 1.1.3.30.11 speed and distance measuring devices "through the water" and "over the ground";
  - (EI) 1.1.3.30.12 transmitting heading device providing heading information to radar, plotting aids and automatic identification system equipment;
  - (EI) 1.1.3.30.13 automatic identification system;

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| (EI) | 1.1.3.30.14 | gyrocompass, including examining the alignment of the master and all repeaters;                                                                                                                                                                                                                             |
| (EI) | 1.1.3.30.15 | rudder angle indicator;                                                                                                                                                                                                                                                                                     |
| (EI) | 1.1.3.30.16 | propeller rate of revolution indicator;                                                                                                                                                                                                                                                                     |
| (EI) | 1.1.3.30.17 | propeller, operational mode, thrust, and pitch indicator;                                                                                                                                                                                                                                                   |
| (EI) | 1.1.3.30.18 | rate-of-turn indicator;                                                                                                                                                                                                                                                                                     |
| (EI) | 1.1.3.30.19 | heading or track control system;                                                                                                                                                                                                                                                                            |
| (EI) | 1.1.3.30.20 | BNWAS;                                                                                                                                                                                                                                                                                                      |
| (EI) | 1.1.3.31    | checking for the provision and operation of the voyage data recorder (SOLAS 74/00 reg.V/20);                                                                                                                                                                                                                |
| (EI) | 1.1.3.32    | checking the record of the voyage data recorder annual performance test (SOLAS 74/00 reg.V/18);                                                                                                                                                                                                             |
| (EI) | 1.1.3.33    | checking navigation bridge visibility (SOLAS 74/00 reg.V/22);                                                                                                                                                                                                                                               |
| (EI) | 1.1.3.34    | checking that a valid conformance test report of the long-range identification and tracking system is available on board (SOLAS 04 reg.V/19-1);                                                                                                                                                             |
| (EI) | 1.1.3.35    | checking the provision and, as appropriate, the deployment or operation of the pilot ladders and hoists/pilot transfer arrangements (SOLAS 74/00 reg.V/23);                                                                                                                                                 |
| (EI) | 1.1.3.36    | checking the provision of means of embarkation and disembarkation from ships for use in port and in port-related operations, such as gangways and accommodation ladders (SOLAS 08 reg.II-1/3-9);                                                                                                            |
| (EI) | 1.1.3.37    | checking, when appropriate, the provision of an appropriate instrument for measuring the concentration of gas or oxygen in the air together with detailed instructions for its use (SOLAS 08 reg.VI/3).                                                                                                     |
| (EI) | 1.1.4       | For the life-saving appliances and the other equipment of cargo ships for the additional requirements for oil tankers the survey during construction and after installation should consist of:                                                                                                              |
| (EI) | 1.1.4.1     | checking the deck foam system, including the supplies of foam concentrate, and testing that the minimum number of jets of water at the required pressure in the fire main is obtained (see (EI) 1.1.3.1) when the system is in operation (SOLAS 74/00 reg.II-2/10.8; FSSC ch.15) (SOLAS 74/88 reg.II-2/61); |
| (EI) | 1.1.4.2     | examining the inert gas system (SOLAS 74/00 reg.II-2/4.5.5; FSSC ch.15) (SOLAS 74/88 reg.II-2/62) and in particular:                                                                                                                                                                                        |
| (EI) | 1.1.4.2.1   | examining externally for any sign of gas or effluent leakage;                                                                                                                                                                                                                                               |
| (EI) | 1.1.4.2.2   | confirming the proper operation of both inert gas blowers;                                                                                                                                                                                                                                                  |
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- (EI) 1.1.4.2.3 observing the operation of the scrubber-room ventilation system;
- (EI) 1.1.4.2.4 checking the deck water seal for automatic filling and draining;
- (EI) 1.1.4.2.5 examining the operation of all remotely operated or automatically controlled valves and, in particular, the flue gas isolating valves;
- (EI) 1.1.4.2.6 observing a test of the interlocking feature of soot blowers;
- (EI) 1.1.4.2.7 observing that the gas pressure-regulating valve automatically closes when the inert gas blowers are secured;
- (EI) 1.1.4.2.8 checking, as far as practicable, the following alarms and safety devices of the inert gas system using simulated conditions where necessary:
  - (EI) 1.1.4.2.8.1 high oxygen content of gas in the inert gas main;
  - (EI) 1.1.4.2.8.2 low gas pressure in the inert gas main;
  - (EI) 1.1.4.2.8.3 low pressure in the supply to the deck water seal;
  - (EI) 1.1.4.2.8.4 high temperature of gas in the inert gas main;
  - (EI) 1.1.4.2.8.5 low water pressure or low water-flow rate;
  - (EI) 1.1.4.2.8.6 accuracy of portable and fixed oxygen-measuring equipment by means of calibration gas;
  - (EI) 1.1.4.2.8.7 high water level in the scrubber;
  - (EI) 1.1.4.2.8.8 failure of the inert gas blowers;
  - (EI) 1.1.4.2.8.9 failure of the power supply to the automatic control system for the gas regulating valve and to the instrumentation for continuous indication and permanent recording of pressure and oxygen content in the inert gas main;
  - (EI) 1.1.4.2.8.10 high pressure of gas in the inert gas main;
- (EI) 1.1.4.2.9 checking the proper operation of the inert gas system on completion of the checks listed above;
- (EI) 1.1.4.3 examining the fixed fire-fighting system for the cargo pump room, confirming that the installation tests have been satisfactorily completed and that its means of operation are clearly marked (SOLAS 74/00 reg.II-2/10.9; FSSC chs.5, 6, 7 and 8, as applicable) and, when appropriate, checking the operation of the remote means for closing the various openings;
- (EI) 1.1.4.4 examining the protection of the cargo pump-rooms and confirming that the installation tests have been satisfactorily completed (SOLAS 74/00 reg.II-2/4.5.10) (SOLAS 74/88 regs.II-2/55 to 58).

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| (EI) | 1.1.5    | For the life-saving appliances and the other equipment of cargo ships the check that the required documentation has been placed on board should consist of:                                                                                                                                                                    |
| (EI) | 1.1.5.1  | confirming that the fire control plans are permanently exhibited or, alternatively, emergency booklets have been provided and that a duplicate of the plans or the emergency booklet are available in a prominently marked enclosure external to the ship's deckhouse (SOLAS 74/00 reg.II-2/15.2.4) (SOLAS 74/88 reg.II-2/20); |
| (EI) | 1.1.5.2  | confirming that maintenance plans have been provided (SOLAS 74/00 regs.II-2/14.2.2 and 14.4);                                                                                                                                                                                                                                  |
| (EI) | 1.1.5.3  | confirming that the training manuals and the fire safety operational booklets have been provided (SOLAS 74/00 regs.II-2/15.2.3, 16.2 and 16.3);                                                                                                                                                                                |
| (EI) | 1.1.5.4  | confirming, where appropriate, that the ship is provided with a document indicating compliance with the special requirement for carrying dangerous goods (SOLAS 74/00, reg.II-2/19.4) (SOLAS 74/88/08 reg.II-2/54(3));                                                                                                         |
| (EI) | 1.1.5.5  | confirming that emergency instructions are available for each person on board, that the muster list is posted in conspicuous places and they are in a language understood by the persons on board (SOLAS 74/00 regs.III/8 and 37);                                                                                             |
| (EI) | 1.1.5.6  | confirming that the training manual and training aids for the life-saving appliances have been provided and are available in the working language of the ship (SOLAS 74/00 reg.III/35);                                                                                                                                        |
| (EI) | 1.1.5.7  | confirming that the instructions for onboard maintenance of the life-saving appliances have been provided (SOLAS 74/88 reg.III/36);                                                                                                                                                                                            |
| (EI) | 1.1.5.8  | confirming that a table or curve of residual deviations for the magnetic compass has been provided, and that a diagram of the radar installations shadow sectors is displayed (SOLAS 74/00 reg.V/19);                                                                                                                          |
| (EI) | 1.1.5.9  | checking that operational and, where appropriate, maintenance manuals for all navigational equipment are provided (SOLAS 74/00 reg.V/16);                                                                                                                                                                                      |
| (EI) | 1.1.5.10 | checking that the charts and nautical publications necessary for the intended voyage are available and have been updated (SOLAS 74/88 reg.V/27);                                                                                                                                                                               |
| (EI) | 1.1.5.11 | checking that the International Code of Signals and a copy of Volume III of the International Aeronautical and Maritime Search and Rescue (IAMSAR) Manual have been provided. (SOLAS 74/00/02 reg.V/21);                                                                                                                       |
| (EI) | 1.1.5.12 | checking that arrangements are provided to maintain records of navigational activities and daily reporting (SOLAS 74/00/03 reg.V/28);                                                                                                                                                                                          |

- (EI) 1.1.5.13 checking that the life-saving signals to be used by ships, aircraft or persons in distress are available (SOLAS 74/00 reg.V/29);
- (EI) 1.1.5.14 confirming that continuous synopsis record is provided (SOLAS 74/02 reg.XI-1/5).
- (EI) 1.1.6 For the life-saving appliances and the other equipment of cargo ships, concerning the additional requirements for oil tankers the check that the required documentation has been placed on board should consist of:
  - (EI) 1.1.6.1 confirming, when appropriate, that the instruction manuals for the inert gas system have been provided (FSSC ch.15 paragraph 2.4.4) (SOLAS 74/88 reg.II-2/62.21).
- (EI) 1.1.7 For the life-saving appliances and the other equipment of cargo ships the completion of the initial survey should consist of:
  - (EI) 1.1.7.1 after a satisfactory survey, the Cargo Ship Safety Equipment Certificate and its associated Record of Equipment (Form E) should be issued.
- (EA) 1.2 **Annual surveys** – see part "General" section 4.2.
- (EA) 1.2.1 For the life-saving appliances and the other equipment of cargo ships the examination of current certificates and other records should consist of:
  - (EA) 1.2.1.1 checking the validity, as appropriate, of the Cargo Ship Safety Equipment Certificate, the Cargo Ship Safety Radio Certificate and the Cargo Ship Safety Construction Certificate or the Cargo Ship Safety Certificate;
  - (EA) 1.2.1.2 checking the validity of the Safety Management Certificate (SMC) and that a copy of the Document of Compliance (DOC) is on board;
  - (EA) 1.2.1.3 checking the validity of the International Ship Security Certificate;
  - (EA) 1.2.1.4 checking the validity of the International Load Line Certificate or International Load Line Exemption Certificate;
  - (EA) 1.2.1.5 checking the validity of the International Oil Pollution Prevention Certificate;
  - (EA) 1.2.1.6 checking the certificates of class, if the ship is classed with a classification society;
  - (EA) 1.2.1.7 checking, when appropriate, the validity of the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk or the Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk;
  - (EA) 1.2.1.8 checking, when appropriate, the validity of the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk;

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|------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (EA) | 1.2.1.9  | checking, when appropriate, the validity of the International Pollution Prevention Certificate for the Carriage of Noxious Liquid Substances in Bulk;                                                                                                                                                                               |
| (EA) | 1.2.1.10 | checking, when appropriate, the validity of the International Sewage Pollution Prevention Certificate;                                                                                                                                                                                                                              |
| (EA) | 1.2.1.11 | checking, when appropriate, the validity of the International Air Pollution Prevention Certificate;                                                                                                                                                                                                                                 |
| (EA) | 1.2.1.12 | checking that the ship's complement complies with the Minimum Safe Manning Document (SOLAS 74/00 reg.V/14);                                                                                                                                                                                                                         |
| (EA) | 1.2.1.13 | checking that the master, officers and ratings are certificated as required by the STCW Convention;                                                                                                                                                                                                                                 |
| (EA) | 1.2.1.14 | checking the manning and supervision of survival craft (SOLAS 74/00 reg.III/10);                                                                                                                                                                                                                                                    |
| (EA) | 1.2.1.15 | confirming that, where applicable, the approved documentation for the alternative design and arrangements is on board (SOLAS 06 reg.III/38);                                                                                                                                                                                        |
| (EA) | 1.2.1.16 | checking whether any new equipment has been fitted and, if so, confirming that it has been approved before installation and that any changes are reflected in the appropriate certificate;                                                                                                                                          |
| (EA) | 1.2.1.17 | confirming that the fire control plans are permanently exhibited or, alternatively, that emergency booklets have been provided and that a duplicate of the plans or the emergency booklet are available in a prominently marked enclosure external to the ship's deckhouse (SOLAS 74/00 reg.II-2/15.2.4) (SOLAS 74/88 reg.II-2/20); |
| (EA) | 1.2.1.18 | confirming that the maintenance plans have been provided (SOLAS 74/00 regs.II-2/14.2.2 and 14.4);                                                                                                                                                                                                                                   |
| (EA) | 1.2.1.19 | confirming that the training manuals and the fire safety operational booklets have been provided (SOLAS 74/00 regs.II-2/15.2.3, 16.2 and 16.3);                                                                                                                                                                                     |
| (EA) | 1.2.1.20 | checking whether any fire has occurred on board necessitating the operation of the fixed fire-extinguishing systems or the portable fire extinguishers since the last survey;                                                                                                                                                       |
| (EA) | 1.2.1.21 | checking, when appropriate, that the ship is provided with a document indicating compliance with the special requirements for carrying dangerous goods (SOLAS 74/00/08 reg.II-2/19.4) (SOLAS 74/88 reg.II-2/54(3));                                                                                                                 |
| (EA) | 1.2.1.22 | confirming, when appropriate, that there is a special list, manifest or stowage plan for the carriage of dangerous goods (SOLAS 74/88 reg.VII/5(3));                                                                                                                                                                                |
| (EA) | 1.2.1.23 | confirming, when appropriate, that the instruction manuals for the inert gas system have been provided and checking from the records of the                                                                                                                                                                                         |

- pressure and oxygen content that the inert gas system is being operated correctly (FSSC ch.15) (SOLAS 74/88 reg.II-2/62);
- (EA) 1.2.1.24 checking that log-book entries are being made (SOLAS 74/00 regs.III/19 and 20) and in particular:
- (EA) 1.2.1.24.1 the date when the last full muster of the crew for boat and fire drill took place;
- (EA) 1.2.1.24.2 the records indicating that the lifeboat equipment was examined at that time and found to be complete;
- (EA) 1.2.1.24.3 the last occasion when the lifeboats were swung out and when each one was lowered into the water;
- (EA) 1.2.1.24.4 the records indicating that crew members have received the appropriate onboard training;
- (EA) 1.2.1.25 confirming that the training manual and training aids for the life-saving appliances are available on board in the working language of the ship (SOLAS 74/00 reg.III/35);
- (EA) 1.2.1.26 confirming that the checklist and instructions for on board maintenance of the life-saving appliances are on board (SOLAS 74/00 reg.III/36);
- (EA) 1.2.1.27 confirming that a table or curve of residual deviations for the magnetic compass has been provided, the compass deviation book has been properly maintained and a diagram of the radar installations shadow sectors is displayed (SOLAS 74/00 reg.V/19);
- (EA) 1.2.1.28 checking that operational and, where appropriate, maintenance manuals for all navigational equipment are provided (SOLAS 74/00 reg.V/16);
- (EA) 1.2.1.29 checking that nautical charts and nautical publications necessary for the intended voyage are available and have been updated, and, where electronic systems are used, the required back-up system is provided (SOLAS 74/00 regs.V/19 and 27);
- (EA) 1.2.1.30 checking that the International Code of Signals and a copy of Volume III of the International Aeronautical and Maritime Search and Rescue (IAMSAR) Manual have been provided. (SOLAS 74/00/02 reg.V/21);
- (EA) 1.2.1.31 checking that the life-saving signals to be used by ships, aircraft or persons in distress are available (SOLAS 74/00 reg.V/29).
- (EA) 1.2.1.32 checking that records of navigational activities and daily reporting have been maintained (SOLAS 74/00/04 reg.V/28);
- (EA) 1.2.1.33 confirming that continuous synopsis record is provided (SOLAS74/02 reg.XI-1/5);
- (EA) 1.2.1.34 confirming the availability of the International Anti-Fouling System Certificate (AFS 2001 Annex 4 reg.2), when applicable.

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- (EA) 1.2.2 For the life-saving appliances and the other equipment of cargo ships the annual survey should consist of:
- (EA) 1.2.2.1 examining the fire pumps, fire main, hydrants, hoses and nozzles and the international shore connection and checking that each fire pump, including the emergency fire pump, can be operated separately so that two jets of water are produced simultaneously from different hydrants at any part of the ship whilst the required pressure is maintained in the fire main (SOLAS 74/00 reg.II-2/10.2; FSSC chs.2 and 12) (SOLAS 74/88 regs.II-2/4 and 19);
- (EA) 1.2.2.2 checking the provision and randomly examining the condition of the portable and non-portable fire extinguishers (SOLAS 74/00 reg.II-2/10.3; FSSC ch.4) (SOLAS 74/88 reg.II-2/6);
- (EA) 1.2.2.3 confirming that the fire fighters' outfits and emergency escape breathing devices (EEBDs) are complete and in good condition and that the cylinders, including the spare cylinders, of any required self-contained breathing apparatus are suitably charged (SOLAS 74/00 regs.II-2/10.10, 13.3.4 and 13.4.3; FSSC ch.3) (SOLAS 74/88 reg.II-2/17) (BCH Code ch.III Part E);
- (EA) 1.2.2.4 checking the operational readiness and maintenance of fire-fighting systems (SOLAS 74/00 reg.II-2/14) (SOLAS 74/88/91 reg.II-2/21);
- (EA) 1.2.2.5 examining the fixed fire-fighting system for the machinery, cargo, vehicle, special category and ro-ro spaces, as appropriate, and confirming that its means of operation is clearly marked (SOLAS 74/00 regs.II-2/10.4, 10.5, 10.7 and 20.6.1; FSSC chs.5 to 7) (SOLAS 74/88 regs.II-2/7 and 53);
- (EA) 1.2.2.6 examining the fire-extinguishing and special arrangements in the machinery spaces and confirming, as far as practicable and as appropriate, the operation of the remote means of control provided for the opening and closing of the skylights, the release of smoke, the closure of the funnel and ventilation openings, the closure of power operated and other doors, the stopping of ventilation and boiler forced and induced draft fans and the stopping of oil fuel and other pumps that discharge flammable liquids (SOLAS 74/00 regs.II-2/5.2, 8.3, 9.5 and 10.5) (SOLAS 74/88 regs.II-2/7 and 11);
- (EA) 1.2.2.7 checking that fixed carbon dioxide fire-extinguishing systems for the protection of machinery spaces and cargo pump-rooms, where applicable, are provided with two separate controls, one for opening of the gas piping and one for discharging the gas from the storage container, each of them located in a release box clearly identified for the particular space (SOLAS 08 reg.II-2/10.4.1.5);
- (EA) 1.2.2.8 examining, as far as possible, and testing, as feasible, any fire detection and alarm system (SOLAS 74/00 regs.II-2/7.2, 7.3, 7.4, 7.5.1, 7.5.5, 19.3.3 and 20.4; FSSC ch.9) (SOLAS 74/88 regs.II-2/11, 13, 14, 53 and 54);

- (EA) 1.2.2.9 examining the fire-extinguishing systems for spaces containing paint and/or flammable liquids and deep-fat cooking equipment in accommodation and service spaces (SOLAS 74/00 regs.II-2/10.6.3 and 10.6.4; FSSC chs.5 to 7) (SOLAS 74/88 reg.II-2/18.7) (BCH Code ch.III Part E);
- (EA) 1.2.2.10 examining the helicopter facilities (SOLAS 74/00 reg.II-2/18) (SOLAS 74/88 reg.II-2/18.8);
- (EA) 1.2.2.11 examining the arrangements for remote closing of valves for oil fuel, lubricating oil and other flammable oils and confirming, as far as practicable and as appropriate, the operation of the remote means of closing the valves on the tanks that contain oil fuel, lubricating oil and other flammable oils (SOLAS 74/00 reg.II-2/4.2.2.3.4) (SOLAS 74/88 reg.II-2/15.2.5);
- (EA) 1.2.2.12 examining and testing of the general emergency alarm system (SOLAS 74/88 reg.III/20);
- (EA) 1.2.2.13 examining the fire protection arrangements in cargo, vehicle and ro-ro spaces and confirming, as far as practicable and as appropriate, the operation of the means of control provided for closing the various openings (SOLAS 74/00 regs.II-2/10.7, 20.2.1, 20.3 and 20.6.2) (SOLAS 74/88 reg.II-2/53);
- (EA) 1.2.2.14 examining, when appropriate, the special arrangements for carrying dangerous goods, including checking the electrical equipment and wiring, the ventilation, the provision of protective clothing and portable appliances and the testing of the water supply, bilge pumping and any water spray system (SOLAS 74/00/08 reg.II-2/19 (except 19.3.8, 19.3.10 and 19.4)) (SOLAS 74/88 reg.II-2/54);
- (EA) 1.2.2.15 checking that emergency instructions are available for each person on board and that copies of the suitably updated muster list are posted in conspicuous places and that they are in a language understood by all persons on board and confirming that there are posters or signs in the vicinity of survival craft and their launching stations (SOLAS 74/00 regs.III/8, 9 and 37);
- (EA) 1.2.2.16 checking that the life-saving appliances are of international or vivid reddish orange, or a comparably highly visible colour on all parts where this will assist detection at sea (LSAC section 1.2.2.6);
- (EA) 1.2.2.17 examining each survival craft, including its equipment and, when fitted, the on-load release and hydrostatic lock and, for inflatable liferafts, the hydrostatic release unit and float-free arrangements. Checking that the hand-held flares are not out of date (SOLAS 74/00 regs.III/20 and 31; LSAC sections 2.5, 3.1 to 3.3);
- (EA) 1.2.2.18 for liferafts provided for easy side to side transfer, verifying that they are less than 185 kg (SOLAS 74/00 reg.III/31.1);
- (EA) 1.2.2.19 checking that the falls used in launching appliances have been periodically inspected and have been renewed as necessary in the past 5 years (SOLAS 74/00 reg.III/20);

- (EA) 1.2.2.20 examining the embarkation arrangements and launching appliances for each survival craft. Each lifeboat should be lowered to the embarkation position or, if the stowage position is the embarkation position, lowered a short distance and, if practicable, one of the survival craft should be lowered to the water. The operation of the launching appliances for davit-launched liferafts should be demonstrated. Checking that a thorough examination of launching appliances, including the dynamic testing of the winch brake, and servicing of lifeboat and rescue boat on-load release gear, including free-fall lifeboat release systems and davit-launched liferaft automatic release hooks, has been carried out (SOLAS 74/00 regs.III/11, 12, 13, 16, 20 and 31; LSAC section 6.1);
- (EA) 1.2.2.21 examining each rescue boat, including its equipment. For inflatable rescue boats, confirming that they are stowed in a fully inflated condition (SOLAS 74/88 reg.III/14 and 31; LSAC sections 2.5 and 5.1);
- (EA) 1.2.2.22 confirming that there are posters or signs in the vicinity of the survival craft, their launching stations and containers, brackets, racks and other similar stowage locations for life-saving equipment (SOLAS 74/00 regs.III/9 and 20);
- (EA) 1.2.2.23 examining the embarkation and recovery arrangements for each rescue boat. If practicable, the rescue boat(s) should be lowered to the water and its recovery demonstrated (SOLAS 74/00 regs.III/14, 17 and 31; LSAC section 6.1);
- (EA) 1.2.2.24 testing that the engine of the rescue boat(s) and of each lifeboat, when so fitted, start satisfactorily and operate both ahead and astern;
- (EA) 1.2.2.25 examining and checking the operation of two-way VHF radiotelephone apparatus and search and rescue locating devices (SOLAS 74/88/08 reg.III/6);
- (EA) 1.2.2.26 examining the line-throwing appliance and checking that its rockets and the ship's distress signals are not out of date, and examining and checking the operation of on board communications equipment and the general emergency alarm system (SOLAS 74/00 regs.II-2/12.2 and III/6 and 18; LSAC sections 3.1, 7.1 and 7.2);
- (EA) 1.2.2.27 examining the provision, disposition, stowage and the condition of the lifebuoys, including those fitted with self-igniting lights, self-activating smoke signals and buoyant lines, lifejacket<sup>6</sup> and their whistles and lights, immersion suits and anti-exposure suits and that their associated batteries are not out of date (SOLAS 74/88/06 regs.III/7 and 32, LSAC sections 2.1 to 2.5);
- (EA) 1.2.2.28 checking that immersion suits designed to be worn in conjunction with a lifejacket are suitably marked (LSAC section 2.3.1);
- (EA) 1.2.2.29 checking the lighting of the muster and embarkation stations and the alleyways, stairways and exits giving access to the muster and

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<sup>6</sup> Regulation III/7.2.1.5 should be considered.



- embarkation stations, including when supplied from the emergency source of power (SOLAS 74/88 regs.II-1/42 or 43 and III/11);
- (EA) 1.2.2.30 checking that the required the navigation lights, shapes and sound signalling equipment are in order (International Regulations for Preventing Collisions at Sea (COLREG) in force, rules 20 to 24, 27 to 30 and 33);
- (EA) 1.2.2.31 checking that the following items of navigation equipment are in working order, as appropriate: daylight signalling lamp, magnetic compass, transmitting heading device, gyro compass, gyro compass repeaters, radar installation(s), electronic plotting aid, automatic tracking aid(s) or automatic radar plotting aid(s), echo-sounding device, speed and distance measuring device(s), rudder angle indicator, propeller rate of revolution indicator, variable-pitch propeller pitch and operational mode indicator, rate-of-turn indicator, heading or track control system, GNSS receiver, terrestrial radio navigation system and sound reception system, means of communication with emergency steering position, a pelorus or compass bearing device, means for correcting heading and bearings, a BNWAS as applicable and ECDIS including back-up arrangements, as applicable. Items that cannot be checked with the ship in port should be verified from records (SOLAS 74/00/09 reg.V/19);
- (EA) 1.2.2.32 checking that the International Code of Signals is available (SOLAS 74/00 reg.V/21);
- (EA) 1.2.2.33 checking the rotational deployment of MES (SOLAS 74/88 reg.III/20.8.2; LSAC section 6.2.2.2);
- (EA) 1.2.2.34 checking the provision, specification, operation and annual performance test of the voyage data recorder, where fitted (SOLAS 74/00/04 reg.V/20);
- (EA) 1.2.2.35 checking the provision, operation and the annual test has been carried out for the automatic identification system, where fitted (SOLAS 74/00/04 reg.V/19);
- (EA) 1.2.2.36 checking that a valid conformance test report of the long-range identification and tracking system is available on board, where fitted (SOLAS 04 reg.V/19-1);
- (EA) 1.2.2.37 checking the provision and specification of the pilot ladders and hoists/pilot transfer arrangements (SOLAS 74/00 reg.V/23);
- (EA) 1.2.2.38 checking that the means of embarkation and disembarkation from ships for use in port and in port-related operations, such as gangways and accommodation ladders, are in satisfactory condition, as applicable (SOLAS 08 reg.II-1/3-9);
- (EA) 1.2.2.39 checking, when appropriate, the provision of an appropriate instrument for measuring the concentration of gas or oxygen in the air together with detailed instructions for its use (SOLAS 08 reg.VI/3).

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| (EA) | 1.2.3        | For the life-saving appliances and the other equipment of cargo ships, concerning the additional requirements for oil tankers the annual survey should consist of:                                                                                                                                         |
| (EA) | 1.2.3.1      | checking the deck foam system, including the supplies of foam concentrate and testing that the minimum number of jets of water at the required pressure in the fire main is obtained (see (EA) 1.2.2.1) when the system is in operation (SOLAS 74/00 reg.II-2/10.8; FSSC ch.14) (SOLAS 74/88 reg.II-2/61); |
| (EA) | 1.2.3.2      | examining the inert gas system (SOLAS 74/00 reg.II-2/4.5.5; FSSC ch.15) (SOLAS 74/88 reg.II-2/62), and in particular:                                                                                                                                                                                      |
| (EA) | 1.2.3.2.1    | examining externally for any sign of gas or effluent leakage;                                                                                                                                                                                                                                              |
| (EA) | 1.2.3.2.2    | confirming the proper operation of both inert gas blowers;                                                                                                                                                                                                                                                 |
| (EA) | 1.2.3.2.3    | observing the operation of the scrubber-room ventilation system;                                                                                                                                                                                                                                           |
| (EA) | 1.2.3.2.4    | checking the deck water seal for automatic filling and draining;                                                                                                                                                                                                                                           |
| (EA) | 1.2.3.2.5    | examining the operation of all remotely operated or automatically controlled valves and, in particular, the flue gas isolating valves;                                                                                                                                                                     |
| (EA) | 1.2.3.2.6    | observing a test of the interlocking feature of soot blowers;                                                                                                                                                                                                                                              |
| (EA) | 1.2.3.2.7    | observing that the gas pressure regulating valve automatically closes when the inert gas blowers are secured;                                                                                                                                                                                              |
| (EA) | 1.2.3.2.8    | checking, as far as practicable, the following alarms and safety devices of the inert gas system using simulated conditions where necessary:                                                                                                                                                               |
| (EA) | 1.2.3.2.8.1  | high oxygen content of gas in the inert gas main;                                                                                                                                                                                                                                                          |
| (EA) | 1.2.3.2.8.2  | low gas pressure in the inert gas main;                                                                                                                                                                                                                                                                    |
| (EA) | 1.2.3.2.8.3  | low pressure in the supply to the deck water seal;                                                                                                                                                                                                                                                         |
| (EA) | 1.2.3.2.8.4  | high temperature of gas in the inert gas main;                                                                                                                                                                                                                                                             |
| (EA) | 1.2.3.2.8.5  | low water pressure or low water-flow rate;                                                                                                                                                                                                                                                                 |
| (EA) | 1.2.3.2.8.6  | accuracy of portable and fixed oxygen-measuring equipment by means of calibration gas;                                                                                                                                                                                                                     |
| (EA) | 1.2.3.2.8.7  | high water level in the scrubber;                                                                                                                                                                                                                                                                          |
| (EA) | 1.2.3.2.8.8  | failure of the inert gas blowers;                                                                                                                                                                                                                                                                          |
| (EA) | 1.2.3.2.8.9  | failure of the power supply to the automatic control system for the gas regulating valve and to the instrumentation for continuous indication and permanent recording of pressure and oxygen content in the inert gas main;                                                                                |
| (EA) | 1.2.3.2.8.10 | high pressure of gas in the inert gas main;                                                                                                                                                                                                                                                                |
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- (EA) 1.2.3.3 checking, when practicable, the proper operation of the inert gas system on completion of the checks listed above (FSSC ch.15) (SOLAS 74/88 reg.II-2/62);
- (EA) 1.2.3.4 examining the fixed fire-fighting system for the cargo pump rooms (SOLAS 74/00 reg.II-2/10.9) (SOLAS 74/88 reg.II-2/63) and confirming, as far as practicable and when appropriate, the operation of the remote means for closing the various openings;
- (EA) 1.2.3.5 checking condition and operation of water spray and air supply systems that are in totally enclosed lifeboats and have self-contained air support systems (LSAC sections 4.4 and 4.6 to 4.9);
- (EA) 1.2.3.6 checking protection of cargo pump room (SOLAS 74/00 reg.II-2/4.5.10), and in particular:
  - (EA) 1.2.3.6.1 checking temperature sensing devices for bulkhead glands and alarms;
  - (EA) 1.2.3.6.2 checking interlock between lighting and ventilation;
  - (EA) 1.2.3.6.3 checking gas detection system;
  - (EA) 1.2.3.6.4 checking bilge level monitoring devices and alarms.
- (EA) 1.2.4 For the life-saving appliances and the other equipment of cargo ships the completion of the annual survey should consist of:
  - (EA) 1.2.4.1 after a satisfactory survey, the Cargo Ship Safety Equipment Certificate should be endorsed;
  - (EA) 1.2.4.2 if a survey shows that the condition of a ship or its equipment is unsatisfactory, see part "General", section 4.8.
- (EP) **1.3 Periodical surveys** – see part "General" section 4.4.
  - (EP) 1.3.1 For the life-saving appliances and the other equipment of cargo ships the examination of current certificates and other records should consist of:
    - (EP) 1.3.1.1 the provisions of (EA) 1.2.1.
  - (EP) 1.3.2 For the life-saving appliances and the other equipment of cargo ships the periodical survey should consist of:
    - (EP) 1.3.2.1 the provisions of (EA) 1.2.2;
    - (EP) 1.3.2.2 confirming, during the examination of the fixed fire-fighting system for the machinery, cargo, vehicle, special category and ro-ro spaces, that, as appropriate, any foam compounds and the CO<sub>2</sub> capacity have been checked and that the distribution pipework has been proved clear (SOLAS 74/00 regs.II-2/10.4, 10.5, 10.7 and 20.6.1; FSSC chs.5 to 7) (SOLAS 74/88 regs.II-2/7 and 53);

- (EP) 1.3.2.3 testing the operation of the remote means of control provided for the opening and closing of the skylights, the release of smoke, the closure of the funnel and ventilation openings, the closure of power operated and other doors, the stopping of ventilation and boiler forced and induced draft fans and the stopping of oil fuel and other pumps that discharge flammable liquids (SOLAS 74/00 regs.II-2/5.2, 8.3, 9.5 and 10.5) (SOLAS 74/88 reg.II-2/11);
- (EP) 1.3.2.4 testing any fire detection and alarm system (SOLAS 74/00 regs.II-2/7.2, 7.3, 7.4, 7.5.5, 19.3.3 and 20.4; FSSC ch.9) (SOLAS 74/88 regs.II-2/11, 13, 14, 53 and 54);
- (EP) 1.3.2.5 testing, as feasible, the fire-extinguishing system for spaces containing paint and/or flammable liquids and deep-fat cooking equipment in accommodation and service spaces (SOLAS 74/00 regs.II-2/10.6.3 and 10.6.4; FSSC chs.5 to 7) (SOLAS 74/88 reg.II-2/18.7);
- (EP) 1.3.2.6 testing the remote closing of valves for oil fuel, lubricating oil and other flammable oils and the operation of the remote means of closing the valves on the tanks that contain oil fuel, lubricating oil and other flammable oils (SOLAS 74/00 reg.II-2/4.2.2.3.4) (SOLAS 74/88 reg.II-2/15.2.5);
- (EP) 1.3.2.7 testing the operation of the means of control provided for closing the various openings for the cargo, vehicle, special category and ro-ro spaces (SOLAS 74/00 regs.II-2/5.2 and 20.3) (SOLAS 74/88 reg.II-2/53);
- (EP) 1.3.2.8 testing, as feasible, the helicopter facilities (SOLAS 74/00 reg.II-2/18) (SOLAS 74/88 reg.II-2/18.8).
- (EP) 1.3.3 For the life-saving appliances and the other equipment for the additional requirements for oil tankers the periodical survey should consist of:
- (EP) 1.3.3.1 the provisions of (EA) 1.2.3;
- (EP) 1.3.3.2 confirming during the examination of the fixed fire-fighting system for the cargo pump rooms that, as appropriate, any foam compounds have been checked and that the distribution pipework has been proved clear (SOLAS 74/00 reg.II-2/10.9; FSSC chs.5 to 7) (SOLAS 74/88 reg.II-2/63) and checking the operation of the remote means for closing the various openings.
- (EP) 1.3.4 For the life-saving appliances and the other equipment of cargo ships the completion of the periodical survey should consist of:
- (EP) 1.3.4.1 after a satisfactory survey, the cargo Ship Safety Equipment Certificate should be endorsed;
- (EP) 1.3.4.2 if a survey shows that the condition of a ship or its equipment is unsatisfactory, see part "General", section 4.8.

- (ER) **1.4**                    **Renewal surveys** – see part "General" section 4.5
- (ER) 1.4.1                    For the life-saving appliances and the other equipment of cargo ships the examination of current certificates and other records should consist of:
- (ER) 1.4.1.1                the provisions of (EA) 1.2.1, except for the validity of the Cargo Ship Safety Equipment Certificate.
- (ER) 1.4.2                    For the life-saving appliances and the other equipment of cargo ships the renewal survey should consist of:
- (ER) 1.4.2.1                the provisions of (EP) 1.3.2.
- (ER) 1.4.3                    For the life-saving appliances and the other equipment of cargo ships, concerning the additional requirements for oil tankers the renewal survey should consist of:
- (ER) 1.4.3.1                the provisions of (EP) 1.3.3;
- (ER) 1.4.3.2                examining the deck water seal for the inert gas system internally and checking the condition of the non-return valve (FSSC ch.15, paragraphs 2.2.4 and 2.3.1.4) (SOLAS 74/88 reg.II-2/62).
- (ER) 1.4.4                    For the life-saving appliances and the other equipment of cargo ships the completion of the renewal survey should consist of:
- (ER) 1.4.4.1                after a satisfactory survey, the Cargo Ship Safety Equipment Certificate should be issued.
- (C) **2**                        **GUIDELINES FOR SURVEYS FOR THE CARGO SHIP SAFETY CONSTRUCTION CERTIFICATE**
- (CI) **2.1**                    **Initial surveys** – see part "General", section 4.1.
- (CI) 2.1.1                    For the hull, machinery and equipment of cargo ships the examination of plans and designs should consist of:
- (CI) 2.1.1.1                examining the plans for the hull (SOLAS 74/88 regs.II-1/11, 12-1, 14, 18 and 19) (SOLAS 06 regs.II-1/9, 10, 11, 12, 13-1, 15, 15-1, 16 and 16-1);
- (CI) 2.1.1.2                examining the plans for the bilge pumping and drainage systems (SOLAS 74/88 reg.II-1/21) (SOLAS 05/08/09 regs.II-1/35-1 and II-2/20.6.1.4);
- (CI) 2.1.1.3                examining the stability information and the damage control plans (SOLAS 74/88/00 regs.II-1/22, 23-1 and 25) (SOLAS 06/08 regs.II-1/5, 5-1 and 19; IS Code chs.1, 2 and 3);
- (CI) 2.1.1.4                examining the plans for the machinery installation (SOLAS 74/88 regs.II-1/26 to 36);
- (CI) 2.1.1.5                examining the plans for the electrical installation (SOLAS 74/88 regs.II-1/40, 41, 43, 44 and 45);

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| (CI) | 2.1.1.6  | examining, where applicable, the approved documentation for the alternative design and arrangements (SOLAS 06 reg.II-1/55);                                                                                                                                                                                                                                                                                                                                           |
| (CI) | 2.1.1.7  | examining the plans for the periodically unattended machinery spaces (SOLAS 74/00 reg.II-2/4.2.5) (SOLAS 74/88 regs.II-1/46 to 53);                                                                                                                                                                                                                                                                                                                                   |
| (CI) | 2.1.1.8  | examining the plans for the structural fire protection, including ventilation systems, in accommodation and service spaces, control stations and machinery spaces and oil fuel and lubricating oil systems (SOLAS 74/00 regs.II-2/4.2.2, 4.2.2.3, 4.2.2.4, 4.2.2.5, 4.4, 5.2, 5.3.1, 5.3.2, 6.2, 6.3, 7.5.5, 7.7, 8.2, 8.4, 9.2.1, 9.2.2, 9.3, 9.5, 9.7.1, 9.7.2, 9.7.3, 9.7.5.2, 11.2, 11.3, 11.4, 11.5 and 17) (SOLAS 74/88 regs.II-2/42 to 52 (except 45 and 51)); |
| (CI) | 2.1.1.9  | examining the plans for the structural fire protection, including ventilation systems, in cargo spaces (SOLAS 74/00 regs.II-2/5.2, 11.2, 11.3, 11.5, 19.3.8, 19.3.10, 20.2.1 and 20.3) (SOLAS 74/88 regs.II-2/42 to 54);                                                                                                                                                                                                                                              |
| (CI) | 2.1.1.10 | examining the plans for the means of escape (SOLAS 74/00 regs.II-2/13.2, 13.3.1, 13.3.3, 13.4.2 and 13.6; FSSC ch.13 paragraph 3) (SOLAS 74/88 reg.II-2/45);                                                                                                                                                                                                                                                                                                          |
| (CI) | 2.1.1.11 | examining the plans for the arrangements for gaseous fuel for domestic purposes (SOLAS 74/00 reg.II-2/4.3) (SOLAS 74/88 reg.II-2/51);                                                                                                                                                                                                                                                                                                                                 |
| (CI) | 2.1.1.12 | examining the arrangements for the openings in the shell plating below the freeboard deck, (SOLAS 06 reg.II-1/15);                                                                                                                                                                                                                                                                                                                                                    |
| (CI) | 2.1.1.13 | examining the plans for helicopter facilities for ships fitted with such facilities (SOLAS 74/00 reg.II-2/18) (SOLAS 74/88 reg.II-2/18.8);                                                                                                                                                                                                                                                                                                                            |
| (CI) | 2.1.1.14 | examining the Cargo Securing Manual for ships carrying cargo units including containers (SOLAS 74/98 reg.VI/5.6);                                                                                                                                                                                                                                                                                                                                                     |
| (CI) | 2.1.1.15 | checking for the loading booklet for carriage of cargoes in bulk (SOLAS 74/00 reg.VI/7);                                                                                                                                                                                                                                                                                                                                                                              |
| (CI) | 2.1.1.16 | examining the loading instrument for bulk carriers of 150 m in length and upwards (SOLAS 74/97/04 reg.XII/11);                                                                                                                                                                                                                                                                                                                                                        |
| (CI) | 2.1.1.17 | confirming that bulk carriers, when appropriate, meet the requirements of damage stability and structural strength with its cargo hold(s) flooded, including other structural requirements(SOLAS 74/97/04 regs.XII/3, 4, 5 and 6);                                                                                                                                                                                                                                    |
| (CI) | 2.1.1.18 | examining the functionality of bilge well alarms to all cargo holds and conveyor tunnels (SOLAS 74/97/04 reg.XII/9);                                                                                                                                                                                                                                                                                                                                                  |
| (CI) | 2.1.1.19 | confirming that the ship is constructed in accordance with the requirements of a recognized classification society, or one with equivalent national standards (SOLAS 74/00 reg.II-1/3-1);                                                                                                                                                                                                                                                                             |
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- (CI) 2.1.1.20 confirming that a corrosion prevention system is fitted, when appropriate, in dedicated seawater ballast tanks arranged in ships and double-side skin spaces arranged in bulk carriers of 150 m in length and upwards (SOLAS 74/04/06 reg.II-1/3-2);
- (CI) 2.1.1.21 examining, for oil tankers and bulk carriers when appropriate, the Ship Structure Access Manual (SOLAS 74/00/02/04 reg.II-1/3-6(4));
- (CI) 2.1.1.22 for bulk carriers, checking the arrangements for hold, ballast and dry space water level detectors and their audible and visual alarms (SOLAS 74/02 reg.XII/12);
- (CI) 2.1.1.23 for bulk carriers, checking the arrangements for availability of draining and pumping systems forward of the collision bulkhead (SOLAS 74/02 reg.XII/13);
- (CI) 2.1.1.24 examining the calculation and drawings for the sufficient safe working load of towing and mooring equipment to enable the safe conduct of all towing and mooring operation in normal operation of the ship (SOLAS 74/04 reg.II-1/3-8);
- (CI) 2.1.1.25 checking the provision of means to prevent blockage of drainage arrangements, for closed vehicle and ro-ro spaces and special category spaces where fixed pressure water-spraying systems are used (SOLAS 08 reg.II-2/20.6.1.5).
- (CI) 2.1.2 For the hull, machinery and equipment of cargo ships, concerning the examination of plans and designs the additional requirements for oil tankers, chemical tankers and gas carriers should consist of:
  - (CI) 2.1.2.1 examining the plans for the steering gear (SOLAS 74/88 reg.II-1/29);
  - (CI) 2.1.2.2 examining the plans for the electrical installation (SOLAS 74/00 reg.II-1/43) (SOLAS 74/88 reg.II-1/45);
  - (CI) 2.1.2.3 examining the plans for the structural fire protection (SOLAS 74/00 regs.II-2/1.6, 4.5.1, 4.5.2, 4.5.9, 9.2.4, 9.3, 9.4, 9.5, 9.6.5 and 11.6) (SOLAS 74/88 regs.II-2/55 to 58);
  - (CI) 2.1.2.4 examining the plans for the cargo tank venting, cargo tank purging and gas-freeing and other ventilation arrangements and protection of the cargo tank structure against pressure or vacuum (SOLAS 74/00 regs.II-2/4.5.3, 4.5.4, 4.5.6, 4.5.8, 11.6 and 16.3) (SOLAS 74/88 reg.II-2/59);
  - (CI) 2.1.2.5 examining the plans of access to bow (SOLAS 74/00/04 reg.II-1/3-3);
  - (CI) 2.1.2.6 examining the plans for emergency towing, for tankers of not less than 20,000 tonnes deadweight (SOLAS 74/00/04 reg.II-1/3-4);
  - (CI) 2.1.2.7 checking the access to spaces in the cargo area of oil tankers (SOLAS 74/88/92/00 reg.II-1/12-2) (SOLAS 04 reg.II-1/3-6).
- (CI) 2.1.3 For the hull, machinery and equipment of cargo ships the survey during construction and after installation should consist of:

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| (CI) | 2.1.3.1   | confirming that the collision bulkhead is watertight up to the freeboard deck, that the valves fitted on the pipes piercing the collision bulkhead are operable from above the freeboard deck and that there are no doors, manholes, ventilation ducts or any other openings (SOLAS 74/88 reg.II-1/11) (SOLAS 06 reg.II-1/12);                                                                                                                                                     |
| (CI) | 2.1.3.2   | confirming that the subdivision bulkheads are constructed and tested as watertight up to the freeboard deck or margin line, as applicable (SOLAS 74/88 reg.II-1/14) (SOLAS 06 reg.II-1/10 and 11);                                                                                                                                                                                                                                                                                 |
| (CI) | 2.1.3.3   | confirming that each watertight door has been tested (SOLAS 74/88 reg.II-1/18) (SOLAS 06 reg.II-1/16);                                                                                                                                                                                                                                                                                                                                                                             |
| (CI) | 2.1.3.4   | confirming that the arrangements for operating any watertight doors are generally in accordance with the requirements for passenger ships and carrying out similar tests, (see (PI) 5.1.2.6 to (PI) 5.1.2.8) (SOLAS 74/88 reg.II-1/15) (SOLAS 06 reg.II-1/13-1);                                                                                                                                                                                                                   |
| (CI) | 2.1.3.5   | confirming by a hose or flooding test the watertightness of watertight decks and trunks, tunnels and ventilators (SOLAS 74/88 reg.II-1/19) (SOLAS 06 reg.II-1/16-1);                                                                                                                                                                                                                                                                                                               |
| (CI) | 2.1.3.6   | confirming that each bilge pump and the bilge pumping system provided for each watertight compartment is working efficiently (SOLAS 74/88 reg.II-1/21) (SOLAS 05 reg.II-1/35-1);                                                                                                                                                                                                                                                                                                   |
| (CI) | 2.1.3.7   | confirming that the drainage system of enclosed cargo spaces situated on the freeboard deck is working efficiently (SOLAS 74/88 reg.II-1/21) (SOLAS 05 reg.II-1/35-1);                                                                                                                                                                                                                                                                                                             |
| (CI) | 2.1.3.7.1 | examining visually the drainage facilities for blockage or other damage and confirming the provision of means to prevent blockage of drainage arrangements, for closed vehicle and ro-ro spaces and special category spaces where fixed pressure water-spraying systems are used (SOLAS 08 reg.II-2/20.6.1.5);                                                                                                                                                                     |
| (CI) | 2.1.3.8   | conducting an inclining test, when this is required (SOLAS 74/88 reg.II-1/22) (SOLAS 06 reg.II-1/5);                                                                                                                                                                                                                                                                                                                                                                               |
| (CI) | 2.1.3.9   | confirming that the machinery, boilers and other pressure vessels, associated piping systems and fittings are installed and protected so as to reduce to a minimum any danger to persons on board, due regard being given to moving parts, hot surfaces and other hazards (SOLAS 74/00 reg.II-2/4.2 (except 4.2.2.3.4 relating to remote closing of valves included in safety equipment)) (SOLAS 74/88 regs.II-1/26, 32, 33 and 34) (SOLAS 74/88/06 reg.II-2/15 (except 15.2.5 )); |
| (CI) | 2.1.3.10  | confirming that the normal operation of the propulsion machinery can be sustained or restored even though one of the essential auxiliaries becomes inoperative (SOLAS 74/88 reg.II-1/26);                                                                                                                                                                                                                                                                                          |
| (CI) | 2.1.3.11  | confirming that means are provided so that the machinery can be brought into operation from the dead ship condition without external aid (SOLAS 74/88 reg.II-1/26);                                                                                                                                                                                                                                                                                                                |
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- (CI) 2.1.3.12 confirming that the boilers, all parts of the machinery, all steam, hydraulic, pneumatic and other systems and their associated fittings which are under internal pressure have been subjected to the appropriate tests, including a pressure test as may be specified in the requirements of the Administration or the classification societies (SOLAS 74/88 reg.II-1/26);
- (CI) 2.1.3.13 confirming that means are provided to ensure that the safe speed is not exceeded where there is the risk of machinery overspeeding (SOLAS 74/88 reg.II-1/27);
- (CI) 2.1.3.14 confirming that, where practicable, means are provided to protect against overpressure in the parts of main, auxiliary and other machinery that are subject to internal pressure and may be subject to dangerous overpressure (SOLAS 74/88 reg.II-1/27);
- (CI) 2.1.3.15 confirming that, when required, crankcase explosion relief devices are fitted to internal combustion engines and that they are arranged so as to minimize the possibility of injury to personnel (SOLAS 74/88 reg.II-1/27);
- (CI) 2.1.3.16 confirming that main turbine propulsion machinery and, where applicable, main internal combustion propulsion machinery and auxiliary machinery are provided with automatic shut-off arrangements in the case of failures, such as lubricating oil supply failure, which could rapidly lead to a complete breakdown, serious damage or explosion (SOLAS 74/88 reg.II-1/27);
- (CI) 2.1.3.17 confirming and recording the ability of the machinery to reverse the direction of the thrust of the propeller in sufficient time and to bring the ship to rest within a reasonable distance, including the effectiveness of any supplementary means of manoeuvring or stopping the ship (SOLAS 74/88 reg.II-1/28);
- (CI) 2.1.3.18 confirming that the main and auxiliary steering gear are so arranged that the failure of one of them does not render the other inoperative (SOLAS 74/88 reg.II-1/29);
- (CI) 2.1.3.19 confirming that, where appropriate, essential components of the steering gear are permanently lubricated or provided with lubrication fittings (SOLAS 74/88 reg.II-1/29);
- (CI) 2.1.3.20 confirming that relief valves are fitted to any part of a steering gear hydraulic system which can be isolated and in which pressure can be generated from the power source or from external forces and that these relief valves are set to a pressure not exceeding the design pressure (SOLAS 74/88 reg.II-1/29);
- (CI) 2.1.3.21 confirming that the main steering gear is capable of steering the ship at maximum ahead service speed and is capable of putting the rudder over from 35° on one side to 35° on the other side with the ship at its deepest seagoing draught and running ahead at maximum ahead service speed and, under the same conditions, from 35° on either side

- to 30° on the other side in not more than 28 s (SOLAS 74/88 reg.II-1/29);
- (CI) 2.1.3.22 confirming that the auxiliary steering gear is capable of steering the ship at navigable speed and of being brought speedily into action in an emergency and that it is capable of putting the rudder over from 15° on one side to 15° on the other side in not more than 60 s with the ship at its deepest seagoing draught and running ahead at one half of the maximum ahead service speed or 7 knots, whichever is the greater (SOLAS 74/88 reg.II-1/29);
- (CI) 2.1.3.23 confirming that the main and auxiliary steering gear power units restart automatically when power is restored after a power failure, that they are capable of being brought into operation from a position on the navigating bridge and that, in the event of a power failure to any one of the steering gear power units, an audible and visual alarm is given on the navigating bridge (SOLAS 74/88 reg.II-1/29);
- (CI) 2.1.3.24 confirming that, where the main steering gear comprises two or more identical power units and an auxiliary steering gear is not fitted, a defect can be isolated so that steering capability can be maintained or speedily regained after a single failure in its piping system or in one of the power units (SOLAS 74/88 reg.II-1/29);
- (CI) 2.1.3.25 confirming that the control systems for the main steering gear from both the navigating bridge and the steering gear compartment are operating satisfactorily (SOLAS 74/88 reg.II-1/29);
- (CI) 2.1.3.26 confirming that, where the main steering gear comprises two or more identical power units and an auxiliary steering gear is not fitted, the two independent control systems from the navigating bridge are operating satisfactorily (SOLAS 74/88 reg.II-1/29);
- (CI) 2.1.3.27 confirming that the control system for the auxiliary steering gear in the steering gear compartment and, if this gear is power operated, from the navigating bridge are operating satisfactorily and that the latter is independent of the control system for the main steering gear (SOLAS 74/88 reg.II-1/29);
- (CI) 2.1.3.28 confirming that the control system for any main and auxiliary steering gear control system operable from the navigating bridge is capable of being brought into operation from a position on the navigating bridge, that means are provided in the steering gear compartment for disconnecting it from the steering gear that it serves and that an audible and visual alarm is given on the navigating bridge in the event of a failure of electrical power supply (SOLAS 74/88 reg.II-1/29);
- (CI) 2.1.3.29 confirming that the electric power circuits and steering gear control systems, together with their associated components, cables and pipes, are separated, as far as practicable, throughout their length (SOLAS 74/88 reg.II-1/29);
- (CI) 2.1.3.30 confirming that the means of communication between the bridge and the steering gear compartment is operating satisfactorily and that, with

- ships having emergency steering positions, a telephone or other means of communication for relaying heading information and supplying visual compass readings to the emergency steering position are provided (SOLAS 74/88 reg.II-1/29) (SOLAS 74/00 reg.V/19);
- (CI) 2.1.3.31 confirming that the angular position of the rudder is indicated independently of the steering control system on the navigating bridge if the main steering gear is power-operated and that this angular position is given in the steering gear compartment (SOLAS 74/88 reg.II-1/29) (SOLAS 74/00 reg.V/19);
- (CI) 2.1.3.32 confirming that with a hydraulic power-operated steering gear the audible and visual low-level alarms on the navigating bridge and in the machinery space for each hydraulic fluid reservoir are operating satisfactorily and that at least one power actuating system including the reservoir can be recharged from a position within the steering gear compartment by means of a fixed storage tank (to which a contents gauge is fitted) with fixed piping (SOLAS 74/88 reg.II-1/29);
- (CI) 2.1.3.33 confirming that the steering gear compartment is readily accessible, that it is separated, as far as practicable, from machinery spaces and is provided with suitable arrangements to ensure working access to steering gear machinery and controls under safe conditions (SOLAS 74/88 reg.II-1/29);
- (CI) 2.1.3.34 confirming that with electric and electro-hydraulic steering gear the means are provided for indicating on the navigating bridge and at a main machinery control position that the motors are running and that the overload alarm and alarm for the loss of a phase in a three-phase supply located at the main machinery control position are operating satisfactorily (SOLAS 74/88 reg.II-1/30);
- (CI) 2.1.3.35 confirming that the main and auxiliary machinery essential for propulsion and the safety of the ship are provided with the effective means for its operation and control (SOLAS 74/88 reg.II-1/31);
- (CI) 2.1.3.36 confirming that appropriate means are provided where it is intended that the propulsion machinery should be remotely controlled from the navigating bridge, including, where necessary, the control, monitoring, reporting, alert and safety actions. (SOLAS 74/00/02 reg.II-1/31);
- (CI) 2.1.3.37 confirming that arrangements to operate main and other machinery from a machinery control room are satisfactory (SOLAS 74/88 reg.II-1/31);
- (CI) 2.1.3.38 confirming that, in general, means are provided for manually overriding automatic controls and that a failure does not prevent the use of the manual override (SOLAS 74/88 reg.II-1/31);
- (CI) 2.1.3.39 confirming that oil-fired and exhaust gas boilers, unfired steam generators, steam pipe systems and air pressure systems are fitted with the appropriate safety features (SOLAS 74/88 regs.II-1/32, 33 and 34);

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| (CI) | 2.1.3.40   | confirming the operation of the ventilation for the machinery spaces (SOLAS 74/88 reg.II-1/35);                                                                                                                                                                      |
| (CI) | 2.1.3.41   | confirming that the measures to prevent noise in machinery spaces are effective (SOLAS 74/88 reg.II-1/36);                                                                                                                                                           |
| (CI) | 2.1.3.42   | confirming that the engine room telegraph giving visual indication of the orders and answers both in the machinery space and on the navigating bridge is operating satisfactorily (SOLAS 74/88, reg.II-1/37);                                                        |
| (CI) | 2.1.3.43   | confirming that the second means of communication between the navigation bridge and machinery space is also operating satisfactorily and that appropriate means are provided to any other positions from which the engines are controlled (SOLAS 74/88 reg.II-1/37); |
| (CI) | 2.1.3.44   | confirming that the engineer's alarm is clearly audible in the engineers' accommodation (SOLAS 74/88 reg.II-1/38);                                                                                                                                                   |
| (CI) | 2.1.3.45   | confirming that precautions, taken to prevent any oil that may escape under pressure from any pump, filter or heater from coming into contact with heated surfaces, are efficient (SOLAS 74/00 reg.II-2/4.2.2.3);                                                    |
| (CI) | 2.1.3.46   | confirming that the means of ascertaining the amount of oil contained in any oil tank are in good working condition (SOLAS 74/00 reg.II-2/4.2.2.3);                                                                                                                  |
| (CI) | 2.1.3.47   | confirming that the devices provided to prevent overpressure in any oil tank or in any part of the oil system, including the filling pipes, are in good working condition (SOLAS 74/00 reg.II-2/4.2.2.4);                                                            |
| (CI) | 2.1.3.48   | confirming that forepeak tanks are not intended for carriage of oil fuel, lubrication oil and other flammable oils (SOLAS 74/00 reg.II-2/4.2.2.3);                                                                                                                   |
| (CI) | 2.1.3.49   | confirming that the electrical installations, including the main source of power and lighting systems, are installed in accordance with the approved plans (SOLAS 74/88 regs.II-1/40 and 41);                                                                        |
| (CI) | 2.1.3.50   | confirming that a self-contained emergency source of electrical power has been provided and that the appropriate systems are satisfactorily supplied (SOLAS 74/88 reg.II-1/43);                                                                                      |
| (CI) | 2.1.3.51   | confirming that the starting arrangements of each emergency generating set are satisfactory (SOLAS 74/88 reg.II-1/44);                                                                                                                                               |
| (CI) | 2.1.3.52   | confirming that precautions have been provided against shock, fire and other hazards of electrical origin (SOLAS 74/88 reg.II-1/45);                                                                                                                                 |
| (CI) | 2.1.3.53   | confirming that the arrangements for periodically unattended machinery spaces are satisfactory (SOLAS 74/88 regs.II-1/46 to 53) and in particular:                                                                                                                   |
| (CI) | 2.1.3.53.1 | checking the fire precautions and testing alarms, as appropriate;                                                                                                                                                                                                    |
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- (CI) 2.1.3.53.2 checking the means for the protection against flooding;
- (CI) 2.1.3.53.3 checking the means to control the propulsion from the navigating bridge;
- (CI) 2.1.3.53.4 ensuring that a means of vocal communication between the main machinery control room or its control position, as appropriate, and the navigating bridge and engineer officer's accommodation is provided and is effective;
- (CI) 2.1.3.53.5 checking that an alarm system is provided with random testing of functions;
- (CI) 2.1.3.53.6 checking that means are provided to automatically shut down machinery or boiler operations in the event of serious malfunction and testing the alarms;
- (CI) 2.1.3.53.7 ensuring that special requirements for the machinery, boiler and electrical installations, as appropriate, are provided;
- (CI) 2.1.3.54 confirming that all aspects of the structural fire protection, including the ventilation systems, in accommodation and service spaces, control stations and machinery spaces are installed in accordance with the approved plans, testing the operation of the means of closing the main inlets and outlets of all ventilation systems and proving that the power ventilation is capable of being stopped from outside the space served (SOLAS 74/00 regs.II-2/4.4, 5.2, 5.3.1, 5.3.2, 6.2, 6.3, 7.5.5, 7.7, 8.2, 8.4, 9.2.1, 9.3, 9.4.2, 9.5, 9.7.1, 9.7.2, 9.7.3, 9.7.5.2, 11.2, 11.3, 11.4 and 11.5) (SOLAS 74/88 regs.II-2/42 to 44, 46 to 50 and 52);
- (CI) 2.1.3.55 confirming that all aspects of the structural fire protection, including the ventilation systems, in cargo spaces are installed in accordance with the approved plans, testing the operation of the means of closing the main inlets and outlets of all ventilation systems and proving that the power ventilation is capable of being stopped from outside the space served (SOLAS 74/00 regs.II-2/5.2.1, 11.2, 11.3, 11.5, 19.3.8, 19.3.10, 20.2.1 and 20.3) (SOLAS 74/88 regs.II-2/42 to 44, 46 to 50 and 52 to 54);
- (CI) 2.1.3.56 confirming that stairways and ladders are so arranged as to provide a means of escape from all accommodation spaces and from spaces in which the crew is normally employed, other than machinery spaces, to the open deck and thence to the lifeboats and liferafts (SOLAS 74/00 regs.II-2/13.2, 13.3.1, 13.3.3 and 13.6; FSSC ch.13 paragraph 3) (SOLAS 74/88 reg.II-2/45) and in particular that:
  - (CI) 2.1.3.56.1 at all levels of accommodation there are provided at least two widely separated means of escape from each restricted space or group of spaces;
  - (CI) 2.1.3.56.2 below the lowest open deck the main means of escape is a stairway (the second being a trunk or a stairway);

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| (CI) | 2.1.3.56.3 | above the lowest open deck the means of escape are stairways or doors to an open deck or a combination of them;                                                                                                                                                                                                                             |
| (CI) | 2.1.3.56.4 | the radiotelegraph station has direct access to the open deck or is provided with two means of access or egress, one of which is a porthole or window of sufficient size;                                                                                                                                                                   |
| (CI) | 2.1.3.57   | confirming that two widely separated means of escape and, when appropriate, a fire shelter from the lower part of the space, are provided from each machinery space of Category A and that suitable escape routes are provided from other machinery spaces (SOLAS 74/00 reg.II-2/13.4.2; FSSC ch.13 paragraph 3) (SOLAS 74/88 reg.II-2/45); |
| (CI) | 2.1.3.58   | examining the arrangements for gaseous fuel for domestic purposes (SOLAS 74/00 reg.II-2/4.3);                                                                                                                                                                                                                                               |
| (CI) | 2.1.3.59   | confirming, when appropriate, that all aspects of the helicopter facilities are installed in accordance with the approved plans (SOLAS 74/00 reg.II-2/18) (SOLAS 74/88 reg.II-2/18.8);                                                                                                                                                      |
| (CI) | 2.1.3.60   | confirming that installed materials do not contain asbestos (SOLAS 74/00/09 reg.II-1/3-5);                                                                                                                                                                                                                                                  |
| (CI) | 2.1.3.61   | confirming, for bulk carriers, that dedicated sea water ballast tanks have an efficient corrosion protection system such as hard coating (SOLAS 74/00 reg.II-1/3-2).                                                                                                                                                                        |
| (CI) | 2.1.3.62   | confirming that dedicated sea water ballast tanks arranged in ships and double side skin spaces arranged in bulk carriers of 150 m in length and upward when appropriate have been coated in accordance with resolution MSC.215(82) (SOLAS 74/00/06 reg.II-1/3-2);                                                                          |
| (CI) | 2.1.3.63   | confirming for oil tankers and bulk carriers, when appropriate, the provision of means of access to cargo and other spaces in accordance with the arrangements in the Ship Structures Access Manual (SOLAS 74/00/02/04 reg.II-1/3-6);                                                                                                       |
| (CI) | 2.1.3.64   | for bulk carriers, examining and testing the hold, ballast and dry space water level detectors and their audible and visual alarms (SOLAS 74/02 reg.XII/12);                                                                                                                                                                                |
| (CI) | 2.1.3.65   | for bulk carriers, checking the arrangements for availability of draining and pumping systems forward of the collision bulkhead (SOLAS 74/02 reg.XII/13);                                                                                                                                                                                   |
| (CI) | 2.1.3.66   | confirming, for bulk carriers, that the loading instrument is on board and functioning (SOLAS 74/97/04 reg.XII/11);                                                                                                                                                                                                                         |
| (CI) | 2.1.3.67   | confirming that ship's identification number is permanently marked (SOLAS 74/02 reg.XI-1/3);                                                                                                                                                                                                                                                |
| (CI) | 2.1.3.68   | confirming that the towing and mooring equipment is properly marked with any restriction associated with its safe operation (SOLAS 74/04 reg.II-1/3-8).                                                                                                                                                                                     |

- (CI) 2.1.4 For the hull, machinery and equipment of cargo ships, concerning the additional requirements for oil tankers the survey during construction and after installation should consist of:
- (CI) 2.1.4.1 confirming, when appropriate, that the main steering gear comprises the necessary two or more identical power units and the requisite arrangements to regain steering capability in the event of the prescribed single failure (SOLAS 74/88 reg.II-1/29);
- (CI) 2.1.4.2 confirming that a hull return system of distribution and earthed distribution system are not used (SOLAS 74/88 reg.II-1/45);
- (CI) 2.1.4.3 confirming that all aspects of the location of spaces and the structural fire protection, including the special arrangements when the ship is a combination carrier, are in accordance with the approved plans (SOLAS 74/00 regs.II-2/1.6, 4.5.1, 4.5.2, 4.5.9, 9.2.4, 9.3 and 9.6.5) (SOLAS 74/88 regs.II-2/55 to 58);
- (CI) 2.1.4.4 confirming that permanent approved gastight lighting enclosures for illuminating cargo pump rooms, having adequate strength and not impairing the integrity and gas tightness of the bulkheads or decks, are fitted in bulkheads and decks separating cargo pump rooms and other spaces (SOLAS 74/00 reg.II-2/4.5.2.5) (SOLAS 74/88 reg.II-2/58.5);
- (CI) 2.1.4.5 confirming that all aspects of the cargo tank venting, cargo tank purging and gas-freeing and other ventilation arrangements and protection of the cargo tank structure against pressure or vacuum are in accordance with the approved plans (SOLAS 74/00 regs.II-2/4.5.3, 4.5.4, 4.5.6, 4.5.8 and 11.6) (SOLAS 74/88 regs.II-2/59 and 62.13.1 to 62.13.3);
- (CI) 2.1.4.6 confirming that access to bow is arranged in accordance with approved plans (SOLAS 74/00/04 reg.II-1/3-3);
- (CI) 2.1.4.7 confirming, for tankers of not less than 20,000 tonnes deadweight, that emergency towing is arranged in accordance with approved plans (SOLAS 74/00/04 reg.II-1/3-4);
- (CI) 2.1.4.8 confirming when appropriate that dedicated sea water ballast tanks have an efficient corrosion protection system such as hard coating (SOLAS 74/00/06 reg.II-1/3-2).
- (CI) 2.1.5 For the hull, machinery and equipment of cargo ships, concerning the additional requirements for chemical tankers and gas carriers, the survey during construction and after installation should consist of:
- (CI) 2.1.5.1 the provisions of (CI) 2.1.4.
- (CI) 2.1.6 For the hull, machinery and equipment of cargo ships the check that the required documentation has been placed on board should consist of:
- (CI) 2.1.6.1 confirming that the stability information and the damage control plans have been provided (SOLAS 74/88 regs.II-1/22 and 23-1) (SOLAS 06 regs.II-1/5-1 and 19);

- (CI) 2.1.6.2 confirming that the manoeuvring booklet has been provided and that the manoeuvring information has been displayed on the navigating bridge (SOLAS 74/88 reg.II-1/28);
- (CI) 2.1.6.3 confirming that the approved Cargo Securing Manual for ships carrying cargo units including containers is provided on board (SOLAS 74/94 reg.VI/5.6);
- (CI) 2.1.6.4 confirming, for oil tankers and bulk carriers when appropriate, that the Ship Structure Access Manual is on board (SOLAS 74/00/02/04 reg.II-1/3-6(4));
- (CI) 2.1.6.5 confirming that a set of as-built construction drawings is available on board (SOLAS 74/04 reg.II-1/3-7);
- (CI) 2.1.6.6 confirming when appropriate that a coating technical file reviewed by the Administration has been provided on board (SOLAS 74/00/06 reg.II-1/3-2);
- (CI) 2.1.6.7 checking the provision of a ship-specific emergency towing procedure (SOLAS 08 reg.II-1/3-4);
- (CI) 2.1.6.8 confirming that, where applicable, the approved documentation for the alternative design and arrangements is on board (SOLAS 00/06 regs.II-1/55 and II-2/17).
- (CI) 2.1.7 For the hull, machinery and equipment of cargo ships the completion of the initial survey should consist of:
  - (CI) 2.1.7.1 after a satisfactory survey, the Cargo Ship Safety Construction Certificate should be issued.
- (CA) **2.2 Annual surveys** – see part "General", section 4.2.
  - (CA) 2.2.1 For the hull, machinery and equipment of cargo ships the examination of current certificates and other records should consist of:
    - (CA) 2.2.1.1 checking the validity, as appropriate, of the Cargo Ship Safety Equipment Certificate, the Cargo Ship Safety Radio Certificate and the Cargo Ship Safety Construction Certificate or the Cargo Ship Safety Certificate;
    - (CA) 2.2.1.2 checking the validity of the Safety Management Certificate (SMC) and that a copy of the Document of Compliance (DOC) is on board;
    - (CA) 2.2.1.3 checking the validity of the International Ship Security Certificate;
    - (CA) 2.2.1.4 checking the validity of the International Load Line Certificate or International Load Line Exemption Certificate;
    - (CA) 2.2.1.5 checking the validity of the International Oil Pollution Prevention Certificate;



- (CA) 2.2.1.6 checking the certificates of class, if the ship is classed with a classification society;
- (CA) 2.2.1.7 checking, when appropriate, the validity of the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk or the Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk;
- (CA) 2.2.1.8 checking, when appropriate, the validity of the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk;
- (CA) 2.2.1.9 checking, when appropriate, the validity of the International Pollution Prevention Certificate for the Carriage of Noxious Liquid Substances in Bulk;
- (CA) 2.2.1.10 checking, when appropriate, the validity of the International Sewage Pollution Prevention Certificate;
- (CA) 2.2.1.11 checking, when appropriate, the validity of the International Air Pollution Prevention Certificate;
- (CA) 2.2.1.12 checking that the ship's complement complies with the Minimum Safe Manning Document (SOLAS 74/00 reg.V/14) (SOLAS 74/88 reg.V/13(b));
- (CA) 2.2.1.13 checking that the master, officers and ratings are certificated as required by the STCW Convention;
- (CA) 2.2.1.14 confirming that, where applicable, the approved documentation for the alternative design and arrangements is on board (SOLAS 00/06 regs.II-1/55 and II-2/17);
- (CA) 2.2.1.15 checking whether any new equipment has been fitted and, if so, confirm that it has been approved before installation and that any changes are reflected in the appropriate certificate;
- (CA) 2.2.1.16 checking the provision of a ship-specific emergency towing procedure (SOLAS 08 reg.II-1/3-4);
- (CA) 2.2.1.17 confirming that the stability information, including damage stability, where applicable, and the damage control plans are on board (SOLAS 74/88 regs.II-1/22, 23 and 25) (SOLAS 06 reg.II-1/5-1 and 19);
- (CA) 2.2.1.18 confirming that the manoeuvring booklet is on board and that the manoeuvring information is displayed on the navigating bridge (SOLAS 74/88 reg.II-1/28);
- (CA) 2.2.1.19 checking by the log-book entries that the testing and the emergency drills of the steering gear have been carried out (SOLAS 74/00 reg.V/26) (SOLAS 74/88 reg.V/19);
- (CA) 2.2.1.20 checking that the routine surveys of the boilers and other pressure vessels, as determined by the Administration, have been carried out as required and that safety devices, such as the boiler safety valves, have been tested;

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|------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (CA) | 2.2.1.21 | checking that, as appropriate, the hull and machinery has been presented for survey in accordance with the continuous survey scheme approved by the Administration or a classification society;                                                                                                                                                                                                                        |
| (CA) | 2.2.1.22 | confirming, when appropriate, that a complete file of the enhanced survey reports and the Condition Evaluation Report are on board <sup>7</sup> ;                                                                                                                                                                                                                                                                      |
| (CA) | 2.2.1.23 | confirming that suitable Material Safety Data Sheets are available on board;                                                                                                                                                                                                                                                                                                                                           |
| (CA) | 2.2.1.24 | confirming, for bulk carriers, that the loading/unloading booklet required in SOLAS regulation VI/7.2 is on board (SOLAS 74/97/04 reg.XII/8.1);                                                                                                                                                                                                                                                                        |
| (CA) | 2.2.1.25 | confirming, that bulk carriers of 150 m in length and upwards of single side skin construction designed to carry solid bulk cargoes having a density of 1,780 kg/m <sup>3</sup> and above, constructed before 1 July 1999, have, after the implementation date given in SOLAS 94/97 reg.XII/3, sufficient stability and strength to withstand flooding of the foremost cargo hold (SOLAS 74/97/04 reg.XII/3, 4 and 6); |
| (CA) | 2.2.1.26 | confirming approved Cargo Securing Manual for ships carrying cargo units including containers is on board (SOLAS 74/94 reg.VI/5.6);                                                                                                                                                                                                                                                                                    |
| (CA) | 2.2.1.27 | confirming that the loading booklet for carriage of cargoes in bulk is on board (SOLAS 74/00 reg.VI/7);                                                                                                                                                                                                                                                                                                                |
| (CA) | 2.2.1.28 | confirming, for oil tankers and bulk carriers when appropriate, that the Ship Structure Access Manual is on board (SOLAS 74/00/02, reg.II-1/3-6(4));                                                                                                                                                                                                                                                                   |
| (CA) | 2.2.1.29 | confirming that structural alterations performed, if any, have been approved by the classification society and reported on the as-built drawings kept on board (SOLAS 74/04 reg.II-1/3-7);                                                                                                                                                                                                                             |
| (CA) | 2.2.1.30 | confirming when appropriate that the coating technical file is available on board (SOLAS 74/00/06 reg.II-1/3-2);                                                                                                                                                                                                                                                                                                       |
| (CA) | 2.2.1.31 | confirming when appropriate that the maintenance of the protective coating is included in the overall ship's maintenance system (SOLAS 74/00/06 reg.II-1/3-2);                                                                                                                                                                                                                                                         |
| (CA) | 2.2.1.32 | confirming the availability of the International Anti-Fouling System Certificate (AFS 2001 Annex 4 reg.2), when applicable.                                                                                                                                                                                                                                                                                            |
| (CA) | 2.2.2    | For the hull <sup>8</sup> , machinery and equipment of cargo ships the annual survey should consist of:                                                                                                                                                                                                                                                                                                                |
| (CA) | 2.2.2.1  | examining, in general and as far as can be seen, the hull and its closing appliances;                                                                                                                                                                                                                                                                                                                                  |

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<sup>7</sup> See the International Code on the Enhanced Programme of Inspections During Surveys of Bulk Carriers and Oil Tankers, 2011 (resolution A...(27)).

<sup>8</sup> See also annex A to the International Code on Enhanced Programme of Inspections During Surveys of Bulk Carriers and Oil Tankers, 2011 (resolution A...(27)).

- (CA) 2.2.2.2 examining the anchoring and mooring equipment as far as can be seen. For ships built after 01/01/2007, confirming that the towing and mooring equipment is properly marked with any restriction associated with its safe operation (SOLAS 74/04 reg.II-1/3-8);
- (CA) 2.2.2.3 examining the collision and the other watertight bulkheads as far as can be seen (SOLAS 74/88 regs.II-1/11 and 14) (SOLAS 06 regs.II-1/10, 11 and 12);
- (CA) 2.2.2.4 examining and testing (locally and remotely) all the watertight doors in watertight bulkheads (SOLAS 74/88 reg.II-1/18) (SOLAS 06 reg.II-1/16);
- (CA) 2.2.2.5 examining the arrangements for closing openings in the shell plating below the freeboard deck (SOLAS 06 reg.II-1/15);
- (CA) 2.2.2.6 examining each bilge pump and confirming that the bilge pumping system for each watertight compartment is satisfactory (SOLAS 74/88 reg.II-1/21) (SOLAS 05 reg.II-1/35-1);
- (CA) 2.2.2.7 confirming that the drainage from enclosed cargo spaces situated on the freeboard deck is satisfactory (SOLAS 74/88 reg.II-1/21) (SOLAS 05 reg.II-1/35-1);
- (CA) 2.2.2.7.1 examining visually the drainage facilities for blockage or other damage and confirming the provision of means to prevent blockage of drainage arrangements, for closed vehicle and ro-ro spaces and special category spaces where fixed pressure water-spraying systems are used (SOLAS 08 reg.II-2/20.6.1.5);
- (CA) 2.2.2.8 confirming that the machinery, boilers and other pressure vessels, associated piping systems and fittings are installed and protected so as to reduce to a minimum any danger to persons on board, due regard being given to moving parts, hot surfaces and other hazards (SOLAS 74/00 reg.II-2/4.2 (except 4.2.2.3.4 relating to remote closing of valves included in safety equipment)) (SOLAS 74/88 regs.II-1/26, 32, 33 and 34) (SOLAS 74/88/06 reg.II-2/15 (except 15.2.5));
- (CA) 2.2.2.9 confirming that the normal operation of the propulsion machinery can be sustained or restored even though one of the essential auxiliaries becomes inoperative (SOLAS 74/88 reg.II-1/26);
- (CA) 2.2.2.10 confirming that means are provided so that the machinery can be brought into operation from the dead ship condition without external aid (SOLAS 74/88 reg.II-1/26);
- (CA) 2.2.2.11 carrying out a general examination of the machinery, the boilers, all steam, hydraulic, pneumatic and other systems and their associated fittings to see whether they are being properly maintained and with particular attention to the fire and explosion hazards (SOLAS 74/88 regs.II-1/26 and 27);
- (CA) 2.2.2.12 examining and testing the operation of main and auxiliary steering arrangements, including their associated equipment and control systems (SOLAS 74/88 reg.II-1/29);

- (CA) 2.2.2.13 confirming that the means of communication between the navigation bridge and steering gear compartment and the means of indicating the angular position of the rudder are operating satisfactorily (SOLAS 74/88 reg.II-1/29) (SOLAS 74/00 reg.V/19);
- (CA) 2.2.2.14 confirming that with ships having emergency steering positions there are means of relaying heading information and, when appropriate, of supplying visual compass readings to the emergency steering position (SOLAS 74/88 reg.II-1/29 and SOLAS 74/00 reg.V/19 or the SOLAS 74/88 text in force prior to 1 July 2002 reg.V/12 as appropriate);
- (CA) 2.2.2.15 confirming that the various alarms required for hydraulic power-operated, electric and electro-hydraulic steering gears are operating satisfactorily and that the re-charging arrangements for hydraulic power-operated steering gears are being maintained (SOLAS 74/88 regs.II-1/29 and 30);
- (CA) 2.2.2.16 examining the means for the operation of the main and auxiliary machinery essential for the propulsion and the safety of the ship, including, when applicable, the means of remotely controlling the propulsion machinery from the navigating bridge (including the control, monitoring, reporting, alert and safety actions) and the arrangements to operate the main and other machinery from a machinery control room (SOLAS 74/88/00/02 reg.II-1/31);
- (CA) 2.2.2.17 confirming the operation of the ventilation for the machinery spaces (SOLAS 74/88 reg.II-1/35);
- (CA) 2.2.2.18 confirming that the measures to prevent noise in machinery spaces are effective (SOLAS 74/88 reg.II-1/36);
- (CA) 2.2.2.19 confirming that the engine room telegraph, the second means of communication between the navigation bridge and the machinery space and the means of communication with any other positions from which the engines are controlled are operating satisfactorily (SOLAS 74/88 reg.II-1/37);
- (CA) 2.2.2.20 confirming that the engineer's alarm is clearly audible in the engineers' accommodation (SOLAS 74/88 reg.II-1/38);
- (CA) 2.2.2.21 examining, as far as practicable, visually and in operation, the electrical installations, including the main source of power and the lighting systems (SOLAS 74/88 regs.II-1/40 and 41);
- (CA) 2.2.2.22 confirming, as far as practicable, the operation of the emergency source(s) of electrical power including their starting arrangements, the systems supplied and, when appropriate, their automatic operation (SOLAS 74/88 regs.II-1/43 and 44);
- (CA) 2.2.2.23 examining, in general, that the precautions provided against shock, fire and other hazards of electrical origin are being maintained (SOLAS 74/88 reg.II-1/45);

- (CA) 2.2.2.24 examining the arrangements for periodically unattended machinery spaces (SOLAS 74/88 regs.II-1/46 to 53) and, in particular, the random testing of alarm, automatic and shutdown functions;
- (CA) 2.2.2.25 confirming, as far as practicable, that no changes have been made in the structural fire protection, examining any manual and automatic fire doors and proving their operation, testing the means of closing the main inlets and outlets of all ventilation systems and testing the means of stopping power ventilation systems from outside the space served (SOLAS 74/00 regs.II-2/4.4, 5.2, 5.3.1, 5.3.2, 6.2, 6.3, 7.5.5, 7.7, 8.2, 8.3, 8.4, 9.2.1, 9.2.3, 9.3, 9.4.2, 9.5, 9.7.1, 9.7.2, 9.7.3, 9.7.5.2, 11.2, 11.3, 11.4, 11.5, 19.3.8, 19.3.10, 20.2.1 and 20.3) (SOLAS 74/88 regs.II-2/42 to 44, 46 to 50 and 52);
- (CA) 2.2.2.26 confirming that the means of escape from accommodation, machinery and other spaces are satisfactory ( SOLAS 74/00 regs.II-2/13.2, 13.3.1, 13.3.3, 13.4.2 and 13.6) (SOLAS 74/88 reg.II-2/45);
- (CA) 2.2.2.27 examining the arrangements for gaseous fuel for domestic purposes (SOLAS 74/00 reg.II-2/4.3) (SOLAS 74/88 reg.II-2/51);
- (CA) 2.2.2.28 examining visually the condition of any expansion joints in seawater systems;
- (CA) 2.2.2.29 confirming, when appropriate and as far as is practicable when examining internal spaces on oil tankers and bulk carriers, that the means of access to cargo and other spaces remain in good condition. (SOLAS 74/00/02 reg.II-1/3-6);
- (CA) 2.2.2.30 confirming that no new materials containing asbestos were installed on board (SOLAS 74/00/04/09 reg.II-1/3-5);
- (CA) 2.2.2.31 examining the functionality of bilge well alarms to all cargo holds and conveyor tunnels (SOLAS 74/97/04 reg.XII/9);
- (CA) 2.2.2.32 for bulk carriers, examining the hold, ballast and dry space water level detectors and their audible and visual alarms. (SOLAS 74/02 reg.XII/12);
- (CA) 2.2.2.33 for bulk carriers, checking the arrangements for availability of draining and pumping systems forward of the collision bulkhead (SOLAS 74/02 reg.XII/13);
- (CA) 2.2.2.34 confirming that ship's identification number is permanently marked (SOLAS 74/02 reg.XI-1/3);
- (CA) 2.2.2.35 for single hull, single hold cargo ships, examining the cargo hold water level detector and its audible and visual alarm (SOLAS 74/04 reg.II-1/23-3) (SOLAS 06 reg.II-1/25);
- (CA) 2.2.2.36 confirming that the coating system in dedicated SWB tanks in ships and double side skin spaces arranged in bulk carriers of 150 m in length and upward when appropriate is maintained and that maintenance, repair and partial recoating are recorded in the coating technical file (SOLAS 74/00/06 reg.II-1/3-2);

- (CA) 2.2.2.37 confirming, for bulk carriers constructed before 1 July 1999 with restrictions imposed with respect to the carriage of cargoes with a density of 1,780 kg/m<sup>3</sup> and above, that a triangle is permanently marked at midship (SOLAS 74/97/04 reg.XII/8.3);
- (CA) 2.2.2.38 confirming, for bulk carriers, that the loading instrument is on board and functioning (SOLAS 74/97/04 reg.XII/11).
- (CA) 2.2.3 For the hull<sup>9</sup>, machinery and equipment of cargo ships, concerning the additional requirements for oil tankers, the annual survey should consist of:
- (CA) 2.2.3.1 confirming, when appropriate, that the requisite arrangements to regain steering capability in the event of the prescribed single failure are being maintained (SOLAS 74/88 reg.II-1/29);
- (CA) 2.2.3.2 examining the cargo tank openings, including gaskets, covers, coamings and screens;
- (CA) 2.2.3.3 examining the cargo tank pressure/vacuum valves and devices to prevent the passage of flame (SOLAS 74/00 reg.II-2/11.6);
- (CA) 2.2.3.4 examining the devices to prevent the passage of flame on vents to all bunker, oily-ballast and oily-slop tanks and void spaces, as far as practicable;
- (CA) 2.2.3.5 examining the cargo tank venting, cargo tank purging and gas-freeing and other ventilation systems (SOLAS 74/00 reg.II-2/4.5.3, 4.5.4, 4.5.6 and 4.5.8) (SOLAS 74/88 reg.II-2/59);
- (CA) 2.2.3.6 examining the cargo, crude oil washing, ballast and stripping systems both on deck and in the cargo pump rooms and the bunker system on deck;
- (CA) 2.2.3.7 confirming that all electrical equipment in dangerous zones is suitable for such locations, is in good condition and is being properly maintained;
- (CA) 2.2.3.8 confirming that potential sources of ignition in or near the cargo pump room are eliminated, such as loose gear, combustible materials, etc., that there are no signs of undue leakage and that access ladders are in good condition;
- (CA) 2.2.3.9 examining all pump room bulkheads for signs of oil leakage or fractures and, in particular, the sealing arrangements of all penetrations of cargo pump room bulkheads;
- (CA) 2.2.3.10 examining, as far as practicable, the cargo, bilge, ballast and stripping pumps for undue gland seal leakage, verification of proper operation of electrical and mechanical remote operating and shutdown devices and operation of cargo pump room bilge system, and checking that pump foundations are intact;

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<sup>9</sup> See also annex B to the International Code on the Enhanced Programme of Inspections During Surveys of Bulk Carriers and Oil Tankers, 2011 (resolution A...(27)).

- (CA) 2.2.3.11 confirming that the pump room ventilation system is operational, ducting intact, dampers are operational and screens clean;
- (CA) 2.2.3.12 verifying that installed pressure gauges on cargo discharge lines and level indicator systems are operational;
- (CA) 2.2.3.13 examining access to bow arrangement (SOLAS 74/00/04 reg.II-1/3-3);
- (CA) 2.2.3.14 examining the towing arrangement for tankers of not less than 20,000 tonnes deadweight (SOLAS 74/00/04 reg.II-1/3-4);
- (CA) 2.2.3.15 confirming that the corrosion prevention system fitted to dedicated ballast water tanks of oil tankers and bulk carriers when appropriate is maintained (SOLAS 74/00 reg.II-1/3-2);
- (CA) 2.2.3.16 examining the emergency lighting in all cargo pump rooms of tankers constructed after 1 July 2002 (SOLAS 74/00 reg.II-1/43).
- (CA) 2.2.4 For the hull, machinery and equipment of cargo ships, concerning the additional requirements for chemical tankers and gas carriers, the annual survey should consist of:
  - (CA) 2.2.4.1 the provisions of (CA) 2.2.3.
- (CA) 2.2.5 For the hull, machinery and equipment of cargo ships the completion of the annual survey should consist of:
  - (CA) 2.2.5.1 after a satisfactory survey, the Cargo Ship Safety Construction Certificate should be endorsed;
  - (CA) 2.2.5.2 if a survey shows that the condition of a ship or its equipment is unsatisfactory, see part "General", section 4.8.
- (CIn) **2.3 Intermediate surveys** – see part "General", section 4.3
  - (CIn) 2.3.1 For the hull, machinery and equipment of cargo ships the examination of current certificates and other records should consist of:
    - (CIn) 2.3.1.1 the provisions of (CA) 2.2.1.
  - (CIn) 2.3.2 For the hull, machinery and equipment of cargo ships the intermediate survey should consist of:
    - (CIn) 2.3.2.1 the provisions of (CA) 2.2.2;
    - (CIn) 2.3.2.2 for ships over 5 years of age, an internal examination of representative spaces used for water ballast;
    - (CIn) 2.3.2.3 for ships over 10 years of age, other than ships engaged in the carriage of dry cargoes only, an internal examination of selected cargo spaces;
    - (CIn) 2.3.2.4 for ships over 15 years of age, engaged in the carriage of dry cargoes only, an internal examination of selected cargo spaces.

- (CIn) 2.3.3 For the hull, machinery and equipment of cargo ships for the additional requirements for oil tankers the intermediate survey should consist of:
- (CIn) 2.3.3.1 the provisions of (CA) 2.2.3;
- (CIn) 2.3.3.2 should there be any doubt as to its condition when examining the various piping systems, the piping may be required to be pressure tested, gauged or both. Particular attention is to be paid to repairs such as welded doublers;
- (CIn) 2.3.3.3 for ships over 10 years of age an internal examination of selected cargo spaces;
- (CIn) 2.3.3.4 testing the insulation resistance of electrical circuits in dangerous zones such as cargo pump rooms and areas adjacent to cargo tanks, but in cases where a proper record of testing is maintained, consideration should be given to accepting recent readings.
- (CIn) 2.3.4 For the hull, machinery and equipment of cargo ships, concerning the additional requirements for chemical tankers and gas carriers, the intermediate survey should consist of:
- (CIn) 2.3.4.1 the provisions of (CA) 2.2.3.
- (CIn) 2.3.5 For the hull, machinery and equipment of cargo ships the completion of the intermediate survey should consist of:
- (CIn) 2.3.5.1 after a satisfactory survey, the Cargo Ship Safety Construction Certificate should be endorsed;
- (CIn) 2.3.5.2 if a survey shows that the condition of a ship or its equipment is unsatisfactory, see part "General", section 4.8.
- (CR) 2.4 **Renewal surveys** – see part "General", section 4.5
- (CR) 2.4.1 For the hull, machinery and equipment of cargo ships the examination of current certificates and other records should consist of:
- (CR) 2.4.1.1 the provisions of (CA) 2.2.1, except for the validity of the Cargo Ship Safety Construction Certificate.
- (CR) 2.4.2 For the hull, machinery and equipment of cargo ships the renewal survey should consist of:
- (CR) 2.4.2.1 the provisions of (CIn) 2.3.2;
- (CR) 2.4.2.2 examination of sea valves and their connections to the hull;
- (CR) 2.4.2.3 examination of anchoring and mooring equipment for which purpose the anchors should be lowered and raised using the windlass.
- (CR) 2.4.3 For the hull, machinery and equipment of cargo ships, concerning the additional requirements for oil tankers, the renewal survey should consist of:
- (CR) 2.4.3.1 the provisions of (CIn) 2.3.3.



- (CR) 2.4.4 For the hull, machinery and equipment of cargo ships, concerning the additional requirements for chemical tankers and gas carriers, the renewal survey should consist of:
- (CR) 2.4.4.1 the provisions of (CA) 2.2.3.
- (CR) 2.4.5 For the hull, machinery and equipment of cargo ships, concerning the additional requirements for bulk carriers the renewal survey should consist of the provisions of (CI) 2.1.3.64 and 2.1.3.66.
- (CR) 2.4.6 after a satisfactory survey, the Cargo Ship Safety Construction Certificate should be issued.
- (B) 3 GUIDELINES FOR THE INSPECTION OF THE OUTSIDE OF THE SHIP'S BOTTOM OF CARGO SHIPS**
- (CB) 3.1 For the inspection of the outside of the ship's bottom of cargo ships the inspection should consist of:
- (CB) 3.1.1 examination of the ship's shell including bottom and bow plating, keel, bilge keels, stem, stern frame and rudder;
- (CB) 3.1.2 noting the clearances measured in the rudder bearings;
- (CB) 3.1.3 examination of the propeller and shaft seals, as far as practicable;
- (CB) 3.1.4 noting the clearance measured in the propeller shafts, as far as practicable;
- (CB) 3.1.5 examination of sea chests and strainers;
- (CB) 3.1.6 the survey of related items inspected at the same time (see part "General" section 5.1).
- (CB) 3.2 For the inspection of the outside of the ship's bottom of cargo ships the completion of the inspection should consist of:
- (CB) 3.2.1 after a satisfactory survey, the Cargo Ship Safety Construction Certificate should be endorsed;
- (CB) 3.2.2 if a survey shows that the condition of a ship or its equipment is unsatisfactory, see part "General" section 4.8.
- (R) 4 GUIDELINES FOR SURVEYS FOR THE CARGO SHIP SAFETY RADIO CERTIFICATE**
- (RI) 4.1 Initial surveys – see part "General" section 4.1**
- (RI) 4.1.1 For the radio installations, including those used in life-saving appliances, of cargo ships the examination of plans and designs should consist of:
- (RI) 4.1.1.1 establishing the sea areas declared for operation, the equipment installed to fulfil the functional requirements for the sea areas of

- operation, the methods adopted to ensure the availability of the functional requirements and the arrangements for supply of an emergency source of energy (if any) (SOLAS 74/88 regs.II-1/43 and IV/1 to 15);
- (RI) 4.1.1.2 establishing which radio equipment is to be surveyed and, if duplication of equipment is used as a means of ensuring the availability of the functional requirements, establishing which is the "basic equipment" and which the "duplicated equipment" (SOLAS 74/88 reg.IV/15) (Additional radiocommunications equipment provided other than for SOLAS compliance should be noted);
- (RI) 4.1.1.3 confirming all SOLAS equipment complies with appropriate performance standards not inferior to those adopted by IMO (SOLAS 74/88 reg.IV/14);
- (RI) 4.1.1.4 examining the plans for the provision and position of the radio installation, including sources of energy and antennas (SOLAS 74/88 regs.II-1/43, IV/6, IV/14 and V/19);
- (RI) 4.1.1.5 examining the plans for the provision and positioning of the radio life-saving appliances (SOLAS 74/88 reg.III/6).
- (RI) 4.1.2 For the radio installations, including radio life-saving appliances, of cargo ships the survey during construction and after installation should consist of:
- (RI) 4.1.2.1 examining the position, physical and electromagnetic protection and illumination of each radio installation (SOLAS 74/88 reg.IV/6);
- (RI) 4.1.2.2 confirming the provision of equipment for the radio installation with due regard to the declared sea areas in which the ship will trade and the declared means of maintaining availability of functional requirements (SOLAS 74/88 regs.III/6, IV/7 to 11, 14 and 15);
- (RI) 4.1.2.3 confirming the ability to initiate the transmission of ship-to-shore distress alerts by at least two separate and independent means, each using a different radiocommunication service, from the position from which the ship is normally navigated (SOLAS 74/88/06 regs.IV/4, 7 to 11);
- (RI) 4.1.2.4 examining all antennas, including:
- (RI) 4.1.2.4.1 visually checking all antennas, including Inmarsat antennas, and feeders for satisfactory siting and absence of defects (SOLAS 74/88 reg.IV/14);
- (RI) 4.1.2.4.2 checking insulation and safety of all antennas;
- (RI) 4.1.2.5 examining the reserve source of energy, including:
- (RI) 4.1.2.5.1 checking there is sufficient capacity to operate the basic or duplicated equipment for 1 hour or 6 hours, as appropriate (SOLAS 74/88 reg.IV/13);

- (RI) 4.1.2.5.2 if the reserve source of energy is a battery:
- (RI) 4.1.2.5.2.1 checking its siting and installation (SOLAS 74/88 reg.IV/13);
- (RI) 4.1.2.5.2.2 where appropriate, checking its condition by specific gravity measurement or voltage measurement;
- (RI) 4.1.2.5.2.3 with the battery off charge, and the maximum required radio installation load connected to the reserve source of energy, checking the battery voltage and discharge current;
- (RI) 4.1.2.5.2.4 checking that the charger(s) are capable of recharging the reserve battery within 10 hours (SOLAS 74/88 reg.IV/13);
- (RI) 4.1.2.5.2.5 checking that information of ship's position is provided continuously and automatically to all two-way communication equipment (SOLAS 74/88 reg.IV/18);
- (RI) 4.1.2.6 examining the VHF transceiver(s), including:
- (RI) 4.1.2.6.1 checking for operation on channels 6, 13 and 16 (SOLAS 74/88 regs.IV/7 and 14);
- (RI) 4.1.2.6.2 checking frequency tolerance, transmission line quality and radio frequency power output (SOLAS 74/88 reg.IV/14);
- (RI) 4.1.2.6.3 checking for correct operation of all controls including priority of control units (SOLAS 74/88 reg.IV/14);
- (RI) 4.1.2.6.4 checking that the equipment operates from the main, emergency (if provided) and reserve sources of energy (SOLAS 74/88 reg.IV/13);
- (RI) 4.1.2.6.5 checking the operation of the VHF control unit(s) or portable VHF equipment provided for navigational safety (SOLAS 74/88 reg.IV/6);
- (RI) 4.1.2.6.6 checking for correct operation by on-air contact with a coast station or other ship;
- (RI) 4.1.2.7 examining the VHF DSC controller and channel 70 DSC watch receiver, including:
- (RI) 4.1.2.7.1 performing an off-air check confirming the correct Maritime Mobile Service Identity is programmed in the equipment (SOLAS 74/88 reg.IV/14);
- (RI) 4.1.2.7.2 checking for correct transmission by means of a routine or test call to a coast station, other ship, on board duplicate equipment or special test equipment;
- (RI) 4.1.2.7.3 checking for correct reception by means of a routine or test call from a coast station, other ship, on board duplicate equipment or special test equipment;
- (RI) 4.1.2.7.4 checking the audibility of the VHF/DSC alarm;

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| (RI) | 4.1.2.7.5  | checking that the equipment operates from the main, emergency (if provided) and reserve sources of energy (SOLAS 74/88 reg.IV/13);                                                                                 |
| (RI) | 4.1.2.8    | examining the MF/HF radiotelephone equipment, including:                                                                                                                                                           |
| (RI) | 4.1.2.8.1  | checking that the equipment operates from the main, emergency (if provided) and reserve sources of energy (SOLAS 74/88 reg.IV/13);                                                                                 |
| (RI) | 4.1.2.8.2  | checking the antenna tuning in all appropriate bands;                                                                                                                                                              |
| (RI) | 4.1.2.8.3  | checking that the equipment is within frequency tolerance on all appropriate bands (SOLAS 74/88 reg.IV/14);                                                                                                        |
| (RI) | 4.1.2.8.4  | checking for correct operation by contact with a coast station and/or measuring transmission line quality and radio frequency output;                                                                              |
| (RI) | 4.1.2.8.5  | checking receiver performance by monitoring known stations on all appropriate bands;                                                                                                                               |
| (RI) | 4.1.2.8.6  | if control units are provided outside the navigating bridge, checking that the control unit on the bridge has first priority for the purpose of initiating distress alerts (SOLAS 74/88 regs.IV/9, 10, 11 and 14); |
| (RI) | 4.1.2.9    | examining the HF radiotelex equipment, including:                                                                                                                                                                  |
| (RI) | 4.1.2.9.1  | checking that the equipment operates from the main, emergency (if provided) and reserve sources of energy (SOLAS 74/88 reg.IV/13);                                                                                 |
| (RI) | 4.1.2.9.2  | confirming that the correct selective calling number is programmed in the equipment;                                                                                                                               |
| (RI) | 4.1.2.9.3  | checking correct operation by inspection of recent hard copy or by a test with a coast radio station (SOLAS 74/88 regs.IV/10 and 11);                                                                              |
| (RI) | 4.1.2.10   | examining the MF/HF DSC controller(s), including:                                                                                                                                                                  |
| (RI) | 4.1.2.10.1 | checking that the equipment operates from the main, emergency (if provided) and reserve sources of energy (SOLAS 74/88 reg.IV/13);                                                                                 |
| (RI) | 4.1.2.10.2 | confirming that the correct Maritime Mobile Service Identity is programmed in the equipment;                                                                                                                       |
| (RI) | 4.1.2.10.3 | checking the off-air self-test program;                                                                                                                                                                            |
| (RI) | 4.1.2.10.4 | checking operation by means of a test call on MF and/or HF to a coast radio station if the rules of the berth permit the use of MF/HF transmissions (SOLAS 74/88 regs.IV/9, 10 and 11);                            |
| (RI) | 4.1.2.10.5 | checking the audibility of the MF/HF DSC alarm;                                                                                                                                                                    |
| (RI) | 4.1.2.11   | examining the MF/HF DSC watch receiver(s), including:                                                                                                                                                              |
| (RI) | 4.1.2.11.1 | confirming that only distress and safety DSC frequencies are being monitored (SOLAS 74/88 regs.IV/9 to 12);                                                                                                        |
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- (RI) 4.1.2.11.2 checking that a continuous watch is being maintained whilst keying MF/HF radio transmitters (SOLAS 74/88 reg.IV/12);
- (RI) 4.1.2.11.3 checking for correct operation by means of a test call from a coast station or other ship;
- (RI) 4.1.2.12 examining the Inmarsat Ship Earth Station(s), including:
  - (RI) 4.1.2.12.1 checking that the equipment operates from the main, emergency (if provided) and reserve sources of energy, and that where an uninterrupted supply of information from the ship's navigational or other equipment is required ensuring such information remains available in the event of failure of the ship's main or emergency source of electrical power (SOLAS 74/88 regs.IV/13 and 14);
  - (RI) 4.1.2.12.2 checking the distress function by means of an approved test procedure where possible (SOLAS 74/88 regs.IV/10, 12 and 14);
  - (RI) 4.1.2.12.3 checking for correct operation by inspection of recent hard copy or by test call;
- (RI) 4.1.2.13 if appropriate, examining the NAVTEX equipment (SOLAS 74/88 regs.IV/7, 12 and 14), including:
  - (RI) 4.1.2.13.1 checking for correct operation by monitoring incoming messages or inspecting recent hard copy;
  - (RI) 4.1.2.13.2 running the self-test program if provided;
- (RI) 4.1.2.14 examining the Enhanced Group Call equipment (SOLAS 74/88 regs.IV/7 and 14), including:
  - (RI) 4.1.2.14.1 checking for correct operation and area by monitoring incoming messages or by inspecting recent hard copy;
  - (RI) 4.1.2.14.2 running the self-test program if provided;
- (RI) 4.1.2.15 if appropriate, examining the radio equipment for receipt of maritime safety information by HF NBDP (SOLAS 74/88 regs.IV/7, 12 and 14), including:
  - (RI) 4.1.2.15.1 checking for correct operation by monitoring incoming messages or inspecting recent hard copy;
  - (RI) 4.1.2.15.2 running the self-test program if provided;
- (RI) 4.1.2.16 examining the 406 MHz satellite EPIRB (SOLAS 74/88 regs.IV/7 and 14), including:
  - (RI) 4.1.2.16.1 checking position and mounting for float free operation;
  - (RI) 4.1.2.16.2 carrying out visual inspection for defects;
  - (RI) 4.1.2.16.3 carrying out the self-test routine;

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| (RI) | 4.1.2.16.4 | checking that the EPIRB ID is clearly marked on the outside of the equipment and, where possible, decoding the EPIRB identity number confirming it is correct;                                                                                         |
| (RI) | 4.1.2.16.5 | checking the battery expiry date;                                                                                                                                                                                                                      |
| (RI) | 4.1.2.16.6 | if provided, checking the hydrostatic release and its expiry date;                                                                                                                                                                                     |
| (RI) | 4.1.2.16.7 | checking the emission on operational frequencies, coding and registration on the 406 MHz signal without transmission of a distress call to the satellite;                                                                                              |
| (RI) | 4.1.2.16.8 | checking that the EPIRB has been subject to maintenance at intervals not exceeding five years at an approved shore-based maintenance facility (SOLAS 74/00 reg.IV/15.9);                                                                               |
| (RI) | 4.1.2.16.9 | if possible, checking the emission on operational frequencies, coding and registration on the 121.5 MHz homing signal without transmission of a distress call to the satellite;                                                                        |
| (RI) | 4.1.2.17   | examining the two-way VHF radiotelephone apparatus (SOLAS 74/88 reg.III/6), including:                                                                                                                                                                 |
| (RI) | 4.1.2.17.1 | checking for correct operation on Channel 16 and one other by testing with another fixed or portable VHF installation (SOLAS 74/88 reg.IV/14);                                                                                                         |
| (RI) | 4.1.2.17.2 | checking the battery charging arrangements where re-chargeable batteries are used;                                                                                                                                                                     |
| (RI) | 4.1.2.17.3 | checking the expiry date of primary batteries where used;                                                                                                                                                                                              |
| (RI) | 4.1.2.17.4 | where appropriate, checking any fixed installation provided in a survival craft;                                                                                                                                                                       |
| (RI) | 4.1.2.18   | examining the search and rescue locating device(s) (SOLAS 74/88/08 regs.III/6, IV/7 and 14), including:                                                                                                                                                |
| (RI) | 4.1.2.18.1 | checking the position and mounting;                                                                                                                                                                                                                    |
| (RI) | 4.1.2.18.2 | monitoring response on ship's 9 GHz radar;                                                                                                                                                                                                             |
| (RI) | 4.1.2.18.3 | checking the battery expiry date;                                                                                                                                                                                                                      |
| (RI) | 4.1.2.19   | examining the test equipment and spares carried to ensure carriage is adequate in accordance with the sea areas in which the ship trades and the declared options for maintaining availability of the functional requirements (SOLAS 74/88 reg.IV/15). |
| (RI) | 4.1.3      | For the radio installations, including those used in life-saving appliances, the check that documentation, etc., has been placed on board should consist of:                                                                                           |

- (RI) 4.1.3.1 checking for a valid radio licence issued by the flag Administration (ITU RR Art.24);
- (RI) 4.1.3.2 checking the radio operator's certificates of competence (SOLAS 74/88 reg.IV/16 and ITU RR Art.56);
- (RI) 4.1.3.3 checking the radio record (log) (SOLAS 74/88 reg.IV/17 and ITU RR App.11);
- (RI) 4.1.3.4 checking the carriage of up-to-date ITU publications (ITU RR App.11);
- (RI) 4.1.3.5 checking the carriage of operating manuals for all equipment (SOLAS 74/88 reg.IV/15);
- (RI) 4.1.3.6 checking the carriage of service manuals for all equipment when at-sea maintenance is the declared option (SOLAS 74/88 reg.IV/15).
- (RI) 4.1.4 For the radio installations, including those used in life-saving appliances, of cargo ships the completion of the initial survey should consist of:
  - (RI) 4.1.4.1 the surveyor preparing and forwarding a survey report, indicating clearly the organization he represents, to the relevant authorities, detailing results of the survey and recording omissions and deficiencies, if satisfied, the relevant authorities should issue a Cargo Ship Safety Radio Certificate and the associated Record of Equipment (form R).
- (RP) **4.2 Periodical surveys** – see part "General" section 4.4
  - (RP) 4.2.1 For radio installations, including radio life-saving appliances, on cargo ships the examination of current certificates and other records should consist of:
    - (RP) 4.2.1.1 checking the validity, as appropriate, of the Cargo Ship Safety Equipment Certificate, the Cargo Ship Safety Radio Certificate and the Cargo Ship Safety Construction Certificate or the Cargo Ship Safety Certificate;
    - (RP) 4.2.1.2 checking the validity, where appropriate, of the Safety Management Certificate (SMC) and that a copy of the Document of Compliance (DOC) is on board;
    - (RP) 4.2.1.3 checking the validity of the International Ship Security Certificate;
    - (RP) 4.2.1.4 checking the validity of the International Load Line Certificate or International Load Line Exemption Certificate;
    - (RP) 4.2.1.5 checking the validity of the International Oil Pollution Prevention Certificate;
    - (RP) 4.2.1.6 checking the certificates of class, if the ship is classed with a classification society;

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| (RP) | 4.2.1.7  | checking, where appropriate, the validity of the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk or the Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk; |
| (RP) | 4.2.1.8  | checking, when appropriate, the validity of the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk;                                                                                    |
| (RP) | 4.2.1.9  | checking, when appropriate, the validity of the International Pollution Prevention Certificate for the Carriage of Noxious Liquid Substances in Bulk;                                                                |
| (RP) | 4.2.1.10 | checking, when appropriate, the validity of the International Sewage Pollution Prevention Certificate;                                                                                                               |
| (RP) | 4.2.1.11 | checking, when appropriate, the validity of the International Air Pollution Prevention Certificate;                                                                                                                  |
| (RP) | 4.2.1.12 | checking that the ship's complement complies with the Minimum Safe Manning Document (SOLAS 74/88 reg.V/13(b));                                                                                                       |
| (RP) | 4.2.1.13 | checking that adequate information is on board to enable the equipment to be properly operated and maintained;                                                                                                       |
| (RP) | 4.2.1.14 | checking that the master, officers and ratings are certificated as required by the STCW Convention;                                                                                                                  |
| (RP) | 4.2.1.15 | confirming that any new equipment has been properly approved before installation and that no changes have been made such as would affect the validity of the certificate;                                            |
| (RP) | 4.2.1.16 | confirming that a record has been kept in the period since the last survey to the satisfaction of the Administration and as required by the Radio Regulations (SOLAS 74/88 reg.IV/17);                               |
| (RP) | 4.2.1.17 | checking documentary evidence that the actual capacity of the battery has been proved in port within the last 12 months (SOLAS 74/88 reg.IV/13);                                                                     |
| (RP) | 4.2.1.18 | confirming that the provisions of (RI) 4.1.3 have been met;                                                                                                                                                          |
| (RP) | 4.2.1.19 | checking that the annual test has been carried out for the Satellite EPIRB and, if applicable, shore-based maintenance has been carried out at intervals not exceeding five years (SOLAS 74/04 reg.IV/15);           |
| (RP) | 4.2.1.20 | confirming the availability of the International Anti-Fouling System Certificate (AFS 2001 Annex 4 reg.2) when applicable.                                                                                           |
| (RP) | 4.2.2    | For radio installations, including radio life-saving appliances, of cargo ships the periodical survey should consist of:                                                                                             |
| (RP) | 4.2.2.1  | the provisions of (RI) 4.1.2.                                                                                                                                                                                        |
| (RP) | 4.2.3    | For radio installations, including those used in radio life-saving appliances, of cargo ships the completion of the periodical survey should consist of:                                                             |



- (RP) 4.2.3.1 after a satisfactory survey, endorsing the Cargo Ship Safety Radio Certificate;
- (RP) 4.2.3.2 if a survey shows that the condition of a ship or its equipment is unsatisfactory, see part "General" section 4.8.
- (RR) **4.3** **Renewal surveys** – see part "General" section 4.5
- (RR) 4.3.1 For the radio installations, including those used in life-saving appliances, of cargo ships the examination of current certificates and other records should consist of:
- (RR) 4.3.1.1 the provisions of (RP) 4.2.1, except for the validity of the Cargo Ship Safety Radio Certificate.
- (RR) 4.3.2 For the radio installations, including those used in radio life-saving appliances, of cargo ships the renewal survey should consist of:
- (RR) 4.3.2.1 the provisions of (RI) 4.1.2.
- (RR) 4.3.3 For the radio installations, including those used in radio life-saving appliances, on cargo ships the completion of the renewal survey should consist of:
- (RR) 4.3.3.1 after a satisfactory survey, issuing the Cargo Ship Safety Radio Certificate as per the provisions of (RI) 4.1.4.
- (P) **5** **GUIDELINES FOR SURVEYS FOR THE PASSENGER SHIP CERTIFICATE**
- (PI) **5.1** **Initial surveys** – see part "General" section 4.1.
- (PI) 5.1.1 For the hull, machinery and equipment of passenger ships the examination of plans and designs should consist of:
- (PI) 5.1.1.1 examining the subdivision and stability (SOLAS 74/88/95 regs.II-1/4 to 8, 8-1, 8-2, 8-3, 13 and 16) (SOLAS 06/08 regs.II-1/5 to 8-1, 14 and 18; IS Code chs.1, 2 and 3);
- (PI) 5.1.1.2 examining the ballasting arrangements (SOLAS 74/88 reg.II-1/9) (SOLAS 06 reg.II-1/20);
- (PI) 5.1.1.3 examining the arrangement of the bulkheads, their construction and the openings therein, including the disposition and means of operation of the watertight doors (SOLAS 74/88 regs.II-1/10, 14, and 15) (SOLAS 06 regs.II-1/10, 11 12 and 13);
- (PI) 5.1.1.4 examining the arrangement of the double bottoms (SOLAS 74/88 reg.II-1/12) (SOLAS 06 reg.II-1/9);
- (PI) 5.1.1.5 examining the arrangements for the openings in the shell plating below the margin line or the bulkhead deck as applicable, the construction of the watertight doors, sidescuttles, watertight decks, trunks, etc., and the watertight integrity above the margin line or the bulkhead deck as

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- applicable (SOLAS 74/88 regs.II-1/17, 18, 19 and 20) (SOLAS 06 regs.II-1/15, 16, 16-1 and 17);
- (PI) 5.1.1.6 examining the plans for the bilge pumping and drainage systems (SOLAS 74/88 regs.II-1/21 and 39) (SOLAS 05/09 reg.II-1/35-1 and SOLAS 08 reg.II-2/20.6.1.4);
- (PI) 5.1.1.7 examining, when appropriate, the means of indicating the status of any bow doors and the leakage there from (SOLAS 74/88 reg.II-1/23-2) (SOLAS 06 reg.II-1/17-1);
- (PI) 5.1.1.8 examining the plans for the machinery installation (SOLAS 74/88 regs.II-1/26 to 36 and 54);
- (PI) 5.1.1.9 examining the plans for the electrical installation (SOLAS 74/88 regs.II-1/39, 40, 41, 42, 44 and 45);
- (PI) 5.1.1.10 checking, when appropriate, the provision of supplementary emergency lighting (SOLAS 74/88 reg.II-1/42-1);
- (PI) 5.1.1.11 examining, where applicable, the approved documentation for the alternative design and arrangements (SOLAS 00/06 regs.II-1/55, II-2/17 and III/38);
- (PI) 5.1.1.12 examining the plans for the fire pumps, fire mains, hydrants, hoses and nozzles and the international shore connection (SOLAS 74/88 reg.II-1/39 and SOLAS 74/00 reg.II-2/10.2; FSSC chs.2 and 12) (SOLAS 74/88 reg.II-1/39 and regs.II-2/4 and 19);
- (PI) 5.1.1.13 examining the plans for the fire-extinguishing arrangements in the machinery spaces (SOLAS 74/00 regs.II-2/10.4 and 10.5; FSSC chs.5, 6 and 7) (SOLAS 74/88 reg.II-2/7);
- (PI) 5.1.1.14 checking the provision and specification of the fire extinguishers and the fire-fighter's (SOLAS 74/88 regs.II-2/6 and 17) (SOLAS 74/00 reg.II-2/10.10);
- (PI) 5.1.1.15 for passenger ship constructed on or after 1 July 2010, checking the provision of a suitably located means for fully recharging breathing air cylinders (SOLAS 08 reg.II-2/10.10.2);
- (PI) 5.1.1.16 examining the plans for the fire extinguishing and special arrangements in the machinery spaces (SOLAS 74/88 regs.II-1/39 and regs.II-2/7 and 11);
- (PI) 5.1.1.17 examining the arrangements for oil fuel, lubricating oil and other flammable oils (SOLAS 74/00 reg.II-2/4.2.3) (SOLAS 74/88 reg.II-2/15);
- (PI) 5.1.1.18 examining the plans for the structural fire protection, including the means of escape (SOLAS 74/00 regs.II-2/4.4.4, 5.2, 5.3, 7.5, 7.8.2, 8.4, 8.5, 9, 10.6, 11, 13, 17, 20; FSSC ch.13 sections 1 and 2) (SOLAS 74/88 regs.II-2/23 to 36);
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| (PI) | 5.1.1.19 | examining the plans for the protection of special category spaces and other cargo spaces (SOLAS 74/88 regs.II-2/37, 38 and 39);                                                                                                                                                           |
| (PI) | 5.1.1.20 | examining the plans for the fixed fire detection and alarm system, the crew alarm and the public address system or other effective means of communication (SOLAS 74/00/06 regs.II-2/7 and 12) (SOLAS 74/88 reg.II-2/40);                                                                  |
| (PI) | 5.1.1.21 | examining the plans for the special arrangements for the carriage of dangerous goods, when appropriate, including water supplies, electrical equipment and wiring, fire detection, bilge pumping and personnel protection (SOLAS 74/88 regs.II-2/41 and 54) (SOLAS 74/00/08 reg.II-2/19); |
| (PI) | 5.1.1.22 | examining the provision and disposition of the survival craft and rescue boats and the arrangements for mustering passengers (SOLAS 74/00 regs.III/11 to 17, 21 and 24);                                                                                                                  |
| (PI) | 5.1.1.23 | examining the design of the survival craft, including their equipment, launching and recovery appliances and embarkation and launching arrangements (SOLAS 74/88/06 regs.III/4, 20 to 24, 36, 38 to 44 and 48);                                                                           |
| (PI) | 5.1.1.24 | examining the design of the rescue boats, including their equipment and launching and recovery appliances and arrangements (SOLAS 74/88 regs.III/16, 20, 47 and 48);                                                                                                                      |
| (PI) | 5.1.1.25 | examining the provision, specification and stowage of two-way VHF radiotelephone apparatus and search and rescue locating devices (SOLAS 74/88/08 reg.III/6.2.2);                                                                                                                         |
| (PI) | 5.1.1.26 | examining the provision, specification and stowage of the distress flares and the line-throwing appliance and the provision of on-board communications equipment and the general alarm system (SOLAS 74/88 regs.III/6, 17, 35, 49 and 50);                                                |
| (PI) | 5.1.1.27 | examining the provision, specification and stowage of the lifebuoys, including those fitted with self-igniting lights, self-activating smoke signals and buoyant lines, lifejackets, immersion suits and thermal protective aids (SOLAS 74/88/06 regs.III/7, 21, 22 and 26);              |
| (PI) | 5.1.1.28 | examining the plans for the lighting of the muster and embarkation stations and the alleyways, stairways and exits giving access to the muster and embarkation stations, including the supply from the emergency source of power (SOLAS 74/88 regs.II-1/42 and III/11);                   |
| (PI) | 5.1.1.29 | examining the plans for the positioning of, and the specification for, the navigation lights, shapes and sound signalling equipment (International Regulations for Preventing Collisions at Sea (COLREG) in force rules 20 to 24, 27 to 30 and 33);                                       |
| (PI) | 5.1.1.30 | examining the plans relating to the bridge design and arrangement of navigational systems and equipment and bridge procedures (SOLAS 74/00 reg.V/15);                                                                                                                                     |

- (PI) 5.1.1.31 checking the provision and specification of the following navigation equipment as appropriate: daylight signalling lamp, magnetic compass, transmitting heading device, gyro compass, gyro compass repeaters, radar installation(s), automatic identification system, electronic plotting aid, automatic tracking aid(s) or automatic radar plotting aid(s), echo-sounding device, speed and distance indicator, rudder angle indicator, propeller rate of revolution indicator, variable pitch propeller pitch and operational mode indicator, rate-of-turn indicator, heading or track control system, GNSS receiver, terrestrial radio navigation system and sound reception system, a pelorus or compass bearing device, means for correcting heading and bearings, a BNWAS as applicable and ECDIS including back-up arrangements as applicable (SOLAS 74/00/09 reg.V/19);
- (PI) 5.1.1.32 checking the provision and specification of the voyage data recorder (SOLAS 74/00 reg.V/20);
- (PI) 5.1.1.33 checking navigation bridge visibility (SOLAS 74/00 reg.V/22);
- (PI) 5.1.1.34 checking for the provision and specification of the long-range identification and tracking system (SOLAS 04 reg.V/19-1);
- (PI) 5.1.1.35 checking the provision and specification of the pilot ladders and hoists/pilot transfer arrangements (SOLAS 74/00 reg.V/23);
- (PI) 5.1.1.36 establishing the sea areas declared for operation, the equipment installed to fulfil the functional requirements for the sea areas of operation, the methods adopted to ensure the availability of the functional requirements and the arrangements for supply of an emergency source of energy (if any) (SOLAS 74/88 regs.II-1/42 and IV/1 to 15);
- (PI) 5.1.1.37 establishing which radio equipment is to be surveyed and, if duplication of equipment is used as a means of ensuring the availability of the functional requirements, establishing which is the "basic equipment" and which the "duplicated equipment" (SOLAS 74/88 reg.IV/15) (Additional radiocommunication equipment provided other than for SOLAS compliance should be noted);
- (PI) 5.1.1.38 confirming that all SOLAS equipment complies with appropriate performance standards not inferior to those adopted by IMO (SOLAS 74/88 reg.IV/14);
- (PI) 5.1.1.39 examining the plans for the provision and positioning of the radio installation including sources of energy and antennas. (SOLAS 74/88 regs.II-1/42, IV/6 and 14);
- (PI) 5.1.1.40 examining the plans for the provision and positioning of the radio life-saving appliances (SOLAS 74/88 reg.III/6);
- (PI) 5.1.1.41 if applicable, checking that a list of all limitations on the operation of a passenger ship is kept on board and updated;

- (PI) 5.1.1.42 checking the provision of means of embarkation and disembarkation from ships for use in port and in port related operations, such as gangways and accommodation ladders (SOLAS 08 reg.II-1/3-9);
- (PI) 5.1.1.43 checking the provision of means to prevent blockage of drainage arrangements, for closed vehicle and ro-ro spaces and special category spaces where fixed pressure water-spraying systems are used (SOLAS 08 reg.II-2/20.6.1.5);
- (PI) 5.1.1.44 for passenger ships constructed on or after 1 July 2010, checking the provision of a safety centre (SOLAS reg.II-2/23) and associated ventilation requirements(SOLAS 06 reg.II-2/8.2);
- (PI) 5.1.1.45 for passenger ships constructed on or after 1 July 2010 and having a length of 120 m or more or having three or more main vertical zones, confirming that design criteria for the ship's safe return to port and for systems to remain operational after a fire casualty have been documented and that safe areas have been designated (SOLAS 06 regs.II-2/21 and 22).
- (PI) 5.1.2 For the hull, machinery and equipment of passenger ships the survey during construction and after installation should consist of:
- (PI) 5.1.2.1 examining the outside of the ship's bottom, including the bottom and bow plating, keel, bilge keels, stem, stern frame, the rudder, sea chests and strainers (SOLAS 74/88 reg.I/7(b)(i));
- (PI) 5.1.2.2 confirming the arrangements for subdivision, including the ship's stability in the damaged condition, and checking the subdivision load lines (SOLAS 74/88 regs.II-1/4 to 8, 13 and 16) (SOLAS 06 regs.II-1/6, 7, 7-1, 7-2, 7-3, 8, 14, 18);
- (PI) 5.1.2.3 checking the ballasting arrangements (SOLAS 74/88 reg.II-1/9) (SOLAS 06 reg.II-1/20);
- (PI) 5.1.2.4 confirming that dedicated sea water ballast tanks have an approved coating system when appropriate (SOLAS 74/00/06 reg.II-1/3-2);
- (PI) 5.1.2.5 confirming the arrangement of the bulkheads, their construction and the openings therein, confirming that the collision bulkhead is watertight up to the freeboard deck, that the valves fitted on the pipes piercing the collision bulkhead are operable from above the freeboard deck and that there are no doors, manholes, ventilation ducts or any other openings, confirming that the other bulkheads, as required for the ship's subdivision, are watertight up to the bulkhead deck and confirming the construction of the watertight doors and that they have been tested (SOLAS 74/88 regs.II-1/10, 14, 15 and 18) (SOLAS 06 regs.II-1/10, 11, 12, 13 and 16);
- (PI) 5.1.2.6 confirming that the watertight integrity has been maintained where pipes, scuppers, etc., pass through subdivision watertight bulkheads (SOLAS 74/88 reg.II-1/15) (SOLAS 06 reg.II-1/13);
- (PI) 5.1.2.7 confirming that a diagram is provided on the navigating bridge showing the location of the watertight doors together with indicators showing whether the doors are open or closed and confirming that the watertight doors and their means of operation have been installed in

accordance with the approved plans (SOLAS 74/88 reg.II-1/15)  
(SOLAS 06 reg.II-1/13);

- (PI) 5.1.2.8 testing the operation of the watertight doors both from the navigating bridge in the event of an emergency and locally at the door itself (SOLAS 74/88 reg.II-1/15) (SOLAS 06 reg.II-1/13) and, in particular, that they are:
  - (PI) 5.1.2.8.1 operable locally from each side of the bulkhead;
  - (PI) 5.1.2.8.2 provided with devices giving an indication of whether the door is open or closed at all remote operating positions;
  - (PI) 5.1.2.8.3 provided with an audible alarm that is distinct from any other alarm in the area and, when appropriate, an intermittent visual signal;
  - (PI) 5.1.2.8.4 provided with control handles on each side of the bulkhead so that a person may hold both handles in the open position and pass safely through the watertight door without accidentally setting the power closing mechanism into operation;
- (PI) 5.1.2.9 confirming that the watertight doors and their indicating devices are operable in the event of a failure of the main and emergency sources of power (SOLAS 74/88 reg.II-1/15) (SOLAS 06 reg.II-1/13);
- (PI) 5.1.2.10 checking, when appropriate, any watertight doors, that are not required to be closed remotely and are fitted in watertight bulkheads dividing 'tween deck spaces, and confirming that a notice is affixed concerning their closure (SOLAS 74/88 reg.II-1/15) (SOLAS 06 reg.II-1/13);
- (PI) 5.1.2.11 confirming that a notice is affixed to any portable plates on bulkheads in machinery spaces concerning their closure and, if appropriate, testing any power operated watertight door fitted in lieu (SOLAS 74/88 reg.II-1/15) (SOLAS 06 reg.II-1/13);
- (PI) 5.1.2.12 confirming the arrangements for closing sidescuttles and their deadlights, also scuppers, sanitary discharges and similar openings and other inlets and discharges in the shell plating below the bulkhead deck (SOLAS 06 reg.II-1/13);
- (PI) 5.1.2.13 confirming that valves for closing the main and auxiliary sea inlets and discharges in the machinery spaces are readily accessible and indicators showing the status of the valves are provided (SOLAS 06 reg.II-1/15);
- (PI) 5.1.2.14 confirming that gangway, cargo and fuelling ports fitted below the bulkhead deck can be effectively closed and that the inboard end of any ash or rubbish chutes are fitted with an effective cover (SOLAS 06 reg.II-1/13);
- (PI) 5.1.2.15 confirming by a hose or flooding test the watertightness of watertight decks and trunks, tunnels and ventilators (SOLAS 74/88 reg.II-1/19) (SOLAS 06 reg.II-1/16-1);
- (PI) 5.1.2.16 confirming the arrangements to maintain the watertight integrity above the bulkhead deck (SOLAS 06 regs.II-1/17 and 17-1);

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| (PI) | 5.1.2.17   | confirming the arrangements for the bilge pumping and that each bilge pump and the bilge pumping system provided for each watertight compartment is working efficiently (SOLAS 74/88 reg.II-1/21) (SOLAS 05, reg.II-1/35-1);                                                                                   |
| (PI) | 5.1.2.18   | confirming that the drainage system of enclosed cargo spaces situated on the freeboard deck is working efficiently (SOLAS 74/88 reg.II-1/21) (SOLAS 05 reg.II-1/35-1);                                                                                                                                         |
| (PI) | 5.1.2.18.1 | examining visually the drainage facilities for blockage or other damage and confirming the provision of means to prevent blockage of drainage arrangements, for closed vehicle and ro-ro spaces and special category spaces where fixed pressure water-spraying systems are used (SOLAS 08 reg.II-2/20.6.1.5); |
| (PI) | 5.1.2.19   | conducting an inclining test (SOLAS 74/88 reg.II-1/22) (SOLAS 06 reg.II-1/5);                                                                                                                                                                                                                                  |
| (PI) | 5.1.2.20   | checking, when appropriate, the means of indicating the status of any bow doors and any leakage there from (SOLAS 74/88 reg.II-1/23-2) (SOLAS 06 reg.II-1/17-1);                                                                                                                                               |
| (PI) | 5.1.2.21   | confirming that the arrangement for monitoring special category spaces or ro-ro spaces, when fitted, is satisfactory (SOLAS 06 reg.II-1/23);                                                                                                                                                                   |
| (PI) | 5.1.2.22   | confirming that the machinery, boilers and other pressure vessels, associated piping systems and fittings are installed and protected so as to reduce to a minimum any danger to persons on board, due regard being given to moving parts, hot surfaces and other hazards (SOLAS 74/88 reg.II-1/26);           |
| (PI) | 5.1.2.23   | confirming that the normal operation of the propulsion machinery can be sustained or restored even though one of the essential auxiliaries becomes inoperative (SOLAS 74/88 reg.II-1/26);                                                                                                                      |
| (PI) | 5.1.2.24   | confirming that means are provided so that the machinery can be brought into operation from the dead ship condition without external aid (SOLAS 74/88 reg.II-1/26);                                                                                                                                            |
| (PI) | 5.1.2.25   | confirming that the boilers, all parts of the machinery, all steam, hydraulic, pneumatic and other systems and their associated fittings which are under internal pressure have been subjected to the appropriate tests, including a pressure test (SOLAS 74/88 reg.II-1/26);                                  |
| (PI) | 5.1.2.26   | confirming that means are provided to ensure that the safe speed is not exceeded where there is the risk of machinery overspeeding (SOLAS 74/88 reg.II-1/27);                                                                                                                                                  |
| (PI) | 5.1.2.27   | confirming that, where practicable, means are provided to protect against overpressure in the parts of main, auxiliary and other machinery that are subject to internal pressure and may be subject to dangerous overpressure (SOLAS 74/88 reg.II-1/27);                                                       |
| (PI) | 5.1.2.28   | confirming that, when required, crankcase explosion relief devices are fitted to internal combustion engines and that they are arranged so as                                                                                                                                                                  |



- to minimize the possibility of injury to personnel (SOLAS 74/88 reg.II-1/27);
- (PI) 5.1.2.29 confirming that main turbine propulsion machinery and, where applicable, main internal combustion propulsion machinery and auxiliary machinery are provided with automatic shut-off arrangements in the case of failures, such as lubricating oil supply failure, which could rapidly lead to a complete breakdown, serious damage or explosion (SOLAS 74/88 reg.II-1/27);
- (PI) 5.1.2.30 confirming and recording the ability of the machinery to reverse the direction of the thrust of the propeller in sufficient time and to bring the ship to rest within a reasonable distance, including the effectiveness of any supplementary means of manoeuvring or stopping the ship (SOLAS 74/88 reg.II-1/28);
- (PI) 5.1.2.31 confirming that the main and auxiliary steering gear are so arranged that the failure of one of them does not render the other inoperative (SOLAS 74/88 reg.II-1/29);
- (PI) 5.1.2.32 confirming that, where appropriate, essential components of the steering gear are permanently lubricated or provided with lubrication fittings (SOLAS 74/88 reg.II-1/29);
- (PI) 5.1.2.33 confirming that relief valves are fitted to any part of a steering gear hydraulic system which can be isolated and in which pressure can be generated from the power source or from external forces and that these relief valves are set to a pressure not exceeding the design pressure (SOLAS 74/88 reg.II-1/29);
- (PI) 5.1.2.34 confirming that the main steering gear is capable of steering the ship at maximum ahead service speed and is capable of putting the rudder over from 35 degrees on one side to 35 degrees on the other side with the ship at its deepest seagoing draught and running ahead at maximum ahead service speed and, under the same conditions, from 35 degrees on either side to 30 degrees on the other side in not more than 28 seconds (SOLAS 74/88 reg.II-1/29);
- (PI) 5.1.2.35 confirming that the auxiliary steering gear is capable of steering the ship at navigable speed and of being brought speedily into action in an emergency and that it is capable of putting the rudder over from 15 degrees on one side to 15 degrees on the other side in not more than 60 seconds with the ship at its deepest seagoing draught and running ahead at one half of the maximum ahead service speed or 7 knots, whichever is the greater (SOLAS 74/88 reg.II-1/29);
- (PI) 5.1.2.36 confirming that the main or auxiliary steering gear power units restart automatically when power is restored after a power failure, that they are capable of being brought into operation from a position on the navigating bridge and that, in the event of a power failure to any one of the steering gear power units, an audible and visual alarm is given on the navigating bridge (SOLAS 74/88 reg.II-1/29);

- (PI) 5.1.2.37 confirming that, where the main steering gear comprises two or more identical power units and an auxiliary steering gear is not fitted, a defect can be isolated so that steering capability can be maintained or speedily regained after a single failure in its piping system or in one of the power units (SOLAS 74/88 reg.II-1/29);
- (PI) 5.1.2.38 confirming that the control systems for the main steering gear from both the navigating bridge and the steering gear compartment are operating satisfactorily (SOLAS 74/88 reg.II-1/29);
- (PI) 5.1.2.39 confirming that, where the main steering gear comprises two or more identical power units and an auxiliary steering gear is not fitted, the two independent control systems from the navigating bridge are operating satisfactorily (SOLAS 74/88 reg.II-1/29);
- (PI) 5.1.2.40 confirming that the control system for the auxiliary steering gear, in the steering gear compartment and, if this gear is power-operated, from the navigating bridge, is operating satisfactorily and that the latter is independent of the control system for the main steering gear (SOLAS 74/88 reg.II-1/29);
- (PI) 5.1.2.41 confirming that the control system for any main and auxiliary steering gear control system operable from the navigating bridge is capable of being brought into operation from a position on the navigating bridge, that means are provided in the steering gear compartment for disconnecting it from the steering gear that it serves and that an audible and visual alarm is given on the navigating bridge in the event of a failure of electrical power supply (SOLAS 74/88 reg.II-1/29);
- (PI) 5.1.2.42 confirming that the electric power circuits and steering gear control system, together with their associated components, cables and pipes, are separated, as far as practicable, throughout their length (SOLAS 74/88 reg.II-1/29);
- (PI) 5.1.2.43 confirming that the means of communication between the bridge and the steering gear is operating satisfactorily and that, with ships having emergency steering positions, a telephone or other means of communication for relaying heading information and supplying visual compass readings to the emergency steering position are provided (SOLAS 74/88 reg.II-1/29) (SOLAS 74/00 reg.V/19);
- (PI) 5.1.2.44 confirming that the angular position of the rudder is indicated independently of the steering control system on the navigating bridge if the main steering gear is power-operated and that this angular position is given in the steering gear compartment (SOLAS 74/88 reg.II-1/29), (SOLAS 74/00 reg.V/19);
- (PI) 5.1.2.45 confirming that with a hydraulic power-operated steering gear the audible and visual low-level alarms on the navigating bridge and in the machinery space for each hydraulic fluid reservoir are operating satisfactorily and that at least one power actuating system including the reservoir can be recharged from a position within the steering gear compartment by means of a fixed storage tank to which a contents gauge is fitted with fixed piping (SOLAS 74/88 reg.II-1/29);

- (PI) 5.1.2.46 confirming that the steering gear compartment is readily accessible, that it is separated, as far as practicable, from machinery spaces and is provided with suitable arrangements to ensure working access to steering gear machinery and controls under safe conditions (SOLAS 74/88 reg.II-1/29);
- (PI) 5.1.2.47 confirming that with electric and electro-hydraulic steering gear, the means for indicating, on the navigating bridge and at a main machinery control position, that the motors are running and that the overload alarm and alarm for the loss of a phase in a three phase supply located at the main machinery control position are operating satisfactorily (SOLAS 74/88 reg.II-1/30);
- (PI) 5.1.2.48 confirming that the main and auxiliary machinery essential for propulsion and the safety of the ship are provided with the effective means for its operation and control (SOLAS 74/88 reg.II-1/31);
- (PI) 5.1.2.49 confirming that appropriate means are provided where it is intended that the propulsion machinery should be remotely controlled from the navigating bridge, including, where necessary, the control, monitoring, reporting, alert and safety actions (SOLAS 74/00/02 reg.II-1/31);
- (PI) 5.1.2.50 confirming that arrangements to operate main and other machinery from a machinery control room are satisfactory (SOLAS 74/88 reg.II-1/31);
- (PI) 5.1.2.51 confirming that, in general, means are provided for manually overriding automatic controls and that a failure does not prevent the use of the manual override (SOLAS 74/88 reg.II-1/31);
- (PI) 5.1.2.52 confirming that oil-fired and exhaust gas boilers, unfired steam generators, steam pipe systems and air pressure systems are fitted with the appropriate safety features (SOLAS 74/88 regs.II-1/32, 33 and 34);
- (PI) 5.1.2.53 confirming the operation of the ventilation for the machinery spaces (SOLAS 74/88 reg.II-1/35);
- (PI) 5.1.2.54 confirming that the measures to prevent noise in machinery spaces are effective (SOLAS 74/88 reg.II-1/36);
- (PI) 5.1.2.55 confirming that the engine room telegraph giving visual indication of the orders and answers both in the machinery space and on the navigating bridge is operating satisfactorily (SOLAS 74/88 regulation II-1/37);
- (PI) 5.1.2.56 confirming that the second means of communication between the navigation bridge and machinery space is also operating satisfactorily and that appropriate means are provided to any other positions from which the engines are controlled (SOLAS 74/88 regulation II-1/37);
- (PI) 5.1.2.57 confirming that the engineer's alarm is clearly audible in the engineers' accommodation (SOLAS 74/88 regulation II-1/38);

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| (PI) | 5.1.2.58   | confirming that precautions, taken to prevent any oil than may escape under pressure from any pump, filter or heater from coming into contact with heated surfaces, are efficient;                                                                                                                                |
| (PI) | 5.1.2.59   | confirming that the means of ascertaining the amount of oil contained in any oil tank are in satisfactory working condition (SOLAS 88 reg.II-2/15) (SOLAS 02 reg.II-2/33);                                                                                                                                        |
| (PI) | 5.1.2.60   | confirming that the devices provided to prevent overpressure in any oil tank or in any part of the oil system, including the filling pipes, are in satisfactory working condition (SOLAS 88 reg.II-2/15) (SOLAS 02 reg.II-2/33);                                                                                  |
| (PI) | 5.1.2.61   | confirming that forepeak tanks are not intended for carriage of oil fuel, lubrication oil and other flammable oils;                                                                                                                                                                                               |
| (PI) | 5.1.2.62   | confirming that the electrical installations, including the main source of power and lighting systems, are installed in accordance with the approved plans (SOLAS 74/88 reg.II-1/40 and 41);                                                                                                                      |
| (PI) | 5.1.2.63   | confirming that a self-contained emergency source of electrical power has been provided and that the appropriate systems are satisfactorily supplied (SOLAS 74/88 reg.II-1/42);                                                                                                                                   |
| (PI) | 5.1.2.64   | confirming that the starting arrangements of each emergency generating set are satisfactory (SOLAS 74/88 reg.II-1/44);                                                                                                                                                                                            |
| (PI) | 5.1.2.65   | checking, when appropriate, the disposition of, and testing, the supplementary emergency lighting (SOLAS 74/88 reg.II-1/42-1);                                                                                                                                                                                    |
| (PI) | 5.1.2.65.1 | for passenger ships, constructed on or after 1 July 2010 <sup>10</sup> , confirming provision of supplementary lighting in all cabins, and checking that such lighting automatically illuminates and remains on for a minimum of 30 min when power to the normal cabin lighting is lost (SOLAS 06 reg.II-1/41.6); |
| (PI) | 5.1.2.65.2 | for passenger ships constructed on or after 1 July 2010, checking the provision of smoke detectors in cabins, which, when activated, are capable of emitting, or cause to be emitted, an audible alarm within the space where they are located (SOLAS 06 regs.II-2/7.5.2 and 7.5.3.1);                            |
| (PI) | 5.1.2.66   | confirming that precautions have been provided against shock, fire and other hazards of electrical origin (SOLAS 74/88 reg.II-1/45);                                                                                                                                                                              |
| (PI) | 5.1.2.67   | confirming, when appropriate, that the arrangements for the machinery spaces being periodically unattended are satisfactory (SOLAS 74/88 reg.II-1/54);                                                                                                                                                            |
| (PI) | 5.1.2.68   | examining the fire pumps and fire main and the disposition of the hydrants, hoses and nozzles and the international shore connection and checking that each fire pump, including the emergency fire pump,                                                                                                         |

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<sup>10</sup> Refer to the Guidance for application of SOLAS regulation II-1/41.6 (MSC.1/Circ.1372).

- can be operated separately so that two jets of water are produced simultaneously from different hydrants at any part of the ship whilst the required pressure is maintained in the fire main (SOLAS 74/88 regs.II-2/4 and 19);
- (PI) 5.1.2.69 examining the provision and disposition of the fire extinguishers (SOLAS 74/00 reg.II-2/10.3; FSSC ch.4) (SOLAS 74/88 reg.II-2/17);
- (PI) 5.1.2.70 examining the fire fighters' outfits and emergency escape breathing devices (EEBDs) and confirming that they are complete and in satisfactory condition and that the cylinders, including the spare cylinders, of the self-contained breathing apparatus, are suitably charged (SOLAS 74/00/08 regs.II-2/10.10, 13.3.4 and 13.4.3; FSSC ch.3) (SOLAS 74/88 reg.II-2/17);
- (PI) 5.1.2.71 checking the operational readiness and maintenance of fire-fighting systems (SOLAS 74/00 reg.II-2/14) (SOLAS 74/88 regs.II-2/6, 17 and 21);
- (PI) 5.1.2.72 examining the fixed fire-fighting system for the machinery cargo, special category and vehicle spaces; as appropriate, and confirming that the installation tests have been satisfactorily completed and that its means of operation are clearly marked (SOLAS 74/00 regs.II-2/10.4, 10.5, 10.7 and 20.6.1; FSSC ch.5 to 7) (SOLAS 74/88 regs.II-2/7 and 53);
- (PI) 5.1.2.73 examining the fire-extinguishing and special arrangements in the machinery spaces and confirming, as far as practicable and as appropriate, the operation of the remote means of control provided for the opening and closing of the skylights, the release of smoke, the closure of the funnel and ventilation openings, the closure of power-operated and other doors, the stopping of ventilation and boiler forced and induced draft fans and the stopping of oil fuel and other pumps that discharge flammable liquids (SOLAS 74/00 regs.II-2/5.2, 8.3, 9.5 and 10.5) (SOLAS 74/88 regs.II-2/7 and 11);
- (PI) 5.1.2.74 checking that fixed carbon dioxide fire-extinguishing systems for the protection of machinery spaces, where applicable, are provided with two separate controls, one for opening of the gas piping and one for discharging the gas from the storage container, each of them located in a release box clearly identified for the particular space (SOLAS 08 reg.II-2/10.4; FSSC ch.5.2.2.2);
- (PI) 5.1.2.75 examining the arrangements for oil fuel, lubricating oil and other flammable oils and confirming, as far as practicable and as appropriate, the operation of the remote means of closing the valves on the tanks that contain oil fuel, lubricating oil and other flammable oils (SOLAS 74/88/06 reg.II-2/15) (SOLAS 74/00 reg.II-2/4.2);
- (PI) 5.1.2.76 examining any fire detection and alarm system and confirming that installation tests have been satisfactorily completed; (SOLAS 74/88 regs.II-2/11, 12, 13, 14, 36 and 41);

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- (PI) 5.1.2.77 confirming that all aspects of installation of the structural fire protection, including the structure, fire integrity, protection of stairways and lifts, cabin balconies, openings in 'A' and 'B' Class divisions, ventilation systems and windows and sidescuttles, and the use of combustible material are in accordance with the approved plans (SOLAS 74/00/04 regs.II-2/4.4.4, 5.2, 5.3, 7.5, 7.8.2, 8.4, 8.5, 9, 10.6, 11, 13, 17, 20 and FSSC ch.13 sections 1 and 2) (SOLAS 74/88 regs.II-2/23 to 35);
- (PI) 5.1.2.78 testing any manual and automatic fire doors, including the means of closing the openings in 'A' and 'B' Class divisions (SOLAS 74/88 regs.II-2/30 and 31);
- (PI) 5.1.2.79 testing the means of closing the main inlets and outlets of all ventilation smoke extraction systems and proving that the power ventilation is capable of being stopped from outside the space served (SOLAS 74/88 reg.II-2/32);
- (PI) 5.1.2.80 confirming that stairways and ladders are so arranged as to provide a means of escape to the lifeboat and liferaft and liferaft embarkation deck from all passenger and crew spaces and from those spaces in which the crew is normally employed (SOLAS 74/00 reg.II- 2/13.7) and in particular that:
- (PI) 5.1.2.80.1 below the bulkhead deck there are two means of escape from each watertight compartment, one being independent of watertight doors;
- (PI) 5.1.2.80.2 above the bulkhead deck there are two means of escape from each vertical zone or similar such area, one leading directly to a stairway forming a vertical escape;
- (PI) 5.1.2.80.3 the radiotelegraph station, if provided, has direct access to the open deck or is provided with two means of access or egress, one of which is a porthole or window of sufficient size;
- (PI) 5.1.2.81 confirming that the means of escape from any special category spaces are generally in accordance with (PI) 5.1.2.80 (SOLAS 74/88 reg.II-2/28);
- (PI) 5.1.2.82 confirming that in the machinery spaces there are two widely separated means of escape leading to the lifeboat and liferaft embarkation decks, including, when from a space below the bulkhead deck, a continuous fire shelter (SOLAS 74/88 reg.II-2/28);
- (PI) 5.1.2.83 confirming the fire protection arrangements for special category spaces and other cargo spaces and testing, as appropriate, the operation of the means for closing the various openings (SOLAS 74/88 regs.II-2/37, 38 and 39);
- (PI) 5.1.2.84 confirming and testing, as appropriate, the fixed fire detection and alarm system, the special alarm and the public address system or other effective means of communication (SOLAS 74/88 reg.II-2/40) (SOLAS 74/00/06 regs.II-2/7 and 12);

- (PI) 5.1.2.85 for passenger ships constructed on or after 1 July 2010, confirming the provision of a fixed fire detection and fire alarm system for passenger ships capable of remotely and individually identifying each detector and manually operated call point (SOLAS 06 reg.II-2/7.2.4);
- (PI) 5.1.2.86 examining, when appropriate, the special arrangements for carrying dangerous goods, including checking the electrical equipment and wiring and boundary insulation, the provision of protective clothing and portable appliances and the testing of the water supply, bilge pumping and any water spray system (SOLAS 74/88 regs.II-2/41 and 54) (SOLAS 74/00/08 reg.II-2/19);
- (PI) 5.1.2.87 checking the provision and disposition of the survival craft and rescue boats and the arrangements for mustering passengers (SOLAS 74/88 regs.III/11 to 16, 20 and 24);
- (PI) 5.1.2.88 examining each survival craft, including its equipment, and that the required number of search and rescue locating devices are fitted in liferafts and those liferafts are clearly marked (SOLAS 74/88/00/02/08 regs.III/20, 21 and 26);
- (PI) 5.1.2.89 examining the embarkation arrangements for each survival craft and the testing of each launching appliance, including overload tests, tests to establish the lowering speed and the lowering of each survival craft to the water with the ship at its lightest sea-going draught, checking the recovery of each lifeboat (SOLAS 74/88 regs.III/11, 12, 13, 15, 20 and 48);
- (PI) 5.1.2.90 deployment of 50% of the MES after installation (LSAC section 5.1 and MSC/Circ.809);
- (PI) 5.1.2.91 examining each rescue boat, including its equipment. For inflatable rescue boats, confirming that they are stowed in a fully inflated condition (SOLAS 74/00/04 regs.III/21 and 26.3; LSAC section 5.1 and MSC/Circ.809);
- (PI) 5.1.2.92 examining the embarkation and recovery arrangements for each rescue boat and testing each launching and recovery appliance, including overload tests, tests to establish the lowering and recovery speeds and ensuring that each rescue boat can be lowered to the water and recovered with the ship at its lightest sea-going draught. The rescue boat(s) should be lowered to the water and its recovery demonstrated while underway at 5 knots (SOLAS 74/88 regs.III/14, 16, 17 and 20);
- (PI) 5.1.2.93 examining the arrangements for mustering passengers (SOLAS 74/88 reg.III/24);
- (PI) 5.1.2.94 testing that the engine of the rescue boat(s) and of each lifeboat, when so fitted, start satisfactorily and operate both ahead and astern (LSAC section 4.4.6.5);
- (PI) 5.1.2.95 confirming that there are posters or signs in the vicinity of survival craft and their launching stations (SOLAS 74/88 reg.III/9);

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| (PI) | 5.1.2.96   | examining the provision and stowage and checking the operation of two-way VHF radiotelephone apparatus and search and rescue locating devices (SOLAS 74/88/08 reg.III/6);                                                                                                                                                                                                                                                                                                                                      |
| (PI) | 5.1.2.97   | examining the provision and stowage of the distress flares and the line-throwing appliance, checking the provision and operation of onboard communications equipment and testing the means of operation of the general alarm system, verifying that the general alarm system is audible in accommodation, normal crew working spaces and on open decks (SOLAS 74/88 reg.III/6);                                                                                                                                |
| (PI) | 5.1.2.98   | examining the provision, disposition and stowage of the lifebuoys, including those fitted with self-igniting lights, self-activating smoke signals and buoyant lines, lifejackets <sup>11</sup> , immersion suits and thermal protective aids (SOLAS 74/88/06 regs.III/7, 21, 22 and 26; LSAC section 2.1-2.5 and 3.3);                                                                                                                                                                                        |
| (PI) | 5.1.2.98.1 | checking that the life-saving appliances are of international or vivid reddish orange, or a comparably highly visible colour on all parts where this will assist detection at sea (LSAC section 1.2.2.6);                                                                                                                                                                                                                                                                                                      |
| (PI) | 5.1.2.98.2 | checking the provision of lifejackets in three sizes (Infant, Child, Adult) and checking that they are marked by either weight or height, or by both weight and height (LSAC section 2.2.1.1). For passenger ships on voyages less than 24 h, checking that the number of infant lifejackets equals to at least 2.5% of the number of passengers on board and for passenger ships on voyages 24 h or greater, checking that infant lifejackets are provided for each infant on board (SOLAS 06 reg.III/7.2.1); |
| (PI) | 5.1.2.98.3 | checking that immersion suits designed to be worn in conjunction with a lifejacket are suitably marked (LSAC section 2.3.1);                                                                                                                                                                                                                                                                                                                                                                                   |
| (PI) | 5.1.2.99   | checking the lighting of the muster and embarkation stations and the alleyways, stairways and exits giving access to the muster and embarkation stations, including when supplied from the emergency source of power (SOLAS 74/88 regs.II-1/42 and III/11);                                                                                                                                                                                                                                                    |
| (PI) | 5.1.2.100  | checking that means of rescue is provided on ro-ro passenger ships (SOLAS 74/00 reg.III/26.4);                                                                                                                                                                                                                                                                                                                                                                                                                 |
| (PI) | 5.1.2.101  | checking that a helicopter pick-up area is provided on ro-ro passenger ships (SOLAS 74/00 reg.III/28);                                                                                                                                                                                                                                                                                                                                                                                                         |
| (PI) | 5.1.2.102  | checking that a decision support system is provided for the Master (SOLAS 74/00 reg.III/29);                                                                                                                                                                                                                                                                                                                                                                                                                   |
| (PI) | 5.1.2.103  | checking the electromagnetic compatibility of electrical and electronic equipment on or in the vicinity of the bridge (SOLAS 74/00 reg.V/17);                                                                                                                                                                                                                                                                                                                                                                  |
| (PI) | 5.1.2.104  | examining the provision and positioning and checking the operation of, as appropriate, the navigation lights, shapes and sound signalling                                                                                                                                                                                                                                                                                                                                                                      |

<sup>11</sup> Refer to the Guidance for application of SOLAS regulation II-1/41.6 (MSC.1/Circ.1372).



- equipment (International Regulations for Preventing Collisions at Sea in force, rules 20 to 24, 27 to 30 and 33);
- (PI) 5.1.2.105 checking the provision and specification of the daylight signalling lamp (SOLAS 74/88 reg.V/11);
- (PI) 5.1.2.106 checking, as appropriate, the provision and operation of the following equipment (SOLAS 74/00 reg.V/19):
- (PI) 5.1.2.106.1 the magnetic compass, including examining the siting, movement, illumination and a pelorus or compass bearing device (SOLAS 74/00 reg.V/19);
- (PI) 5.1.2.106.2 that nautical charts and nautical publications necessary for the intended voyage are available and have been updated and, where an electronic chart display and information system (ECDIS) is used, that the electronic charts have been updated and the required back-up system is provided and updated (SOLAS 74/00/09 reg.V/19);
- (PI) 5.1.2.106.3 global navigation satellite receiver or terrestrial radionavigation system;
- (PI) 5.1.2.106.4 sound reception system, when bridge is totally enclosed;
- (PI) 5.1.2.106.5 means of communication to emergency steering position, where provided;
- (PI) 5.1.2.106.6 spare magnetic compass;
- (PI) 5.1.2.106.7 daylight signalling lamp;
- (PI) 5.1.2.106.8 echo sounding device, including examining the display for good access, viewing and lighting;
- (PI) 5.1.2.106.9 radar(s), including examining the waveguide and cable runs for routing and protection and the display unit confirming lighting, plotting facilities, correct operation of all controls, functions and the true-motion facility if provided;
- (PI) 5.1.2.106.10 electronic plotting aid, automatic tracking aid or automatic radar plotting aid as appropriate, using the appropriate test facilities;
- (PI) 5.1.2.106.11 speed and distance measuring device;
- (PI) 5.1.2.106.12 transmitting heading device providing heading information to radar, plotting aids and automatic identification system equipment and distance devices;
- (PI) 5.1.2.106.13 heading or track control system;
- (PI) 5.1.2.106.14 BNWAS;
- (PI) 5.1.2.107 checking for the provision, specification, operation and annual performance test of the voyage data recorder (SOLAS 74/00/04 reg.V/20);

- (PI) 5.1.2.108 checking that a valid conformance test report of the long-range and identification tracking system is available on board (SOLAS 04 reg.V/19-1);
- (PI) 5.1.2.109 checking that the International Code of Signals and a copy of Volume III of the International Aeronautical and Maritime Search and Rescue (IAMSAR) Manual have been provided. (SOLAS 74/00/02 reg.V/21);
- (PI) 5.1.2.110 checking the provision and, as appropriate, the deployment or operation of the pilot ladders and hoists/pilot transfer arrangements (SOLAS 74/00 reg.V/23);
- (PI) 5.1.2.111 examining the position, physical and electromagnetic protection and illumination of each radio installation (SOLAS 74/88 reg.IV/6);
- (PI) 5.1.2.112 confirming the provision of equipment for the radio installation with due regard to the declared sea areas in which the ship will trade and the declared means of maintaining availability of functional requirements (SOLAS 74/88 regs.III/6, IV/7 to 11, 14 and 15);
- (PI) 5.1.2.113 confirming the ability to initiate the transmission of ship-to-shore distress alerts by at least two separate and independent means, each using a different radio communication service, from the position from which the ship is normally navigated (SOLAS 74/88/06 regs.IV/4, 7 to 11);
- (PI) 5.1.2.114 examining all antennas, including:
  - (PI) 5.1.2.114.1 visually checking all antennas, including Inmarsat antennas, and feeders for satisfactory siting and absence of defects (SOLAS 74/88 reg.IV/14);
  - (PI) 5.1.2.114.2 checking insulation and safety of all antennas;
- (PI) 5.1.2.115 examining the reserve source of energy, including:
  - (PI) 5.1.2.115.1 checking there is sufficient capacity to operate the basic or duplicated equipment for 1 hour or 6 hours, as appropriate (SOLAS 74/88 reg.IV/13);
  - (PI) 5.1.2.115.2 and, if the reserve source of energy is a battery:
    - (PI) 5.1.2.115.2.1 checking its siting and installation (SOLAS 74/88 reg.IV/13);
    - (PI) 5.1.2.115.2.2 where appropriate, checking its condition by specific gravity measurement or voltage measurement;
    - (PI) 5.1.2.115.2.3 with the battery off charge, and the maximum required radio installation load connected to the reserve source of energy, checking the battery voltage and discharge current;
    - (PI) 5.1.2.115.2.4 checking that the charger(s) are capable of recharging the reserve battery within 10 hours (SOLAS 74/88 reg.IV/13);

- (PI) 5.1.2.116 examining the VHF transceiver(s), including:
  - (PI) 5.1.2.116.1 checking for operation on channels 6, 13 and 16 (SOLAS 74/88 regs.IV/7 and 14);
  - (PI) 5.1.2.116.2 checking frequency tolerance, transmission line quality and radio frequency power output (SOLAS 74/88 reg.IV/14);
  - (PI) 5.1.2.116.3 checking for correct operation of all controls including priority of control units (SOLAS 74/88 reg.IV/14);
  - (PI) 5.1.2.116.4 checking that the equipment operates from the main, emergency (if provided) and reserve sources of energy (SOLAS 74/88 reg.IV/13);
  - (PI) 5.1.2.116.5 checking the operation of the VHF control unit(s) or portable VHF equipment provided for navigational safety (SOLAS 74/88 reg.IV/6);
  - (PI) 5.1.2.116.6 checking for correct operation by on-air contact with a coast station or other ship;
- (PI) 5.1.2.117 examining the VHF DSC controller and channel 70 DSC watch receiver, including:
  - (PI) 5.1.2.117.1 performing an off-air check confirming the correct Maritime Mobile Service Identity is programmed in the equipment (SOLAS 74/88 reg.IV/14);
  - (PI) 5.1.2.117.2 checking for correct transmission by means of a routine or test call to a coast station, other ship, onboard duplicate equipment or special test equipment;
  - (PI) 5.1.2.117.3 checking for correct reception by means of a routine or test call from a coast station, other ship, onboard duplicate equipment or special test equipment;
  - (PI) 5.1.2.117.4 checking the audibility of the VHF/DSC alarm;
  - (PI) 5.1.2.117.5 checking that the equipment operates from the main, emergency (if provided) and reserve sources of energy (SOLAS 74/88 reg.IV/13);
- (PI) 5.1.2.118 examining the MF/HF radiotelephone equipment, including:
  - (PI) 5.1.2.118.1 checking that the equipment operates from the main, emergency (if provided) and reserve sources of energy (SOLAS 74/88 reg.IV/13);
  - (PI) 5.1.2.118.2 checking the antenna tuning in all appropriate bands;
  - (PI) 5.1.2.118.3 checking the equipment is within frequency tolerance on all appropriate bands (SOLAS 74/88 reg.IV/14);
  - (PI) 5.1.2.118.4 checking for correct operation by contact with a coast station and/or measuring transmission line quality and radio frequency output;

- (PI) 5.1.2.118.5 checking receiver performance by monitoring known stations on all appropriate bands;
- (PI) 5.1.2.118.6 if control units are provided outside the navigating bridge, checking the control unit on the bridge has first priority for the purpose of initiating distress alerts (SOLAS 74/88 regs.IV/9, 10, 11 and 14);
- (PI) 5.1.2.118.7 checking the correct operation of the radiotelephone alarm signal generating device on a frequency other than 2182 kHz;
- (PI) 5.1.2.119 examining the HF radiotelex equipment, including:
  - (PI) 5.1.2.119.1 checking that the equipment operates from the main, emergency (if provided) and reserve sources of energy (SOLAS 74/88 reg.IV/13);
  - (PI) 5.1.2.119.2 confirming that the correct selective calling number is programmed in the equipment;
  - (PI) 5.1.2.119.3 checking correct operation by inspection of recent hard copy or by a test with a coast radio station (SOLAS 74/88 regs.IV/10 and 11);
- (PI) 5.1.2.120 examining the MF/HF DSC controller(s), including:
  - (PI) 5.1.2.120.1 checking that the equipment operates from the main, emergency (if provided) and reserve sources of energy (SOLAS 74/88 reg.IV/13);
  - (PI) 5.1.2.120.2 confirming that the correct Maritime Mobile Service Identity is programmed in the equipment;
  - (PI) 5.1.2.120.3 checking the off-air self-test program;
  - (PI) 5.1.2.120.4 checking operation by means of a test call on MF and/or HF to a coast radio station if the rules of the berth permit the use of MF/HF transmissions (SOLAS 74/88 regs.IV/9 to 11);
  - (PI) 5.1.2.120.5 checking the audibility of the MF/HF DSC alarm;
- (PI) 5.1.2.121 examining the MF/HF DSC watch receiver(s), including:
  - (PI) 5.1.2.121.1 confirming that only distress and safety DSC frequencies are being monitored (SOLAS 74/88 regs.IV/9 to 12);
  - (PI) 5.1.2.121.2 checking that a continuous watch is being maintained whilst keying MF/HF radio transmitters (SOLAS 74/88 reg.IV/12);
  - (PI) 5.1.2.121.3 checking for correct operation by means of a test call from a coast station or other ship;
- (PI) 5.1.2.122 examining the Inmarsat ship earth station(s), including:
  - (PI) 5.1.2.122.1 checking that the equipment operates from the main, emergency (if provided) and reserve sources of energy, and that where an uninterrupted supply of information from the ship's navigational or other equipment is required ensuring such information remains available in

- the event of failure of the ship's main or emergency source of electrical power. (SOLAS 74/88 regs.IV/13 and 14);
- (PI) 5.1.2.122.2 checking the distress function by means of an approved test procedure where possible (SOLAS 74/88 regs.IV/10, 12 and 14);
- (PI) 5.1.2.122.3 checking for correct operation by inspection of recent hard copy or by test call;
- (PI) 5.1.2.123 if appropriate, examining the NAVTEX equipment (SOLAS 74/88 regs.IV/7, 12 and 14), including:
- (PI) 5.1.2.123.1 checking for correct operation by monitoring incoming messages or inspecting recent hard copy;
- (PI) 5.1.2.123.2 running the self-test program if provided;
- (PI) 5.1.2.124 examining the enhanced group call equipment (SOLAS 74/88 regs.IV/7 and 14), including:
- (PI) 5.1.2.124.1 checking for correct operation and area by monitoring incoming messages or by inspecting recent hard copy;
- (PI) 5.1.2.124.2 running the self-test program if provided;
- (PI) 5.1.2.125 if appropriate, examining the radio equipment for receipt of maritime safety information by HF NBDP (SOLAS 74/88 regs.IV/7, 12 and 14), including:
- (PI) 5.1.2.125.1 checking for correct operation by monitoring incoming messages or inspecting recent hard copy;
- (PI) 5.1.2.125.2 running the self-test program if provided;
- (PI) 5.1.2.126 examining the 406 MHz satellite EPIRB (SOLAS 74/88 regs.IV/7 and 14), including:
- (PI) 5.1.2.126.1 checking position and mounting for float-free operation;
- (PI) 5.1.2.126.2 carrying out visual inspection for defects;
- (PI) 5.1.2.126.3 carrying out the self-test routine;
- (PI) 5.1.2.126.4 checking that the EPIRB ID is clearly marked on the outside of the equipment and, where possible, decoding the EPIRB identity number confirming it is correct;
- (PI) 5.1.2.126.5 checking the battery expiry date;
- (PI) 5.1.2.126.6 if provided, checking the hydrostatic release and its expiry date;
- (PI) 5.1.2.127 examining the two-way VHF radiotelephone apparatus (SOLAS 74/88 reg.III/6), including:

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| (PI) | 5.1.2.127.1 | checking for correct operation on channel 16 and one other by testing with another fixed or portable VHF installation (SOLAS 74/88 reg.IV/14);                                                                                                         |
| (PI) | 5.1.2.127.2 | checking the battery charging arrangements where rechargeable batteries are used (SOLAS 74/88 reg.IV/14);                                                                                                                                              |
| (PI) | 5.1.2.127.3 | checking the expiry date of primary batteries where used (SOLAS 74/88 reg.IV/14);                                                                                                                                                                      |
| (PI) | 5.1.2.127.4 | where appropriate, checking any fixed installation provided in a survival craft (SOLAS 74/88 reg.IV/14);                                                                                                                                               |
| (PI) | 5.1.2.128   | examining the search and rescue locating device(s) (SOLAS 74/88/08 reg.III/6 and regs.IV/7 and 14), including:                                                                                                                                         |
| (PI) | 5.1.2.128.1 | checking the position and mounting;                                                                                                                                                                                                                    |
| (PI) | 5.1.2.128.2 | monitoring response on ship's 9 GHz radar;                                                                                                                                                                                                             |
| (PI) | 5.1.2.128.3 | checking the battery expiry date;                                                                                                                                                                                                                      |
| (PI) | 5.1.2.129   | examining the test equipment and spares carried to ensure carriage is adequate in accordance with the sea areas in which the ship trades and the declared options for maintaining availability of the functional requirements (SOLAS 74/88 reg.IV/15); |
| (PI) | 5.1.2.130   | checking the distress panel installed at the conning position; or, where applicable, an additional EPIRB is placed near the conning position (SOLAS 74/88 reg.IV/6);                                                                                   |
| (PI) | 5.1.2.131   | checking that positional information is provided continuously and automatically to all communications equipment included in the initial distress alert (SOLAS 74/88 reg.IV/6);                                                                         |
| (PI) | 5.1.2.132   | checking the distress alarm panel installed at the conning position and its visual and aural indications of received distress alerts (SOLAS 74/88 reg.IV/6);                                                                                           |
| (PI) | 5.1.2.133   | checking the provision and operation of the means for two-way on-scene communication for search and rescue purposes and its operation on 121.5 MHz and 123.1 MHz from the position from which the ship is normally navigated (SOLAS 74/88 reg.IV/7);   |
| (PI) | 5.1.2.134   | confirming that the ship's identification number is permanently marked. (SOLAS 74/02 reg.XI-1/3);                                                                                                                                                      |
| (PI) | 5.1.2.135   | checking that the provision, operation and the annual test has been carried out for the automatic identification system (SOLAS 74/00/04 reg.V/19);                                                                                                     |
| (PI) | 5.1.2.136   | for passenger ships carrying more than 36 passengers constructed on or after 1 July 2010, checking the provision of a suitably located means                                                                                                           |

- for fully recharging breathing air cylinders, free from contamination (SOLAS 08 reg.II-2/10.10.2.6);
- (PI) 5.1.2.137 confirming that installed materials do not contain asbestos (SOLAS 09 reg.II-1/3-5);
- (PI) 5.1.2.138 confirming the provision of means of embarkation and disembarkation from ships for use in port and in port-related operations, such as gangways and accommodation ladders (SOLAS 08 reg.II-1/3-9);
- (PI) 5.1.2.139 for passenger ships constructed on or after 1 July 2010 and having a length of 120 m or more or having three or more main vertical zones, checking the designation of safe areas (SOLAS 06 reg.II-2/21);
- (PI) 5.1.2.140 for passenger ships constructed on or after 1 July 2010, checking the provision of a safety centre (SOLAS reg.II-2/23) and associated ventilation requirements(SOLAS 06 reg.II-2/8.2).
- (PI) 5.1.3 For the hull, machinery and equipment of passenger ships the check that the required documentation has been placed on board should consist of:
- (PI) 5.1.3.1 confirming that the stability information and damage control plans have been provided (SOLAS 74/88 regs.II-1/22 and 23) (SOLAS 06 regs.II-1/5-1 and 19);
- (PI) 5.1.3.2 checking the provision of a ship-specific emergency towing procedure (SOLAS 08 reg.II-1/3-4);
- (PI) 5.1.3.3 confirming that the manoeuvring booklet has been provided and that the manoeuvring information has been displayed on the navigating bridge (SOLAS 74/88 reg.II-1/28);
- (PI) 5.1.3.4 confirming that documented operating procedures for closing and securing the openings in special category spaces and ro-ro spaces are available on board (SOLAS 06 reg.II-1/23);
- (PI) 5.1.3.5 confirming that, where applicable, the approved documentation for the alternative design and arrangements is on board (SOLAS 00/06 regs.II-1/55, II-2/17 and III/38);
- (PI) 5.1.3.6 confirming that the fire control plans are permanently exhibited or, alternatively, that emergency booklets have been provided to each officer and a duplicate of the plans or the emergency booklet are available in a prominently marked enclosure external to the ship's deckhouse (SOLAS 74/00 regs.II-2/15.2.4 and 15.3.2) (SOLAS 74/88 reg.II-2/20). The fire control plan is in the language required by the Administration;
- (PI) 5.1.3.7 confirming that the maintenance plans have been provided (SOLAS 74/88 regs.II-2/14.2.2 and 14.3);
- (PI) 5.1.3.8 confirming that the training manuals and the fire safety operational booklets have been provided (SOLAS 74/88 regs.II-2/15.2.3 and 16.2);

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| (PI) | 5.1.3.9  | confirming, when appropriate, that the ship is provided with a document indicating compliance with the special requirements for carrying dangerous goods (SOLAS 74/00/08 reg.II-2/19.4) (SOLAS 74/88 regs.II-2/41 and 54.3);             |
| (PI) | 5.1.3.10 | confirming that emergency instructions are available for each person on board, that the muster list is posted in conspicuous places, and that they are in a language understood by the persons on board (SOLAS 74/00 regs.III/8 and 53); |
| (PI) | 5.1.3.11 | confirming that the training manual for the life-saving appliances has been provided and is available in the working language of the ship (SOLAS 74/00/04 reg.III/35);                                                                   |
| (PI) | 5.1.3.12 | confirming that the checklist and instructions for MES, if provided, and onboard maintenance of the life-saving appliances have been provided (SOLAS 74/00 reg.III/36);                                                                  |
| (PI) | 5.1.3.13 | confirming that a table or curve of residual deviations for the magnetic compass have been provided, and that a diagram of the radar installations shadow sectors is displayed (SOLAS 74/00 reg.V/19);                                   |
| (PI) | 5.1.3.14 | checking that operational and, where appropriate, maintenance manuals for all navigational equipment are provided (SOLAS 74/00 regs.V/16 and 19);                                                                                        |
| (PI) | 5.1.3.15 | checking that the charts and nautical publications necessary for the intended voyage are available and have been updated (SOLAS 74/00 regs.V/19 and 27);                                                                                 |
| (PI) | 5.1.3.16 | checking that the International Code of Signals is available where the ship is required to carry a radio installation (SOLAS 74/88 reg.V/21);                                                                                            |
| (PI) | 5.1.3.17 | confirming that a list showing the operational limitations imposed to the ship is kept on board (SOLAS 74/00 reg.V/30);                                                                                                                  |
| (PI) | 5.1.3.18 | checking that the life-saving signals to be used by ships, aircraft or persons in distress (SOLAS 74/00 reg.V/29);                                                                                                                       |
| (PI) | 5.1.3.19 | checking the carriage of operating manuals for all equipment (SOLAS 74/88 reg.IV/15);                                                                                                                                                    |
| (PI) | 5.1.3.20 | checking the carriage of service manuals for all equipment when at-sea maintenance is the declared option (SOLAS 74/88 reg.IV/15);                                                                                                       |
| (PI) | 5.1.3.21 | checking for a valid radio licence issued by the flag Administration (ITU RR Art.24);                                                                                                                                                    |
| (PI) | 5.1.3.22 | checking the radio operators' certificates of competence (ITU RR Art.55);                                                                                                                                                                |
| (PI) | 5.1.3.23 | checking the emission on operational frequencies, coding and registration on the 406 MHz signal without transmission of a distress call to the satellite;                                                                                |
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- (PI) 5.1.3.24 checking the radio log (SOLAS 74/88 text in force prior to 1 February 1992 reg.IV/19 and ITU RR App.11);
- (PI) 5.1.3.25 checking the carriage of up-to-date ITU publications (ITU RR App.11);
- (PI) 5.1.3.26 checking that the EPIRB has been subject to maintenance at intervals not exceeding five years at an approved shore-based maintenance facility;
- (PI) 5.1.3.27 if possible, checking the emission on operational frequencies, coding and registration on the 121.5 MHz homing signal without transmission of the distress call to satellite system;
- (PI) 5.1.3.28 confirming that a continuous synopsis record is provided (SOLAS 74/02 reg.XI-1/5);
- (PI) 5.1.3.29 checking that arrangements are provided to maintain records of navigational activities and daily reporting (SOLAS 74/00/03 reg.V/28).
- (PI) 5.1.4 For the hull, machinery and equipment of passenger ships the completion of the initial survey should consist of:
  - (PI) 5.1.4.1 after a satisfactory survey, issuing the Passenger Ship Safety Certificate and its associated Record of Equipment (Form P).
- (PR) **5.2** **Renewal surveys** – see part "General", section 4.5
  - (PR) 5.2.1 For the hull, machinery and equipment of passenger ships the examination of current certificates and other records should consist of:
    - (PR) 5.2.1.1 checking the validity of the International Load Line Certificate or International Load Line Exemption Certificate;
    - (PR) 5.2.1.2 checking the validity of the Safety Management Certificate (SMC) and that a copy of the Document of Compliance (DOC) is on board;
    - (PR) 5.2.1.3 checking the validity of the International Ship Security Certificate;
    - (PR) 5.2.1.4 checking the validity of the International Oil Pollution Prevention Certificate;
    - (PR) 5.2.1.5 checking the certificates of class, if the ship is classed with a classification society;
    - (PR) 5.2.1.6 checking, when appropriate, the validity of the International Pollution Prevention Certificate for the Carriage of Noxious Liquid Substances in Bulk;
    - (PR) 5.2.1.7 checking, when appropriate, the validity of the International Sewage Pollution Prevention Certificate;
    - (PR) 5.2.1.8 checking, when appropriate, the validity of the International Air Pollution Prevention Certificate;

- (PR) 5.2.1.9 checking that the ship's complement complies with the Minimum Safe Manning Document (SOLAS 74/00 reg.V/14) (SOLAS 74/88 reg.V/13(b));
- (PR) 5.2.1.10 checking that the master, officers and ratings are certificated as required by the STCW Convention;
- (PR) 5.2.1.11 confirming that, where applicable, the approved documentation for the alternative design and arrangements is on board (SOLAS 00/06 regs.II-1/55, II-2/17 and III/38);
- (PR) 5.2.1.12 checking whether any new equipment has been fitted and, if so, confirming that it has been approved before installation and that any changes are reflected in the appropriate certificate;
- (PR) 5.2.1.13 checking that the routine surveys of the boilers and other pressure vessels, as determined by the Administration, have been carried out as required and that safety devices, such as the boiler safety valves, have been tested;
- (PR) 5.2.1.14 checking that, as appropriate, the hull and machinery has been presented for survey in accordance with the continuous survey scheme approved by the Administration or a classification society;
- (PR) 5.2.1.15 confirming that the opening and the closing and locking of side scuttles positioned below the margin line or the bulkhead deck, as applicable, are being recorded in the log-book (SOLAS 74/88 reg.II-1/17), (SOLAS 74/06 reg.II-1/15);
- (PR) 5.2.1.16 confirming that the closure of the cargo loading doors and the opening and closing of any doors at sea required for the operation of the ship or the embarking and disembarking of passengers are being recorded in the log-book (SOLAS 74/88 reg.II-1/20-1) (SOLAS 06 reg.II-1/22);
- (PR) 5.2.1.17 confirming that the stability information and damage control plans are readily available (SOLAS 74/88 regs.II-1/22 and 23) (SOLAS 06 regs.II-1/5-1 and 19);
- (PR) 5.2.1.18 confirming from the log-book entries that the openings required to be closed at sea are being kept closed and that the required drills and inspections of watertight doors, etc., are being carried out (SOLAS 74/88 regs.II-1/24 and 25) (SOLAS 06 regs.II-1/21 and 22);
- (PR) 5.2.1.19 confirming that documented operating procedures for closing and securing the openings in special category spaces and ro-ro spaces are available on board (SOLAS 06 reg.II-1/23);
- (PR) 5.2.1.20 confirming that the manoeuvring booklet is readily available and that the manoeuvring information is displayed on the navigating bridge (SOLAS 74/88 reg.II-1/28);
- (PR) 5.2.1.21 confirming that the fire control plans are permanently exhibited or, alternatively, that emergency booklets have been provided and a duplicate of the plans or the emergency booklet is available in a prominently marked enclosure external to the ship's deckhouse (SOLAS 74/88 reg.II-2/20);

- (PR) 5.2.1.22 confirming that the maintenance plans have been provided (SOLAS 74/00 regs.II-2/14.2.2 and 14.3);
- (PR) 5.2.1.23 confirming that the training manuals and the fire safety operational booklets have been provided (SOLAS 74/00 regs.II-2/15.2.3 and 16.2);
- (PR) 5.2.1.24 checking whether any fire has occurred on board necessitating the operation of the fixed fire-extinguishing systems or the portable fire extinguishers since the last survey and the entries into the ship's log-book;
- (PR) 5.2.1.25 checking, when appropriate, that the ship is provided with a document indicating compliance with the special requirements for carrying dangerous goods (SOLAS 74/00/08 reg.II-2/19.4) (SOLAS 74/88 reg.II-2/54.3);
- (PR) 5.2.1.26 confirming, when appropriate, that there is a special list, manifest or stowage plan for the carriage of dangerous goods (SOLAS 74/88 reg.VII/5);
- (PR) 5.2.1.27 confirming that emergency instructions are available for each person on board, that the muster list is posted in conspicuous places, and that they are in a language understood by the persons on board (SOLAS 74/00 regs.III/8 and 37);
- (PR) 5.2.1.28 checking that log-book entries are being made (SOLAS 74/00 regs.III/19 and 20), in particular:
- (PR) 5.2.1.28.1 the date when the last full muster of the passengers and crew for boat and fire drill took place;
- (PR) 5.2.1.28.2 the records indicating that the lifeboat equipment was examined at that time and found to be complete;
- (PR) 5.2.1.28.3 the last occasion when the lifeboats were swung out and when each one was lowered into the water;
- (PR) 5.2.1.28.4 the records indicating that crew members have received the appropriate onboard training;
- (PR) 5.2.1.29 confirming that the training manual and training aids for the life-saving appliances are available on board in the working language of the ship (SOLAS 74/00/04 reg.III/35);
- (PR) 5.2.1.30 confirming that the instructions for on board maintenance of the life-saving appliances is on board (SOLAS 74/00 reg.III/36);
- (PR) 5.2.1.31 checking by the log-book entries that the testing and the emergency drills of the steering gear have been carried out (SOLAS 74/00 reg.V/26);
- (PR) 5.2.1.32 confirming that a table or curve of residual deviations for the magnetic compass is available and that a diagram of the radar installations shadow sectors is displayed (SOLAS 74/00 reg.V/19);

- (PR) 5.2.1.33 checking that operational and, where appropriate, maintenance manuals for all navigational equipment are provided (SOLAS 74/00 reg.V/16);
- (PR) 5.2.1.34 checking that the charts and nautical publications necessary for the intended voyage are available and have been updated (SOLAS 74/00 reg.V/27);
- (PR) 5.2.1.35 checking that the compass deviation book is properly maintained (SOLAS 74/00 reg.V/19);
- (R) 5.2.1.36 confirming that a list showing the operational limitations imposed on the ship is kept on board (SOLAS 74/00 reg.V/30);
- (R) 5.2.1.37 checking the life-saving signals to be used by ships, aircraft or persons in distress (SOLAS 74/00 reg.V/29);
- (PR) 5.2.1.38 confirming the provisions of (PI) 5.1.3.14 to (PI) 5.1.3.19;
- (PR) 5.2.1.39 confirming that a record has been kept in the period since the last survey to the satisfaction of the Administration and as required by the Radio Regulations (SOLAS 74/88 reg.IV/17);
- (PR) 5.2.1.40 checking documentary evidence that the actual capacity of the battery has been proved in port within the last 12 months (SOLAS 74/88 reg.IV/13);
- (PR) 5.2.1.41 if applicable, checking that a list of all limitations on the operation of a passenger ship is kept on board and updated;
- (PR) 5.2.1.42 confirming that continuous synopsis record is provided (SOLAS 74/02 reg.XI-1/5);
- (PR) 5.2.1.43 checking that the annual test has been carried out for the Satellite EPIRB and, if applicable, that shore-based maintenance has been carried out at intervals not exceeding five years (SOLAS 74/04 reg.IV/15);
- (PR) 5.2.1.44 checking that arrangements are provided to maintain records of navigational activities and daily reporting (SOLAS 74/00/03 reg.V/28);
- (PR) 5.2.1.45 confirming the availability of the International Anti-Fouling System Certificate (AFS 2001 Annex 4 reg.2), when applicable.
- (PR) 5.2.2 For the hull, machinery and equipment of passenger ships the renewal survey should consist of:
  - (PR) 5.2.2.1 examining the outside of the ship's bottom, including the bottom and bow plating, keel, bilge keels, stem, stern frame, the rudder, sea chests and strainers, noting the clearance measured in the rudder bearings, examining the propeller and shaft seals, as far as practicable, and noting the clearance measured in the propeller shafts (SOLAS 74/88 reg.I/7(b)(ii));

- (PR) 5.2.2.2 examining the arrangements for subdivision, including the ship's stability in the damaged condition, and checking the subdivision load lines (SOLAS 74/88 regs.II-1/4 to 8, 13 and 16) (SOLAS 06 regs.II-1/8, 14 and 18);
- (PR) 5.2.2.3 checking the ballasting arrangements (SOLAS 74/88 reg.II-1/9) (SOLAS 06 reg.II-1/20);
- (PR) 5.2.2.4 confirming that dedicated sea water ballast tanks have been coated in accordance with resolution MSC.215(82) when appropriate (SOLAS 74/00/06 reg.II-1/3-2);
- (PR) 5.2.2.5 confirming when appropriate that the maintenance of the protective coating is included in the overall ship's maintenance system (SOLAS 74/00/06 reg.II-1/3-2);
- (PR) 5.2.2.6 examining the collision and other watertight bulkheads required for the ship's subdivision (SOLAS 74/88 regs.II-1/10, 14, 15 and 18) (SOLAS 06, regs.II-1/10, 11, 12, 13 and 16);
- (PR) 5.2.2.7 confirming that the watertight integrity has been maintained where pipes, scuppers, etc., pass through subdivision watertight bulkheads (SOLAS 74/88 reg.II-1/15) (SOLAS 06 reg.II-1/13);
- (PR) 5.2.2.8 confirming that a diagram is provided on the navigating bridge showing the location of the watertight doors together with indicators showing whether the doors are open or closed (SOLAS 74/88 reg.II-1/15) (SOLAS 06 reg.II-1/13);
- (PR) 5.2.2.9 testing the operation of the watertight doors both from the navigating bridge in the event of an emergency and locally at the door itself (SOLAS 74/88 reg.II-1/15) (SOLAS 06 reg.II-1/13) and, in particular, that they are:
  - (PR) 5.2.2.9.1 operable locally from each side of the bulkhead;
  - (PR) 5.2.2.9.2 provided with devices giving an indication of whether the door is open or closed at all remote operating positions;
  - (PR) 5.2.2.9.3 provided with an audible alarm that is distinct from any other alarm in the area and, when appropriate, an intermittent visual signal;
  - (PR) 5.2.2.9.4 provided with control handles on each side of the bulkhead so that a person may hold both handles in the open position and pass safely through the watertight door without accidentally setting the power closing mechanism into operation;
- (PR) 5.2.2.10 confirming that the watertight doors and their indicating devices are operable in the event of a failure of the main and emergency sources of power (SOLAS 74/88 reg.II-1/15) (SOLAS 06 reg.II-1/13);
- (PR) 5.2.2.11 checking, when appropriate, any watertight doors that are not required to be closed remotely, fitted in watertight bulkheads dividing 'tween deck spaces, and confirming that a notice is affixed concerning their closure (SOLAS 74/88 reg.II-1/15) (SOLAS 06 reg.II-1/13);

- (PR) 5.2.2.12 confirming that a notice is affixed to any portable plates on bulkheads in machinery spaces concerning their closure and, if appropriate, testing any power-operated watertight door fitted in lieu (SOLAS 74/88 reg.II-1/15) (SOLAS 06 reg.II-1/13);
- (PR) 5.2.2.13 examining the arrangements for closing side scuttles and their deadlights, also scuppers, sanitary discharges and similar openings and other inlets and discharges in the shell plating below the margin line (SOLAS 74/88 reg.II-1/17);
- (PR) 5.2.2.14 examining the arrangements for closing side scuttles and their deadlights, also scuppers, sanitary discharges and similar openings and other inlets and discharges in the shell plating below the bulkhead deck (SOLAS 06 reg.II-1/15);
- (PR) 5.2.2.15 confirming that valves for closing the main and auxiliary sea inlets and discharges in the machinery spaces are readily accessible and indicators showing the status of the valves are provided (SOLAS 74/88 reg.II-1/17) (SOLAS 06 reg.II-1/15);
- (PR) 5.2.2.16 confirming that gangway, cargo and coaling ports fitted below the margin line may be effectively closed and that the inboard ends of any ash or rubbish chutes are fitted with an effective cover (SOLAS 74/88 reg.II-1/17);
- (PR) 5.2.2.17 confirming that gangway, cargo and fuelling ports fitted below the bulkhead deck may be effectively closed and that the inboard ends of any ash or rubbish chutes are fitted with an effective cover (SOLAS 06 reg.II-1/15);
- (PR) 5.2.2.18 examining the arrangements to maintain the watertight integrity above the margin line or the bulkhead deck as applicable (SOLAS 74/88 reg.II-1/20) (SOLAS 06 reg.II-1/17);
- (PR) 5.2.2.19 examining the arrangements for the bilge pumping and confirming that each bilge pump and the bilge pumping system provided for each watertight compartment is working efficiently (SOLAS 74/88 reg.II-1/21) (SOLAS 05 reg.II-1/35-1);
- (PR) 5.2.2.20 confirming that the drainage system of enclosed cargo spaces situated on the freeboard deck is working efficiently (SOLAS 74/88 reg.II-1/21) (SOLAS 05 reg.II-1/35-1);
- (PR) 5.2.2.21 examining visually the drainage facilities for blockage or other damage and confirming the provision of means to prevent blockage of drainage arrangements, for closed vehicle and ro-ro spaces and special category spaces where fixed pressure water-spraying systems are used (SOLAS 08 reg.II-2/20.6.1.5);
- (PR) 5.2.2.22 examining, when appropriate, the means of indicating the status of any bow doors and any leakage there from (SOLAS 74/88 reg.II-1/23-2);

- (PR) 5.2.2.23 confirming, that the arrangement for monitoring special category spaces or ro-ro spaces, when fitted, is satisfactory (SOLAS 06 reg.II-1/23);
- (PR) 5.2.2.24 confirming that the machinery, boilers and other pressure vessels, associated piping systems and fittings are being maintained so as to reduce to a minimum any danger to persons on board, due regard being given to moving parts, hot surfaces and other hazards (SOLAS 74/88 reg.II-1/26);
- (PR) 5.2.2.25 confirming that normal operation of the propulsion machinery can be sustained or restored even though one of the essential auxiliaries becomes inoperative (SOLAS 74/88 reg.II-1/26);
- (PR) 5.2.2.26 confirming that means are provided so that the machinery can be brought into operation from the dead ship condition without external aid (SOLAS 74/88 reg.II-1/26);
- (PR) 5.2.2.27 examining, where practicable, the means provided to protect against overpressure in the parts of main, auxiliary and other machinery that is subject to internal pressure and may be subject to dangerous overpressure (SOLAS 74/88 reg.II-1/27);
- (PR) 5.2.2.28 examining, when appropriate, the crankcase explosion relief devices fitted to internal combustion engines and confirming that they are arranged so as to minimize the possibility of injury to personnel (SOLAS 74/88 reg.II-1/27);
- (PR) 5.2.2.29 confirming that the automatic shut-off arrangements fitted to the main turbine propulsion machinery and, where applicable, main internal combustion propulsion machinery and auxiliary machinery are being properly maintained (SOLAS 74/88 reg.II-1/27);
- (PR) 5.2.2.30 confirming, as far as practicable, the ability of the machinery to reverse the direction of the thrust of the propeller in sufficient time, including the effectiveness of any supplementary means of manoeuvring or stopping the ship (SOLAS 74/88 reg.II-1/28);
- (PR) 5.2.2.31 confirming that the main and auxiliary steering gear are being properly maintained, are arranged so that the failure of one does not render the other inoperative and that the auxiliary steering gear is capable of being brought speedily into action in an emergency (SOLAS 74/88 reg.II-1/29);
- (PR) 5.2.2.32 confirming that, where appropriate, essential components of the steering gear are permanently lubricated or provided with lubrication fittings (SOLAS 74/88 reg.II-1/29);
- (PR) 5.2.2.33 confirming that relief valves fitted to the steering gear hydraulic system which can be isolated, and in which pressure can be generated from the power source or from external forces, are being maintained and are set to a pressure not exceeding the design pressure (SOLAS 74/88 reg.II-1/29);

- (PR) 5.2.2.34 confirming that the main or auxiliary steering gear power units restart automatically when power is restored after a power failure, that they are capable of being brought into operation from a position on the navigating bridge and that, in the event of a power failure to any one of the steering gear power units, an audible and visual alarm is given on the navigating bridge (SOLAS 74/88 reg.II-1/29);
- (PR) 5.2.2.35 confirming that the control systems for the main steering gear from both the navigating bridge and the steering gear compartment are operating satisfactorily (SOLAS 74/88 reg.II-1/29);
- (PR) 5.2.2.36 confirming that, where the main steering gear comprises two or more identical power units and an auxiliary steering gear is not fitted, the two independent control systems from the navigating bridge are operating satisfactorily (SOLAS 74/88 reg.II-1/29);
- (PR) 5.2.2.37 confirming that the control system for the auxiliary steering gear, in the steering gear compartment and, if this gear is power-operated, from the navigating bridge, are operating satisfactorily and that the latter is independent of the control system for the main steering gear (SOLAS 74/88 reg.II-1/29);
- (PR) 5.2.2.38 confirming that an audible and visual alarm is given on the navigating bridge in the event of a failure of electrical power supply (SOLAS 74/88 reg.II-1/29);
- (PR) 5.2.2.39 confirming that the means of communication between the bridge and the steering gear is operating satisfactorily and that, with ships having emergency steering positions, a telephone or other means of communication for relaying heading information and supplying visual compass readings to the emergency steering position is provided (SOLAS 74/00 regs.II-1/29 and V/19);
- (PR) 5.2.2.40 confirming that the angular position of the rudder is indicated independently of the steering control system on the navigating bridge if the main steering gear is power-operated and that this angular position is given in the steering gear compartment (SOLAS 74/00 reg.II-1/29 and reg.V/19);
- (PR) 5.2.2.41 confirming that with a hydraulic power-operated steering gear the audible and visual low-level alarms on the navigating bridge and in the machinery space for each hydraulic fluid reservoir are operating satisfactorily and that at least one power-actuating system including the reservoir can be recharged from a position within the steering gear compartment by means of a fixed storage tank to which a contents gauge is fitted with fixed piping (SOLAS 74/88 reg.II-1/29);
- (PR) 5.2.2.42 confirming that the steering gear compartment is readily accessible and is provided with suitable arrangements to ensure working access to steering gear machinery and controls under safe conditions (SOLAS 74/88 reg.II-1/29);
- (PR) 5.2.2.43 confirming that, with electric and electro-hydraulic steering gear, the means for indicating on the navigating bridge and at a main machinery control position that the motors are running and, as far as practicable, that the overload alarm and alarm for the loss of a phase in a three



- phase supply located at the main machinery control position are operating satisfactorily (SOLAS 74/88 reg.II-1/30);
- (PR) 5.2.2.44 confirming that the effective means of operation and control of the main and auxiliary machinery essential for the propulsion and the safety of the ship are being maintained, including, when appropriate, any means for remotely controlling the propulsion machinery from the navigating bridge (including the control, monitoring, reporting, alert and safety actions) (SOLAS 74/88/00/02 reg.II-1/31);
- (PR) 5.2.2.45 confirming that arrangements to operate main and other machinery from a machinery control room are satisfactory (SOLAS 74/88 reg.II-1/31);
- (PR) 5.2.2.46 confirming that the means provided for manually overriding automatic controls are being maintained and that a failure does not prevent the use of the manual override (SOLAS 74/88 reg.II-1/31);
- (PR) 5.2.2.47 confirming that the appropriate safety features fitted to the oil-fired and exhaust gas boilers, unfired steam generators, steam pipe systems and air pressure systems are being maintained (SOLAS 74/88 regs.II-1/32, 33 and 34);
- (PR) 5.2.2.48 confirming the operation of the ventilation for the machinery spaces (SOLAS 74/88 reg.II-1/35);
- (PR) 5.2.2.49 confirming that the measures to prevent noise in machinery spaces are effective (SOLAS 74/88 reg.II-1/36);
- (PR) 5.2.2.50 confirming that the engine room telegraph giving visual indication of the orders and answers both in the machinery space and on the navigation bridge is operating satisfactorily (SOLAS 74/88 reg.II-1/37);
- (PR) 5.2.2.51 confirming that the second means of communication between the navigation bridge and machinery space is also operating satisfactorily, including any appropriate means provided to any other positions from which the engines are controlled (SOLAS 74/88 reg.II-1/37);
- (PR) 5.2.2.52 confirming that the engineer's alarm is clearly audible in the engineers' accommodation (SOLAS 74/88 reg.II-1/38);
- (PR) 5.2.2.53 confirming that precautions taken to prevent any oil that may escape under pressure from any pump, filter or heater from coming into contact with heated surfaces, are efficient;
- (PR) 5.2.2.54 confirming that the means of ascertaining the amount of oil contained in any oil tank are in satisfactory working condition (SOLAS 88 reg.II-2/15) (SOLAS 02 reg.II-2/33);
- (PR) 5.2.2.55 confirming that the devices provided to prevent overpressure in any oil tank or in any part of the oil system, including the filling pipes, are in satisfactory working condition (SOLAS 88 reg.II-2/15) (SOLAS 02 reg.II-2/33);

- (PR) 5.2.2.56 confirming that the electrical installations, including the main source of power and lighting systems, are being maintained (SOLAS 74/88 regs.II-1/40 and 41);
- (PR) 5.2.2.57 confirming that the self-contained emergency source of electrical power and its associated systems are operating satisfactorily (SOLAS 74/88 reg.II-1/42);
- (PR) 5.2.2.58 confirming that the starting arrangements of each emergency generating set are satisfactory (SOLAS 74/88 reg.II-1/44);
- (PR) 5.2.2.59 checking, when appropriate, the disposition of and testing the supplementary emergency lighting (SOLAS 74/88 reg.II-1/42-1);
- (PR) 5.2.2.60 for passenger ships constructed on or after 1 July 2010, checking the provision of supplementary lighting in all cabins, and checking that such lighting automatically illuminates and remains on for a minimum of 30 min when power to the normal cabin lighting is lost (SOLAS 06 reg.II-1/41.6);
- (PR) 5.2.2.61 confirming that precautions provided against shock, fire and other hazards of electrical origin are being maintained (SOLAS 74/88 reg.II-1/45);
- (PR) 5.2.2.62 confirming, when appropriate, that the arrangements for the machinery spaces being periodically unattended are satisfactory (SOLAS 74/88 reg.II-1/54);
- (PR) 5.2.2.63 examining the fire pumps and fire main and the disposition of the hydrants, hoses and nozzles and the international shore connection and checking that each fire pump, including the emergency fire pump, can be operated separately so that two jets of water are produced simultaneously from different hydrants at any part of the ship whilst the required pressure is maintained in the fire main (SOLAS 74/00 reg.II-2/10.2; FSSC chs.2 and 12) (SOLAS 74/88 regs.II-2/4 and 19);
- (PR) 5.2.2.64 examining the provision and randomly examining the condition of the portable and non-portable fire extinguishers (SOLAS 74/00 reg.II-2/10.3; FSSC ch.4) (SOLAS 74/88 reg.II-2/6);
- (PR) 5.2.2.65 examining the fixed fire extinguishing system for the machinery spaces and confirming that its means of operation are clearly marked (SOLAS 74/00 regs.II-2/10.4 and 10.5; FSSC chs.2 and 12) (SOLAS 74/88 regs.II-2/5, 7, 9, 10 and 53);
- (PR) 5.2.2.66 examining the special arrangements in the machinery spaces and confirming, as far as practicable and as appropriate, the operation of the remote means of control provided for the opening and closing of the skylights, the release of smoke, the closure of the funnel and ventilation openings, the closure of power-operated and other doors, the stopping of ventilation and boiler forced and induced draft fans and the stopping of oil fuel and other pumps that discharge flammable liquids (SOLAS 74/00 regs.II-2/5.2, 8.3 and 9.5) (SOLAS 74/88 reg.II-2/11);

- (PR) 5.2.2.67 checking that fixed carbon dioxide fire-extinguishing systems for the protection of machinery spaces, where applicable, are provided with two separate controls, one for opening of the gas piping and one for discharging the gas from the storage container, each of them located in a release box clearly identified for the particular space (SOLAS 08 reg.II-2/10.4, FSSC Ch.5.2.2.2);
- (PR) 5.2.2.68 examining the fire-extinguishing arrangements in control stations, accommodation and service spaces (SOLAS 74/00 reg.II-2/10.6.1; FSSC ch.8) (SOLAS 74/88 reg.II-2/36);
- (PR) 5.2.2.69 examining, when applicable, the fire-extinguishing arrangements in cabin balconies (SOLAS 74/00 reg.II-2/10.6.1);
- (PR) 5.2.2.70 examining the provision of fire-extinguishing systems for the spaces containing paint and/or flammable liquids and deep-fat cooking equipment in accommodation and service spaces (SOLAS 74/00 regs.II-2/10.6.3 and 10.6.4; FSSC chs.5, 6 and 7) (SOLAS 74/88 reg.II-2/15.2.5));
- (PR) 5.2.2.71 examining the arrangements for oil fuel, lubricating oil and other flammable oils and confirming, as far as practicable and as appropriate, the operation of the remote means of closing the valves on the tanks that contain oil fuel, lubricating oil and other flammable oils (SOLAS 74/00 reg.II-2/4.2) (SOLAS 74/88 reg.II-2/15);
- (PR) 5.2.2.72 examining and testing, as far as practicable, any fire detection and fire alarm arrangements in machinery spaces, if applicable, accommodation and service spaces and control spaces (SOLAS 74/00 reg.II-2/7 (except 7.5.5, 7.6 and 7.9); FSSC ch.9) (SOLAS 74/88 regs.II-2/11, 12, 13, 13-1, 14, 36 and 41);
- (PR) 5.2.2.72.1 examining and testing, where applicable, any fire detection and fire alarm arrangements on cabin balconies. (SOLAS 74/00 reg.II-2/7.10);
- (PR) 5.2.2.72.2 for passenger ships constructed on or after 1 July 2010, confirming the smoke detectors in cabins, when activated, are emitting, or cause to emit, an audible alarm within the space where they are located (SOLAS 06 reg.II-2/7.5.2 and 7.5.3.1);
- (PR) 5.2.2.72.3 for passenger ships constructed on or after 1 July 2010, confirming detectors and manually operated call points of a fixed fire detection and fire alarm system can be remotely and individually identified (SOLAS 06 reg.II-2/7.2.4);
- (PR) 5.2.2.73 confirming that the firefighters' outfits and the emergency escape breathing devices (EEBDs) are complete and in good condition and that the cylinders, including the spare cylinders, of the self-contained breathing apparatus, are suitably charged (SOLAS 74/00 regs.II-2/10.10, 13.3.4 and 13.4.3; FSSC ch.3) (SOLAS 74/88 reg.II-2/17);
- (PR) 5.2.2.74 checking the operational readiness and maintenance of firefighting systems (SOLAS 74/00 reg.II-2/14) (SOLAS 74/88/91 reg.II- 2/21);

- (PR) 5.2.2.75 confirming, as far as practicable, that no changes have been made in the structural fire protection, including the structure, fire integrity, protection of stairways and lifts, cabin balconies, openings in 'A' and 'B' Class divisions, ventilation systems and windows and side scuttles, and the use of combustible material (SOLAS 74/00/04 regs.II-2/5.2, 5.3, 6, 8.2, 8.5, 9.2.1, 9.2.2, 9.3, 9.4.1, 9.5, 9.6 (except 9.6.5), 9.7 and 11 (except 11.6)) (SOLAS 74/88 regs.II-2/11, 16, 18, 23 to 35 and 37);
- (PR) 5.2.2.76 confirming, as far as practicable, that no changes have been made in the structural fire protection in cargo spaces intended for the carriage of dangerous goods (SOLAS 74/00 regs.II-2/19.3.8 and 19.3.10) (SOLAS 74/88 regs.II-2/4, 54.2.8, 54.2.10 and 54.2.11);
- (PR) 5.2.2.77 examining and testing any manual and automatic fire doors including the means of closing the openings in 'A' and 'B' Class divisions (SOLAS 74/00 reg.II-2/9.4.1) (SOLAS 74/88 regs.II-2/30 and 31);
- (PR) 5.2.2.78 examining and testing the main inlets and outlets of all ventilation systems and checking that the power ventilation is capable of being stopped from outside the space served (SOLAS 74/00 reg.II-2/5.2.1) (SOLAS 74/88 regs.II-2/16 and 32);
- (PR) 5.2.2.79 confirming that the stairways and ladders, including the low-location lighting system, arranged to provide a means of escape to the lifeboat and liferaft embarkation deck from all passenger and crew spaces and from those spaces in which the crew is normally employed are being maintained (SOLAS 74/00 regs.II-2/13.2, 13.3.1, 13.3.2 and 13.7; FSSC chs.11 and 13 (except paragraph 3)) (SOLAS 74/88 reg.II-2/28);
- (PR) 5.2.2.80 confirming that the means of escape from any special category spaces and ro-ro spaces are satisfactory (SOLAS 74/00 regs.II-2/13.5 and 13.6) (SOLAS 74/88 reg.II-2/28);
- (PR) 5.2.2.81 confirming that the means of escape from the machinery spaces are satisfactory (SOLAS 74/00 reg.II-2/13.4.1) (SOLAS 74/88 reg.II-2/28);
- (PR) 5.2.2.82 examining the fire-extinguishing arrangements including fire detection in cargo spaces for general cargo and dangerous goods and testing, as far as practicable and as appropriate, the operation of the means for closing the various openings (SOLAS 74/00 regs.II-2/7.6 and 10.7; FSSC ch.5) (SOLAS 74/88 reg.II-2/39);
- (PR) 5.2.2.83 examining the fire-extinguishing arrangements including fire detection in vehicle, special category and ro-ro spaces and testing, as far as practicable and as appropriate, the operation of the means for closing the various openings (SOLAS 74/00 reg.II-2/20 (except 20.5); FSSC chs.5, 6, 7, 9 and 10) (SOLAS 74/88 regs.II-2/37, 38 and 38-1);
- (PR) 5.2.2.84 examining and testing, as appropriate and as far as practicable, the crew alarm and the public address system or other effective means of communication (SOLAS 74/00 regs.II-2/7.9 and 12; LSAC ch.7) (SOLAS 74/88 reg.II-2/40);

- (PR) 5.2.2.85 examining, when appropriate, the special arrangements for carrying dangerous goods, including checking the electrical equipment and wiring, fire detection, ventilation, the provision of personnel protection clothing and portable appliances and testing, as far as practicable, the water supply, bilge pumping and any water spray system (SOLAS 74/00/08 reg.II-2/19 (except 19.3.8, 19.3.10 and 19.4); FSSC chs.3, 4, 7, 9 and 10) (SOLAS 74/88 regs.II-2/41 and 54);
- (PR) 5.2.2.86 examining, when appropriate, the helicopter facilities (SOLAS 74/00 regs.II-2/18, III/28) (SOLAS 74/88 reg.II-2/18.8);
- (PR) 5.2.2.87 checking the requirement for passenger ships carrying more than 36 passengers and constructed before 1 October 1994 (SOLAS 74/88/92 regs.II-2/41-1 and 41-2);
- (PR) 5.2.2.88 for passenger ships constructed on or after 1 July 2010 and having a length of 120 m or more or having three or more main vertical zones, checking the designation of safe areas (SOLAS 06 reg.II-2/21);
- (PR) 5.2.2.89 for passenger ships constructed on or after 1 July 2010, checking the provision of a safety centre (SOLAS 06 reg.II-2/23) and its associated ventilation system (SOLAS 06 reg.II-2/8.2);
- (PR) 5.2.2.90 checking that emergency instructions are available for each person on board, the muster list is posted in conspicuous places and there are signs or posters in the vicinity of survival craft and their launching stations (SOLAS 74/96 regs.III/8, 9 and 37);
- (PR) 5.2.2.91 checking that the falls used in launching have been periodically inspected and have been renewed in the past 5 years (SOLAS 74/96/04 reg.III/20);
- (PR) 5.2.2.92 examining each survival craft, including its equipment and, when fitted, the on-load release and hydrostatic lock, and for inflatable liferafts the hydrostatic release unit and float free arrangements, including the date of servicing or replacement. Checking that the hand-flares are not out of date and that the required number of search and rescue locating devices are fitted in liferafts and those liferafts are clearly marked (SOLAS 74/96/00/02/08 regs.III/20, 21, 23, 24 and 26; LSAC sections 2.3 to 2.5, 3.2 and 4.1 to 4.6);
- (PR) 5.2.2.93 checking that the life-saving appliances are of international or vivid reddish orange, or a comparably highly visible colour on all parts where this will assist detection at sea (LSAC section 1.2.2.6);
- (PR) 5.2.2.94 examining the embarkation arrangements and launching appliances for each survival craft. Each lifeboat should be lowered to the embarkation position or, if the stowage position is the embarkation position, lowered a short distance and, if practicable, one of the survival craft should be lowered to the water. The operation of the launching appliances for davit launched liferafts should be demonstrated. Checking that a thorough examination of launching appliances, including the dynamic testing of the winch brake, and servicing of lifeboat and rescue boat on-load release gear and

- davit-launched liferaft automatic release hooks have been carried out (SOLAS 74/96/04 regs.III/11, 12, 13, 15, 16, 20, 21 and 23; LSAC sections 6.1 and 6.2);
- (PR) 5.2.2.95 checking the rotational deployment of MES (SOLAS 74/88 reg.III/20.8.2; LSAC section 6.2.2.2);
- (PR) 5.2.2.96 examining each rescue boat, including its equipment. For inflatable rescue boats, confirming that they are stowed in a fully inflated condition (SOLAS 74/88/04 regs.III/14, 17, 21, 26.3 and 34);
- (PR) 5.2.2.97 examining the embarkation and recovery arrangements for each rescue boat;
- (PR) 5.2.2.98 checking the arrangements for mustering passengers (SOLAS 74/96 regs.III/11, 24 and 25);
- (PR) 5.2.2.99 confirming that a means of rescue is provided on ro-ro passenger ships (SOLAS 74/00 regs.III/11, 26.4);
- (PR) 5.2.2.100 confirming that a helicopter pick-up area is provided on ro-ro passenger ships (SOLAS 74/00 reg.III/28);
- (PR) 5.2.2.101 confirming that a decision support system is provided for the Master (SOLAS 74/88 reg.III/29);
- (PR) 5.2.2.102 testing that the engine of the rescue boat(s) and of each lifeboat, when so fitted, start satisfactorily and operate both ahead and astern;
- (PR) 5.2.2.103 examining and checking the operation of two-way VHF radiotelephone apparatus and search and rescue locating devices (SOLAS 74/88/08 regs.III/6, IV/7 and 14);
- (PR) 5.2.2.104 examining the line-throwing appliance and checking that its rockets and the ship's distress signals are not out of date (SOLAS 74/96 regs.III/6, 18 and 35; LSAC sections 3.1 and 7.1);
- (PR) 5.2.2.105 examining and checking the operation of onboard communications equipment and verifying that the general alarm system is audible in accommodation, normal crew working spaces and on open decks (SOLAS 74/96 regs.III/6, 18 and 35; LSAC sections 3.1 and 7.1);
- (PR) 5.2.2.106 examining the provision, disposition, stowage and condition of the lifebuoys, including those fitted with self-igniting lights, self-activating smoke signals and buoyant lines, lifejackets<sup>12</sup>, immersion suits, anti-exposure suits and thermal protective aids and that their associated batteries are not out of date (SOLAS 74/88/06 regs.III/7, 21, 22 and 26; LSAC sections 2.1 to 2.5 and 3.1 to 3.3);
- (PR) 5.2.2.106.1 checking the provision of lifejackets in three sizes (Infant, Child, Adult) and checking that they are marked by either weight or height, or by both weight and height (LSAC section 2.2.1.1). For passenger ships

<sup>12</sup> Refer to the Guidance for application of SOLAS regulation II-1/41.6 (MSC.1/Circ.1372).

- on voyages less than 24 h, checking that the number of infant lifejackets equals to at least 2.5% of the number of passengers on board and for passenger ships on voyages 24 h or greater, checking that infant lifejackets are provided for each infant on board (SOLAS 06 reg.III/7.2.1);
- (PR) 5.2.2.106.2 checking that immersion suits designed to be worn in conjunction with a lifejacket are suitably marked (LSAC section 2.3.1);
- (PR) 5.2.2.107 checking the lighting of the muster and embarkation stations and the alleyways, stairways and exits giving access to the muster and embarkation stations, including when supplied from the emergency source of power (SOLAS 74/88 regs.II-1/42 and III/11);
- (PR) 5.2.2.108 checking that the required navigation lights, shapes and sound signalling equipment are in order (International Regulations for Preventing Collisions at Sea in force (COLREG), rules 20 to 24, 27 to 30 and 33);
- (PR) 5.2.2.109 checking the provision and specification of the following navigation equipment as appropriate: daylight signalling lamp, magnetic compass, transmitting heading device, gyro compass, gyro compass repeaters, radar installation(s), electronic plotting aid, automatic tracking aid(s) or automatic radar plotting aid(s), echo-sounding device, speed and distance indicator, rudder angle indicator, propeller rate-of-revolution indicator, variable pitch propeller pitch and operational mode indicator, rate-of-turn indicator, heading or track control system, GNSS receiver, terrestrial radio navigation system and sound reception system, a pelorus or compass bearing device, means for correcting heading and bearings, a BNWAS, as applicable, and ECDIS including back-up arrangements, as applicable. Items that cannot be checked with the ship in port should be verified from records (SOLAS 74/00/09 reg.V/19);
- (PR) 5.2.2.110 checking for the provision, specification operation and annual performance test of the voyage data recorder (SOLAS 74/00/04 reg.V/20);
- (PR) 5.2.2.111 checking that the International Code of Signals and a copy of Volume III of the International Aeronautical and Maritime Search and Rescue (IAMSAR) Manual have been provided. (SOLAS 74/00/02 reg.V/21);
- (PR) 5.2.2.112 checking that a valid conformance test report of the long-range identification and tracking system is available on board, where fitted (SOLAS 04 reg.V/19-1);
- (PR) 5.2.2.113 checking the provision, operation and that the annual test has been carried out, for the automatic identification system, where fitted (SOLAS 74/00/04 reg.V/19);
- (PR) 5.2.2.114 checking the provision and specification of the pilot ladders and hoists/pilot transfer arrangements (SOLAS 74/00 reg.V/17);

- (PR) 5.2.2.115 checking the provisions of (PI) 5.1.2.111 to (PI) 5.1.2.136 and (PI) 5.1.2.139 to (PI) 5.1.2.140;
- (PR) 5.2.2.116 confirming that no new materials containing asbestos were installed on board (SOLAS 74/00/05/09 reg.II-1/3-5);
- (PR) 5.2.2.117 checking that the means of embarkation and disembarkation from ships for use in port and in port related operations, such as gangways and accommodation ladders, are in satisfactory condition, as applicable (SOLAS 08 reg.II-1/3-9).
- (PR) 5.2.3 For the hull, machinery and equipment of passenger ships the completion of the renewal survey should consist of:
- (PR) 5.2.3.1 after a satisfactory survey, issuing the Passenger Ship Safety Certificate and its associated Record of Equipment (Form P).



ANNEX 2

**SURVEY GUIDELINES UNDER THE 1966 LOAD LINE CONVENTION,  
AS MODIFIED BY THE 1988 PROTOCOL RELATING THERETO**

- (L)     **1**                   **GUIDELINES FOR SURVEYS FOR THE INTERNATIONAL LOAD LINE  
CERTIFICATE OR INTERNATIONAL LOAD LINE EXEMPTION CERTIFICATE**
- (LI)    **1.1**               **Initial surveys** – see part "General" section 4.1
- (LI)    1.1.1           For the load line the examination of plans and designs should  
consist of:
- (LI)    1.1.1.1       examining the structural strength at the draft corresponding to the  
assigned freeboard (LLC 66/88 reg.1);
- (LI)    1.1.1.2       examining the intact stability, and, where applicable, the damaged  
stability information and the loading and ballasting information that is to  
be supplied to the master, and, where not dispensed by the  
Administration, inclining experimental data (LLC 66/88/08 regs.1  
and 10; IS Code chs.1, 2 and 3);
- (LI)    1.1.1.3       determining the freeboard, including specifying and the consideration  
of the conditions of assignment for the freeboard (LLC 66/88/03  
regs.11 to 45).
- (LI)    1.1.2           For the load line the survey during construction and after installation  
should consist of:
- (LI)    1.1.2.1       checking that, as far as its strength is concerned, the ship has been  
constructed in accordance with the approved plans (LLC 66/88 reg.1);
- (LI)    1.1.2.2       confirming that the deck line and load line mark are properly positioned  
(LLC 66/88 regs.4 to 9);
- (LI)    1.1.2.3       witnessing the inclining experiment or lightweight survey  
(LLC 66/88/03 reg.10);
- (LI)    1.1.2.4       examining the superstructure end bulkheads and the openings therein  
(LLC 66/88 regs.11 and 12);
- (LI)    1.1.2.5       examining the means of securing the weathertightness of cargo  
hatchways, other hatchways and other openings on the freeboard and  
superstructure decks (LLC 66/88 regs.13 to 18);
- (LI)    1.1.2.6       examining the ventilators and air pipes, including their coamings and  
closing appliances (LLC 66/88 regs.19 and 20);
- (LI)    1.1.2.7       examining the watertight integrity of the closures to any openings in  
the ship's side below the freeboard deck (LLC 66/88 reg.21);

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- (LI) 1.1.2.8 examining the scuppers, inlets and discharges (LLC 66/88/03 reg.22);
  - (LI) 1.1.2.9 examining the garbage chutes (LLC 66/88/03 reg.22-1);
  - (LI) 1.1.2.10 examining the spurling pipes and cable lockers (LLC 66/88/03 reg.22-2);
  - (LI) 1.1.2.11 examining the side scuttles and deadlights (LLC 66/88 reg.23);
  - (LI) 1.1.2.12 examining the bulwarks including the provision of freeing ports, special attention being given to any freeing ports fitted with shutters (LLC 66/88/03 regs.24 and 25);
  - (LI) 1.1.2.13 examining the guardrails, gangways, walkways and other means provided for the protection of the crew and means for safe passage of crew (LLC 66/88/03 regs.25 and 25-1);
  - (LI) 1.1.2.14 special requirements for ships permitted to sail with type "A" or type "B-minus" freeboards (LLC 66/88/03 regs.26 and 27);
  - (LI) 1.1.2.15 checking, when applicable, of the fittings and appliances for timber deck cargoes (LLC 66/88 regs.42 to 45).
  - (LI) 1.1.3 For the load line the check that certificates, etc., have been placed on board should consist of:
    - (LI) 1.1.3.1 checking that the loading and ballasting information has been supplied to the master (LLC 66/88 reg.10).
  - (LI) 1.1.4 For the load line the completion of the initial survey should consist of:
    - (LI) 1.1.4.1 after a satisfactory survey, issuing the International Load Line Certificate or International Load Line Exemption Certificate.
  - (LA) **1.2 Annual surveys** – see part "General", section 4.2
    - (LA) 1.2.1 For the load line the examination of current certificates and other records should consist of:
      - (LA) 1.2.1.1 checking the validity, as appropriate, of the Cargo Ship Safety Equipment Certificate, the Cargo Ship Safety Radio Certificate and the Cargo Ship Safety Construction Certificate or the Cargo Ship Safety Certificate;
      - (LA) 1.2.1.2 checking the validity of the Safety Management Certificate (SMC) and that a copy of the Document of Compliance (DOC) is on board;
      - (LA) 1.2.1.3 checking the validity of the International Ship Security Certificate;
      - (LA) 1.2.1.4 checking the validity of the International Load Line Certificate or International Load Line Exemption Certificate;
      - (LA) 1.2.1.5 checking the validity of the International Oil Pollution Prevention Certificate;

- (LA) 1.2.1.6 checking the certificate of class, if the ship is classed with a classification society;
- (LA) 1.2.1.7 checking, when appropriate, the validity of the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk or the Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk;
- (LA) 1.2.1.8 checking, when appropriate, the validity of the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk;
- (LA) 1.2.1.9 checking, when appropriate, the validity of the International Pollution Prevention Certificate for the Carriage of Noxious Liquid Substances in Bulk;
- (LA) 1.2.1.10 checking, when appropriate, the validity of the International Sewage Pollution Prevention Certificate;
- (LA) 1.2.1.11 checking, when appropriate, the validity of the International Air Pollution Prevention Certificate;
- (LA) 1.2.1.12 checking that the ship's complement complies with the Minimum Safe Manning Document (SOLAS 74/88 reg.V/14);
- (LA) 1.2.1.13 checking that the master, officers and ratings are certificated as required by the STCW Convention;
- (LA) 1.2.1.14 checking whether any new equipment has been fitted and, if so, confirm that it has been approved before installation and that any changes are reflected in the appropriate certificate;
- (LA) 1.2.1.15 checking that the stability and, where applicable, the loading and ballasting information is available (LLC 66/88/08 regs.1 and 10; IS Code chs.1, 2 and 3);
- (LA) 1.2.1.16 confirming the availability of the International Anti-Fouling System Certificate (AFS 2001 Annex 4 reg.2), when applicable.
- (LA) 1.2.2 For the load line the annual survey should consist of:
  - (LA) 1.2.2.1 checking, in general, that there has been no deterioration in the strength of the hull (LLC 66/88 reg.1);
  - (LA) 1.2.2.2 checking of the positions of the deck line and load line which, if necessary, are to be re-marked and re-painted (LLC 66/88 regs.4 to 9);
  - (LA) 1.2.2.3 checking that no alterations have been made to the hull or superstructures that would affect the calculations determining the position of the load lines (LLC 66/88 regs.11 to 45);
  - (LA) 1.2.2.4 examining the superstructure end bulkheads and the openings therein (LLC 66/88 regs.11 and 12);

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- (LA) 1.2.2.5 examining the means of securing the weathertightness of cargo hatchways, other hatchways and other openings on the freeboard and superstructure decks (LLC 66/88 regs.13 to 18);
- (LA) 1.2.2.6 examining the ventilators and air pipes, including their coamings and closing appliances (LLC 66/88 regs.19 and 20);
- (LA) 1.2.2.7 examining the watertight integrity of the closures to any openings in the ship's side below the freeboard deck (LLC 66/88 reg.21);
- (LA) 1.2.2.8 examining the scuppers, inlets and discharges (LLC 66/88 reg.22);
- (LA) 1.2.2.9 examining the garbage chutes (LLC 66/88/03 reg.22-1);
- (LA) 1.2.2.10 examining the means provided to minimize water ingress through the spurling pipes and chain lockers (LLC 66/88/03 reg.22-2);
- (LA) 1.2.2.11 examining the side scuttles and deadlights (LLC 66/88 reg.23);
- (LA) 1.2.2.12 examining the bulwarks including the provision of freeing ports, special attention being given to any freeing ports fitted with shutters (LLC 66/88/03 regs.24 and 25);
- (LA) 1.2.2.13 examining the guardrails, gangways, walkways and other means provided for the protection of the crew and means for safe passage of crew (LLC 66/88/03 regs.25 and 25-1);
- (LA) 1.2.2.14 examining the special requirements for ships permitted to sail with type "A" or type "B-minus" freeboards (LLC 66/88/03 regs.26 and 27);
- (LA) 1.2.2.15 checking, when applicable, the fittings and appliances for timber deck cargoes (LLC 66/88 regs.42 to 45).
- (LA) 1.2.3 For the load line the completion of the annual survey should consist of:
- (LA) 1.2.3.1 after a satisfactory survey, endorsement of the International Load Line Certificate or International Load Line Exemption Certificate;
- (LA) 1.2.3.2 if a survey shows that the condition of a ship or its equipment is unsatisfactory, see part "General", section 4.8.
- (LR) **1.3** **Renewal surveys** – see part "General", section 4.5
- (LR) 1.3.1 For the load line the examination of current certificates and other records should consist of:
- (LR) 1.3.1.1 the provisions of (LA) 1.2.1, except for the validity of the International Load Line Certificate or International Load Line Exemption Certificate.
- (LR) 1.3.2 For the load line the renewal survey should consist of:
- (LR) 1.3.2.1 the provisions of (LA) 1.2.2;
- (LR) 1.3.2.2 examining the hull to ensure that its strength is sufficient for the draft corresponding to the freeboard assigned (LLC 66/88 reg.1).

- (LR) 1.3.3 For the load line the completion of the renewal survey should consist of:
- (LR) 1.3.3.1 after a satisfactory survey, issuing the International Load Line Certificate or International Load Line Exemption Certificate.

## ANNEX 3

### SURVEY GUIDELINES UNDER THE MARPOL CONVENTION

- (O) **1** **GUIDELINES FOR SURVEYS FOR THE INTERNATIONAL OIL POLLUTION PREVENTION CERTIFICATE**
- (OI) **1.1** **Initial surveys** – see part "General", section 4.1
- (OI) **1.1.1** For oil pollution prevention the examination of plans and designs should consist of:
  - (OI) **1.1.1.1** examining the arrangements for the control of the discharge of oil and examining the plans and designs of the oil discharge monitoring and control system and oily-water separating and oil filtering equipment; confirming that pollution prevention equipment is type approved in accordance with the relevant Resolution (MARPOL 90/04 Annex I regs.14 & 15);
  - (OI) **1.1.1.2** examining the arrangements for operation in special areas (MARPOL 90/04 Annex I reg.15);
  - (OI) **1.1.1.3** examining the arrangements for the segregation of oil and water ballast and the prohibition of carriage of oil in the forepeak tanks or in spaces forward of the collision bulkhead (MARPOL 90/04 Annex I reg.16);
  - (OI) **1.1.1.4** examining the oil residue (sludge) tank and standard discharge arrangements (MARPOL 90/04/09 Annex I regs.12 and 13);
  - (OI) **1.1.1.5** examining oil fuel tank protection arrangements (MARPOL 90/04 Annex I reg.12A);
  - (OI) **1.1.1.6** confirming that requirements regarding capacity and protection of oil fuel tanks are complied with (MARPOL 90/04 Annex I reg.12A);
  - (OI) **1.1.1.7** examining the shipboard oil pollution emergency plan or in the case of a chemical/product tanker the shipboard marine pollution emergency plan (MARPOL 90/04 Annex I reg.37).
- (OI) **1.1.2** For oil pollution prevention, concerning the additional requirements for oil tankers the examination of plans and designs should consist of:
  - (OI) **1.1.2.1** examining the ODME Manual and the arrangements for the control of the discharge of oil and for the retention of oil on board. Verifying that the ODME is type-approved in accordance with the relevant Resolution (MARPOL 90/04 Annex I regs.29, 31 and 34);
  - (OI) **1.1.2.2** examining the arrangements for operation in special areas (MARPOL 90/04 Annex I reg.34);
  - (OI) **1.1.2.3** examining the arrangements for the segregated ballast tanks, checking their capacity and ascertaining whether the draft and trim conditions will be met (MARPOL 90/04 Annex I reg.18);

- (OI) 1.1.2.4 examining the arrangements for crude oil washing, including shadow diagrams and the Operations and Equipment Manual, checking that an inert gas system is to be fitted (MARPOL 90/04 Annex I regs.33 and 35);
- (OI) 1.1.2.5 examining, as appropriate, the arrangements for the prevention of oil pollution in the event of collision or stranding (MARPOL 90/04 Annex I regs.19 to 22);
- (OI) 1.1.2.6 examining the protective location of the segregated ballast spaces and the arrangements for minimizing pollution due to side and bottom damages (MARPOL 90/04 Annex I regs.18, and 24 to 26);
- (OI) 1.1.2.7 confirming, as appropriate, that arrangements are made for the maintenance and inspection of wing and double bottom tanks or spaces (MARPOL 90/04 Annex I reg.19);
- (OI) 1.1.2.8 examining the arrangements for cargo pump-room bottom protection (double bottom where required) (MARPOL 90/04 Annex I reg.22);
- (OI) 1.1.2.9 examining the pumping, piping and discharge arrangements (MARPOL 90/04 Annex I reg.30);
- (OI) 1.1.2.10 examining the arrangements of the oil/water interface detector (MARPOL 90/04 Annex I reg.32);
- (OI) 1.1.2.11 examining, for oil tanker of 5,000 tonnes deadweight and above delivered after 1 February 2002, the intact stability (MARPOL 90/04 Annex I reg.27);
- (OI) 1.1.2.12 examining, for oil tanker of 150 gross tonnage and above delivered after 31 December 1979, the subdivision and damage stability (MARPOL 90/04 Annex I reg.28);
- (OI) 1.1.2.13 examining the accidental oil outflow performance, as applicable (MARPOL 90/04 Annex I reg.23).
- (OI) 1.1.3 For the oil pollution prevention the survey during construction and after installation should consist of:
  - (OI) 1.1.3.1 confirming the satisfactory installation and operation of, as appropriate, oil filtering equipment and when appropriate the operation of the automatic means provided to stop the discharge of effluent and the satisfactory operation of the alarm – or other installation (MARPOL 90/04 Annex I regs.14 and 15);
  - (OI) 1.1.3.2 confirming, when applicable, that the oil content meter and its recording device are operable and that there is a sufficient supply of consumables for the recording device on board (MARPOL 90/04 Annex I regs.14 and 15);
  - (OI) 1.1.3.3 testing, where fitted, the automatic stopping device required for discharges in Special Areas (MARPOL 90/04 Annex I reg.15);

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- (OI) 1.1.3.4 confirming the segregation of the oil fuel and water ballast system and the non-carriage of oil in forepeak tanks (MARPOL 90/04 Annex I reg.16);
- (OI) 1.1.3.5 confirming that the oil residue (sludge) tank and its discharge arrangements are satisfactory and, when the size of the sludge tank is approved on the basis of such installations, confirming the satisfactory operation of homogenizers, sludge incinerators or other recognized means for the control of sludge (MARPOL 90/04/09 Annex I reg.12);
- (OI) 1.1.3.6 confirming the provision of the standard discharge connection (MARPOL 90/04 Annex I reg.13);
- (OI) 1.1.3.7 confirming oil fuel tank protection arrangements (MARPOL 90/04 Annex I reg.12A).
- (OI) 1.1.4 For oil pollution prevention, concerning the additional requirements for oil tankers the survey during construction and after installation should consist of:
- (OI) 1.1.4.1 confirming that the arrangements of slop tanks or cargo tanks designated as slop tanks, and associated piping systems, are satisfactory (MARPOL 90/04 Annex I regs.29 and 34);
- (OI) 1.1.4.2 confirming the satisfactory installation and operation of the oil discharge monitoring and control system, including any audible or visual alarms, the automatic and manual means to stop the discharge of effluent, the starting interlock, the accuracy of the flow meter and the applicable resolution's requirements for installation survey<sup>13</sup> (MARPOL 90/04 Annex I regs.31 and 34);
- (OI) 1.1.4.3 confirming that the oil content meter and its recording device are operable and that there is a sufficient supply of consumables for the recording device on board (MARPOL 90/04 Annex I regs.31 and 34);
- (OI) 1.1.4.4 confirming that the approved oil/water interface detectors are on board and are operational (MARPOL 90/04 Annex I reg.32);
- (OI) 1.1.4.5 confirming that the arrangements of pumps, pipes and valves are in accordance with the requirements for segregated ballast systems and that there are no cross-connections between the cargo and segregated ballast systems (MARPOL 90/04 Annex I reg.18);
- (OI) 1.1.4.6 where a portable spool piece is provided for the emergency discharge of segregated ballast by connecting the segregated ballast system to a cargo pump, confirming that non-return valves are fitted on the segregated ballast connections and that the spool piece is mounted in a conspicuous position in the pump room with a permanent notice restricting its use (MARPOL 90/04 Annex I reg.18);
- (OI) 1.1.4.7 testing ballast pipelines that pass through cargo tanks and those cargo pipelines that pass through ballast tanks to ensure there is no cross contamination (MARPOL 90/04 Annex I reg.18);

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<sup>13</sup> Resolution A.586(14) or MEPC.108(49), as applicable.



- (OI) 1.1.4.8 confirming that the crude oil washing system is installed in accordance with the approved plans (MARPOL 90/04 Annex I regs.18 & 33) and, in particular:
- (OI) 1.1.4.8.1 examining crude oil washing piping, pumps, valves and deck-mounted washing machines for signs of leakage and to check that all anchoring devices for crude oil washing piping are intact and secure;
- (OI) 1.1.4.8.2 carrying out pressure testing of the crude oil washing system to 1.5 times the working pressure;
- (OI) 1.1.4.8.3 confirming in those cases where drive units are not integral with the tank washing machines, that the number of operational drive units specified in the Manual are on board;
- (OI) 1.1.4.8.4 checking that, when fitted, steam heaters for water washing can be properly isolated during crude oil washing operations, either by double shut-off valves or by clearly identifiable blanks;
- (OI) 1.1.4.8.5 checking that the prescribed means of communication between the deck watch keeper and the cargo control position is operational;
- (OI) 1.1.4.8.6 confirming that an overpressure relief device (or other approved arrangement) is fitted to the pumps supplying the crude oil washing system;
- (OI) 1.1.4.8.7 verifying that flexible hoses for supply of oil to the washing machines on combination carriers are of an approved type, are properly stored and are in good condition;
- (OI) 1.1.4.9 verifying the effectiveness of the crude oil washing system (MARPOL 90/04 Annex I reg.33) and, in particular:
- (OI) 1.1.4.9.1 checking that the crude oil washing machines are operable and observing the proper operation of the washing machines by means of the movement indicators and/or sound patterns or other approved methods;
- (OI) 1.1.4.9.2 checking the effectiveness of the stripping system in appropriate cargo tanks by observing the monitoring equipment and by hand-dipping or other approved means;
- (OI) 1.1.4.9.3 verifying by internal tank inspection after crude oil washing that the installation and operational procedures laid down in the Operations and Equipment Manual are satisfactory;
- (OI) 1.1.4.10 confirming that, where there is a crude oil washing system, an inert gas system has been installed and tested in accordance with the requirements of SOLAS 74/88/2000 (see (EI) 1.1.4.2 in Annex 1);
- (OI) 1.1.4.11 confirming, as appropriate, that the arrangements for the prevention of oil pollution in the event of collision or stranding are in accordance with the approved plans (MARPOL 90/04 Annex I regs.19 to 22);

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- (OI) 1.1.4.12 confirming that the piping systems associated with the discharge of dirty ballast water or oil-contaminated water are satisfactory (MARPOL 90/04 Annex I reg.30);
- (OI) 1.1.4.13 confirming that the observation and discharge control positions for visually observing the discharge of oil-contaminated water, including the testing of the communication system between the two positions are satisfactory (MARPOL 90/04 Annex I reg.30);
- (OI) 1.1.4.14 confirming that the means of draining cargo pumps and cargo lines, including the provision of a stripping device and the connections for pumping to the slop or cargo tanks or ashore, are satisfactory (MARPOL 90/04 Annex I reg.30);
- (OI) 1.1.4.15 confirming that closing devices installed in the cargo transfer system and cargo piping, as appropriate, are satisfactory (MARPOL 90/04 Annex I regs.23 and 26);
- (OI) 1.1.4.16 confirming that the subdivision and stability arrangements, in addition to the provision of (OI) 1.1.4.15, to prevent progressive flooding are satisfactory (MARPOL 90/04 Annex I regs.23 and 26);
- (OI) 1.1.4.17 confirming the arrangements for cargo pump-room bottom protection (double bottom where required) (MARPOL 90/04 Annex I reg.22).
- (OI) 1.1.5 For the oil pollution prevention the check that the documentation has been placed on board cargo ships should consist of:
- (OI) 1.1.5.1 confirming that certificates for type approval for the oil filtering equipment and oil content meters are available (MARPOL 90/04 Annex I reg.14);
- (OI) 1.1.5.2 confirming that the Oil Record Book (Part I) has been provided (MARPOL 90/04 Annex I reg.17);
- (OI) 1.1.5.3 confirming that the shipboard oil pollution emergency plan or, in the case of a chemical/product tanker, a shipboard marine pollution emergency plan has been provided (MARPOL 90/04 Annex I reg.37);
- (OI) 1.1.5.4 confirming, as appropriate, that the Operating and Maintenance manuals for the 15 ppm bilge separator and 15ppm bilge alarm are available.
- (OI) 1.1.6 For the oil pollution prevention the check that the documentation has been placed on board oil tankers should additionally consist of:
- (OI) 1.1.6.1 confirming that, if applicable, a Ship to Ship (STS) operations Plan approved by the Administration has been provided (MARPOL Annex I Reg.41);
- (OI) 1.1.6.2 confirming that, if applicable, a Crude Oil Washing Operations and Equipment Manual has been provided (MARPOL 90/04 Annex I reg.35);

- (OI) 1.1.6.3 confirming that an operations manual for the oil discharge monitoring and control system has been provided together with any other documentation requested by the applicable resolution<sup>14</sup> (MARPOL 90/04 Annex I reg.31);
- (OI) 1.1.6.4 confirming that certificates for type approval for the oil content meters, oil discharge monitoring and control system and oil/water interface detectors are available (MARPOL 90/04 Annex I regs.31 and 32);
- (OI) 1.1.6.5 confirming that the Oil Record Book (Part II) has been provided (MARPOL 90/04 Annex I reg.36);
- (OI) 1.1.6.6 confirming that the information and data concerning the subdivision and damage stability has been provided (MARPOL 90/04 Annex I reg.28);
- (OI) 1.1.6.7 confirming that the shipboard oil pollution emergency plan or in the case of a chemical/product tanker a shipboard marine pollution emergency plan has been provided (MARPOL 90/04 Annex I reg.37);
- (OI) 1.1.6.8 confirming, for oil tankers of 5,000 tonnes deadweight and above delivered on/after 1 February 2002, that the intact stability has been approved (MARPOL 90/04 Annex I reg.27);
- (OI) 1.1.6.9 confirming, for oil tankers of 5,000 tonnes deadweight and above, that arrangements are in place to provide prompt access to shore-based damage stability and residual structural strength computerized calculation programs (MARPOL 90/04 Annex I reg.37.4).
- (OI) 1.1.7 For oil pollution prevention the completion of the initial survey should consist of:
- (OI) 1.1.7.1 after satisfactory survey, issuing the International Oil Pollution Prevention Certificate.
- (OA) **1.2** **Annual surveys** – see part "General", section 4.2
- (OA) 1.2.1 For oil pollution prevention the examination of current certificates and other records should consist of:
- (OA) 1.2.1.1 checking the validity, as appropriate, of the Cargo Ship Safety Equipment Certificate, the Cargo Ship Safety Radio Certificate and the Cargo Ship Safety Construction Certificate or the Cargo Ship Safety Certificate;
- (OA) 1.2.1.2 checking the validity of the International Load Line Certificate or International Load Line Exemption Certificate;
- (OA) 1.2.1.3 checking the validity of the International Oil Pollution Prevention Certificate;
- (OA) 1.2.1.4 checking the certificates of class, if the ship is classed with a classification society;

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<sup>14</sup> Resolution A.586(14) or MEPC.108(49), as applicable.

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|------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (OA) | 1.2.1.5  | checking, when appropriate, the validity of the International Sewage Pollution Prevention Certificate;                                                                                                                 |
| (OA) | 1.2.1.6  | checking, when appropriate, the validity of the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk or the Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk;    |
| (OA) | 1.2.1.7  | checking, when appropriate, the validity of the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk;                                                                                      |
| (OA) | 1.2.1.8  | checking, when appropriate, the validity of the International Pollution Prevention Certificate for the Carriage of Noxious Liquid Substances in Bulk;                                                                  |
| (OA) | 1.2.1.9  | checking, when appropriate, the validity of the International Air Pollution Prevention Certificate;                                                                                                                    |
| (OA) | 1.2.1.10 | checking, when appropriate, the validity of the Safety Management Certificate (SMC) and that a copy of the Document of Compliance (DOC) is on board;                                                                   |
| (OA) | 1.2.1.11 | checking the validity of the International Ship Security Certificate;                                                                                                                                                  |
| (OA) | 1.2.1.12 | checking that the ship's complement complies with the Minimum Safe Manning Document (SOLAS 74/88/2000 reg.V/14.2);                                                                                                     |
| (OA) | 1.2.1.13 | checking that the master, officers and ratings are certificated as required by the STCW Convention;                                                                                                                    |
| (OA) | 1.2.1.14 | checking whether any new equipment has been fitted and, if so, confirming that it has been approved before installation and that any changes are reflected in the appropriate certificate;                             |
| (OA) | 1.2.1.15 | checking the certificates for the type approval of the oil filtering equipment (MARPOL 90/04 Annex I regs.14 and 15);                                                                                                  |
| (OA) | 1.2.1.16 | checking, when appropriate, that the Operating and Maintenance manuals for the 15ppm bilge separator and 15ppm bilge alarm are available on board;                                                                     |
| (OA) | 1.2.1.17 | verifying, if applicable, that the 15ppm bilge alarm has been calibrated by the manufacturer or a person authorized by the manufacturer and that a valid calibration certificate is available on board <sup>15</sup> ; |
| (OA) | 1.2.1.18 | checking whether the appropriate entries have been made in Part I of the Oil Record Book (MARPOL 90/04 Annex I reg.17);                                                                                                |
| (OA) | 1.2.1.19 | confirming the availability of the International Anti-Fouling System Certificate (AFS 2001 Annex 4 reg.2), when applicable;                                                                                            |
| (OA) | 1.2.1.20 | confirming that the oil pollution emergency plan or, in the case of a chemical/product tanker, a shipboard marine pollution emergency plan, is on board (MARPOL 90/04 Annex I reg.37).                                 |

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<sup>15</sup> For installations complying with resolution MEPC.107(49).

- (OA) 1.2.2 For oil pollution prevention the examination of current certificates and other records for oil tankers should additionally consist of:
- (OA) 1.2.2.1 confirming that the approved Dedicated Clean Ballast Tank Operation Manual, and/or the approved Operations and Equipment Manual for the Crude Oil Washing Systems, as appropriate, is/are on board (MARPOL 90/04 Annex I regs.18 and 35);
- (OA) 1.2.2.2 confirming, when appropriate, that a CAS Statement of Compliance together with the CAS Final Report<sup>16</sup> are on board (MARPOL 90/04 Annex I, regs. 20.6, 20.7 and 21.6);
- (OA) 1.2.2.3 confirming that the Operating and Maintenance manual for the oil discharge monitoring and control system, is on board (MARPOL 90/04 Annex I reg.31);
- (OA) 1.2.2.4 confirming that a valid calibration certificate for the oil discharge monitoring equipment is available on board<sup>17</sup>;
- (OA) 1.2.2.5 checking whether the appropriate entries have been made in Part II of the Oil Record Book (MARPOL 90/04 Annex I reg.36);
- (OA) 1.2.2.6 confirming that for oil tankers of 5,000 tonnes deadweight and above delivered on/after 1 February 2002 the loading conditions and intact stability information, in an approved form, is on board (MARPOL 90/04 Annex I reg.27);
- (OA) 1.2.2.7 confirming that subdivision and damage stability information in an approved form, where applicable, is on board (MARPOL 90/04 Annex I reg.28);
- (OA) 1.2.2.8 checking the certificates for the type approval of the oil pollution prevention equipment, such as the oil content meters and oil/water interface detectors, and sighting the records of the various oil discharge monitoring equipment, as applicable (MARPOL 90/04 Annex I reg.31);
- (OA) 1.2.2.9 checking that the ship is allowed continued operation according to the phase-out scheme of MARPOL 90/04 Annex I reg.20);
- (OA) 1.2.2.10 confirming that, if applicable, a Ship to Ship (STS) operations Plan approved by the Administration has been provided (MARPOL Annex I reg.41).
- (OA) 1.2.3 For the oil pollution prevention the annual survey should consist of:
- (OA) 1.2.3.1 examining externally the oil filtering equipment and confirming, as far as practicable, its satisfactory operation including, when appropriate, testing the operation of the automatic means provided to stop the discharge of effluent and the alarm for the oil filtering equipment (MARPOL 90/04 Annex I regs.14 and 15);

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<sup>16</sup> Refer to the Condition Assessment Scheme (resolution MEPC.94(46), as amended).

<sup>17</sup> For installations complying with resolution MEPC.108(49).

- (OA) 1.2.3.2 testing, where fitted, the oil filtering equipment required for discharge in special areas (MARPOL 90/04 Annex I reg.15);
- (OA) 1.2.3.3 confirming the segregation of oil fuel and water ballast systems and that the arrangements prohibit the carriage of oil in forepeak tanks or in spaces forward of the collision bulkhead (MARPOL 90/04 Annex I reg.16);
- (OA) 1.2.3.4 checking that the arrangement of oil residue (sludge) tank and its discharge arrangements are satisfactory and confirming that, where applicable, homogenizers, sludge incinerators or other recognized means for the control of sludge are satisfactory (MARPOL 90/04/09 Annex I reg.12);
- (OA) 1.2.3.5 confirming that a standard discharge connection is provided (MARPOL 90/04 Annex I reg.13).
- (OA) 1.2.4 For oil pollution prevention the annual survey of the additional requirements for oil tankers should consist of:
  - (OA) 1.2.4.1 examining the oil discharge monitoring and control system and its associated equipment (MARPOL 90/04 Annex I reg.31) and, in particular:
    - (OA) 1.2.4.1.1 examining externally the system and equipment and, if applicable, verifying that the instrument is properly sealed;
    - (OA) 1.2.4.1.2 confirming, as far as practicable, the satisfactory operation of the oil discharge monitoring and control system including the oil content meter and, where applicable, the automatic and manual means provided to stop the discharge of effluent and the starting interlock;
    - (OA) 1.2.4.1.3 observing that indicators and recording devices are operable and verifying that sufficient supply of consumables for the recorders are on board;
    - (OA) 1.2.4.1.4 testing, as far as practicable, any audible or visual alarms fitted to the oil discharge monitoring and control system;
  - (OA) 1.2.4.2 examining, as far as practicable, the oil/water interface detectors (MARPOL 90/04 Annex I reg.32);
  - (OA) 1.2.4.3 confirming that no cross-connections have been fitted between the cargo and segregated ballast systems (MARPOL 90/04 Annex I reg.18);
  - (OA) 1.2.4.4 where a portable spool piece is provided for the emergency discharge of segregated ballast by connecting the segregated ballast system to a cargo pump, confirming that non-return valves are fitted on the segregated ballast connections and that the spool piece is mounted in a conspicuous position in the pump room with a permanent notice restricting its use (MARPOL 90/04 Annex I reg.18);

- (OA) 1.2.4.5 confirming by sighting that there has been no contamination with oil in the segregated ballast tanks (MARPOL 90/04 Annex I reg.18);
- (OA) 1.2.4.6 confirming, as far as practicable, that the dedicated clean ballast tank arrangement remains satisfactory (MARPOL 90/04 Annex I reg.18);
- (OA) 1.2.4.7 confirming by sighting that there has been no contamination with oil in the dedicated clean ballast tanks (MARPOL 90/04 Annex I reg.18);
- (OA) 1.2.4.8 confirming, as far as practicable, that the crude oil washing system remains satisfactory (MARPOL 90/04 Annex I reg.33) and, in particular:
  - (OA) 1.2.4.8.1 examining externally the crude oil washing piping, pumps, valves and deck mounted washing machines for signs of leakage and checking that all anchoring devices for crude oil washing piping are intact and secure;
  - (OA) 1.2.4.8.2 confirming, in those cases where drive units are not integral with the tank cleaning machines, that the number of operational drive units as specified in the Manual are on board;
  - (OA) 1.2.4.8.3 checking that, when fitted, steam heaters for water washing can be properly isolated during crude oil washing operations, either by double shut-off valves or clearly identifiable blanks;
  - (OA) 1.2.4.8.4 checking that the prescribed means of communications between the deck watch keeper and the cargo control position is operational;
  - (OA) 1.2.4.8.5 confirming that an overpressure relief device (or other approved arrangement) is fitted to the pumps supplying the crude oil washing systems;
  - (OA) 1.2.4.8.6 confirming that flexible hoses for supply of oil to the washing machines on combination carriers, are of an approved type, are properly stored and are in good condition;
- (OA) 1.2.4.9 verifying, where applicable and as far as practicable, the effectiveness of the crude-oil washing system (MARPOL 90/04 Annex I reg.33) and, in particular:
  - (OA) 1.2.4.9.1 checking tanks containing departure and/or arrival ballast water, as applicable, to confirm the effectiveness of the cleaning and stripping;
  - (OA) 1.2.4.9.2 checking, as far as practicable, that the crude oil washing machines are operable and, when the survey is carried out during crude oil washing operations, observing the proper operation of the washing machines by means of the movement indicators and/or sound patterns or other approved methods;
  - (OA) 1.2.4.9.3 checking, as far as practicable, the effectiveness of the stripping system in appropriate cargo tanks by observing the monitoring equipment and by hand-dipping or other approved means;

- (OA) 1.2.4.10 confirming that on those existing tankers operating with special ballast arrangements, the arrangements are as approved and are satisfactory (MARPOL 90/04 Annex I reg.18);
- (OA) 1.2.4.11 confirming, as appropriate and as practicable, that the arrangements for the prevention of oil pollution in the event of collision or stranding are approved and are satisfactory (MARPOL 90/04 Annex I regs.19 to 22);
- (OA) 1.2.4.12 examining the piping systems associated with the discharge of dirty ballast or oil-contaminated water including the part flow system, if fitted (MARPOL 90/04 Annex I reg.30);
- (OA) 1.2.4.13 testing the communication system between the observation and discharge control positions (MARPOL 90/04 Annex I reg.30);
- (OA) 1.2.4.14 examining the means of draining cargo pumps and cargo lines, including the stripping device and the connections for pumping to the slop or cargo tanks or ashore (MARPOL 90/04 Annex I reg.30);
- (OA) 1.2.4.15 confirming for oil tankers of 5,000 tonnes deadweight and above that arrangements are in place to provide prompt access to shore-based damage stability and residual structural strength computerized calculation programs(MARPOL 90/04 Annex I reg.37.4).
- (OA) 1.2.5 For oil pollution prevention the completion of the annual survey should consist of:
- (OA) 1.2.5.1 after a satisfactory survey, endorsing the International Oil Pollution Prevention Certificate;
- (OA) 1.2.5.2 if a survey shows that the condition of a ship or its equipment is unsatisfactory, see part "General", section 4.8.
- (OIn) **1.3 Intermediate surveys** – see part "General", section 4.3
- (OIn) 1.3.1 For oil pollution prevention the examination of current certificates and other records should consist of:
- (OIn) 1.3.1.1 the provisions of (OA) 1.2.1.
- (OIn) 1.3.2 For oil pollution prevention the examination of current certificates and other records for oil tankers should additionally consist of:
- (OIn) 1.3.2.1 the provisions of (OA) 1.2.2.
- (OIn) 1.3.3 For oil pollution prevention the intermediate survey should consist of:
- (OIn) 1.3.3.1 the provisions of (OA) 1.2.3;
- (OIn) 1.3.3.2 examining the oily-water separating equipment or oil filtering equipment or process unit, where fitted, including associated pumps, piping and fittings for wear and corrosion (MARPOL 90/04 Annex I regs.14 and 15);



- (OIn) 1.3.3.3 examining the oil content meter (15 ppm alarm and bilge monitor) for obvious defects, deterioration or damage and checking the record of calibration of the meter when done in accordance with the manufacturer's operational and instruction manual (MARPOL 90/04 Annex I reg.14).
- (OIn) 1.3.4 For oil pollution prevention the intermediate survey of the additional requirements for oil tankers should consist of:
  - (OIn) 1.3.4.1 the provisions of (OA) 1.2.4;
  - (OIn) 1.3.4.2 examining the oil discharge monitoring and control system and the oil content meter for obvious defects, deterioration or damage, and checking the record of calibration of the meter when done in accordance with the manufacturer's operational and instruction manual (MARPOL 90/04 Annex I reg.31);
  - (OIn) 1.3.4.3 confirming the satisfactory operation of the oil/water interface detectors (MARPOL 90/04 Annex I reg.32);
  - (OIn) 1.3.4.4 for the crude oil washing system (MARPOL 90/04 Annex I reg.33):
    - (OIn) 1.3.4.4.1 examining the crude oil washing piping outside the cargo tanks. If upon examination there is any doubt as to its condition, the piping may be required to be pressure tested, gauged or both. Particular attention should be paid to any repairs such as welded doublers;
    - (OIn) 1.3.4.4.2 confirming the satisfactory operation of the isolation valves to steam heaters for washing water, when fitted;
    - (OIn) 1.3.4.4.3 examining at least two selected cargo tanks for the express purpose of verifying the continued effectiveness of the installed crude oil washing and stripping systems. If the tank cannot be gas-freed for the safe entry of the surveyor, an internal examination should not be conducted. In this case this examination may be conducted in conjunction with the internal examination of cargo tanks required in (CIn) 2.3.3.3 in Annex 1;
  - (OIn) 1.3.4.5 examining the manual and/or remote operation of the individual tank valves (or other similar closing devices) to be kept closed at sea (MARPOL 90/04 Annex I regs.23 and 26).
- (OIn) 1.3.5 For the oil pollution prevention the completion of the intermediate survey should consist of:
  - (OIn) 1.3.5.1 after a satisfactory survey, endorsing the International Oil Pollution Prevention Certificate;
  - (OIn) 1.3.5.2 if a survey shows that the condition of a ship or its equipment is unsatisfactory; see part "General", section 4.8.
- (OR) 1.4 **Renewal surveys** – see part "General" section 4.5

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- (OR) 1.4.1 For oil pollution prevention the examination of current certificates and other records should consist of:
- (OR) 1.4.1.1 the provisions of (OA) 1.2.1, except for the validity of the International Oil Pollution Prevention Certificate;
- (OR) 1.4.1.2 verifying that, if applicable, the 15ppm bilge alarm has been calibrated by the manufacturer or a person authorized by the manufacturer and that a valid calibration certificate is available on board<sup>18</sup>.
- (OR) 1.4.2 For oil pollution prevention the examination of current certificates and other records for tankers should additionally consist of:
- (OR) 1.4.2.1 the provisions of (OA) 1.2.2;
- (OR) 1.4.2.2 verifying that, if applicable, the oil discharge monitoring equipment has been calibrated and that a valid calibration certificate is available on board<sup>19</sup>.
- (OR) 1.4.3 For oil pollution prevention the renewal survey should consist of:
- (OR) 1.4.3.1 the provisions of (OIn) 1.3.3;
- (OR) 1.4.3.2 confirming, if necessary by simulated test or equivalent, the satisfactory operation of the oily-water separating equipment or oil filtering equipment (MARPOL 90/04 Annex I reg.14);
- (OR) 1.4.3.3 confirming, if necessary by simulated test or equivalent, the satisfactory operation of the oil discharge monitoring and control system, including where practicable the automatic and manual operation of the means provided to stop the discharge of effluent (MARPOL 90/04 Annex I reg.31);
- (OR) 1.4.3.4 confirming the satisfactory operation of the alarm for the oil filtering system (MARPOL 90/04 Annex I reg.14);
- (OR) 1.4.3.5 confirming the satisfactory operation of homogenizers, sludge incinerators or other recognized means for the control of sludge when the size of oil residue (sludge) tank is approved on the basis of such installations (MARPOL 90/04 Annex I reg.12).
- (OR) 1.4.4 For oil pollution prevention the renewal survey of the additional requirements for oil tankers should consist of:
- (OR) 1.4.4.1 the provisions of (OIn) 1.3.4;
- (OR) 1.4.4.2 confirming that the arrangements of slop tanks or cargo tanks designated as slop tanks and associated piping systems are satisfactory (MARPOL 90/04 Annex I regs.29 and 34);
- (OR) 1.4.4.3 confirming, if necessary by simulated test or equivalent, the satisfactory operation of the oil discharge monitoring and control system and its

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<sup>18</sup> For installations complying with resolution MEPC.107(49).

<sup>19</sup> For installations complying with resolution MEPC.108(49).

- (OR) 1.4.4.4 associated equipment, including the oil/water interface detectors (MARPOL 90/04 Annex I regs.31 and 32);  
confirming that the arrangements of pumps, pipes and valves are in accordance with the requirements for SBT systems (MARPOL 90/04 Annex I reg.18);
- (OR) 1.4.4.5 confirming that the arrangements of pumps, pipes and valves are in accordance with the Revised Specifications for Oil Tankers with Dedicated Clean Ballast Tanks (MARPOL 90/04 Annex I reg.18);
- (OR) 1.4.4.6 confirming that the crude oil washing system is in accordance with the requirements for such systems (MARPOL 90/04 Annex I reg.33) and, in particular:
- (OR) 1.4.4.6.1 carrying out pressure testing of the crude oil washing system to at least the working pressure;
- (OR) 1.4.4.6.2 examining the cargo tanks for the express purpose of verifying the continued effectiveness of the installed crude oil washing and stripping systems;
- (OR) 1.4.4.6.3 examining internally, when fitted, the isolation valves for any steam heaters;
- (OR) 1.4.4.7 verifying, by internal tank inspection or by another alternative method acceptable to the Administration, the effectiveness of the crude oil washing system. If the tank cannot be gas-freed for the safe entry of the surveyor, an internal inspection should not be conducted. An acceptable alternative would be satisfactory results during the surveys required by (OA) 1.2.4.9 (MARPOL 90/04 Annex I reg.33);
- (OR) 1.4.4.8 confirming that there is no leakage from those ballast pipelines passing through cargo tanks and those cargo pipelines passing through ballast tanks (MARPOL 90/04 Annex I regs.18 and 33);
- (OR) 1.4.4.9 confirming that the pumping, piping and discharge arrangements are satisfactory (MARPOL 90/04 Annex I reg.30) and, in particular:
- (OR) 1.4.4.9.1 confirming that the piping systems associated with the discharge of dirty ballast water or oil contaminated water are satisfactory;
- (OR) 1.4.4.9.2 confirming that the means of draining cargo pumps and cargo lines, including the stripping device and the connections for pumping to the slop or cargo tanks or ashore are satisfactory;
- (OR) 1.4.4.9.3 confirming that the arrangements for the part flow system, where fitted, are satisfactory;
- (OR) 1.4.4.10 confirming that closing devices installed in the cargo transfer system and cargo piping as appropriate are satisfactory (MARPOL 90/04 Annex I regs.23 and 26);
- (OR) 1.4.4.11 confirming, as appropriate and as practicable, that the arrangements for the prevention of oil pollution in the event of collision or stranding are satisfactory (MARPOL 73/78/90 Annex I regs.19 to 22);

- (OR) 1.4.4.12 confirming for oil tankers of 5,000 tonnes deadweight and above that arrangements are in place to provide prompt access to shore based damage stability and residual structural strength computerized calculation programs (MARPOL 90/04 Annex I reg.37.4).
- (OR) 1.4.5 For oil pollution prevention the completion of the renewal survey should consist of:
- (OR) 1.4.5.1 after a satisfactory survey, issuing the International Oil Pollution Prevention Certificate.
- (N) 2 **GUIDELINES FOR SURVEYS FOR THE INTERNATIONAL POLLUTION PREVENTION CERTIFICATE FOR THE CARRIAGE OF NOXIOUS LIQUID SUBSTANCES IN BULK**
- (NI) 2.1 **Initial surveys** – see part General section 4.1
- (NI) 2.1.1 For the carriage of noxious liquid substances in bulk the examination of plans and designs (as applicable to the cargoes the ship is to be certified to carry) should consist of:
- (NI) 2.1.1.1 drawing up the list of noxious liquid substances it is proposed the ship will be certified to carry (MARPOL 90/04 Annex II reg.6);
- (NI) 2.1.1.2 examining the pumping system (MARPOL 90/04 Annex II reg.12);
- (NI) 2.1.1.3 examining the stripping system (MARPOL 90/04 Annex II reg.12);
- (NI) 2.1.1.4 examining the tank washing system and equipment (MARPOL 90/04 Annex II reg.14 and App.4);
- (NI) 2.1.1.5 examining the underwater discharge arrangements (MARPOL 90/04 Annex II reg.12);
- (NI) 2.1.1.6 examining the ventilation equipment for residue removal (MARPOL 90/04 Annex II reg.13 and App.7);
- (NI) 2.1.1.7 examining the heating system for solidifying and high viscosity substances (MARPOL 90/04 Annex II reg.14 and App.4);
- (NI) 2.1.1.8 examining the Procedures and Arrangements Manual (including cargo carriage requirements to meet Annex II regulations) (MARPOL 90/04 Annex II reg.14 and App.4);
- (NI) 2.1.1.9 examining the shipboard marine pollution emergency plan (MARPOL 90/04 Annex II reg.17);
- (NI) 2.1.1.10 examining if applicable the construction and arrangements of a ship certified to carry individually identified vegetable oils under exemption from the carriage requirements (MARPOL 90/04 Annex II reg.4.3).
- (NI) 2.1.2 For the carriage of noxious liquid substances in bulk, the survey during construction and after installation (as applicable to the cargoes the ship is to be certified to carry) should consist of:

- (NI) 2.1.2.1 confirming that the pumping and stripping systems are satisfactory and that portable pipes or bends in sufficient number, if required, are on board (MARPOL 90/04 Annex II reg.12);
- (NI) 2.1.2.2 conducting the water test for assessing the stripping quantity, as required (MARPOL 73/78/90/04 Annex II reg.12 and App.5);
- (NI) 2.1.2.3 confirming that the tank washing machines provided on board are in working order, are those described in the Procedures and Arrangements Manual and are installed in accordance with the approved plans (MARPOL 90/04 Annex II reg.14 and App.4);
- (NI) 2.1.2.4 confirming that the wash water heating system, if required, is installed in accordance with the approved plans (MARPOL 90/04 Annex II reg.14 and App.4);
- (NI) 2.1.2.5 confirming that the number and position of tank cleaning openings for portable machines are in accordance with the approved plans (MARPOL 90/04 Annex II reg.14 and App.4);
- (NI) 2.1.2.6 confirming that the underwater discharge outlet(s) are in accordance with the approved plans (MARPOL 90/04 Annex II reg.12);
- (NI) 2.1.2.7 verifying by actual test that the discharge rate of the pumps, where a variable rate type is used, can be controlled as specified in the Procedures and Arrangements Manual (MARPOL 90/04 Annex II reg.14 and App.4);
- (NI) 2.1.2.8 confirming that the ventilation equipment for residue removal is installed in accordance with the approved plan and is in working order and that the pressure in the driving medium for portable fans for ventilation equipment for residue removal can be achieved to give the required fan capacity (MARPOL 90/04 Annex II reg.13 and App.7);
- (NI) 2.1.2.9 confirming that the heating system for solidifying and high viscosity substances is installed in accordance with the approved plan (MARPOL 90/04 Annex II reg.14 and App.4);
- (NI) 2.1.2.10 confirming if applicable the construction and arrangements of a ship certified to carry individually identified vegetable oils under exemption from the carriage requirements (MARPOL 90/04 Annex II reg.4.3).
- (NI) 2.1.3 For the carriage of noxious liquid substances in bulk the check that the required documentation has been placed on board cargo ships (as applicable to the cargoes the ship is to be certified to carry) should consist of:
  - (NI) 2.1.3.1 confirming that Procedures and Arrangements Manual has been provided (MARPOL 90/04 Annex II reg.14);
  - (NI) 2.1.3.2 confirming that the Cargo Record Book has been provided (MARPOL 90/04 Annex II reg.15);

- (NI) 2.1.3.3 confirming that the shipboard marine pollution emergency plan is provided (MARPOL 90/04 Annex II reg.17).
- (NI) 2.1.4 For the carriage of noxious liquid substances in bulk the completion of initial survey should consist of:
- (NI) 2.1.4.1 after satisfactory survey, issuing the International Certificate for the Carriage of Noxious Liquid Substances in Bulk.
- (NA) **2.2 Annual surveys** – see part "General", section 4.2
- (NA) 2.2.1 For the carriage of noxious liquid substances in bulk the examination of current certificates and other records should consist of:
- (NA) 2.2.1.1 checking the validity, as appropriate, of the Cargo Ship Safety Equipment Certificate, the Cargo Ship Safety Radio Certificate and the Cargo Ship Safety Construction Certificate or the Cargo Ship Safety Certificate;
- (NA) 2.2.1.2 checking the validity of the International Load Line Certificate or International Load Line Exemption Certificate;
- (NA) 2.2.1.3 checking the validity of the International Oil Pollution Prevention Certificate;
- (NA) 2.2.1.4 checking the certificates of class, if the ship is classed with a classification society;
- (NA) 2.2.1.5 checking, when appropriate, the validity of the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk;
- (NA) 2.2.1.6 checking the validity of the International Pollution Prevention Certificate for the Carriage of Noxious Liquid Substances in Bulk;
- (NA) 2.2.1.7 checking, when appropriate, the validity of the International Air Pollution Prevention Certificate;
- (NA) 2.2.1.8 checking, when appropriate, the validity of the International Sewage Pollution Prevention Certificate;
- (NA) 2.2.1.9 checking, when appropriate, the validity of the Safety Management Certificate (SMC) and that a copy of the Document of Compliance (DOC) is on board;
- (NA) 2.2.1.10 checking the validity of the International Ship Security Certificate;
- (NA) 2.2.1.11 checking that the ship's complement complies with the Minimum Safe Manning Document (SOLAS 74/88/2000 reg.V/14.2);
- (NA) 2.2.1.12 checking that the master, officers and ratings are certificated as required by the STCW Convention;

- (NA) 2.2.1.13 checking whether any new equipment has been fitted and, if so, confirming that it has been approved before installation and that any changes are reflected in the appropriate certificate;
- (NA) 2.2.1.14 confirming that the Procedures and Arrangements Manual is on board (MARPOL 90/04 Annex II reg.14);
- (NA) 2.2.1.15 confirming that the Cargo Record Book is being correctly used (MARPOL 90/04 Annex II reg.15);
- (NA) 2.2.1.16 confirming that the shipboard marine pollution emergency plan is on board (MARPOL 90/04 Annex II reg.17);
- (NA) 2.2.1.17 confirming the availability of the International Anti-Fouling System Certificate (AFS 2001 Annex 4 reg.2), when applicable.
- (NA) 2.2.2 For the carriage of noxious liquid substances in bulk the annual survey should consist of:
  - (NA) 2.2.2.1 examining externally and confirming that the pumping and piping systems, including a stripping system if fitted, and associated equipment remain as approved (MARPOL 90/04 Annex II reg.12);
  - (NA) 2.2.2.2 examining externally the tank washing piping and confirming that the type, capacity, number, and arrangement of the tank washing machines are as approved (MARPOL 90/04 Annex II reg.14 and App.4);
  - (NA) 2.2.2.3 examining externally the wash water heating system (MARPOL 90/04 Annex II reg.14 and App.4);
  - (NA) 2.2.2.4 examining externally, as far as practicable, the underwater discharge arrangements (MARPOL 90/04 Annex II reg.12);
  - (NA) 2.2.2.5 confirming that the means of controlling the rate of discharge of the residue is as approved (MARPOL 90/04 Annex II reg.14 and App.4);
  - (NA) 2.2.2.6 confirming that the ventilation equipment for residue removal is as approved (MARPOL 90/04 Annex II reg.13 and App.7);
  - (NA) 2.2.2.7 examining externally, as far as is accessible, the heating system required for solidifying and high viscosity substances (MARPOL 90/04 Annex II reg.14 and App.4);
  - (NA) 2.2.2.8 examining any additional requirements listed on the International Certificate for the Carriage of Noxious Liquid Substances in Bulk.
- (NA) 2.2.3 For the carriage of noxious liquid substances in bulk the completion of annual survey should consist of:
  - (NA) 2.2.3.1 after satisfactory survey, endorsing the International Certificate for the Carriage of Noxious Liquid Substances in Bulk;

- (NA) 2.2.3.2 if a survey shows that the condition of a ship or its equipment is unsatisfactory, see part "General" section 4.8.
- (NIn) **2.3 Intermediate surveys** – see part "General", section 4.3
- (NIn) 2.3.1 For the carriage of noxious liquid substances in bulk the examination of current certificates and other records should consist of:
- (NIn) 2.3.1.1 the provisions of (NA) 2.2.1.
- (NIn) 2.3.2 For the carriage of noxious liquid substances in bulk the intermediate survey should consist of:
- (NIn) 2.3.2.1 the provisions of (NA) 2.2.2;
- (NIn) 2.3.2.2 verifying from the cargo record book that the pumping and stripping arrangements have been emptying the tanks efficiently and are all in working order (MARPOL 90/04 Annex II regs.12 and 15);
- (NIn) 2.3.2.3 confirming, if possible, that the discharge outlet(s) are in good condition (MARPOL 90/04 Annex II P & A Standards);
- (NIn) 2.3.2.4 confirming that the ventilation equipment for residue removal is satisfactory and that the pressure in the driving medium for portable fans for ventilation equipment for residue removal can be achieved to give the required fan capacity (MARPOL 90/04 Annex II App.7).
- (NIn) 2.3.3 For the carriage of noxious liquid substances in bulk the completion of intermediate survey should consist of:
- (NIn) 2.3.3.1 after satisfactory survey, endorsing the International Certificate for the Carriage of Noxious Liquid Substances in Bulk; should be endorsed;
- (NIn) 2.3.3.2 if a survey shows that the condition of a ship or its equipment is unsatisfactory, see part "General", section 4.8.
- (NR) **2.4 Renewal surveys** – see part "General", section 4.4
- (NR) 2.4.1 For the carriage of noxious liquid substances in bulk the examination of current certificates and other records should consist of:
- (NR) 2.4.1.1 the provisions of (NA) 2.2.1, except for the validity of the International Certificate for the Carriage of Noxious Liquid Substances in Bulk.
- (NR) 2.4.2 For the carriage of noxious liquid substances in bulk the renewal survey should consist of:
- (NR) 2.4.2.1 the provisions of (NIn) 2.3.2;
- (NR) 2.4.2.2 confirming that the pumping and stripping systems are satisfactory and that portable pipes or bends in sufficient number, if required, are on board (MARPOL 73/78/90/04 Annex II reg.12);



- (NR) 2.4.2.3 conducting the water test for assessing the stripping quantity, as required (MARPOL 73/78/90/04 Annex II reg.12 and App.5);
- (NR) 2.4.2.4 confirming that the tank washing machines provided on board are in working order, are those described in the Procedures and Arrangements Manual and are installed in accordance with the approved plans (MARPOL 90/04 Annex II reg.14 and App.4);
- (NR) 2.4.2.5 confirming that the wash water heating system, if required, is installed in accordance with the approved plans and is in working order (MARPOL 90/04 Annex II reg.14 and App.4);
- (NR) 2.4.2.6 confirming that the number and position of tank cleaning openings for portable machines are in accordance with the approved plans (MARPOL 90/04 Annex II reg.14 and App.4);
- (NR) 2.4.2.7 confirming that the underwater discharge outlet(s) are in good condition and are in accordance with the approved plans (MARPOL 90/04 Annex II regs.12, 14 and App.4);
- (NR) 2.4.2.8 verifying by actual test that the discharge rate of the pumps, where a variable rate type is used, can be controlled as specified in the Procedures and Arrangements Manual (MARPOL 90/04 Annex II reg.14 and App.4);
- (NR) 2.4.2.9 confirming that the ventilation equipment for residue removal is installed in accordance with the approved plan and is in working order (MARPOL 90/04 Annex II regs.12, 14 and App.4);
- (NR) 2.4.2.10 confirming that the heating system for solidifying and high viscosity substances is installed in accordance with the approved plan and is in working order (MARPOL 90/04 Annex II regs.12, 14 and App.4).
- (NR) 2.4.3 For the carriage of noxious liquid substances in bulk the completion of renewal survey should consist of:
- (NR) 2.4.3.1 after satisfactory survey, issuing the International Certificate for the Carriage of Noxious Liquid Substances in Bulk.
- (S) **3** **GUIDELINES FOR SURVEYS FOR THE INTERNATIONAL SEWAGE POLLUTION PREVENTION CERTIFICATE**
- (SI) **3.1** **Initial surveys** – see part "General", section 4.1
- (SI) 3.1.1 For sewage pollution prevention the examination of plans and designs should consist of:
- (SI) 3.1.1.1 examining as appropriate the arrangements for the provision of a sewage treatment plant, or of a sewage comminuting and disinfecting system, or of a sewage holding tank (MARPOL Annex IV reg.9);
- (SI) 3.1.1.2 if a sewage treatment plant is fitted, checking that it is type approved by the administration in accordance with the appropriate resolution (MARPOL Annex IV reg.9.1);

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|------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (SI) | 3.1.1.3    | if a sewage comminuting and disinfecting system is fitted, checking that it is approved by the administration and that facilities for the temporary storage of sewage are provided (MARPOL Annex IV reg.9.2);                                                |
| (SI) | 3.1.1.4    | if a sewage holding tank is fitted, checking its capacity having regard to the number of persons on board (MARPOL Annex IV reg.9.1.3);                                                                                                                       |
| (SI) | 3.1.1.5    | examining the arrangements for the provision of a standard discharge connection (MARPOL Annex IV reg.10);                                                                                                                                                    |
| (SI) | 3.1.1.6    | examining the arrangements for the provision of a pipeline for the discharge of sewage to a reception facility (MARPOL Annex IV reg.10).                                                                                                                     |
| (SI) | 3.1.2      | For sewage pollution prevention the survey during construction and after installation should consist of:                                                                                                                                                     |
| (SI) | 3.1.2.1    | checking externally, as applicable, the sewage treatment plant or the sewage comminuting and disinfecting system, and confirming their operation (MARPOL Annex IV regs.4.1.1 and 9);                                                                         |
| (SI) | 3.1.2.2    | if a sewage holding tank is fitted, checking that it has been constructed in a satisfactory manner, and checking that the holding tank has a means to indicate visually the amount of its contents (MARPOL Annex IV reg.9.1.3);                              |
| (SI) | 3.1.2.3    | confirming that a standard discharge connection is provided (MARPOL Annex IV reg.10);                                                                                                                                                                        |
| (SI) | 3.1.2.4    | confirming that a pipeline for the discharge of sewage to a reception facility is provided (MARPOL Annex IV reg.10).                                                                                                                                         |
| (SR) | <b>3.2</b> | <b>Renewal surveys</b> – See "General", section 4.5                                                                                                                                                                                                          |
| (SR) | 3.2.1      | For sewage pollution prevention the examination of current certificates and other records should consist of:                                                                                                                                                 |
| (SR) | 3.2.1.1    | checking the validity, as appropriate, of the Cargo Ship Safety Equipment Certificate, the Cargo Ship Safety Radio Certificate and the Cargo Ship Safety Construction Certificate or the Cargo Ship Safety Certificate or Passenger Ship Safety Certificate; |
| (SR) | 3.2.1.2    | checking the validity of the International Load Line Certificate or International Load Line Exemption Certificate;                                                                                                                                           |
| (SR) | 3.2.1.3    | checking the validity of the International Oil Pollution Prevention Certificate;                                                                                                                                                                             |
| (SR) | 3.2.1.4    | checking the validity of the International Air Pollution Prevention Certificate;                                                                                                                                                                             |
| (SR) | 3.2.1.5    | checking the validity of the International Ship Security Certificate;                                                                                                                                                                                        |
| (SR) | 3.2.1.6    | checking the certificates of class, if the ship is classed with a classification society;                                                                                                                                                                    |
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- (SR) 3.2.1.7 checking, when appropriate, the validity of the International Certificate of Fitness for the Carriage of Dangerous Chemicals in bulk or the Certificate of Fitness for the Carriage of Dangerous Chemical in Bulk;
- (SR) 3.2.1.8 checking, when appropriate, the validity of the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk;
- (SR) 3.2.1.9 checking when appropriate the validity of the International Pollution Prevention Certificate for the Carriage of Noxious Liquid Substances in Bulk;
- (SR) 3.2.1.10 checking that the ship's complement complies with the Minimum Safe Manning Document (SOLAS 74/88 reg.V/13(b));
- (SR) 3.2.1.11 checking that the master, officers and ratings are certificated as required by the STCW Convention;
- (SR) 3.2.1.12 checking the validity of the Safety management certificate(SMC) and that a copy of the Document of Compliance (DOC) is on board, where applicable;
- (SR) 3.2.1.13 checking whether any new equipment has been fitted and, if so, confirm that it has been approved before installation and that any changes are reflected in the certificate;
- (SR) 3.2.1.14 confirming the availability of the International Anti-Fouling System Certificate (AFS 2001 Annex 4 reg.2), when applicable.
- (SR) 3.2.2 For sewage pollution prevention the renewal survey should consist of:
- (SR) 3.2.2.1 confirming that no change has been made nor any new equipment installed which would affect the validity of the certificate (MARPOL Annex IV reg.4.8);
- (SR) 3.2.2.2 examining externally the sewage pollution prevention system and confirming, as far as practicable its satisfactory operation;
- (SR) 3.2.2.3 confirming that a procedure for discharge of animal effluent is implemented on board (MARPOL 73/78/07 Annex IV reg.11.1.1).
- (SR) 3.2.3 For sewage pollution prevention the completion of the renewal survey should consist of:
- (SR) 3.2.3.1 after satisfactory survey the International Sewage Prevention Certificate should be issued.
- (A) 4 **GUIDELINES FOR THE SURVEYS FOR THE INTERNATIONAL AIR POLLUTION PREVENTION CERTIFICATE AND THE NO<sub>x</sub> TECHNICAL CODE**
- (AI) 4.1 **Initial surveys** – see part "General", section 4.1
- (AI) 4.1.1 For air pollution prevention the examination of plans and designs should consist of:

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- (AI) 4.1.1.1 examining the arrangements for systems using ozone-depleting substances (MARPOL Annex VI reg.12);
  - (AI) 4.1.1.2 examining the arrangements for NO<sub>x</sub> emission control, if applicable (MARPOL Annex VI reg.13);
  - (AI) 4.1.1.3 examining the arrangements for SO<sub>x</sub> and particulate matter control, if applicable (MARPOL Annex VI reg.14);
  - (AI) 4.1.1.4 examining the arrangements for vapour collection systems, if applicable (MARPOL Annex VI reg.15 and MSC/Circ.585);
  - (AI) 4.1.1.5 examining the arrangements for shipboard incinerators, if applicable (MARPOL Annex VI reg.16).
  - (AI) 4.1.2 For air pollution prevention the survey should consist of:
    - (AI) 4.1.2.1 Ozone-depleting substances (MARPOL Annex VI reg.12):
      - (AI) 4.1.2.1.1 confirming, if applicable, the satisfactory installation and operation of systems using ozone depleting substances;
      - (AI) 4.1.2.1.2 confirming that no installation or equipment containing ozone depleting substances has been installed after 19 May 2005, other than hydro-chlorofluorocarbons (MARPOL Annex VI reg.12.3.1);
      - (AI) 4.1.2.1.3 confirming that no installation or equipment containing hydro-chlorofluorocarbons are fitted after 1 January 2020 (MARPOL Annex VI reg.12.3.2);
    - (AI) 4.1.2.2 Nitrogen oxide emissions from marine diesel engines (MARPOL Annex VI reg.13):
      - (AI) 4.1.2.2.1 confirming that all marine diesel engines which are required to be certified are pre-certified in accordance with section 2.2 of the NO<sub>x</sub> Technical Code to the required Tier and installed in accordance with the approved duty cycle.
        - (AI) 4.1.2.2.1.1 If engine parameter check method is used:
          - (AI) 4.1.2.2.1.1.1 an onboard verification survey in accordance with section 6.2 of the NO<sub>x</sub> Technical Code;
        - (AI) 4.1.2.2.1.2 If the simplified method is used:
          - (AI) 4.1.2.2.1.2.1 an onboard verification survey in accordance with section 6.3 of the NO<sub>x</sub> Technical Code;
        - (AI) 4.1.2.2.1.3 If direct measurement and monitoring method is used (for existing ships only):
          - (AI) 4.1.2.2.1.3.1 an onboard verification survey, in accordance with section 6.4 of the NO<sub>x</sub> Technical Code;
        - (AI) 4.1.2.2.1.4 For marine diesel engines of an output more than 5,000 kW and a per cylinder displacement at or above 90 litres/cylinder installed on ships constructed between 1 January 1990 and 31 December 1999, check whether:
          - .1 an approved method exists;

- .2 an approved method is not commercially available; or
  - .3 that an approved method is installed and where this is the case, that there is an approved method file,
- and apply the verification procedures as given in the approved method file;
- (AI) 4.1.2.3 Sulphur Oxides and Particulate Matter (MARPOL Annex VI reg.14):
- (AI) 4.1.2.3.1 confirming, if appropriate, that:
  - .1 satisfactory arrangements are in place for using compliant fuel as required; or
  - .2 satisfactory installation and operation of the fuel switching arrangements are in place when tanks are provided for different grades of fuel; or
  - .3 satisfactory installation and operation of the exhaust gas cleaning system or other technological methods are examined, (regulation 4 of Annex VI);
- (AI) 4.1.2.4 Volatile Organic Compounds (MARPOL Annex VI reg.15) (if applicable):
- (AI) 4.1.2.4.1 confirming the satisfactory installation of the vapour collection piping;
- (AI) 4.1.2.4.2 confirming the satisfactory installation and operation of the means provided to eliminate the collection of condensation in the system, such as drains in low points of the line end;
- (AI) 4.1.2.4.3 confirming the satisfactory installation and operation of the isolation valves at the vapour manifolds;
- (AI) 4.1.2.4.4 confirming that the ends of each line are properly identified as vapour collection lines;
- (AI) 4.1.2.4.5 confirming that the vapour collection flanges are in accordance with the IMO guidelines and industrial standards;
- (AI) 4.1.2.5 Shipboard Incinerators (MARPOL Annex VI reg.16) (installed on or after 1 January 2000):
- (AI) 4.1.2.5.1 confirming the satisfactory installation and operation of each incinerator;
- (AI) 4.1.2.5.2 confirming that the manufacturer's name, incinerator model number/type and capacity in heat units per hour is permanently marked on the incinerator.
- (AI) 4.1.3 For air pollution prevention the check that certificates and other relevant documentation have been placed on board should consist of:

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- (AI) 4.1.3.1 the provision of (AA) 4.2.2.2 as applicable except (AA) 4.2.2.2.14.
- (AI) 4.1.4 For air pollution prevention the completion of the initial survey should consist of:
- (AI) 4.1.4.1 after satisfactory survey, issuing the International Air Pollution Prevention Certificate.
- (AA) **4.2 Annual surveys** – see "General", section 4.2
- (AA) 4.2.1 For air pollution prevention the examination of current certificates and other records should consist of:
- (AA) 4.2.1.1 checking the validity, as appropriate, of the Cargo Ship Safety Equipment Certificate, the Cargo Ship Safety Radio Certificate and the Cargo Ship Safety Construction Certificate or the Cargo Ship Safety Certificate;
- (AA) 4.2.1.2 checking the validity of the Safety Management Certificate (SMC) and that a copy of the Document of Compliance (DOC) is on board, where applicable;
- (AA) 4.2.1.3 checking the validity of the International Load Line Certificate or International Load Line Exemption Certificate;
- (AA) 4.2.1.4 checking the validity of the International Oil Pollution Prevention Certificate;
- (AA) 4.2.1.5 checking the certificates of class, if the ship is classed with a classification society;
- (AA) 4.2.1.6 checking, when appropriate, the validity of the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk or the Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk;
- (AA) 4.2.1.7 checking that the ship's complement complies with the Minimum Safe Manning Document (SOLAS 74/88, regulation V/13(b));
- (AA) 4.2.1.8 checking that the master, officers and ratings are certificated as required by the STCW Convention;
- (AA) 4.2.1.9 checking whether any new equipment has been fitted and, if so, confirming that it has been approved before installation and that any changes are reflected in the appropriate certificate.
- (AA) 4.2.2 For air pollution prevention the annual survey should consist of the following:
- (AA) 4.2.2.1 General:
- (AA) 4.2.2.1.1 confirm that no changes have been made or any new equipment installed which would affect the validity of the certificate;
- (AA) 4.2.2.2 Documentation:
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- (AA) 4.2.2.2.1 confirm that there is an Ozone Depleting Substances Record Book, if applicable (MARPOL Annex VI reg.12.6);
- (AA) 4.2.2.2.2 confirm that there are Engine International Air Pollution Prevention (EIAPP) Certificates for each marine diesel engine, required to be certified, as described in chapter 2.1 of the NO<sub>x</sub> Technical Code;
- (AA) 4.2.2.2.3 confirm that there is on board an approved Technical File for each marine diesel engine required to be certified;
- (AA) 4.2.2.2.4 confirm that there is a record book of engine parameters for each marine diesel engine required to be certified in the case where the engine parameter check method is used as a means of onboard NO<sub>x</sub> verification (NO<sub>x</sub> Technical Code para. 6.2.3);
- (AA) 4.2.2.2.5 confirm that there is an approved onboard monitoring manual for each marine diesel engine required to be certified in the case where the direct measurement and monitoring method is to be used as a means of onboard NO<sub>x</sub> verification (NO<sub>x</sub> Technical Code para. 6.4.17.1);
- (AA) 4.2.2.2.6 confirm that there are written procedures covering fuel change over, where applicable;
- (AA) 4.2.2.2.7 confirm that there is a record of fuel changeover, where applicable, and that this record should take the form of a log-book as prescribed by the Administration (MARPOL Annex VI reg.14.6)<sup>20</sup>;
- (AA) 4.2.2.2.8 confirm that there is for each Exhaust Gas Cleaning System (EGCS)-SO<sub>x</sub> either a SO<sub>x</sub> Emission Control Area (SECA<sup>21</sup>) Compliance Certificate for the EGCS-SO<sub>x</sub>, or an Onboard Monitoring Manual (OMM) as appropriate, plus in either cases a SECA Compliance Plan (regulation 4 of Annex VI) or approved documentation in respect of other technological means of achieving compliance;
- (AA) 4.2.2.2.9 confirm that there is a VOC Management Plan, if required (MARPOL Annex VI reg.15.6);
- (AA) 4.2.2.2.10 confirm that there is a transfer procedure, if required, for the VOC collection system;
- (AA) 4.2.2.2.11 confirm that there is, if required, an IMO Type Approval Certificate for each incinerator on board (MARPOL Annex VI reg.16.6.1);
- (AA) 4.2.2.2.12 confirm that there is an instruction manual for each incinerator if required (MARPOL Annex VI reg.16.7);
- (AA) 4.2.2.2.13 confirm that records documenting training of the crew in operating each incinerator, if required;

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<sup>20</sup> When not prescribed by the Administration, this information could be contained in the engine-room log-book, the deck log-book, the official log-book, the oil record book or a separate log-book solely for this purpose.

<sup>21</sup> This will need to be updated when the exhaust gas cleaning system guidelines are updated to take into account the revised Annex VI, for consistency with the terminology used in the revised guideline.

- (AA) 4.2.2.2.14 confirm that there are the required bunker delivery notes on board and the required fuel oil samples are kept under the ships control (MARPOL Annex VI reg.18) or other relevant documentation;
- (AA) 4.2.2.3 Systems containing ozone-depleting substances, if fitted:
- (AA) 4.2.2.3.1 confirm that no new installation or equipment containing ozone depleting substances except those covered by (AA) 4.2.2.3.2 have been fitted to the ship after 19 May 2005 (MARPOL Annex VI reg.12.3.1);
- (AA) 4.2.2.3.2 confirm that no installations containing hydro-chlorofluocarbons have been fitted after 1 January 2020 (MARPOL Annex VI reg.12.3.2);
- (AA) 4.2.2.3.3 examine externally any installation or equipment as far as practicable to ensure satisfactory maintenance and that there are no emissions of ozone-depleting substances;
- (AA) 4.2.2.3.4 confirm through documentary evidence that there has been no deliberate emission of ozone-depleting substance;
- (AA) 4.2.2.4 Nitrogen oxide emissions from each diesel marine diesel engine:
- (AA) 4.2.2.4.1 confirm that each marine diesel engine has been operated as required in accordance with its applicable NO<sub>x</sub> emission limit(s);
- (AA) 4.2.2.4.2 confirm that no marine diesel engine been subject to major conversion in the intervening period;
- (AA) 4.2.2.4.3 if engine parameter check method is used:
- (AA) 4.2.2.4.3.1 review engine documentation contained in the Technical File and the record book of engine parameters to check, as far as practicable, engine rating, duty and limitation/restrictions as given in the Technical File;
- (AA) 4.2.2.4.3.2 confirm that the engine has not undergone any modifications or adjustments outside the options and ranges permitted in the Technical File since the last survey;
- (AA) 4.2.2.4.3.3 conduct survey as detailed in the Technical File;
- (AA) 4.2.2.4.4 if the simplified method is used:
- (AA) 4.2.2.4.4.1 review engine documentation contained in the Technical File;
- (AA) 4.2.2.4.4.2 confirm that the test procedure is acceptable to the Administration;
- (AA) 4.2.2.4.4.3 confirm that the analysers, engine performance sensors, ambient condition measurement equipment, span check gases and other test equipment are the correct type and have been calibrated in accordance with the NO<sub>x</sub> Technical Code;
- (AA) 4.2.2.4.4.4 confirm that the correct test cycle, as defined in the engine's Technical File, is used for this onboard confirmation test measurements;



- (AA) 4.2.2.4.4.5 ensure that a fuel sample is taken during the test and submitted for analysis;
- (AA) 4.2.2.4.4.6 witness the test and confirm that a copy of the test report has been submitted for approval on completion of the test;
- (AA) 4.2.2.4.5 if the direct measurement and monitoring method is used:
- (AA) 4.2.2.4.5.1 review the Technical File and the onboard monitoring manual that the arrangements are as approved;
- (AA) 4.2.2.4.5.2 the procedures to be checked in the direct monitoring and measure method and the data obtained as given in the approved onboard monitoring manual should be followed (NO<sub>x</sub> Technical Code para.6.4.16.1);
- (AA) 4.2.2.4.6 for a marine diesel engine with an output of more than 5,000 kW and a per cylinder displacement at or above 90 litres/cylinder installed on ships constructed between 1 January 1990 and 31 December 1999, check whether:
- .1 an approved method exists;
  - .2 an approved method is not commercially available; or
  - .3 that an approved method is installed and where this is the case, that there is an approved method file,
- and apply the verification procedures as given in the approved method file;
- (AA) 4.2.2.5 Sulphur Oxides and Particulate Matter:
- confirming, if appropriate, that:
- .1 satisfactory arrangements are in place for using compliant fuel as required; or
  - .2 satisfactory installation and operation of the fuel switching arrangements are in place when tanks are provided for different grades of fuel, including records of the changeover to and from low sulphur fuel during transit through an emission control area established for SO<sub>x</sub> and particulate matter control; or
  - .3 satisfactory installation and operation of the exhaust gas cleaning system or other technological methods are examined, (MARPOL Annex VI reg.4);
- (AA) 4.2.2.6 Volatile Organic Compounds (VOCs):
- (AA) 4.2.2.6.1 confirm that the vapour collect system, if required, is maintained in accordance with its approved arrangement;

- (AA) 4.2.2.6.2 for ships carrying crude oil, confirm the VOC management plan has been implemented as appropriate;
- (AA) 4.2.2.7 Incineration:
  - (AA) 4.2.2.7.1 confirm that prohibited materials have not been incinerated;
  - (AA) 4.2.2.7.2 confirm that shipboard incineration of sewage sludge or sludge oil in boilers or marine power plants is not undertaken while the ship is inside ports, harbours or estuaries;
- (AA) 4.2.2.8 Incinerators (installed on or after 1 January 2000):
  - (AA) 4.2.2.8.1 confirm that operators have been trained as required;
  - (AA) 4.2.2.8.2 confirm from an external examination that each incinerator is in a generally satisfactory condition and free from leaks of gas or smoke;
  - (AA) 4.2.2.8.3 confirm that combustion chamber outlet temperatures have been maintained as required;
  - (AA) 4.2.2.8.4 confirm that each incinerator is maintained according to its approved arrangement.
- (AA) 4.2.3 Fuel Oil Quality:
  - (AA) 4.2.3.1 confirm that Bunker Delivery Notes as required conform to the requirements of MARPOL Annex VI, Appendix V;
  - (AA) 4.2.3.2 confirm that MARPOL samples as required are retained on board and labels duly completed or otherwise retained under the ship's control;
  - (AA) 4.2.3.3 confirm that documentation in lieu of that required by (AA) 4.2.3.1 or 4.2.3.2 is available on board.
- (AA) 4.2.4 For air pollution prevention the completion of the annual survey should consist of:
  - (AA) 4.2.4.1 after a satisfactory survey, endorsing the International Air Pollution Prevention certificate;
  - (AA) 4.2.4.2 if a survey shows that the condition of the ship or its equipment is unsatisfactory – see "General", section 4.8.

- (Aln) **4.3** **Intermediate surveys** – see "General", section 4.3
- (Aln) 4.3.1 For air pollution prevention the examination of current certificates and other records should consist of:
- (Aln) 4.3.1.1 the provisions of (AA) 4.2.1.
- (Aln) 4.3.2 For air pollution prevention the intermediate survey should consist of:
- (Aln) 4.3.2.1 the provisions of (AA) 4.2.2.
- (Aln) 4.3.3 For air pollution prevention the completion of the intermediate survey should consist of:
- (Aln) 4.3.3.1 after a satisfactory survey, endorsing the International Air Pollution Prevention Certificate;
- (Aln) 4.3.3.2 if a survey shows that the condition of the ship or its equipment is unsatisfactory see "General", section 4.8.
- (AR) **4.4** **Renewal surveys** – see "General", section 4.5
- (AR) 4.4.1 For air pollution prevention the examination of current certificates and other records should consist of:
- (AR) 4.4.1.1 the provisions of (AA) 4.2.1 except the validity of the International Air Pollution Prevention Certificate.
- (AR) 4.4.2 For air pollution prevention the renewal survey should consist of:
- (AR) 4.4.2.1 the provisions of (AA) 4.2.2;
- (AR) 4.4.2.2 for each incinerator the renewal survey should consist of;
- (AR) 4.4.2.2.1 confirming, if necessary by simulated test or equivalent, the satisfactory operation of the following alarms and safety devices.
- (AR) 4.4.3 For air pollution prevention the completion of the renewal survey should consist of:
- (AR) 4.4.3.1 after satisfactory survey the International Air Pollution prevention Certificate should be issued.

ANNEX 4

**SURVEY GUIDELINES UNDER MANDATORY CODES**

- (D)    **1**                   **GUIDELINES FOR THE SURVEYS FOR THE INTERNATIONAL CERTIFICATE OF FITNESS FOR THE CARRIAGE OF DANGEROUS CHEMICALS IN BULK AND THE CERTIFICATE OF FITNESS FOR THE CARRIAGE OF DANGEROUS CHEMICALS IN BULK**
- (DI)    **1.1**               **Initial surveys** – see part "General" section 4.1
- (DI)    1.1.1           For compliance with the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk the examination of plans and designs of the structure, equipment, fittings, arrangements and materials should consist of:
- (DI)    1.1.1.1       determining the products that it is intended that the ship will be permitted to carry and noting the corresponding minimum special requirements (IBC Code 83/90/00 ch.17) and any other special requirements (IBC Code 83/90/00 ch.15);
- (DI)    1.1.1.2       examining the plans for the ship type, location of the cargo tanks, cargo containment, materials of construction, cargo temperature control, cargo tank vent systems, continuous monitoring of the concentration of flammable vapours, environmental control, electrical installations, fire protection and fire extinction, instrumentation and the provision, specification and stowage of the equipment for personnel protection (IBC Code 83/90/00 chs.2, 4, 6, 7, 8, 9, 10, 11, 13 and 14);
- (DI)    1.1.1.3       examining the plans for the freeboard and intact stability, discharges below the bulkhead deck and survival capability (IBC Code 83/90/00 ch.2);
- (DI)    1.1.1.4       examining the plans for the ship arrangements IBC Code 83/90/00 ch.3);
- (DI)    1.1.1.5       examining the plans for the cargo transfer IBC Code 83/90/00 ch.5);
- (DI)    1.1.1.6       examining the plans for the mechanical ventilation in the cargo area (IBC Code 83/90/00 ch.12);
- (DI)    1.1.1.7       the provisions of (NI) 2.1.1 in Annex 3.
- (DI)    1.1.2           For compliance with the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk, the survey during construction and after installation of the structure, equipment, fittings, arrangements and materials should consist of:
- (DI)    1.1.2.1       confirming that tanks containing cargo or residues of cargo are suitably segregated from accommodation, service and machinery spaces and from drinking water and stores for human consumption, that cargo piping does not pass through any accommodation, service or machinery space other than cargo pump rooms or pump rooms and that cargoes are not to be carried in either the fore or the aft peak tank (IBC Code 83/90/00 ch.3);

- (DI) 1.1.2.2 examining the air intakes and openings into the accommodation, service and machinery spaces in relation to the cargo piping and vent systems and their entrances, air inlets and openings in relation to the cargo area (IBC Code 83/90/00 ch.3);
- (DI) 1.1.2.3 examining the arrangements of the cargo pump rooms (IBC Code 83/90/00 ch.3);
- (DI) 1.1.2.4 examining the accesses to spaces in the cargo area (IBC Code 83/90/00 ch.3);
- (DI) 1.1.2.5 examining the bilge and ballast arrangements and confirming that pumps and pipelines are identified (IBC Code 83/90/00 ch.3);
- (DI) 1.1.2.6 examining, when applicable, the bow or stern loading and unloading arrangements with particular reference to the air inlets and entrances to the accommodation, machinery and service spaces, the electrical equipment, fire-fighting arrangements and means of communication and testing the remote shut down for the cargo pumps (IBC Code 83/90/00 ch.3);
- (DI) 1.1.2.7 confirming that the cargo tank types are arranged and installed in accordance with the approved plans, internally examining the cargo tanks, water ballast tanks and other spaces in the cargo area and pressure testing the boundaries (IBC Code 83/90/00 ch.4);
- (DI) 1.1.2.8 examining the cargo transfer arrangements and confirming that any hoses are suitable for their intended purpose and, where appropriate, type-approved or marked with date of testing (IBC Code 83/90/00 ch.5);
- (DI) 1.1.2.9 examining and testing any cargo heating and cooling systems (IBC Code 83/90/00 ch.7);
- (DI) 1.1.2.10 confirming that the cargo tank vent systems have been installed in accordance with the approved plans (IBC Code 83/90/00 ch.8);
- (DI) 1.1.2.11 confirming that high-level alarms, or overflow control systems or spill valves or other equivalent means provided to control possible liquid rising in the venting system, are operating satisfactorily (IBC Code 83/90/00 ch.8);
- (DI) 1.1.2.12 confirming that suitable provision is made for drainage of vent lines and that no shut-off valves or other means of stoppage, including spectacle or blank flanges, are fitted either to the individual vents or to the header, if the vents are combined or either above or below pressure/vacuum relief valves with closed vent systems (IBC Code 83/90/00 ch.8);
- (DI) 1.1.2.13 confirming that suitable provisions are made for primary and secondary means (or alternative measures) for controlled tank venting (IBC Code 83/90/00 ch.8);

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|------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (DI) | 1.1.2.14   | examining the location of the vent outlets in respect of the height above the weather deck or the fore and aft gangway, from the nearest air intakes or openings to accommodation, service and machinery spaces and ignition sources and confirming that any high velocity vents are of the approved type (IBC Code 83/90/00 ch.8); |
| (DI) | 1.1.2.15   | examining the arrangements for environmental control, including the means of storing or generating and drying an inert gas (IBC Code 83/90/00 ch.9);                                                                                                                                                                                |
| (DI) | 1.1.2.16   | examining the electrical installations and confirming that, when appropriate, special materials have been used and that the electrical equipment installed in hazardous locations, as permitted, is certified by a recognized authority for the cargoes to be carried (IBC Code 83/90/00 ch.10);                                    |
| (DI) | 1.1.2.17   | confirming that independent cargo tanks are electrically bonded to the hull and that all gasketed cargo pipe joints and hose connections are electrically bonded (IBC Code 83/90/00 ch.10);                                                                                                                                         |
| (DI) | 1.1.2.18   | examining the arrangements for the fire protection and fire extinction (IBC Code 83/90/00 ch.11);                                                                                                                                                                                                                                   |
| (DI) | 1.1.2.19   | examining the fixed fire fighting system for the cargo pump room and confirming that the installation tests have been satisfactorily completed and that its means of operation are clearly marked (IBC Code 83/90/00 ch.11);                                                                                                        |
| (DI) | 1.1.2.20   | checking the deck foam system for the cargo area, including the supplies of foam concentrate, and testing that the minimum number of jets of water at the required pressure in the fire main is obtained (see (EI) 1.1.3.1 in Annex 1), when the system is in operation (IBC Code 83/90/00 ch.11);                                  |
| (DI) | 1.1.2.21   | examining the system for continuous monitoring of the concentration of flammable vapours and confirming that the installation tests have been satisfactorily completed (IBC Code 83/90/00 ch.11);                                                                                                                                   |
| (DI) | 1.1.2.22   | confirming that suitable portable fire extinguishing equipment for the cargoes to be carried is provided in the cargo area (IBC Code 83/90/00, ch.11);                                                                                                                                                                              |
| (DI) | 1.1.2.23   | examining, and confirming the satisfactory operation of, the arrangements for the mechanical ventilation of spaces in the cargo area normally entered during cargo handling operations (IBC Code 83/90/00, ch.12) and checking in particular that:                                                                                  |
| (DI) | 1.1.2.23.1 | it may be controlled from outside the space;                                                                                                                                                                                                                                                                                        |
| (DI) | 1.1.2.23.2 | warning notices concerning its use have been posted;                                                                                                                                                                                                                                                                                |
| (DI) | 1.1.2.23.3 | it is of the extraction type, with extraction from below the floor plates, unless the space houses electrical motors driving cargo pumps when it should be of the positive pressure type;                                                                                                                                           |

- (DI) 1.1.2.23.4 the ducting does not pass through accommodation, machinery and service spaces and that the exhaust ducts are clear of the ventilation inlets and openings to such spaces;
- (DI) 1.1.2.23.5 the electric motors driving ventilation fans are positioned outside the ventilation ducts and the ventilation fans and the ducts, in way of the fans only, are of non-sparking construction in hazardous locations;
- (DI) 1.1.2.24 examining, and confirming the satisfactory operation of, the arrangements for the mechanical ventilation of spaces normally entered, other than those covered by (DI) 1.1.2.23 (IBC Code 83/90/00, ch.12);
- (DI) 1.1.2.25 confirming that double bottoms, cofferdams, duct keels, pipe tunnels, hold spaces and other spaces where cargo may accumulate are capable of being efficiently ventilated to ensure a safe environment when entry into the space is necessary and that, when appropriate, permanent ducting is provided and any ventilation fans comply with (DI) 1.1.2.23.5 (IBC Code 83/90/00, ch.12);
- (DI) 1.1.2.26 examining the intrinsically safe systems and circuits used for measurement, monitoring, control and communication purposes in all hazardous locations (IBC Code 83/90/00, ch.13);
- (DI) 1.1.2.27 checking the provision of equipment for personnel protection (IBC Code 83/90/00, ch.14) and in particular that:
  - (DI) 1.1.2.27.1 suitable protective clothing is available for the crew engaged in loading and discharging operations and that suitable storage is provided;
  - (DI) 1.1.2.27.2 the required safety equipment and associated breathing apparatus and air supplies and, when appropriate, emergency-escape respiratory and eye protection, are provided and are properly stowed;
  - (DI) 1.1.2.27.3 medical first-aid equipment, including stretchers and oxygen resuscitation equipment are provided;
  - (DI) 1.1.2.27.4 arrangements have been made for the antidotes for the cargoes actually carried to be on board;
  - (DI) 1.1.2.27.5 decontamination arrangements and eyewashes are operational;
  - (DI) 1.1.2.27.6 the required gas detection instruments are on board and that arrangements have been made for the supply of the appropriate vapour detection tubes;
  - (DI) 1.1.2.27.7 the stowage for cargo samples is satisfactory;
- (DI) 1.1.2.28 the provisions of (NI) 2.1.2 in annex 3;
- (DI) 1.1.2.29 confirming that sampling points or detector heads are located in suitable positions in order that potentially dangerous leakages are readily detected (IBC Code 07 ch.11.1.4, BCH Code ch.IIIE 3.13).

- (DI) 1.1.3 For compliance with the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk the check that all the required documentation has been placed on board the ship should consist of:
- (DI) 1.1.3.1 confirming that a loading and stability information booklet, containing details of typical service and ballast conditions, provisions for evaluating other conditions of loading, a summary of the ship's survival capabilities and sufficient information to ensure that the ship is loaded and operated in a safe and seaworthy manner, is available on board (IBC Code 83/90/00 ch.2);
- (DI) 1.1.3.2 confirming that damage survival capability information is supplied on the basis of loading information for all anticipated conditions of loading and variations in draught and trim (IBC Code 83/90/00 ch.2);
- (DI) 1.1.3.3 confirming that a table giving the filling ratios for the cargo tanks at various densities has been provided (IBC Code 83/90/00 ch.16);
- (DI) 1.1.3.4 confirming that a copy of the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk, or the equivalent national regulations, has been provided (IBC Code 83/90/00 ch.16);
- (DI) 1.1.3.5 confirming that information relating to the chemical and physical properties of the products to be carried has been provided and that provision has been made for the measures to be taken in an accident (IBC Code 83/90/00 ch.16);
- (DI) 1.1.3.6 confirming that a manual covering procedures for cargo transfer, tank cleaning, gas freeing, ballasting, etc., has been provided (IBC Code 83/90/00 ch.16);
- (DI) 1.1.3.7 the provisions of (NI) 2.1.3 in annex 3;
- (DI) 1.1.3.8 confirming that compatibility information as to material of construction, protective linings and coating is provided on board. (IBC Code 83/04 ch.6).
- (DI) 1.1.4 For compliance with the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk the completion of the initial survey should consist of:
- (DI) 1.1.4.1 after a satisfactory survey issuing the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk.
- (DA) 1.2 **Annual surveys** – see part "General" section 4.2
- (DA) 1.2.1 For compliance with the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk and the Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk the examination of current certificates and other records should consist of:



- (DA) 1.2.1.1 checking the validity, as appropriate, of the Cargo Ship Safety Equipment Certificate, the Cargo Ship Safety Radio Certificate and the Cargo Ship Safety Construction Certificate or the Cargo Ship Safety Certificate;
- (DA) 1.2.1.2 checking the validity of the Safety Management Certificate (SMC) and that a copy of the Document of Compliance (DOC) is on board;
- (DA) 1.2.1.3 checking the validity of the International Ship Security Certificate;
- (DA) 1.2.1.4 checking the validity of the International Load Line Certificate or International Load Line Exemption Certificate;
- (DA) 1.2.1.5 checking the validity of the International Oil Pollution Prevention Certificate;
- (DA) 1.2.1.6 checking the certificates of class, if the ship is classed with a classification society;
- (DA) 1.2.1.7 checking, when appropriate, the validity of the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk or the Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk;
- (DA) 1.2.1.8 checking, when appropriate, the validity of the International Sewage Pollution Prevention Certificate;
- (DA) 1.2.1.9 checking, when appropriate, the validity of the International Air Pollution Prevention Certificate;
- (DA) 1.2.1.10 checking that the ship's complement complies with the Minimum Safe Manning Document (SOLAS 74/88 reg.V/13(b));
- (DA) 1.2.1.11 checking that the master, officers and ratings are certificated as required by the STCW Convention;
- (DA) 1.2.1.12 checking whether any new equipment has been fitted and, if so, confirming that it has been approved before installation and that any changes are reflected in the appropriate certificate;
- (DA) 1.2.1.13 confirming that the loading and stability information booklet, containing details of typical service and ballast conditions, provisions for evaluating other conditions of loading, a summary of the ship's survival capabilities and sufficient information to ensure that the ship is loaded and operated in a safe and seaworthy manner, is available on board (IBC Code 83/90/00, ch.2) (No BCH Code 85/90/00 reference);
- (DA) 1.2.1.14 confirming that damage survival capability information is supplied on the basis of loading information for all anticipated conditions of loading and variations in draught and trim (IBC Code 83/90/00 ch.2) (No BCH Code 85/90/00 reference);
- (DA) 1.2.1.15 confirming that a table giving the filling ratios for the cargo tanks at various densities has been provided (IBC Code 83/90/00 ch.16) (BCH Code 85/90/00 ch.IIIG);

- (DA) 1.2.1.16 confirming that a copy of the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk or the Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk, or the equivalent national regulations, has been provided (IBC Code 83/90/00 ch.16) (BCH Code 85/90/00 ch.V);
- (DA) 1.2.1.17 confirming that information relating to the chemical and physical properties of the products to be carried has been provided, and that provision has been made for the measures to be taken in an accident (IBC Code 83/90/00 ch.16) (BCH Code 85/90/00 ch.V);
- (DA) 1.2.1.18 confirming that a manual covering procedures for cargo transfer, tank cleaning, gas freeing, ballasting, etc., has been provided (IBC Code 83/90/00 ch.16) (BCH Code 85/90/00 ch.V);
- (DA) 1.2.1.19 confirming that the Procedures and Arrangements Manual is on board (IBC Code 83/90/00 ch.16A) (BCH Code 85/90/00 ch.VA);
- (DA) 1.2.1.20 confirming that the Shipboard marine pollution emergency plan is on board (MARPOL 73/78/02 Annex II reg.16);
- (DA) 1.2.1.21 confirming that the Cargo Record Book is on board and being correctly used (MARPOL 73/78/91/97/02 Annex II reg.9);
- (DA) 1.2.1.22 confirming that compatibility information as to material of construction, protective linings and coating is provided onboard. (IBC Code 83/04 Ch.6) (BCH Code 85/90/00 ch.IIG);
- (DA) 1.2.1.23 confirming the availability of the International Anti-Fouling System Certificate (AFS 2001 Annex 4 reg.2), when applicable.
- (DA) 1.2.2 For compliance with the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk and the Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk the annual survey of the structure, equipment, fittings, arrangements and materials should consist of:
- (DA) 1.2.2.1 confirming that wheelhouse doors and windows, sidescuttles and windows in superstructure and deckhouse ends facing the cargo area are in a satisfactory condition (IBC Code 83/90/00 ch.3) (BCH Code 85/90/00 ch.IIC);
- (DA) 1.2.2.2 confirming that potential sources of ignition in or near the cargo pump room are eliminated, such as loose gear, combustible materials, etc., that there are no signs of undue leakage and that access ladders are in a satisfactory condition (IBC Code 83/90/00 ch.3) (BCH Code 85/90/00 ch.IIC);
- (DA) 1.2.2.3 confirming that removable pipe lengths or other approved equipment necessary for cargo separation are available in the pump room and are in a satisfactory condition (IBC Code 83/90/00 ch.3) (BCH Code 85/90/00 ch.IIC);

- (DA) 1.2.2.4 examining all pump room bulkheads for signs of cargo leakage or fractures and, in particular, the sealing arrangements of all penetrations of pump room bulkheads (IBC Code 83/90/00 ch.3) (BCH Code 85/90/00 ch.IIC);
- (DA) 1.2.2.5 confirming that the remote operation of the cargo pump bilge system is satisfactory (IBC Code 83/90/00 ch.3) (BCH Code 85/90 ch.IIC);
- (DA) 1.2.2.6 examining the bilge and ballast arrangements and confirming that pumps and pipelines are identified (IBC Code 83/90/00 ch.3) (No BCH Code 85/90/00 reference);
- (DA) 1.2.2.7 confirming, when applicable, that the bow or stern loading and unloading arrangements are in order and testing the means of communication and the remote shut down for the cargo pumps (IBC Code 83/90/00 ch.3) (No BCH Code 85/90/00 reference);
- (DA) 1.2.2.8 examining the cargo transfer arrangements and confirming that any hoses are suitable for their intended purpose and, where appropriate, type-approved or marked with date of testing (IBC Code 83/90/00 ch.5) (BCH Code 85/90/00 ch.IID);
- (DA) 1.2.2.9 examining, when applicable, the cargo heating or cooling systems, including any sampling arrangements, and confirming that the means for measuring the temperature and associated alarms are operating satisfactorily (IBC Code 83/90/00 ch.7) (BCH Code 85/90/00 ch.IIF);
- (DA) 1.2.2.10 examining, as far as practicable, the cargo tank vent system, including the pressure/vacuum valves and secondary means to prevent over- or under-pressure and devices to prevent the passage of flame (IBC Code 83/90/00 ch.8, MSC.102(73) and MEPC.79(43)) (BCH Code 85/90/00 ch.IIE and MEPC.80(43));
- (DA) 1.2.2.11 examining the gauging devices, high-level alarms and valves associated with overflow control (IBC Code 83/90/00 ch.8) (BCH Code 85/90/00 ch.IIE);
- (DA) 1.2.2.12 confirming that arrangements for sufficient gas to be carried or generated to compensate for normal losses, and that the means provided for monitoring ullage spaces, are satisfactory (IBC Code 83/90/00 ch.9) (BCH Code 85/90/00 ch.IIH);
- (DA) 1.2.2.13 confirming that arrangements are made for sufficient medium to be carried where drying agents are used on air inlets to cargo tanks (IBC Code 83/90/00 ch.9) (BCH Code 85/90/00 ch.IIH);
- (DA) 1.2.2.14 confirming that all electrical equipment in dangerous zones is suitable for such locations, is in satisfactory condition and has been properly maintained (IBC Code 83/90/00 ch.10) (BCH Code 85/90/00 ch.IIIB);
- (DA) 1.2.2.15 examining the fixed fire-fighting system for the cargo pump room and the deck foam system for the cargo area and confirming that their means of operation are clearly marked (IBC Code 83/90/00 ch.11) (BCH Code 85/90/00 ch.IIIIE);

- (DA) 1.2.2.16 confirming that the condition of the portable fire extinguishing equipment for the cargoes to be carried in the cargo area is satisfactory (IBC Code 83/90/00 ch.11) (BCH Code 85/90/00 ch.IIIE);
- (DA) 1.2.2.17 confirming that the system for continuous monitoring of the concentration of flammable vapours is satisfactory (IBC Code 83/90/00 ch.11);
- (DA) 1.2.2.18 examining, as far as practicable, and confirming the satisfactory operation of, the arrangements for the ventilation of spaces normally entered during cargo handling operations and other spaces in the cargo area (IBC Code 83/90/00 ch.12) (BCH Code 85/90/00 ch.IIIA);
- (DA) 1.2.2.19 confirming, as far as practicable, that the intrinsically safe systems and circuits used for measurement, monitoring, control and communication purposes in all hazardous locations are being properly maintained (IBC Code 83/90/00 ch.13) (BCH Code 85/90/00 ch.IIIC);
- (DA) 1.2.2.20 examining the equipment for personnel protection (IBC Code 83/90/00, ch.14) (BCH Code 85/90/00 ch.IIIF) and in particular that:
- (DA) 1.2.2.20.1 the protective clothing for crew engaged in loading and discharging operations and its stowage is in a satisfactory condition;
- (DA) 1.2.2.20.2 the required safety equipment and associated breathing apparatus and associated air supplies and, when appropriate, emergency-escape respiratory and eye protection, are in a satisfactory condition and are properly stowed;
- (DA) 1.2.2.20.3 medical first-aid equipment, including stretchers and oxygen resuscitation equipment are in a satisfactory condition;
- (DA) 1.2.2.20.4 arrangements have been made for the antidotes for the cargoes actually carried to be on board;
- (DA) 1.2.2.20.5 decontamination arrangements and eyewashes are operational;
- (DA) 1.2.2.20.6 the required gas detection instruments are on board and arrangements have been made for the supply of the appropriate vapour detection tubes;
- (DA) 1.2.2.20.7 the arrangements for the stowage of cargo samples are satisfactory;
- (DA) 1.2.2.21 the provisions of (NA) 2.2.2 in annex 3;
- (DA) 1.2.2.22 confirming that sampling points or detector heads are located in suitable positions in order that potentially dangerous leakages are readily detected (IBC Code 07 ch.11.1.4)( BCH Code ch.IIIE 3.13).
- (DA) 1.2.3 For compliance with the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk and the Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk the completion of the annual survey should consist of:

- (DA) 1.2.3.1 after a satisfactory survey, endorsing the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk or the Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk;
- (DA) 1.2.3.2 if a survey shows that the condition of a ship or its equipment is unsatisfactory – see part "General" section 4.8.
- (DIn) **1.3** **Intermediate surveys** – see part "General", section 4.3
- (DIn) 1.3.1 For compliance with the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk and the Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk the examination of current certificates and other records should consist of:
- (DIn) 1.3.1.1 the provisions of (DA) 1.2.1.
- (DIn) 1.3.2 For compliance with the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk and the Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk the intermediate survey of the structure, equipment, fittings, arrangements and materials should consist of:
- (DIn) 1.3.2.1 the provisions of (DA) 1.2.2;
- (DIn) 1.3.2.2 examination of vent line drainage arrangements (IBC Code 83/90/00 ch.8) (BCH Code 85/90/00 ch.IIE);
- (DIn) 1.3.2.3 confirmation, where applicable, that pipelines and independent cargo tanks are electrically bonded to the hull (IBC Code 83/90/00 ch.10) (BCH Code 85/90/00 ch.IIIB);
- (DIn) 1.3.2.4 generally examining the electrical equipment and cables in dangerous zones such as cargo pump rooms and areas adjacent to cargo tanks to check for defective equipment, fixtures and wiring. The insulation resistance of the circuits should be tested and in cases where a proper record of testing is maintained, consideration should be given to accepting recent readings (IBC Code 83/90/00 ch.10) (BCH Code 85/90/00 ch.IIIB);
- (DIn) 1.3.2.5 confirmation that spares are provided for cargo area mechanical ventilation fans (IBC Code 83/90/00 ch.12) (BCH Code 85/90/00 ch.IIIA);
- (DIn) 1.3.2.6 the provisions of (NIn) 2.3.2 in annex 3.
- (DIn) 1.3.3 For compliance with the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk and the Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk the completion of the intermediate survey should consist of:

- (DIn) 1.3.3.1 after a satisfactory survey endorsing the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk or the Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk;
- (DIn) 1.3.3.2 if a survey shows that the condition of a ship or its equipment is unsatisfactory – see part "General", section 4.8.
- (DR) **1.4** **Renewal surveys** – see part "General" section 4.4
- (DR) 1.4.1 For compliance with the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk and the Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk the examination of current certificates and other records should consist of:
- (DR) 1.4.1.1 the provisions of (DA) 1.2.1, except the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk or the Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk.
- (DR) 1.4.2 For compliance with the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk and the Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk the renewal survey of the structure, equipment, fittings, arrangements and materials should consist of:
- (DR) 1.4.2.1 the provisions of (DIn) 1.3.3;
- (DR) 1.4.2.2 the provisions of (NR) 2.4.2 in annex 3.
- (DR) 1.4.3 For compliance with the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk and the Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk the completion of the renewal survey should consist of:
- (DR) 1.4.3.1 after a satisfactory survey, issuing the International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk or the Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk.
- (G) 2** **GUIDELINES FOR SURVEYS FOR THE INTERNATIONAL CERTIFICATE OF FITNESS FOR THE CARRIAGE OF LIQUEFIED GASES IN BULK**
- (GI) **2.1** **Initial surveys** – see part "General", section 4.1.
- (GI) 2.1.1 For compliance with the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk the examination of plans and designs of the structure, equipment, fittings, arrangements and materials should consist of:
- (GI) 2.1.1.1 determining the products that it is intended that the ship will be permitted to carry and noting the corresponding minimum special requirements (IGC Code 83/90/00 ch.19);

- (GI) 2.1.1.2 examining the plans for the ship type, cargo containment, control of vapour space within the cargo tanks, vapour detection, gauging, personnel protection, filling limits for cargo tanks and other special requirements (IGC Code 83/90/00 chs.2, 4, 6, 13, 14, 15, and 17);
- (GI) 2.1.1.3 examining the plans for the freeboard and intact stability, discharges below the bulkhead deck and survival capability (IGC Code 83/90/00 ch.2);
- (GI) 2.1.1.4 examining the plans for the ship arrangements (IGC Code 83/90/00 ch.3);
- (GI) 2.1.1.5 examining the plans for the process pressure vessels and liquid, vapour and pressure piping systems (IGC Code 83/90/00 chs.5 and 6);
- (GI) 2.1.1.6 examining the plans for the cargo pressure/temperature control (IGC Code 83/90/00 ch.7);
- (GI) 2.1.1.7 examining the plans for the cargo tank ventilation systems (IGC Code 83/90/00 ch.8);
- (GI) 2.1.1.8 examining the plans for the environmental control (IGC Code 83/90/00 ch.9);
- (GI) 2.1.1.9 examining the plans for the electrical installations (IGC Code 83/90/00 ch.10);
- (GI) 2.1.1.10 examining the plans for fire protection and fire extinction equipment (IGC Code 83/90/00 ch.11);
- (GI) 2.1.1.11 examining the plans for the mechanical ventilation in the cargo area (IGC Code 83/90/00 ch.12);
- (GI) 2.1.1.12 examining the plans for the instrumentation (gauging, gas detection) (IGC Code 83/90/00 ch.13);
- (GI) 2.1.1.13 examining, when applicable, the plans for the use of cargo as fuel (IGC Code 83/90/00 ch.16).
- (GI) 2.1.2 For compliance with the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk the survey during construction and after installation of the structure, equipment, fittings, arrangements and materials should consist of:
  - (GI) 2.1.2.1 confirming that the segregation in the cargo area and the arrangement of the accommodation, service and machinery spaces are in accordance with the approved plans (IGC Code 83/90/00 ch.3);
  - (GI) 2.1.2.2 examining the arrangements of the cargo pump rooms and cargo compressor rooms (IGC Code 83/90/00 ch.3);
  - (GI) 2.1.2.3 confirming that the manually operated emergency shutdown system together with the automatic shutdown of the cargo pumps and compressors are satisfactory (IGC Code 83/90/00 ch.3);

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| (GI) | 2.1.2.4    | examining the arrangement of the cargo control room (IGC Code 83/90/00 ch.3);                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| (GI) | 2.1.2.5    | examining the accesses to spaces in the cargo area (IGC Code 83/90/00 ch.3);                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| (GI) | 2.1.2.6    | confirming the arrangements for the air locks (IGC Code 83/90/00 ch.3);                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| (GI) | 2.1.2.7    | examining the bilge, ballast and oil fuel arrangements (IGC Code 83/90/00 ch.3);                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| (GI) | 2.1.2.8    | examining, when applicable, the bow or stern loading and unloading arrangements with particular reference to the air inlets and entrances to the accommodation, machinery and service spaces, the electrical equipment, fire-fighting arrangements and means of communication between the cargo control room and the shore location (IGC Code 83/90/00 ch.3);                                                                                                                                                                                                    |
| (GI) | 2.1.2.9    | confirming that the cargo tanks are arranged and installed in accordance with the approved plans, internally examining the cargo tanks, water ballast tanks and other spaces in the cargo area, ensuring that the appropriate non-destructive and pressure testing are carried out (IGC Code 83/90/00 ch.4);                                                                                                                                                                                                                                                     |
| (GI) | 2.1.2.10   | for containment systems with glued secondary barriers, confirming that a tightness test has been carried out in accordance with the approved procedures of the system manufacturer before and after the initial cool down. Where significant differences in the results before and after cool down for each tanks or between tanks have been observed, confirming that an investigation has been carried out including additional testing, such as differential pressure, thermo graphic or acoustic emission testing, where necessary (IGC Code 83/90/00 ch.4); |
| (GI) | 2.1.2.11   | examining during the initial cool down, loading and discharging of the first cargo, the overall performance of the cargo containment system and confirming that the system is in compliance with the design parameters. For vessels carrying liquefied natural gas, the examination includes witnessing the satisfactory operation of the following systems, if fitted:                                                                                                                                                                                          |
| (GI) | 2.1.2.11.1 | Gas detection system;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| (GI) | 2.1.2.11.2 | Cargo control and monitoring systems such as level gauging; equipment, temperature sensors, pressure gauges, cargo pump room and compressors, and proper control of cargo heat exchanges, if operating;                                                                                                                                                                                                                                                                                                                                                          |
| (GI) | 2.1.2.11.3 | Nitrogen generating plant or inert gas generator;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| (GI) | 2.1.2.11.4 | Nitrogen pressure control systems for insulation, interbarrier and annular spaces;                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| (GI) | 2.1.2.11.5 | Re-liquefaction plant;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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- (GI) 2.1.2.11.6 Equipment fitted for the burning of cargo vapours, such as boilers or engines gas combustion units;
- (GI) 2.1.2.11.7 Cofferdam heating systems;
- (GI) 2.1.2.11.8 On-deck cargo piping systems including expansion and supporting arrangements;
- (GI) 2.1.2.11.9 High level alarms, by witnessing topping-off process for cargo tanks (IGC Code 83/90/00 ch.4);
- (GI) 2.1.2.12 examining the hull for cold spots following the first loaded voyage (IGC Code 83/90/00 ch.4);
- (GI) 2.1.2.13 examining the cargo and process piping, including the expansion arrangements, insulation from the hull structure, pressure relief and drainage arrangements and carrying out a leak detection test (IGC Code 83/90/00 ch.5);
- (GI) 2.1.2.14 confirming that the cargo system valving arrangements are in accordance with the approved plans (IGC Code 83/90/00 ch.5);
- (GI) 2.1.2.15 confirming that any liquid and vapour hoses are suitable for their intended purpose and, where appropriate, type-approved or marked with date of testing (IGC Code 83/90/00 ch.5);
- (GI) 2.1.2.16 examining the arrangements for the cargo pressure/temperature control including, when fitted, any refrigeration system and confirming that any associated alarms are satisfactory (IGC Code 83/90/00 ch.7);
- (GI) 2.1.2.17 confirming that the cargo tank vent systems, including, when appropriate, any additional pressure relieving system for liquid level control and vacuum pressure systems, have been installed in accordance with the approved plans (IGC Code 83/90/00 ch.8);
- (GI) 2.1.2.18 examining the arrangements for the environmental control, including the means of storing or generating and drying an inert gas (IGC Code 83/90/00 ch.9);
- (GI) 2.1.2.19 examining the electrical installations with particular reference to the certified safe type equipment fitted in gas-dangerous spaces and zones (IGC Code 83/90/00 ch.10);
- (GI) 2.1.2.20 examining the arrangements for the fire protection and fire extinction (IGC Code 83/90/00 ch.11);
- (GI) 2.1.2.21 examining the fixed fire-fighting system for the cargo pump room and confirming that the installation tests have been satisfactorily completed and that its means of operation is clearly marked (IGC Code 83/90/00 ch.11);
- (GI) 2.1.2.22 examining the fire water main with particular reference to the provision of hydrants and isolation arrangements, checking that the two jets of water reach all areas of the cargo and containment area at the required pressure and testing the remote means of starting one main fire pump (IGC Code 83/90/00 ch.11);

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| (GI) | 2.1.2.23   | examining and testing the water spray system for cooling, fire protection and crew protection and confirming that its means of operation is clearly marked (IGC Code 83/90/00 ch.11);                                                                                                          |
| (GI) | 2.1.2.24   | examining the dry chemical powder fire-extinguishing system for the cargo area, seeing that the fixed piping has been properly installed and has been proved clear and confirming that its means of operation is clearly marked (IGC Code 83/90/00 ch.11);                                     |
| (GI) | 2.1.2.25   | examining the carbon dioxide system for the cargo compressor and pump rooms and confirming that the installation tests have been satisfactorily completed and that its means of operation is clearly marked (IGC Code 83/90/00 ch.11);                                                         |
| (GI) | 2.1.2.26   | confirming the provision and examining the disposition of the firefighters outfits (IGC Code 83/90/00 ch.11);                                                                                                                                                                                  |
| (GI) | 2.1.2.27   | examining, and confirming the satisfactory operation of, the arrangements for the mechanical ventilation of spaces in the cargo area normally entered during cargo handling operations (IGC Code 83/90/00 ch.12) and checking in particular that:                                              |
| (GI) | 2.1.2.27.1 | it may be controlled from outside the space;                                                                                                                                                                                                                                                   |
| (GI) | 2.1.2.27.2 | warning notices concerning its use have been posted;                                                                                                                                                                                                                                           |
| (GI) | 2.1.2.27.3 | it is fixed and is of the negative pressure type, permitting extraction from either the upper or lower parts of the space or from both the upper and lower parts when appropriate, for cargo compressor and pump rooms and for cargo control rooms when considered to be gas-dangerous spaces; |
| (GI) | 2.1.2.27.4 | it is of the positive pressure type for spaces containing electric motors driving cargo compressors or pumps and other gas-safe spaces within the cargo area, except those containing inert gas generators;                                                                                    |
| (GI) | 2.1.2.27.5 | exhaust ducts are clear of the ventilation inlets and openings to accommodation spaces, service spaces, control stations and other gas-safe spaces;                                                                                                                                            |
| (GI) | 2.1.2.27.6 | intakes are arranged to minimize the recycling or hazardous vapours;                                                                                                                                                                                                                           |
| (GI) | 2.1.2.27.7 | ducts from gas-dangerous spaces are not led through accommodation, service and machinery spaces and control stations, except when (GI) 2.1.2.33 applies;                                                                                                                                       |
| (GI) | 2.1.2.27.8 | the electric motors driving ventilation fans are positioned outside the ventilation ducts when the carriage of flammable products is intended and the ventilation fans and the ducts, in way of the fans only, are of non-sparking construction in gas-dangerous spaces;                       |

- (GI) 2.1.2.28 examining, and confirming the satisfactory operation of, the arrangements for the mechanical ventilation of spaces normally entered other than those covered by (GI) 2.1.2.27 (IGC Code 83/90/00 ch.12);
- (GI) 2.1.2.29 examining, and testing as appropriate, the liquid level indicators, overflow control, pressure gauges, high pressure and, when applicable, low pressure alarms, and temperature indicating devices for the cargo tanks (IGC Code 83/90/00 ch.13);
- (GI) 2.1.2.30 examining, and testing as appropriate, the gas detection equipment (IGC Code 83/90/00 ch.13);
- (GI) 2.1.2.31 confirming that two sets of portable gas detection equipment suitable for the cargoes to be carried and a suitable instrument for measuring oxygen levels have been provided (IGC Code 83/90/00 ch.13);
- (GI) 2.1.2.32 checking the provision of equipment for personnel protection (IGC Code 83/90/00 ch.14) and in particular that:
  - (GI) 2.1.2.32.1 two complete sets of safety equipment each permitting personnel to enter and work in a gas-filled space are provided and are properly stowed;
  - (GI) 2.1.2.32.2 the requisite supply of compressed air is provided and examining, when applicable, the arrangements for any special air compressor and low-pressure air line system;
  - (GI) 2.1.2.32.3 medical first-aid equipment, including stretchers and oxygen resuscitation equipment and antidotes, when available, for the products to be carried are provided;
  - (GI) 2.1.2.32.4 respiratory and eye protection suitable for emergency escape purposes are provided;
  - (GI) 2.1.2.32.5 decontamination arrangements and eyewashes are operational;
  - (GI) 2.1.2.32.6 when applicable, personnel are protected against the effects of a major cargo release by a special suitably designed and equipped space within the accommodation area;
  - (GI) 2.1.2.32.7 when applicable, the cargo control room is of the gas-safe type;
- (GI) 2.1.2.33 examining, when applicable, the arrangements for the use of cargo as fuel and testing that the gas supply to the machinery space is cut off should the exhaust ventilation not be functioning correctly and that the master gas fuel valve may be remotely closed from within the machinery space (IGC Code 83/90/00 ch.16).
- (GI) 2.1.3 For compliance with the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk the check that all the required documentation has been placed on board the ship should consist of:

- (GI) 2.1.3.1 confirming that a loading and stability information booklet, containing details of typical service and ballast conditions, provisions for evaluating other conditions of loading, a summary of the ship's survival capabilities and sufficient information to ensure that the ship is loaded and operated in a safe and seaworthy manner, is available on board (IGC Code 83/90/00 ch.2);
- (GI) 2.1.3.2 confirming that damage survival capability information is supplied on the basis of loading information for all anticipated conditions of loading and variations in draught and trim (IGC Code 83/90/00 ch.2);
- (GI) 2.1.3.3 confirming that necessary information for the safe carriage of the products to be carried has been provided (IGC Code 83/90/00 ch.18);
- (GI) 2.1.3.4 confirming that a copy of the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk, or the equivalent national regulations, has been provided (IGC Code 83/90/00 ch.18).
- (GI) 2.1.4 For compliance with the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk the completion of the initial survey should consist of:
  - (GI) 2.1.4.1 after a satisfactory survey, issuing the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk.
- (GA) **2.2** **Annual surveys** – see part "General", section 4.2.
- (GA) 2.2.1 For compliance with the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk the examination of current certificates and other records should consist of:
  - (GA) 2.2.1.1 checking the validity, as appropriate, of the Cargo Ship Safety Equipment Certificate, the Cargo Ship Safety Radio Certificate and the Cargo Ship Safety Construction Certificate or the Cargo Ship Safety Certificate;
  - (GA) 2.2.1.2 checking the validity of the Safety Management Certificate (SMC) and that a copy of the Document of Compliance (DOC) is on board;
  - (GA) 2.2.1.3 checking the validity of the International Ship Security Certificate;
  - (GA) 2.2.1.4 checking the validity of the International Load Line Certificate or International Load Line Exemption Certificate;
  - (GA) 2.2.1.5 checking the validity of the International Oil Pollution Prevention Certificate;
  - (GA) 2.2.1.6 checking the certificates of class, if the ship is classed with a classification society;
  - (GA) 2.2.1.7 checking the validity of the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk;

- (GA) 2.2.1.8 checking, when appropriate, the validity of the International Sewage Pollution Prevention Certificate;
- (GA) 2.2.1.9 checking, when appropriate, the validity of the International Air Pollution Prevention Certificate;
- (GA) 2.2.1.10 checking that the ship's complement complies with the Minimum Safe Manning Document (SOLAS 74/88 reg.V/13(b));
- (GA) 2.2.1.11 checking that the master, officers and ratings are certificated as required by the STCW Convention;
- (GA) 2.2.1.12 checking whether any new equipment has been fitted and, if so, confirming that it has been approved before installation and that any changes are reflected in the appropriate certificate;
- (GA) 2.2.1.13 confirming that the loading and stability information booklet, containing details of typical service and ballast conditions, provisions for evaluating other conditions of loading, a summary of the ship's survival capabilities and sufficient information to ensure that the ship is loaded and operated in a safe and seaworthy manner, is available on board (IGC Code 83/90/00 ch.2);
- (GA) 2.2.1.14 confirming that damage survival capability information is supplied on the basis of loading information for all anticipated conditions of loading and variations in draught and trim (IGC Code 83/90/00 ch.2);
- (GA) 2.2.1.15 confirming that necessary information for the safe carriage of the products to be carried has been provided (IGC Code 83/90/00 ch.18);
- (GA) 2.2.1.16 confirming that a copy of the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk, or the equivalent national regulations, has been provided (IGC Code 83/90/00 ch.18);
- (GA) 2.2.1.17 confirming that there are records of the performance of the cargo containment system (IGC Code 83/90/00 ch.4);
- (GA) 2.2.1.18 confirming the availability of the International Anti-Fouling System Certificate (AFS 2001 Annex 4 reg.2), when applicable.
- (GA) 2.2.2 For compliance with the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk the annual survey of the structure, equipment, fittings, arrangements and materials should consist of:
  - (GA) 2.2.2.1 confirming that any special arrangements to survive conditions of damage are in order (IGC Code 83/90/00 ch.2);
  - (GA) 2.2.2.2 confirming that the wheelhouse doors and windows, sidescuttles and windows in superstructure and deckhouse ends in the cargo area are in a satisfactory condition (IGC Code 83/90/00 ch.3);

- (GA) 2.2.2.3 examining the cargo pump rooms and cargo compressor rooms (IGC Code 83/90/00 ch.3);
- (GA) 2.2.2.4 confirming that the manually operated emergency shutdown system together with the automatic shutdown of the cargo pumps and compressors are satisfactory (IGC Code 83/90/00 ch.3);
- (GA) 2.2.2.5 examining the cargo control room (IGC Code 83/90/00 ch.3);
- (GA) 2.2.2.6 examining the gas detection arrangements for cargo control rooms and the measures taken to exclude ignition sources where such spaces are not gas-safe (IGC Code 83/90/00 ch.3);
- (GA) 2.2.2.7 confirming the arrangements for the air locks are being properly maintained (IGC Code 83/90/00 ch.3);
- (GA) 2.2.2.8 examining, as far as practicable, the bilge, ballast and oil fuel arrangements (IGC Code 83/90/00 ch.3);
- (GA) 2.2.2.9 examining, when applicable, the bow or stern loading and unloading arrangements with particular reference to the electrical equipment, fire-fighting arrangements and means of communication between the cargo control room and the shore location (IGC Code 83/90/00 ch.3);
- (GA) 2.2.2.10 confirming that the sealing arrangements at the gas domes are satisfactory (IGC Code 83/90/00 ch.4);
- (GA) 2.2.2.11 confirming that portable or fixed drip trays or deck insulation for cargo leakage is in order (IGC Code 83/90/00 ch.4);
- (GA) 2.2.2.12 examining the cargo and process piping, including the expansion arrangements, insulation from the hull structure, pressure relief and drainage arrangements (IGC Code 83/90/00 ch.5);
- (GA) 2.2.2.13 confirming that the cargo tank and interbarrier space pressure and relief valves, including safety systems and alarms, are satisfactory (IGC Code 83/90/00 ch.5);
- (GA) 2.2.2.14 confirming that any liquid and vapour hoses are suitable for their intended purpose and, where appropriate, type-approved or marked with date of testing (IGC Code 83/90/00 ch.5);
- (GA) 2.2.2.15 examining the arrangements for the cargo pressure/temperature control including, when fitted, any refrigeration system and confirming that any associated alarms are satisfactory (IGC Code 83/90/00 ch.7);
- (GA) 2.2.2.16 examining the cargo, bunker, ballast and vent piping systems, including vent masts and protective screens, as far as practicable (IGC Code 83/90/00 ch.8);
- (GA) 2.2.2.17 confirming that arrangements are made for sufficient inert gas to be carried to compensate for normal losses and that means are provided for monitoring the spaces (IGC Code 83/90/00 ch.9);

- (GA) 2.2.2.18 confirming that the use of inert gas has not increased beyond that needed to compensate for normal losses by examining records of inert gas usage (IGC Code 83/90/00 ch.9);
- (GA) 2.2.2.19 confirming that any air-drying system and any interbarrier and hold space purging inert gas system are satisfactory (IGC Code 83/90/00 ch.9);
- (GA) 2.2.2.20 confirming that electrical equipment in gas-dangerous spaces and zones is in a satisfactory condition and is being properly maintained (IGC Code 83/90/00 ch.10);
- (GA) 2.2.2.21 examining the arrangements for the fire protection and fire extinction and testing the remote means of starting one main fire pump (IGC Code 83/90/00 ch.11);
- (GA) 2.2.2.22 examining the fixed fire-fighting system for the cargo pump room and confirming that its means of operation is clearly marked (IGC Code 83/90/00 ch.11);
- (GA) 2.2.2.23 examining the water spray system for cooling, fire protection and crew protection and confirming that its means of operation is clearly marked (IGC Code 83/90/00 ch.11);
- (GA) 2.2.2.24 examining the dry chemical powder fire-extinguishing system for the cargo area and confirming that its means of operation is clearly marked (IGC Code 83/90/00 ch.11);
- (GA) 2.2.2.25 examining the fixed installation for the gas-dangerous spaces and confirming its means of operation is clearly marked (IGC Code 83/90/00 ch.11);
- (GA) 2.2.2.26 confirming the provision and examining the condition of the firefighters outfits (IGC Code 83/90/00 ch.11);
- (GA) 2.2.2.27 examining, as far as practicable, and confirming the satisfactory operation of, the arrangements for the mechanical ventilation of spaces in the cargo area normally entered during cargo handling operations (IGC Code 83/90/00 ch.12);
- (GA) 2.2.2.28 examining, and confirming the satisfactory operation of, the arrangements for the mechanical ventilation of spaces normally entered other than those covered by (GI) 2.1.2.27 (IGC Code 83/90/00 ch.12);
- (GA) 2.2.2.29 examining, and testing as appropriate and as far as practicable, the liquid level indicators, overflow control, pressure gauges, high pressure and, when applicable, low pressure alarms, and temperature indicating devices for the cargo tanks (IGC Code 83/90/00 ch.13);
- (GA) 2.2.2.30 examining, and testing as appropriate, the gas detection equipment (IGC Code 83/90/00 ch.13);

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- (GA) 2.2.2.31 confirming that two sets of portable gas detection equipment suitable for the cargoes to be carried and a suitable instrument for measuring oxygen levels have been provided (IGC Code 83/90/00 ch.13);
- (GA) 2.2.2.32 checking the provision of equipment for personnel protection (IGC Code 83/90/00 ch.14) and in particular that:
- (GA) 2.2.2.32.1 two complete sets of safety equipment each permitting personnel to enter and work in a gas-filled space are provided and are properly stowed;
- (GA) 2.2.2.32.2 the requisite supply of compressed air is provided and examining, when applicable, the arrangements for any special air compressor and low-pressure air line system;
- (GA) 2.2.2.32.3 medical first-aid equipment, including stretchers and oxygen resuscitation equipment and antidotes, when available, for the products to be carried, are provided;
- (GA) 2.2.2.32.4 respiratory and eye protection suitable for emergency escape purposes are provided;
- (GA) 2.2.2.32.5 decontamination arrangements and eyewashes are operational;
- (GA) 2.2.2.32.6 examining, when applicable, the arrangements to protect personnel against the effects of a major cargo release by a special suitably designed and equipped space within the accommodation area;
- (GA) 2.2.2.33 examining, when applicable, the arrangements for the use of cargo as fuel and testing, as far as practicable, that the gas supply to the machinery space is cut off should the exhaust ventilation not be functioning correctly and that master gas fuel valve may be remotely closed from within the machinery space (IGC Code 83/90/00 ch.16).
- (GA) 2.2.3 For compliance with the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk the completion of the annual survey should consist of:
- (GA) 2.2.3.1 after a satisfactory survey, endorsing the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk;
- (GA) 2.2.3.2 if a survey shows that the condition of a ship or its equipment is unsatisfactory - see part "General" section 4.8.
- (GIn) **2.3 Intermediate surveys** – see part "General", section 4.3.
- (GIn) 2.3.1 For compliance with the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk the examination of current certificates and other records should consist of:
- (GIn) 2.3.1.1 the provisions of (GA) 2.2.1.
- (GIn) 2.3.2 For compliance with the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk the intermediate survey of the structure, equipment, fittings, arrangements and materials should consist of:



- (GIn) 2.3.2.1 the provisions of (GA) 2.2.2;
- (GIn) 2.3.2.2 confirming, where applicable, that pipelines and independent cargo tanks are electrically bonded to the hull (IGC Code 83/90/00 ch.10);
- (GIn) 2.3.2.3 generally examining the electrical equipment and cables in dangerous zones such as cargo pump rooms and areas adjacent to cargo tanks to check for defective equipment, fixtures and wiring. The insulation resistance of the circuits should be tested and in cases where a proper record of testing is maintained consideration should be given to accepting recent readings (IGC Code 83/90/00 ch.10);
- (GIn) 2.3.2.4 confirming that spares are provided for cargo area mechanical ventilation fans (IGC Code 83/90/00 ch.12);
- (GIn) 2.3.2.5 confirming that the heating arrangements, if any, for steel structures are satisfactory.
- (GIn) 2.3.3 For compliance with the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk the completion of the intermediate survey should consist of:
  - (GIn) 2.3.3.1 after a satisfactory survey, endorsing the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk;
  - (GIn) 2.3.3.2 if a survey shows that the condition of a ship or its equipment is unsatisfactory – see part "General" section 4.8.
- (GR) **2.4** **Renewal surveys** – see part "General", section 4.4.
- (GR) 2.4.1 For compliance with the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk the examination of current certificates and other records should consist of:
  - (GR) 2.4.1.1 the provisions of (GA) 2.2.1, except the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk.
- (GR) 2.4.2 For compliance with the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk the renewal survey of the structure, equipment, fittings, arrangements and materials should consist of:
  - (GR) 2.4.2.1 the provisions of (GIn) 2.3.3;
  - (GR) 2.4.2.2 examining the insulation and means of support of the cargo tanks and confirming that the secondary barrier remains effective (IGC Code 83/90/00 ch.4).
- (GR) 2.4.3 For compliance with the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk the completion of the renewal survey should consist of:
  - (GR) 2.4.3.1 after a satisfactory survey, issuing the International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk.

## APPENDIX 1

### **SUMMARY OF AMENDMENTS TO MANDATORY INSTRUMENTS REFLECTED IN THE SURVEY GUIDELINES UNDER THE HSSC**

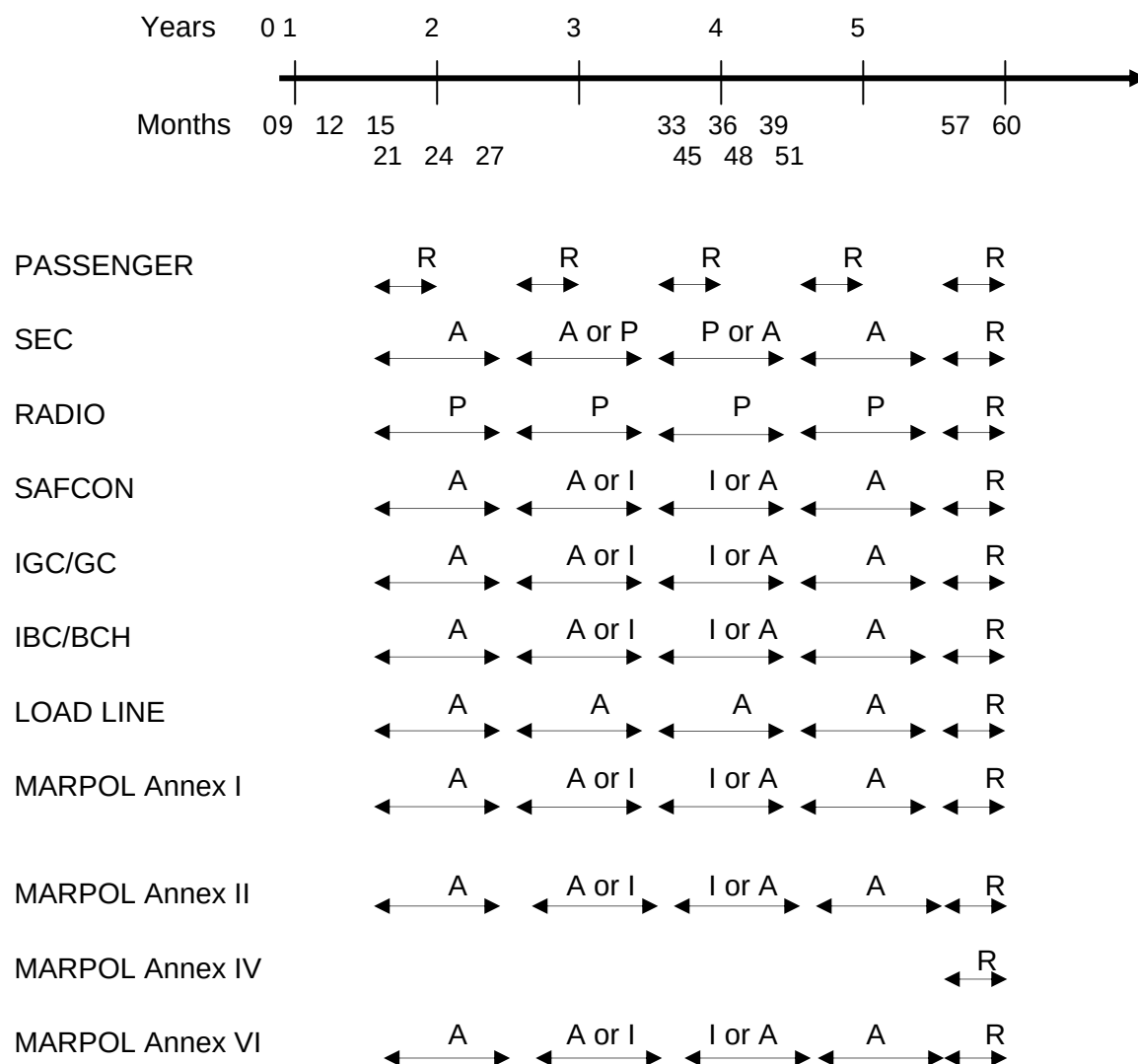
The amendments to mandatory instruments reflected in Annexes 1 to 4 are summarized below to facilitate amendments to the Survey Guidelines under the HSSC in the future:

|                 |                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------|
| SOLAS 1974      | up to and including the 2009 amendments, (resolution MSC.282(86))                                    |
| SOLAS PROT 1988 | up to and including the 2009 amendments (resolution MSC.283(86))                                     |
| LL 1966         | up to and including the 2005 amendments (resolution A.972(24))                                       |
| LL PROT 1988    | up to and including 2008 amendments (resolution MSC.270(85))                                         |
| MARPOL          | up to and including the 2010 amendments (resolution MEPC.190(60))                                    |
| NOx Code        | up to and including the 2008-amendments (resolution MEPC.177(58))                                    |
| IBC Code        | up to and including the 2006/2007 amendments<br>(resolution MEPC.166(56)/resolution MSC.219(82))     |
| IGC Code        | up to and including the 2006/2007 amendments<br>(resolution MSC.220(82) and resolution MEPC.166(56)) |
| BCH Code        | up to and including the 2006 amendments<br>(resolution MSC.212(81)/resolution MEPC.144(54))          |

## APPENDIX 2

### THE HARMONIZED SYSTEM OF SURVEY AND CERTIFICATION

#### DIAGRAMMATIC ARRANGEMENT



Code of types of survey:

|    |   |              |
|----|---|--------------|
| I  | - | Initial      |
| R  | - | Renewal      |
| P  | - | Periodical   |
| In | - | Intermediate |
| A  | - | Annual       |

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**ANNEX 26**

**DRAFT IMO INSTRUMENTS IMPLEMENTATION CODE (III CODE)**

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## **PART 1 – COMMON AREAS**

### **Objective**

1 The objective of this Code is to enhance global maritime safety and protection of the marine environment and assist States in the implementation of instruments of the Organization.

2 Different States will view this Code according to their own circumstances and will be bound only for the implementation of those instruments to which they are Contracting Governments or Parties. By virtue of geography and circumstance, some States may have a greater role as a flag State than as a port State or as a coastal State, whilst others may have a greater role as a coastal State or port State than as a flag State.

### **Strategy**

3 In order to meet the objective of this Code, a State should:

- .1 develop an overall strategy to ensure that its international obligations and responsibilities as a flag, port and coastal State are met;
- .2 establish a methodology to monitor and assess that the strategy ensures effective implementation and enforcement of relevant international mandatory instruments; and
- .3 continuously review the strategy to achieve, maintain and improve the overall organizational performance and capability as a flag, port and coastal State.

### **General**

4 Under the general provisions of treaty law and of IMO conventions, States are responsible for promulgating laws and regulations and for taking all other steps which may be necessary to give those instruments full and complete effect so as to ensure safety of life at sea and protection of the marine environment.

5 In taking measures to prevent, reduce and control pollution of the marine environment, States shall act so as not to transfer, directly or indirectly, damage or hazards from one area to another or transform one type of pollution into another.

### **Scope**

6 The Code seeks to address those aspects necessary for a Contracting Government or Party to give full and complete effect to the provisions of the applicable international instruments to which it is a Contracting Government or Party, pertaining to:

- .1 safety of life at sea;
- .2 prevention of pollution from ships;
- .3 standards of training, certification and watchkeeping for seafarers;
- .4 load lines;

- .5 tonnage measurement of ships; and
- .6 regulations for preventing collisions at sea.

7 The following areas are to be considered and addressed in the development of policies, legislation, associated rules and regulations and administrative procedures for the implementation and enforcement of those obligations and responsibilities by the State:

- .1 jurisdiction;
- .2 organization and authority;
- .3 legislation, rules and regulations;
- .4 promulgation of the applicable international mandatory instruments, rules and regulations;
- .5 enforcement arrangements;
- .6 control, survey, inspection, audit, verification, approval and certification functions;
- .7 selection, recognition, authorization, empowerment and monitoring of recognized organizations, as appropriate, and of nominated surveyors;
- .8 investigations required to be reported to the Organization; and
- .9 reporting to the Organization and other Administrations.

#### **Initial actions**

8 When a new or amended instrument of the Organization enters into force for a State, the Government of that State shall be in a position to implement and enforce its provisions through appropriate national legislation and to provide the necessary implementation and enforcement infrastructure. This means that the Government of the State shall have:

- .1 the ability to promulgate laws, which permit effective jurisdiction and control in administrative, technical and social matters over ships flying its flag and, in particular, provide the legal basis for general requirements for registries, the inspection of ships, safety and pollution-prevention laws applying to such ships and the making of associated regulations;
- .2 a legal basis for the enforcement of its national laws and regulations including the associated investigative and penal processes; and
- .3 the availability of sufficient personnel with maritime expertise to assist in the promulgation of the necessary national laws and to discharge all the responsibilities of the State, including reporting as required by the respective conventions.

## **Communication of information**

9 The State shall communicate its strategy, as referred to in paragraph 3, including information on its national legislation to all concerned.

## **Records**

10 Records, as appropriate, shall be established and maintained to provide evidence of conformity to requirements and of the effective operation of the State. Records shall remain legible, readily identifiable and retrievable. A documented procedure shall be established to define the controls needed for the identification, storage, protection, retrieval, retention time and disposition of records.

## **Improvement**

11 States shall continually improve the adequacy of the measures which are taken to give effect to those conventions and protocols which they have accepted. Improvement shall be made through rigorous and effective application and enforcement of national legislation, as appropriate, and monitoring of compliance.

12 The State shall stimulate a culture which provides opportunities for improvement of performance in maritime safety and environmental protection activities, which may include, *inter alia*:

- .1 continual training programmes relating to safety and pollution prevention;
- .2 regional and national drills on safety and pollution prevention, which engage a broad spectrum of maritime related national, regional and international organizations and companies and seafarers; and
- .3 using reward and incentive mechanisms for shipping companies and seafarers, regarding improving safety and pollution prevention.

13 Further, the State shall take action to identify and eliminate the cause of any non-conformities in order to prevent recurrence, including:

- .1 review and analysis of non-conformities;
- .2 implementation of necessary corrective action; and
- .3 review of the corrective action taken.

14 The State shall determine action needed to eliminate the causes of potential non-conformities in order to prevent their occurrence.

## **PART 2 – FLAG STATES**

### **Implementation**

15 In order to effectively discharge their responsibilities and obligations, flag States shall:

- .1 implement policies through the issuance of national legislation and guidance, which will assist in the implementation and enforcement of the requirements of all safety and pollution prevention conventions and protocols to which they are parties; and

- .2 assign responsibilities within their Administrations to update and revise any relevant policies adopted, as necessary.

16 A flag State shall establish resources and processes capable of administering a safety and environmental protection programme, which, as a minimum, shall consist of the following:

- .1 administrative instructions to implement applicable international rules and regulations as well as develop and disseminate any interpretative national regulations that may be needed including certificates issued by a classification society, which is recognized by the flag State in accordance with the provisions of SOLAS regulation XI-1/1, and which certificate is required by the flag State to demonstrate compliance with structural, mechanical, electrical, and/or other requirements of an international convention to which the flag State is a party or a requirement of the flag State's national regulations;
- .2 compliance with the requirements of the applicable international instruments, using an audit and inspection programme, independent of any administrative bodies issuing the required certificates and relevant documentation and/or of any entity which has been delegated authority by the State to issue the required certificates and relevant documentation;
- .3 compliance with the requirements related to international standards of training, certification and watchkeeping of seafarers. This includes, *inter alia*,
  - .1 training, assessment of competence and certification of seafarers;
  - .2 certificates and endorsements that accurately reflect the competencies of the seafarers, using the appropriate terminology as well as terms which are identical to those used in any safe manning document issued to the ship;
  - .3 impartial investigation to be held of any reported failure, whether by act or omission, that may pose a direct threat to safety of life or property at sea or to the marine environment, by the holders of certificates or endorsements issued by the State;
  - .4 that certificates or endorsements issued by the State can be effectively withdrawn, suspended or cancelled when warranted, and when necessary to prevent fraud; and
  - .5 administrative arrangements, including those involving training, assessment and certification activities conducted under the purview of another State, are such that the flag State accepts its responsibility for ensuring the competence of masters, officers and other seafarers serving on ships entitled to fly its flag;
- .4 the conduct of investigations into casualties and adequate and timely handling of cases of ships with identified deficiencies; and



- .5 the development, documentation and provision of guidance concerning those requirements that are to the satisfaction of the Administration, found in the relevant international instruments.

17 A flag State shall ensure that ships entitled to fly its flag are sufficiently and efficiently manned, taking into account relevant and existing measures such as the Principles of Safe Manning adopted by the Organization.

### **Delegation of authority**

18 With regard only to ships entitled to fly its flag a flag State authorizing a recognized organization to act on its behalf, in conducting the surveys, inspections and audits, issuing of certificates and documents, marking of ships and other statutory work required under the conventions of the Organization or under their national legislation, must regulate such authorization(s) in accordance with the applicable requirements of the international mandatory instruments to:

- .1 determine that the recognized organization has adequate resources in terms of technical, managerial and research capabilities to accomplish the tasks being assigned, in accordance with the "Minimum Standards for Recognized Organizations Acting on Behalf of the Administration" set out in the relevant guidelines of the Organization<sup>1</sup>;
- .2 have as its basis a formal written agreement between the Administration and the recognized organization which, as a minimum, includes the elements set out in the relevant guidelines of the Organization<sup>2</sup>, or equivalent legal arrangements, and which may be based on the model agreement for the authorization of recognized organizations acting on behalf of the Administration<sup>3</sup>;
- .3 issue specific instructions detailing actions to be followed in the event that a ship is found unfit to proceed to sea without danger to the ship or persons on board, or is found to present an unreasonable threat of harm to the marine environment;
- .4 provide the recognized organization with all appropriate instruments of national law and interpretations thereof giving effect to the provisions of the conventions and specify, only for application to ships entitled to fly its flag, whether any additional Administration's standards go beyond convention requirements in any respect; and
- .5 require that the recognized organization shall maintain records, which will provide the Administration with data to assist in interpretation of requirements contained in the applicable international instruments.

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<sup>1</sup> Appendix 1 of the Guidelines for the authorization of organizations acting on behalf of the Administration (resolution A.739(18)).

<sup>2</sup> Appendix 2 of the Guidelines for the authorization of organizations acting on behalf of the Administration (resolution A.739(18)).

<sup>3</sup> MSC/Circ.710-MEPC/Circ.307.

19 No flag State shall mandate its recognized organizations to apply to ships, other than those entitled to fly its flag, any requirement pertaining to their classification rules, requirements, procedures or performance of other statutory certification processes, beyond convention requirements and the mandatory instruments of the Organization.

20 The flag State shall establish or participate in an oversight programme with adequate resources for monitoring of, and communication with, its recognized organization(s) in order to ensure that its international obligations are fully met, by:

- .1 exercising its authority to conduct supplementary surveys to ensure that ships entitled to fly its flag in fact comply with the requirements of the applicable international instruments;
- .2 conducting supplementary surveys as it deems necessary to ensure that ships entitled to fly its flag comply with national requirements, which supplement the international mandatory requirements; and
- .3 providing staff who have a good knowledge of the rules and regulations of the flag State and the recognized organizations and who are available to carry out effective oversight of the recognized organizations.

21 A flag State nominating surveyor(s) for the purpose of carrying out surveys, audits and inspections on its behalf shall regulate such nominations, as appropriate, in accordance with the guidance provided in paragraph 18, in particular subparagraphs .3 and .4.

## Enforcement

22 A flag State shall take all necessary measures to secure observance of international rules and standards by ships entitled to fly its flag and by entities and persons under their jurisdiction so as to ensure compliance with their international obligations. Such measures shall include, *inter alia*:

- .1 prohibiting ships entitled to fly their flag from sailing until such ships can proceed to sea in compliance with the requirements of international rules and standards;
- .2 the periodic inspection of ships entitled to fly its flag to verify that the actual condition of the ship and its crew is in conformity with the certificates it carries;
- .3 the surveyor ensuring, during the periodic inspection referred to in subparagraph .2, that seafarers assigned to the ships are familiar with:
  - .1 their specific duties; and
  - .2 ship arrangements, installations, equipments and procedures;
- .4 ensuring that the ship's complement, as a whole, can effectively coordinate their activities in an emergency situation and in performing functions vital to safety or to the prevention or mitigation of pollution;
- .5 providing, in national laws and regulations, for penalties of adequate severity to discourage violation of international rules and standards by ships entitled to fly its flag;

- .6 instituting proceedings – after an investigation has been conducted – against ships entitled to fly its flag, which have violated international rules and standards, irrespective of where the violation has occurred;
- .7 providing, in national laws and regulations, for penalties of adequate severity to discourage violations of international rules and standards by individuals issued with certificates or endorsements under their authority; and
- .8 instituting proceedings – after an investigation has been conducted – against individuals holding certificates or endorsements who have violated international rules and standards, irrespective of where the violation has occurred.

23 A flag State shall develop and implement a control and monitoring programme, as appropriate, in order to:

- .1 provide for prompt and thorough casualty investigations, with reporting to the Organization as appropriate;
- .2 provide for the collection of statistical data, so that trend analyses can be conducted to identify problem areas; and
- .3 provide for a timely response to deficiencies and alleged pollution incidents reported by port or coastal States.

24 Furthermore, the flag State shall:

- .1 ensure compliance with the applicable international instruments through national legislation;
- .2 provide an appropriate number of qualified personnel to implement and enforce the national legislation referred to in subparagraph 15.1, including personnel for performing investigations and surveys;
- .3 provide a sufficient number of qualified flag State personnel to investigate incidents where ships entitled to fly its flag have been detained by port States;
- .4 provide a sufficient number of qualified flag State personnel to investigate incidents where the validity of a certificate or endorsement or competence of individuals holding certificates or endorsements issued under its authority are questioned by port States; and
- .5 ensure the training and oversight of the activities of flag State surveyors and investigators.

25 When a State is informed that a ship entitled to fly its flag has been detained by a port State, the flag State shall oversee that appropriate corrective measures to bring the ship in question into immediate compliance with the applicable international instruments are taken.

26 A flag State, or a recognized organization acting on its behalf, shall only issue or endorse an international certificate to a ship after it has determined that the ship meets all applicable requirements.

27 A flag State shall only issue an international certificate of competency or endorsement to a person after it has determined that the person meets all applicable requirements.

### **Flag State surveyors**

28 The flag State shall define and document the responsibilities, authority and interrelation of all personnel who manage, perform and verify work relating to and affecting safety and pollution prevention.

29 Personnel responsible for, or performing, surveys, inspections and audits on ships and companies covered by the relevant international mandatory instruments shall have as a minimum the following:

- .1 appropriate qualifications from a marine or nautical institution and relevant seagoing experience as a certificated ship officer holding or having held a valid management level certificate of competency and have maintained their technical knowledge of ships and their operation since gaining their certificate of competency; or
- .2 a degree or equivalent from a tertiary institution within a relevant field of engineering or science recognized by the State; or
- .3 accreditation as a surveyor through a formalized training programme that leads to the same standard of surveyor's experience and competency as that required in paragraphs 29.1, 29.2 and 32.

30 Personnel qualified under paragraph 29.1 shall have served for a period of not less than three years at sea as an officer in the deck or engine department.

31 Personnel qualified under paragraph 29.2 shall have worked in a relevant capacity for at least three years.

32 In addition, such personnel shall have appropriate practical and theoretical knowledge of ships, their operation and the provisions of the relevant national and international instruments necessary to perform their duties as flag State surveyors obtained through documented training programmes.

33 Other personnel assisting in the performance of such work shall have education, training and supervision commensurate with the tasks they are authorized to perform.

34 Previous relevant experience in the field of expertise should be considered an advantage; in case of no previous experience, the Administration shall provide appropriate field training.

35 The flag State shall implement a documented system for qualification of personnel and continuous updating of their knowledge as appropriate to the tasks they are authorized to undertake.

- 36 Depending on the function(s) to be performed, the qualifications shall encompass:
- .1 knowledge of applicable, international and national, rules and regulations for ships, their companies, their crew, their cargo and their operation;
  - .2 knowledge of the procedures to be applied in survey, certification, control, investigative and oversight functions;
  - .3 understanding of the goals and objectives of the international and national instruments dealing with maritime safety and protection of the marine environment, and of related programmes;
  - .4 understanding of the processes both on board and ashore, internal as well as external;
  - .5 possession of professional competency necessary to perform the given tasks effectively and efficiently;
  - .6 full safety awareness in all circumstances, also for one's own safety; and
  - .7 training or experience in the various tasks to be performed and, preferably, also in the functions to be assessed.
- 37 The flag State shall issue an identification document for the surveyor to carry when performing his/her tasks.

#### **Flag State investigations**

38 Marine safety investigations should be conducted by qualified, impartial and objective investigators, knowledgeable in matters relating to the casualty. Subject to any agreement on which the State or States will be the marine safety investigating State(s), the flag State shall provide qualified investigators for this purpose, irrespective of the location of the casualty or incident.

39 The flag State should ensure that individual investigators have working knowledge and practical experience in those subject areas pertaining to their normal duties. Additionally, to assist individual investigators in performing duties outside their normal assignments, the flag State should ensure ready access to expertise in the following areas, as necessary:

- .1 navigation and the Collision Regulations;
- .2 flag State regulations on certificates of competency;
- .3 causes of marine pollution;
- .4 interviewing techniques;
- .5 evidence gathering; and
- .6 evaluation of the effects of the human element.

40 Any accidents involving personal injury necessitating absence from duty of three days or more and any deaths resulting from occupational accidents and casualties to ships of the flag State should be investigated, and the results of such investigations made public.

41 Ship casualties are to be investigated and reported in accordance with the relevant international instruments, taking into account the Casualty Investigation Code, as may be amended, and guidelines developed by the Organization<sup>4</sup>. The report on the investigation shall be forwarded to the Organization together with the flag State's observations, in accordance with the guidelines referred to above.

### **Evaluation and review**

42 A flag State shall, on a periodic basis, evaluate its performance with respect to the implementation of administrative processes, procedures and resources necessary to meet its obligations as required by the international instruments to which it is a party.

43 Measures to evaluate the performance of flag States shall include, *inter alia*, port State control detention rates, flag State inspection results, casualty statistics, communication and information processes, annual loss statistics (excluding constructive total losses (CTLs)), and other performance indicators as may be appropriate, to determine whether staffing, resources and administrative procedures are adequate to meet its flag State obligations.

44 Areas to be regularly reviewed may include, *inter alia*:

- .1 fleet loss and accident ratios to identify trends over selected time periods;
- .2 the number of verified cases of detained ships in relation to the size of the fleet;
- .3 the number of verified cases of incompetence or wrongdoing by individuals holding certificates or endorsements issued under its authority;
- .4 responses to port State deficiency reports or interventions;
- .5 investigations into very serious and serious casualties and lessons learned from them;
- .6 technical and other resources committed;
- .7 results of inspections, surveys and controls of the ships in the fleet;
- .8 investigation of occupational accidents;
- .9 the number of incidents and violations that occur under the applicable international maritime pollution prevention regulations; and
- .10 the number of suspensions or withdrawals of certificates, endorsements, approvals, etc.

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<sup>4</sup> Refer to the Code for the Investigation of Marine Casualties and Incidents, adopted by the Organization by resolution A.849(20), as amended by resolution A.884(21), and the mandatory Code of the International Standards and Recommended Practices for a Safety Investigation into a Marine Casualty or Marine Incident (Casualty Investigation Code) adopted by the Organization by resolution MSC.255(84).

## **PART 3 – COASTAL STATES<sup>5</sup>**

### **Implementation**

45 Coastal States have certain rights and obligations under various international instruments. When exercising their rights under those instruments coastal States incur additional obligations.

46 In order to effectively meet its obligations, a coastal State shall:

- .1 implement policies through the issuance of national legislation and guidance, which will assist in the implementation and enforcement of the requirements of all safety and pollution prevention conventions and protocols to which it is a party; and
- .2 assign responsibilities to update and revise any relevant policies adopted, as necessary.

47 A coastal State shall ensure that its legislation, guidance and procedures are established for the consistent implementation and verification of its rights, obligations and responsibilities contained in the relevant international instruments to which it is a party.

48 Those rights, obligations and responsibilities may include, *inter alia*:

- .1 radiocommunication services;
- .2 meteorological services and warnings;
- .3 search and rescue services;
- .4 hydrographic services;
- .5 ship routeing;
- .6 ship reporting systems;
- .7 vessel traffic services; and
- .8 aids to navigation.

### **Enforcement**

49 A coastal State shall take all necessary measures to ensure their observance of international rules when exercising their rights and fulfilling their obligations.

50 A coastal State shall consider, develop and implement a control and monitoring programme, as appropriate, in order to:

- .1 provide for the allocation of statistical data so that trend analyses can be conducted to identify problem areas;

---

<sup>5</sup> The requirements contained in this section apply to the extent that ships subject to IMO mandatory instruments can access the ports of the Contracting Government.

- .2 establish mechanisms for timely response to pollution incidents in its waters; and
- .3 co-operate with flag States and/or port States, as appropriate, in investigations of maritime casualties.

### **Evaluation and review**

51 A coastal State shall periodically evaluate its performance in respect of exercising its rights and meeting its obligations under the applicable international instruments.

## **PART 4 – PORT STATES<sup>6</sup>**

### **Implementation**

52 Port States have certain rights and obligations under various international instruments. When exercising their rights under those instruments, port States incur additional obligations.

53 Port States can play an integral role in the achievement of maritime safety and environmental protection, including pollution prevention. The role and responsibilities of the port State with respect to maritime safety and environmental protection is derived from a combination of international treaties, conventions, national laws, as well as in some instances, bilateral and multilateral agreements.

54 In order to effectively meet its obligations, a port State shall:

- .1 implement policies through the issuance of national legislation and guidance, which will assist in the implementation and enforcement of the requirements of all safety and pollution prevention conventions and protocols to which it is a party; and
- .2 assign responsibilities to update and revise any relevant policies adopted, as necessary.

55 A port State shall ensure that its legislation, guidance and procedures are established for the consistent implementation and verification of its rights, obligations and responsibilities contained in the relevant international instruments to which it is a party.

56 Those rights, obligations and responsibilities include, *inter alia*:

- .1 provision of appropriate reception facilities or capability to accept all waste streams regulated under the instruments of the Organization;
- .2 port State control<sup>7</sup>; and
- .3 keeping a register of fuel oil suppliers.

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<sup>6</sup> The requirements contained in this section apply to the extent that ships subject to IMO mandatory instruments can access the ports of the Contracting Government.

<sup>7</sup> Refer to the Procedures for Port State Control (resolution A...(27)).



## **Enforcement**

57 Port States shall take all necessary measures to ensure their observance of international rules when exercising their rights and fulfilling their obligations.

58 Several international maritime instruments on safety and maritime pollution prevention contain specific provisions that permit port State control.

59 Also, a number of those instruments obligate port States to treat non-parties to those conventions no more favourably than those that are parties. This means that port States are obliged to impose the conditions of those instruments on parties, as well as on non-parties.

60 When exercising its right to carry out port State control, a port State shall establish processes to administer a port State control programme consistent with the relevant resolution adopted by the Organization<sup>8</sup>.

61 Port State control shall be carried out only by authorized and qualified port State control officers in accordance with the relevant procedures adopted by the Organization\*.

62 Port State control officers and persons assisting them shall be free from any commercial, financial, and other pressures and have no commercial interest, either in the port of inspection or the ships inspected, in ship repair facilities or any support services in the port or elsewhere nor shall the port State control officers be employed by or undertake work on behalf of recognized organizations or classification societies. Further procedures shall be implemented to ensure that persons or organizations external to the port State cannot influence the results of port State inspection and control carried out.

## **Evaluation and review**

63 A port State shall periodically evaluate its performance in respect of exercising its rights and meeting its obligations under the applicable instruments of the Organization.

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<sup>8</sup> Refer to the Procedures for Port State Control (resolution A...(27)).